VOLUME I: FOUNDATIONS

The Quantum Theory of Fields

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ABSTRACT: Weinberg's QFT, typeset in nicer notation. Typesetting done entirely by coding agents, namely GPT 5.6.

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Preface to Volume I

Why another book on quantum field theory? Today the student of quantum field theory can choose from among a score of excellent books, several of them quite up-to-date. Another book will be worth while only if it offers something new in content or perspective.

wave function of a multi-electron system is antisymmetric in the coordinates, orbital and spin, of all the electrons. This principle was incorporated into statistical mechanics by Enrico Fermi [26] and Dirac [27], and for this reason particles obeying the exclusion principle are generally called 'fermions,' just as particles like photons for which the wave function is symmetric and which obey the statistics of Bose and Einstein are called 'bosons.' The exclusion principle has played a fundamental role in the theory of metals, white dwarf and neutron stars, etc., as well as in chemistry and atomic physics, but a discussion of these matters would take us too far afield here.

Dirac's proposal was that the positive energy electrons cannot fall down into negative energy states because 'all the states of negative energy are occupied except perhaps a few of small velocity.' The few vacant states, or 'holes,' in the sea of negative energy electrons behave like particles with opposite quantum numbers: positive energy and positive charge. The only particle with positive charge that was known at that time was the proton, and as Dirac later recalled [27a], 'the whole climate of opinion at that time was against new particles' so Dirac identified his holes as protons; in fact, the title of his 1930 article [22] was 'A Theory of Electrons and Protons.'

The hole theory faced a number of immediate difficulties. One obvious problem was raised by the infinite charge density of the ubiquitous negative-energy electrons: where is their electric field? Dirac proposed to reinterpret the charge density appearing in Maxwell's equations as 'the departure from the normal state of electrification of the world.' Another problem has to do with the huge dissimilarity between the observed masses and interactions of the electrons and protons. Dirac hoped that Coulomb interactions between electrons would somehow account for these differences but Hermann Weyl [28] showed that the hole theory was in fact entirely symmetric between negative and positive charge. Finally, Dirac [22] predicted the existence of an electron–proton annihilation process in which a positive-energy electron meets a hole in the sea of negative-energy electrons and falls down into the unoccupied level, emitting a pair of gamma ray photons. By itself this would not have created difficulties for the hole theory; it was even hoped by some that this would provide an explanation, then lacking, of the energy source of the stars. However, it was soon pointed out [29] by Julius Robert Oppenheimer and Igor Tamm that electron–proton annihilation in atoms would take place at much too fast a rate to be consistent with the observed stability of ordinary matter. For these reasons, by 1931 Dirac had changed his mind, and decided that the holes would have to appear not as protons but as a new sort of positively charged particle, of the same mass as the electron [29a].

The second and third of these problems were eliminated by the discovery of the positron by Carl D. Anderson [30], who apparently did not know of this prediction by Dirac. On August 2, 1932, a peculiar cosmic ray track was observed in a Wilson cloud chamber subjected to a 15 kG magnetic field. The track was observed to curve in a direction that would be expected for a positively charged particle, and yet its range was at least ten times greater than the expected range of a proton! Both the range and the specific ionization of the track were consistent with the hypothesis that this was a new particle which differs from the electron only in the sign of its charge, as would be expected for one of Dirac's holes. (This discovery had been made earlier by P.M.S. Blackett, but not immediately published

by him. Anderson quotes press reports of evidence for light positive particles in cosmic ray tracks, obtained by Blackett and Giuseppe Occhialini.) Thus it appeared that Dirac was wrong only in his original identification of the hole with the proton.

The discovery of the more-or-less predicted positron, together with the earlier successes of the Dirac equation in accounting for the magnetic moment of the electron and the fine structure of hydrogen, gave Dirac's theory a prestige that it has held for over six decades. However, although there seems little doubt that Dirac's theory will survive in some form in any future physical theory, there are serious reasons for being dissatisfied with its original rationale:

(i) Dirac's analysis of the problem of negative probabilities in Schrödinger's relativistic wave equation would seem to rule out the existence of any particle of zero spin. Yet even in the 1920s particles of zero spin were known—for instance, the hydrogen atom in its ground state, and the helium nucleus. Of course, it could be argued that hydrogen atoms and alpha particles are not elementary, and therefore do not need to be described by a relativistic wave equation, but it was not (and still is not) clear how the idea of elementarity is incorporated in the formalism of relativistic quantum mechanics. Today we know of a large number of spin zero particles—π mesons, K mesons, and so on—that are no less elementary than the proton and neutron. We also know of spin one particles—the $W^\pm$ and Z^0—which seem as elementary as the electron or any other particle. Further, apart from effects of the strong interactions, we would today calculate the fine structure of 'mesonic atoms,' consisting of a spinless negative π or K meson bound to an atomic nucleus, from the stationary solutions of the relativistic Klein–Gordon–Schrödinger equation! Thus, it is difficult to agree that there is anything fundamentally wrong with the relativistic equation for zero spin that *forced* the development of the Dirac equation—the problem simply is that the electron happens to have spin $\hbar/2$, not zero.

(ii) As far as we now know, for *every* kind of particle there is an 'antiparticle' with the same mass and opposite charge. (Some purely neutral particles, such as the photon, are their own antiparticles.) But how can we interpret the antiparticles of charged *bosons*, such as the $\pi^\pm$ mesons or $W^\pm$ particles, as holes in a sea of negative energy states? For particles quantized according to the rules of Bose–Einstein statistics, there is no exclusion principle, and hence nothing to keep positive-energy particles from falling down into the negative-energy states, occupied or not. And if the hole theory does not work for bosonic antiparticles, why should we believe it for fermions? I asked Dirac in 1972 how he then felt about this point; he told me that he did not regard bosons like the pion or $W^\pm$ as 'important.' In a lecture [27a] a few years later, Dirac referred to the fact that for bosons 'we no longer have the picture of a vacuum with negative energy states filled up', and remarked that in this case 'the whole theory becomes more complicated.' The next section will show how the development of quantum field theory made the interpretation of antiparticles as holes unnecessary, even though unfortunately it lingers on in many textbooks. To quote Julian Schwinger [30a], 'The picture of an infinite sea of negative energy electrons is now best regarded as a historical curiosity, and forgotten.'

(iii) One of the great successes of the Dirac theory was its correct prediction of the mag-

netic moment of the electron. This was particularly striking, as the magnetic moment (1.1.8) is twice as large as would be expected for the orbital motion of a charged point particle with angular momentum $\hbar/2$; this factor of 2 had remained mysterious until Dirac's theory. However, there is really nothing in Dirac's line of argument that leads unequivocally to this particular value for the magnetic moment. At the point where we brought electric and magnetic fields into the wave equation (1.1.23), we could just as well have added a 'Pauli term' [31]

$$\kappa e\beta[\gamma^\mu,\gamma^\nu]\psi F_{\mu\nu} \tag{1.1.32}$$

with arbitrary coefficient κ. (Here $F_{\mu\nu}$ is the usual electromagnetic field strength tensor, with $F^{12}=B_3$, $F^{01}=E_1$, etc.) This term could be obtained by first adding a term to the free-field equations proportional to $[\gamma^\mu,\gamma^\nu](\partial^2/\partial x^\mu\partial x^\nu)\psi$, which of course equals zero, and then making the substitutions (1.1.22) as before. A more modern approach would be simply to remark that the term (1.1.32) is consistent with all accepted invariance principles, including Lorentz invariance and gauge invariance, and so there is no reason why such a term should not be included in the field equations. (See Section 12.3.) This term would give an additional contribution proportional to κ to the magnetic moment of the electron, so apart from the possible demand for a purely formal simplicity, there was no reason to expect any particular value for the magnetic moment of the electron in Dirac's theory.

As we shall see in this book, these problems were all eventually to be solved (or at least clarified) through the development of quantum field theory.

1.2 The Birth of Quantum Field Theory

The photon is the only particle that was known as a field before it was detected as a particle. Thus it is natural that the formalism of quantum field theory should have been developed in the first instance in connection with radiation and only later applied to other particles and fields.

In 1926, in one of the central papers on matrix mechanics, Born, Heisenberg, and Jordan [32] applied their new methods to the free radiation field. For simplicity, they ignored the polarization of electromagnetic waves and worked in one space dimension, with coordinate x running from 0 to L; the radiation field $u(x,t)$ if constrained to vanish at these endpoints thus has the same behavior as the displacement of a string with ends fixed at $x=0$ and $x=L$. By analogy with either the case of a string or the full electromagnetic field, the Hamiltonian was taken to have the form

$$H=\frac{1}{2}\int_0^L\left\{\left(\frac{\partial u}{\partial t}\right)^2+c^2\left(\frac{\partial u}{\partial x}\right)^2\right\}dx. \tag{1.2.1}$$

In order to reduce this expression to a sum of squares, the field u was expressed as a sum of Fourier components with $u=0$ at both $x=0$ and $x=L$:

$$u(x,t)=\sum_{k=1}^{\infty}q_k(t)\sin\left(\frac{\omega_k x}{c}\right), \tag{1.2.2}$$

$$\omega_k=k\pi c/L, \tag{1.2.3}$$

so that

$$H=\frac{L}{4}\sum_{k=1}^{\infty}\left\{\dot q_k^2(t)+\omega_k^2 q_k^2(t)\right\}. \tag{1.2.4}$$

Thus the string or field behaves like sum of independent harmonic oscillators with angular frequencies ω_k, as had been anticipated 20 years earlier by Paul Ehrenfest [32a].

In particular, the 'momentum' $p_k(t)$ canonically conjugate to $q_k(t)$ is determined, as in particle mechanics, by the condition that if H is expressed as a function of the ps and qs, then

$$\dot q_k(t)=\frac{\partial}{\partial p_k(t)}H(p(t),q(t)).$$

This yields a 'momentum'

$$p_k(t)=\frac{L}{2}\dot q_k(t), \tag{1.2.5}$$

so the canonical commutation relations may be written

$$[\dot q_k(t),q_j(t)]=\frac{2}{L}[p_k(t),q_j(t)]=\frac{-2i\hbar}{L}\delta_{kj}, \tag{1.2.6}$$

$$[q_k(t),q_j(t)]=0. \tag{1.2.7}$$

Also, the time-dependence of $q_k(t)$ is governed by the Hamiltonian equation of motion

$$\ddot q_k(t)=\frac{2}{L}\dot p_k(t)=-\frac{2}{L}\frac{\partial H}{\partial q_k(t)}=-\omega_k^2 q_k(t). \tag{1.2.8}$$

The form of the matrices defined by Eqs. (1.2.6)–(1.2.8) was already known to Born, Heisenberg, and Jordan through previous work on the harmonic oscillator. The q-matrix is given by

$$q_k(t)=\sqrt{\frac{\hbar}{L\omega_k}}\left[a_k\exp(-i\omega_k t)+a_k^\dagger\exp(+i\omega_k t)\right] \tag{1.2.9}$$

with a_k a time-independent matrix and $a_k^\dagger$ its Hermitian adjoint, satisfying the commutation relations

$$[a_k,a_j^\dagger]=\delta_{kj}, \tag{1.2.10}$$

$$[a_k,a_j]=0. \tag{1.2.11}$$

The rows and columns of these matrices are labelled with a set of positive integers $n_1,n_2,\ldots$, one for each normal mode. The matrix elements are

$$(a_k)_{n'_1,n'_2,\ldots;n_1,n_2,\ldots}=\sqrt{n_k}\,\delta_{n'_k,n_k-1}\prod_{j\neq k}\delta_{n'_j,n_j}, \tag{1.2.12}$$

$$(a_k^\dagger)_{n'_1,n'_2,\ldots;n_1,n_2,\ldots}=\sqrt{n_k+1}\,\delta_{n'_k,n_k+1}\prod_{j\neq k}\delta_{n'_j,n_j}. \tag{1.2.13}$$

For a single normal mode, these matrices may be written explicitly as

$$a=\begin{bmatrix}0&\sqrt1&0&0&\cdots\\0&0&\sqrt2&0&\cdots\\0&0&0&\sqrt3&\cdots\\0&0&0&0&\cdots\\\vdots&\vdots&\vdots&\vdots&\ddots\end{bmatrix},\qquad a^\dagger=\begin{bmatrix}0&0&0&0&\cdots\\\sqrt1&0&0&0&\cdots\\0&\sqrt2&0&0&\cdots\\0&0&\sqrt3&0&\cdots\\\vdots&\vdots&\vdots&\vdots&\ddots\end{bmatrix}.$$

It is straightforward to check that (1.2.12) and (1.2.13) do satisfy the commutation relations (1.2.10) and (1.2.11).

The physical interpretation of a column vector with integer components $n_1,n_2,\ldots$ is that it represents a state with n_k quanta in each normal mode k. The matrix a_k or $a_k^\dagger$ acting on such a column vector will respectively lower or raise n_k by one unit, leaving all n_ℓ with $\ell\neq k$ unchanged; they may therefore be interpreted as operators which annihilate or create one quantum in the kth normal mode. In particular, the vector with all n_k equal to zero represents the vacuum; it is annihilated by any a_k.

This interpretation is further borne out by inspection of the Hamiltonian. Using (1.2.9) and (1.2.10) in (1.2.4) gives

$$H=\sum_k\hbar\omega_k\left(a_k^\dagger a_k+\frac{1}{2}\right). \tag{1.2.14}$$

The Hamiltonian is then diagonal in the n-representation

$$(H)_{n'_1,n'_2,\ldots;n_1,n_2,\ldots}=\sum_k\hbar\omega_k\left(n_k+\frac{1}{2}\right)\prod_j\delta_{n'_j,n_j}. \tag{1.2.15}$$

We see that the energy of the state is just the sum of energies $\hbar\omega_k$ for each quantum present in the state, plus an infinite zero-point energy $E_0=\frac{1}{2}\sum_k\hbar\omega_k$. Applied to the radiation field, this formalism justified the Bose method of counting radiation states according to the numbers n_k of quanta in each normal mode.

Born, Heisenberg, and Jordan used this formalism to derive an expression for the r.m.s. energy fluctuations in black-body radiation. (For this purpose they actually only used the commutation relations (1.2.6)–(1.2.7).) However, this approach was soon applied to a more urgent problem, the calculation of the rates for spontaneous emission of radiation.

In order to appreciate the difficulties here, it is necessary to go back in time a bit. In one of the first papers on matrix mechanics, Born and Jordan [33] had assumed in effect that an atom, in dropping from a state β to a lower state α, would emit radiation just like a classical charged oscillator with displacement

$$\mathbf r(t)=\mathbf r_{\beta\alpha}\exp(-2\pi i\nu t)+\mathbf r_{\beta\alpha}^*\exp(2\pi i\nu t), \tag{1.2.16}$$

where

$$h\nu=E_\beta-E_\alpha \tag{1.2.17}$$

and $\mathbf r_{\beta\alpha}$ is the β,α element of the matrix associated with the electron position. The energy E of such an oscillator is

$$E=\frac{1}{2}m\left\{\dot{\mathbf r}^2+(2\pi\nu)^2\mathbf r^2\right\}=8\pi^2m\nu^2|\mathbf r_{\beta\alpha}|^2. \tag{1.2.18}$$

A straightforward classical calculation then gives the radiated power, and dividing by the energy $h\nu$ per photon gives the rate of photon emission

$$A(\beta\to\alpha)=\frac{16\pi^3e^2\nu^3}{3hc^3}|\mathbf r_{\beta\alpha}|^2. \tag{1.2.19}$$

However, it was not at all clear why the formulas for emission of radiation by a classical dipole should be taken over in this manner in dealing with spontaneous emission.

A little later a more convincing though even less direct derivation was given by Dirac [34]. By considering the behavior of quantized atomic states in an oscillating classical electromagnetic field with energy density per frequency interval u at frequency (1.2.17), he was able to derive formulas for the rates $uB(\alpha\to\beta)$ and $uB(\beta\to\alpha)$ for absorption or induced emission:

$$B(\alpha\to\beta)=B(\beta\to\alpha)\simeq\frac{2\pi^2e^2}{3\hbar^2}|\mathbf r_{\beta\alpha}|^2. \tag{1.2.20}$$

(Note that the expression on the right is symmetric between states α and β, because $\mathbf r_{\alpha\beta}$ is just $\mathbf r^*_{\beta\alpha}$.) Einstein [34a] had already shown in 1917 that the possibility of thermal equilibrium between atoms and black-body radiation imposes a relation between the rate $A(\beta\to\alpha)$ of spontaneous emission and the rates uB for induced emission or absorption:

$$A(\beta\to\alpha)=\left(\frac{8\pi h\nu^3}{c^3}\right)B(\beta\to\alpha). \tag{1.2.21}$$

Using (1.2.20) in this relation immediately yields the Born–Jordan result (1.2.19) for the rate of spontaneous emission. Nevertheless, it still seemed unsatisfactory that thermodynamic arguments should be needed to derive formulas for processes involving a single atom.

Finally, in 1927 Dirac [35] was able to give a thoroughly quantum mechanical treatment of spontaneous emission. The vector potential $\mathbf A(\mathbf x,t)$ was expanded in normal modes, as in Eq. (1.2.2), and the coefficients were shown to satisfy commutation relations like (1.2.6). In consequence, each state of the free radiation field was specified by a set of integers n_k, one for each normal mode, and the matrix elements of the electromagnetic interaction $e\dot{\mathbf r}\cdot\mathbf A$ took the form of a sum over normal modes, with matrix coefficients proportional to the matrices a_k and $a_k^\dagger$ defined in Eqs. (1.2.10)–(1.2.13). The crucial result here is the factor $\sqrt{n_k+1}$ in Eq. (1.2.13); the probability for a transition in which the number of photons in a normal mode k rises from n_k to n_k+1 is proportional to the square of this factor, or n_k+1. But in a radiation field with n_k photons in a normal mode k, the energy density u per frequency interval is

$$u(\nu_k)=\left(\frac{8\pi\nu_k^2}{c^3}\right)n_k\cdot h\nu_k,$$

so the rate for emission of radiation in normal mode k is proportional to

$$n_k+1=\frac{c^3u(\nu_k)}{8\pi h\nu_k^3}+1.$$

The first term is interpreted as the contribution of induced emission, and the second term as the contribution of spontaneous emission. Hence, without any appeal to thermodynamics, Dirac could conclude that the ratio of the rates uB for induced emission and A for spontaneous emission is given by the Einstein relation, Eq. (1.2.21). Using his earlier result (1.2.20) for B, Dirac was thus able to rederive the Born–Jordan formula [33] (1.2.19) for spontaneous emission rate A. A little later, similar methods were used by Dirac to give a quantum mechanical treatment of the scattering of radiation and the lifetime of excited atomic states [36], and by Victor Weisskopf and Eugene Wigner to make a detailed study of spectral line shapes [36a]. Dirac in his work was separating the electromagnetic potential into a radiation field $\mathbf A$ and a static Coulomb potential A^0, in a manner which did not preserve the manifest Lorentz and gauge invariance of classical electrodynamics. These matters were put on a firmer foundation a little later by Enrico Fermi [36b]. Many physicists in the 1930s learned their quantum electrodynamics from Fermi's 1932 review.

The use of canonical commutation relations for q and p or a and $a^\dagger$ also raised a question as to the Lorentz invariance of the quantized theory. Jordan and Pauli [37] in 1928 were able to show that the commutators of fields at different spacetime points were in fact Lorentz-invariant. (These commutators are calculated in Chapter 5.) Somewhat later, Bohr and Leon Rosenfeld [38] used a number of ingenious thought experiments to show that these commutation relations express limitations on our ability to measure fields at spacetime points separated by time-like intervals.

It was not long after the successful quantization of the electromagnetic field that these techniques were applied to other fields. At first this was regarded as a 'second quantization'; the fields to be quantized were the wave functions used in one-particle quantum mechanics, such as the Dirac wave function of the electron. The first step in this direction seems to have been taken in 1927 by Jordan [39]. In 1928 an essential element was supplied by Jordan and Wigner [40]. They recognized that the Pauli exclusion principle prevents the occupation number n_k of electrons in any normal mode k (counting spin as well as position variables) from taking any values other than 0 or 1. The electron field therefore cannot be expanded as a superposition of operators satisfying the commutation relations (1.2.10), (1.2.11), because these relations require n_k to take all integer values from 0 to ∞. Instead, they proposed that the electron field should be expanded in a sum of operators $a_k,a_k^\dagger$ satisfying the *anticommutation* relations

$$a_ka_j^\dagger+a_j^\dagger a_k=\delta_{jk}, \tag{1.2.22}$$

$$a_ka_j+a_ja_k=0. \tag{1.2.23}$$

The relations can be satisfied by matrices labelled by a set of integers $n_1,n_2,\ldots$, one for each normal mode, each integer taking just the values zero and one:

$$(a_k)_{n'_1,n'_2,\ldots;n_1,n_2,\ldots}=\begin{cases}1 & n'_k=0,\ n_k=1,\ n'_j=n_j\text{ for }j\neq k,\\0 & \text{otherwise},\end{cases} \tag{1.2.24}$$

$$(a_k^\dagger)_{n'_1,n'_2,\ldots;n_1,n_2,\ldots}=\begin{cases}1 & n'_k=1,\ n_k=0,\ n'_j=n_j\text{ for }j\neq k,\\0 & \text{otherwise}.\end{cases} \tag{1.2.25}$$

For instance, for a single normal mode we have just two rows and two columns, corresponding to the values unity and zero of n' and n; the a and $a^\dagger$ matrices take the form

$$a=\begin{bmatrix}0&0\\1&0\end{bmatrix},\qquad a^\dagger=\begin{bmatrix}0&1\\0&0\end{bmatrix}.$$

The reader may check that (1.2.24) and (1.2.25) do satisfy the anticommutation relations (1.2.22) and (1.2.23).

The interpretation of a column vector characterized by integers $n_1,n_2,\ldots$ is that it represents a state with n_k quanta in each normal mode k, just as for bosons. The difference is, of course, that since each n_k takes only the values 0 and 1, there can be at most one quantum in each normal mode, as required by the Pauli exclusion principle. Again, a_k destroys a quantum in normal mode k if there is one there already, and otherwise gives zero; also, $a_k^\dagger$ creates a quantum in normal mode k unless there is one there already, in which case it gives zero. Much later it was shown by Fierz and Pauli [40a] that the choice between commutation and anticommutation relations is dictated solely by the particle's spin: commutators must be used for particles with integer spin like the photon, and anticommutators for particles with half-integer spin like the electron. (This will be shown in a different way in Chapter 5.)

The theory of general quantum fields was first laid out in 1929, in a pair of comprehensive articles by Heisenberg and Pauli [41]. The starting point of their work was the application of the canonical formalism to the fields themselves, rather than to the coefficients of the normal modes appearing in the fields. Heisenberg and Pauli took the Lagrangian L as the space-integral of a local function of fields and spacetime derivatives of fields; the field equations were then determined from the principle that the action $\int L\,dt$ should be stationary when the fields are varied; and the commutation relations were determined from the assumption that the variational derivative of the Lagrangian with respect to any field's time-derivative behaves like a 'momentum' conjugate to that field (except that commutation relations become anticommutation relations for fermion fields). They also went on to apply this general formalism to the electromagnetic and Dirac fields, and explored the various invariance and conservation laws, including the conservation of charge, momentum, and energy, and Lorentz and gauge invariance.

The Heisenberg–Pauli formalism is essentially the same as that described in our Chapter 7, and so for the present we can limit ourselves to a single example which will turn out to be useful later in this section. For a free complex scalar field $\phi(x)$ the Lagrangian is taken as

$$L=\int d^3x\left[\dot\phi^\dagger\dot\phi-c^2(\nabla\phi)^\dagger\cdot(\nabla\phi)-\left(\frac{mc^2}{\hbar}\right)^2\phi^\dagger\phi\right]. \tag{1.2.26}$$

If we subject $\phi(x)$ to an infinitesimal variation $\delta\phi(x)$, the Lagrangian is changed by the

amount

$$\delta L=\int d^3x\left[\dot\phi^\dagger\delta\dot\phi+\dot\phi\,\delta\dot\phi^\dagger-c^2\nabla\phi^\dagger\cdot\nabla\delta\phi-c^2\nabla\phi\cdot\nabla\delta\phi^\dagger-\left(\frac{mc^2}{\hbar}\right)^2\phi^\dagger\delta\phi-\left(\frac{mc^2}{\hbar}\right)^2\phi\,\delta\phi^\dagger\right]. \tag{1.2.27}$$

It is assumed in using the principle of stationary action that the variation in the fields should vanish on the boundaries of the spacetime region of integration. Thus, in computing the change in the action $\int L\,dt$, we can immediately integrate by parts, and write

$$\delta\int L\,dt=c^2\int d^4x\left[\delta\phi^\dagger\left(\Box-\left(\frac{mc}{\hbar}\right)^2\right)\phi+\delta\phi\left(\Box-\left(\frac{mc}{\hbar}\right)^2\right)\phi^\dagger\right].$$

But this must vanish for any $\delta\phi$ and $\delta\phi^\dagger$, so ϕ must satisfy the familiar relativistic wave equation

$$\left[\Box-\left(\frac{mc}{\hbar}\right)^2\right]\phi=0 \tag{1.2.28}$$

and its adjoint. The 'momenta' canonically conjugate to the fields ϕ and $\phi^\dagger$ are given by the variational derivatives of L with respect to $\dot\phi$ and $\dot\phi^\dagger$, which we can read off from (1.2.27) as

$$\pi=\frac{\delta L}{\delta\dot\phi}=\dot\phi^\dagger, \tag{1.2.29}$$

$$\pi^\dagger=\frac{\delta L}{\delta\dot\phi^\dagger}=\dot\phi. \tag{1.2.30}$$

These field variables satisfy the usual canonical commutation relations, with a delta function in place of a Kronecker delta

$$[\pi(\mathbf x,t),\phi(\mathbf y,t)]=[\pi^\dagger(\mathbf x,t),\phi^\dagger(\mathbf y,t)]=-i\hbar\delta^3(\mathbf x-\mathbf y), \tag{1.2.31}$$

$$[\pi(\mathbf x,t),\phi^\dagger(\mathbf y,t)]=[\pi^\dagger(\mathbf x,t),\phi(\mathbf y,t)]=0, \tag{1.2.32}$$

$$[\pi(\mathbf x,t),\pi(\mathbf y,t)]=[\pi^\dagger(\mathbf x,t),\pi^\dagger(\mathbf y,t)]=[\pi(\mathbf x,t),\pi^\dagger(\mathbf y,t)]=0, \tag{1.2.33}$$

$$[\phi(\mathbf x,t),\phi(\mathbf y,t)]=[\phi^\dagger(\mathbf x,t),\phi^\dagger(\mathbf y,t)]=[\phi(\mathbf x,t),\phi^\dagger(\mathbf y,t)]=0. \tag{1.2.34}$$

The Hamiltonian here is given (just as in particle mechanics) by the 'sum' of all canonical momenta times the time-derivatives of the corresponding fields, minus the Lagrangian:

$$H=\int d^3x\left[\pi\dot\phi+\pi^\dagger\dot\phi^\dagger\right]-L \tag{1.2.35}$$

or, using (1.2.26), (1.2.29), and (1.2.30):

$$H=\int d^3x\left[\pi^\dagger\pi+c^2(\nabla\phi)^\dagger\cdot(\nabla\phi)+\left(\frac{m^2c^4}{\hbar^2}\right)\phi^\dagger\phi\right]. \tag{1.2.36}$$

After the papers by Heisenberg and Pauli one element was still missing before quantum field theory could reach its final pre-war form: a solution to the problem of the negative-energy states. We saw in the last section that in 1930, at just about the time of the

Heisenberg–Pauli papers, Dirac had proposed that the negative-energy states of the electron were all filled, but with only the holes in the negative-energy sea observable, rather than the negative-energy electrons themselves. After Dirac's idea was seemingly confirmed by the discovery of the positron in 1932, his 'hole theory' was used to calculate a number of processes to the lowest order of perturbation theory, including electron–positron pair production and scattering.

At the same time, a great deal of work was put into the development of a formalism whose Lorentz invariance would be explicit. The most influential effort was the 'many-time' formalism of Dirac, Vladimir Fock, and Boris Podolsky [42], in which the state vector was represented by a wave function depending on the spacetime and spin coordinates of all electrons, positive-energy and negative-energy. In this formalism, the total number of electrons of either positive or negative energy is conserved; for instance, production of an electron–positron pair is described as the excitation of a negative-energy electron to a positive-energy state, and the annihilation of an electron and positron is described as the corresponding deexcitation. This many-time formalism had the advantage of manifest Lorentz invariance, but it had a number of disadvantages; In particular, there was a profound difference between the treatment of the photon, described in terms of a quantized electromagnetic field, and that of the electron and positron. Not all physicists felt this to be a disadvantage; the electron field unlike the electromagnetic field did not have a classical limit, so there were doubts about its physical significance. Also, Dirac [42a] conceived of fields as the means by which we observe particles, so that he did not expect particles and fields to be described in the same terms. Though I do not know whether it bothered anyone at the time, there was a more practical disadvantage of the many-time formalism: it would have been difficult to use it to describe a process like nuclear beta decay, in which an electron and antineutrino are created without an accompanying positron or neutrino. The successful calculation by Fermi [43] of the electron energy distribution in beta decay deserves to be counted as one of the early triumphs of quantum field theory.

The essential idea that was needed to demonstrate the equivalence of the Dirac hole theory with a quantum field theory of the electron was provided by Fock [43a] and by Wendell Furry and Oppenheimer [44] in 1933–4. To appreciate this idea from a more modern standpoint, suppose we try to construct an electron field in analogy with the electromagnetic field or the Born–Heisenberg–Jordan field (1.2.2). Since electrons carry a charge, we would not like to mix annihilation and creation operators, so we might try to write the field as

$$\psi(x)=\sum_k u_k(\mathbf x)e^{-i\omega_kt}a_k, \tag{1.2.37}$$

where $u_k(\mathbf x)e^{-i\omega_kt}$ are a complete set of orthonormal plane-wave solutions of the Dirac equation (1.1.13) (with k now labelling the three-momentum, spin, and sign of the energy):

$$\mathcal H u_k=\hbar\omega_k u_k, \tag{1.2.38}$$

$$\mathcal H\equiv-i\hbar c\boldsymbol\alpha\cdot\nabla+\beta mc^2, \tag{1.2.39}$$

$$\int u_k^\dagger u_\ell\,d^3x=\delta_{k\ell}. \tag{1.2.40}$$

and a_k are the corresponding annihilation operators, satisfying the Jordan–Wigner anticommutation relations (1.2.22)–(1.2.23). According to the ideas of 'second quantization' or the canonical quantization procedure of Heisenberg and Pauli [41], the Hamiltonian is formed by calculating the 'expectation value' of $\mathcal H$ with a 'wave function' replaced by the quantized field (1.2.37)

$$H=\int d^3x\,\psi^\dagger\mathcal H\psi=\sum_k\hbar\omega_ka_k^\dagger a_k. \tag{1.2.41}$$

The trouble is, of course, that this is not a positive operator—half the ω_k are negative while the operators $a_k^\dagger a_k$ take only the positive eigenvalues 1 and 0. (See Eqs. (1.2.24) and (1.2.25).) In order to cure this disease, Furry and Oppenheimer picked up Dirac's idea [42] that the positron is the absence of a negative-energy electron; the anticommutation relations are symmetric between creation and annihilation operators, so they defined the positron creation and annihilation operators as the corresponding annihilation and creation operators for negative-energy electrons

$$b_k^\dagger\equiv a_k,\qquad b_k\equiv a_k^\dagger\qquad(\text{for }\omega_k<0) \tag{1.2.42}$$

where the label k on b denotes a positive-energy positron mode with momenta and spin opposite to those of the electron mode k. The Dirac field (1.2.37) may then be written

$$\psi(x)=\sum_k^{(+)}a_ku_k(x)+\sum_k^{(-)}b_k^\dagger u_k(x), \tag{1.2.43}$$

where $(+)$ and $(-)$ indicate sums over normal modes k with $\omega_k>0$ and $\omega_k<0$, respectively, and $u_k(x)\equiv u_k(\mathbf x)e^{-i\omega_kt}$. Similarly, using the anticommutation relations for the bs, we can rewrite the energy operator (1.2.41) as

$$H=\sum_k^{(+)}\hbar\omega_ka_k^\dagger a_k+\sum_k^{(-)}\hbar|\omega_k|b_k^\dagger b_k+E_0, \tag{1.2.44}$$

where E_0 is the infinite c-number

$$E_0=-\sum_k^{(-)}\hbar|\omega_k|. \tag{1.2.45}$$

In order for this redefinition to be more than a mere formality, it is necessary also to specify that the physical vacuum is a state $|\Omega\rangle$ containing no positive-energy electrons or positrons:

$$a_k|\Omega\rangle=0\qquad(\omega_k>0), \tag{1.2.46}$$

$$b_k|\Omega\rangle=0\qquad(\omega_k<0). \tag{1.2.47}$$

Hence (1.2.44) gives the energy of the vacuum as just E_0. If we measure all energies relative to the vacuum energy E_0, then the physical energy operator is $H-E_0$; and Eq. (1.2.44) shows that this is a positive operator.

The problem of negative-energy states for a charged spin zero particle was also resolved in 1934, by Pauli and Weisskopf [45], in a paper written in part to challenge Dirac's picture of filled negative-energy states. Here the creation and annihilation operators satisfy commutation rather than anticommutation relations, so it is not possible to interchange the roles of these operators freely, as was the case for fermions. Instead we must return to the Heisenberg–Pauli canonical formalism [41] to decide which coefficients of the various normal modes are creation or annihilation operators.

Pauli and Weisskopf expanded the free charged scalar field in plane waves in a cube of spatial volume $V=L^3$:

$$\phi(\mathbf x,t)=\frac{1}{\sqrt V}\sum_{\mathbf k}q(\mathbf k,t)e^{i\mathbf k\cdot\mathbf x} \tag{1.2.48}$$

with the wave numbers restricted by the periodicity condition, that the quantities $k_jL/2\pi$ for $j=1,2,3$ should be a set of three positive or negative integers. Similarly the canonically conjugate variable (1.2.29) was expanded as

$$\pi(\mathbf x,t)=\frac{1}{\sqrt V}\sum_{\mathbf k}p(\mathbf k,t)e^{-i\mathbf k\cdot\mathbf x}. \tag{1.2.49}$$

The minus sign is put into the exponent here so that (1.2.29) now becomes:

$$p(\mathbf k,t)=\dot q^\dagger(\mathbf k,t). \tag{1.2.50}$$

The Fourier inversion formula gives

$$q(\mathbf k,t)=\frac{1}{\sqrt V}\int d^3x\,\phi(\mathbf x,t)e^{-i\mathbf k\cdot\mathbf x}, \tag{1.2.51}$$

$$p(\mathbf k,t)=\frac{1}{\sqrt V}\int d^3x\,\pi(\mathbf x,t)e^{+i\mathbf k\cdot\mathbf x}, \tag{1.2.52}$$

and therefore the canonical commutation relations (1.2.31)–(1.2.34) yield for the qs and ps:

$$[p(\mathbf k,t),q(\mathbf l,t)]=\frac{-i\hbar}{V}\int d^3x\,e^{i\mathbf k\cdot\mathbf x}e^{-i\mathbf l\cdot\mathbf x}=-i\hbar\delta_{\mathbf{kl}} \tag{1.2.53}$$

$$[p(\mathbf k,t),q^\dagger(\mathbf l,t)]=[p(\mathbf k,t),p(\mathbf l,t)]=[p(\mathbf k,t),p^\dagger(\mathbf l,t)]$$
$$=[q(\mathbf k,t),q(\mathbf l,t)]=[q(\mathbf k,t),q^\dagger(\mathbf l,t)]=0 \tag{1.2.54}$$

together with other relations that may be derived from these by taking their Hermitian adjoints. By inserting (1.2.48) and (1.2.49) in the formula (1.2.36) for the Hamiltonian, we can also write this operator in terms of ps and qs:

$$H=\sum_{\mathbf k}\left[p^\dagger(\mathbf k,t)p(\mathbf k,t)+\omega_{\mathbf k}^2q^\dagger(\mathbf k,t)q(\mathbf k,t)\right], \tag{1.2.55}$$

where

$$\omega_{\mathbf k}^2=c^2\mathbf k^2+\left(\frac{mc^2}{\hbar}\right)^2. \tag{1.2.56}$$

The time-derivatives of the ps are then given by the Hamiltonian equation

$$\dot p(\mathbf k,t)=-\frac{\partial H}{\partial q(\mathbf k,t)}=-\omega_{\mathbf k}^2q^\dagger(\mathbf k,t) \tag{1.2.57}$$

(and its adjoint), a result which in the light of Eq. (1.2.50) is just equivalent to the Klein–Gordon–Schrödinger wave equation (1.2.28).

We see that, just as in the case of the 1926 model of Born, Heisenberg, and Jordan [4], the free field behaves like an infinite number of coupled harmonic oscillators. Pauli and Weisskopf could construct p and q operators which satisfy the commutation relations (1.2.53)–(1.2.54) and the 'equations of motion' (1.2.50) and (1.2.57), by introducing annihilation and creation operators $a,b,a^\dagger,b^\dagger$ of *two* different kinds, corresponding to particles and antiparticles:

$$q(\mathbf k,t)=i\sqrt{\frac{\hbar}{2\omega_{\mathbf k}}}\left[a_{\mathbf k}\exp(-i\omega_{\mathbf k}t)-b_{\mathbf k}^\dagger\exp(i\omega_{\mathbf k}t)\right], \tag{1.2.58}$$

$$p(\mathbf k,t)=\sqrt{\frac{\hbar\omega_{\mathbf k}}{2}}\left[b_{\mathbf k}\exp(-i\omega_{\mathbf k}t)+a_{\mathbf k}^\dagger\exp(+i\omega_{\mathbf k}t)\right] \tag{1.2.59}$$

where

$$[a_{\mathbf k},a_{\mathbf l}^\dagger]=[b_{\mathbf k},b_{\mathbf l}^\dagger]=\delta_{\mathbf{kl}}, \tag{1.2.60}$$

$$[a_{\mathbf k},a_{\mathbf l}]=[b_{\mathbf k},b_{\mathbf l}]=0, \tag{1.2.61}$$

$$[a_{\mathbf k},b_{\mathbf l}]=[a_{\mathbf k},b_{\mathbf l}^\dagger]=[a_{\mathbf k}^\dagger,b_{\mathbf l}]$$
$$=[a_{\mathbf k}^\dagger,b_{\mathbf l}^\dagger]=0. \tag{1.2.62}$$

It is straightforward to check that these operators do satisfy the desired relations (1.2.53), (1.2.54), (1.2.50), and (1.2.57). The field (1.2.48) may be written

$$\phi(\mathbf x,t)=\frac{i}{\sqrt V}\sum_{\mathbf k}\sqrt{\frac{\hbar}{2\omega_{\mathbf k}}}\left[a_{\mathbf k}\exp(i\mathbf k\cdot\mathbf x-i\omega_{\mathbf k}t)-b_{-\mathbf k}^\dagger\exp(-i\mathbf k\cdot\mathbf x+i\omega_{\mathbf k}t)\right] \tag{1.2.63}$$

and the Hamiltonian (1.2.55) takes the form

$$H=\sum_{\mathbf k}\frac{1}{2}\hbar\omega_{\mathbf k}\left[b_{\mathbf k}^\dagger b_{\mathbf k}+b_{\mathbf k}b_{\mathbf k}^\dagger+a_{\mathbf k}^\dagger a_{\mathbf k}+a_{\mathbf k}a_{\mathbf k}^\dagger\right]$$

or, using (1.2.60)–(1.2.62)

$$H=\sum_{\mathbf k}\hbar\omega_{\mathbf k}\left[b_{\mathbf k}^\dagger b_{\mathbf k}+a_{\mathbf k}^\dagger a_{\mathbf k}\right]+E_0. \tag{1.2.64}$$

where E_0 is the infinite c-number

$$E_0=\sum_{\mathbf k}\hbar\omega_{\mathbf k}. \tag{1.2.65}$$

The existence of two different kinds of operators a and b, which appear in precisely the same way in the Hamiltonian, shows that this is a theory with two kinds of particles with the same mass. As emphasized by Pauli and Weisskopf, these two varieties can be identified as particles and the corresponding antiparticles, and if charged have opposite charges. Thus, as we stressed above, bosons of spin zero as well as fermions of spin 1/2 can have distinct antiparticles, which for bosons cannot be identified as holes in a sea of negative energy particles.

We now can tell whether a and b or $a^\dagger$ and $b^\dagger$ are the annihilation operators by taking the expectation values of commutation relations in the vacuum state $|\Omega\rangle$. For instance, if $a_{\mathbf k}^\dagger$ were an annihilation operator it would give zero when applied to the vacuum state, so the vacuum expectation value of (1.2.60) would give

$$-\|a_{\mathbf k}|\Omega\rangle\|^2=\langle\Omega|[a_{\mathbf k},a_{\mathbf k}^\dagger]|\Omega\rangle=+1 \tag{1.2.66}$$

in conflict with the requirement that the left-hand side must be negative-definite. In this way we can conclude that it is $a_{\mathbf k}$ and $b_{\mathbf k}$ that are the annihilation operators, and therefore

$$a_{\mathbf k}|\Omega\rangle=b_{\mathbf k}|\Omega\rangle=0. \tag{1.2.67}$$

This is consistent with all commutation relations. Thus, the canonical formalism forces the coefficient of the $e^{+i\omega_kt}$ in the field (1.2.58) to be a creation operator, as it also is in the Furry–Oppenheimer formalism [44] for spin 1/2.

Equations (1.2.64) and (1.2.67) now tell us that E_0 is the energy of the vacuum state. If we measure all energies relative to E_0, then the physical energy operator is $H-E_0$, and (1.2.64) shows that this again is positive.

What about the problem that served Dirac as a starting point, the problem of negative probabilities? As Dirac had recognized, the only probability density ρ, which can be formed from solutions of the Klein–Gordon–Schrödinger free scalar wave equation (1.2.28), and which satisfies a conservation law of the form (1.1.10), must be proportional to the quantity

$$\rho=2\,\mathrm{Im}\left[\phi^\dagger\frac{\partial\phi}{\partial t}\right] \tag{1.2.68}$$

and therefore is not necessarily a positive quantity. Similarly, in the 'second-quantized' theory, where ϕ is given by Eq. (1.2.63), ρ is not a positive operator. Since $\phi^\dagger(x)$ does not commute with $\phi(x)$ here, we can write (1.2.68) in various forms, which differ by infinite c-numbers; it proves convenient to write it as

$$\rho=\frac{i}{\hbar}\left[\frac{\partial\phi}{\partial t}\phi^\dagger-\frac{\partial\phi^\dagger}{\partial t}\phi\right]. \tag{1.2.69}$$

The space-integral of this operator is then easily calculated to be

$$N\equiv\int\rho\,d^3x=\sum_{\mathbf k}\left(a_{\mathbf k}^\dagger a_{\mathbf k}-b_{\mathbf k}^\dagger b_{\mathbf k}\right) \tag{1.2.70}$$

and clearly has eigenvalues of either sign.

However, in a sense this problem appears in quantum field theory for spin 1/2 as well as spin zero. The density operator $\psi^\dagger\psi$ of Dirac is indeed a positive operator, but in order to construct a physical density we ought to subtract the contribution of the filled electron states. In particular, using the plane-wave decomposition (1.2.43), we may write the total number operator as

$$N\equiv\int d^3x\,\psi^\dagger\psi=\sum_{\mathbf k}^{(+)}a_{\mathbf k}^\dagger a_{\mathbf k}+\sum_{\mathbf k}^{(-)}b_{\mathbf k}b_{\mathbf k}^\dagger.$$

The anticommutation relations for the bs allow us to rewrite this as

$$N-N_0=\sum_{\mathbf k}^{(+)}a_{\mathbf k}^\dagger a_{\mathbf k}-\sum_{\mathbf k}^{(-)}b_{\mathbf k}^\dagger b_{\mathbf k}, \tag{1.2.71}$$

where N_0 is the infinite constant

$$N_0=\sum_{\mathbf k}^{(-)}1. \tag{1.2.72}$$

According to Eqs. (1.2.46) and (1.2.47), N_0 is the number of particles in the vacuum, so Furry and Oppenheimer reasoned that the physical number operator is $N-N_0$, and this now has both negative and positive eigenvalues, just as for spin zero.

The solution to this problem provided by quantum field theory is that neither the ψ of Furry and Oppenheimer nor the ϕ of Pauli and Weisskopf are probability amplitudes, which would have to define conserved positive probability densities. Instead, the physical Hilbert space is spanned by states defined as containing definite numbers of particles and/or antiparticles in each mode. If $|\Phi_n\rangle$ are a complete orthonormal set of such states, then a measurement of particle numbers in an arbitrary state $|\Psi\rangle$ will yield a probability for finding the system in state $|\Phi_n\rangle$, given by

$$P_n=|(\Phi_n|\Psi)|^2, \tag{1.2.73}$$

where $(\Phi_n|\Psi)$ is the usual Hilbert space scalar product. Hence, no question as to the possibility of negative probabilities will arise for any spin. The wave fields ϕ, ψ, etc., are not probability amplitudes at all, but *operators* which create or destroy particles in the various normal modes. It would be a good thing if the misleading expression 'second quantization' were permanently retired.

In particular, the operators N and $N-N_0$ of Eqs. (1.2.70) and (1.2.71) are not to be interpreted as total probabilities, but as number operators: specifically, the number of particles *minus* the number of antiparticles. For charged particles, the conservation of charge forces the charge operators to be proportional to these number operators, so the minus signs in (1.2.70) and (1.2.71) allow us immediately to conclude that particles and antiparticles have opposite charge. In this field-theoretic formalism, interactions contribute terms to the Hamiltonian which are of third, fourth, or higher order in field variables, and the rates of various processes are given by using these interaction operators in a time-dependent perturbation theory. The conceptual framework described in the above brief remarks will serve as the basis for much of the work in this book.

Despite its apparent advantages, quantum field theory did not immediately supplant hole theory; rather, the two points of view coexisted for a while, and various combinations of field-theoretic and hole-theoretic ideas were used in calculations of physical reaction rates. This period saw a number of calculations of cross sections to lowest order in powers of e^2 for various processes, such as $e^-+\gamma\to e^-+\gamma$ in 1929 by Klein and Nishina [46]; $e^++e^-\to2\gamma$ in 1930 by Dirac [47]; $e^-+e^-\to e^-+e^-$ in 1932 by Møller [48]; $e^-+Z\to e^-+\gamma+Z$ and $\gamma+Z\to e^++e^-+Z$ (where Z denotes the Coulomb field of a heavy atom) in 1934 by Bethe and Heitler [49]; and $e^++e^-\to e^++e^-$ in 1936 by Bhabha [50]. (Rules for the calculation of such processes are given in Chapter 8, and worked out in detail there for the case of electron–photon scattering.) These lowest-order calculations gave finite results, in reasonable agreement with the experimental data.

Nevertheless, a general feeling of dissatisfaction with quantum field theory (whether or not in the form of hole theory) persisted throughout the 1930s. One of the reasons for this was the apparent failure of quantum electrodynamics to account for the penetrating power of the charged particles in cosmic ray showers, noted in 1936 by Oppenheimer and Franklin Carlson [50a]. Another cause of dissatisfaction that turned out to be related to the first was the steady discovery of new kinds of particles and interactions. We have already mentioned the electron, photon, positron, neutrino, and, of course, the nucleus of hydrogen, the proton. Throughout the 1920s it was generally believed that heavier nuclei are composed of protons and electrons, but it was hard to see how a light particle like the electron could be confined in the nucleus. Another severe difficulty with this picture was pointed out in 1931 by Ehrenfest and Oppenheimer [51]: the nucleus of ordinary nitrogen, N^{14}, in order to have atomic number 7 and atomic weight 14, would have to be composed of 14 protons and 7 electrons, and would therefore have to be a fermion, in conflict with the result of molecular spectroscopy [52] that N^{14} is a boson. This problem (and others) were solved in 1932 with the discovery of the neutron [53], and by Heisenberg's subsequent suggestion [54] that nuclei are composed of protons and neutrons, not protons and electrons. It was clear that a strong non-electromagnetic force of short range would have to operate between neutrons and protons to hold nuclei together.

After the success of the Fermi theory of beta decay, several authors [54a] speculated that nuclear forces might be explained in this theory as due to the exchange of electrons and neutrinos. A few years later, in 1935, Hideki Yukawa proposed a quite different quantum field theory of the nuclear force [55]. In an essentially classical calculation, he found that the interaction of a scalar field with nucleons (protons or neutrons) would produce a nucleon–nucleon potential, with a dependence on the nucleon separation r given by

$$V(r)\propto\frac{1}{r}\exp(-\lambda r) \tag{1.2.74}$$

instead of the $1/r$ Coulomb potential produced by electric fields. The quantity λ was introduced as a parameter in Yukawa's scalar field equation, and when this equation was quantized, Yukawa found that it described particles of mass $\hbar\lambda/c$. The observed range of the strong interactions within nuclei led Yukawa to estimate that $\hbar\lambda/c$ is of the order of 200 electron masses. In 1937 such 'mesons' were discovered in cloud chamber experiments [56]

by Seth Neddermeyer and Anderson and by Jabez Curry Street and Edward Carl Stevenson, and it was generally believed that these were the hypothesized particles of Yukawa.

The discovery of mesons revealed that the charged particles in cosmic ray showers are not all electrons, and thus cleared up the problem with these showers that had bothered Oppenheimer and Carlson. At the same time, however, it created new difficulties. Lothar Nordheim [56a] pointed out in 1939 that the same strong interactions by which the mesons are copiously produced at high altitudes (and which are required in Yukawa's theory) should have led to the mesons' absorption in the atmosphere, a result contradicted by their copious appearance at lower altitudes. In 1947 it was shown in an experiment by Marcello Conversi, Ettore Pancini, and Oreste Piccioni [57] that the mesons which predominate in cosmic rays at low altitude actually interact weakly with nucleons, and therefore could not be identified with Yukawa's particle. This puzzle was cleared up by a theoretical suggestion [58], and its subsequent experimental confirmation [59] by Cesare Lattes, Occhialini, and Cecil Powell—there are two kinds of mesons with slightly different masses: the heavier (now called the π meson or pion) has strong interactions and plays the role in nuclear force envisaged by Yukawa; the lighter (now called the μ meson, or muon) has only weak and electromagnetic interactions, and predominates in cosmic rays at sea level, being produced by the decay of π mesons. In the same year, 1947, entirely new kinds of particles (now known as K mesons and hyperons) were found in cosmic rays by George Rochester and Clifford Butler [60]. From 1947 until the present particles have continued to be discovered in a bewildering variety, but to pursue this story would take us outside the bounds of our present survey. These discoveries showed clearly that any conceptual framework which was limited to photons, electrons, and positrons would be far too narrow to be taken seriously as a fundamental theory. But an even more important obstacle was presented by a purely theoretical problem—the problem of infinities.

1.3 The Problem of Infinities

Quantum field theory deals with fields $\psi(x)$ that destroy and create particles at a spacetime point x. Earlier experience with classical electron theory provided a warning that a point electron will have infinite electromagnetic self-mass; this mass is $e^2/6\pi ac^2$ for a surface distribution of charge with radius a, and therefore blows up for $a\to0$. Disappointingly this problem appeared with even greater severity in the early days of quantum field theory, and although greatly ameliorated by subsequent improvements in the theory, it remains with us to the present day.

The problem of infinities in quantum field theory was apparently first noted in the 1929–30 papers of Heisenberg and Pauli [41]. Soon after, the presence of infinities was confirmed in calculations of the electromagnetic self-energy of a bound electron by Oppenheimer [61], and of a free electron by Ivar Waller [62]. They used ordinary second-order perturbation theory, with an intermediate state consisting of an electron and a photon: for instance, the shift of the energy E_n of an electron in the nth energy level of hydrogen is given by

$$\Delta E_n=\sum_{m,\lambda}\int d^3k\,\frac{|(m;\mathbf k,\lambda|H'|n)|^2}{E_n-E_m-|\mathbf k|c}, \tag{1.3.1}$$

where the sums and integral are over all intermediate electron states m, photon helicities λ, and photon momenta $\mathbf k$, and H' is the term in the Hamiltonian representing the interaction of radiation and electrons. This calculation gave a self-energy that is formally infinite; further, if this infinity is removed by discarding all intermediate states with photon wave numbers greater than $1/a$, then the self-energy behaves like $1/a^2$ as $a\to0$. Infinities of this sort are often called ultraviolet divergences, because they arise from intermediate states containing particles of very short wavelength.

These calculations treated the electron according to the rules of the original Dirac theory, without filled negative-electron states. A few years later Weisskopf repeated the calculation of the electron self-mass in the new hole theory, with all negative-energy states full. In this case another term appears in second-order perturbation theory, which in a non-hole-theory language can be described as arising from processes in which the electron in its final state first appears out of the vacuum together with a photon and a positron which then annihilate along with the initial electron. Initially Weisskopf found a $1/a^2$ dependence on the photon wave-number cutoff $1/a$. The same calculation was being carried out (at the suggestion of Bohr) at that time by Carlson and Furry. After seeing Weisskopf's results, Furry realized that while Weisskopf had included an electrostatic term that he and Carlson had neglected, Weisskopf had made a new mistake in the calculation of the magnetic self-energy. After hearing from Furry and correcting his own error, Weisskopf found that the $1/a^2$ terms in the total mass shift cancelled! However, despite this cancellation, an infinity remained: with a wave-number cutoff $1/a$, the self-mass was found to be [63]

$$m_{em}=\frac{3\alpha}{2\pi}m\ln\left(\frac{\hbar}{mca}\right). \tag{1.3.2}$$

The weakening of the cut-off dependence, to $\ln a$ as compared with the classical $1/a$ or the early quantum $1/a^2$, was mildly encouraging at the time and turned out to be of great importance later, in the development of renormalization theory.

An infinity of quite a different kind was encountered in 1933, apparently first by Dirac [64]. He considered the effect of an external static nearly uniform charge density $\varepsilon(\mathbf x)$ on the vacuum, i.e., on the negative-energy electrons in the filled energy levels of hole theory. The Coulomb interaction between $\varepsilon(\mathbf x)$ and the charge density of the negative-energy electrons produces a "vacuum polarization," with induced charge density

$$\delta\varepsilon=A\varepsilon+B\left(\frac{\hbar}{mc}\right)^2\nabla^2\varepsilon+\cdots. \tag{1.3.3}$$

The constant B is finite, and of order α. On the other hand, A is logarithmically divergent, of order $\alpha\ln a$, where $1/a$ is the wave-number cutoff.

Infinities also seemed to occur in a related problem, the scattering of light by light. Hans Euler, Bernard Kockel, and Heisenberg [65] showed in 1935–6 that these infinities could be eliminated by using a more-or-less arbitrary prescription suggested earlier by Dirac [66] and Heisenberg [67]. They calculated an effective Lagrangian density for the non-linear electrodynamic effects produced by virtual electron–positron pairs:

$$\mathcal L=\frac{1}{2}\left(\mathbf E^2-\mathbf B^2\right)+\frac{e^4\hbar}{360\pi^2m_e^4c^7}\left[\left(\mathbf E^2-\mathbf B^2\right)^2+7\left(\mathbf E\cdot\mathbf B\right)^2\right]+\cdots, \tag{1.3.4}$$

valid for frequencies $\nu\ll m_ec^2/h$. Soon after, Nicholas Kemmer and Weisskopf [68] presented arguments that in this case the infinities are spurious, and that Eq. (1.3.4) can be derived without any subtraction prescription.

One bright spot in the struggle with infinities was the successful treatment of *infrared* divergences, those that arise from the low-energy rather than the high-energy part of the range of integration. In 1937 it was shown by Felix Bloch and Arne Nordsieck [68a] that these infinities cancel provided one includes processes in which arbitrary numbers of low-energy photons are produced. This will be discussed in modern terms in Chapter 13.

Yet another infinity turned up in a calculation by Sidney Michael Dancoff [69] in 1939 of the radiative corrections to the scattering of electrons by the static Coulomb field of an atom. The calculation contained a mistake (one of the terms was omitted), but this was not realized until later [69a].

Throughout the 1930s, these various infinities were seen not merely as failures of specific calculations. Rather, they seemed to indicate a gap in the understanding of relativistic quantum field theory on the most fundamental level, an opinion reinforced by the problems with cosmic rays mentioned in the previous section.

One of the symptoms of this uneasy pessimism was the continued exploration throughout the 1930s and 1940s of alternative formalisms. As Julian Schwinger [69b] later recalled, "The preoccupation of the majority of involved physicists was not with analyzing and carefully applying the known relativistic theory of coupled electron and electromagnetic fields but with changing it." Thus in 1938 Heisenberg [70] proposed the existence of a fundamental length L, analogous to the fundamental action $\hbar$ and fundamental velocity c. Field theory was supposed to work only for distances larger than L, so that all divergent integrals would effectively be cut off at distances L, or momenta $\hbar/L$. Several specific proposals [70a] were made for giving field theory a non-local structure. Some theorists began to suspect that the formalism of state-vectors and quantum fields should be replaced by one based solely on observable quantities, such as the S-matrix introduced by John Archibald Wheeler [71] in 1937 and Heisenberg [72] in 1943, whose elements are the amplitudes for various scattering processes. As we shall see, the concept of the S-matrix has now become a vital part of modern quantum field theory, and for some theorists a pure S-matrix theory became an ideal, especially as a possible solution to the problems of the strong interactions [73]. In yet another direction, Wheeler and Richard Feynman [74] in 1945 attempted to eliminate the electromagnetic field, deriving electromagnetic interactions in terms of an interaction at a distance. They were able to show that a pure retarded (or pure advanced) potential could be obtained by taking into account the interaction not only between source and test charges, but also between these charges and all the other charges in the universe. Perhaps the most radical modification of quantum mechanics suggested during this period was the introduction by Dirac [75] of states of negative probability, as a means of cancelling infinities in sums over states. This idea, of an "indefinite metric" in Hilbert space, has also flourished in quantum field theory, though not in the form originally suggested.

A more conservative idea for dealing with the infinities was also in the air during the 1930s. Perhaps these infinities could all be absorbed into a redefinition, a "renormalization" of the parameters of the theory. For instance, it was already known that in any Lorentz-

invariant classical theory the electromagnetic self-energy and self-momentum of an electron *must* take the form of corrections to the mass of the electron; hence the infinities in these quantities can be cancelled by a negative infinity in the "bare" non-electromagnetic mass of the electron, leaving a finite measurable "renormalized" mass. Also, Eq. (1.3.3) shows that the vacuum polarization changes the charge of the electron, from $e=\int d^3x\,\varepsilon$, to

$$e_{\rm TOTAL}=\int d^3x\,(\varepsilon+\delta\varepsilon)=(1+A)e. \tag{1.3.5}$$

Vacuum polarization gives finite results in lowest order if observables like scattering cross-sections are expressed in terms of $e_{\rm TOTAL}$ rather than e. The question was, whether all infinities in quantum field theory could be dealt with in this way. In 1936 Weisskopf [76] suggested that this is the case, and verified that known infinities could be eliminated by renormalization of physical parameters in a variety of sample calculations. However, it was impossible with the calculational techniques then available to show that infinities could always be eliminated in this way, and Dancoff's calculation [69] seemed to show that they could not.

Another effect of the appearance of infinities was a tendency to believe that any effect which turned out to be infinite in quantum field theory was actually not there at all. In particular, the 1928 Dirac theory had predicted complete degeneracy of the $2s_{1/2}$–$2p_{1/2}$ levels of hydrogen to all orders in α; any attempt at a quantum electromagnetic calculation of the splitting of these two levels ran into the problem of the infinite self-energy of a bound electron; therefore the existence of such a splitting was generally not taken seriously. Later Bethe [80] recalled that "This shift comes out infinite in all existing theories, and has therefore always been ignored." This attitude persisted even in the late 1930s, when spectroscopic experiments [77] began to indicate the presence of a $2s_{1/2}$–$2p_{1/2}$ splitting of order 1000 MHz. One notable exception was Edwin Albrecht Uehling [78], who realized that the vacuum polarization effect mentioned earlier would produce a $2s_{1/2}$–$2p_{1/2}$ splitting; unfortunately, as we shall see in Chapter 14, this contribution to the splitting is much smaller than 1000 MHz, and of the wrong sign.

The gloom surrounding quantum field theory began to lift soon after World War II. On June 1–4, 1947, the Conference on the Foundations of Quantum Mechanics at Shelter Island, NY brought theoretical physicists who had been working on the problems of quantum field theory through the 1930s together with a younger generation of theorists who had started scientific work during the war, and — of crucial importance — a few experimental physicists. The discussion leaders were Hans Kramers, Oppenheimer, and Weisskopf. One of the experimentalists (or rather theorist turned experimentalist), Willis Lamb, described a decisive measurement [79] of the $2s_{1/2}$–$2p_{1/2}$ shift in hydrogen. A beam of hydrogen atoms from an oven, many in $2s$ and $2p$ states, was aimed at a detector sensitive only to atoms in excited states. The atoms in $2p$ states can decay very rapidly to the $1s$ ground state by one-photon (Lyman α) emission, while the $2s$ states decay only very slowly by two-photon emission, so in effect the detector was measuring the number of atoms in the metastable $2s$ state. The beam was passed through a magnetic field, which added a known Zeeman splitting to any $2s_{1/2}$–$2p_{1/2}$ splitting naturally present. The beam was also exposed to a

microwave-frequency electromagnetic field, with a fixed frequency $\nu \sim 10$ GHz. At a certain magnetic field strength the detector signal was observed to be quenched, indicating that the microwave field was producing resonant transitions from the metastable $2s$ state to the $2p$ state and thence by a rapid Lyman α emission to the ground state. The total (Zeeman plus intrinsic) $2s$–$2p$ splitting at this value of the magnetic field strength would have to be just $h\nu$, from which the intrinsic splitting could be inferred. A preliminary value of 1000 MHz was announced, in agreement with the earlier spectroscopic measurements [77]. The impact of this discovery can be summarized in a saying that was current in Copenhagen when I was a graduate student there in 1954: "Just because something is infinite does not mean it is zero!"

The discovery of the Lamb shift aroused intense interest among the theorists at Shelter Island, many of whom had already been working on improved formalisms for calculation in quantum electrodynamics. Kramers described his work on mass renormalization in the classical electrodynamics of an extended electron [79a], which showed that the difficulties associated with the divergence of the self-energy in the limit of zero radius do not appear explicitly if the theory is reexpressed so that the mass parameter in the formalism is identified with the experimental electron mass. Schwinger and Weisskopf (who had already heard rumors of Lamb's result, and discussed the matter on the trip to Shelter Island) suggested that since the inclusion of intermediate states involving positrons was known to reduce the divergence in energy level shifts from $1/a^2$ to $\ln a$, perhaps the *differences* of the shifts in atomic energy levels might turn out to be finite when these intermediate states were taken into account. (In fact, in 1946, before he learned of Lamb's experiment, Weisskopf had already assigned this problem to a graduate student, Bruce French.) Almost immediately after the conference, during a train ride to Schenectady, Hans Bethe [80] carried out a non-relativistic calculation, still without including the effects of intermediate states containing positrons, but using a simple cutoff at virtual photon momenta of order m_ec^2 to eliminate infinities. He obtained the encouraging approximate value of 1040 MHz. Fully relativistic calculations using the renormalization idea to eliminate infinities were soon thereafter carried out by a number of other authors [81], with excellent agreement with experiment.

Another exciting experimental result was reported at Shelter Island by Isidor I. Rabi. Measurements in his laboratory of the hyperfine structure of hydrogen and deuterium had suggested [82] that the magnetic moment of the electron is larger than the Dirac value $e\hbar/2mc$ by a factor of about 1.0013, and subsequent measurements of the gyromagnetic ratios in sodium and gallium had given a precise value [83]

$$\mu = \frac{e\hbar}{2mc}[1.00118 \pm 0.00003].$$

Learning of these results, Gregory Breit suggested [83a] that they arose from an order α radiative correction to the electron magnetic moment. At Shelter Island, both Breit and Schwinger described their efforts to calculate this correction. Shortly after the conference Schwinger completed a successful calculation of the anomalous magnetic moment of the electron [84]

$$\mu = \frac{e\hbar}{2mc}\left[1 + \frac{\alpha}{2\pi}\right] = \frac{e\hbar}{2mc}[1.001162]$$

in excellent agreement with observation. This, together with Bethe's calculation of the Lamb shift, at last convinced physicists of the reality of radiative corrections.

The mathematical methods used in this period presented a bewildering variety of concepts and formalisms. One approach developed by Schwinger [85] was based on operator methods and the action principle, and was presented by him at a conference at Pocono Manor in 1948, the successor to the Shelter Island Conference. Another Lorentz-invariant operator formalism had been developed earlier by Sin-Itiro Tomonaga [86] and his co-workers in Japan, but their work was not at first known in the West. Tomonaga had grappled with infinities in Yukawa's meson theory in the 1930s. In 1947 he and his group were still out of the loop of scientific communication; they learned about Lamb's experiment from an article in *Newsweek*.

An apparently quite different approach was invented by Feynman [87], and described briefly by him at the Pocono Conference. Instead of introducing quantum field operators, Feynman represented the S-matrix as a functional integral of $\exp(iW)$, where W is the action integral for a set of Dirac particles interacting with a *classical* electromagnetic field, integrated over all Dirac particle trajectories satisfying certain initial and final conditions for $t \to \pm\infty$. One result of great practical importance that came out of Feynman's work was a set of graphical rules for calculating S-matrix elements to any desired order of perturbation theory. Unlike the old perturbation theory of the 1920s and 1930s, these Feynman rules automatically lumped together particle creation and antiparticle annihilation processes, and thereby gave results that were Lorentz-invariant at every stage. We have already seen in Weisskopf's early calculation [63] of the electron self-energy, that it is only in such calculations, including particles and antiparticles on the same footing, that the nature of the infinities becomes transparent.

Finally, in a pair of papers in 1949, Freeman Dyson [88] showed that the operator formalisms of Schwinger and Tomonaga would yield the same graphical rules that had been found by Feynman. Dyson also carried out an analysis of the infinities in general Feynman diagrams, and outlined a proof that these are always precisely the sort which could be removed by renormalization. One of the most striking results that could be inferred from Dyson's analysis was a criterion for deciding which quantum field theories are "renormalizable," in the sense that all infinities can be absorbed into a redefinition of a *finite* number of coupling constants and masses. In particular, an interaction like the Pauli term (1.1.32), which would have changed the predicted magnetic moment of the electron, would spoil the renormalizability of quantum electrodynamics. With the publication of Dyson's papers, there was at last a general and systematic formalism that physicists could easily learn to use, and that would provide a common language for the subsequent applications of quantum field theory to the problems of physics.

I cannot leave the infinities without taking up a puzzling aspect of this story. Oppenheimer [61] in 1930 had already noticed that most of the ultraviolet divergence in the self-energy of a bound electron cancels when one takes the difference between the shifts of two atomic energy levels, and Weisskopf [63] in 1934 had found that most of the divergence in the self-energy of a free electron cancels when one includes intermediate states containing positrons. It would have been natural even in 1934 to guess that including positron

intermediate states *and* subtracting the energy shifts of pairs of atomic states would eliminate the ultraviolet divergence in their relative energy shift.[2] There was even experimental evidence [77] for a $2s_{1/2}$–$2p_{1/2}$ energy difference of order 1000 MHz. So why did no one before 1947 attempt a numerical estimate of this energy difference?

Strictly speaking, there was one such attempt [88a] in 1939, but it focused on the wrong part of the problem, the charge radius of the *proton*, which has only a tiny effect on hydrogen energy levels. The calculation gave a result in rough agreement with the early experiments [77]. This was a mistake, as shown in 1939 by Lamb [88b].

A fully relativistic calculation of the Lamb shift including positrons in intermediate states could have been attempted during the 1930s, using the old non-relativistic perturbation theory. As long as one keeps all terms up to a given order, old-fashioned non-relativistic perturbation theory gives the same results as the manifestly relativistic formalisms of Feynman, Schwinger, and Tomonaga. In fact, after Bethe's work, the first precise calculations [81] of the Lamb shift in the USA by French and Weisskopf and Norman Kroll and Lamb were done in just this way, though Tomonaga's group [81] in Japan was already using covariant methods to solve this and other problems.

The one missing element was confidence in renormalization as a means of dealing with infinities. As we have seen, renormalization was widely discussed in the late 1930s. But it had become accepted wisdom in the 1930s, and a point of view especially urged by Oppenheimer [89], that quantum electrodynamics could not be taken seriously at energies of more than about 100 MeV, and that the solution to its problems could be found only in really adventurous new ideas.

Several things happened at Shelter Island to change this expectation. One was news that the problems concerning cosmic rays discussed in the previous section were beginning to be resolved; Robert Marshak presented the hypothesis [58] that there were two types of "meson" with similar masses; the muons that had actually been observed, and the pions responsible for nuclear forces. More important was the fact that now there were reliable experimental values for the Lamb shift and the anomalous magnetic moment that forced physicists to think carefully about radiative corrections. Probably equally important was the fact that the conference brought together theorists who had in their own individual ways been thinking about renormalization as a solution to the problem of infinities. When the revolution came in the late 1940s, it was made by physicists who though mostly young were playing a conservative role, turning away from the search by their predecessors for a radical solution.

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2 Relativistic Quantum Mechanics

The point of view of this book is that quantum field theory is the way it is because (with certain qualifications) this is the only way to reconcile quantum mechanics with special relativity. Therefore our first task is to study how symmetries like Lorentz invariance appear in a quantum setting.

2.1 Quantum Mechanics

First, some good news: quantum field theory is based on the same quantum mechanics that was invented by Schrödinger, Heisenberg, Pauli, Born, and others in 1925–26, and has been used ever since in atomic, molecular, nuclear, and condensed matter physics. The reader is assumed to be already familiar with quantum mechanics; this section provides only the briefest of summaries of quantum mechanics, in the generalized version of Dirac [1].

(i) Physical states are represented by rays in Hilbert space. A Hilbert space is a kind of complex vector space; that is, if $|\Phi\rangle$ and $|\Psi\rangle$ are vectors in the space (often called 'state-vectors') then so is $\xi|\Phi\rangle + \eta|\Psi\rangle$, for arbitrary complex numbers ξ, η. It has a norm[3]: for any pair of vectors there is a complex number $(\Phi|\Psi)$, such that

$$(\Phi|\Psi) = (\Psi|\Phi)^*, \tag{2.1.1}$$

$$(\Phi \mid \xi_1\Psi_1 + \xi_2\Psi_2) = \xi_1(\Phi|\Psi_1) + \xi_2(\Phi|\Psi_2), \tag{2.1.2}$$

$$(\eta_1\Phi_1 + \eta_2\Phi_2 \mid \Psi) = \eta_1^*(\Phi_1|\Psi) + \eta_2^*(\Phi_2|\Psi). \tag{2.1.3}$$

The norm $(\Psi|\Psi)$ also satisfies a positivity condition: $(\Psi|\Psi) \geq 0$, and vanishes if and only if $|\Psi\rangle = 0$. (There are also certain technical assumptions that allow us to take limits of vectors within Hilbert space.) A ray is a set of normalized vectors (i.e., $(\Psi|\Psi) = 1$) with $|\Psi\rangle$ and $|\Psi'\rangle$ belonging to the same ray if $|\Psi'\rangle = \xi|\Psi\rangle$, where ξ is an arbitrary complex number with $|\xi| = 1$.

(ii) Observables are represented by Hermitian operators. These are mappings $|\Psi\rangle \to A|\Psi\rangle$ of Hilbert space into itself, linear in the sense that

$$A\big(\xi|\Psi\rangle + \eta|\Phi\rangle\big) = \xi A|\Psi\rangle + \eta A|\Phi\rangle \tag{2.1.4}$$

and satisfying the reality condition $A^\dagger = A$, where for any linear operator A the adjoint $A^\dagger$ is defined by

$$(\Phi|A^\dagger|\Psi) \equiv (A\Phi|\Psi) = \big((\Psi|A|\Phi)\big)^*. \tag{2.1.5}$$

(There are also technical assumptions about the continuity of $A|\Psi\rangle$ as a function of $|\Psi\rangle$.)
A state represented by a ray $\mathcal{R}$ has a definite value α for the observable represented by an operator A if vectors $|\Psi\rangle$ belonging to this ray are eigenvectors of A with eigenvalue α:

$$A|\Psi\rangle = \alpha|\Psi\rangle \quad \text{for } |\Psi\rangle \text{ in } \mathcal{R}. \tag{2.1.6}$$

An elementary theorem tells us that for A Hermitian, α is real, and eigenvectors with different αs are orthogonal.

[3] We shall often use the Dirac bra-ket notation: instead of $(\Psi_1|\Psi_2)$, we may write $\langle 1|2\rangle$.

(iii) If a system is in a state represented by a ray $\mathcal{R}$, and an experiment is done to test whether it is in any one of the different states represented by mutually orthogonal rays $\mathcal{R}_1, \mathcal{R}_2, \ldots$ (for instance, by measuring one or more observables) then the probability of finding it in the state represented by $\mathcal{R}_n$ is

$$P(\mathcal{R} \to \mathcal{R}_n) = |(\Psi|\Psi_n)|^2, \tag{2.1.7}$$

where $|\Psi\rangle$ and $|\Psi_n\rangle$ are any vectors belonging to rays $\mathcal{R}$ and $\mathcal{R}_n$, respectively. (A pair of rays is said to be orthogonal if the state-vectors from the two rays have vanishing scalar products.) Another elementary theorem gives a total probability unity:

$$\sum_n P(\mathcal{R} \to \mathcal{R}_n) = 1 \tag{2.1.8}$$

if the state-vectors $|\Psi_n\rangle$ form a complete set.

2.2 Symmetries

A symmetry transformation is a change in our point of view that does not change the results of possible experiments. If an observer O sees a system in a state represented by a ray $\mathcal{R}$ or $\mathcal{R}_1$ or $\mathcal{R}_2, \ldots$, then an equivalent observer O' who looks at the *same* system will observe it in a different state, represented by a ray $\mathcal{R}'$ or $\mathcal{R}_1'$ or $\mathcal{R}_2', \ldots$, respectively, but the two observers must find the same probabilities

$$P(\mathcal{R} \to \mathcal{R}_n) = P(\mathcal{R}' \to \mathcal{R}_n'). \tag{2.2.1}$$

(This is only a necessary condition for a ray transformation to be a symmetry; further conditions are discussed in the next chapter.) An important theorem proved by Wigner [2] in the early 1930s tells us that for any such transformation $\mathcal{R} \to \mathcal{R}'$ of rays we may define an operator U on Hilbert space, such that if $|\Psi\rangle$ is in ray $\mathcal{R}$ then $U|\Psi\rangle$ is in the ray $\mathcal{R}'$, with U either *unitary* and *linear*

$$(U\Phi|U\Psi) = (\Phi|\Psi), \tag{2.2.2}$$

$$U\big(\xi|\Phi\rangle + \eta|\Psi\rangle\big) = \xi U|\Phi\rangle + \eta U|\Psi\rangle \tag{2.2.3}$$

or else *antiunitary* and *antilinear*

$$(U\Phi|U\Psi) = (\Phi|\Psi)^*, \tag{2.2.4}$$

$$U\big(\xi|\Phi\rangle + \eta|\Psi\rangle\big) = \xi^* U|\Phi\rangle + \eta^* U|\Psi\rangle. \tag{2.2.5}$$

Wigner's proof omits some steps. A more complete proof is given at the end of this chapter in Appendix A.

As already mentioned, the adjoint of a linear operator L is defined by

$$(\Phi|L^\dagger|\Psi) \equiv (L\Phi|\Psi). \tag{2.2.6}$$

This condition cannot be satisfied for an antilinear operator, because in this case the right-hand side of Eq. (2.2.6) would be linear in $|\Phi\rangle$, while the left-hand side is antilinear in $|\Phi\rangle$. Instead, the adjoint of an antilinear operator A is defined by

$$(\Phi|A^\dagger|\Psi) \equiv (A\Phi|\Psi)^* = (\Psi|A|\Phi). \tag{2.2.7}$$

With this definition, the conditions for unitarity or antiunitarity both take the form

$$U^\dagger = U^{-1}. \tag{2.2.8}$$

There is always a trivial symmetry transformation $\mathcal{R} \to \mathcal{R}$, represented by the identity operator $U = \mathbf{1}$. This operator is, of course, unitary and linear. Continuity then demands that any symmetry (like a rotation or translation or Lorentz transformation) that can be made trivial by a continuous change of some parameters (like angles or distances or velocities) must be represented by a linear unitary operator U rather than one that is antilinear and antiunitary. (Symmetries represented by antiunitary antilinear operators are less prominent in physics; they all involve a reversal in the direction of time's flow. See Section 2.6.)

In particular, a symmetry transformation that is infinitesimally close to being trivial can be represented by a linear unitary operator that is infinitesimally close to the identity:

$$U = \mathbf{1} + i\epsilon t \tag{2.2.9}$$

with ϵ a real infinitesimal. For this to be unitary and linear, t must be Hermitian and linear, so it is a candidate for an observable. Indeed, most (and perhaps all) of the observables of physics, such as angular momentum or momentum, arise in this way from symmetry transformations.

The set of symmetry transformations has certain properties that define it as a *group*. If T_1 is a transformation that takes rays $\mathcal{R}_n$ into $\mathcal{R}_n'$, and T_2 is another transformation that takes $\mathcal{R}_n'$ into $\mathcal{R}_n''$, then the result of performing both transformations is another symmetry transformation, which we write T_2T_1, that takes $\mathcal{R}_n$ into $\mathcal{R}_n''$. Also, a symmetry transformation T which takes rays $\mathcal{R}_n$ into $\mathcal{R}_n'$ has an inverse, written T^{-1}, which takes $\mathcal{R}_n'$ into $\mathcal{R}_n$, and there is an identity transformation, $T = \mathbf{1}$, which leaves rays unchanged.

The unitary or antiunitary operators $U(T)$ corresponding to these symmetry transformations have properties that mirror this group structure, but with a complication due to the fact that, unlike the symmetry transformations themselves, the operators $U(T)$ act on vectors in the Hilbert space, rather than on rays. If T_1 takes $\mathcal{R}_n$ into $\mathcal{R}_n'$, then acting on a vector $|\Psi_n\rangle$ in the ray $\mathcal{R}_n$, $U(T_1)$ must yield a vector $U(T_1)|\Psi_n\rangle$ in the ray $\mathcal{R}_n'$, and if T_2 takes this ray into $\mathcal{R}_n''$, then acting on $U(T_1)|\Psi_n\rangle$ it must yield a vector $U(T_2)U(T_1)|\Psi_n\rangle$ in the ray $\mathcal{R}_n''$. But $U(T_2T_1)|\Psi_n\rangle$ is also in this ray, so these vectors can differ only by a phase $\phi_n(T_2, T_1)$

$$U(T_2)U(T_1)|\Psi_n\rangle = e^{i\phi_n(T_2,T_1)}U(T_2T_1)|\Psi_n\rangle. \tag{2.2.10}$$

Furthermore, with one significant exception, the linearity (or antilinearity) of $U(T)$ tells us that these phases are independent of the state $|\Psi_n\rangle$. Here is the proof. Consider any two different vectors $|\Psi_A\rangle, |\Psi_B\rangle$, which are not proportional to each other. Then, applying Eq. (2.2.10) to the state $|\Psi_{AB}\rangle = |\Psi_A\rangle + |\Psi_B\rangle$, we have

$$\begin{aligned} e^{i\phi_{AB}}U(T_2T_1)\big(|\Psi_A\rangle + |\Psi_B\rangle\big) &= U(T_2)U(T_1)\big(|\Psi_A\rangle + |\Psi_B\rangle\big) \\ &= U(T_2)U(T_1)|\Psi_A\rangle + U(T_2)U(T_1)|\Psi_B\rangle \\ &= e^{i\phi_A}U(T_2T_1)|\Psi_A\rangle + e^{i\phi_B}U(T_2T_1)|\Psi_B\rangle. \end{aligned} \tag{2.2.11}$$

Any unitary or antiunitary operator has an inverse (its adjoint) which is also unitary or antiunitary. Multiplying (2.2.11) on the left with $U^{-1}(T_2T_1)$, we have then

$$e^{\pm i\phi_{AB}}\big(|\Psi_A\rangle + |\Psi_B\rangle\big) = e^{\pm i\phi_A}|\Psi_A\rangle + e^{\pm i\phi_B}|\Psi_B\rangle, \tag{2.2.12}$$

the upper and lower signs referring to $U(T_2T_1)$ unitary or antiunitary, respectively. Since $|\Psi_A\rangle$ and $|\Psi_B\rangle$ are linearly independent, this is only possible if

$$e^{i\phi_{AB}} = e^{i\phi_A} = e^{i\phi_B}. \tag{2.2.13}$$

So as promised, the phase in Eq. (2.2.10) is independent of the state-vector $|\Psi_n\rangle$, and therefore this can be written as an operator relation

$$U(T_2)U(T_1) = e^{i\phi(T_2,T_1)}U(T_2T_1). \tag{2.2.14}$$

For $\phi = 0$, this would say that the $U(T)$ furnish a representation of the group of symmetry transformations. For general phases $\phi(T_2, T_1)$, we have what is called a projective representation, or a representation 'up to a phase'. The structure of the Lie group cannot by itself tell us whether physical state-vectors furnish an ordinary or a projective representation, but as we shall see, it can tell us whether the group has any intrinsically projective representations at all.

The exception to the argument that led to Eq. (2.2.14) is that it may not be possible to prepare the system in a state represented by $|\Psi_A\rangle + |\Psi_B\rangle$. For instance, it is widely believed to be impossible to prepare a system in a superposition of two states whose total angular momenta are integers and half-integers, respectively. In such cases, we say that there is a 'superselection rule' between different classes of states [3], and the phases $\phi(T_2, T_1)$ may depend on which of these classes of states the operators $U(T_2)U(T_1)$ and $U(T_2T_1)$ act upon. We will have more to say about these phases and projective representations in Section 2.7. As we shall see there, any symmetry group with projective representations can always be enlarged (without otherwise changing its physical implications) in such a way that its representations can all be defined as non-projective, with $\phi = 0$. Until Section 2.7, we will just assume that this has been done, and take $\phi = 0$ in Eq. (2.2.14).

There is a kind of group, known as a *connected Lie group*, of special importance in physics. These are groups of transformations $T(\theta)$ that are described by a finite set of real continuous parameters, say θ^a, with each element of the group connected to the identity by a path within the group. The group multiplication law then takes the form

$$T(\bar\theta)T(\theta) = T\big(f(\bar\theta, \theta)\big) \tag{2.2.15}$$

with $f^a(\bar\theta, \theta)$ a function of the $\bar\theta$s and θs. Taking $\theta^a = 0$ as the coordinates of the identity, we must have

$$f^a(\theta, 0) = f^a(0, \theta) = \theta^a. \tag{2.2.16}$$

As already mentioned, the transformations of such continuous groups must be represented on the physical Hilbert space by unitary (rather than antiunitary) operators $U(T(\theta))$. For

a Lie group, these operators can be represented in at least a finite neighborhood of the identity by a power series

$$U(T(\theta)) = \mathbf{1} + i\theta^a t_a + \frac{1}{2}\theta^b\theta^c t_{bc} + \cdots, \tag{2.2.17}$$

where t_a, $t_{bc} = t_{cb}$, etc. are Hermitian operators independent of the θs. Suppose that the $U(T(\theta))$ form an ordinary (i.e., not projective) representation of this group of transformations, i.e.,

$$U(T(\bar\theta))U(T(\theta)) = U\big(T(f(\bar\theta, \theta))\big). \tag{2.2.18}$$

Let us see what this condition looks like when expanded in powers of θ^a and $\bar\theta^a$. According to Eq. (2.2.16), the expansion of $f^a(\bar\theta, \theta)$ to second order must take the form

$$f^a(\bar\theta, \theta) = \theta^a + \bar\theta^a + f^a{}_{bc}\bar\theta^b\theta^c + \cdots \tag{2.2.19}$$

with real coefficients $f^a{}_{bc}$. (The presence of any terms of order θ^2 or $\bar\theta^2$ would violate Eq. (2.2.16).) Then Eq. (2.2.18) reads

$$\begin{aligned} &\left[\mathbf{1} + i\bar\theta^a t_a + \frac{1}{2}\bar\theta^b\bar\theta^c t_{bc} + \cdots\right]\left[\mathbf{1} + i\theta^a t_a + \frac{1}{2}\theta^b\theta^c t_{bc} + \cdots\right] \\ &= \mathbf{1} + i\big(\theta^a + \bar\theta^a + f^a{}_{bc}\bar\theta^b\theta^c + \cdots\big)t_a \\ &\quad + \frac{1}{2}\big(\theta^b + \bar\theta^b + \cdots\big)\big(\theta^c + \bar\theta^c + \cdots\big)t_{bc} + \cdots. \end{aligned} \tag{2.2.20}$$

The terms of order 1, θ, $\bar\theta$, θ^2, and $\bar\theta^2$ automatically match on both sides of Eq. (2.2.20), but from the $\bar\theta\theta$ terms we obtain a non-trivial condition

$$t_{bc} = -t_b t_c - i f^a{}_{bc} t_a. \tag{2.2.21}$$

This shows that if we are given the structure of the group, i.e., the function $f(\theta, \bar\theta)$, and hence its quadratic coefficient $f^a{}_{bc}$, we can calculate the second-order terms in $U(T(\theta))$ from the generators t_a appearing in the first-order terms. However, there is a consistency condition: the operator t_{bc} must be *symmetric* in b and c (because it is the second derivative of $U(T(\theta))$ with respect to θ^b and θ^c) so Eq. (2.2.21) requires that

$$[t_b, t_c] = iC^a{}_{bc}t_a, \tag{2.2.22}$$

where $C^a{}_{bc}$ are a set of real constants known as *structure constants*

$$C^a{}_{bc} \equiv -f^a{}_{bc} + f^a{}_{cb}. \tag{2.2.23}$$

Such a set of commutation relations is known as a *Lie algebra*. In Section 2.7 we will prove in effect that the commutation relation (2.2.22) is the single condition needed to ensure that this process can be continued: the complete power series for $U(T(\theta))$ may be calculated from an infinite sequence of relations like Eq. (2.2.21), provided we know the first-order terms, the generators t_a. This does not necessarily mean that the operators $U(T(\theta))$ are uniquely determined for all θ^a if we know the t_a, but it does mean that the $U(T(\theta))$ are uniquely

determined in at least a finite neighborhood of the coordinates $\theta^a = 0$ of the identity, in such a way that Eq. (2.2.15) is satisfied if θ, $\bar\theta$, and $f(\theta, \bar\theta)$ are in this neighborhood. The extension to all θ^a is discussed in Section 2.7.

There is a special case of some importance, that we will encounter again and again. Suppose that the function $f(\theta, \bar\theta)$ (perhaps just for some subset of the coordinates θ^a) is simply additive

$$f^a(\theta, \bar\theta) = \theta^a + \bar\theta^a. \tag{2.2.24}$$

This is the case for instance for translations in spacetime, or for rotations about any one fixed axis (though *not* for both together). Then the coefficients $f^a{}_{bc}$ in Eq. (2.2.19) vanish, and so do the structure constants (2.2.23). The generators then all commute

$$[t_b, t_c] = 0. \tag{2.2.25}$$

Such a group is called *Abelian*. In this case, it is easy to calculate $U(T(\theta))$ for all θ^a. From Eqs. (2.2.18) and (2.2.24), we have for any integer N

$$U(T(\theta)) = \left[U\left(T\left(\frac{\theta}{N}\right)\right)\right]^N.$$

Letting $N \to \infty$, and keeping only the first-order term in $U(T(\theta/N))$, we have then

$$U(T(\theta)) = \lim_{N\to\infty}\left[\mathbf{1} + \frac{i}{N}\theta^a t_a\right]^N$$

and hence

$$U(T(\theta)) = \exp(it_a\theta^a). \tag{2.2.26}$$

2.3 Quantum Lorentz Transformations

Einstein's principle of relativity states the equivalence of certain 'inertial' frames of reference. It is distinguished from the Galilean principle of relativity, obeyed by Newtonian mechanics, by the transformation connecting coordinate systems in different inertial frames. If x^μ are the coordinates in one inertial frame (with $x^0 = t$ a time coordinate and x^1, x^2, x^3 Cartesian space coordinates, the speed of light being set equal to unity) then in any other inertial frame, the coordinates x'^μ must satisfy

$$\eta_{\mu\nu}\,dx'^\mu dx'^\nu = \eta_{\mu\nu}\,dx^\mu dx^\nu \tag{2.3.1}$$

or equivalently

$$\eta_{\mu\nu}\frac{\partial x'^\mu}{\partial x^\rho}\frac{\partial x'^\nu}{\partial x^\sigma} = \eta_{\rho\sigma}. \tag{2.3.2}$$

Here $\eta_{\mu\nu}$ is the diagonal matrix, with elements

$$\eta_{00} = -1, \qquad \eta_{11} = \eta_{22} = \eta_{33} = +1, \tag{2.3.3}$$

and the summation convention is in force: we sum over any index like μ and ν in Eq. (2.3.2), which appears twice in the same term, once upstairs and once downstairs. These transformations have the special property that the speed of light is the same (in our units, equal to

unity) in all inertial frames;[4] a light wave travelling at unit speed satisfies $|d\mathbf{x}/dt| = 1$, or in other words $\eta_{\mu\nu}dx^\mu dx^\nu = d\mathbf{x}^2 - dt^2 = 0$, from which it follows that also $\eta_{\mu\nu}dx'^\mu dx'^\nu = 0$, and hence $|d\mathbf{x}'/dt'| = 1$.

Any coordinate transformation $x^\mu \to x'^\mu$ that satisfies Eq. (2.3.2) is *linear* [3a]

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu \tag{2.3.4}$$

with a^μ arbitrary constants, and $\Lambda^\mu{}_\nu$ a constant matrix satisfying the conditions

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}. \tag{2.3.5}$$

For some purposes, it is useful to write the Lorentz transformation condition in a different way. The matrix $\eta_{\mu\nu}$ has an inverse, written $\eta^{\mu\nu}$, which happens to have the same components: it is diagonal, with $\eta^{00} = -1$, $\eta^{11} = \eta^{22} = \eta^{33} = +1$. Multiplying Eq. (2.3.5) with $\eta^{\sigma\tau}\Lambda^\kappa{}_\tau$ and inserting parentheses judiciously, we have

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho\big(\Lambda^\nu{}_\sigma\Lambda^\kappa{}_\tau\eta^{\sigma\tau}\big) = \Lambda^\kappa{}_\rho = \eta_{\mu\nu}\eta^{\nu\kappa}\Lambda^\mu{}_\rho.$$

Multiplying with the inverse of the matrix $\eta_{\mu\nu}\Lambda^\mu{}_\rho$ then gives

$$\Lambda^\nu{}_\sigma\Lambda^\kappa{}_\tau\eta^{\sigma\tau} = \eta^{\nu\kappa}. \tag{2.3.6}$$

These transformations form a group. If we first perform a Lorentz transformation (2.3.4), and then a second Lorentz transformation $x'^\mu \to x''^\mu$, with

$$x''^\mu = \bar\Lambda^\mu{}_\rho x'^\rho + \bar a^\mu = \bar\Lambda^\mu{}_\rho\big(\Lambda^\rho{}_\nu x^\nu + a^\rho\big) + \bar a^\mu$$

then the effect is the same as the Lorentz transformation $x^\mu \to x''^\mu$, with

$$x''^\mu = \big(\bar\Lambda^\mu{}_\rho\Lambda^\rho{}_\nu\big)x^\nu + \big(\bar\Lambda^\mu{}_\rho a^\rho + \bar a^\mu\big). \tag{2.3.7}$$

(Note that if $\Lambda^\mu{}_\nu$ and $\bar\Lambda^\mu{}_\nu$ both satisfy Eq. (2.3.5), then so does $\bar\Lambda^\mu{}_\rho\Lambda^\rho{}_\nu$, so this is a Lorentz transformation. The bar is used here just to distinguish one Lorentz transformation from the other.) The transformations $T(\Lambda, a)$ induced on physical states therefore satisfy the composition rule

$$T(\bar\Lambda, \bar a)T(\Lambda, a) = T(\bar\Lambda\Lambda, \bar\Lambda a + \bar a). \tag{2.3.8}$$

Taking the determinant of Eq. (2.3.5) gives

$$(\operatorname{Det}\Lambda)^2 = 1 \tag{2.3.9}$$

so $\Lambda^\mu{}_\nu$ has an inverse, $(\Lambda^{-1})^\nu{}_\mu$, which we see from Eq. (2.3.5) takes the form

$$(\Lambda^{-1})^\nu{}_\mu = \Lambda_\mu{}^\nu \equiv \eta_{\mu\rho}\eta^{\nu\sigma}\Lambda^\rho{}_\sigma. \tag{2.3.10}$$

[4] There is a larger class of coordinate transformations, known as *conformal transformations*, for which $\eta_{\mu\nu}dx'^\mu dx'^\nu$ is proportional though generally not equal to $\eta_{\mu\nu}dx^\mu dx^\nu$, and which therefore also leave the speed of light invariant. Conformal invariance in two dimensions has proved enormously important in string theory and statistical mechanics, but the physical relevance of these conformal transformations in four spacetime dimensions is not yet clear.

The inverse of the transformation $T(\Lambda, a)$ is seen from Eq. (2.3.8) to be $T(\Lambda^{-1}, -\Lambda^{-1}a)$, and, of course, the identity transformation is $T(\mathbf{1}, 0)$.

In accordance with the discussion in the previous section, the transformations $T(\Lambda, a)$ induce a unitary linear transformation on vectors in the physical Hilbert space

$$|\Psi\rangle \longrightarrow U(\Lambda, a)|\Psi\rangle.$$

The operators U satisfy a composition rule

$$U(\bar\Lambda, \bar a)U(\Lambda, a) = U(\bar\Lambda\Lambda, \bar\Lambda a + \bar a). \tag{2.3.11}$$

(As already mentioned, to avoid the appearance of a phase factor on the right-hand side of Eq. (2.3.11), it is, in general, necessary to enlarge the Lorentz group. The appropriate enlargement is described in Section 2.7.)

The whole group of transformations $T(\Lambda, a)$ is properly known as the *inhomogeneous Lorentz group*, or *Poincaré group*. It has a number of important subgroups. First, those transformations with $a^\mu = 0$ obviously form a subgroup, with

$$T(\bar\Lambda, 0)T(\Lambda, 0) = T(\bar\Lambda\Lambda, 0), \tag{2.3.12}$$

known as the *homogeneous Lorentz group*. Also, we note from Eq. (2.3.9) that either $\operatorname{Det}\Lambda = +1$ or $\operatorname{Det}\Lambda = -1$; those transformations with $\operatorname{Det}\Lambda = +1$ obviously form a subgroup of either the homogeneous or the inhomogeneous Lorentz groups. Further, from the 00-components of Eqs. (2.3.5) and (2.3.6), we have

$$(\Lambda^0{}_0)^2 = 1 + \Lambda^i{}_0\Lambda^i{}_0 = 1 + \Lambda^0{}_i\Lambda^0{}_i, \tag{2.3.13}$$

with i summed over the values 1, 2, and 3. We see that either $\Lambda^0{}_0 \geq +1$ or $\Lambda^0{}_0 \leq -1$. Those transformations with $\Lambda^0{}_0 \geq +1$ form a subgroup. Note that if $\Lambda^\mu{}_\nu$ and $\bar\Lambda^\mu{}_\nu$ are two such Λs, then

$$(\bar\Lambda\Lambda)^0{}_0 = \bar\Lambda^0{}_0\Lambda^0{}_0 + \bar\Lambda^0{}_1\Lambda^1{}_0 + \bar\Lambda^0{}_2\Lambda^2{}_0 + \bar\Lambda^0{}_3\Lambda^3{}_0.$$

But Eq. (2.3.13) shows that the three-vector $(\Lambda^1{}_0, \Lambda^2{}_0, \Lambda^3{}_0)$ has length $\sqrt{(\Lambda^0{}_0)^2 - 1}$, and similarly the three-vector
$(\bar\Lambda^0{}_1, \bar\Lambda^0{}_2, \bar\Lambda^0{}_3)$ has length $\sqrt{(\bar\Lambda^0{}_0)^2 - 1}$, so the scalar product of these two three-vectors is bounded by

$$|\bar\Lambda^0{}_1\Lambda^1{}_0 + \bar\Lambda^0{}_2\Lambda^2{}_0 + \bar\Lambda^0{}_3\Lambda^3{}_0| \leq \sqrt{(\Lambda^0{}_0)^2 - 1}\sqrt{(\bar\Lambda^0{}_0)^2 - 1}, \tag{2.3.14}$$

and so

$$(\bar\Lambda\Lambda)^0{}_0 \geq \bar\Lambda^0{}_0\Lambda^0{}_0 - \sqrt{(\Lambda^0{}_0)^2 - 1}\sqrt{(\bar\Lambda^0{}_0)^2 - 1} \geq 1.$$

The subgroup of Lorentz transformations with $\operatorname{Det}\Lambda = +1$ and $\Lambda^0{}_0 \geq +1$ is known as the *proper orthochronous Lorentz group*. Since it is not possible by a continuous change of parameters to jump from $\operatorname{Det}\Lambda = +1$ to $\operatorname{Det}\Lambda = -1$, or from $\Lambda^0{}_0 \geq +1$ to $\Lambda^0{}_0 \leq -1$, any Lorentz transformation that can be obtained from the identity by a continuous change of parameters must have $\operatorname{Det}\Lambda$ and $\Lambda^0{}_0$ of the same sign as for the identity, and hence must belong to the proper orthochronous Lorentz group.

Any Lorentz transformation is either proper and orthochronous, or may be written as the product of an element of the proper orthochronous Lorentz group with one of the discrete transformations $\mathcal{P}$ or $\mathcal{T}$ or $\mathcal{PT}$, where $\mathcal{P}$ is the space inversion, whose non-zero elements are

$$\mathcal{P}^0{}_0 = 1, \qquad \mathcal{P}^1{}_1 = \mathcal{P}^2{}_2 = \mathcal{P}^3{}_3 = -1, \tag{2.3.15}$$

and $\mathcal{T}$ is the time-reversal matrix, whose non-zero elements are

$$\mathcal{T}^0{}_0 = -1, \qquad \mathcal{T}^1{}_1 = \mathcal{T}^2{}_2 = \mathcal{T}^3{}_3 = 1. \tag{2.3.16}$$

Thus the study of the whole Lorentz group reduces to the study of its proper orthochronous subgroup, plus space inversion and time-reversal. We will consider space inversion and time-reversal separately in Section 2.6. Until then, we will deal only with the homogeneous or inhomogeneous proper orthochronous Lorentz group.

2.4 The Poincaré Algebra

As we saw in Section 2.2, much of the information about any Lie symmetry group is contained in properties of the group elements near the identity. For the inhomogeneous Lorentz group, the identity is the transformation $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu$, $a^\mu = 0$, so we want to study those transformations with

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu\,, \qquad a^\mu = \epsilon^\mu\,, \tag{2.4.1}$$

both $\omega^\mu{}_\nu$ and ϵ^μ being taken infinitesimal. The Lorentz condition (2.3.5) reads here

$$\eta_{\rho\sigma} = \eta_{\mu\nu}(\delta^\mu_\rho + \omega^\mu{}_\rho)(\delta^\nu_\sigma + \omega^\nu{}_\sigma)$$
$$= \eta_{\rho\sigma} + \omega_{\sigma\rho} + \omega_{\rho\sigma} + O(\omega^2)\,.$$

We are here using the convention, to be used throughout this book, that indices may be lowered or raised by contraction with $\eta_{\mu\nu}$ or $\eta^{\mu\nu}$

$$\omega_{\sigma\rho} = \eta_{\mu\sigma}\omega^\mu{}_\rho, \qquad \omega^\mu{}_\rho = \eta^{\mu\sigma}\omega_{\sigma\rho}\,.$$

Keeping only the terms of first order in ω in the Lorentz condition (2.3.5), we see that this condition now reduces to the antisymmetry of $\omega_{\mu\nu}$

$$\omega_{\mu\nu} = -\omega_{\nu\mu}\,. \tag{2.4.2}$$

An antisymmetric second-rank tensor in four dimensions has $(4\cdot 3)/2 = 6$ independent components, so including the four components of ϵ^μ, an inhomogeneous Lorentz transformation is described by $6 + 4 = 10$ parameters.

Since $U(\mathbf{1}, 0)$ carries any ray into itself, it must be proportional to the unit operator,[5] and by a choice of phase may be made equal to it. For an infinitesimal Lorentz transformation (2.4.1), $U(\mathbf{1} + \omega, \epsilon)$ must then equal $\mathbf{1}$ plus terms linear in $\omega_{\rho\sigma}$ and ϵ_ρ. We write

[5] In the absence of superselection rules, the possibility that the proportionality factor may depend on the state on which $U(\mathbf{1},0)$ acts can be ruled out by the same reasoning that we used in Section 2.2 to rule out the possibility that the phases in projective representations of symmetry groups may depend on the states on which the symmetries act. Where superselection rules apply, it may be necessary to redefine $U(\mathbf{1},0)$ by phase factors that depend on the sector on which it acts.

this as

$$U(\mathbf{1} + \omega, \epsilon) = \mathbf{1} + \frac{i}{2}\omega_{\rho\sigma}J^{\rho\sigma} - i\epsilon_\rho P^\rho + \cdots\,. \tag{2.4.3}$$

Here $J^{\rho\sigma}$ and P^ρ are ω- and ϵ-independent operators, and the dots denote terms of higher order in ω and/or ϵ. In order for $U(\mathbf{1} + \omega, \epsilon)$ to be unitary, the operators $J^{\rho\sigma}$ and P^ρ must be Hermitian

$$J^{\rho\sigma\dagger} = J^{\rho\sigma}\,, \qquad P^{\rho\dagger} = P^\rho\,. \tag{2.4.4}$$

Since $\omega_{\rho\sigma}$ is antisymmetric, we can take its coefficient $J^{\rho\sigma}$ to be antisymmetric also

$$J^{\rho\sigma} = -J^{\sigma\rho}\,. \tag{2.4.5}$$

As we shall see, P^1, P^2, and P^3 are the components of the momentum operators, J^{23}, J^{31}, and J^{12} are the components of the angular momentum vector, and P^0 is the energy operator, or Hamiltonian.[6]

We now examine the Lorentz transformation properties of $J^{\rho\sigma}$ and P^ρ. We consider the product

$$U(\Lambda, a)\,U(\mathbf{1} + \omega, \epsilon)\,U^{-1}(\Lambda, a)\,,$$

where $\Lambda^\mu{}_\nu$ and a^μ are here the parameters of a new transformation, unrelated to ω and ϵ. According to Eq. (2.3.11), the product $U(\Lambda^{-1}, -\Lambda^{-1}a)U(\Lambda, a)$ equals $U(\mathbf{1}, 0)$, so $U(\Lambda^{-1}, -\Lambda^{-1}a)$ is the inverse of $U(\Lambda, a)$. It follows then from (2.3.11) that

$$\begin{aligned} &U(\Lambda, a)U(\mathbf{1} + \omega, \epsilon)U^{-1}(\Lambda, a) \\ &= U\big(\Lambda(\mathbf{1} + \omega)\Lambda^{-1}, \Lambda\epsilon - \Lambda\omega\Lambda^{-1}a\big)\,. \end{aligned} \tag{2.4.6}$$

To first order in ω and ϵ, we have then

$$\begin{aligned} &U(\Lambda, a)\left[\frac{1}{2}\omega_{\rho\sigma}J^{\rho\sigma} - \epsilon_\rho P^\rho\right]U^{-1}(\Lambda, a) \\ &= \frac{1}{2}(\Lambda\omega\Lambda^{-1})_{\mu\nu}J^{\mu\nu} - (\Lambda\epsilon - \Lambda\omega\Lambda^{-1}a)_\mu P^\mu\,. \end{aligned} \tag{2.4.7}$$

Equating coefficients of $\omega_{\rho\sigma}$ and ϵ_ρ on both sides of this equation (and using (2.3.10)), we find

$$U(\Lambda, a)J^{\rho\sigma}U^{-1}(\Lambda, a) = \Lambda_\mu{}^\rho\Lambda_\nu{}^\sigma\left(J^{\mu\nu} - a^\mu P^\nu + a^\nu P^\mu\right), \tag{2.4.8}$$

$$U(\Lambda, a)P^\rho U^{-1}(\Lambda, a) = \Lambda_\mu{}^\rho P^\mu\,. \tag{2.4.9}$$

For homogeneous Lorentz transformations (with $a^\mu = 0$), these transformation rules simply say that $J^{\mu\nu}$ is a tensor and P^μ is a vector. For pure translations (with $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu$), they tell us that P^ρ is translation-invariant, but $J^{\rho\sigma}$ is not. In particular, the change of the space-space components of $J^{\rho\sigma}$ under a spatial translation is just the usual change of the angular momentum under a change of the origin relative to which the angular momentum is calculated.

[6] We will see that this identification of the angular-momentum generators is forced on us by the commutation relations of the $J^{\mu\nu}$. On the other hand, the commutation relations do not allow us to distinguish between P^μ and $-P^\mu$, so the sign for the $\epsilon_\rho P^\rho$ term in (2.4.3) is a matter of convention. The consistency of the choice in (2.4.3) with the usual definition of the Hamiltonian P^0 is shown in Section 3.1.

Next, let's apply rules (2.4.8), (2.4.9) to a transformation that is itself infinitesimal, i.e., $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega'^\mu{}_\nu$ and $a^\mu = \epsilon'^\mu$, with infinitesimals $\omega'^\mu{}_\nu$ and ϵ'^μ unrelated to the previous ω and ϵ. Using Eq. (2.4.3), and keeping only terms of first order in $\omega'^\mu{}_\nu$ and ϵ'^μ, Eqs. (2.4.8) and (2.4.9) now become

$$i\left[\frac{1}{2}\omega'_{\mu\nu}J^{\mu\nu} - \epsilon'_\mu P^\mu, J^{\rho\sigma}\right] = \omega'^\rho{}_\mu J^{\mu\sigma} + \omega'^\sigma{}_\nu J^{\rho\nu} - \epsilon'^\rho P^\sigma + \epsilon'^\sigma P^\rho\,, \tag{2.4.10}$$

$$i\left[\frac{1}{2}\omega'_{\mu\nu}J^{\mu\nu} - \epsilon'_\mu P^\mu, P^\rho\right] = \omega'^\rho{}_\mu P^\mu\,. \tag{2.4.11}$$

Equating coefficients of $\omega'_{\mu\nu}$ and ϵ'_μ on both sides of these equations, we find the commutation rules

$$i[J^{\mu\nu}, J^{\rho\sigma}] = \eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - \eta^{\sigma\mu}J^{\rho\nu} + \eta^{\sigma\nu}J^{\rho\mu}\,, \tag{2.4.12}$$

$$i[P^\mu, J^{\rho\sigma}] = \eta^{\mu\rho}P^\sigma - \eta^{\mu\sigma}P^\rho\,, \tag{2.4.13}$$

$$[P^\mu, P^\rho] = 0\,. \tag{2.4.14}$$

This is the Lie algebra of the Poincaré group.

In quantum mechanics a special role is played by those operators that are *conserved*, i.e., that commute with the energy operator $H = P^0$. Inspection of Eqs. (2.4.13) and (2.4.14) shows that these are the momentum three-vector

$$\mathbf{P} = \{P^1, P^2, P^3\} \tag{2.4.15}$$

and the angular-momentum three-vector

$$\mathbf{J} = \{J^{23}, J^{31}, J^{12}\} \tag{2.4.16}$$

and, of course, the energy P^0 itself. The remaining generators form what is called the 'boost' three-vector

$$\mathbf{K} = \{J^{10}, J^{20}, J^{30}\}\,. \tag{2.4.17}$$

These are *not* conserved, which is why we do not use the eigenvalues of $\mathbf{K}$ to label physical states. In a three-dimensional notation, the commutation relations (2.4.12), (2.4.13), (2.4.14) may be written

$$[J_i, J_j] = i\epsilon_{ijk}J_k\,, \tag{2.4.18}$$
$$[J_i, K_j] = i\epsilon_{ijk}K_k\,, \tag{2.4.19}$$
$$[K_i, K_j] = -i\epsilon_{ijk}J_k\,, \tag{2.4.20}$$
$$[J_i, P_j] = i\epsilon_{ijk}P_k\,, \tag{2.4.21}$$
$$[K_i, P_j] = iH\delta_{ij}\,, \tag{2.4.22}$$
$$[J_i, H] = [P_i, H] = [H, H] = 0\,, \tag{2.4.23}$$
$$[K_i, H] = iP_i\,, \tag{2.4.24}$$

where i, j, k, etc. run over the values 1, 2, and 3, and ϵ_{ijk} is the totally antisymmetric quantity with $\epsilon_{123} = +1$. The commutation relation (2.4.18) will be recognized as that of the angular-momentum operator.

The pure translations $T(\mathbf{1}, a)$ form a subgroup of the inhomogeneous Lorentz group with a group multiplication rule given by (2.3.7) as

$$T(\mathbf{1}, \bar{a})T(\mathbf{1}, a) = T(\mathbf{1}, \bar{a} + a)\,. \tag{2.4.25}$$

This is additive in the same sense as (2.2.24), so by using (2.4.3) and repeating the same arguments that led to (2.2.26), we find that finite translations are represented on the physical Hilbert space by

$$U(\mathbf{1}, a) = \exp(-iP^\mu a_\mu)\,. \tag{2.4.26}$$

In exactly the same way, we can show that a rotation $R_{\boldsymbol\theta}$ by an angle $|\boldsymbol\theta|$ around the direction of $\boldsymbol\theta$ is represented on the physical Hilbert space by

$$U(R_{\boldsymbol\theta}, 0) = \exp(i\mathbf{J}\cdot\boldsymbol\theta)\,. \tag{2.4.27}$$

It is interesting to compare the Poincaré algebra with the Lie algebra of the symmetry group of Newtonian mechanics, the Galilean group. We could derive this algebra by starting with the transformation rules of the Galilean group and then following the same procedure that was used here to derive the Poincaré algebra. However, since we already have Eqs. (2.4.18)–(2.4.24), it is easier to obtain the Galilean algebra as a low-velocity limit of the Poincaré algebra, by what is known as an *Inönü–Wigner contraction* [4,5]. For a system of particles of typical mass m and typical velocity v, the momentum and the angular-momentum operators are expected to be of order $J \sim 1$, $P \sim mv$. On the other hand, the energy operator is $H = M + W$ with a total mass M and non-mass energy W (kinetic plus potential) of order $M \sim m$, $W \sim mv^2$. Inspection of Eqs. (2.4.18)–(2.4.24) shows that these commutation relations have a limit for $v \ll 1$ of the form

$$[J_i, J_j] = i\epsilon_{ijk}J_k\,, \qquad [J_i, K_j] = i\epsilon_{ijk}K_k\,, \qquad [K_i, K_j] = 0\,,$$
$$[J_i, P_j] = i\epsilon_{ijk}P_k\,, \qquad [K_i, P_j] = iM\delta_{ij}\,,$$
$$[J_i, W] = [P_i, W] = 0\,, \qquad [K_i, W] = iP_i\,,$$
$$[J_i, M] = [P_i, M] = [K_i, M] = [W, M] = 0\,,$$

with K of order $1/v$. Note that the product of a translation $\mathbf{x} \to \mathbf{x} + \mathbf{a}$ and a 'boost' $\mathbf{x} \to \mathbf{x} + \mathbf{v}t$ should be the transformation $\mathbf{x} \to \mathbf{x} + \mathbf{v}t + \mathbf{a}$, but this is not true for the action of these operators on Hilbert space:

$$\exp(i\mathbf{K}\cdot\mathbf{v})\exp(-i\mathbf{P}\cdot\mathbf{a}) = \exp(iM\mathbf{a}\cdot\mathbf{v}/2)\exp(i(\mathbf{K}\cdot\mathbf{v} - \mathbf{P}\cdot\mathbf{a}))\,.$$

The appearance of the phase factor $\exp(iM\mathbf{a}\cdot\mathbf{v}/2)$ shows that this is a projective representation, with a superselection rule forbidding the superposition of states of different mass. In this respect, the mathematics of the Poincaré group is simpler than that of the Galilean group. However, there is nothing to prevent us from formally enlarging the Galilean group, by adding one more generator to its Lie algebra, which commutes with all the other generators, and whose eigenvalues are the masses of the various states. In this case physical states provide an ordinary rather than a projective representation of the expanded symmetry group. The difference appears to be a mere matter of notation, except that with this reinterpretation of the Galilean group there is no need for a mass superselection rule.

2.5 One-Particle States

We now consider the classification of one-particle states according to their transformation under the inhomogeneous Lorentz group.

The components of the energy-momentum four-vector all commute with each other, so it is natural to express physical state-vectors in terms of eigenvectors of the four-momentum. Introducing a label σ to denote all other degrees of freedom, we thus consider state-vectors $|p, \sigma\rangle$ with

$$P^\mu|p, \sigma\rangle = p^\mu|p, \sigma\rangle\,. \tag{2.5.1}$$

For general states, consisting for instance of several unbound particles, the label σ would have to be allowed to include continuous as well as discrete labels. We take as part of the definition of a *one-particle* state, that the label σ is purely discrete, and will limit ourselves here to that case. (However, a specific bound state of two or more particles, such as the lowest state of the hydrogen atom, is to be considered as a one-particle state. It is not an *elementary* particle, but the distinction between composite and elementary particles is of no relevance here.)

Eqs. (2.5.1) and (2.4.26) tell us how the states $|p, \sigma\rangle$ transform under translations:

$$U(\mathbf{1}, a)|p, \sigma\rangle = e^{-ip\cdot a}|p, \sigma\rangle\,.$$

We must now consider how these states transform under homogeneous Lorentz transformations.

Using (2.4.9), we see that the effect of operating on $|p, \sigma\rangle$ with a quantum homogeneous Lorentz transformation $U(\Lambda, 0) \equiv U(\Lambda)$ is to produce an eigenvector of the four-momentum with eigenvalue Λp

$$\begin{aligned} P^\mu U(\Lambda)|p, \sigma\rangle &= U(\Lambda)\left[U^{-1}(\Lambda)P^\mu U(\Lambda)\right]|p, \sigma\rangle \\ &= U(\Lambda)(\Lambda^{-1\mu}{}_\rho P^\rho)|p, \sigma\rangle \\ &= \Lambda^\mu{}_\rho p^\rho U(\Lambda)|p, \sigma\rangle\,. \end{aligned} \tag{2.5.2}$$

Hence $U(\Lambda)|p, \sigma\rangle$ must be a linear combination of the state-vectors $|\Lambda p, \sigma'\rangle$:

$$U(\Lambda)|p, \sigma\rangle = \sum_{\sigma'} C_{\sigma'\sigma}(\Lambda, p)|\Lambda p, \sigma'\rangle\,. \tag{2.5.3}$$

In general, it may be possible by using suitable linear combinations of the $|p, \sigma\rangle$ to choose the σ labels in such a way that the matrix $C_{\sigma'\sigma}(\Lambda, p)$ is block-diagonal; in other words, so that the $|p, \sigma\rangle$ with σ within any one block by themselves furnish a representation of the inhomogeneous Lorentz group. It is natural to identify the states of a specific particle type with the components of a representation of the inhomogeneous Lorentz group which is irreducible, in the sense that it cannot be further decomposed in this way.[7] Our task now

[7] Of course, different particle species may correspond to representations that are isomorphic, i.e., that have matrices $C_{\sigma'\sigma}(\Lambda, p)$ that are either identical, or identical up to a similarity transformation. In some cases it may be convenient to define particle types as irreducible representations of larger groups that contain the inhomogeneous proper orthochronous Lorentz group as a subgroup; for instance, as we shall see, for massless particles whose interactions respect the symmetry of space inversion it is customary to treat all the components of an irreducible representation of the inhomogeneous Lorentz group including space inversion as a single particle type.

is to work out the structure of the coefficients $C_{\sigma'\sigma}(\Lambda, p)$ in irreducible representations of the inhomogeneous Lorentz group.

For this purpose, note that the only functions of p^μ that are left invariant by all proper orthochronous Lorentz transformations $\Lambda^\mu{}_\nu$ are the invariant square $p^2 \equiv \eta_{\mu\nu}p^\mu p^\nu$, and for $p^2 \le 0$, also the sign of p^0. Hence, for each value of p^2, and (for $p^2 \le 0$) each sign of p^0, we can choose a 'standard' four-momentum, say k^μ, and express any p^μ of this class as

$$p^\mu = L^\mu{}_\nu(p)k^\nu\,, \tag{2.5.4}$$

where $L^\mu{}_\nu$ is some standard Lorentz transformation that depends on p^μ, and also implicitly on our choice of the standard k^μ. We can then *define* the states $|p, \sigma\rangle$ of momentum p by

$$|p, \sigma\rangle \equiv N(p)U(L(p))|k, \sigma\rangle\,, \tag{2.5.5}$$

where $N(p)$ is a numerical normalization factor, to be chosen later. Up to this point, we have said nothing about how the σ labels are related for different momenta; Eq. (2.5.5) now fills that gap.

Operating on (2.5.5) with an arbitrary homogeneous Lorentz transformation $U(\Lambda)$, we now find

$$\begin{aligned} U(\Lambda)|p, \sigma\rangle &= N(p)U(\Lambda L(p))|k, \sigma\rangle \\ &= N(p)U(L(\Lambda p))U(L^{-1}(\Lambda p)\Lambda L(p))|k, \sigma\rangle\,. \end{aligned} \tag{2.5.6}$$

The point of this last step is that the Lorentz transformation $L^{-1}(\Lambda p)\Lambda L(p)$ takes k to $L(p)k = p$, and then to Λp, and then back to k, so it belongs to the subgroup of the homogeneous Lorentz group consisting of Lorentz transformations $W^\mu{}_\nu$ that leave k^μ invariant:

$$W^\mu{}_\nu k^\nu = k^\mu\,. \tag{2.5.7}$$

This subgroup is called the *little group* [5]. For any W satisfying Eq. (2.5.7), we have

$$U(W)|k, \sigma\rangle = \sum_{\sigma'} D_{\sigma'\sigma}(W)|k, \sigma'\rangle\,. \tag{2.5.8}$$

The coefficients $D(W)$ furnish a representation of the little group; that is, for any elements $\bar{W}, W$ we have

$$\begin{aligned} \sum_{\sigma'} D_{\sigma'\sigma}(\bar{W}W)|k, \sigma'\rangle &= U(\bar{W}W)|k, \sigma\rangle = U(\bar{W})U(W)|k, \sigma\rangle \\ &= U(\bar{W})\sum_{\sigma''} D_{\sigma''\sigma}(W)|k, \sigma''\rangle \\ &= \sum_{\sigma'\sigma''} D_{\sigma'\sigma''}(\bar{W})D_{\sigma''\sigma}(W)|k, \sigma'\rangle \end{aligned}$$

and so

$$D_{\sigma'\sigma}(\bar{W}W) = \sum_{\sigma''} D_{\sigma'\sigma''}(\bar{W})D_{\sigma''\sigma}(W)\,. \tag{2.5.9}$$

In particular, we may apply Eq. (2.5.8) to the little-group transformation

$$W(\Lambda, p) \equiv L^{-1}(\Lambda p)\Lambda L(p) \tag{2.5.10}$$

and then Eq. (2.5.6) takes the form

$$U(\Lambda)|p, \sigma\rangle = N(p)\sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p))U(L(\Lambda p))|k, \sigma'\rangle\,,$$

or, recalling the definition (2.5.5):

$$U(\Lambda)|p, \sigma\rangle = \left(\frac{N(p)}{N(\Lambda p)}\right)\sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p))|\Lambda p, \sigma'\rangle\,. \tag{2.5.11}$$

Apart from the question of normalization, the problem of determining the coefficients $C_{\sigma'\sigma}$ in the transformation rule (2.5.3) has been reduced to the problem of finding the representations of the little group. This approach, of deriving representations of a group like the inhomogeneous Lorentz group from the representations of a little group, is called the method of induced representations [6].

Table 2.1 gives a convenient choice of the standard momentum k^μ and the corresponding little group for the various classes of four-momenta.

Of these six classes of four-momenta, only (a), (c), and (f) have any known interpretations in terms of physical states. Not much needs to be said here about case (f)—$p^\mu = 0$; it describes the vacuum, which is simply left invariant by $U(\Lambda)$. In what follows we will consider only cases (a) and (c), which cover particles of mass $M > 0$ and mass zero, respectively.

This is a good place to pause, and say something about the normalization of these states. By the usual orthonormalization procedure of quantum mechanics, we may choose the states with standard momentum k^μ to be orthonormal, in the sense that

$$\langle k', \sigma'|k, \sigma\rangle = \delta^3(\mathbf{k}' - \mathbf{k})\delta_{\sigma'\sigma}\,. \tag{2.5.12}$$

(The delta function appears here because $|k, \sigma\rangle$ and $|k', \sigma'\rangle$ are eigenstates of a Hermitian operator with eigenvalues $\mathbf{k}$ and $\mathbf{k}'$, respectively.) This has the immediate consequence that the representation of the little group in Eqs. (2.5.8) and (2.5.11) must be unitary[8]

$$D^\dagger(W) = D^{-1}(W)\,. \tag{2.5.13}$$

[8] The little groups $SO(2,1)$ and $SO(3,1)$ for $p^2 > 0$ and $p^\mu = 0$ have no non-trivial finite-dimensional unitary representations, so if there were any states with a given momentum p^μ with $p^2 > 0$ or $p^\mu = 0$ that transform non-trivially under the little group, there would have to be an infinite number of them.

Table 2.1. Standard momenta and the corresponding little group for various classes of four-momenta. Here κ is an arbitrary positive energy, say 1 eV. The little groups are mostly pretty obvious: $SO(3)$ is the ordinary rotation group in three dimensions (excluding space inversions), because rotations are the only proper orthochronous Lorentz transformations that leave at rest a particle with zero momentum, while $SO(2,1)$ and $SO(3,1)$ are the Lorentz groups in (2 + 1)- and (3 + 1)-dimensions, respectively. The group $ISO(2)$ is the group of Euclidean geometry, consisting of rotations and translations in two dimensions. Its appearance as the little group for $p^2 = 0$ is explained below.

	Standard k^μ	Little Group
(a) $p^2 = -M^2 < 0,\ p^0 > 0$	$(M, 0, 0, 0)$	$SO(3)$
(b) $p^2 = -M^2 < 0,\ p^0 < 0$	$(-M, 0, 0, 0)$	$SO(3)$
(c) $p^2 = 0,\ p^0 > 0$	$(\kappa, 0, 0, \kappa)$	$ISO(2)$
(d) $p^2 = 0,\ p^0 < 0$	$(-\kappa, 0, 0, \kappa)$	$ISO(2)$
(e) $p^2 = N^2 > 0$	$(0, 0, 0, N)$	$SO(2,1)$
(f) $p^\mu = 0$	$(0, 0, 0, 0)$	$SO(3,1)$

Now, what about scalar products for arbitrary momenta? Using the unitarity of the operator $U(\Lambda)$ in Eqs. (2.5.5) and (2.5.11), we find for the scalar product:

$$\begin{aligned} \langle p', \sigma'|p, \sigma\rangle &= N(p)\langle U^{-1}(L(p))p', \sigma'|k, \sigma\rangle \\ &= N(p)N^*(p')D_{\sigma\sigma'}(W(L^{-1}(p), p'))^*\delta^3(\mathbf{k}' - \mathbf{k})\,, \end{aligned}$$

where $k' \equiv L^{-1}(p)p'$. Since also $k = L^{-1}(p)p$, the delta function $\delta^3(\mathbf{k} - \mathbf{k}')$ is proportional to $\delta^3(\mathbf{p} - \mathbf{p}')$. For $p' = p$, the little-group transformation here is trivial, $W(L^{-1}(p), p) = \mathbf{1}$, and so the scalar product is

$$\langle p', \sigma'|p, \sigma\rangle = |N(p)|^2\delta_{\sigma'\sigma}\delta^3(\mathbf{k}' - \mathbf{k})\,. \tag{2.5.14}$$

It remains to work out the proportionality factor relating $\delta^3(\mathbf{k} - \mathbf{k}')$ and $\delta^3(\mathbf{p} - \mathbf{p}')$. Note that the Lorentz-invariant integral of an arbitrary scalar function $f(p)$ over four-momenta with $-p^2 = M^2 \ge 0$ and $p^0 > 0$ (i.e., cases (a) or (c)) may be written

$$\begin{aligned} \int d^4p\,\delta(p^2 + M^2)\theta(p^0)f(p) &= \int d^3\mathbf{p}\,dp^0\,\delta((p^0)^2 - \mathbf{p}^2 - M^2)\theta(p^0)f(\mathbf{p}, p^0) \\ &= \int d^3\mathbf{p}\,\frac{f(\mathbf{p}, \sqrt{\mathbf{p}^2 + M^2})}{2\sqrt{\mathbf{p}^2 + M^2}}\,. \end{aligned}$$

($\theta(p^0)$ is the step function: $\theta(x) = 1$ for $x \ge 0$, $\theta(x) = 0$ for $x < 0$.) We see that when integrating on the 'mass shell' $p^2 + M^2 = 0$, the invariant volume element is

$$d^3\mathbf{p}/\sqrt{\mathbf{p}^2 + M^2}\,. \tag{2.5.15}$$

The delta function is defined by

$$\begin{aligned} F(\mathbf{p}) &= \int F(\mathbf{p}')\delta^3(\mathbf{p} - \mathbf{p}')d^3\mathbf{p}' \\ &= \int F(\mathbf{p}')\left[\sqrt{\mathbf{p}'^2 + M^2}\,\delta^3(\mathbf{p}' - \mathbf{p})\right]\frac{d^3\mathbf{p}'}{\sqrt{\mathbf{p}'^2 + M^2}}\,, \end{aligned}$$

so we see that the invariant delta function is

$$\sqrt{\mathbf{p}^2 + M^2}\,\delta^3(\mathbf{p}' - \mathbf{p}) = p^0\delta^3(\mathbf{p}' - \mathbf{p})\,. \tag{2.5.16}$$

Since p' and p are related to k' and k respectively by a Lorentz transformation, $L(p)$, we have then

$$p^0\delta^3(\mathbf{p}' - \mathbf{p}) = k^0\delta^3(\mathbf{k}' - \mathbf{k})$$

and therefore

$$\langle p', \sigma'|p, \sigma\rangle = |N(p)|^2\delta_{\sigma'\sigma}\left(\frac{p^0}{k^0}\right)\delta^3(\mathbf{p}' - \mathbf{p})\,. \tag{2.5.17}$$

The normalization factor $N(p)$ is sometimes chosen to be just $N(p) = 1$, but then we would need to keep track of the p^0/k^0 factor in scalar products. Instead, I will here adopt the more usual convention that

$$N(p) = \sqrt{k^0/p^0} \tag{2.5.18}$$

for which

$$\langle p', \sigma'|p, \sigma\rangle = \delta_{\sigma'\sigma}\delta^3(\mathbf{p}' - \mathbf{p})\,. \tag{2.5.19}$$

We now consider the two cases of physical interest: particles of mass $M > 0$, and particles of zero mass.

Mass Positive-Definite

The little group here is the three-dimensional rotation group. Its unitary representations can be broken up into a direct sum of irreducible unitary representations [7] $D^{(j)}_{\sigma'\sigma}(R)$ of dimensionality $2j+1$, with $j = 0, \frac{1}{2}, 1, \ldots$. These can be built up from the standard matrices for infinitesimal rotations $R_{ik} = \delta_{ik} + \Theta_{ik}$, with $\Theta_{ik} = -\Theta_{ki}$ infinitesimal:

$$D^{(j)}_{\sigma'\sigma}(\mathbf{1} + \Theta) = \delta_{\sigma'\sigma} + \frac{i}{2}\Theta_{ik}(J^{(j)}_{ik})_{\sigma'\sigma}\,, \tag{2.5.20}$$

$$\begin{aligned} (J^{(j)}_{23} \pm iJ^{(j)}_{31})_{\sigma'\sigma} &= (J^{(j)}_1 \pm iJ^{(j)}_2)_{\sigma'\sigma} \\ &= \delta_{\sigma',\sigma\pm 1}\sqrt{(j \mp \sigma)(j \pm \sigma + 1)}\,, \end{aligned} \tag{2.5.21}$$

$$(J^{(j)}_{12})_{\sigma'\sigma} = (J^{(j)}_3)_{\sigma'\sigma} = \sigma\delta_{\sigma'\sigma}\,, \tag{2.5.22}$$

with σ running over the values $j, j-1, \ldots, -j$. For a particle of mass $M > 0$ and spin j, Eq. (2.5.11) now becomes

$$U(\Lambda)|p, \sigma\rangle = \sqrt{\frac{(\Lambda p)^0}{p^0}}\sum_{\sigma'} D^{(j)}_{\sigma'\sigma}(W(\Lambda, p))|\Lambda p, \sigma'\rangle\,, \tag{2.5.23}$$

with the little-group element $W(\Lambda, p)$ (the Wigner rotation [5]) given by Eq. (2.5.10):

$$W(\Lambda, p) = L^{-1}(\Lambda p)\Lambda L(p)\,.$$

To calculate this rotation, we need to choose a 'standard boost' $L(p)$ which carries the four-momentum from $k^\mu = (M, \mathbf{0})$ to p^μ. This is conveniently chosen as

$$\begin{aligned} L^i{}_k(p) &= \delta_{ik} + (\gamma - 1)\hat{p}_i\hat{p}_k\,, \\ L^i{}_0(p) &= L^0{}_i(p) = \hat{p}_i\sqrt{\gamma^2 - 1}\,, \\ L^0{}_0(p) &= \gamma\,, \end{aligned} \tag{2.5.24}$$

where

$$\hat{p}_i \equiv p_i/|\mathbf{p}|\,, \qquad \gamma \equiv \sqrt{\mathbf{p}^2 + M^2}/M\,.$$

It is very important that when $\Lambda^\mu{}_\nu$ is an arbitrary three-dimensional rotation $\mathcal{R}$, the Wigner rotation $W(\Lambda, p)$ is the same as $\mathcal{R}$ for all p. To see this, note that the boost (2.5.24) may be expressed as

$$L(p) = R(\hat{\mathbf{p}})B(|\mathbf{p}|)R^{-1}(\hat{\mathbf{p}})\,,$$

where $R(\hat{\mathbf{p}})$ is a rotation (to be defined in a standard way below, in Eq. (2.5.47)) that takes the three-axis into the direction of $\mathbf{p}$, and

$$B(|\mathbf{p}|) = \begin{pmatrix} \gamma & 0 & 0 & \sqrt{\gamma^2 - 1} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sqrt{\gamma^2 - 1} & 0 & 0 & \gamma \end{pmatrix}\,.$$

Then for an arbitrary rotation $\mathcal{R}$

$$W(\mathcal{R}, p) = R(\mathcal{R}\hat{\mathbf{p}})B^{-1}(|\mathbf{p}|)R^{-1}(\mathcal{R}\hat{\mathbf{p}})\mathcal{R}R(\hat{\mathbf{p}})B(|\mathbf{p}|)R^{-1}(\hat{\mathbf{p}})\,.$$

But the rotation $R^{-1}(\mathcal{R}\hat{\mathbf{p}})\mathcal{R}R(\hat{\mathbf{p}})$ takes the three-axis into the direction $\hat{\mathbf{p}}$, and then into the direction $\mathcal{R}\hat{\mathbf{p}}$, and then back to the three-axis, so it must be just a rotation by some angle θ around the three-axis

$$R^{-1}(\mathcal{R}\hat{\mathbf{p}})\mathcal{R}R(\hat{\mathbf{p}}) = R(\theta) \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\,.$$

Since $R(\theta)$ commutes with $B(|\mathbf{p}|)$, this now gives

$$\begin{aligned} W(\mathcal{R}, p) &= R(\mathcal{R}\hat{\mathbf{p}})B^{-1}(|\mathbf{p}|)R(\theta)B(|\mathbf{p}|)R^{-1}(\hat{\mathbf{p}}) \\ &= R(\mathcal{R}\hat{\mathbf{p}})R(\theta)R^{-1}(\hat{\mathbf{p}}) \end{aligned}$$

and hence

$$W(\mathcal{R}, p) = \mathcal{R}$$

as was to be shown. Thus states of a moving massive particle (and, by extension, multi-particle states) have the same transformation under rotations as in non-relativistic quantum mechanics. This is another piece of good news—the whole apparatus of spherical harmonics, Clebsch–Gordan coefficients, etc. can be carried over wholesale from non-relativistic to relativistic quantum mechanics.

Mass Zero

First, we have to work out the structure of the little group. Consider an arbitrary little-group element $W^\mu{}_\nu$, with $W^\mu{}_\nu k^\nu = k^\mu$, where k^μ is the standard four-momentum for this case, $k^\mu = (1, 0, 0, 1)$. Acting on a time-like four-vector $t^\mu = (1, 0, 0, 0)$, such a Lorentz transformation must yield a four-vector Wt whose length and scalar product with $Wk = k$ are the same as those of t:

$$(Wt)^\mu(Wt)_\mu = t^\mu t_\mu = -1\,, \qquad (Wt)^\mu k_\mu = t^\mu k_\mu = -1\,.$$

Any four-vector that satisfies the second condition may be written

$$(Wt)^\mu = (1 + \zeta, \alpha, \beta, \zeta)$$

and the first condition then yields the relation

$$\zeta = (\alpha^2 + \beta^2)/2\,. \tag{2.5.25}$$

It follows that the effect of $W^\mu{}_\nu$ on t^ν is the same as that of the Lorentz transformation

$$S^\mu{}_\nu(\alpha, \beta) = \begin{pmatrix} 1 + \zeta & \alpha & \beta & -\zeta \\ \alpha & 1 & 0 & -\alpha \\ \beta & 0 & 1 & -\beta \\ \zeta & \alpha & \beta & 1 - \zeta \end{pmatrix}\,. \tag{2.5.26}$$

This does not mean that W equals $S(\alpha, \beta)$, but it does mean that $S^{-1}(\alpha, \beta)W$ is a Lorentz transformation that leaves the time-like four-vector $(1, 0, 0, 0)$ invariant, and is therefore a pure rotation. Also, $S^\mu{}_\nu$ like $W^\mu{}_\nu$ leaves the light-like four-vector $(1, 0, 0, 1)$ invariant, so $S^{-1}(\alpha, \beta)W$ must be a rotation by some angle θ around the three-axis

$$S^{-1}(\alpha, \beta)W = R(\theta)\,, \tag{2.5.27}$$

where

$$R^\mu{}_\nu(\theta) \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\,.$$

The most general element of the little group is therefore of the form

$$W(\theta, \alpha, \beta) = S(\alpha, \beta)R(\theta)\,. \tag{2.5.28}$$

What group is this? We note that the transformations with $\theta = 0$ or with $\alpha = \beta = 0$ form subgroups:

$$S(\bar\alpha, \bar\beta)S(\alpha, \beta) = S(\bar\alpha + \alpha, \bar\beta + \beta) \tag{2.5.29}$$

$$R(\bar\theta)R(\theta) = R(\bar\theta + \theta)\,. \tag{2.5.30}$$

These subgroups are *Abelian*—that is, their elements all commute with each other. Furthermore, the subgroup with $\theta = 0$ is *invariant*, in the sense that its elements are transformed into other elements of the same subgroup by any member of the group

$$R(\theta)S(\alpha, \beta)R^{-1}(\theta) = S(\alpha\cos\theta + \beta\sin\theta, -\alpha\sin\theta + \beta\cos\theta)\,. \tag{2.5.31}$$

From Eqs. (2.5.29)–(2.5.31) we can work out the product of any group elements. The reader will recognize these multiplication rules as those of the group $ISO(2)$, consisting of translations (by a vector (α, β)) and rotations (by an angle θ) in two dimensions.

Groups that do *not* have invariant Abelian subgroups have certain simple properties, and for this reason are called *semi-simple*. As we have seen, the little group $ISO(2)$ like the inhomogeneous Lorentz group is *not* semi-simple, and this leads to interesting complications. First, let's take a look at the Lie algebra of $ISO(2)$. For θ, α, β infinitesimal, the general group element is

$$W(\theta, \alpha, \beta)^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu\,, \qquad \omega_{\mu\nu} = \begin{pmatrix} 0 & -\alpha & -\beta & 0 \\ \alpha & 0 & \theta & -\alpha \\ \beta & -\theta & 0 & -\beta \\ 0 & \alpha & \beta & 0 \end{pmatrix}\,.$$

From (2.4.3), we see then that the corresponding Hilbert space operator is

$$U(W(\theta, \alpha, \beta)) = \mathbf{1} + i\alpha A + i\beta B + i\theta J_3\,, \tag{2.5.32}$$

where A and B are the Hermitian operators

$$A = -J^{13} + J^{10} = J_2 + K_1\,, \tag{2.5.33}$$

$$B = -J^{23} + J^{20} = -J_1 + K_2\,, \tag{2.5.34}$$

and, as before, $J_3 = J_{12}$. Either from (2.4.18)–(2.4.20), or directly from Eqs. (2.5.29)–(2.5.31), we see that these generators have the commutators

$$[J_3, A] = +iB\,, \tag{2.5.35}$$
$$[J_3, B] = -iA\,, \tag{2.5.36}$$
$$[A, B] = 0\,. \tag{2.5.37}$$

Since A and B are commuting Hermitian operators they (like the momentum generators of the inhomogeneous Lorentz group) can be simultaneously diagonalized by states $|k, a, b\rangle$

$$A|k, a, b\rangle = a|k, a, b\rangle\,, \qquad B|k, a, b\rangle = b|k, a, b\rangle\,.$$

The problem is that if we find one such set of non-zero eigenvalues of A, B, then we find a whole continuum. From Eq. (2.5.31), we have

$$U[R(\theta)]AU^{-1}[R(\theta)] = A\cos\theta - B\sin\theta\,,$$
$$U[R(\theta)]BU^{-1}[R(\theta)] = A\sin\theta + B\cos\theta\,,$$

and so, for arbitrary θ,

$$A|k, a, b\rangle^\theta = (a\cos\theta - b\sin\theta)|k, a, b\rangle^\theta\,,$$
$$B|k, a, b\rangle^\theta = (a\sin\theta + b\cos\theta)|k, a, b\rangle^\theta\,,$$

where

$$|k, a, b\rangle^\theta \equiv U^{-1}(R(\theta))|k, a, b\rangle\,.$$

Massless particles are not observed to have any continuous degree of freedom like θ; to avoid such a continuum of states, we must require that physical states (now called $|k, \sigma\rangle$) are eigenvectors of A and B with $a = b = 0$:

$$A|k, \sigma\rangle = B|k, \sigma\rangle = 0\,. \tag{2.5.38}$$

These states are then distinguished by the eigenvalue of the remaining generator

$$J_3|k, \sigma\rangle = \sigma|k, \sigma\rangle\,. \tag{2.5.39}$$

Since the momentum $\mathbf{k}$ is in the three-direction, σ gives the component of angular momentum in the direction of motion, or *helicity*.

We are now in a position to calculate the Lorentz transformation properties of general massless particle states. First note that by use of the general arguments of Section 2.2, Eq. (2.5.32) generalizes for finite α and β to

$$U(S(\alpha, \beta)) = \exp(i\alpha A + i\beta B) \tag{2.5.40}$$

and for finite θ to

$$U(R(\theta)) = \exp(iJ_3\theta)\,. \tag{2.5.41}$$

An arbitrary element W of the little group can be put in the form (2.5.28), so that

$$U(W)|k, \sigma\rangle = \exp(i\alpha A + i\beta B)\exp(i\theta J_3)|k, \sigma\rangle = \exp(i\theta\sigma)|k, \sigma\rangle$$

and therefore Eq. (2.5.8) gives

$$D_{\sigma'\sigma}(W) = \exp(i\theta\sigma)\delta_{\sigma'\sigma}\,,$$

where θ is the angle defined by expressing W as in Eq. (2.5.28). The Lorentz transformation rule for a massless particle of arbitrary helicity is now given by Eqs. (2.5.11) and (2.5.18) as

$$U(\Lambda)|p, \sigma\rangle = \sqrt{\frac{(\Lambda p)^0}{p^0}}\exp(i\sigma\theta(\Lambda, p))|\Lambda p, \sigma\rangle \tag{2.5.42}$$

with $\theta(\Lambda, p)$ defined by

$$W(\Lambda, p) \equiv L^{-1}(\Lambda p)\Lambda L(p) \equiv S(\alpha(\Lambda, p), \beta(\Lambda, p))R(\theta(\Lambda, p))\,. \tag{2.5.43}$$

We shall see in Section 5.9 that electromagnetic gauge invariance arises from the part of the little group parameterized by α and β.

At this point we have not yet encountered any reason that would forbid the helicity σ of a massless particle from being an arbitrary real number. As we shall see in Section 2.7, there are topological considerations that restrict the allowed values of σ to integers and half-integers, just as for massive particles.

To calculate the little-group element (2.5.43) for a given Λ and p, (and also to enable us to calculate the effect of space or time inversion on these states in the next section) we need to fix a convention for the standard Lorentz transformation that takes us from $k^\mu = (\kappa, 0, 0, \kappa)$ to p^μ. This may conveniently be chosen to have the form

$$L(p) = R(\hat{\mathbf{p}})B(|\mathbf{p}|/\kappa) \tag{2.5.44}$$

where $B(u)$ is a pure boost along the three-direction:

$$B(u) \equiv \begin{pmatrix} (u^2 + 1)/2u & 0 & 0 & (u^2 - 1)/2u \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ (u^2 - 1)/2u & 0 & 0 & (u^2 + 1)/2u \end{pmatrix} \tag{2.5.45}$$

and $R(\hat{\mathbf{p}})$ is a pure rotation that carries the three-axis into the direction of the unit vector $\hat{\mathbf{p}}$. For instance, suppose we take $\hat{\mathbf{p}}$ to have polar and azimuthal angles θ and ϕ:

$$\hat{\mathbf{p}} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)\,. \tag{2.5.46}$$

Then we can take $R(\hat{\mathbf{p}})$ as a rotation by angle θ around the two-axis, which takes $(0, 0, 1)$ into $(\sin\theta, 0, \cos\theta)$, followed by a rotation by angle ϕ around the three-axis:

$$U(R(\hat{\mathbf{p}})) = \exp(i\phi J_3)\exp(i\theta J_2)\,, \tag{2.5.47}$$

where $0 \le \theta \le \pi$, $0 \le \phi < 2\pi$. (We give $U(R(\hat{\mathbf{p}}))$ rather than $R(\hat{\mathbf{p}})$, together with a specification of the range of ϕ and θ, because shifting θ or ϕ by 2π would give the same rotation $R(\hat{\mathbf{p}})$, but a different sign for $U(R(\hat{\mathbf{p}}))$ when acting on half-integer spin states.) Since (2.5.47) is a rotation, and does take the three-axis into the direction (2.5.46), any other choice of such an $R(\hat{\mathbf{p}})$ would differ from this one by at most an initial rotation around the three-axis, corresponding to a mere redefinition of the phase of the one-particle states.

Note that the helicity is Lorentz-invariant; a massless particle of a given helicity σ looks the same (aside from its momentum) in all inertial frames. Indeed, we would be justified in thinking of massless particles of each different helicity as different species of particles. However, as we shall see in the next section, particles of opposite helicity are related by the symmetry of space inversion. Thus, because electromagnetic and gravitational forces obey space inversion symmetry, the massless particles of helicity ± 1 associated with electromagnetic phenomena are both called *photons*, and the massless particles of helicity ± 2 that are believed to be associated with gravitation are both called *gravitons*. On the other hand, the supposedly massless particles of helicity $\pm 1/2$ that are emitted in nuclear beta decay have no interactions (apart from gravitation) that respect the symmetry of space inversion, so these particles are given different names: *neutrinos* for helicity $+1/2$, and *antineutrinos* for helicity $-1/2$.

Even though the helicity of a massless particle is Lorentz-invariant, the state itself is not. In particular, because of the helicity-dependent phase factor $\exp(i\sigma\theta)$ in Eq. (2.5.42), a state formed as a linear superposition of one-particle states with opposite helicities will be changed by a Lorentz transformation into a different superposition. For instance, a general one-photon state of four-momenta may be written

$$|p; \alpha\rangle = \alpha_+|p, +1\rangle + \alpha_-|p, -1\rangle\,,$$

where

$$|\alpha_+|^2 + |\alpha_-|^2 = 1\,.$$

The generic case is one of *elliptic polarization*, with $|\alpha_\pm|$ both non-zero and unequal. *Circular polarization* is the limiting case where either α_+ or α_- vanishes, and *linear polarization* is the opposite extreme, with $|\alpha_+| = |\alpha_-|$. The overall phase of α_+ and α_- has no physical significance, and for linear polarization may be adjusted so that $\alpha_- = \alpha_+^*$, but the relative phase is still important. Indeed, for linear polarizations with $\alpha_- = \alpha_+^*$, the phase of α_+ may be identified as the angle between the plane of polarization and some fixed reference direction perpendicular to $\mathbf{p}$. Eq. (2.5.42) shows that under a Lorentz transformation $\Lambda^\mu{}_\nu$, this angle rotates by an amount $\theta(\Lambda, p)$. Plane polarized gravitons can be defined in a similar way, and here Eq. (2.5.42) has the consequence that a Lorentz transformation Λ rotates the plane of polarization by an angle $2\theta(\Lambda, p)$.

2.6 Space Inversion and Time-Reversal

We saw in Section 2.3 that any homogeneous Lorentz transformation is either proper and orthochronous (i.e., $\mathrm{Det}\,\Lambda = +1$ and $\Lambda^0{}_0 \ge +1$) or else equal to a proper orthochronous transformation times either $\mathcal{P}$ or $\mathcal{T}$ or $\mathcal{PT}$, where $\mathcal{P}$ and $\mathcal{T}$ are the space inversion and time-reversal transformations

$$\mathcal{P}^\mu{}_\nu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}\,, \qquad \mathcal{T}^\mu{}_\nu = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\,.$$

It used to be thought self-evident that the fundamental multiplication rule of the Poincaré group

$$U(\bar\Lambda, \bar a)U(\Lambda, a) = U(\bar\Lambda\Lambda, \bar\Lambda a + \bar a)$$

would be valid even if Λ and/or $\bar\Lambda$ involved factors of $\mathcal{P}$ or $\mathcal{T}$ or $\mathcal{PT}$. In particular, it was believed that there are operators corresponding to $\mathcal{P}$ and $\mathcal{T}$ themselves:

$$P \equiv U(\mathcal{P}, 0), \qquad T \equiv U(\mathcal{T}, 0)$$

such that

$$PU(\Lambda, a)P^{-1} = U(\mathcal{P}\Lambda\mathcal{P}^{-1}, \mathcal{P}a), \tag{2.6.1}$$

$$TU(\Lambda, a)T^{-1} = U(\mathcal{T}\Lambda\mathcal{T}^{-1}, \mathcal{T}a) \tag{2.6.2}$$

for any proper orthochronous Lorentz transformation $\Lambda^\mu{}_\nu$, and translation a^μ. These transformation rules incorporate most of what is meant when we say that P or T are 'conserved'.

In 1956–57 it became understood [8] that this is true for P only in the approximation in which one ignores the effects of weak interactions, such as those that produce nuclear beta decay. Time-reversal survived for a while, but in 1964 there appeared indirect evidence [9] that these properties of T are also only approximately satisfied. (See Section 3.3.) In what follows, we will make believe that operators P and T satisfying Eqs. (2.6.1) and (2.6.2) actually exist, but it should be kept in mind that this is only an approximation.

Let us apply Eqs. (2.6.1) and (2.6.2) in the case of an infinitesimal transformation, i.e.,

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \qquad a^\mu = \epsilon^\mu$$

with $\omega_{\mu\nu} = -\omega_{\nu\mu}$ and ϵ_μ both infinitesimal. Using (2.4.3), and equating coefficients of $\omega_{\rho\sigma}$ and ϵ_ρ in Eqs. (2.6.1) and (2.6.2), we obtain the P and T transformation properties of the Poincaré generators

$$PiJ^{\rho\sigma}P^{-1} = i\mathcal{P}_\mu{}^\rho\mathcal{P}_\nu{}^\sigma J^{\mu\nu}, \tag{2.6.3}$$
$$PiP^\rho P^{-1} = i\mathcal{P}_\mu{}^\rho P^\mu, \tag{2.6.4}$$
$$TiJ^{\rho\sigma}T^{-1} = i\mathcal{T}_\mu{}^\rho\mathcal{T}_\nu{}^\sigma J^{\mu\nu}, \tag{2.6.5}$$
$$TiP^\rho T^{-1} = i\mathcal{T}_\mu{}^\rho P^\mu. \tag{2.6.6}$$

This is much like Eqs. (2.4.8) and (2.4.9), except that we have not cancelled factors of i on both sides of these equations, because at this point we have not yet decided whether P and T are linear and unitary or antilinear and antiunitary.

The decision is an easy one. Setting $\rho = 0$ in Eq. (2.6.4) gives

$$PiHP^{-1} = iH,$$

where $H \equiv P^0$ is the energy operator. If P were antiunitary and antilinear then it would anticommute with i, so $PHP^{-1} = -H$. But then for any state $|\Psi\rangle$ of energy $E > 0$, there would have to be another state $P^{-1}|\Psi\rangle$ of energy $-E < 0$. There are no states of negative energy (energy less than that of the vacuum), so we are forced to choose the other alternative: *P is linear and unitary, and commutes rather than anticommutes with H.*

On the other hand, setting $\rho = 0$ in Eq. (2.6.6) yields

$$TiHT^{-1} = -iH.$$

If we supposed that T is linear and unitary then we could simply cancel the is, and find $THT^{-1} = -H$, with the again disastrous conclusion that for any state $|\Psi\rangle$ of energy E there is another state $T^{-1}|\Psi\rangle$ of energy $-E$. To avoid this, we are forced here to conclude that *T is antilinear and antiunitary.*

Now that we have decided that P is linear and T is antilinear, we can conveniently rewrite Eqs. (2.6.3)–(2.6.6) in terms of the generators (2.4.15)–(2.4.17) in a three-dimensional notation

$$P\mathbf{J}P^{-1} = +\mathbf{J}, \tag{2.6.7}$$
$$P\mathbf{K}P^{-1} = -\mathbf{K}, \tag{2.6.8}$$
$$P\mathbf{P}P^{-1} = -\mathbf{P}, \tag{2.6.9}$$
$$T\mathbf{J}T^{-1} = -\mathbf{J}, \tag{2.6.10}$$
$$T\mathbf{K}T^{-1} = +\mathbf{K}, \tag{2.6.11}$$
$$T\mathbf{P}T^{-1} = -\mathbf{P}, \tag{2.6.12}$$

and, as shown before,

$$PHP^{-1} = THT^{-1} = H. \tag{2.6.13}$$

It is physically sensible that P should preserve the sign of $\mathbf{J}$, because at least the orbital part is a vector product $\mathbf{r}\cdot\mathbf{p}$ of two vectors, both of which change sign under an inversion of the spatial coordinate system. On the other hand, T reverses $\mathbf{J}$, because after time-reversal an observer will see all bodies spinning in the opposite direction. Note by the way that Eq. (2.6.10) is consistent with the angular-momentum commutation relations $\mathbf{J}\cdot\mathbf{J} = i\mathbf{J}$, because T reverses not only $\mathbf{J}$, but also i. The reader can easily check that Eqs. (2.6.7)–(2.6.13) are consistent with all the commutation relations (2.4.18)–(2.4.24).

Let us now consider what P and T do to one-particle states:

$P : M > 0$

The one-particle states $|k, \sigma\rangle$ are defined as eigenvectors of $\mathbf{P}$, H, and J_3 with eigenvalues 0, M, and σ, respectively. From Eqs. (2.6.7), (2.6.9), and (2.6.13), we see that the same must be true of the state $P|k, \sigma\rangle$, and therefore (barring degeneracies) these states can only differ by a phase

$$P|k, \sigma\rangle = \eta_\sigma|k, \sigma\rangle$$

with a phase factor ($|\eta_\sigma| = 1$) that may or may not depend on the spin σ. To see that η_σ is σ-independent, we note from (2.5.8), (2.5.20), and (2.5.21) that

$$(J_1 \pm iJ_2)|k, \sigma\rangle = \sqrt{(j \mp \sigma)(j \pm \sigma + 1)}|k, \sigma \pm 1\rangle, \tag{2.6.14}$$

where j is the particle's spin. Operating on both sides with P, we find

$$\eta_\sigma = \eta_{\sigma\pm 1}$$

and so η_σ is actually independent of σ. We therefore write

$$P|k, \sigma\rangle = \eta|k, \sigma\rangle \tag{2.6.15}$$

with η a phase, known as the *intrinsic parity*, that depends only on the species of particle on which P acts.

To get to finite momentum states, we must apply the unitary operator $U(L(p))$ corresponding to the 'boost' (2.5.24):

$$|p, \sigma\rangle = \sqrt{M/p^0}\,U(L(p))|k, \sigma\rangle.$$

We note that

$$\mathcal{P}L(p)\mathcal{P}^{-1} = L(\mathcal{P}p), \qquad \mathcal{P}p = (p^0, -\mathbf{p}),$$

so using Eqs. (2.6.1) and (2.6.15), we have

$$P|p, \sigma\rangle = \sqrt{M/p^0}\,U(L(\mathcal{P}p))\eta|k, \sigma\rangle$$

or in other words

$$P|p, \sigma\rangle = \eta|\mathcal{P}p, \sigma\rangle. \tag{2.6.16}$$

$T : M > 0$

From Eqs. (2.6.10), (2.6.12), and (2.6.13), we see that the effect of T on the zero-momentum one-particle state $|k, \sigma\rangle$ is to yield a state with

$$\mathbf{P}(T|k, \sigma\rangle) = 0,$$
$$H(T|k, \sigma\rangle) = M(T|k, \sigma\rangle),$$
$$J_3(T|k, \sigma\rangle) = -\sigma(T|k, \sigma\rangle),$$

and so

$$T|k, \sigma\rangle = \zeta_\sigma|k, -\sigma\rangle,$$

where ζ_σ is a phase factor. Applying the operator T to (2.6.14), and recalling that T anticommutes with $\mathbf{J}$ *and* i, we find

$$(-J_1 \pm iJ_2)\zeta_\sigma|k, -\sigma\rangle = \sqrt{(j \mp \sigma)(j \pm \sigma + 1)}\,\zeta_{\sigma\pm 1}|k, -\sigma \mp 1\rangle.$$

Using Eq. (2.6.14) again on the left, we see that the square-root factors cancel, and so

$$-\zeta_\sigma = \zeta_{\sigma\pm 1}.$$

We write the solution as $\zeta_\sigma = \zeta(-1)^{j-\sigma}$, with ζ another phase that depends only on the species of particle:

$$T|k, \sigma\rangle = \zeta(-1)^{j-\sigma}|k, -\sigma\rangle. \tag{2.6.17}$$

However, unlike the 'intrinsic parity' η, the time-reversal phase ζ has no physical significance. This is because we can redefine the one-particle states by a change of phase

$$|k, \sigma\rangle \longrightarrow |k, \sigma\rangle' = \zeta^{1/2}|k, \sigma\rangle$$

in such a way that the phase ζ is eliminated from the transformation rule

$$T|k, \sigma\rangle' = \zeta^{*1/2}T|k, \sigma\rangle = \zeta^{*1/2}\zeta(-1)^{j-\sigma}|k, -\sigma\rangle = (-1)^{j-\sigma}|k, -\sigma\rangle'.$$

In what follows we will keep the arbitrary phase ζ in Eq. (2.6.17), just to keep open our options in choosing the phase of the one-particle states, but it should be kept in mind that this phase is of no real importance.

To deal with states of finite momentum, we again apply the 'boost' (2.5.24). Note that

$$\mathcal{T}L(p)\mathcal{T}^{-1} = L(\mathcal{P}p), \qquad \mathcal{P}p = (p^0, -\mathbf{p}).$$

(That is, changing the sign of each element of $L^\mu{}_\nu$ with an odd number of *time*-indices is the same as changing the signs of elements with an odd number of *space*-indices.) Using Eqs. (2.6.2) and (2.5.5), we have then

$$T|p, \sigma\rangle = \zeta(-1)^{j-\sigma}|\mathcal{P}p, -\sigma\rangle. \tag{2.6.18}$$

$P : M = 0$

Acting on a state $|k, \sigma\rangle$ that is defined as an eigenvector of P^μ with eigenvalue $k^\mu = (\kappa, 0, 0, \kappa)$ and an eigenvector of J_3 with eigenvalue σ, the parity operator P yields a state with four-momentum $(\mathcal{P}k)^\mu = (\kappa, 0, 0, -\kappa)$ and J_3 equal to σ. Thus it takes a state of helicity (the component of spin along the direction of motion) σ into one of helicity $-\sigma$. As mentioned earlier, this shows that the existence of a space-inversion symmetry requires that any species of massless particle with non-zero helicity must be accompanied with another of opposite helicity. Because P does not leave the standard momentum invariant, it is convenient to consider instead the operator $U(R_2^{-1})P$, where R_2 is a rotation that also takes k to $\mathcal{P}k$, conveniently chosen as a rotation by $-180°$ around the two-axis

$$U(R_2) = \exp(-i\pi J_2). \tag{2.6.19}$$

Since $U(R_2^{-1})$ reverses the sign of J_3, we have

$$U(R_2^{-1})P|k, \sigma\rangle = \eta_\sigma|k, -\sigma\rangle \tag{2.6.20}$$

with η_σ a phase factor. Now, $R_2^{-1}\mathcal{P}$ commutes with the Lorentz 'boost' (2.5.45), and $\mathcal{P}$ commutes with the rotation $R(\hat{\mathbf{p}})$ which takes the three-direction into the direction of $\mathbf{p}$, so by operating on (2.5.5) with P, we find for a general four-momentum p^μ

$$\begin{aligned} P|p, \sigma\rangle &= \sqrt{\frac{\kappa}{p^0}}\,U\left(R(\hat{\mathbf{p}})R_2B\left(\frac{|\mathbf{p}|}{\kappa}\right)\right)U(R_2^{-1})P|k, \sigma\rangle \\ &= \sqrt{\frac{\kappa}{p^0}}\,\eta_\sigma U\left(R(\hat{\mathbf{p}})R_2B\left(\frac{|\mathbf{p}|}{\kappa}\right)\right)|k, -\sigma\rangle. \end{aligned}$$

Note that $R(\hat{\mathbf{p}})R_2$ is a rotation that takes the three-axis into the direction of $-\hat{\mathbf{p}}$, but $U(R(\hat{\mathbf{p}})R_2)$ is not quite equal to $U(R(-\hat{\mathbf{p}}))$. According to (2.5.47),

$$U(R(-\hat{\mathbf{p}})) = \exp\big(i(\phi \pm \pi)J_3\big)\exp\big(i(\pi - \theta)J_2\big)$$

with azimuthal angle chosen as $\phi + \pi$ or $\phi - \pi$ according to whether $0 \le \phi < \pi$ or $\pi \le \phi < 2\pi$, so that it remains in the range of 0 to 2π. Then

$$\begin{aligned} U^{-1}(R(-\hat{\mathbf{p}}))U(R(\hat{\mathbf{p}})R_2) &= \exp\big(-i(\pi - \theta)J_2\big) \\ &\quad\cdot\exp\big(-i(\phi \pm \pi)J_3\big)\exp(i\phi J_3)\exp(i\theta J_2)\exp(-i\pi J_2) \\ &= \exp\big(-i(\pi - \theta)J_2\big)\exp(\mp i\pi J_3)\exp\big(-i(\pi - \theta)J_2\big). \end{aligned}$$

But a rotation of $\pm 180°$ around the three-axis reverses the sign of J_2, so

$$U(R(\hat{\mathbf{p}})R_2) = U(R(-\hat{\mathbf{p}}))\exp(\pm i\pi J_3). \tag{2.6.21}$$

Also, $R(-\hat{\mathbf{p}})B(|\mathbf{p}|/\kappa)$ is just the standard boost $L(\mathcal{P}p)$ in the direction $\mathcal{P}p = (p^0, -\mathbf{p})$. We have then finally

$$P|p, \sigma\rangle = \eta_\sigma\exp(\mp i\pi\sigma)|\mathcal{P}p, -\sigma\rangle \tag{2.6.22}$$

with the phase $-\pi\sigma$ or $+\pi\sigma$ according to whether the two-component of $\mathbf{p}$ is positive or negative, respectively. This peculiar change of sign in the operation of parity for massless particles of half-integer spin is due to the convention adopted in Eq. (2.5.47) for the rotation used to define massless particle states of arbitrary momentum. Because the rotation group is not simply connected, some discontinuity of this sort is unavoidable.

$T : M = 0$

Acting on the state $|k, \sigma\rangle$, which has values $k^\mu = (\kappa, 0, 0, \kappa)$ and σ for P^μ and J_3, the time-reversal operator T yields a state which has values $(\mathcal{P}k)^\mu = (\kappa, 0, 0, -\kappa)$ and $-\sigma$ for P^μ and J_3. Thus T does not change the helicity $\mathbf{J}\cdot\hat{\mathbf{k}}$, and by itself has nothing to say about whether massless particles of one helicity σ are accompanied with others of helicity $-\sigma$. Because T like P does not leave the standard four-momentum k invariant, it is convenient to consider the generator $U(R_2^{-1})T$, where R_2 is the rotation (2.6.19), which also takes k into $\mathcal{P}k$. This commutes with J_3, so

$$U(R_2^{-1})T|k, \sigma\rangle = \zeta_\sigma|k, \sigma\rangle \tag{2.6.23}$$

with ζ_σ another phase. Since $R_2^{-1}\mathcal{T}$ commutes with the boost (2.5.45), and $\mathcal{T}$ commutes with the rotation $R(\hat{\mathbf{p}})$, operating with T on the state (2.5.5) gives

$$T|p, \sigma\rangle = \sqrt{\frac{\kappa}{p^0}}\,U\left[R(\hat{\mathbf{p}})R_2B\left(\frac{|\mathbf{p}|}{\kappa}\right)\right]\zeta_\sigma|k, \sigma\rangle. \tag{2.6.24}$$

Using Eq. (2.6.21), this yields finally

$$T|p, \sigma\rangle = \zeta_\sigma\exp(\pm i\pi\sigma)|\mathcal{P}p, \sigma\rangle. \tag{2.6.25}$$

Again, the top or bottom sign applies according to whether the two-component of $\mathbf{p}$ is positive or negative, respectively.

* * *

It is interesting that the square T^2 of the time-reversal operator has a very simple action on both massive and massless one-particle states. Using Eq. (2.6.18), and recalling that T is antiunitary, we see that for massive one-particle states:

$$\begin{aligned} T^2|p, \sigma\rangle &= T\zeta(-1)^{j-\sigma}|\mathcal{P}p, -\sigma\rangle \\ &= \zeta^*(-1)^{j-\sigma}\zeta(-1)^{j+\sigma}|p, \sigma\rangle \end{aligned}$$

or in other words

$$T^2|p, \sigma\rangle = (-1)^{2j}|p, \sigma\rangle. \tag{2.6.26}$$

We get the same result for massless particles. If the two-component of $\mathbf{p}$ is positive then the two-component of $\mathcal{P}\mathbf{p}$ is negative, and vice-versa, so Eq. (2.6.25) gives

$$\begin{aligned} T^2|p,\sigma\rangle &= T\zeta_\sigma \exp(\pm i\pi\sigma)|\mathcal{P}p,\sigma\rangle \\ &= \zeta_\sigma^* \exp(\mp i\pi\sigma)\zeta_\sigma \exp(\mp i\pi\sigma)|p,\sigma\rangle \\ &= \exp(\mp 2\pi i\sigma)|p,\sigma\rangle. \end{aligned}$$

As long as σ is an integer or half-integer, this can be written

$$T^2|p,\sigma\rangle = (-1)^{2|\sigma|}|p,\sigma\rangle. \tag{2.6.27}$$

By the 'spin' of a massless particle, we usually mean the absolute value of the helicity, so Eq. (2.6.27) is the same as Eq. (2.6.26).

This result has an interesting consequence. When T^2 acts on any state $|\Psi\rangle$ of a system of non-interacting particles, either massive or massless, it yields a factor $(-1)^{2j}$ or $(-1)^{2|\sigma|}$ for each particle. Hence if the state contains an odd number of particles of half-integer spin or helicity (plus any number of particles of integer spin or helicity), we get an overall change of sign

$$T^2|\Psi\rangle = -|\Psi\rangle. \tag{2.6.28}$$

If we now 'turn on' various interactions, this result will be preserved, provided these interactions respect invariance under time-reversal, even if they do not respect rotational invariance. (For instance, these arguments will apply even if our system is subjected to arbitrary static gravitational and electric fields.) Now, suppose that $|\Psi\rangle$ is an eigenstate of the Hamiltonian. Since T commutes with the Hamiltonian, $T|\Psi\rangle$ will also be an eigenstate of the Hamiltonian. Is it the same state? If so, then $T|\Psi\rangle$ can differ from $|\Psi\rangle$ only by a phase

$$T|\Psi\rangle = \zeta|\Psi\rangle,$$

but then

$$T^2|\Psi\rangle = T(\zeta|\Psi\rangle) = \zeta^* T|\Psi\rangle = |\zeta|^2|\Psi\rangle = |\Psi\rangle,$$

in contradiction with Eq. (2.6.28). We see that any energy eigenstate $|\Psi\rangle$ satisfying Eq. (2.6.28) must be degenerate, with another eigenstate of the same energy. This is known as a 'Kramers degeneracy' [10]. Of course, this conclusion is trivial if the system is in a rotationally invariant environment, because the total angular-momentum j of any state of this system would have to be a half-integer, and there would therefore be $2j+1 = 2, 4, \ldots$ degenerate states. The surprising result is that at least a two-fold degeneracy persists even if rotational invariance is perturbed by external fields, such as electrostatic fields, as long as these fields are invariant under T. In particular, if any particle had an electric or gravitational dipole moment then the degeneracy among its $2j+1$ spin states would be entirely removed in a static electric or gravitational field, so such dipole moments are forbidden by time-reversal invariance.

For the sake of completeness, it should be mentioned that P and T can have more complicated effects on multiplets of particles with the same mass. This possibility will be considered in Appendix C of this chapter. No physically relevant examples are known.

2.7 Projective Representations[9]

We now return to the possibility mentioned in Section 2.2, that a group of symmetries may be represented projectively on physical states; that is, the elements T, $\bar{T}$, etc. of the symmetry group may be represented on the physical Hilbert space by unitary operators $U(T)$, $U(\bar{T})$, etc., which satisfy the composition rule

$$U(T)U(\bar{T}) = \exp\big(i\phi(T,\bar{T})\big)U(T\bar{T}) \tag{2.7.1}$$

with ϕ a real phase. (A bar is used here just to distinguish one symmetry operator from another.) The basic requirement that any phase ϕ in Eq. (2.7.1) would have to satisfy is the associativity condition

$$U(T_3)\big(U(T_2)U(T_1)\big) = \big(U(T_3)U(T_2)\big)U(T_1),$$

which imposes on ϕ the corresponding condition

$$\phi(T_2,T_1) + \phi(T_3,T_2T_1) = \phi(T_3,T_2) + \phi(T_3T_2,T_1). \tag{2.7.2}$$

Of course, any phase of the form

$$\phi(T,\bar{T}) = \alpha(T\bar{T}) - \alpha(T) - \alpha(\bar{T}) \tag{2.7.3}$$

will automatically satisfy Eq. (2.7.2), but a projective representation with such a phase can be replaced with an ordinary representation by replacing $U(T)$ with

$$\tilde{U}(T) \equiv U(T)\exp\big(i\alpha(T)\big)$$

for which

$$\tilde{U}(T)\tilde{U}(\bar{T}) = \tilde{U}(T\bar{T}).$$

Any set of functions $\phi(T,\bar{T})$ that satisfy Eq. (2.7.2), and that differ only by functions $\Delta\phi(T,\bar{T})$ of the form (2.7.3), is called a 'two-cocycle'. A trivial cocycle is one that contains the function $\phi = 0$, and hence consists of functions of the form (2.7.3), which can be eliminated by a redefinition of $U(T)$. We are interested here in whether a symmetry group allows any non-trivial two-cocyles; that is, whether it may have a representation on the physical Hilbert space that is *intrinsically* projective, in the sense that the phase $\phi(T,\bar{T})$ *cannot* be eliminated in this way.

In order to answer this question, it is useful first to consider the effect of a phase ϕ in Eq. (2.7.1) on the commutation relations of the generators of infinitesimal transformations. When either $\bar{T}$ or T is the identity, the phase ϕ must clearly vanish

$$\phi(T,\mathbf{1}) = \phi(\mathbf{1},\bar{T}) = 0. \tag{2.7.4}$$

[9] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

When both T and $\bar{T}$ are *near* the identity the phase must be small. Using coordinates θ^a to parameterize group elements (as in Section 2.2), with $T(0) = \mathbf{1}$, Eq. (2.7.4) tells us that the expansion of $\phi[T(\theta),T(\bar{\theta})]$ around $\theta = \bar{\theta} = 0$ must start with terms of order $\theta\bar{\theta}$:

$$\phi\big(T(\theta),T(\bar{\theta})\big) = f_{ab}\theta^a\bar{\theta}^b + \cdots, \tag{2.7.5}$$

where f_{ab} are real numerical constants. Inserting this expansion in the power series expansion of Eq. (2.7.1), and repeating the steps that led to (2.2.22), we now have

$$[t_b,t_c] = iC^a{}_{bc}t_a + iC_{bc}\mathbf{1}, \tag{2.7.6}$$

where C_{bc} is the antisymmetric coefficient

$$C_{bc} = -f_{bc} + f_{cb}. \tag{2.7.7}$$

The appearance of terms on the right-hand side of the commutation relation proportional to the unit element (so-called *central charges*) is the counterpart for the Lie algebra of the presence of phases in a projective representation of a group.

The constants C_{bc} as well as $C^a{}_{bc}$ are subject to an important constraint, which follows from the Jacobi identity. Taking the commutator of (2.7.6) with t_d, and adding the same expressions with b,c,d replaced with c,d,b and d,b,c, the sum of the three double-commutators on the left-hand side vanishes identically, and so

$$C^a{}_{bc}C^e{}_{ad} + C^a{}_{cd}C^e{}_{ab} + C^a{}_{db}C^e{}_{ac} = 0 \tag{2.7.8}$$

and also

$$C^a{}_{bc}C_{ad} + C^a{}_{cd}C_{ab} + C^a{}_{db}C_{ac} = 0. \tag{2.7.9}$$

Eq. (2.7.9) always has one obvious class of non-zero solutions for C_{ab}:

$$C_{ab} = C^e{}_{ab}\phi_e, \tag{2.7.10}$$

where ϕ_e is an arbitrary set of real constants. For these solutions, we can eliminate the central charges from Eq. (2.7.6) by a redefinition of the generators

$$t_a \longrightarrow \tilde{t}_a \equiv t_a + \phi_a. \tag{2.7.11}$$

The new generators then satisfy the commutation relations without central charges

$$[\tilde{t}_b,\tilde{t}_c] = iC^a{}_{bc}\tilde{t}_a. \tag{2.7.12}$$

A given Lie algebra may or may not allow solutions of Eq. (2.7.9) other than Eq. (2.7.10).

We can now state the key theorem that governs the occurrence of intrinsically projective representations. The phase of any representation $U(T)$ of a given group can be chosen so that $\phi = 0$ in Eq. (2.7.1), if two conditions are met:

(a) The generators of the group in this representation can be redefined (as in Eq. (2.7.11)), so as to eliminate all central charges from the Lie algebra.

(b) The group is simply connected, i.e., any two group elements may be connected by a path lying within the group, and any two such paths may be continuously transformed into one another. (An equivalent statement is that any loop that starts and ends at the same group element may be shrunk continuously to a point.)

This theorem is proved in Appendix B of this chapter, which also offers comments about the case of groups that are not simply connected. It shows that there are just two (not exclusive) ways that intrinsically projective representations may arise: either algebraically, because the group is represented projectively even near the identity, or topologically, because the group is not simply connected, and hence a path from $\mathbf{1}$ to T and then from T to $\bar{T}$ may not be continuously deformable into some other path from $\mathbf{1}$ to $T\bar{T}$. In the latter case, the phase ϕ in Eq. (2.7.1) depends on the particular choice of standard paths, leading from the origin to the various group elements, that are used to define the corresponding U-operators.

Let's now consider each of these possibilities in turn for the special case of the inhomogeneous Lorentz group.

(A) Algebra

With central charges, the commutation relations of the generators of the inhomogeneous Lorentz group would read

$$i[J^{\mu\nu},J^{\rho\sigma}] = \eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - \eta^{\sigma\mu}J^{\rho\nu} + \eta^{\sigma\nu}J^{\rho\mu} + C^{\rho\sigma,\mu\nu}, \tag{2.7.13}$$

$$i[P^\mu,J^{\rho\sigma}] = \eta^{\mu\rho}P^\sigma - \eta^{\mu\sigma}P^\rho + C^{\rho\sigma,\mu}, \tag{2.7.14}$$

$$i[J^{\mu\nu},P^\rho] = \eta^{\nu\rho}P^\mu - \eta^{\mu\rho}P^\nu + C^{\rho,\mu\nu}, \tag{2.7.15}$$

$$i[P^\mu,P^\rho] = C^{\rho,\mu} \tag{2.7.16}$$

in place of Eqs. (2.4.12)–(2.4.14). We see that the Cs also satisfy the antisymmetry conditions

$$C^{\rho\sigma,\mu\nu} = -C^{\mu\nu,\rho\sigma}, \tag{2.7.17}$$

$$C^{\rho\sigma,\mu} = -C^{\mu,\rho\sigma}, \tag{2.7.18}$$

$$C^{\rho,\mu} = -C^{\mu,\rho}. \tag{2.7.19}$$

We will now show that all these constants have additional algebraic properties that allow them to be eliminated by shifting the definitions of $J^{\mu\nu}$ and P^μ by constant terms. (This corresponds to redefining the phase of the operators $U(\Lambda,a)$.) To derive these properties, we apply the Jacobi identities

$$[J^{\mu\nu},[P^\rho,P^\sigma]] + [P^\sigma,[J^{\mu\nu},P^\rho]] + [P^\rho,[P^\sigma,J^{\mu\nu}]] = 0, \tag{2.7.20}$$

$$[J^{\lambda\eta},[J^{\mu\nu},P^\rho]] + [P^\rho,[J^{\lambda\eta},J^{\mu\nu}]] + [J^{\mu\nu},[P^\rho,J^{\lambda\eta}]] = 0, \tag{2.7.21}$$

$$[J^{\lambda\eta},[J^{\mu\nu},J^{\rho\sigma}]] + [J^{\rho\sigma},[J^{\lambda\eta},J^{\mu\nu}]] + [J^{\mu\nu},[J^{\rho\sigma},J^{\lambda\eta}]] = 0. \tag{2.7.22}$$

[illegible]

Problems

References

3 Scattering Theory

3.1 'In' and 'Out' States

3.2 The S-Matrix

3.3 Symmetries of the S-Matrix

Lorentz Invariance

Internal Symmetries

Parity

values of α, β, and γ, then its inverse square root must also be a member of this group, say I_P, with $I_P^2 P^2 = \mathbf{1}$ and $[I_P, P] = 0$. (For instance, if $P^2 = \exp(i\alpha B + \cdots)$, then take $I_P = \exp(-\frac{1}{2}i\alpha B + \cdots)$.) We can then define a new parity operator $P' \equiv P I_P$ with $P'^2 = \mathbf{1}$. This is conserved to the same extent as P, so there is no reason why we should not call this *the* parity operator, in which case the intrinsic parities can only take the values ± 1.

The only sort of theory in which it is not necessarily possible to define parity so that all intrinsic parities have the values ± 1 is one in which there is some discrete internal symmetry which is not a member of any continuous symmetry group of phase transformations [10]. For instance, it is a consequence of angular-momentum conservation that the total number F of all particles of half-integer spin can only change by even numbers, so the internal symmetry operator $(-1)^F$ is conserved. All known particles of half-integer spin have odd values of the sum $B + L$ of baryon number and lepton number, so as far as we know, $(-1)^F = (-1)^{B+L}$. If this is true, then $(-1)^F$ is part of a continuous symmetry group, consisting of the operators $\exp(i\alpha(B+L))$ with arbitrary real α, and has an inverse square root $\exp(-i\pi(B+L)/2)$. In this case, if $P^2 = (-1)^F$ then P can be redefined so that all intrinsic parities are ± 1. However, if there were to be discovered a particle of half-integer spin and an even value of $B + L$ (such as a so-called Majorana neutrino, with $j = \frac{1}{2}$ and $B + L = 0$), then it would be possible to have $P^2 = (-1)^F$ without our being able to redefine the parity operator itself to have eigenvalues ± 1. In this case, of course, we would have $P^4 = \mathbf{1}$, so all particles would have intrinsic parities either ± 1 or (like the Majorana neutrino) $\pm i$.

It follows from Eq. (3.3.42) that if the product of intrinsic parities in the final state is equal to the product of intrinsic parities in the initial state, or equal to minus this product, then the S-matrix must be respectively even or odd overall in the three-momenta. For instance, it was observed [11] in 1951 that a pion can be absorbed by a deuteron from the $\ell = 0$ ground state of the $\pi^- d$ atom, in the reaction $\pi^- + d \to n + n$. (As discussed in Section 3.7, the orbital angular-momentum quantum number ℓ can be used in relativistic physics in the same way as in non-relativistic wave mechanics.) The initial state has total angular-momentum $j = 1$ (the pion and deuteron having spins zero and one, respectively), so the final state must have orbital angular-momentum $\ell = 1$ and total neutron spin $s = 1$. (The other possibilities $\ell = 1$, $s = 0$; $\ell = 0$, $s = 1$; and $\ell = 2$, $s = 1$, which are allowed by angular-momentum conservation, are forbidden by the requirement that the final state be antisymmetric in the two neutrons.) Because the final state has $\ell = 1$, the matrix element is odd under reversal of the direction of all three-momenta, so we can conclude that the intrinsic parities of the particles in this reaction must be related by:

$$\eta_d \eta_{\pi^-} = -\eta_n^2 .$$

The deuteron is known to be a bound state of a proton and neutron with even orbital angular-momentum (chiefly $\ell = 0$), and as we have seen we can take the neutron and proton to have the same intrinsic parity, so $\eta_d = \eta_n^2$, and we can conclude that $\eta_{\pi^-} = -1$; that is, the negative pion is a *pseudoscalar* particle. The π^+ and π^0 have also been found to

have negative parity, as would be expected from the symmetry (isospin invariance) among these three particles.

The negative parity of the pion has some striking consequences. A spin zero particle that decays into three pions must have intrinsic parity $\eta_\pi^3 = -1$, because in the Lorentz frame in which the decaying particle is at rest, rotational invariance only allows the matrix element to depend on scalar products of the pion momenta with each other, all of which are even under reversal of all momenta. (The triple scalar product $\mathbf{p}_1 \cdot (\mathbf{p}_2 \cdot \mathbf{p}_3)$ formed from the three pion momenta vanishes because $\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 = 0$.) For the same reason, a spin zero particle that decays into two pions must have intrinsic parity $\eta_\pi^2 = +1$. In particular, among the strange particles discovered in the late 1940s there seemed to be two different particles of zero spin (inferred from the angular distribution of their decay products): one, the τ, was identified by its decay into three pions, and hence was assigned a parity -1, while the other, the θ, was identified by its decay into two pions and was assigned a parity $+1$. The trouble with all this was that as the τ and θ were studied in greater detail, they seemed increasingly to have identical masses and lifetimes. After many suggested solutions of this puzzle, Lee and Yang in 1956 finally cut the Gordian knot, and proposed that the τ and θ are the same particle, (now known as the $K^\pm$) and that parity is simply not conserved in the weak interactions that lead to its decay [12].

As we shall see in detail in the next section of this chapter, the rate for a physical process $\alpha \to \beta$ (with $\alpha \neq \beta$) is proportional to $|S_{\beta\alpha}|^2$, with proportionality factors that are invariant under reversal of all three-momenta. As long as the states α and β contain definite numbers of particles of each type, the phase factors in Eq. (3.3.42) have no effect on $|S_{\beta\alpha}|^2$, so Eq. (3.3.42) would imply that the rate for $\alpha \to \beta$ is invariant under the reversal of direction of all three-momenta. As we have seen, this is a trivial consequence of rotational invariance for the decays of a K meson into two or three pions, but it is a non-trivial restriction on rates in more complicated processes. For example, following theoretical suggestions by Lee and Yang, Wu together with a group at the National Bureau of Standards measured the angular distribution of the electron in the final state of the beta decay $Co^{60} \to Ni^{60} + e^- + \bar{\nu}$ with a polarized cobalt source [13]. (No attempt was made in this experiment to measure the momentum of the antineutrino or nickel nucleus.) The electrons were found to be preferentially emitted in a direction opposite to that of the spin of the decaying nucleus, which would, of course, be impossible if the decay rate were invariant under a reversal of all three-momenta. A similar result was found in the decay of a positive muon (polarized in its production in the process $\pi^+ \to \mu^+ + \nu$) into a positron, neutrino, and antineutrino [14]. In this way, it became clear that parity is indeed not conserved in the weak interactions responsible for these decays. Nevertheless, for reasons discussed in Section 12.5, parity *is* conserved in the strong and electromagnetic interactions, and therefore continues to play an important part in theoretical physics.

Time-Reversal

We saw in Section 2.6 that the time-reversal operator T acting on a one-particle state $|p, \sigma, n\rangle$ gives a state $|\mathcal{P}p, -\sigma, n\rangle$ with reversed spin and momentum, times a phase $\zeta_n(-1)^{j-\sigma}$. A multi-particle state transforms as usual as a direct product of one-particle states, except

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eigenstates of the strong interaction S-matrix (with $\delta_\alpha = \delta_\beta = 0$). Thus if time-reversal invariance is respected, the differential rate for a beta decay process should be unchanged if we reverse both the momenta and the spin z-components σ of all particles. This prediction was *not* contradicted in the 1956 experiments [13,14] that discovered the non-conservation of parity; for instance, time-reversal invariance is consistent with the observation that electrons from the decay $Co^{60} \to Ni^{60} + e^- + \bar{\nu}$ are preferentially emitted in a direction opposite to that of the Co^{60} spin. As described below, indirect evidence against time-reversal invariance did emerge in 1964, but it remains a useful approximate symmetry in weak as well as strong and electromagnetic interactions.

In some cases we can use a basis of states for which $\Theta\alpha = \alpha$ and $\Theta\beta = \beta$, for which Eq. (3.3.54) reads

$$S^{(1)}_{\beta\alpha} = -e^{2i(\delta_\alpha+\delta_\beta)} S^{(1)*}_{\beta\alpha} , \tag{3.3.55}$$

which just says that $iS^{(1)}_{\beta\alpha}$ has the phase $\delta_\alpha + \delta_\beta$ mod π. This is known as *Watson's theorem* [15]. The phases in Eqs. (3.3.54) or (3.3.55) may be measured in processes where there is interference between different final states. For instance, in the decay of the spin 1/2 hyperon Λ into a nucleon and a pion, the final state can only have orbital angular-momentum $\ell = 0$ or $\ell = 1$; the angular distribution of the pion relative to the Λ spin involves the interference between these states, and hence according to Watson's theorem depends on the difference $\delta_s - \delta_p$ of their phase shifts.

PT

Although the 1957 experiments on parity violation did not rule out time-reversal invariance, they did show immediately that the product PT is not conserved. If conserved this operator would have to be antiunitary for the same reasons as for T, so in processes like nuclear beta decay its consequences would take the form of relations like Eq. (3.3.54):

$$S^{(1)}_{\beta\alpha} = -e^{i(\delta_\alpha+\delta_\beta)} S^{(1)*}_{\mathcal{P}\Theta\beta,\mathcal{P}\Theta\alpha} ,$$

where $\mathcal{P}\Theta$ reverses the signs of all spin z-components but *not* any momenta. Neglecting the final-state Coulomb interaction, it would then follow that there could be no preference for the electron in the decay $Co^{60} \to Ni^{60} + e^- + \bar{\nu}$ to be emitted in the same or the opposite direction to the Co^{60} spin, in contradiction with what was observed.

C*, *CP*, and *CPT

As already mentioned, there is an internal symmetry transformation, known as charge-conjugation, which interchanges particles and antiparticles. Formally, this entails the existence of a unitary operator C, whose effect on multi-particle states is:

$$C|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \ldots\rangle_{\text{in/out}} = \xi_{n_1}\xi_{n_2}\cdots|p_1, \sigma_1, n_1^c; p_2, \sigma_2, n_2^c; \ldots\rangle_{\text{in/out}} , \tag{3.3.56}$$

where n^c is the antiparticle of particle type n, and ξ_n is yet another phase. If this is true [illegible]

As with other internal symmetries, the S-matrix will satisfy this condition if the operator C_0 that is defined to act as stated in Eq. (3.3.56) on free-particle states commutes with the interaction H_{int} as well as H_0; in this case, we take $C = C_0$.

The phases ξ_n are called charge-conjugation parities. Just as for the ordinary parities η_n, the ξ_n are in general not uniquely defined, because for any operator C that is defined to satisfy Eq. (3.3.56), we can find another such operator with different ξ_n by multiplying C with any internal symmetry phase transformation, such as $\exp(i\alpha B + i\beta L + i\gamma Q)$; the only particles whose charge-conjugation parities are individually measurable are those completely neutral particles like the photon or the neutral pion that carry no conserved quantum numbers and are their own antiparticles. In reactions involving only completely neutral particles, Eq. (3.3.57) tells us that the product of the charge-conjugation parities in the initial and final states must be equal; for instance, as we shall see the photon is required by quantum electrodynamics to have charge-conjugation parity $\eta_\gamma = -1$, so the observation of the neutral pion decay $\pi^0 \to 2\gamma$ requires that $\eta_{\pi^0} = +1$; it then follows that the process $\pi^0 \to 3\gamma$ should be forbidden, as is in fact known to be the case. For these two particles, the charge-conjugation parities are real, either $+1$ or -1. Just as for ordinary parity, this will always be the case if all internal phase transformation symmetries are members of continuous groups of phase transformations, because then we can redefine C by multiplying by the inverse square root of the internal symmetry equal to C^2, with the result that the new C satisfies $C^2 = \mathbf{1}$.

For general reactions, Eq. (3.3.57) requires that the [illegible]

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The differential scattering cross-section is $|f|^2$. We will take the direction of the initial momentum $\mathbf{k}$ to be along the three-direction, in which case

$$Y_\ell^m(\hat{\mathbf{k}}) = \delta_{m0}\sqrt{\frac{2\ell+1}{4\pi}}. \tag{3.7.9}$$

Integrating $|f|^2$ over the direction of the final momentum $\mathbf{k}'$ and respectively summing and averaging over final and initial spin three-components gives the total cross-section[24] for transitions from channel n to n':

$$\sigma(n \to n'; E) = \frac{\pi}{k^2(2s_1+1)(2s_2+1)} \sum_{j\ell s\ell' s'} (2j+1) \cdot \left|\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell's'n',\ell sn}(E)\right|^2. \tag{3.7.10}$$

Summing (3.7.10) over all two-body channels gives the total cross-section for all elastic or inelastic two-body reactions:

$$\sum_{n'} \sigma(n \to n'; E) = \frac{\pi}{k^2(2s_1+1)(2s_2+1)} \sum_{j\ell s}(2j+1) \cdot \left[(\mathbf{1} - S^j(E))^\dagger(\mathbf{1} - S^j(E))\right]_{\ell sn,\ell sn}. \tag{3.7.11}$$

For comparison, Eqs. (3.7.8), (3.7.9), (3.7.4), and the Clebsch–Gordan sum rules give the spin-averaged forward scattering amplitude as

$$f(n;E) = \frac{i}{2k(2s_1+1)(2s_2+1)} \sum_{j\ell s}(2j+1)[\mathbf{1} - S^j(E)]_{\ell sn,\ell sn}.$$

The optical theorem (3.6.10) then gives the total cross-section as

$$\sigma_{\text{total}}(n;E) = \frac{2\pi}{k^2(2s_1+1)(2s_2+1)} \sum_{j\ell s}(2j+1)\,\text{Re}[\mathbf{1} - S^j(E)]_{\ell sn,\ell sn}. \tag{3.7.12}$$

If only two-body channels can be reached from the channel n at energy E, then the matrix $S^j(E)$ (or at least some submatrix that includes channel n) is unitary, and thus

$$\left[(\mathbf{1} - S^j(E))^\dagger(\mathbf{1} - S^j(E))\right]_{\ell sn,\ell sn} = 2\,\text{Re}[\mathbf{1} - S^j(E)]_{\ell sn,\ell sn}, \tag{3.7.13}$$

[24] In deriving this result, we use standard sum rules [9] for the Clebsch–Gordan coefficients: first

$$\sum_{\sigma_1\sigma_2} C_{s_1s_2}(s,\mu;\sigma_1,\sigma_2)C_{s_1s_2}(\bar{s},\bar{\mu};\sigma_1,\sigma_2) = \delta_{s\bar{s}}\delta_{\mu\bar{\mu}},$$

and the same with primes; then

$$\sum_{m\sigma} C_{\ell s}(j,\sigma;m,\sigma)C_{\ell' s}(\bar{j},\sigma;m,\sigma) = \delta_{j\bar{j}}\delta_{\sigma\bar{\sigma}},$$

and finally

$$\sum_{\sigma\mu} C_{\ell s}(j,\sigma;0,\mu)C_{\ell' s}(j,\sigma;0,\mu) = \frac{2j+1}{2\ell+1}\delta_{\ell\ell'}.$$

so (3.7.12) and (3.7.11) are equal. On the other hand, if channels involving three or more particles are open, then the difference of (3.7.12) and (3.7.11) gives the total cross-section for producing extra particles:

$$\sigma_{\text{production}}(n;E) = \frac{\pi}{k^2(2s_1+1)(2s_2+1)} \sum_{j\ell s}(2j+1)[\mathbf{1} - S^j(E)^\dagger S^j(E)]_{\ell sn,\ell sn}, \tag{3.7.14}$$

and this must be positive.

The partial wave expansion is particularly useful when applied to processes where the relevant part of the S-matrix is diagonal. This is the case, for instance, if the initial channel n contains just two spinless particles, and no other channels are open at this energy, as in $\pi^+ - \pi^+$ or $\pi^+ - \pi^0$ scattering at energies below the threshold for producing extra pions (provided one ignores weak and electromagnetic interactions). For a pair of spinless particles we have $j = \ell$, and angular momentum conservation keeps the S-matrix diagonal. It is also possible for the S-matrix to be diagonal in certain processes involving particles with spin; for instance in pion–nucleon scattering we can have $j = \ell + \frac{1}{2}$ or $j = \ell - \frac{1}{2}$, but for a given j these two states have opposite parity, so they cannot be connected by non-zero S-matrix elements. In any case, if for some n and E the S-matrix elements $S^j_{N',\ell sn}(E, \mathbf{0})$ all vanish unless N' is the two-body state j, ℓ, s, n, then unitarity requires that

$$S^j_{\ell's'n',\ell sn}(E) = \exp[2i\delta_{j\ell sn}(E)]\delta_{\ell'\ell}\delta_{s's}\delta_{n'n}, \tag{3.7.15}$$

where $\delta_{j\ell sn}(E)$ is a real phase, commonly known as the *phase shift*. This formula is also often used where the two-body part of the S-matrix is diagonal but channels containing three or more particles are also open; in such cases the phase shift must have a positive imaginary part, to keep (3.7.14) positive. For real phase shifts, the elastic and total cross-sections are then given by Eq. (3.7.10) or Eq. (3.7.12) as:

$$\sigma(n \to n; E) = \sigma_{\text{total}}(n;E) = \frac{4\pi}{k^2(2s_1+1)(2s_2+1)} \sum_{j\ell s}(2j+1)\sin^2\delta_{j\ell sn}(E). \tag{3.7.16}$$

This familiar result is usually derived in non-relativistic quantum mechanics by studying the coordinate space wave function for a particle in a potential. The derivation given here is offered both to show that the partial wave expansion applies for elastic scattering even at relativistic velocities, and also to emphasize that it depends on no particular dynamical assumptions, only on unitarity and invariance principles.

It is also often useful to introduce phase shifts in dealing with problems where several channels are open, forming a few irreducible representations of some internal symmetry group. The classic example of such an internal symmetry is isotopic spin symmetry, for which the channel index n includes a specification of the isospins T_1, T_2 of the two particles together with their three-components t_1, t_2; the states in channel n may be expressed as linear combinations of the tth components of irreducible representations T, with coefficients given by the familiar Clebsch–Gordan coefficients $C_{T_1T_2}(T,t;t_1,t_2)$. Suppose that for the channels and energy of interest the S-matrix is diagonal in ℓ and s as well as j, T, and t. Unitarity and isospin symmetry then allow us to write the S-matrix as

$$S^j_{\ell's'T't',\ell sTt} = \exp[2i\delta_{j\ell sT}(E)]\delta_{\ell'\ell}\delta_{s's}\delta_{T'T}\delta_{t't}, \tag{3.7.17}$$

with $\delta_{j\ell sT}(E)$ a real phase shift, t-independent according to the Wigner–Eckart theorem. The partial cross-sections can again be calculated from Eq. (3.7.10), and the total cross-section is given by Eq. (3.7.12) as

$$\sigma_{\text{total}}(t_1,t_2;E) = \frac{4\pi}{k^2(2s_1+1)(2s_2+1)} \cdot \sum_{j\ell sTt}(2j+1)C_{T_1T_2}(T,t;t_1,t_2)^2\sin^2\delta_{j\ell sT}(E). \tag{3.7.18}$$

For instance, in pion–pion scattering we have phase shifts $\delta_{\ell\ell 0T}(E)$ with $T = 0$ or $T = 2$ for each even ℓ and $T = 1$ for each odd ℓ, while for pion–nucleon scattering we have phase shifts $\delta_{j,j\pm\frac{1}{2},\frac{1}{2},T}$ with $T = \frac{1}{2}$ or $T = \frac{3}{2}$.

We can gain some useful insight about the threshold behavior of the scattering amplitudes and phase shifts from considerations of analyticity that are nearly independent of any dynamical assumptions. Unless there are special circumstances that would produce singularities in momentum space, we would expect the matrix element $M_{\mathbf{k}'\sigma_1',-\mathbf{k}'\sigma_2',n';\mathbf{k}\sigma_1,-\mathbf{k}\sigma_2,n}$ to be an analytic function[25] of the three-momenta $\mathbf{k}$ and $\mathbf{k}'$ near $k = 0$ or $k' = 0$ or (for elastic scattering) $k = k' = 0$. Turning to the partial wave expansion (3.7.8) for M, we note that $k^\ell Y_\ell^m(\hat{\mathbf{k}})$ is a simple polynomial function of the three-vector $\mathbf{k}$, so in order for the matrix element to be an analytic function of the three-momenta $\mathbf{k}$ and $\mathbf{k}'$ near $k = 0$ or $k' = 0$, the coefficients $M^j_{\ell's'n',\ell sn}$ or equivalently $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell's'n',\ell sn}$ must go as $k^{\ell+1/2}k'^{\ell'+1/2}$ when k and/or k' go to zero. Hence for small k and/or k', it is only the lowest partial waves in the initial and/or final state that contribute appreciably to the scattering amplitude. We have three possible cases:

Exothermic reactions

Here k' approaches a finite value as $k \to 0$, and in this limit $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell's'n',\ell sn}$ goes as $k^{\ell+1/2}$. The cross-section (3.7.11) in this case goes as $k^{2\ell-1}$, where ℓ is here the *lowest* orbital angular momentum that can lead to the reaction. In the most usual case $\ell = 0$, so the reaction cross-section goes as $1/k$. (This is the case, for instance, in the absorption of slow neutrons by complex nuclei, or for the annihilation of electron–positron pairs into photons at low energy, aside from the higher-order effects of Coulomb forces.) The reaction rate is the cross-section times the flux, which goes like k, so the rate for an exothermic reaction behaves like a constant for $k \to 0$. However, it is the cross-section rather than the reaction rate that determines the probability of absorption when a beam crosses a given thickness of target material, and the factor $1/k$ makes this probability very high for slow neutrons in an absorbing material like boron.

Endothermic Reactions

[25] For instance, in the Born approximation (3.2.8), M is proportional to the Fourier transform of the coordinate space matrix elements of the interaction, and hence is analytic at zero momenta as long as these matrix elements fall off sufficiently rapidly at large separations. The chief exception is for scattering involving long-distance forces, such as the Coulomb force.

Here the reaction is forbidden until k reaches a finite threshold value, where $k' = 0$. Just above this threshold $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell's'n',\ell sn}$ goes as $(k')^{\ell'+1/2}$. The cross-section (3.7.11) in this case goes as $(k')^{2\ell'+1}$, where ℓ' is here the *lowest* orbital angular momentum that can be produced at threshold. In the most usual case $\ell' = 0$, so the reaction cross-section rises above threshold like k', and hence like $\sqrt{E - E_{\text{threshold}}}$. (This is the case, for instance, in the associated production of strange particles, or the production of electron–positron pairs in the scattering of photons.)

Elastic Reactions

Here $k = k'$, so k and k' go to zero together. (This is the case where $n' = n$, or where n' consists of particles in the same isotopic spin multiplets as are those in n.) In elastic scattering the partial waves with $\ell = \ell' = 0$ are always present, so in the limit $k \to 0$ the scattering amplitude (3.7.8) becomes a constant:

$$f(\mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n \to \mathbf{k}'\sigma_1', -\mathbf{k}'\sigma_2', n') \to \sum_{s\sigma} C_{s_1s_2}(s,\sigma;\sigma_1,\sigma_2)C_{s_1's_2'}(s,\sigma;\sigma_1',\sigma_2')a_s(n \to n'), \tag{3.7.19}$$

where a is a constant, known as the *scattering length*, defined by the limit

$$S^s_{0sn',0sn} \longrightarrow \delta_{n'n} + 2ika_s(n \to n') \qquad (k = k' \to 0). \tag{3.7.20}$$

Summing $4\pi|f|^2$ over final spins and averaging over initial spins gives the total cross-section for the transition $n \to n'$ at $k = k' = 0$:

$$\sigma(n \to n'; k = 0) = \frac{4\pi}{(2s_1+1)(2s_2+1)} \sum_s (2s+1)|a_s(n \to n')|^2. \tag{3.7.21}$$

The classic instance of the use of this formula is in neutron–proton scattering, where there are two scattering lengths, with the spin singlet length a_0 considerably larger than the spin triplet length a_1.

The partial wave expansion can also be used to make a crude guess about the behavior of cross-sections at high energy. With decreasing wavelengths, we may expect scattering to be described more or less classically: a particle of momentum k and orbital angular momentum ℓ would have an impact parameter ℓ/k, and will therefore strike a disk of radius R if $\ell \leq kR$. This can be interpreted as a statement about S-matrix elements:

$$S^j_{\ell's'n',\ell sn} \longrightarrow \begin{cases} 0, & \ell \ll kR_n, \\ 1, & \ell \gg kR_n, \end{cases} \tag{3.7.22}$$

where R_n is some sort of interaction radius for channel n. For a given $\ell \gg s$, there are $2s+1$ values of j, all close enough to ℓ to approximate $2j+1 \simeq 2\ell+1$, so the sum over j and s in Eq. (3.7.12) merely gives a factor of order

$$\sum_{js}(2j+1) = (2\ell+1)\sum_s(2s+1) = (2\ell+1)(2s_1+1)(2s_2+1).$$

[illegible]

to have the scalar products (3.7.6).

[illegible]

with an upper or lower sign if n is a boson or a fermion, respectively, and dots representing other particles that may be present in the state. (Equivalently, this could be stated as a condition on the 'wave functions,' the coefficients of these multi-particle basis vectors in physically allowable state-vectors.) These two cases are often referred to as Bose or Fermi 'statistics.' We will see in the next chapter that Bose and Fermi statistics are only possible for particles that have integer or half-integer spins, respectively, but we shall not need this information in the present chapter. In this section we shall offer a non-rigorous argument that all particles must be either bosons or fermions, and then set up normalization conditions for multi-boson or multi-fermion states.

First note that if two particles with spins and momenta $\mathbf{p}, \sigma$ and $\mathbf{p}', \sigma'$ belong to identical species n, then the state-vectors $|\ldots \mathbf{p}\sigma n \ldots \mathbf{p}'\sigma' n \ldots\rangle$ and $|\ldots \mathbf{p}'\sigma' n \ldots \mathbf{p}\sigma n \ldots\rangle$ represent the same physical state; if this were not the case then the particles would be distinguished by their order in the labelling of the state-vector, and the first listed would not be identical with the second. Since the two state-vectors are physically indistinguishable, they must belong to the same ray, and so

$$|\ldots \mathbf{p}\sigma n \ldots \mathbf{p}'\sigma' n \ldots\rangle = \alpha_n|\ldots \mathbf{p}'\sigma' n \ldots \mathbf{p}\sigma n \ldots\rangle, \tag{4.1.2}$$

where α_n is a complex number of unit absolute value. We may regard this as part of the definition of what we mean by identical particles.

The crux of the matter is to decide on what the phase factor α_n may depend. *If* it depends only on the species index n, then we are nearly done. Interchanging the two particles in Eq. (4.1.2) again, we find

$$|\ldots \mathbf{p}\sigma n \ldots \mathbf{p}'\sigma' n \ldots\rangle = \alpha_n^2|\ldots \mathbf{p}\sigma n \ldots \mathbf{p}'\sigma' n \ldots\rangle$$

so that $\alpha_n^2 = 1$, yielding Eq. (4.1.1) as the only two possibilities.

On what else could α_n depend? It might depend on the numbers and species of the other particles in the state (indicated by dots in Eqs. (4.1.1) and (4.1.2)), but this would lead to

[illegible]

The sum here is over all permutations $\mathcal{P}$ of the integers $1, 2, \ldots, N$. (For instance, in the first term in Eq. (4.1.6), $\mathcal{P}$ is the identity, $\mathcal{P}1 = 1$, $\mathcal{P}2 = 2$, while in the second term $\mathcal{P}1 = 2$, $\mathcal{P}2 = 1$.) Also, $\delta_{\mathcal{P}}$ is a sign factor equal to -1 if $\mathcal{P}$ involves an odd permutation of fermions (an odd number of fermion interchanges) and $+1$ otherwise. It is easy to see that Eq. (4.1.7) has the desired symmetry or antisymmetry properties under interchange of the q_i, and also under interchange of the q'_j.

4.2 Creation and Annihilation Operators

Creation and annihilation operators may be defined in terms of their effect on the normalized multi-particle states discussed in the previous section. The *creation operator* $a^\dagger_q$ (or in more detail, $a^\dagger_{\mathbf{p},\sigma,n}$) is defined as the operator that simply adds a particle with quantum numbers q at the front of the list of particles in the state

$$a^\dagger_q |q_1 q_2 \cdots q_N\rangle = |q q_1 q_2 \cdots q_N\rangle. \qquad (4.2.1)$$

In particular, the N-particle state can be obtained by acting on the vacuum with N creation operators

$$a^\dagger_{q_1} a^\dagger_{q_2} \cdots a^\dagger_{q_N} |\Omega\rangle = |q_1 \cdots q_N\rangle. \qquad (4.2.2)$$

It is conventional for this operator to be called $a^\dagger_q$; its adjoint, which is then called a_q, may be calculated from Eq. (4.1.7). As we shall now show, a_q removes a particle from any state on which it acts, and is therefore known as an *annihilation operator*. In particular, when the particles $q, q_1, \ldots, q_N$ are either all bosons or all fermions, we have

$$a_q |q_1 q_2 \cdots q_N\rangle = \sum_{r=1}^{N} (\pm)^{r+1} \delta(q - q_r) |q_1 \cdots q_{r-1} q_{r+1} \cdots q_N\rangle, \qquad (4.2.3)$$

with a $+1$ or -1 sign for bosons or fermions, respectively. (Here is the proof. We want to calculate the scalar product of $a_q|q_1 q_2 \cdots q_N\rangle$ with an arbitrary state $|q'_1 \cdots q'_M\rangle$. Using Eq. (4.2.1), this is

$$\langle q'_1 \cdots q'_M | a_q | q_1 \cdots q_N\rangle = \left(a^\dagger_q |q'_1 \cdots q'_M\rangle\right)^\dagger |q_1 \cdots q_N\rangle = \langle q q'_1 \cdots q'_M | q_1 \cdots q_N\rangle.$$

We now use Eq. (4.1.7). The sum over permutations $\mathcal{P}$ of $1, 2, \ldots, N$ can be written as a sum over the integer r that is permuted into the first place, i.e. $\mathcal{P}r = 1$, and over mappings $\overline{\mathcal{P}}$ of the remaining integers $1, \ldots, r-1, r+1, \ldots, N$ into $1, \ldots, N-1$. Furthermore, the sign factor is

$$\delta_{\mathcal{P}} = (\pm)^{r-1} \delta_{\overline{\mathcal{P}}}$$

with upper and lower signs for bosons and fermions, respectively. Hence, using Eq. (4.1.7) twice,

$$\begin{aligned}\langle q'_1 \cdots q'_M | a_q | q_1 \cdots q_N\rangle &= \delta_{N,M+1} \sum_{r=1}^{N} \sum_{\overline{\mathcal{P}}} (\pm)^{r-1} \delta_{\overline{\mathcal{P}}} \delta(q - q_r) \prod_{i=1}^{M} \delta(q'_i - q_{\overline{\mathcal{P}}i}) \\ &= \delta_{N,M+1} \sum_{r=1}^{N} (\pm)^{r-1} \langle q'_1 \cdots q'_M | q_1 \cdots q_{r-1} q_{r+1} \cdots q_N\rangle.\end{aligned}$$

Both sides of Eq. (4.2.3) thus have the same matrix element with any state $|q'_1 \cdots q'_M\rangle$, and are therefore equal, as was to be shown.) As a special case of Eq. (4.2.3), we note that for both bosons and fermions, a_q annihilates the vacuum

$$a_q |\Omega\rangle = 0. \qquad (4.2.4)$$

As defined here, the creation and annihilation operators satisfy an important commutation or anticommutation relation. Applying the operator $a_{q'}$ to Eq. (4.2.1) and using Eq. (4.2.3) gives

$$\begin{aligned}a_{q'} a^\dagger_q |q_1 \cdots q_N\rangle &= \delta(q' - q) |q_1 \cdots q_N\rangle \\ &\quad + \sum_{r=1}^{N} (\pm)^{r+2} \delta(q' - q_r) |q q_1 \cdots q_{r-1} q_{r+1} \cdots q_N\rangle.\end{aligned}$$

(The sign in the second term is $(\pm)^{r+2}$ because q_r is in the $(r+1)$-th place in $|q q_1 \cdots q_N\rangle$.) On the other hand, applying the operator $a^\dagger_q$ to Eq. (4.2.3) gives

$$a^\dagger_q a_{q'} |q_1 \cdots q_N\rangle = \sum_{r=1}^{N} (\pm)^{r+1} \delta(q' - q_r) |q q_1 \cdots q_{r-1} q_{r+1} \cdots q_N\rangle.$$

Subtracting or adding, we have then

$$\left[a_{q'} a^\dagger_q \mp a^\dagger_q a_{q'}\right] |q_1 \cdots q_N\rangle = \delta(q' - q) |q_1 \cdots q_N\rangle.$$

But this holds for all states $|q_1 \cdots q_N\rangle$ (and may easily be seen to hold also for states containing both bosons and fermions) and therefore implies the operator relation

$$a_{q'} a^\dagger_q \mp a^\dagger_q a_{q'} = \delta(q' - q). \qquad (4.2.5)$$

In addition, Eq. (4.2.2) gives immediately

$$a^\dagger_{q'} a^\dagger_q \mp a^\dagger_q a^\dagger_{q'} = 0 \qquad (4.2.6)$$

and so also

$$a_{q'} a_q \mp a_q a_{q'} = 0. \qquad (4.2.7)$$

As always, the top and bottom signs apply for bosons and fermions, respectively. According to the conventions discussed in the previous section, the creation and/or annihilation operators for particles of two different species commute if either particle is a boson, and anticommute if both are fermions.

The above discussion could have been presented in reverse order (and in most textbooks usually is). That is, we could have started with the commutation or anticommutation relations Eqs. (4.2.5)–(4.2.7), derived from the canonical quantization of some given field theory. Multi-particle states would have then been defined by Eq. (4.2.2), and their scalar products Eq. (4.1.7) derived from the commutation or anticommutation relations. In fact, as discussed in Chapter 1, such a treatment would be much closer to the way that this

formalism developed historically. We have followed an unhistorical approach here because we want to free ourselves from any dependence on pre-existing field theories, and rather wish to understand why field theories are the way they are.

We will now prove the fundamental theorem quoted at the beginning of this chapter: *any operator $\mathcal{O}$ may be expressed as a sum of products of creation and annihilation operators*

$$\begin{aligned}\mathcal{O} = \sum_{N=0}^{\infty} \sum_{M=0}^{\infty} \int & dq'_1 \cdots dq'_N \, dq_1 \cdots dq_M \\ & \cdot a^\dagger_{q'_1} \cdots a^\dagger_{q'_N} a_{q_M} \cdots a_{q_1} \\ & \cdot C_{NM}(q'_1 \cdots q'_N q_1 \cdots q_M).\end{aligned} \qquad (4.2.8)$$

That is, we want to show that the C_{NM} coefficients can be chosen to give the matrix elements of this expression any desired values. We do this by mathematical induction. First, it is trivial that by choosing C_{00} properly, we can give $\langle\Omega|\mathcal{O}|\Omega\rangle$ any desired value, irrespective of the values of C_{NM} with $N > 0$ and/or $M > 0$. We need only use Eq. (4.2.4) to see that Eq. (4.2.8) has the vacuum expectation value

$$\langle\Omega|\mathcal{O}|\Omega\rangle = C_{00}.$$

Now suppose that the same is true for all matrix elements of $\mathcal{O}$ between N- and M-particle states, with $N < L, M \le K$ or $N \le L, M < K$; that is, that these matrix elements have been given some desired values by an appropriate choice of the corresponding coefficients C_{NM}. To see that the same is then also true of matrix elements of $\mathcal{O}$ between any L- and K-particle states, use Eq. (4.2.8) to evaluate

$$\begin{aligned}\langle q'_1 \cdots q'_L | \mathcal{O} | q_1 \cdots q_K\rangle &= L!K! C_{LK}(q'_1 \cdots q'_L q_1 \cdots q_K) \\ &\quad + \text{terms involving } C_{NM} \text{ with } N < L, M \le K \text{ or } N \le L, M < K.\end{aligned}$$

Whatever values have already been given to C_{NM} with $N < L, M \le K$ or $N \le L, M < K$, there is clearly some choice of C_{LK} which gives this matrix element any desired value.

Of course, an operator need not be expressed in the form (4.2.8), with all creation operators to the left of all annihilation operators. (This is often called the 'normal' order of the operators.) However, if the formula for some operator has the creation and annihilation operators in some other order, we can always bring the creation operators to the left of the annihilation operators by repeated use of the commutation or anticommutation relations, picking up new terms from the delta function in Eq. (4.2.5).

For instance, consider any sort of additive operator F (like momentum, charge, etc.) for which

$$F|q_1 \cdots q_N\rangle = \left(f(q_1) + \cdots + f(q_N)\right) |q_1 \cdots q_N\rangle. \qquad (4.2.9)$$

Such an operator can be written as in Eq. (4.2.8), but using only the term with $N = M = 1$:

$$F = \int dq \, a^\dagger_q a_q f(q). \qquad (4.2.10)$$

In particular, the free-particle Hamiltonian is always

$$H_0 = \int dq \, a^\dagger_q a_q E(q) \qquad (4.2.11)$$

where $E(q)$ is the single-particle energy

$$E(\mathbf{p}, \sigma, n) = \sqrt{\mathbf{p}^2 + m_n^2}.$$

We will need the transformation properties of the creation and annihilation operators for various symmetries. First, let's consider inhomogeneous proper orthochronous Lorentz transformations. Recall that the N-particle states have the Lorentz transformation property

$$\begin{aligned}U_0(\Lambda, a) |\mathbf{p}_1 \sigma_1 n_1; \mathbf{p}_2 \sigma_2 n_2; \ldots\rangle &= e^{-i(\Lambda p_1)\cdot a} e^{-i(\Lambda p_2)\cdot a} \cdots \sqrt{\frac{(\Lambda p_1)^0 (\Lambda p_2)^0 \cdots}{p_1^0 p_2^0 \cdots}} \\ &\quad \cdot \sum_{\bar\sigma_1 \bar\sigma_2 \cdots} D^{(j_1)}_{\bar\sigma_1 \sigma_1}(W(\Lambda, p_1)) D^{(j_2)}_{\bar\sigma_2 \sigma_2}(W(\Lambda, p_2)) \cdots \\ &\quad \cdot |\mathbf{p}_{1\Lambda} \bar\sigma_1 n_1; \mathbf{p}_{2\Lambda} \bar\sigma_2 n_2; \ldots\rangle.\end{aligned}$$

Here $\mathbf{p}_\Lambda$ is the three-vector part of Λp, $D^{(j)}_{\bar\sigma\sigma}(R)$ is the same unitary spin-j representation of the three-dimensional rotation group as used in Section 2.5, and $W(\Lambda, p)$ is the particular rotation

$$W(\Lambda, p) = L^{-1}(\Lambda p) \Lambda L(p),$$

where $L(p)$ is the standard 'boost' that takes a particle of mass m from rest to four-momentum p^μ. (Of course, m and j depend on the species label n. This is all for $m \neq 0$; we will return to the massless particle case in the following chapter.) Now, these states can be expressed as in Eq. (4.2.2)

$$|\mathbf{p}_1 \sigma_1 n_1; \mathbf{p}_2 \sigma_2 n_2; \ldots\rangle = a^\dagger_{\mathbf{p}_1, \sigma_1, n_1} a^\dagger_{\mathbf{p}_2, \sigma_2, n_2} \cdots |\Omega\rangle,$$

where $|\Omega\rangle$ is the Lorentz-invariant vacuum state

$$U_0(\Lambda, a) |\Omega\rangle = |\Omega\rangle.$$

In order that the state (4.2.2) should transform properly, it is necessary and sufficient that the creation operator have the transformation rule

$$\begin{aligned}U_0(\Lambda, a) a^\dagger_{\mathbf{p},\sigma,n} U_0^{-1}(\Lambda, a) = e^{-i(\Lambda p)\cdot a} \sqrt{\frac{(\Lambda p)^0}{p^0}} \sum_{\bar\sigma} D^{(j)}_{\bar\sigma\sigma}(W(\Lambda, p)) a^\dagger_{\mathbf{p}_\Lambda, \bar\sigma, n}.\end{aligned} \qquad (4.2.12)$$

In the same way, the operators C, P, and T, that induce charge-conjugation, space inversion, and time-reversal transformations on free particle states[29] transform the creation operators

[29] We omit the subscript '0' on these operators, because in virtually all cases where C, P, and/or T are conserved, the operators that induce these transformations on 'in' and 'out' states are the same as those defined by their action on free-particle states. This is not the case for continuous Lorentz transformations, for which it is necessary to distinguish between the operators $U(\Lambda, a)$ and $U_0(\Lambda, a)$.

as:

$$C a^\dagger_{\mathbf{p},\sigma,n} C^{-1} = \xi_n a^\dagger_{\mathbf{p},\sigma,n^c}, \qquad (4.2.13)$$

$$P a^\dagger_{\mathbf{p},\sigma,n} P^{-1} = \eta_n a^\dagger_{-\mathbf{p},\sigma,n}, \qquad (4.2.14)$$

$$T a^\dagger_{\mathbf{p},\sigma,n} T^{-1} = \zeta_n (-1)^{j-\sigma} a^\dagger_{-\mathbf{p},-\sigma,n}. \qquad (4.2.15)$$

As mentioned in the previous section, although we have been dealing with operators that create and annihilate particles in free-particle states, the whole formalism can be applied to 'in' and 'out' states, in which case we would introduce operators $a_{\rm in}$ and $a_{\rm out}$ defined in the same way by their action on these states. These operators satisfy a Lorentz transformation rule just like Eq. (4.2.12), but with the true Lorentz transformation operator $U(\Lambda, a)$ instead of the free-particle operator $U_0(\Lambda, a)$.

4.3 Cluster Decomposition and Connected Amplitudes

It is one of the fundamental principles of physics (indeed, of all science) that experiments that are sufficiently separated in space have unrelated results. The probabilities for various collisions measured at Fermilab should not depend on what sort of experiments are being done at CERN at the same time. If this principle were not valid, then we could never make any predictions about any experiment without knowing everything about the universe.

In S-matrix theory, the cluster decomposition principle states that if multi-particle processes $\alpha_1 \to \beta_1$, $\alpha_2 \to \beta_2$, …, $\alpha_{\mathcal{N}} \to \beta_{\mathcal{N}}$ are studied in $\mathcal{N}$ very distant laboratories, then the S-matrix element for the overall process factorizes. That is,[30]

$$S_{\beta_1+\beta_2+\cdots+\beta_{\mathcal{N}},\, \alpha_1+\alpha_2+\cdots+\alpha_{\mathcal{N}}} \longrightarrow S_{\beta_1\alpha_1} S_{\beta_2\alpha_2} \cdots S_{\beta_{\mathcal{N}}\alpha_{\mathcal{N}}} \qquad (4.3.1)$$

if for all $i \neq j$, *all* of the particles in states α_i and β_i are at a great spatial distance from *all* of the particles in states α_j and β_j. This factorization of S-matrix elements will ensure a factorization of the corresponding transition probabilities, corresponding to uncorrelated experimental results.

There is a combinatoric trick that allows us to rewrite Eq. (4.3.1) in a more transparent

[30] We are here returning to the notation used in Chapter 3; Greek letters α or β stand for a collection of particles, including for each particle a specification of its momentum, spin, and species. Also, $\alpha_1 + \alpha_2 + \cdots + \alpha_{\mathcal{N}}$ is the state formed by combining all the particles in the states $\alpha_1, \alpha_2, \ldots,$ and $\alpha_{\mathcal{N}}$, and likewise for $\beta_1 + \beta_2 + \cdots + \beta_{\mathcal{N}}$.

way. Suppose we define the *connected* part of the S-matrix, $S^C_{\beta\alpha}$, by the formula[31]

$$S_{\beta\alpha} = \sum_{\rm PART} (\pm) S^C_{\beta_1\alpha_1} S^C_{\beta_2\alpha_2} \cdots. \qquad (4.3.2)$$

Here the sum is over all different ways of partitioning the particles in the state α into clusters $\alpha_1, \alpha_2, \ldots$, and likewise a sum over all ways of partitioning the particles in the state β into clusters $\beta_1, \beta_2, \ldots$; not counting as different those that merely arrange particles within a given cluster or permute whole clusters. The sign is $+$ or $-$ according to whether the rearrangements $\alpha \to \alpha_1\alpha_2\cdots$ and $\beta \to \beta_1\beta_2\cdots$ involve altogether an even or an odd number of fermion interchanges, respectively. The term 'connected' is used because of the interpretation of $S^C_{\beta\alpha}$ in terms of diagrams representing different contributions in perturbation theory, to be discussed in the next section.

This is a recursive definition. For each α and β, the sum on the right-hand side of Eq. (4.3.2) consists of a term $S^C_{\beta\alpha}$, plus a sum $\sum'$ over products of two or more S^C-matrix elements, with a total number of particles in each of the states α_j and β_j that is *less* than the number of particles in the states α and β

$$S_{\beta\alpha} = S^C_{\beta\alpha} + \sum_{\rm PART}{}' (\pm) S^C_{\beta_1\alpha_1} S^C_{\beta_2\alpha_2} \cdots.$$

Suppose that the S^C-matrix elements in this sum have already been chosen in such a way that Eq. (4.3.2) is satisfied for states β, α containing together fewer than, say, N particles. Then no matter what values are found in this way for the S^C-matrix elements appearing in the sum $\sum'$, we can always choose the remaining term $S^C_{\beta\alpha}$ so that Eq. (4.3.2) is also satisfied for states α, β containing a total of N particles.[32] Thus Eq. (4.3.2) contains no information in itself; it is merely a definition of S^C.

If the states α and β each consist of just a single particle, say with quantum numbers q and q' respectively, then the only term on the right-hand-side of Eq. (4.3.2) is just $S^C_{\beta\alpha}$ itself, so for one-particle states

$$S^C_{q'q} = S_{q'q} = \delta(q' - q). \qquad (4.3.3)$$

[31] This decomposition has been used in classical statistical mechanics by Ursell, Mayer, and others, and in quantum statistical mechanics by Lee and Yang and others [3]. It has also been used to calculate many-body ground state energies by Goldstone [4] and Hugenholtz [5]. In all of these applications the purpose of isolating the connected parts of Green's functions, partition functions, resolvents, etc., is to deal with objects with a simple volume dependence. This is essentially our purpose too, because as we shall see, the crucial property of the connected parts of the S-matrix is that they are proportional to a single momentum-conservation delta function, and in a box the delta function becomes a Kronecker delta times the volume. The cluster decomposition is also the same formal device as that used in the theory of noise [6] to decompose the correlation function of several random variables into its 'cumulants'; if the random variable receives contributions from a large number N of independent fluctuations, then each cumulant is proportional to N.

[32] A technicality should be mentioned here. This argument works only if we neglect the possibility that for one or more of the connected S-matrix elements in Eq. (4.3.2), the states α_j and β_j both contain no particles at all. We must therefore define the connected vacuum-vacuum element $S^C_{0,0}$ to be zero. We do not use Eq. (4.3.2) for the vacuum-vacuum S-matrix $S_{0,0}$, which in the absence of time-varying external fields is simply defined to be unity, $S_{0,0} = 1$. We will have more to say about the vacuum-vacuum amplitude in the presence of external fields in Volume II.

(Apart from possible degeneracies, the fact that $S_{q'q}$ is proportional to $\delta(q' - q)$ follows from conservation laws. The absence of any proportionality factor in Eq. (4.3.3) is based on a suitable choice of the relative phase of 'in' and 'out' states.) We are here assuming that single-particle states are stable, so that there are no transitions between single-particle states and any others, such as the vacuum.

For transitions between two-particle states, Eq. (4.3.2) reads

$$\begin{aligned}S_{q'_1 q'_2, q_1 q_2} = S^C_{q'_1 q'_2, q_1 q_2} &+ \delta(q'_1 - q_1)\delta(q'_2 - q_2) \\ &\pm \delta(q'_1 - q_2)\delta(q'_2 - q_1).\end{aligned} \qquad (4.3.4)$$

(We are here using Eq. (4.3.3).) The sign $\pm$ is $-$ if both particles are fermions, and otherwise $+$. We recognize that the two delta function terms just add up to the norm (4.1.6), so here $S^C_{\beta\alpha}$ is just $(S - \mathbf{1})_{\beta\alpha}$. But the general case is more complicated.

For transitions between three-particle or four-particle states, Eq. (4.3.2) reads

$$\begin{aligned}S_{q'_1 q'_2 q'_3, q_1 q_2 q_3} = S^C_{q'_1 q'_2 q'_3, q_1 q_2 q_3} &+ \delta(q'_1 - q_1) S^C_{q'_2 q'_3, q_2 q_3} \pm \text{permutations} \\ &+ \delta(q'_1 - q_1)\delta(q'_2 - q_2)\delta(q'_3 - q_3) \pm \text{permutations}\end{aligned} \qquad (4.3.5)$$

and

$$\begin{aligned}S_{q'_1 q'_2 q'_3 q'_4, q_1 q_2 q_3 q_4} = S^C_{q'_1 q'_2 q'_3 q'_4, q_1 q_2 q_3 q_4} &+ S^C_{q'_1 q'_2, q_1 q_2} S^C_{q'_3 q'_4, q_3 q_4} \pm \text{permutations} \\ &+ \delta(q'_1 - q_1) S^C_{q'_2 q'_3 q'_4, q_2 q_3 q_4} \pm \text{permutations} \\ &+ \delta(q'_1 - q_1)\delta(q'_2 - q_2) S^C_{q'_3 q'_4, q_3 q_4} \pm \text{permutations} \\ &+ \delta(q'_1 - q_1)\delta(q'_2 - q_2)\delta(q'_3 - q_3)\delta(q'_4 - q_4) \pm \text{permutations}.\end{aligned} \qquad (4.3.6)$$

(Taking account of all permutations, there are a total of $1 + 9 + 6 = 16$ terms in Eq. (4.3.5) and $1 + 18 + 16 + 72 + 24 = 131$ terms in Eq. (4.3.6). If we had not assumed that one-particle states are stable, there would be even more terms.) As explained previously, the definition of $S^C_{\beta\alpha}$ is recursive: we use Eq. (4.3.4) to define $S^C_{\beta\alpha}$ for two-particle states, then use this definition in Eq. (4.3.5) when we define $S^C_{\beta\alpha}$ for three-particle states, then use both of these definitions in Eq. (4.3.6) to obtain the definition of $S^C_{\beta\alpha}$ for four-particle states, and so on.

The point of this definition of the connected part of the S-matrix is that the cluster decomposition principle is equivalent to the requirement that $S^C_{\beta\alpha}$ must vanish when any one or more of the particles in the states β and/or α are far away in space from the others.[33] To see this, suppose that the particles in the states β and α are grouped into clusters $\beta_1, \beta_2, \ldots$ and $\alpha_1, \alpha_2, \ldots$, and that all particles in the set $\alpha_i + \beta_i$ are far from all particles in the set $\alpha_j + \beta_j$ for any $j \neq i$. Then if $S^C_{\beta'\alpha'}$ vanishes if any particles in β' or α' are far from the others, it vanishes if any particles in these states are in different clusters, so the definition

[33] In order to give a meaning to 'far,' we will have to Fourier transform S^C, so that each three-momentum label $\mathbf{p}$ is replaced with a spatial coordinate three-vector $\mathbf{x}$.

Eq. (4.3.2) yields

$$\begin{aligned}S_{\beta\alpha} \longrightarrow \sum_{\rm PART}^{(1)} (\pm) S^C_{\beta_{11}\alpha_{11}} S^C_{\beta_{12}\alpha_{12}} \cdots \\ \cdot \sum_{\rm PART}^{(2)} (\pm) S^C_{\beta_{21}\alpha_{21}} S^C_{\beta_{22}\alpha_{22}} \cdots \cdots,\end{aligned} \qquad (4.3.7)$$

where $\sum^{(j)}$ is a sum over all different ways of partitioning the clusters β_j and α_j into subclusters $\beta_{j1}, \beta_{j2}, \ldots$ and $\alpha_{j1}, \alpha_{j2}, \ldots$. But referring back to Eq. (4.3.2), this is just the desired factorization property (4.3.1).

For instance, suppose that in the four-particle reaction $1234 \to 1'2'3'4'$, we let particles $1, 2, 1'$, and $2'$ be very far from $3, 4, 3'$, and $4'$. Then if $S^C_{\beta\alpha}$ vanishes when any particles in β and/or α are far from the others, the only terms in Eq. (4.3.6) that survive (in an even more abbreviated notation) are

$$\begin{aligned}S_{1'2'3'4',1234} \longrightarrow S^C_{1'2',12} S^C_{3'4',34} &+ (\delta_{1'1}\delta_{2'2} \pm \delta_{1'2}\delta_{2'1}) S^C_{3'4',34} \\ &+ (\delta_{3'3}\delta_{4'4} \pm \delta_{3'4}\delta_{4'3}) S^C_{1'2',12} \\ &+ (\delta_{1'1}\delta_{2'2} \pm \delta_{1'2}\delta_{2'1})(\delta_{3'3}\delta_{4'4} \pm \delta_{3'4}\delta_{4'3}).\end{aligned}$$

Comparison with Eq. (4.3.4) shows that this is just the required factorization condition (4.3.1)

$$S_{1'2'3'4',1234} \longrightarrow S_{1'2',12} S_{3'4',34}.$$

We have formulated the cluster decomposition principle in coordinate space, as the condition that $S^C_{\beta\alpha}$ vanishes if any particles in the states β or α are far from any others. It is convenient for us to reexpress this in momentum space. The coordinate space matrix elements are defined as a Fourier transform

$$\begin{aligned}S^C_{\mathbf{x}'_1\mathbf{x}'_2\cdots,\mathbf{x}_1\mathbf{x}_2\cdots} \equiv \int d^3p'_1 \, d^3p'_2 \cdots d^3p_1 \, d^3p_2 \cdots S^C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\mathbf{p}_1\mathbf{p}_2\cdots} \\ \cdot e^{i\mathbf{p}'_1\cdot\mathbf{x}'_1} e^{i\mathbf{p}'_2\cdot\mathbf{x}'_2} \cdots e^{-i\mathbf{p}_1\cdot\mathbf{x}_1} e^{-i\mathbf{p}_2\cdot\mathbf{x}_2} \cdots.\end{aligned} \qquad (4.3.8)$$

(We are here temporarily dropping spin and species labels, which just go along with the momentum or coordinate labels.) If $|S^C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\mathbf{p}_1\mathbf{p}_2\cdots}|$ were sufficiently well behaved (to be specific, if it were Lebesgue integrable) then according to the Riemann–Lebesgue theorem [7] the integral (4.3.8) would vanish when any combination of spatial coordinates goes to infinity. Now, this is certainly too strong a requirement. Translational invariance tells us that the connected part of the S-matrix, like the S-matrix itself, can only depend on differences of coordinate vectors, and therefore does not change at all if all of the x_i and x'_j vary together, with their differences held constant. This requires that the elements of S^C in a momentum basis must, like those of S, be proportional to a three-dimensional delta function that ensures momentum conservation (and makes $|S^C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\mathbf{p}_1\mathbf{p}_2\cdots}|$ *not* Lebesgue integrable), as well as the energy-conservation delta function required by scattering theory.

That is, we can write

$$\begin{aligned}S^C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\mathbf{p}_1\mathbf{p}_2\cdots} = \delta^3(\mathbf{p}'_1 + \mathbf{p}'_2 + \cdots - \mathbf{p}_1 - \mathbf{p}_2 - \cdots) \\ \cdot \delta(E'_1 + E'_2 + \cdots - E_1 - E_2 - \cdots) \\ \cdot C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\mathbf{p}_1\mathbf{p}_2\cdots}.\end{aligned} \qquad (4.3.9)$$

This is no problem: the cluster decomposition principle only requires that Eq. (4.3.8) vanish when the *differences* among some of the x_i and/or x'_i become large. However, if C itself in Eq. (4.3.9) contained additional delta functions of linear combinations of the three-momenta, then this principle would not be satisfied. For instance, suppose that there were a delta function in C that required that the sum of the $\mathbf{p}'_i$ and $-\mathbf{p}_i$ for some subset of the particles vanished. Then Eq. (4.3.8) would not vary if all of the $\mathbf{x}'_i$ and $\mathbf{x}_i$ for the particles in that subset moved together (with constant differences) away from all the other $\mathbf{x}'_k$ and $\mathbf{x}_k$, in contradiction to the cluster decomposition principle. Loosely speaking then, the cluster decomposition principle simply says that *the connected part of the S-matrix, unlike the S-matrix itself, contains just a single momentum-conservation delta function.*

In order to put this a bit more precisely, we can say that the coefficient function $C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\mathbf{p}_1\mathbf{p}_2\cdots}$ in Eq. (4.3.9) is a smooth function of its momentum labels. But how smooth? It would be most straightforward if we could simply require that $C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\mathbf{p}_1\mathbf{p}_2\cdots}$ be analytic in all of the momenta at $\mathbf{p}'_1 = \mathbf{p}'_2 = \cdots = \mathbf{p}_1 = \mathbf{p}_2 = \cdots = 0$. This requirement would indeed guarantee that $S^C_{\mathbf{x}'_1\mathbf{x}'_2\cdots,\mathbf{x}_1\mathbf{x}_2\cdots}$ vanishes exponentially fast when any of the $\mathbf{x}$ and $\mathbf{x}'$ is very distant from any of the other $\mathbf{x}$ and $\mathbf{x}'$. However, an exponential fall-off of S^C is not an essential part of the cluster decomposition principle, and, in fact, the requirement of analyticity is not met in all theories. Most notably, in theories with massless particles, S^C can have poles at certain values of the $\mathbf{p}$ and $\mathbf{p}'$. For instance, as we will see in Chapter 10, if a massless particle can be emitted in the transition $1 \to 3$ and absorbed in the transition $2 \to 4$, then $S^C_{34,12}$ will have a term proportional to $1/(\mathbf{p}_1 - \mathbf{p}_3)^2$. After Fourier transforming, such poles yield terms in $S^C_{\mathbf{x}'_1\mathbf{x}'_2\cdots,\mathbf{x}_1\mathbf{x}_2\cdots}$ that fall off only as negative powers of coordinate differences [1]. There is no need to formulate the cluster decomposition principle so stringently that such behavior is ruled out. Thus the 'smoothness' condition on S^C should be understood to allow various poles and branch-cuts at certain values of the $\mathbf{p}$ and $\mathbf{p}'$, but not singularities as severe as delta functions.

4.4 Structure of the Interaction

We now ask, what sort of Hamiltonian will yield an S-matrix that satisfies the cluster decomposition principle? It is here that the formalism of creation and annihilation operators comes into its own. The answer is contained in the theorem that the S-matrix satisfies the cluster decomposition principle if (and as far as I know, only if) the Hamiltonian can be

expressed as in Eq. (4.2.8):

$$\begin{aligned}H = \sum_{N=0}^{\infty} \sum_{M=0}^{\infty} \int & dq'_1 \cdots dq'_N \, dq_1 \cdots dq_M \\ & \cdot a^\dagger_{q'_1} \cdots a^\dagger_{q'_N} a_{q_M} \cdots a_{q_1} \\ & \cdot h_{NM}(q'_1 \cdots q'_N, q_1 \cdots q_M).\end{aligned} \qquad (4.4.1)$$

with coefficient functions h_{NM} that contain just a *single* three-dimensional momentum-conservation delta function (returning here briefly to a more explicit notation)

$$\begin{aligned}h_{NM}(\mathbf{p}'_1\sigma'_1 n'_1 \cdots \mathbf{p}'_N\sigma'_N n'_N, \mathbf{p}_1\sigma_1 n_1 \cdots \mathbf{p}_M\sigma_M n_M) \\ = \delta^3(\mathbf{p}'_1 + \cdots + \mathbf{p}'_N - \mathbf{p}_1 - \cdots - \mathbf{p}_N) \\ \cdot \tilde{h}_{NM}(\mathbf{p}'_1\sigma'_1 n'_1 \cdots \mathbf{p}'_N\sigma'_N n'_N, \mathbf{p}_1\sigma_1 n_1 \cdots \mathbf{p}_M\sigma_M n_M),\end{aligned} \qquad (4.4.2)$$

where $\tilde{h}_{NM}$ contains no delta function factors. Note that Eq. (4.4.1) by itself has no content—we saw in Section 4.2 that *any* operator can be put in this form. It is only Eq. (4.4.1) combined with the requirement that h_{NM} has only the single delta function shown in Eq. (4.4.2) that guarantees that the S-matrix satisfies the cluster decomposition principle.

The validity of this theorem in perturbation theory will become obvious when we develop the Feynman diagram formalism in Chapter 6. The trusting reader may prefer to skip the rest of the present chapter, and move on to consider the implications of this theorem in Chapter 5. However, the proof has some instructive features, and will help to clarify in what sense the field theory of the next chapter is inevitable.

To prove this theorem, we make use of perturbation theory in its time-dependent form. (One of the advantages of time-dependent perturbation theory is that it makes the combinatorics underlying the cluster decomposition principle much more transparent; if E is a sum of one-particle energies then e^{-iEt} is a product of functions of the individual energies, while $[E - E_\alpha + i\epsilon]^{-1}$ is not.) The S-matrix is given by Eq. (3.5.10) as[34]

$$S_{\beta\alpha} = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dt_1 \cdots dt_n \, \langle\beta| T\{H_I(t_1) \cdots H_I(t_n)\} |\alpha\rangle, \qquad (4.4.3)$$

where the Hamiltonian is split into a free-particle part H_0 and an interaction $H_{\rm int}$, and

$$H_I(t) \equiv \exp(iH_0 t) H_{\rm int} \exp(-iH_0 t). \qquad (4.4.4)$$

Now, the states $|\alpha\rangle$ and $|\beta\rangle$ may be expressed as in Eq. (4.2.2) as products of creation operators acting on the vacuum $|\Omega\rangle$, and $H_I(t)$ is itself a sum of products of creation and annihilation operators, so each term in the sum (4.4.3) may be written as a sum of vacuum expectation values of products of creation and annihilation operators. By using the commutation or anticommutation relations (4.2.5) we may move each annihilation operator

[34] We are now adopting the convention that for $n = 0$, the time-ordered product in Eq. (4.4.3) is taken as the unit operator, so the $n = 0$ term in the sum just yields the term $\delta(\beta - \alpha)$ in $S_{\beta\alpha}$.

in turn to the right past all the creation operators. For each annihilation operator moved to the right past a creation operator we have two terms, as shown by writing Eq. (4.2.5) in the form

$$a_{q'} a^\dagger_q = \pm a^\dagger_q a_{q'} + \delta(q' - q).$$

Moving other creation operators past the annihilation operator in the first term generates yet more terms. But Eq. (4.2.4) shows that any annihilation operator that moves all the way to the right and acts on $|\Omega\rangle$ gives zero, so in the end all we have left is the delta functions. In this way, the vacuum expectation value of a product of creation and annihilation operators is given by a sum of different terms, each term equal to a product of delta functions and $\pm$ signs from the commutators or anticommutators. It follows that each term in Eq. (4.4.3) may be expressed as a sum of terms, each term equal to a product of delta functions and $\pm$ signs from the commutators or anticommutators and whatever factors are contributed by $H_I(t)$, integrated over all the times and integrated and summed over the momenta, spins, and species in the arguments of the delta functions.

Each of the terms generated in this way may be symbolized by a diagram. (This is not yet the full Feynman diagram formalism, because we are not yet going to associate numerical quantities with the ingredients in the diagrams; we are using the diagrams here only as a way of keeping track of three-momentum delta functions.) Draw n points, called *vertices*, one for each $H_I(t)$ operator. For each delta function produced when an annihilation operator in one of these $H_I(t)$ operators moves past a creation operator in the initial state $|\alpha\rangle$, draw a line coming into the diagram from below that ends at the corresponding vertex. For each delta function produced when an annihilation operator in the adjoint of the final state $\langle\beta|$ moves past a creation operator in one of the $H_I(t)$, draw a line from the corresponding vertex upwards out of the diagram. For each delta function produced when an annihilation operator in one $H_I(t)$ moves past a creation operator in another $H_I(t)$ draw a line between the two corresponding vertices. Finally, for each delta function produced when an annihilation operator in the adjoint of the final state moves past a creation operator in the initial state, draw a line from bottom to top, right through the diagrams. Each of the delta functions associated with one of these lines enforces the equality of the momentum arguments of the pair of creation and annihilation operators represented by the line. There is also at least one delta function contributed by each of the vertices, which enforces the conservation of the total three-momentum at the vertex.

Such a diagram may be connected (every point connected to every other by a set of lines) and if not connected, it breaks up into a number of connected pieces. The $H_I(t)$ operator associated with a vertex in one connected component effectively commutes with the $H_I(t)$ associated with any vertex in any other connected component, because for this diagram, we are not including any terms in which an annihilation operator in one vertex destroys a particle that is produced by a creation operator in the other vertex—if we did, the two vertices would be in the same connected component. Thus the matrix element in Eq. (4.4.3) can be expressed as a sum over *products* of contributions, one from each

connected component:

$$\begin{aligned}\langle\beta| T\{H_I(t_1) \cdots H_I(t_n)\} |\alpha\rangle \\ = \sum_{\rm clusterings} (\pm) \prod_{j=1}^{\nu} \Big(\langle\beta_j| T\{H_I(t_{j1}) \cdots H_I(t_{jn_j})\} |\alpha_j\rangle\Big)_C.\end{aligned} \qquad (4.4.5)$$

Here the sum is over all ways of splitting up the incoming and outgoing particles and $H_I(t)$ operators into ν clusters (including a sum over ν from 1 to n) with the n_j operators $H_I(t_{j1}) \cdots H_I(t_{jn_j})$ and the subsets of initial particles α_j and final particles β_j all in the jth cluster. Of course, this means that

$$n = n_1 + \cdots + n_\nu$$

and also the set α is the union of all the particles in the subsets $\alpha_1, \alpha_2, \ldots, \alpha_\nu$, and likewise for the final state. Some of the clusters in Eq. (4.4.5) may contain no vertices at all, i.e., $n_j = 0$; for these factors, we must take the matrix element factor in Eq. (4.4.5) to vanish unless β_j and α_j are both one-particle states (in which case it is just a delta function $\delta(\alpha_j - \beta_j)$), because the only connected diagrams without vertices consist of a single line running through the diagram from bottom to top. Most important, the subscript C in Eq. (4.4.5) means that we exclude any contributions corresponding to disconnected diagrams, that is, any contributions in which any $H_I(t)$ operator or any initial or final particle is not connected to every other by a sequence of particle creations and annihilations.

Now let us use Eq. (4.4.5) in the sum (4.4.3). Every time variable is integrated from $-\infty$ to $+\infty$, so it makes no difference which of the $t_1, \ldots, t_n$ are sorted out into each cluster. The sum over clusterings therefore yields a factor $n!/n_1! n_2! \cdots n_\nu!$, equal to the number of ways of sorting out n vertices into ν clusters, each containing $n_1, n_2, \ldots$ vertices:

$$\begin{aligned}\int_{-\infty}^{\infty} dt_1 \cdots dt_n \, \langle\beta| T\{H_I(t_1) \cdots H_I(t_n)\} |\alpha\rangle \\ = \frac{n!}{n_1! n_2! \cdots n_\nu!} \sum_{\rm PART} (\pm) \sum_{n_1 + \cdots + n_\nu = n} \prod_{j=1}^{\nu} \int_{-\infty}^{\infty} dt_{j1} \cdots dt_{jn_j} \\ \cdot \Big(\langle\beta_j| T\{H_I(t_{j1}) \cdots H_I(t_{jn_j})\} |\alpha_j\rangle\Big)_C.\end{aligned}$$

The first sum here is over all ways of partitioning the particles in the initial and final states into clusters $\alpha_1 \cdots \alpha_\nu$ and $\beta_1 \cdots \beta_\nu$ (including a sum over the number ν of clusters). The factor $n!$ here cancels the $1/n!$ in Eq. (4.4.3), and the factor $(-i)^n$ in the perturbation series for (4.4.5) can be written as a product $(-i)^{n_1} \cdots (-i)^{n_\nu}$, so instead of summing over n and then summing separately over $n_1, \ldots, n_\nu$, constrained by $n_1 + \cdots + n_\nu = n$, we can simply sum independently over each $n_1, \ldots, n_\nu$. This gives finally

$$\begin{aligned}S_{\beta\alpha} = \sum_{\rm PART} (\pm) \prod_{j=1}^{\nu} \sum_{n_j=0}^{\infty} \frac{(-i)^{n_j}}{n_j!} \int_{-\infty}^{\infty} dt_{j1} \cdots dt_{jn_j} \\ \cdot \Big(\langle\beta_j| T\{H_I(t_{j1}) \cdots H_I(t_{jn_j})\} |\alpha_j\rangle\Big)_C.\end{aligned}$$

Comparing this with the definition (4.2.2) of the connected matrix elements $S^C_{\beta\alpha}$, we see that these matrix elements are just given by the factors in the product here

$$S^C_{\beta\alpha} = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dt_1 \cdots dt_n \Big(\langle\beta| T\{H_I(t_1) \cdots H_I(t_n)\} |\alpha\rangle\Big)_C. \qquad (4.4.6)$$

(The subscript j is dropped on all the ts and ns, as these are now mere integration and summation variables.) We see that $S^C_{\beta\alpha}$ is calculated by a very simple prescription: $S^C_{\beta\alpha}$ is *the sum of all contributions to the S-matrix that are connected, in the sense that we drop all terms in which any initial or final particle or any operator $H_I(t)$ is not connected to all the others by a sequence of particle creations and annihilations.* This justifies the adjective 'connected' for S^C.

As we have seen, momentum is conserved at each vertex and along every line, so the connected parts of the S-matrix individually conserve momentum: $S^C_{\beta\alpha}$ contains a factor $\delta^3(\mathbf{p}_\beta - \mathbf{p}_\alpha)$. What we want to prove is that $S^C_{\beta\alpha}$ contains no other delta functions.

We now make the assumption that the coefficient fractions h_{NM} in the expansion (4.4.1) of the Hamiltonian in terms of creation and annihilation operators are proportional to a *single* three-dimensional delta function, that ensures momenta conservation. This is automatically true for the free-particle Hamiltonian H_0, so it is also then true separately for the interaction $H_{\rm int}$. Returning to the graphical interpretation of the matrix elements that we have been using, this means that each vertex contributes one three-dimensional delta function. (The other delta functions in matrix elements $(H_{\rm int})_{\gamma\delta}$ simply keep the momentum of any particle that is not created or annihilated at the corresponding vertex unchanged.) Now, most of these delta functions simply go to fix the momentum of intermediate particles. The only momenta that are left unfixed by such delta functions are those that circulate in loops of internal lines. (Any line which if cut leaves the diagram disconnected carries a momentum that is fixed by momentum conservation as some linear combination of the momenta of the lines coming into or going out of the diagram. If the diagram has L lines that can all be cut at the same time without the diagram becoming disconnected, then we say it has L independent loops, and there are L momenta that are not fixed by momentum conservation.) With V vertices, I internal lines, and L loops, there are V delta functions, of which $I - L$ go to fix internal momenta, leaving $V - I + L$ delta functions relating the momenta of incoming or outgoing particles. But a well-known topological identity[35] tells that for any graph consisting of C connected pieces, the numbers of vertices, internal lines, and loops are related by

$$V - I + L = C. \qquad (4.4.7)$$

Hence for a connected matrix element like $S^C_{\beta\alpha}$, which arises from graphs with $C = 1$, we find just a single three-dimensional delta function $\delta^3(\mathbf{p}_\beta - \mathbf{p}_\alpha)$, as was to be proved.

[35] A graph consisting of a single vertex has $V = 1$, $L = 0$, and $C = 1$. If we add $V - 1$ vertices with just enough internal lines to keep the graph connected, we have $I = V - 1$, $L = 0$, and $C = 1$. Any additional internal lines attached (without new vertices) to the same connected graph produce an equal number of loops, so $I = V + L - 1$ and $C = 1$. If a disconnected graph consists of C such connected parts, the sums of I, V, and L in each connected part will then satisfy $\sum I = \sum V + \sum L - C$.

It was not important in the above argument that the time variables were integrated from $-\infty$ to $+\infty$. Thus exactly the same arguments can be used to show that if the coefficients $h_{N,M}$ in the Hamiltonian contain just single delta functions, then $U(t, t_0)$ can also be decomposed into connected parts, each containing a single momentum-conservation delta function factor. On the other hand, the connected part of the S-matrix also contains an energy-conservation delta function, and when we come to Feynman diagrams in Chapter 6 we shall see that $S^C_{\beta\alpha}$ contains only a single energy-conservation delta function, $\delta(E_\beta - E_\alpha)$, while $U(t, t_0)$ contains no energy-conservation delta functions at all.

It should be emphasized that the requirement that h_{NM} in Eq. (4.4.1) should have only a single three-dimensional momentum conservation delta function factor is very far from trivial, and has far-reaching implications. For instance, assume that $H_{\rm int}$ has non-vanishing matrix elements between two-particle states. Then Eq. (4.4.1) must contain a term with $N = M = 2$, and coefficient

$$v_{2,2}(\mathbf{p}'_1\mathbf{p}'_2, \mathbf{p}_1\mathbf{p}_2) = (H_{\rm int})_{\mathbf{p}'_1\mathbf{p}'_2; \mathbf{p}_1\mathbf{p}_2}. \qquad (4.4.8)$$

(We are here temporarily dropping spin and species labels.) But then the matrix element of the interaction between three-particle states is

$$\begin{aligned}(H_{\rm int})_{\mathbf{p}'_1\mathbf{p}'_2\mathbf{p}'_3; \mathbf{p}_1\mathbf{p}_2\mathbf{p}_3} = v_{3,3}(\mathbf{p}'_1\mathbf{p}'_2\mathbf{p}'_3, \mathbf{p}_1\mathbf{p}_2\mathbf{p}_3) \\ + v_{2,2}(\mathbf{p}'_1\mathbf{p}'_2, \mathbf{p}_1\mathbf{p}_2) \, \delta^3(\mathbf{p}'_3 - \mathbf{p}_3) \pm \text{permutations}.\end{aligned} \qquad (4.4.9)$$

As mentioned at the beginning of this chapter, we might try to make a relativistic quantum theory that is not a field theory by choosing $v_{2,2}$ so that the two-body S-matrix is Lorentz-invariant, and adjusting the rest of the Hamiltonian so that there is no scattering in states containing three or more particles. We would then have to take $v_{3,3}$ to cancel the other terms in Eq. (4.4.9)

$$v_{3,3}(\mathbf{p}'_1\mathbf{p}'_2\mathbf{p}'_3, \mathbf{p}_1\mathbf{p}_2\mathbf{p}_3) = -v_{2,2}(\mathbf{p}'_1\mathbf{p}'_2, \mathbf{p}_1\mathbf{p}_2) \, \delta^3(\mathbf{p}'_3 - \mathbf{p}_3) \mp \text{permutations}. \qquad (4.4.10)$$

However, this would mean that each term in $v_{3,3}$ contains *two* delta function factors (recall that $v_{2,2}(\mathbf{p}'_1\mathbf{p}'_2, \mathbf{p}_1\mathbf{p}_2)$ has a factor $\delta^3(\mathbf{p}'_1 + \mathbf{p}'_2 - \mathbf{p}_1 - \mathbf{p}_2)$) and this would violate the cluster decomposition principle. Thus in a theory satisfying the cluster decomposition principle, the existence of scattering processes involving two particles makes processes involving three or more particles inevitable.

* * *

When we set out to solve three-body problems in quantum theories that satisfy the cluster decomposition principle, the term $v_{3,3}$ in Eq. (4.4.9) gives no particular trouble, but the extra delta function in the other terms makes the Lippmann–Schwinger equation difficult to solve directly. The problem is that these delta functions make the kernel $[E_\alpha - E_\beta + i\epsilon]^{-1}(H_{\rm int})_{\beta\alpha}$ of this equation not square-integrable, even after we factor out an overall momentum conservation delta function. In consequence, it cannot be approximated by a finite matrix, even one of very large rank. To solve problems involving three or more

particles, it is necessary to replace the Lippmann–Schwinger equation with one that has a connected right-hand side. Such equations have been developed for the scattering of three or more particles [8,9], and in non-relativistic scattering problems they can be solved recursively, but they have not turned out to be useful in relativistic theories and so will not be described in detail here.

However, recasting the Lippmann–Schwinger equation in this manner is useful in another way. Our arguments in this section have so far relied on perturbation theory. I do not know of any non-perturbative proof of the main theorem of this section, but it has been shown [9] that these reformulated non-perturbative dynamical equations are *consistent* with the requirement that $U^C(t, t_0)$ (and hence S^C) should also contain only a single momentum-conservation delta function, as required by the cluster decomposition principle, provided that the Hamiltonian satisfies our condition that the coefficient functions $h_{N,M}$ each contain only a single momentum-conservation delta function.

Problems

1. Define generating functionals for the S-matrix and its connected part:

$$\begin{aligned}F[\nu] \equiv 1 + \sum_{N=1}^{\infty} \sum_{M=1}^{\infty} \frac{1}{N!M!} \int \nu^*(q'_1) \cdots \nu^*(q'_N) \nu(q_1) \cdots \nu(q_M) \\ \cdot S_{q'_1 \cdots q'_N, q_1 \cdots q_M} \, dq'_1 \cdots dq'_N \, dq_1 \cdots dq_M,\end{aligned}$$

$$\begin{aligned}F^C[\nu] \equiv \sum_{N=1}^{\infty} \sum_{M=1}^{\infty} \frac{1}{N!M!} \int \nu^*(q'_1) \cdots \nu^*(q'_N) \nu(q_1) \cdots \nu(q_M) \\ \cdot S^C_{q'_1 \cdots q'_N, q_1 \cdots q_M} \, dq'_1 \cdots dq'_N \, dq_1 \cdots dq_M.\end{aligned}$$

   Derive a formula relating $F[\nu]$ and $F^C[\nu]$. (You may consider the purely bosonic case.)

2. Consider an interaction

$$\begin{aligned}H_{\rm int} = g \int d^3\mathbf{p}_1 \, d^3\mathbf{p}_2 \, d^3\mathbf{p}_3 \, d^3\mathbf{p}_4 \, \delta^3(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}_3 - \mathbf{p}_4) \\ \cdot a^\dagger_{\mathbf{p}_1} a^\dagger_{\mathbf{p}_2} a_{\mathbf{p}_3} a_{\mathbf{p}_4},\end{aligned}$$

   where g is a real constant and $a_{\mathbf{p}}$ is the annihilation operator of a spinless boson of mass $M > 0$. Use perturbation theory to calculate the S-matrix element for scattering of these particles in the center-of-mass frame to order g^2. What is the corresponding differential cross-section?

3. A coherent state $|\lambda\rangle$ is defined to be an eigenstate of the annihilation operators a_q with eigenvalues $\lambda(q)$. Construct such a state as a superposition of the multi-particle states $|q_1, q_2, \ldots, q_N\rangle$.

References

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5 Quantum Fields and Antiparticles

We now have all the pieces needed to motivate the introduction of quantum fields [1]. In the course of this construction, we shall encounter some of the most remarkable and universal consequences of the union of relativity with quantum mechanics: the connection between spin and statistics, the existence of antiparticles, and various relationships between particles and antiparticles, including the celebrated CPT theorem.

5.1 Free Fields

We have seen in Chapter 3 that the S-matrix will be Lorentz-invariant if the interaction can be written as

$$H_I(t) = \int d^3x \, \mathcal{H}_I(\mathbf{x}, t), \qquad (5.1.1)$$

where $\mathcal{H}_I$ is a scalar, in the sense that

$$U_0(\Lambda, a) \mathcal{H}_I(x) U_0^{-1}(\Lambda, a) = \mathcal{H}_I(\Lambda x + a), \qquad (5.1.2)$$

and satisfies the additional condition:

$$[\mathcal{H}_I(x), \mathcal{H}_I(x')] = 0 \quad \text{for} \quad (x - x')^2 \ge 0. \qquad (5.1.3)$$

As we shall see, there are more general possibilities, but none of them are very different from this. (For the present we are leaving it as an open question whether Λ here is restricted to a proper orthochronous Lorentz transformation, or can also include space inversions.) In order to facilitate also satisfying the cluster decomposition principle we are going to construct $\mathcal{H}_I(x)$ out of creation and annihilation operators, but here we face a problem: as shown by Eq. (4.2.12), under Lorentz transformations each such operator is multiplied by a matrix that depends on the momentum carried by that operator. How can we couple such operators together to make a scalar? The solution is to build $\mathcal{H}_I(x)$ out of fields—both annihilation fields $\psi^{(+)}_\ell(x)$ and creation fields $\psi^{(-)}_\ell(x)$:

$$\psi^{(+)}_\ell(x) = \sum_{\sigma n} \int d^3p \, u_\ell(x; \mathbf{p}, \sigma, n) a_{\mathbf{p},\sigma,n}, \qquad (5.1.4)$$

$$\psi^{(-)}_\ell(x) = \sum_{\sigma n} \int d^3p \, v_\ell(x; \mathbf{p}, \sigma, n) a^\dagger_{\mathbf{p},\sigma,n}, \qquad (5.1.5)$$

with coefficients[36] $u_\ell(x; \mathbf{p}, \sigma, n)$ and $v_\ell(x; \mathbf{p}, \sigma, n)$ chosen so that under Lorentz transformations each field is multiplied with a position-independent matrix:

$$U_0(\Lambda, a) \psi^{(+)}_\ell(x) U_0^{-1}(\Lambda, a) = \sum_{\bar\ell} D_{\ell\bar\ell}(\Lambda^{-1}) \psi^{(+)}_{\bar\ell}(\Lambda x + a), \qquad (5.1.6)$$

$$U_0(\Lambda, a) \psi^{(-)}_\ell(x) U_0^{-1}(\Lambda, a) = \sum_{\bar\ell} D_{\ell\bar\ell}(\Lambda^{-1}) \psi^{(-)}_{\bar\ell}(\Lambda x + a). \qquad (5.1.7)$$

[36] A reminder: the labels n and σ run over all different particle species and spin z-components, respectively.

(We might, in principle, have different transformation matrices $D^\pm$ for the annihilation and creation fields, but as we shall see, it is always possible to choose the fields so that these matrices are the same.) By applying a second Lorentz transformation $\bar\Lambda$, we find that

$$D(\Lambda^{-1}) D(\bar\Lambda^{-1}) = D((\bar\Lambda\Lambda)^{-1}),$$

so taking $\Lambda_1 = (\Lambda)^{-1}$ and $\Lambda_2 = (\bar\Lambda)^{-1}$, we see that the D-matrices furnish a *representation* of the homogeneous Lorentz group:

$$D(\Lambda_1) D(\Lambda_2) = D(\Lambda_1 \Lambda_2). \qquad (5.1.8)$$

There are many such representations, including the scalar $D(\Lambda) = \mathbf{1}$, the vector $D(\Lambda)^\mu{}_\nu = \Lambda^\mu{}_\nu$, and a host of tensor and spinor representations. These particular representations are irreducible, in the sense that it is not possible by a choice of basis to reduce all $D(\Lambda)$ to the same block-diagonal form, with two or more blocks. However, we do not require at this point that $D(\Lambda)$ be irreducible; in general it is a block-diagonal matrix with an arbitrary array of irreducible representations in the blocks. That is, the index ℓ here includes a label that runs over the types of particle described and the irreducible representations in the different blocks, as well as another that runs over the components of the individual irreducible representations. Later we will separate these fields into irreducible fields that each describe only a single particle species (and its antiparticle) and transform irreducibly under the Lorentz group.

Once we have learned how to construct fields satisfying the Lorentz transformation rules (5.1.6) and (5.1.7), we will be able to construct the interaction density as

$$\begin{aligned}\mathcal{H}_I(x) = \sum_{NM} \sum_{\ell'_1 \cdots \ell'_N} \sum_{\ell_1 \cdots \ell_M} g_{\ell'_1 \cdots \ell'_N, \ell_1 \cdots \ell_M} \\ \cdot \psi^{(-)}_{\ell'_1}(x) \cdots \psi^{(-)}_{\ell'_N}(x) \psi^{(+)}_{\ell_1}(x) \cdots \psi^{(+)}_{\ell_M}(x),\end{aligned} \qquad (5.1.9)$$

and this will be a scalar in the sense of Eq. (5.1.2) if the constant coefficients $g_{\ell'_1 \cdots \ell'_N, \ell_1 \cdots \ell_M}$ are chosen to be Lorentz covariant, in the sense that for all Λ:

$$\begin{aligned}\sum_{\ell'_1 \cdots \ell'_N} \sum_{\ell_1 \cdots \ell_M} D_{\ell'_1 \bar\ell'_1}(\Lambda^{-1}) \cdots D_{\ell'_N \bar\ell'_N}(\Lambda^{-1}) D_{\ell_1 \bar\ell_1}(\Lambda^{-1}) \cdots D_{\ell_M \bar\ell_M}(\Lambda^{-1}) \\ \cdot g_{\ell'_1 \cdots \ell'_N, \ell_1 \cdots \ell_M} = g_{\bar\ell'_1 \cdots \bar\ell'_N, \bar\ell_1 \cdots \bar\ell_M}.\end{aligned} \qquad (5.1.10)$$

(Note that we do not include derivatives here, because we regard the derivatives of components of these fields as just additional sorts of field components.) The task of finding coefficients $g_{\ell'_1 \cdots \ell'_N, \ell_1 \cdots \ell_M}$ that satisfy Eq. (5.1.10) is no different in principle (and not much more difficult in practice) than that of using Clebsch–Gordan coefficients to couple together various representations of the three-dimensional rotation group to form rotational scalars. Later we will be able to combine creation and annihilation fields so that this density also commutes with itself at space-like separations.

Now, what shall we take as the coefficient functions $u_\ell(x; \mathbf{p}, \sigma, n)$ and $v_\ell(x; \mathbf{p}, \sigma, n)$? Eq. (4.2.12) and its adjoint give the transformation rules[37] for the annihilation and creation

[37] This is for massive particles. The case of zero mass will be taken up in Section 5.9.

operators

$$\begin{aligned}U_0(\Lambda, b) a_{\mathbf{p},\sigma,n} U_0^{-1}(\Lambda, b) = \exp\big(i(\Lambda p)\cdot b\big) \sqrt{\frac{(\Lambda p)^0}{p^0}} \\ \cdot \sum_{\bar\sigma} D^{(j_n)}_{\sigma\bar\sigma}(W^{-1}(\Lambda, p)) a_{\mathbf{p}_\Lambda, \bar\sigma, n},\end{aligned} \qquad (5.1.11)$$

$$\begin{aligned}U_0(\Lambda, b) a^\dagger_{\mathbf{p},\sigma,n} U_0^{-1}(\Lambda, b) = \exp\big(-i(\Lambda p)\cdot b\big) \sqrt{\frac{(\Lambda p)^0}{p^0}} \\ \cdot \sum_{\bar\sigma} D^{(j_n)*}_{\sigma\bar\sigma}(W^{-1}(\Lambda, p)) a^\dagger_{\mathbf{p}_\Lambda, \bar\sigma, n},\end{aligned} \qquad (5.1.12)$$

where j_n is the spin of particles of species n, and $\mathbf{p}_\Lambda$ is the three-vector part of Λp. (We have used the unitarity of the rotation matrices $D^{(j_n)}$ to put both Eqs. (5.1.11) and (5.1.12) in the form shown here.) Also, as we saw in Section 2.5 the volume element d^3p/p^0 is Lorentz-invariant, so we can replace d^3p in Eqs. (5.1.4) and (5.1.5) with $d^3(\Lambda p) p^0/(\Lambda p)^0$. Putting this all together, we find

$$\begin{aligned}U_0(\Lambda, b) \psi^{(+)}_\ell(x) U_0^{-1}(\Lambda, b) = \sum_{\sigma\bar\sigma n} \int d^3(\Lambda p) \, u_\ell(x; \mathbf{p}, \sigma, n) \\ \cdot \exp\big(i(\Lambda p)\cdot b\big) D^{(j_n)}_{\sigma\bar\sigma}(W^{-1}(\Lambda, p)) \sqrt{\frac{p^0}{(\Lambda p)^0}} a_{\mathbf{p}_\Lambda, \bar\sigma, n}\end{aligned}$$

and

$$\begin{aligned}U_0(\Lambda, b) \psi^{(-)}_\ell(x) U_0^{-1}(\Lambda, b) = \sum_{\sigma\bar\sigma n} \int d^3(\Lambda p) \, v_\ell(x; \mathbf{p}, \sigma, n) \\ \cdot \exp\big(-i(\Lambda p)\cdot b\big) D^{(j_n)*}_{\sigma\bar\sigma}(W^{-1}(\Lambda, p)) \sqrt{\frac{p^0}{(\Lambda p)^0}} a^\dagger_{\mathbf{p}_\Lambda, \bar\sigma, n}.\end{aligned}$$

We see that in order for the fields to satisfy the Lorentz transformation rules (5.1.6) and (5.1.7), it is necessary and sufficient that

$$\begin{aligned}\sum_{\bar\ell} D_{\ell\bar\ell}(\Lambda^{-1}) u_{\bar\ell}(\Lambda x + b; \mathbf{p}_\Lambda, \bar\sigma, n) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \sum_{\sigma} D^{(j_n)}_{\sigma\bar\sigma}(W^{-1}(\Lambda, p)) \\ \cdot \exp\big(+i(\Lambda p)\cdot b\big) u_\ell(x; \mathbf{p}, \sigma, n)\end{aligned}$$

and

$$\begin{aligned}\sum_{\bar\ell} D_{\ell\bar\ell}(\Lambda^{-1}) v_{\bar\ell}(\Lambda x + b; \mathbf{p}_\Lambda, \bar\sigma, n) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \sum_{\sigma} D^{(j_n)*}_{\sigma\bar\sigma}(W^{-1}(\Lambda, p)) \\ \cdot \exp\big(-i(\Lambda p)\cdot b\big) v_\ell(x; \mathbf{p}, \sigma, n),\end{aligned}$$

or somewhat more conveniently

$$\begin{aligned}\sum_{\bar\sigma} u_{\bar\ell}(\Lambda x + b; \mathbf{p}_\Lambda, \bar\sigma, n) D^{(j_n)}_{\bar\sigma\sigma}(W(\Lambda, p)) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \\ \cdot \sum_{\ell} D_{\bar\ell\ell}(\Lambda) \exp\big(i(\Lambda p)\cdot b\big) u_\ell(x; \mathbf{p}, \sigma, n),\end{aligned} \qquad (5.1.13)$$

and

$$\begin{aligned}\sum_{\bar\sigma} v_{\bar\ell}(\Lambda x + b; \mathbf{p}_\Lambda, \bar\sigma, n) D^{(j_n)*}_{\bar\sigma\sigma}(W(\Lambda, p)) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \\ \cdot \sum_{\ell} D_{\bar\ell\ell}(\Lambda) \exp\big(-i(\Lambda p)\cdot b\big) v_\ell(x; \mathbf{p}, \sigma, n).\end{aligned} \qquad (5.1.14)$$

These are the fundamental requirements that will allow us to calculate the u_ℓ and v_ℓ coefficient functions in terms of a finite number of free parameters.

Translations

We will use Eqs. (5.1.13) and (5.1.14) in three steps, considering in turn three different types of proper orthochronous Lorentz transformation. First we consider Eqs. (5.1.13) and (5.1.14) with $\Lambda = \mathbf{1}$ and b arbitrary. We see immediately that $u_\ell(x; \mathbf{p}, \sigma, n)$ and $v_\ell(x; \mathbf{p}, \sigma, n)$ must take the form

$$u_\ell(x; \mathbf{p}, \sigma, n) = (2\pi)^{-3/2} e^{ip\cdot x} u_\ell(\mathbf{p}, \sigma, n), \qquad (5.1.15)$$

$$v_\ell(x; \mathbf{p}, \sigma, n) = (2\pi)^{-3/2} e^{-ip\cdot x} v_\ell(\mathbf{p}, \sigma, n), \qquad (5.1.16)$$

so the fields are Fourier transforms:

$$\psi^{(+)}_\ell(x) = \sum_{\sigma, n} (2\pi)^{-3/2} \int d^3p \, u_\ell(\mathbf{p}, \sigma, n) e^{ip\cdot x} a_{\mathbf{p},\sigma,n}, \qquad (5.1.17)$$

and

$$\psi^{(-)}_\ell(x) = \sum_{\sigma, n} (2\pi)^{-3/2} \int d^3p \, v_\ell(\mathbf{p}, \sigma, n) e^{-ip\cdot x} a^\dagger_{\mathbf{p},\sigma,n}. \qquad (5.1.18)$$

(The factors $(2\pi)^{-3/2}$ could be absorbed into the definition of u_ℓ and v_ℓ, but it is conventional to show them explicitly in these Fourier integrals.) Using Eqs. (5.1.15) and (5.1.16), we see that Eqs. (5.1.13) and (5.1.14) are satisfied if and only if

$$\sum_{\bar\sigma} u_{\bar\ell}(\mathbf{p}_\Lambda, \bar\sigma, n) D^{(j_n)}_{\bar\sigma\sigma}(W(\Lambda, p)) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \sum_{\ell} D_{\bar\ell\ell}(\Lambda) u_\ell(\mathbf{p}, \sigma, n) \qquad (5.1.19)$$

and

$$\sum_{\bar\sigma} v_{\bar\ell}(\mathbf{p}_\Lambda, \bar\sigma, n) D^{(j_n)*}_{\bar\sigma\sigma}(W(\Lambda, p)) = \sqrt{\frac{p^0}{(\Lambda p)^0}} \sum_{\ell} D_{\bar\ell\ell}(\Lambda) v_\ell(\mathbf{p}, \sigma, n) \qquad (5.1.20)$$

for arbitrary homogeneous Lorentz transformations Λ.

Boosts

Next take $\mathbf{p} = 0$ in Eqs. (5.1.19) and (5.1.20), and let Λ be the standard boost $L(q)$ that takes a particle of mass m from rest to some four-momentum q^μ. Then $L(p) = \mathbf{1}$, and

$$W(\Lambda, p) \equiv L^{-1}(\Lambda p)\Lambda L(p) = L^{-1}(q)L(q) = \mathbf{1}.$$

Hence in this special case, Eqs. (5.1.19) and (5.1.20) give

$$u_\ell(\mathbf{q}, \sigma, n) = (m/q^0)^{1/2} \sum_{\bar\ell} D_{\ell\bar\ell}(L(q)) u_{\bar\ell}(0, \sigma, n) \tag{5.1.21}$$

and

$$v_\ell(\mathbf{q}, \sigma, n) = (m/q^0)^{1/2} \sum_{\bar\ell} D_{\ell\bar\ell}(L(q)) v_{\bar\ell}(0, \sigma, n). \tag{5.1.22}$$

In other words, if we know the quantities $u_\ell(0, \sigma, n)$ and $v_\ell(0, \sigma, n)$ for zero momentum, then for a given representation $D(\Lambda)$ of the homogeneous Lorentz group, we know the functions $u_\ell(\mathbf{p}, \sigma, n)$ and $v_\ell(\mathbf{p}, \sigma, n)$ for all $\mathbf{p}$. (Explicit formulas for the matrices $D_{\ell\bar\ell}(L(q))$ will be given for arbitrary representations of the homogeneous Lorentz group in Section 5.7.)

Rotations

Next, take $\mathbf{p} = 0$, but this time let Λ be a Lorentz transformation with $\mathbf{p}_\Lambda = 0$; that is, take Λ as a rotation R. Here obviously $W(\Lambda, p) = R$, and so Eqs. (5.1.19) and (5.1.20) read

$$\sum_{\bar\sigma} u_\ell(0, \bar\sigma, n) D^{(j_n)}_{\bar\sigma\sigma}(R) = \sum_{\bar\ell} D_{\ell\bar\ell}(R) u_{\bar\ell}(0, \sigma, n) \tag{5.1.23}$$

and

$$\sum_{\bar\sigma} v_\ell(0, \bar\sigma, n) D^{(j_n)*}_{\bar\sigma\sigma}(R) = \sum_{\bar\ell} D_{\ell\bar\ell}(R) v_{\bar\ell}(0, \sigma, n), \tag{5.1.24}$$

or equivalently

$$\sum_{\bar\sigma} u_\ell(0, \bar\sigma, n) \mathbf{J}^{(j_n)}_{\bar\sigma\sigma} = \sum_{\bar\ell} \boldsymbol{\mathcal{J}}_{\ell\bar\ell} u_{\bar\ell}(0, \sigma, n) \tag{5.1.25}$$

and

$$\sum_{\bar\sigma} v_\ell(0, \bar\sigma, n) \mathbf{J}^{(j_n)*}_{\bar\sigma\sigma} = -\sum_{\bar\ell} \boldsymbol{\mathcal{J}}_{\ell\bar\ell} v_{\bar\ell}(0, \sigma, n), \tag{5.1.26}$$

where $\mathbf{J}^{(j)}$ and $\boldsymbol{\mathcal{J}}$ are the angular-momentum matrices in the representations $D^{(j)}(R)$ and $D(R)$, respectively. Any representation $D(\Lambda)$ of the homogeneous Lorentz group obviously yields a representation of the rotation group when Λ is restricted to rotations R; Eqs. (5.1.25) and (5.1.26) tell us that if the field $\psi^{(\pm)}_\ell(x)$ is to describe particles of some particular spin j, then this representation $D(R)$ must contain among its irreducible components the spin-j representation $D^{(j)}(R)$, with the coefficients $u_\ell(0, \sigma, n)$ and $v_\ell(0, \sigma, n)$ simply describing how the spin-j representation of the rotation group is embedded in $D(R)$. We shall see in Section 5.5 that each irreducible representation of the proper orthochronous Lorentz group contains any given irreducible representation of the rotation group at most once, so that if the fields $\psi^{(+)}_\ell(x)$ and $\psi^{(-)}_\ell(x)$ transform irreducibly, then they are unique

up to overall scale. More generally, the number of free parameters in the annihilation or creation fields (including their overall scales) is equal to the number of irreducible representations in the field.

It is straightforward to show that coefficient functions $u_\ell(\mathbf{p}, \sigma, n)$ and $v_\ell(\mathbf{p}, \sigma, n)$ given by Eqs. (5.1.21) and (5.1.22), with $u_\ell(0, \sigma, n)$ and $v_\ell(0, \sigma, n)$ satisfying Eqs. (5.1.23) and (5.1.24), will automatically satisfy the more general requirements (5.1.19) and (5.1.20). This is left as an exercise for the reader.

Let us now return to the cluster decomposition principle. Inserting Eqs. (5.1.17) and (5.1.18) in Eq. (5.1.9) and integrating over $\mathbf{x}$, the interaction Hamiltonian is

$$\begin{aligned} H_{\rm int} = \sum_{NM} \int d^3p'_1 \cdots d^3p'_N\, d^3p_1 \cdots d^3p_M \sum_{\sigma'_1\cdots\sigma'_N} \sum_{\sigma_1\cdots\sigma_M} \sum_{n'_1\cdots n'_N} \sum_{n_1\cdots n_M} \\ \cdot a^\dagger_{\mathbf{p}'_1,\sigma'_1,n'_1} \cdots a^\dagger_{\mathbf{p}'_N,\sigma'_N,n'_N} a_{\mathbf{p}_M,\sigma_M,n_M} \cdots a_{\mathbf{p}_1,\sigma_1,n_1} \\ \cdot \mathcal{V}_{NM}(\mathbf{p}'_1\sigma'_1n'_1 \cdots \mathbf{p}'_N\sigma'_Nn'_N,\, \mathbf{p}_1\sigma_1n_1 \cdots \mathbf{p}_M\sigma_Mn_M) \end{aligned} \tag{5.1.27}$$

with coefficient functions given by

$$\begin{aligned} \mathcal{V}_{NM}(\mathbf{p}'_1\sigma'_1n'_1 \cdots, \mathbf{p}_1\sigma_1n_1 \cdots) = \delta^3(\mathbf{p}'_1 + \cdots - \mathbf{p}_1 - \cdots) \\ \cdot \widetilde{\mathcal{V}}_{NM}(\mathbf{p}'_1\sigma'_1n'_1 \cdots, \mathbf{p}_1\sigma_1n_1 \cdots), \end{aligned} \tag{5.1.28}$$

where

$$\begin{aligned} \widetilde{\mathcal{V}}_{NM}(\mathbf{p}'_1\sigma'_1n'_1 \cdots \mathbf{p}'_N\sigma'_Nn'_N,\, \mathbf{p}_1\sigma_1n_1 \cdots \mathbf{p}_M\sigma_Mn_M) = (2\pi)^{3-3N/2-3M/2} \\ \cdot \sum_{\ell'_1\cdots\ell'_N} \sum_{\ell_1\cdots\ell_M} g_{\ell'_1\cdots\ell'_N,\ell_1\cdots\ell_M} v_{\ell'_1}(\mathbf{p}'_1, \sigma'_1, n'_1) \cdots v_{\ell'_N}(\mathbf{p}'_N, \sigma'_N, n'_N) \\ \cdot u_{\ell_1}(\mathbf{p}_1, \sigma_1, n_1) \cdots u_{\ell_M}(\mathbf{p}_M, \sigma_M, n_M). \end{aligned} \tag{5.1.29}$$

This interaction is manifestly of the form that will guarantee that the S-matrix satisfies the cluster decomposition principle: $\mathcal{V}_{NM}$ has a single delta function factor, with a coefficient $\widetilde{\mathcal{V}}_{NM}$ that (at least for a finite number of field types) has at most branch point singularities at zero particle momenta. In fact, we could turn this argument around; any operator can be written as in Eq. (5.1.27), and the cluster decomposition principle requires that the coefficient $\mathcal{V}_{NM}$ may be written as in Eq. (5.1.28) as the product of a single momentum-conservation delta function times a smooth coefficient function. Any sufficiently smooth function (but not one containing additional delta functions) can be expressed as in Eq. (5.1.29).[38] *The cluster decomposition principle together with Lorentz invariance thus makes it natural that the interaction density should be constructed out of the annihilation and creation fields.*

If all we needed were to construct a scalar interaction density that satisfied the cluster decomposition principle, then we could combine annihilation and creation operators in

[38]For general functions the indices ℓ and ℓ' may have to run over an infinite range. The reasons for restricting ℓ and ℓ' to a finite range have to do with the principle of renormalizability, discussed in Chapter 12.

in five spacetime dimensions. For the particular $4\cdot 4$ representation (5.4.17) of the γ-matrices, the matrix γ^5 is

$$\gamma^5 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}. \tag{5.4.28}$$

This representation is convenient because it reduces $\mathcal{J}^{\rho\sigma}$ and γ^5 to block-diagonal form. As we shall see, this makes it particularly useful for dealing with particles in the ultra-relativistic limit, $v \to c$. (It is not, however, the representation described in Section 1.1 that was originally introduced by Dirac, because Dirac was mostly interested in electrons in atoms where $v \ll c$, and in this case it is more convenient to adopt a representation for which γ^0 rather than γ^5 is diagonal.)

The representation of the homogeneous Lorentz group we have constructed here is not unitary, because the generators $\mathcal{J}^{\rho\sigma}$ are not all represented by Hermitian matrices. In particular, in the representation (5.4.17) we have $\mathcal{J}^{ij}$ Hermitian, but $\mathcal{J}^{i0}$ is anti-Hermitian. Such reality conditions can conveniently be written in a manifestly Lorentz-invariant fashion by introducing the matrix $\beta = i\gamma^0$ of Eq. (5.4.13), which in the representation (5.4.17) takes the form

$$\beta = \begin{bmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{bmatrix}. \tag{5.4.29}$$

Inspection of Eq. (5.4.17) shows that

$$\beta\gamma^{\mu\dagger}\beta = -\gamma^\mu \tag{5.4.30}$$

and it follows then that

$$\beta\mathcal{J}^{\rho\sigma\dagger}\beta = \mathcal{J}^{\rho\sigma}. \tag{5.4.31}$$

Hence, though not unitary, the matrices $D(\Lambda)$ satisfy the pseudounitarity relation

$$\beta D(\Lambda)^\dagger\beta = D(\Lambda)^{-1}. \tag{5.4.32}$$

Also, γ^5 is Hermitian and anticommutes with β so

$$\beta(\gamma^5)^\dagger\beta = -\gamma^5 \tag{5.4.33}$$

and it follows that

$$\beta(\gamma^5\gamma_\mu)^\dagger\beta = -\gamma^5\gamma_\mu. \tag{5.4.34}$$

The Dirac and related matrices also have important symmetry properties. Inspection of Eqs. (5.4.17) and (5.4.18) shows that γ_μ is symmetric for $\mu = 0, 2$ and antisymmetric for $\mu = 1, 3$, so

$$\gamma_\mu^{\mathrm{T}} = -\mathcal{C}\gamma_\mu\mathcal{C}^{-1}, \tag{5.4.35}$$

where T denotes a transpose, and

$$\mathcal{C} \equiv \gamma_2\beta = -i\begin{bmatrix} \sigma_2 & 0 \\ 0 & -\sigma_2 \end{bmatrix}. \tag{5.4.36}$$

It follows immediately that

$$\mathcal{J}_{\mu\nu}^{\mathrm{T}} = -\mathcal{C}\mathcal{J}_{\mu\nu}\mathcal{C}^{-1}, \tag{5.4.37}$$

$$(\gamma^5)^{\mathrm{T}} = +\mathcal{C}\gamma^5\mathcal{C}^{-1}, \tag{5.4.38}$$

$$(\gamma^5\gamma_\mu)^{\mathrm{T}} = +\mathcal{C}\gamma^5\gamma_\mu\mathcal{C}^{-1}. \tag{5.4.39}$$

These signs will prove significant when we consider the charge-conjugation properties of various currents in the next section. Of course, we can combine our results for adjoints and transposes to obtain the complex conjugates of the Dirac and allied matrices:

$$\gamma_\mu^* = \beta\mathcal{C}\gamma_\mu\mathcal{C}^{-1}\beta, \tag{5.4.40}$$

$$\mathcal{J}_{\mu\nu}^* = -\beta\mathcal{C}\mathcal{J}_{\mu\nu}\mathcal{C}^{-1}\beta, \tag{5.4.41}$$

$$(\gamma^5)^* = -\beta\mathcal{C}\gamma^5\mathcal{C}^{-1}\beta, \tag{5.4.42}$$

$$(\gamma^5\gamma_\mu)^* = -\beta\mathcal{C}\gamma^5\gamma_\mu\mathcal{C}^{-1}\beta. \tag{5.4.43}$$

5.5 Causal Dirac Fields

We now want to construct particle annihilation and antiparticle creation fields that transform under the Lorentz group according to the Dirac representation of this group, discussed in the previous section. In general these take the form given in Eqs. (5.1.17) and (5.1.18):

$$\psi_\ell^{(+)}(x) = (2\pi)^{-3/2}\sum_\sigma \int d^3p\, u_\ell(\mathbf{p},\sigma)e^{ip\cdot x}a_{\mathbf{p},\sigma} \tag{5.5.1}$$

and

$$\psi_\ell^{(-c)}(x) = (2\pi)^{-3/2}\sum_\sigma \int d^3p\, v_\ell(\mathbf{p},\sigma)e^{-ip\cdot x}a^{c\dagger}_{\mathbf{p},\sigma}, \tag{5.5.2}$$

with the particle species label omitted here. In order to calculate the coefficient functions $u_\ell(\mathbf{p},\sigma)$ and $v_\ell(\mathbf{p},\sigma)$ appearing in these formulas, we must first use Eqs. (5.1.25) and (5.1.26) to find u_ℓ and v_ℓ for zero momentum, and then apply Eqs. (5.1.21) and (5.1.22) to calculate them for arbitrary momenta, with $D_{\ell\bar\ell}(\Lambda)$ in both cases taken as the $4\cdot 4$ Dirac representation of the homogeneous Lorentz group discussed in the previous section.

Using Eq. (5.4.19), the zero-momentum conditions (5.1.25) and (5.1.26) read[43]

$$\sum_{\bar\sigma} u_{\bar m,\pm}(\mathbf{0},\bar\sigma)\mathbf{J}^{(j)}_{\bar\sigma\sigma} = \sum_m \frac{1}{2}\boldsymbol{\sigma}_{\bar m m}u_{m,\pm}(\mathbf{0},\sigma)$$

and

$$-\sum_{\bar\sigma} v_{\bar m,\pm}(\mathbf{0},\bar\sigma)\mathbf{J}^{(j)*}_{\bar\sigma\sigma} = \sum_m \frac{1}{2}\boldsymbol{\sigma}_{\bar m m}v_{m,\pm}(\mathbf{0},\sigma).$$

[43] We are here dropping the species label n, and replacing the four-component index ℓ with a pair of indices, one 2-valued index m labelling the rows and columns of the submatrices in Eqs. (5.4.19) and (5.4.20), and a second index taking values $\pm$, labelling the rows and columns of the supermatrix in Eqs. (5.4.19) and (5.4.20).

In other words, if we regard $u_{m,\pm}(\mathbf{0},\sigma)$ and $v_{m,\pm}(\mathbf{0},\sigma)$ as the m,σ elements of matrices $U_\pm$ and $V_\pm$, we have in matrix notation

$$U_\pm\mathbf{J}^{(j)} = \frac{1}{2}\boldsymbol{\sigma}U_\pm \tag{5.5.3}$$

and

$$-V_\pm\mathbf{J}^{(j)*} = \frac{1}{2}\boldsymbol{\sigma}V_\pm. \tag{5.5.4}$$

Now, the $(2j+1)$-dimensional matrices $\mathbf{J}^{(j)}$ and $-\mathbf{J}^{(j)*}$ and the $2\cdot 2$ matrices $\frac{1}{2}\boldsymbol{\sigma}$ all provide irreducible representations of the Lie algebra of the rotation group. A general theorem of group theory known as Schur's lemma [6] tells us that when a matrix like $U_\pm$ or $V_\pm$ connects two such representations as in Eqs. (5.5.3) and (5.5.4), the matrix must either vanish (a possibility of no interest here) or else be square and non-singular. Hence the Dirac field can only describe particles of spin $j = \frac{1}{2}$ (so that $2j+1 = 2$) and the matrices $\mathbf{J}^{(1/2)}$ and $-\mathbf{J}^{(1/2)*}$ must be the same as $\frac{1}{2}\boldsymbol{\sigma}$ up to a similarity transformation. In fact, in the standard representation (2.5.21), (2.5.22) of the rotation generators, we have $\mathbf{J}^{(1/2)} = \frac{1}{2}\boldsymbol{\sigma}$ and $-\mathbf{J}^{(1/2)*} = \frac{1}{2}\sigma_2\boldsymbol{\sigma}\sigma_2$. It follows then that $U_\pm$ and $V_\pm\sigma_2$ must commute with $\boldsymbol{\sigma}$, and hence must be proportional to the unit matrix:

$$u_{m,\pm}(\mathbf{0},\sigma) = c_\pm\delta_{m\sigma}, \qquad v_{m,\pm}(\mathbf{0},\sigma) = id_\pm(\sigma_2)_{m\sigma}. \tag{5.5.5}$$

In other words

$$u(\mathbf{0},\tfrac{1}{2}) = \begin{bmatrix} c_+ \\ 0 \\ c_- \\ 0 \end{bmatrix}, \quad u(\mathbf{0},-\tfrac{1}{2}) = \begin{bmatrix} 0 \\ c_+ \\ 0 \\ c_- \end{bmatrix},$$

$$v(\mathbf{0},\tfrac{1}{2}) = \begin{bmatrix} 0 \\ d_+ \\ 0 \\ d_- \end{bmatrix}, \quad v(\mathbf{0},-\tfrac{1}{2}) = -\begin{bmatrix} d_+ \\ 0 \\ d_- \\ 0 \end{bmatrix},$$

and the spinors at finite momentum are

$$u(\mathbf{p},\sigma) = \sqrt{m/p^0}\,D(L(\mathbf{p}))u(\mathbf{0},\sigma), \tag{5.5.6}$$

$$v(\mathbf{p},\sigma) = \sqrt{m/p^0}\,D(L(\mathbf{p}))v(\mathbf{0},\sigma). \tag{5.5.7}$$

It now only remains to say something about the constants $c_\pm$ and $d_\pm$. In general, these are quite arbitrary—we could even choose c_- and d_- or c_+ and d_+ to be zero if we liked, so that the Dirac field would have only two non-vanishing components. The only physical principle that could tell us anything about the relative values of the $c_\pm$ or the $d_\pm$ is the conservation of parity. We recall that under a space inversion, the particle annihilation and antiparticle creation operators undergo the transformations:

$$Pa_{\mathbf{p},\sigma}P^{-1} = \eta^* a_{-\mathbf{p},\sigma} \tag{5.5.8}$$

$$Pa^{c\dagger}_{\mathbf{p},\sigma}P^{-1} = \eta^c a^{c\dagger}_{-\mathbf{p},\sigma} \tag{5.5.9}$$

and so

$$P\psi_\ell^{(+)}(x)P^{-1} = \eta^*(2\pi)^{-3/2}\sum_\sigma\int d^3p\,u_\ell(-\mathbf{p},\sigma)e^{ip\cdot\mathcal{P}x}a_{\mathbf{p},\sigma}, \tag{5.5.10}$$

$$P\psi_\ell^{(-c)}(x)P^{-1} = \eta^c(2\pi)^{-3/2}\sum_\sigma\int d^3p\,v_\ell(-\mathbf{p},\sigma)e^{-ip\cdot\mathcal{P}x}a^{c\dagger}_{\mathbf{p},\sigma}. \tag{5.5.11}$$

Also, Eqs. (5.4.16), (5.1.21), and (5.1.22) give

$$u(-\mathbf{p},\sigma) = \sqrt{m/p^0}\,\beta D(L(\mathbf{p}))\beta u(\mathbf{0},\sigma) \tag{5.5.12}$$

$$v(-\mathbf{p},\sigma) = \sqrt{m/p^0}\,\beta D(L(\mathbf{p}))\beta v(\mathbf{0},\sigma). \tag{5.5.13}$$

(Since $\beta^2 = \mathbf{1}$, we are no longer making a distinction between β and β^{-1}.) In order that the parity operator should transform the annihilation and creation fields at the point x into something proportional to these fields at $\mathcal{P}x$, it is necessary that $\beta u(\mathbf{0},\sigma)$ and $\beta v(\mathbf{0},\sigma)$ be proportional to $u(\mathbf{0},\sigma)$ and $v(\mathbf{0},\sigma)$, respectively:

$$\beta u(\mathbf{0},\sigma) = b_u u(\mathbf{0},\sigma), \qquad \beta v(\mathbf{0},\sigma) = b_v v(\mathbf{0},\sigma), \tag{5.5.14}$$

where b_u and b_v are sign factors, $b_u^2 = b_v^2 = 1$. In this case, the fields have the simple space-inversion properties:

$$P\psi^{(+)}(x)P^{-1} = \eta^* b_u\beta\psi^{(+)}(\mathcal{P}x), \tag{5.5.15}$$

$$P\psi^{(-c)}(x)P^{-1} = \eta^c b_v\beta\psi^{(-c)}(\mathcal{P}x). \tag{5.5.16}$$

By adjusting the overall scales of the fields, we can choose the coefficient functions at zero momentum to have the form:

$$u(\mathbf{0},\tfrac{1}{2}) = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 0 \\ b_u \\ 0 \end{bmatrix}, \qquad u(\mathbf{0},-\tfrac{1}{2}) = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 1 \\ 0 \\ b_u \end{bmatrix}, \tag{5.5.17}$$

$$v(\mathbf{0},\tfrac{1}{2}) = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 1 \\ 0 \\ b_v \end{bmatrix}, \qquad v(\mathbf{0},-\tfrac{1}{2}) = -\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 0 \\ b_v \\ 0 \end{bmatrix}. \tag{5.5.18}$$

Now let's try to put together the annihilation and creation fields in a linear combination

$$\psi(x) = \kappa\psi^{(+)}(x) + \lambda\psi^{(-c)}(x) \tag{5.5.19}$$

that commutes or anticommutes with itself and its adjoint at space-like separations. A straightforward calculation gives

$$[\psi_\ell(x), \psi_{\bar\ell}^\dagger(y)]_\mp = \int d^3p\left[|\kappa|^2 N_{\ell\bar\ell}(\mathbf{p})e^{ip\cdot(x-y)} \mp |\lambda|^2 M_{\ell\bar\ell}(\mathbf{p})e^{-ip\cdot(x-y)}\right], \tag{5.5.20}$$

where

$$N_{\ell\bar\ell}(\mathbf{p}) \equiv \sum_\sigma u_\ell(\mathbf{p},\sigma)u^*_{\bar\ell}(\mathbf{p},\sigma), \tag{5.5.21}$$

$$M_{\ell\bar\ell}(\mathbf{p}) \equiv \sum_\sigma v_\ell(\mathbf{p},\sigma)v^*_{\bar\ell}(\mathbf{p},\sigma). \tag{5.5.22}$$

By using either the eigenvalue conditions (5.5.14) or the explicit formulas (5.5.17) and (5.5.18), we find at zero momentum:

$$N(\mathbf{0}) = \frac{\mathbf{1} + b_u\beta}{2(2\pi)^3}, \qquad M(\mathbf{0}) = \frac{\mathbf{1} + b_v\beta}{2(2\pi)^3}. \tag{5.5.23}$$

From Eqs. (5.5.6) and (5.5.7) we have then

$$N(\mathbf{p}) = \frac{m}{2p^0}D(L(\mathbf{p}))[\mathbf{1} + b_u\beta]D^\dagger(L(\mathbf{p})), \tag{5.5.24}$$

$$M(\mathbf{p}) = \frac{m}{2p^0}D(L(\mathbf{p}))[\mathbf{1} + b_v\beta]D^\dagger(L(\mathbf{p})). \tag{5.5.25}$$

The pseudounitarity condition (5.4.32) yields

$$D(L(\mathbf{p}))\beta D^\dagger(L(\mathbf{p})) = \beta$$

and

$$D(L(\mathbf{p}))D^\dagger(L(\mathbf{p})) = D(L(\mathbf{p}))\beta D^{-1}(L(\mathbf{p}))\beta.$$

We also recall that $\beta = -i\gamma^0$, so by using the Lorentz transformation rule (5.4.8) we have

$$D(L(\mathbf{p}))\beta D^{-1}(L(\mathbf{p})) = -iL_\mu{}^0(\mathbf{p})\gamma^\mu = -ip^\mu\gamma_\mu/m. \tag{5.5.26}$$

Putting this together, we find[44]

$$N(\mathbf{p}) = \frac{1}{2p^0}[-ip^\mu\gamma_\mu + b_u m]\beta, \tag{5.5.27}$$

$$M(\mathbf{p}) = \frac{1}{2p^0}[-ip^\mu\gamma_\mu + b_v m]\beta. \tag{5.5.28}$$

Using this in Eq. (5.5.20) yields finally

$$[\psi_\ell(x), \psi^\dagger_{\bar\ell}(y)]_\mp = \left([|\kappa|^2[-\gamma^\mu\partial_\mu + b_u m]\beta\Delta_+(x-y)\right.$$
$$\left.\mp |\lambda|^2[-\gamma^\mu\partial_\mu + b_v m]\beta\Delta_+(y-x)\right)_{\ell\bar\ell}, \tag{5.5.29}$$

where Δ_+ is the function introduced in Section 5.2:

$$\Delta_+(x) \equiv \int\frac{d^3p}{2p^0(2\pi)^3}e^{ip\cdot x}.$$

[44] Sometimes an extra factor $\sqrt{p^0/m}$ is included in the Dirac spinors, so that m appears in place of p^0 in the denominators of the spin sums (5.5.27) and (5.5.28). The normalization convention used here has the advantage that it goes smoothly over to the case $m = 0$.

We saw in Section 5.2 that for $x - y$ space-like $\Delta_+(x-y)$ is an even function of $x-y$ and, of course, this implies that its first derivatives are odd functions of $x - y$. Hence, in order that both the derivative and the non-derivative terms in the commutator or anticommutator should vanish at space-like separations, it is necessary and sufficient that

$$|\kappa|^2 = \mp|\lambda|^2 \tag{5.5.30}$$

and

$$|\kappa|^2 b_u = \pm|\lambda|^2 b_v. \tag{5.5.31}$$

Clearly Eq. (5.5.30) is only possible if we choose the bottom sign, $\mp = +$; that is, *the particles described by a Dirac field must be fermions*. It is also then necessary that $|\kappa|^2 = |\lambda|^2$ and $b_u = -b_v$. Just as for scalars we have the freedom to redefine the relative phase of the creation and annihilation operators to make the ratio κ/λ real, in which case $\kappa = \lambda$, and by adjusting the overall scale and phase of the field ψ we may then take

$$\kappa = \lambda = 1. \tag{5.5.32}$$

Finally if we like we can replace ψ with $\gamma^5\psi$, which changes the sign of both b_u and b_v, so we can always take

$$b_u = -b_v = +1. \tag{5.5.33}$$

For future use, we record here that the Dirac field is now

$$\psi_\ell(x) = (2\pi)^{-3/2}\sum_\sigma\int d^3p\left[u_\ell(\mathbf{p},\sigma)e^{ip\cdot x}a_{\mathbf{p},\sigma} + v_\ell(\mathbf{p},\sigma)e^{-ip\cdot x}a^{c\dagger}_{\mathbf{p},\sigma}\right] \tag{5.5.34}$$

while the coefficient functions at zero momentum are

$$u(\mathbf{0},\tfrac{1}{2}) = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \qquad u(\mathbf{0},-\tfrac{1}{2}) = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \tag{5.5.35}$$

$$v(\mathbf{0},\tfrac{1}{2}) = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \qquad v(\mathbf{0},-\tfrac{1}{2}) = -\frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}. \tag{5.5.36}$$

The spin sums are

$$N(\mathbf{p}) = \frac{1}{2p^0}[-ip^\mu\gamma_\mu + m]\beta, \tag{5.5.37}$$

$$M(\mathbf{p}) = \frac{1}{2p^0}[-ip^\mu\gamma_\mu - m]\beta, \tag{5.5.38}$$

so the anticommutator is given by Eq. (5.5.20) as

$$[\psi_\ell(x), \psi^\dagger_{\bar\ell}(y)]_+ = \{[-\gamma^\mu\partial_\mu + m]\beta\}_{\ell\bar\ell}\Delta(x-y). \tag{5.5.39}$$

Now let's return to the requirement that under a space inversion the field $\psi(x)$ must transform into something proportional to $\psi(\mathcal{P}x)$. For this to be possible the phases in Eqs. (5.5.15) and (5.5.16) must be equal, and so the intrinsic parities of particles and their antiparticles must be related by

$$\eta^c = -\eta^*. \tag{5.5.40}$$

That is, *the intrinsic parity $\eta\eta^c$ of a state consisting of a spin $\frac{1}{2}$ particle and its antiparticle is odd.* It is for this reason that negative parity mesons like the ρ^0 and J/ψ can be interpreted as s-wave bound states of quark–antiquark pairs. Eqs. (5.5.15) and (5.5.16) now give the transformation of the causal Dirac field under space inversion as

$$P\psi(x)P^{-1} = \eta^*\beta\psi(\mathcal{P}x). \tag{5.5.41}$$

Before going on to the other inversions, this is a good place to mention that Eqs. (5.5.14), (5.5.33), and (5.5.26) show that $u(\mathbf{p},\sigma)$ and $v(\mathbf{p},\sigma)$ are eigenvectors of $-ip^\mu\gamma_\mu/m$ with eigenvalues $+1$ and -1, respectively:

$$(ip^\mu\gamma_\mu + m)u(\mathbf{p},\sigma) = 0, \qquad (-ip^\mu\gamma_\mu + m)v(\mathbf{p},\sigma) = 0. \tag{5.5.42}$$

It follows then that the field (5.5.33) satisfies the differential equation

$$(\gamma^\mu\partial_\mu + m)\psi(x) = 0. \tag{5.5.43}$$

This is the celebrated Dirac equation for a free particle of spin $\frac{1}{2}$. From the point of view adopted here, the free-particle Dirac equation is nothing but a Lorentz-invariant record of the convention that we have used in putting together the two irreducible representations of the proper orthochronous Lorentz group to form a field that transforms simply also under space inversion.

In order to work out the charge-conjugation and time-reversal properties of the Dirac field, we will need expressions for the complex-conjugates of the u and v coefficient functions. These functions are real for zero momentum, but to obtain the coefficient functions at finite momentum we have to multiply with the complex matrix $D(L(\mathbf{p}))$. From Eq. (5.4.41) we see that for general real $\omega_{\mu\nu}$:

$$[\exp(\tfrac{1}{2}i\mathcal{J}^{\mu\nu}\omega_{\mu\nu})]^* = \beta\mathcal{C}\exp(\tfrac{1}{2}i\mathcal{J}^{\mu\nu}\omega_{\mu\nu})\mathcal{C}^{-1}\beta$$

and so in particular

$$D(L(\mathbf{p}))^* = \beta\mathcal{C}D(L(\mathbf{p}))\mathcal{C}^{-1}\beta.$$

We also note that $\mathcal{C}^{-1}\beta u(\mathbf{0},\sigma) = -v(\mathbf{0},\sigma)$ and $\mathcal{C}^{-1}\beta v(\mathbf{0},\sigma) = -u(\mathbf{0},\sigma)$, so

$$u^*_\ell(\mathbf{p},\sigma) = -(\beta\mathcal{C}v(\mathbf{p},\sigma))_\ell, \tag{5.5.44}$$

$$v^*_\ell(\mathbf{p},\sigma) = -(\beta\mathcal{C}u(\mathbf{p},\sigma))_\ell. \tag{5.5.45}$$

In order for the field to transform under charge-conjugation into another field with which it commutes at space-like separations, it is necessary again that the charge conjugation parities of the particle and antiparticle be related by

$$\xi^c = \xi^*. \tag{5.5.46}$$

In this case, the field transforms as

$$C\psi(x)C^{-1} = -\xi^*\beta\mathcal{C}\psi^*(x). \tag{5.5.47}$$

(We are calling the Hermitian adjoint of the field on the right-hand side ψ^* instead of $\psi^\dagger$ to emphasize that this is still a column vector, not a row.)

Although we have been distinguishing particles from their antiparticles, we have not ruled out the possibility that the two are actually identical. Such spin $\frac{1}{2}$ particles are called *Majorana fermions*. Following the same reasoning that led to Eq. (5.5.47), the Dirac field of such a particle must satisfy the reality condition

$$\psi(x) = -\beta\mathcal{C}\psi^*(x). \tag{5.5.48}$$

For Majorana fermions the intrinsic space-inversion parity must be imaginary, $\eta = \pm i$, while the charge-conjugation parity must be real, $\xi = \pm 1$.

There is an important difference between fermions and bosons in the intrinsic charge-conjugation phase of states consisting of a particle and its antiparticle. Such a state may be written

$$|\Phi\rangle = \sum_{\sigma,\sigma'}\int d^3p\int d^3p'\,\chi(\mathbf{p},\sigma;\mathbf{p}',\sigma')a^\dagger_{\mathbf{p},\sigma}a^{c\dagger}_{\mathbf{p}',\sigma'}|\Omega\rangle,$$

where $|\Omega\rangle$ is the vacuum state. Under charge-conjugation, this state is transformed into

$$C|\Phi\rangle = \xi^*\xi^{c*}\sum_{\sigma,\sigma'}\int d^3p\int d^3p'\,\chi(\mathbf{p},\sigma;\mathbf{p}',\sigma')a^{c\dagger}_{\mathbf{p},\sigma}a^\dagger_{\mathbf{p}',\sigma'}|\Omega\rangle.$$

Interchanging the variables of integration and summation and using the anticommutation of the creation operators and Eq. (5.5.46), we can rewrite this as

$$C|\Phi\rangle = -\sum_{\sigma,\sigma'}\int d^3p\int d^3p'\,\chi(\mathbf{p}',\sigma';\mathbf{p},\sigma)a^\dagger_{\mathbf{p},\sigma}a^{c\dagger}_{\mathbf{p}',\sigma'}|\Omega\rangle.$$

That is, *the intrinsic charge-conjugation parity of a state consisting of a particle described by a Dirac field and its antiparticle is odd*, in the sense that if the wave function χ of the state is even or odd under interchange of the momenta and spins of the particle and antiparticle, then the charge-conjugation operator applied to such a state gives a sign -1 or $+1$, respectively. The classic example here is positronium, the bound state of an electron and a positron. The two lowest states are a pair of nearly degenerate s-wave states with total spin $s = 0$ and $s = 1$, known respectively as para- and ortho-positronium. The wave function for these two states is even under interchange of momenta and odd or even respectively under interchange of spin z-components, so para- and ortho-positronium have $C = +1$ and $C = -1$, respectively. These values are dramatically confirmed in the decay modes of positronium: para-positronium decays rapidly into a pair of photons (each of which has $C = -1$), while ortho-positronium can only decay much more slowly into three or more photons. In the same way, single ρ^0 and ω^0 mesons are produced as resonances in high-energy electron–positron annihilation through a one-photon intermediate state, so

they must have $C = -1$, which is consistent with their interpretation as quark–antiquark bound states with orbital angular momentum zero and total quark spin one.

Now we come to time-reversal. Recall the transformation properties of the particle annihilation and antiparticle creation operators given by Eq. (4.2.15):

$$Ta_{\mathbf{p},\sigma}T^{-1} = \zeta^*(-1)^{1/2-\sigma}a_{-\mathbf{p},-\sigma}, \tag{5.5.49}$$

$$Ta^{c\dagger}_{\mathbf{p},\sigma}T^{-1} = \zeta^c(-1)^{1/2-\sigma}a^{c\dagger}_{-\mathbf{p},-\sigma}. \tag{5.5.50}$$

Time-reversal of the field (5.5.34) thus gives

$$T\psi_\ell(x)T^{-1} = (2\pi)^{-3/2}\sum_\sigma\int d^3p\,(-1)^{1/2-\sigma}$$
$$\cdot\left[\zeta^*u^*_\ell(\mathbf{p},\sigma)e^{-ip\cdot x}a_{-\mathbf{p},-\sigma} + \zeta^c v^*_\ell(\mathbf{p},\sigma)e^{ip\cdot x}a^{c\dagger}_{-\mathbf{p},-\sigma}\right].$$

In order to put this back in the form given for ψ, we shall redefine the variables of integration and summation as $-\mathbf{p}$ and $-\sigma$, so we need formulas for $u^*(-\mathbf{p},-\sigma)$ and $v^*(-\mathbf{p},-\sigma)$ in terms of $u(\mathbf{p},\sigma)$ and $v(\mathbf{p},\sigma)$, respectively. For this purpose, we can use the fact that $\mathcal{J}^{i0}$ anticommutes with β and commutes with γ^5 together with our former result for $D(L(\mathbf{p}))^*$ to write

$$D^*(L(-\mathbf{p})) = \gamma^5\beta D^*(L(\mathbf{p}))\beta\gamma^5 = \gamma^5\mathcal{C}D(L(\mathbf{p}))\mathcal{C}^{-1}\gamma^5.$$

Also, Eqs. (5.4.36) and (5.5.35)–(5.5.36) give

$$\gamma^5\mathcal{C}^{-1}u(\mathbf{0},-\sigma) = (-1)^{1/2-\sigma}u(\mathbf{0},\sigma),$$

$$\gamma^5\mathcal{C}^{-1}v(\mathbf{0},-\sigma) = (-1)^{1/2-\sigma}v(\mathbf{0},\sigma),$$

so

$$(-1)^{1/2+\sigma}u^*(-\mathbf{p},-\sigma) = -\gamma^5\mathcal{C}u(\mathbf{p},\sigma), \tag{5.5.51}$$

$$(-1)^{1/2+\sigma}v^*(-\mathbf{p},-\sigma) = -\gamma^5\mathcal{C}v(\mathbf{p},\sigma). \tag{5.5.52}$$

We see then that in order for time-reversal to take the Dirac field into something proportional to itself at the time-reversed point (with which it would anticommute at space-like separations) it is necessary that the intrinsic time-reversal phases be related by

$$\zeta^c = \zeta^*. \tag{5.5.53}$$

and in this case

$$T\psi(x)T^{-1} = -\zeta^*\gamma^5\mathcal{C}\psi(-\mathcal{T}x). \tag{5.5.54}$$

Now let us consider how to construct scalar interaction densities out of the Dirac fields and their adjoints. As already mentioned the Dirac representation is not unitary, so $\psi^\dagger\psi$ is not a scalar. To deal with this complication it is convenient to define a new sort of adjoint:

$$\bar\psi \equiv \psi^\dagger\beta. \tag{5.5.55}$$

Using the pseudounitarity condition (5.4.32), we see that the fermion bilinears constructed with $\bar\psi$ have the Lorentz transformation property

$$U_0(\Lambda)[\bar\psi(x)M\psi(x)]U_0^{-1}(\Lambda) = \bar\psi(\Lambda x)D^{-1}(\Lambda)MD(\Lambda)\psi(\Lambda x). \tag{5.5.56}$$

Also, under a space inversion

$$P[\bar\psi(x)M\psi(x)]P^{-1} = \bar\psi(\mathcal{P}x)\beta M\beta\psi(\mathcal{P}x). \tag{5.5.57}$$

Taking the matrix M as $\mathbf{1}$, γ^μ, $\mathcal{J}^{\mu\nu}$, $\gamma^5\gamma^\mu$, or γ^5 yields a bilinear $\bar\psi M\psi$ that transforms as a scalar, vector, tensor, axial vector, and pseudoscalar, respectively. (The terms 'axial' and 'pseudo' indicate that these have space-inversion properties opposite to those of ordinary vectors and scalars: a pseudoscalar has negative parity, while the space and time components of an axial vector have positive and negative parity, respectively.) These results apply also when the two fermion fields in the bilinear refer to different particle species, except that in this case a space inversion also yields a ratio of the intrinsic parities.

For instance, the original Fermi theory of beta decay involved an interaction density proportional to $\bar\psi_p\gamma^\mu\psi_n\bar\psi_e\gamma_\mu\psi_\nu$. Later it was realized that the most general Lorentz-invariant and parity-conserving non-derivative beta decay interaction takes the form of a linear combination of products like this, with γ_μ replaced with any one of the five covariant types of $4\cdot 4$ matrices $\mathbf{1}$, γ^μ, $\mathcal{J}^{\mu\nu}$, $\gamma^5\gamma^\mu$, or γ^5. (As discussed in Chapter 2, we are defining the space-inversion operator so that the proton, neutron, and electron all have parity $+1$. If the neutrino is massless then its parity may also be defined as $+1$, if necessary by replacing the neutrino field with $\gamma^5\psi_\nu$.) When Lee and Yang [7] called parity conservation into question in 1956, they expanded the list of possible non-derivative interactions to include ten terms proportional to $\bar\psi_p M\psi_n\bar\psi_e M\psi_\nu$ and also $\bar\psi_p M\psi_n\bar\psi_e M\gamma^5\psi_\nu$, with M running over the matrices $\mathbf{1}$, γ^μ, $\mathcal{J}^{\mu\nu}$, $\gamma^5\gamma^\mu$, or γ^5.

It is also of some interest to study the charge-conjugation properties of these bilinears. Using Eqs. (5.5.47) and (5.4.35)–(5.4.39), we have

$$C(\bar\psi M\psi)C^{-1} = (\beta\mathcal{C}\psi)^{\mathrm{T}}\beta M(\beta\mathcal{C}\psi^*) = -(\beta\mathcal{C}\psi^*)^{\mathrm{T}}M^{\mathrm{T}}\mathcal{C}\psi$$
$$= \bar\psi\mathcal{C}^{-1}M^{\mathrm{T}}\mathcal{C}\psi = \pm\bar\psi M\psi, \tag{5.5.58}$$

the sign in the last expression being $+$ for the matrices $\mathbf{1}$, $\gamma^5\gamma_\mu$, and γ^5, and $-$ for γ_μ and $\mathcal{J}_{\mu\nu}$. (The minus sign in the first line arises from Fermi statistics. We ignore a c-number anticommutator.) A boson field that interacts with the current $\bar\psi M\psi$ must therefore have $C = +1$ for scalars, pseudoscalars, or axial vectors, and $C = -1$ for vectors or antisymmetric tensors. This is one way of seeing that the π^0 (which couples to pseudoscalar or axial-vector nucleon currents) has $C = +1$, while the photon has $C = -1$.

5.6 General Irreducible Representations of the Homogeneous Lorentz Group[45]

[45] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

We shall now generalize from the special cases of vector and Dirac fields to the case of a field that transforms according to a general irreducible representation of the homogeneous Lorentz group. All fields may be constructed as direct sums of these irreducible fields.

A general representation of the proper orthochronous homogeneous Lorentz group (or, more properly, its infinitesimal part) is provided by a set of matrices $\mathcal{J}_{\mu\nu}$ satisfying the same commutation relations (5.4.4) as the generators of the group

$$[\mathcal{J}_{\mu\nu}, \mathcal{J}_{\rho\sigma}] = i(\mathcal{J}_{\rho\mu}\eta_{\sigma\nu} + \mathcal{J}_{\mu\sigma}\eta_{\rho\nu} - \mathcal{J}_{\sigma\nu}\eta_{\mu\rho} - \mathcal{J}_{\nu\rho}\eta_{\sigma\mu}). \tag{5.6.1}$$

(Of course, $\mathcal{J}_{\mu\nu} = -\mathcal{J}_{\nu\mu}$, and indices on $\mathcal{J}_{\mu\nu}$ are as usual raised or lowered by contraction with $\eta^{\mu\nu}$ or $\eta_{\mu\nu}$.) To see how to construct such matrices, first divide the six independent components of $\mathcal{J}_{\mu\nu}$ into two three-vectors: an angular momentum matrix

$$\mathcal{J}_1 = \mathcal{J}_{23}, \qquad \mathcal{J}_2 = \mathcal{J}_{31}, \qquad \mathcal{J}_3 = \mathcal{J}_{12} \tag{5.6.2}$$

and a boost

$$\mathcal{K}_1 = \mathcal{J}_{10}, \qquad \mathcal{K}_2 = \mathcal{J}_{20}, \qquad \mathcal{K}_3 = \mathcal{J}_{30}. \tag{5.6.3}$$

Eq. (5.6.1) then reads

$$[\mathcal{J}_i, \mathcal{J}_j] = i\epsilon_{ijk}\mathcal{J}_k, \tag{5.6.4}$$

$$[\mathcal{J}_i, \mathcal{K}_j] = i\epsilon_{ijk}\mathcal{K}_k, \tag{5.6.5}$$

$$[\mathcal{K}_i, \mathcal{K}_j] = -i\epsilon_{ijk}\mathcal{J}_k, \tag{5.6.6}$$

where i, j, k run over the values $1, 2, 3$, and ϵ_{ijk} is the totally antisymmetric quantity with $\epsilon_{123} = +1$. Eq. (5.6.4) just says that the matrices $\mathcal{J}$ generate a representation of the rotation subgroup of the Lorentz group, and Eq. (5.6.5) just represents the fact that $\mathcal{K}$ is a three-vector. The minus sign in the right-hand side of Eq. (5.6.6) arises from the fact that $\eta_{00} = -1$, and plays a crucial role in what follows.

It is very convenient to replace the matrices $\mathcal{J}$ and $\mathcal{K}$ with two decoupled spin three-vectors, writing

$$\mathcal{A} \equiv \frac{1}{2}(\mathcal{J} + i\mathcal{K}), \tag{5.6.7}$$

$$\mathcal{B} \equiv \frac{1}{2}(\mathcal{J} - i\mathcal{K}). \tag{5.6.8}$$

It is easy to see that the commutation relations (5.6.4)–(5.6.6) are equivalent to

$$[\mathcal{A}_i, \mathcal{A}_j] = i\epsilon_{ijk}\mathcal{A}_k, \tag{5.6.9}$$

$$[\mathcal{B}_i, \mathcal{B}_j] = i\epsilon_{ijk}\mathcal{B}_k, \tag{5.6.10}$$

$$[\mathcal{A}_i, \mathcal{B}_j] = 0. \tag{5.6.11}$$

We find matrices satisfying Eqs. (5.6.9)–(5.6.11) in the same way that we find matrices representing the spins of a pair of uncoupled particles—as a direct sum. That is, we label the rows and columns of these matrices with a pair of integers and/or half-integers a, b, running over the values

$$a = -A, -A+1, \ldots, +A, \tag{5.6.12}$$

$$b = -B, -B+1, \ldots, +B \tag{5.6.13}$$

and take[46]

$$(\mathcal{A})_{a'b',ab} = \delta_{b'b}\mathbf{J}^{(A)}_{a'a}, \tag{5.6.14}$$

$$(\mathcal{B})_{a'b',ab} = \delta_{a'a}\mathbf{J}^{(B)}_{b'b}, \tag{5.6.15}$$

where $\mathbf{J}^{(A)}$ and $\mathbf{J}^{(B)}$ are the standard spin matrices for spins A or B:

$$(\mathbf{J}^{(A)}_3)_{a'a} = a\delta_{a'a}, \tag{5.6.16}$$

$$(\mathbf{J}^{(A)}_1 \pm i\mathbf{J}^{(A)}_2)_{a'a} = \delta_{a',a\pm 1}\sqrt{(A \mp a)(A \pm a + 1)}, \tag{5.6.17}$$

and likewise for $\mathbf{J}^{(B)}$. The representation is labelled by the values of the positive integers and/or half-integers A and B. We see that the (A, B) representation has dimensionality $(2A+1)(2B+1)$.

The finite-dimensional representations of the homogeneous Lorentz group are not unitary, because $\mathcal{A}$ and $\mathcal{B}$ are Hermitian, and therefore $\mathcal{J}$ is Hermitian but $\mathcal{K}$ is *anti-Hermitian*. This is because of the i in Eqs. (5.6.7) and (5.6.8), which is required by the minus sign in (5.6.6), and hence stems from the fact that the homogeneous Lorentz group is not the same as the four-dimensional rotation group SO(4), a compact group, but instead is the non-compact group known as SO(3,1). It is only compact groups that can have finite-dimensional unitary representations (aside from representations in which the non-compact part is represented trivially, by the identity). There is no problem in working with non-unitary representations, because the objects we are now concerned with are fields, not wave functions, and do not need to have a Lorentz-invariant positive norm.

In contrast, the rotation group is represented unitarily, with its generators represented by the Hermitian matrices

$$\mathcal{J} = \mathcal{A} + \mathcal{B}. \tag{5.6.18}$$

By the usual rules of vector addition, we can see that a field that transforms according to the (A, B) representation of the homogeneous Lorentz group has components that rotate like objects of spin j, with

$$j = A + B, A + B - 1, \ldots, |A - B|.$$

This is enough to identify the (A, B) representations with the perhaps more familiar tensors and spinors. For instance, a $(0, 0)$ field is obviously scalar, with only a single $j = 0$ component. A $(1/2, 0)$ or $(0, 1/2)$ field can only have $j = 1/2$; these are the top (i.e., $\gamma^5 = +1$) and bottom ($\gamma^5 = -1$) two components of the Dirac spinor. A $(1/2, 1/2)$ field has components with $j = 1$ and $j = 0$, corresponding to the spatial part $\mathbf{v}$ and time-component v^0 of a four-vector v^μ. More generally, an (A, A) field contains terms with only

[46] There is an alternative formalism [8], based on the fact that the spin j representation of the rotation group can be written as the symmetrized direct product of $2j$ spin 1/2 representations—i.e., as a symmetric SU(2) tensor with $2j$ two-valued indices. We can therefore write fields belonging to the (A, B) representation with $2A$ two-valued $(1/2, 0)$ indices and $2B$ two-valued $(0, 1/2)$ indices, the latter written with dots to distinguish them from the former.

integer spins $2A, 2A-1, \ldots, 0$, and corresponds to a traceless symmetric tensor of rank $2A$. (Note that the number of independent components of a symmetric tensor of rank $2A$ in four dimensions is

$$\frac{4\cdot 5\cdots(4+2A-1)}{(2A)!} = \frac{(3+2A)!}{6(2A)!}$$

and the tracelessness condition reduces this to

$$\frac{(3+2A)!}{6(2A)!} - \frac{(1+2A)!}{6(2A-2)!} = (2A+1)^2$$

as expected for an (A, A) field.) One more example: a $(1, 0)$ or $(0, 1)$ field can only have $j = 1$, and corresponds to an antisymmetric tensor $F^{\mu\nu}$ that satisfies the further irreducibility 'duality' conditions

$$F^{\mu\nu} = \pm\frac{i}{2}\epsilon^{\mu\nu\lambda\rho}F_{\lambda\rho}$$

for $(1, 0)$ and $(0, 1)$ fields, respectively. Of course, it is only in four dimensions that an antisymmetric two-index tensor $F^{\mu\nu}$ can be divided into such 'self-dual' and 'anti-self-dual' parts.

A general tensor of rank N transforms as the direct product of N $(1/2, 1/2)$ four-vector representations. It can therefore be decomposed (by suitable symmetrizations and antisymmetrizations and extracting traces) into irreducible terms (A, B) with $A = N/2, N/2-1, \ldots$ and $B = N/2, N/2-1, \ldots$. In this way, we can construct any irreducible representation (A, B) for which $A+B$ is an integer. The spin representations, for which $A+B$ is half an odd integer, can similarly be constructed from the direct product of these tensor representations and the Dirac representation $(1/2, 0) \oplus (0, 1/2)$.

For instance, taking the direct product of the vector $(1/2, 1/2)$ representation and the Dirac $(1/2, 0) \oplus (0, 1/2)$ representation gives a spinor-vector ψ^μ, that transforms according to the reducible representation

$$(\tfrac{1}{2}, \tfrac{1}{2}) \otimes [(\tfrac{1}{2}, 0) \oplus (0, \tfrac{1}{2})] = (\tfrac{1}{2}, 1) \oplus (\tfrac{1}{2}, 0) \oplus (1, \tfrac{1}{2}) \oplus (0, \tfrac{1}{2}).$$

The quantity $\gamma_\mu\psi^\mu$ would transform as an ordinary $(1/2, 0) \oplus (0, 1/2)$ Dirac field, so we can isolate the $(1/2, 1) \oplus (1, 1/2)$ representation[47] by requiring that $\gamma_\mu\psi^\mu = 0$. This is the *Rarita–Schwinger field* [9].

So far in this section we have only considered the representations of the proper orthochronous Lorentz group. In any representation of the Lorentz group including space inversion, there must be a matrix β which reverses the signs of tensors with odd numbers of space indices, and in particular

$$\beta\mathcal{J}\beta^{-1} = +\mathcal{J}, \qquad \beta\mathcal{K}\beta^{-1} = -\mathcal{K}. \tag{5.6.19}$$

In terms of the matrices (5.6.7) and (5.6.8), this is

$$\beta\mathcal{A}\beta^{-1} = \mathcal{B}, \qquad \beta\mathcal{B}\beta^{-1} = \mathcal{A}. \tag{5.6.20}$$

[47] According to Eq. (5.6.18), such a field transforms under ordinary rotations as a direct sum of two $j = 3/2$ and two $j = 1/2$ components. The doubling is eliminated by imposing the Dirac equation $[\gamma^\nu\partial_\nu + m]\psi^\mu = 0$, and the remaining $j = 1/2$ component is eliminated by requiring that $\partial_\mu\psi^\mu = 0$. With these conditions, the field describes a single particle of spin $j = 3/2$.

Thus an irreducible (A, B) representation of the proper orthochronous homogeneous Lorentz group does not provide a representation of the Lorentz group including space inversion unless $A = B$. As we have seen, these (A, A) representations are the scalar, the vector, and the symmetric traceless tensors. For $A \neq B$, the irreducible representations of the Lorentz group including space inversion are the direct sums $(A, B) \oplus (B, A)$, of dimensionality $2(2A+1)(2B+1)$. One of these is the $(1/2, 0) \oplus (0, 1/2)$ Dirac representation discussed in Section 5.4. The $4\cdot 4$ matrix (5.4.29) provides the β-matrix for this representation. Another familiar example is the $(1, 0) \oplus (0, 1)$ representation, which as we have seen is just the antisymmetric tensor of second rank, including both self-dual and anti-self-dual parts.

5.7 General Causal Fields[48]

We now proceed to construct causal fields that transform according to the general irreducible (A, B) representations described in the previous section. The index ℓ is replaced here with a pair of indices a, b, running over the ranges (5.6.12), (5.6.13), so the fields are now written as

$$\psi_{ab}(x) = (2\pi)^{-3/2}\sum_\sigma\int d^3p\left[\kappa a_{\mathbf{p},\sigma}e^{ip\cdot x}u_{ab}(\mathbf{p},\sigma)\right.$$
$$\left. + \lambda a^{c\dagger}_{\mathbf{p},\sigma}e^{-ip\cdot x}v_{ab}(\mathbf{p},\sigma)\right]. \tag{5.7.1}$$

with κ and λ arbitrary constants. We are here leaving open the possibility that this particle is its own antiparticle, in which case $a^c_{\mathbf{p},\sigma} = a_{\mathbf{p},\sigma}$.

Our first task is to find the zero-momentum coefficient functions $u_{ab}(0,\sigma)$ and $v_{ab}(0,\sigma)$. The fundamental conditions (5.1.25)–(5.1.26) on $u(0,\sigma)$ and $v(0,\sigma)$ read here

$$\sum_{\bar\sigma}u_{\bar a\bar b}(0,\bar\sigma)\mathbf{J}^{(j)}_{\bar\sigma\sigma} = \sum_{a,b}\mathcal{J}_{\bar a\bar b,ab}u_{ab}(0,\sigma),$$

$$-\sum_{\bar\sigma}v_{\bar a\bar b}(0,\bar\sigma)\mathbf{J}^{(j)*}_{\bar\sigma\sigma} = \sum_{a,b}\mathcal{J}_{\bar a\bar b,ab}v_{ab}(0,\sigma),$$

or using Eqs. (5.6.14)–(5.6.15)

$$\sum_{\bar\sigma}u_{\bar a\bar b}(0,\bar\sigma)\mathbf{J}^{(j)}_{\bar\sigma\sigma} = \sum_a\mathbf{J}^{(A)}_{\bar aa}u_{a\bar b}(0,\sigma) + \sum_b\mathbf{J}^{(B)}_{\bar bb}u_{\bar ab}(0,\sigma), \tag{5.7.2}$$

$$-\sum_{\bar\sigma}v_{\bar a\bar b}(0,\bar\sigma)\mathbf{J}^{(j)*}_{\bar\sigma\sigma} = \sum_a\mathbf{J}^{(A)}_{\bar aa}v_{a\bar b}(0,\sigma) + \sum_b\mathbf{J}^{(B)}_{\bar bb}v_{\bar ab}(0,\sigma). \tag{5.7.3}$$

But Eq. (5.7.2) is the defining condition for the Clebsch–Gordan coefficients $C_{AB}(j\sigma;ab)$! These coefficients are defined by the requirement that if Ψ_{ab} are states that under an infinitesimal rotation transform as

$$\delta\Psi_{\bar a\bar b} = i\sum_a\boldsymbol{\theta}\cdot\mathbf{J}^{(A)}_{\bar aa}\Psi_{a\bar b} + i\sum_b\boldsymbol{\theta}\cdot\mathbf{J}^{(B)}_{\bar bb}\Psi_{\bar ab},$$

[48] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

then, under the same rotation, the state

$$\Psi^j_\sigma \equiv \sum_{ab}C_{AB}(j\sigma;ab)\Psi_{ab}$$

transforms as

$$\delta\Psi^j_{\bar\sigma} = i\sum_\sigma\boldsymbol{\theta}\cdot\mathbf{J}^{(j)}_{\bar\sigma\sigma}\Psi^j_\sigma.$$

Inspection of Eq. (5.7.2) shows that this requirement is satisfied by the coefficients $u_{ab}(0,\sigma)$, and therefore, up to a possible proportionality factor, $u_{ab}(0,\sigma)$ is just $C_{AB}(j\sigma;ab)$. This constant is conventionally chosen so that

$$u_{ab}(0,\sigma) = (2m)^{-1/2}C_{AB}(j\sigma;ab). \tag{5.7.4}$$

This result is unique because each irreducible (A, B) representation of the homogeneous Lorentz group contains a given spin j representation of the rotation group at most *once*. Similarly, inspection of Eqs. (5.6.16)–(5.6.17) shows that the complex conjugates of the angular momentum matrices are

$$-\mathbf{J}^{(j)*}_{\sigma\sigma'} = (-1)^{\sigma-\sigma'}\mathbf{J}^{(j)}_{-\sigma,-\sigma'}. \tag{5.7.5}$$

Therefore if we write Eq. (5.7.3) in terms of $(-1)^{j-\sigma}v_{ab}(\mathbf{p},-\sigma)$, it takes the same form as Eq. (5.7.2). With a suitable adjustment of a constant factor, the unique solution for $v(0,\sigma)$ is

$$v_{ab}(0,\sigma) = (-1)^{j+\sigma}u_{ab}(0,-\sigma). \tag{5.7.6}$$

We must now perform a boost to calculate the coefficient functions for finite momentum. For a fixed direction $\hat{\mathbf{p}} \equiv \mathbf{p}/|\mathbf{p}|$, we can write the boost (2.5.24) as a function of a parameter θ defined by

$$\cosh\theta = \sqrt{\mathbf{p}^2 + m^2}/m, \qquad \sinh\theta = |\mathbf{p}|/m \tag{5.7.7}$$

and write $L^\mu{}_\nu(\theta)$ in place of $L^\mu{}_\nu(p)$, where

$$L^i{}_k(\theta) = \delta_{ik} + (\cosh\theta - 1)\hat p_i\hat p_k,$$
$$L^i{}_0(\theta) = L^0{}_i(\theta) = \hat p_i\sinh\theta, \tag{5.7.8}$$
$$L^0{}_0(\theta) = \cosh\theta.$$

The advantage of this parameterization is that

$$L(\bar\theta)L(\theta) = L(\bar\theta + \theta). \tag{5.7.9}$$

For infinitesimal θ, we have $[L(\theta)]^\mu{}_\nu \to \delta^\mu{}_\nu + \omega^\mu{}_\nu$, where $\omega^i{}_0 = \omega^0{}_i = \hat p_i\theta$ and $\omega^i{}_j = \omega^0{}_0 = 0$. Following the same reasoning that led from Eq. (2.2.24) to Eq. (2.2.26), it follows then that

$$D(L(p)) = \exp(-i\hat{\mathbf{p}}\cdot\mathcal{K}\theta). \tag{5.7.10}$$

This is for any representation of the homogeneous Lorentz group; for the irreducible (A, B) representations, Eqs. (5.6.7) and (5.6.8) give

$$i\mathcal{K} = \mathcal{A} - \mathcal{B} \tag{5.7.11}$$

and since $\mathcal{A}$ and $\mathcal{B}$ are commuting matrices

$$D(L(p)) = \exp(-\hat{\mathbf{p}}\cdot\mathcal{A}\theta)\exp(+\hat{\mathbf{p}}\cdot\mathcal{B}\theta). \tag{5.7.12}$$

In more detail, using Eqs. (5.6.14) and (5.6.15)

$$D(L(p))_{a'b',ab} = \left(\exp(-\hat{\mathbf{p}}\cdot\mathbf{J}^{(A)}\theta)\right)_{a'a}\left(\exp(+\hat{\mathbf{p}}\cdot\mathbf{J}^{(B)}\theta)\right)_{b'b}. \tag{5.7.13}$$

Eqs. (5.7.4) and (5.7.6) then give the coefficient functions at finite momentum as

$$u_{ab}(\mathbf{p},\sigma) = \frac{1}{\sqrt{2p^0}}\sum_{a'b'}\left(\exp(-\hat{\mathbf{p}}\cdot\mathbf{J}^{(A)}\theta)\right)_{aa'}\left(\exp(+\hat{\mathbf{p}}\cdot\mathbf{J}^{(B)}\theta)\right)_{bb'}$$
$$\cdot C_{AB}(j\sigma;a'b') \tag{5.7.14}$$

and

$$v_{ab}(\mathbf{p},\sigma) = (-1)^{j+\sigma}u_{ab}(\mathbf{p},-\sigma). \tag{5.7.15}$$

These results give the field explicitly for a given transformation type (A, B), so the field (5.1.31) of this type is unique up to the choice of the constant factors κ and λ.

It is very easy in this formalism to construct Lorentz scalar interaction densities. The (A, B) representation of the homogeneous Lorentz group is just the direct product of the $(A, 0)$ and $(0, B)$ representations, so the general Lorentz transformation rules (5.1.6), (5.1.7) read here

$$U_0(\Lambda)\psi_{ab}(x)U_0^{-1}(\Lambda) = \sum_{a'b'}D^{A0}_{aa'}(\Lambda^{-1})D^{0B}_{bb'}(\Lambda^{-1})\psi_{a'b'}(\Lambda x). \tag{5.7.16}$$

Furthermore, Eqs. (5.6.14) and (5.6.15) show that the matrix generators of the $(A, 0)$ and $(0, B)$ representations are just the spin matrices for spin A and B, respectively. Thus we can construct scalars of the form

$$\sum_{a_1a_2\cdots a_n}\sum_{b_1b_2\cdots b_n}g_{a_1a_2\cdots a_n;b_1b_2\cdots b_n}\psi^{(1)}_{a_1b_1}(x)\psi^{(2)}_{a_2b_2}(x)\cdots\psi^{(n)}_{a_nb_n}(x) \tag{5.7.17}$$

by simply taking $g_{a_1a_2\cdots a_n;b_1b_2\cdots b_n}$ as the product of a coefficient for coupling spins $A_1, A_2, \ldots, A_n$ to make a scalar and a coefficient for coupling spins $B_1, B_2, \ldots, B_n$ to make a scalar. (Even though we do not explicitly consider interactions involving derivatives, we will in this way obtain the most general interaction involving n fields, because the derivative of a field of type (A, B) can always be decomposed into fields of other types without derivatives.) For instance, the most general Lorentz scalar formed from a product of three fields of transformation types (A_1, B_1), (A_2, B_2), and (A_3, B_3) is

$$g\sum_{a_1a_2a_3}\sum_{b_1b_2b_3}\begin{pmatrix} A_1 & A_2 & A_3 \\ a_1 & a_2 & a_3 \end{pmatrix}\begin{pmatrix} B_1 & B_2 & B_3 \\ b_1 & b_2 & b_3 \end{pmatrix}\psi^{(1)}_{a_1b_1}\psi^{(2)}_{a_2b_2}\psi^{(3)}_{a_3b_3}. \tag{5.7.18}$$

with a single free parameter g. This is the most general three-field interaction. (The brackets in (5.7.17) denote the Wigner "three-j" symbols [10]

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \equiv \sum_{m_3'}C_{j_1j_2}(j_3m_3';m_1m_2)C_{j_3j_3}(00;m_3'm_3)$$

which describe the coupling of three spins to make a rotational scalar.)

For the S-matrix to be Lorentz-invariant it is not enough that the interaction density $\mathcal{H}_I(x)$ be a scalar like (5.7.18); it is also necessary that $\mathcal{H}_I(x)$ should commute with $\mathcal{H}_I(y)$ at space-like separations $x - y$. To see how to satisfy this condition, consider the commutator or anticommutator of two fields for the same particle species, a field ψ of type (A, B), and the adjoint $\tilde\psi^\dagger$ of a field $\tilde\psi$ of type $(\tilde A, \tilde B)$. We find

$$[\psi_{ab}(x), \tilde\psi^\dagger_{\tilde a\tilde b}(y)]_\mp = (2\pi)^{-3}\int d^3p\,(2p^0)^{-1}\pi_{ab,\tilde a\tilde b}(\mathbf{p})$$
$$\cdot\left[\kappa\tilde\kappa^* e^{ip\cdot(x-y)} \mp \lambda\tilde\lambda^* e^{-ip\cdot(x-y)}\right], \tag{5.7.19}$$

where $\pi(\mathbf{p})$ is the spin sum

$$(2p^0)^{-1}\pi_{ab,\tilde a\tilde b}(\mathbf{p}) = \sum_\sigma u_{ab}(\mathbf{p},\sigma)\tilde u^*_{\tilde a\tilde b}(\mathbf{p},\sigma) = \sum_\sigma v_{ab}(\mathbf{p},\sigma)\tilde v^*_{\tilde a\tilde b}(\mathbf{p},\sigma) \tag{5.7.20}$$

and as usual, the top and bottom signs are for bosons and fermions, respectively. (We allow here for different coefficients $\tilde\kappa$ and $\tilde\lambda$ in the $\tilde\psi$ field.) In more detail

$$\pi_{ab,\tilde a\tilde b}(\mathbf{p}) = \sum_{a'b'}\sum_{\tilde a'\tilde b'}\sum_\sigma C_{AB}(j\sigma;a'b')C_{\tilde A\tilde B}(j\sigma;\tilde a'\tilde b')$$
$$\cdot\left(\exp(-\hat{\mathbf{p}}\cdot\mathbf{J}^{(A)}\theta)\right)_{aa'}\left(\exp(+\hat{\mathbf{p}}\cdot\mathbf{J}^{(B)}\theta)\right)_{bb'}$$
$$\cdot\left(\exp(-\hat{\mathbf{p}}\cdot\mathbf{J}^{(\tilde A)}\theta)\right)^*_{\tilde a\tilde a'}\left(\exp(+\hat{\mathbf{p}}\cdot\mathbf{J}^{(\tilde B)}\theta)\right)^*_{\tilde b\tilde b'}. \tag{5.7.21}$$

The function $\pi(\mathbf{p})$ has been calculated explicitly [11]. What concerns us here is the fact that it turns out to be the mass-shell value of a *polynomial* function P of $\mathbf{p}$ and p^0:

$$\pi_{ab,\tilde a\tilde b}(\mathbf{p}) = P_{ab,\tilde a\tilde b}(\mathbf{p}, \sqrt{\mathbf{p}^2 + m^2}) \tag{5.7.22}$$

and that P is even or odd according to whether $2A + 2\tilde B$ is an even or odd integer

$$P(-\mathbf{p}, -p^0) = (-1)^{2A+2\tilde B}P(\mathbf{p}, p^0). \tag{5.7.23}$$

We shall check this here for just one particular direction of $\mathbf{p}$. Taking $\mathbf{p}$ in the three-direction, (5.7.21) gives

$$\pi_{ab,\tilde a\tilde b}(\mathbf{p}) = \sum_\sigma C_{AB}(j\sigma;ab)C_{\tilde A\tilde B}(j\sigma;\tilde a\tilde b)\exp((-a + b - \tilde a + \tilde b)\theta).$$

The Clebsch–Gordan coefficients vanish unless $\sigma = a + b$ and $\sigma = \tilde a + \tilde b$, so we can replace

$$-a + b - \tilde a + \tilde b = -2a + \sigma + 2\tilde b - \sigma = 2\tilde b - 2a.$$

We can write $\exp(\pm\theta)$ as $(p^0 \pm p^3)/m$, so here

$$\pi_{ab,\tilde a\tilde b}(\mathbf{p}) = \sum_\sigma C_{AB}(j\sigma;ab)C_{\tilde A\tilde B}(j\sigma;\tilde a\tilde b)\begin{cases} [(p^0 + p^3)/m]^{2\tilde b - 2a}, & \tilde b \geq a, \\ [(p^0 - p^3)/m]^{2a - 2\tilde b}, & a \geq \tilde b. \end{cases}$$

where $p^0 \equiv \sqrt{\mathbf{p}^2 + m^2}$. We see that $\pi(\mathbf{p})$ can indeed be written as the mass-shell value of a polynomial $P(\mathbf{p}, p^0)$. Also, $2\tilde b - 2a$ equals $2\tilde B + 2A$ minus an even integer, so the polynomial satisfies the reflection condition (5.7.23).

Any polynomial in $\mathbf{p}$ and $\sqrt{\mathbf{p}^2 + m^2}$ can be written in a form *linear* in $\sqrt{\mathbf{p}^2 + m^2}$ (by expressing even powers of $\sqrt{\mathbf{p}^2 + m^2}$ in terms of $\mathbf{p}$) so $\pi(\mathbf{p})$ can be written

$$\pi_{ab,\tilde a\tilde b}(\mathbf{p}) = P_{ab,\tilde a\tilde b}(\mathbf{p}) + 2\sqrt{\mathbf{p}^2 + m^2}\,Q_{ab,\tilde a\tilde b}(\mathbf{p}), \tag{5.7.24}$$

where P and Q are now polynomials in $\mathbf{p}$ alone, with

$$P(-\mathbf{p}) = (-1)^{2A+2\tilde B}P(\mathbf{p}), \tag{5.7.25}$$

$$Q(-\mathbf{p}) = -(-1)^{2A+2\tilde B}Q(\mathbf{p}). \tag{5.7.26}$$

For $x - y$ space-like, we can adopt a Lorentz frame in which $x^0 = y^0$, and write Eq. (5.7.19) as

$$[\psi_{ab}(x), \tilde\psi^\dagger_{\tilde a\tilde b}(y)]_\mp = [\kappa\tilde\kappa^* \mp (-1)^{2A+2\tilde B}\lambda\tilde\lambda^*]P_{ab,\tilde a\tilde b}(-i\nabla)\Delta_+(\mathbf{x} - \mathbf{y}, 0)$$
$$+ [\kappa\tilde\kappa^* \pm (-1)^{2A+2\tilde B}\lambda\tilde\lambda^*]Q_{ab,\tilde a\tilde b}(-i\nabla)\delta^3(\mathbf{x} - \mathbf{y}).$$

In order that this should vanish when $\mathbf{x} \neq \mathbf{y}$, we must have

$$\kappa\tilde\kappa^* = \pm(-1)^{2A+2\tilde B}\lambda\tilde\lambda^*. \tag{5.7.27}$$

Now let us consider the special case where ψ and $\tilde\psi$ are the same, so in particular $A = \tilde A$ and $B = \tilde B$. (It is unavoidable that such commutators or anticommutators will appear in $[\mathcal{H}_I(x), \mathcal{H}_I(y)]$, because the hermiticity of the Hamiltonian requires that if $\mathcal{H}_I(x)$ involves ψ, it also involves $\psi^\dagger$.) In this case, Eqs. (5.7.27) gives

$$|\kappa|^2 = \pm(-1)^{2A+2B}|\lambda|^2.$$

This is possible if and only if

$$\pm(-1)^{2A+2B} = +1 \tag{5.7.28}$$

and

$$|\kappa|^2 = |\lambda|^2. \tag{5.7.29}$$

Of course, $2A + 2B$ differs from $2j$ by an *even* integer, so Eq. (5.7.28) says that *our particle is a boson or fermion according to whether $2j$ is even or odd.* This is the general relation between spin and statistics [12], of which we have already seen special examples for particles described by scalar, vector, or Dirac fields.

Now let's return to the general case, where the fields ψ and $\tilde\psi$ may be different. Using Eq. (5.7.27), and dividing both sides by $|\tilde\kappa|^2 = |\tilde\lambda|^2$, we have

$$\frac{\kappa}{\tilde\kappa} = (-1)^{2B+2\tilde B}\frac{\lambda}{\tilde\lambda}.$$

It follows that, for any field,

$$\lambda = (-1)^{2B}c\kappa, \tag{5.7.30}$$

where c is the same factor for all fields of a given particle. Furthermore, Eq. (5.7.29) shows that c is just a phase, $|c| = 1$. We can therefore eliminate c for all fields by a redefinition of the relative phase of the operators $a_{\mathbf{p},\sigma}$ and $a^{c\dagger}_{\mathbf{p},\sigma}$, so that $c = 1$, and hence $\lambda = (-1)^{2B}\kappa$. Also, the factor κ for each field type may be eliminated by a redefinition of the over-all scale of the field. We emerge from all this with a formula for the (A, B) field of a given particle, that is unique up to overall scale

$$\psi_{ab}(x) = (2\pi)^{-3/2}\sum_\sigma\int d^3p\left[u_{ab}(\mathbf{p},\sigma)a_{\mathbf{p},\sigma}e^{ip\cdot x}\right.$$
$$\left. + (-1)^{2B}u_{ab}(\mathbf{p},\sigma)a^{c\dagger}_{\mathbf{p},-\sigma}e^{-ip\cdot x}\right]. \tag{5.7.31}$$

The different fields for a given particle do not really represent possibilities that are physically distinct. For instance, the possible fields for $j = 0$ are those of type (A, A) (because the triangle inequality $|A - B| \leq j \leq A + B$ here requires $A = B$). Starting with a $(0, 0)$ scalar field ϕ, we can easily construct such (A, A) fields from the $2A$th derivative

$$\{\partial_{\mu_1}\cdots\partial_{\mu_{2A}}\}\phi, \tag{5.7.32}$$

where $\{\}$ here denotes the traceless part; for instance

$$\{\partial_\mu\partial_\nu\} \equiv \frac{\partial^2}{\partial x^\mu\partial x^\nu} - \frac{1}{4}\eta_{\mu\nu}\Box.$$

(Recall that a traceless symmetric tensor of rank N transforms according to the $(N/2, N/2)$ representation.) But Eq. (5.7.31) represents the *unique* causal (A, B) field for a given particle of spin j, so the (A, A) fields (5.7.31) for $j = 0$ can be nothing but linear combinations of the $2A$th derivatives (5.7.32) of a scalar field.

More generally, *any* field (A, B) for a given particle of spin j can be expressed as a differential operator of rank $2B$ acting on the field[49] $\phi_\sigma(x)$ of type $(j, 0)$ (or a differential operator of rank $2A$ acting on the field of type $(0, j)$). To see this, consider the field

$$\{\partial_{\mu_1}\cdots\partial_{\mu_{2B}}\}\phi_\sigma. \tag{5.7.33}$$

This transforms as the direct product of the representations (B, B) and $(j, 0)$, and hence by the usual rules of vector addition, it can be decomposed into fields transforming according to all the irreducible representations (A, B) with $|j - B| \leq A \leq j + B$, or equivalently $|A - B| \leq j \leq A + B$. Since Eq. (5.7.31) represents the unique field of type (A, B) for a given particle of spin j, it can be nothing but the (A, B) field obtained from the derivatives (5.7.33).

[49] The only possible flaw in this argument would be if some of the (A, B) fields obtained in this way actually vanished. But in this case, the $(j, 0)$ field ϕ_σ would satisfy a field equation $\sum_\sigma M_\sigma(i\partial/\partial x)\phi_\sigma(x) = 0$ and hence, for each σ, $\sum_\sigma M_\sigma(ip)u_\sigma(\mathbf{p},\sigma) = 0$. For the $(j, 0)$ representation the Clebsch–Gordan coefficient $C_{j0}(j\sigma;\sigma 0)$ is just the Kronecker symbol $\delta_{\sigma\sigma}$, so this would require $\sum_\sigma M_\sigma(ip)D_{\sigma\sigma'}(L(p)) = 0$, which is impossible unless all the $M_\sigma(ip)$ vanish, since $D(\Lambda)$ has an inverse $D(\Lambda^{-1})$. The $(j, 0)$ fields $\phi_\sigma(x)$ thus satisfy no field equation other than the Klein–Gordon equation $(\Box - m^2)\phi_\sigma(x) = 0$, and therefore none of the (A, B) fields obtained from (5.7.33) can vanish.

Now let us consider the behavior of these fields under inversions, beginning with space inversion. Using the results of Section 4.2, the space-inversion properties of the particle annihilation and antiparticle creation operators are

$$Pa_{\mathbf{p},\sigma}P^{-1} = \eta^* a_{-\mathbf{p},\sigma}, \tag{5.7.34}$$

$$Pa^{c\dagger}_{\mathbf{p},\sigma}P^{-1} = \eta^c a^{c\dagger}_{-\mathbf{p},\sigma}, \tag{5.7.35}$$

where η and η^c are the intrinsic parities for the particle and antiparticle, respectively. The general causal (A, B) field (5.7.31) thus transforms under the parity operator P into

$$P\psi^{AB}_{ab}(x)P^{-1} = (2\pi)^{-3/2}\sum_\sigma\int d^3p\left[\eta^* a_{-\mathbf{p},\sigma}e^{ip\cdot x}u^{AB}_{ab}(\mathbf{p},\sigma)\right.$$
$$\left. + \eta^c(-1)^{2B}a^{c\dagger}_{-\mathbf{p},\sigma}e^{-ip\cdot x}v^{AB}_{ab}(\mathbf{p},\sigma)\right]. \tag{5.7.36}$$

We want to change the integration variable from $\mathbf{p}$ to $-\mathbf{p}$, and for this purpose we will need to evaluate $u_{ab}(-\mathbf{p},\sigma)$ and $v_{ab}(-\mathbf{p},\sigma)$. To do this, we need only glance back at (5.7.14) and (5.7.15), and use the symmetry property of the Clebsch–Gordan coefficient [14]

$$C_{AB}(j\sigma;ab) = (-1)^{A+B-j}C_{BA}(j\sigma;ba). \tag{5.7.37}$$

This gives

$$u^{AB}_{ab}(-\mathbf{p},\sigma) = (-1)^{A+B-j}u^{BA}_{ba}(\mathbf{p},\sigma), \tag{5.7.38}$$

$$v^{AB}_{ab}(-\mathbf{p},\sigma) = (-1)^{A+B-j}v^{BA}_{ba}(\mathbf{p},\sigma), \tag{5.7.39}$$

so

$$P\psi^{AB}_{ab}(x)P^{-1} = (2\pi)^{-3/2}\sum_\sigma\int d^3p\,(-1)^{A+B-j}$$
$$\cdot\left[\eta^* a_{\mathbf{p},\sigma}e^{ip\cdot\mathcal{P}x}u^{BA}_{ba}(\mathbf{p},\sigma)\right.$$
$$\left. + \eta^c(-1)^{2B}a^{c\dagger}_{\mathbf{p},\sigma}e^{-ip\cdot\mathcal{P}x}v^{BA}_{ba}(\mathbf{p},\sigma)\right], \tag{5.7.40}$$

where, as before, $\mathcal{P}x \equiv (-\mathbf{x}, x^0)$. This is the causal field ψ^{BA}_{ba} evaluated at $\mathcal{P}x$, except that the coefficients of the annihilation and creation terms may not be the same as called for in Eq. (5.7.31). But these coefficients *must* be the same up to an overall constant factor as in Eq. (5.7.31) because, aside from scale, Eq. (5.7.31) is the unique causal field of any type. Hence the ratio of the coefficients of the two terms in Eq. (5.7.40) must be the same as in Eq. (5.7.31) (but with B replaced with A because this is supposed to be a (B, A) field):

$$\eta^c(-1)^{2B}/\eta^* = (-1)^{2A}. \tag{5.7.41}$$

But $A - B$ differs from the spin j by only an integer, so this gives

$$\eta^c = \eta^*(-1)^{2j}. \tag{5.7.42}$$

We saw special cases of this result in Sections 5.2, 5.3, and 5.5, where $j = 0$, $j = 1$, and $j = \frac{1}{2}$, respectively. We now see that the result is general; *the intrinsic parity $\eta^c\eta$ of a*

particle–antiparticle pair is $+1$ *for bosons, and* -1 *for fermions.* Using Eq. (5.7.42) in Eq. (5.7.40), our final result for space inversion is

$$P\psi^{AB}_{ab}(x)P^{-1} = \eta^*(-1)^{A+B-j}\psi^{BA}_{ba}(-\mathbf{x}, x^0). \tag{5.7.43}$$

Let's see how this applies to the Dirac field. For the top $(\frac{1}{2},0)$ and bottom $(0,\frac{1}{2})$ components of the Dirac field, the sign $(-1)^{A+B-j}$ is just $+1$, so the parity operator simply takes $\mathbf{x}$ into $-\mathbf{x}$; reverses the top and bottom components; and multiplies the field with η^*. The reversal of the top and bottom components of the Dirac field is accomplished by the matrix β in (5.5.41).

Now let us consider charge-conjugation. Its effect on the particle annihilation and antiparticle creation operators is

$$C a_{\mathbf{p},\sigma} C^{-1} = \xi^* a^c_{\mathbf{p},\sigma}, \tag{5.7.44}$$
$$C a^{c\dagger}_{\mathbf{p},\sigma} C^{-1} = \xi^c a^\dagger_{\mathbf{p},\sigma}, \tag{5.7.45}$$

where ξ and ξ^c are the charge-conjugation parities of the particle and antiparticle, respectively. Applying this transformation to the field (5.7.31), we find

$$C\psi^{AB}_{ab}(x)C^{-1} = (2\pi)^{-3/2}\sum_\sigma \int d^3p\, u^{AB}_{ab}(\mathbf{p},\sigma) \cdot \left[\xi^* a^c_{\mathbf{p},\sigma}e^{ip\cdot x} + \xi^c(-1)^{2B}a^\dagger_{\mathbf{p},-\sigma}(-1)^{j-\sigma}e^{-ip\cdot x}\right]. \tag{5.7.46}$$

It is useful to compare this formula for the charge-conjugate of an (A,B) field with the adjoint of the (B,A) field for the same particle:

$$\psi^{BA\dagger}_{ba}(x) = (2\pi)^{-3/2}\sum_\sigma \int d^3p\, u^{BA*}_{ba}(\mathbf{p},\sigma) \cdot \left[(-1)^{2A}(-1)^{j-\sigma}a^c_{\mathbf{p},-\sigma}e^{ip\cdot x} + a^\dagger_{\mathbf{p},\sigma}e^{-ip\cdot x}\right]. \tag{5.7.47}$$

To calculate the u^*, we use our previous result

$$\mathbf{J}^{(j)*} = -\mathcal{C}_j\mathbf{J}^{(j)}\mathcal{C}_j^{-1}, \qquad (\mathcal{C}_j)_{\bar\sigma\sigma} \propto (-1)^{j-\sigma}\delta_{\bar\sigma,-\sigma}.$$

The Clebsch–Gordan coefficient in Eq. (5.7.14) is real, so

$$u^{BA}_{ba}(\mathbf{p},\sigma)^* = \frac{1}{\sqrt{2p^0}}\sum_{a'b'}\left(\exp(-\hat{\mathbf{p}}\cdot\mathbf{J}^{(A)}\theta)\right)_{-a,-a'}\left(\exp(+\hat{\mathbf{p}}\cdot\mathbf{J}^{(B)}\theta)\right)_{-b,-b'} \cdot (-1)^{a'-a}(-1)^{b'-b}C_{BA}(j\sigma;b'a').$$

We use the reflection property of the Clebsch–Gordan coefficients [14]

$$C_{BA}(j,-\sigma;-b',-a') = C_{AB}(j\sigma;a'b') \tag{5.7.48}$$

and the fact that those coefficients vanish unless $a'+b'=\sigma$, to write

$$u^{BA}_{-b,-a}(\mathbf{p},-\sigma)^* = (-1)^{a+b-\sigma}u^{AB}_{ab}(\mathbf{p},\sigma). \tag{5.7.49}$$

The field adjoint (5.7.47) is then (replacing $a\to -a$, $b\to -b$, $\sigma\to-\sigma$)

$$\psi^{BA\dagger}_{-b,-a}(x) = (2\pi)^{-3/2}\sum_\sigma\int d^3p\,(-1)^{a+b-\sigma}u^{AB}_{ab}(\mathbf{p},\sigma) \cdot \left[(-1)^{2A}(-1)^{j+\sigma}a^c_{\mathbf{p},\sigma}e^{ip\cdot x} + a^\dagger_{\mathbf{p},-\sigma}e^{-ip\cdot x}\right].$$

Using the sign relation $(-1)^{-2A-j} = (-1)^{2B+j}$, this is

$$(-1)^{-2A-a-b-j}\psi^{BA\dagger}_{-b,-a}(x) = (2\pi)^{-3/2}\sum_\sigma\int d^3p\, u^{AB}_{ab}(\mathbf{p},\sigma) \cdot \left[a^c_{\mathbf{p},\sigma}e^{ip\cdot x} + (-1)^{j-\sigma+2B}a^\dagger_{\mathbf{p},-\sigma}e^{-ip\cdot x}\right]. \tag{5.7.50}$$

In order that $C\psi^{AB}_{ab}(x)C^{-1}$ should commute or anticommute with all ordinary fields at space-like separations, it is necessary that it be proportional to $\psi^{BA\dagger}_{-b,-a}(x)$, because this is the adjoint of the unique causal field of transformation type (B,A). Comparing Eq. (5.7.50) with Eq. (5.7.46), we see that this is only possible if the charge-conjugation parities are related by

$$\xi^* = \xi^c, \tag{5.7.51}$$

in which case

$$C\psi^{AB}_{ab}(x)C^{-1} = \xi^*(-1)^{-2A-a-b-j}\psi^{BA\dagger}_{-b,-a}(x). \tag{5.7.52}$$

We have already encountered the relation (5.7.51) for spins 0, 1, and $\frac{1}{2}$ in Sections 5.2, 5.3, and 5.5, and noted some of its implications for electron–positron and quark–antiquark states in Section 5.5.

In particular, for a particle that is its own antiparticle, Eq. (5.7.52) is satisfied without any charge-conjugation operator on the left-hand side or phase ξ^* on the right:

$$\psi^{AB}_{ab}(x) = (-1)^{-2A-a-b-j}\psi^{BA\dagger}_{-b,-a}(x). \tag{5.7.53}$$

We have already seen an example of this sort of reality condition for Majorana spin $\frac{1}{2}$ particles in Section 5.5.

Finally we come to time-reversal. Applied to particle annihilation and antiparticle creation operators, this gives

$$T a_{\mathbf{p},\sigma} T^{-1} = \zeta^*(-1)^{j-\sigma}a_{-\mathbf{p},-\sigma}, \tag{5.7.54}$$
$$T a^{c\dagger}_{\mathbf{p},\sigma} T^{-1} = \zeta^c(-1)^{j-\sigma}a^{c\dagger}_{-\mathbf{p},-\sigma}. \tag{5.7.55}$$

The irreducible field (5.7.31) thus has the transformation property

$$T\psi^{AB}_{ab}(x)T^{-1} = (2\pi)^{-3/2}\sum_\sigma\int d^3p\, u^{AB*}_{ab}(\mathbf{p},\sigma)(-1)^{j-\sigma} \cdot \left[\zeta^* a_{-\mathbf{p},-\sigma}e^{-ip\cdot x} + \zeta^c(-1)^{2B}a^{c\dagger}_{-\mathbf{p},-\sigma}e^{ip\cdot x}\right]. \tag{5.7.56}$$

To calculate the complex conjugate of the coefficient function, we use Eq. (5.7.14) and the standard formula [14]

$$C_{AB}(j\sigma;a,b) = (-1)^{A+B-j}C_{AB}(j,-\sigma;-a,-b) \tag{5.7.57}$$

and find:

$$u^{AB*}_{ab}(-\mathbf{p},-\sigma) = (-1)^{a+b+\sigma+A+B-j}u^{AB}_{-a,-b}(\mathbf{p},\sigma). \tag{5.7.58}$$

Changing the variables of integration and summation in Eq. (5.7.56) to $-\mathbf{p}$ and $-\sigma$, we find that in order for an (A,B) field to be transformed by time-reversal into something proportional to another (A,B) field, it is necessary that

$$\zeta^c = \zeta^*, \tag{5.7.59}$$

in which case

$$T\psi^{AB}_{ab}(x)T^{-1} = (-1)^{a+b+A+B-2j}\zeta^*\psi^{AB}_{-a,-b}(\mathbf{x},-x^0). \tag{5.7.60}$$

* * *

It should be mentioned that from time to time various difficulties have been reported [15] in the field theory of particles with spin $j \geq 3/2$. Generally, these are encountered in the study of the propagation of a higher spin field in the presence of a c-number external field. Depending on the details of the theory, the difficulties encountered include non-causality, inconsistency, unphysical mass states, and violation of unitarity. I will not go into details about these problems here, because it seems to me that they are not relevant to the calculational scheme described in this chapter, for the following reasons:

(1) The fields $\psi_{ab}(x)$ have been constructed here directly from the creation and annihilation operators for physical particles, so no question of inconsistency or unphysical mass states can arise. These are free fields, but by incorporating them into an interaction Hamiltonian density in the interaction picture, we can use perturbation theory to calculate S-matrix elements that automatically satisfy the cluster decomposition principle. As long as the interaction Hamiltonian is Hermitian, there can be no difficulty with unitarity. Lorentz invariance is guaranteed in perturbation theory as long as we add appropriate local but non-covariant terms in the Hamiltonian density; though a rigorous proof is lacking, there is no reason to doubt that this is always possible. Thus any difficulties with higher spin can only arise when we try to go beyond perturbation theory.

(2) As discussed in Section 13.6, the solution of field equations in the presence of a c-number background field (the context where all the problems with higher spin have been found) does go beyond perturbation theory, in that the results correspond to summing an infinite subset of terms in the perturbation series. This partial summation is justified, even for weak external fields, if the fields are sufficiently slowly varying, the smallness of energy denominators making up for the weakness of the fields. But the results obtained in this way depend on all the details of the interaction of the high-spin particle with the external fields: not only the multipole moments of the particle but also possible terms in the interaction that are non-linear in the external fields. The problems reported [15] with higher spin have been encountered only for higher-spin particles that have been arbitrarily assumed to have only very simple interactions with external fields. No one has shown that the problems persist for arbitrary interactions, and as we shall see in Chapter 12, particles of higher spin are expected to have interactions of all possible types allowed by symmetry principles.

(3) In fact, there are good reasons to believe that the problems with higher spin disappear if the interaction with external fields is sufficiently complicated. For one thing, there is no doubt about the existence of higher-spin *particles*, including various stable nuclei and hadronic resonances. If there is any problem with higher spin, it can only be for 'point' particles, that is, those whose interactions with external fields are particularly simple. It should be kept in mind that the requirement of simplicity depends on the choice of which field we choose to represent the higher-spin particle. Remember that any free field types for a given particle can be expressed as a derivative operator acting on any other field type, so in the interaction picture any interaction with external fields may be written in terms of any field types we like, but interactions that are simple when expressed in terms of a field of one type may look complicated when expressed in terms of a field of another type. So the requirement of simplicity does not seem to have any objective content.

(4) Also, both higher-dimensional 'Kaluza–Klein' theories and string theories provide examples of consistent theories of a charged massive particles of spin two interacting with a electromagnetic background field [16]. (It was found that the consistency of the theory depends on the assumption of realistic external fields that satisfy the field equations, a point generally neglected in earlier work.) Reformulating this work in the interaction picture, the spin two particle is represented by a $(1,1)$ free field, but as mentioned above, the interactions may be reexpressed in the interaction picture in terms of any field type (A,B) that contains the $j=2$ representation of the rotation group.

5.8 The CPT Theorem

We have seen that the demands of relativity combined with quantum mechanics require the existence of antiparticles. Not only is it necessary that every particle have an antiparticle (which may for a purely neutral particle be itself); there is a precise relation between the properties of particles and antiparticles, that can be summarized in the statement that *for an appropriate choice of inversion phases, the product CPT of all the inversions is conserved.* This is the celebrated CPT theorem.[50]

As a first step in the proof, let us work out the effect of the product CPT on free fields of various types. For a scalar, vector, or Dirac field the results of Sections 5.2, 5.3, and 5.5 give

$$CPT\,\phi(x)[CPT]^{-1} = \zeta^*\xi^*\eta^*\phi^\dagger(-x), \tag{5.8.1}$$
$$CPT\,\phi_\mu(x)[CPT]^{-1} = -\zeta^*\xi^*\eta^*\phi^\dagger_\mu(-x), \tag{5.8.2}$$
$$CPT\,\psi(x)[CPT]^{-1} = -\zeta^*\xi^*\eta^*\gamma^5\psi^*(-x). \tag{5.8.3}$$

(Of course, the phases ζ, ξ, and η depend on the species of particle described by each field.) We are going to choose the phases so that for all particles

$$\zeta\xi\eta = 1. \tag{5.8.4}$$

[50] The original proofs of this theorem were by Lüders and Pauli [17]. It has been proved rigorously in axiomatic field theory [18], by using commutativity assumptions to extend the Lorentz invariance of the theory to the complex Lorentz group, then using complex Lorentz transformations to prove a reflection property of vacuum expectation values of products of fields, and then using this reflection property to infer the existence of an antiunitary operator that induces CPT transformations on the fields.

Then any tensor $\phi_{\mu_1\cdots\mu_n}$ formed from any set of scalar and vector fields and their derivatives transforms into

$$CPT\,\phi_{\mu_1\cdots\mu_n}(x)[CPT]^{-1} = (-1)^n\phi^\dagger_{\mu_1\cdots\mu_n}(-x). \tag{5.8.5}$$

(Any complex numerical coefficient appearing in these tensors is transformed into its complex conjugate because CPT is antiunitary.) We can easily see that the same transformation rule applies to tensors formed from bilinear combinations of Dirac fields. Applying Eq. (5.8.3) to such a bilinear gives

$$CPT[\bar\psi_1(x)M\psi_2(x)][CPT]^{-1} = \psi_1^T(-x)\gamma^5\beta M^*\gamma^5\psi_2^*(-x) = [\bar\psi_1(-x)\gamma^5 M\gamma^5\psi_2(-x)]^\dagger. \tag{5.8.6}$$

(A minus sign from the anticommutation of β and γ^5 is cancelled by the minus sign from the anticommutation of fermionic operators.) If the bilinear is a tensor of rank n, then M is a product of n modulo 2 Dirac matrices, so $\gamma^5 M\gamma^5 = (-1)^n M$, and the bilinear therefore satisfies Eq. (5.8.5).

A Hermitian scalar interaction density $\mathcal{H}_I(x)$ must be formed from tensors with an *even* total number of spacetime indices, and therefore

$$CPT\,\mathcal{H}_I(x)[CPT]^{-1} = \mathcal{H}_I(-x). \tag{5.8.7}$$

More generally (and somewhat more easily) we can see that the same is true for Hermitian scalars formed from the fields $\psi^{AB}_{ab}(x)$ belonging to one or more of the general irreducible representations of the homogeneous Lorentz group. Putting together our results in the previous section for the effects of inversions on such fields, we find

$$CPT\,\psi^{AB}_{ab}(x)[CPT]^{-1} = (-1)^{2B}\psi^{AB\dagger}_{ba}(-x). \tag{5.8.8}$$

(For the Dirac field the factor $(-1)^{2B}$ is supplied by the matrix γ^5 in Eq. (5.8.3).) In order to couple together a product $\psi^{A_1B_1}_{a_1b_1}(x)\psi^{A_2B_2}_{a_2b_2}(x)\cdots$ to form a scalar $\mathcal{H}_I(x)$, it is necessary that both $A_1+A_2+\cdots$ and $B_1+B_2+\cdots$ be integers, so $(-1)^{2B_1+2B_2+\cdots}=1$, and so a Hermitian scalar $\mathcal{H}_I(x)$ will automatically satisfy Eq. (5.8.7).

From Eq. (5.8.7) it follows immediately that CPT commutes with the interaction $H_{\rm int} \equiv \int d^3x\,\mathcal{H}_I(\mathbf{x},0)$:

$$CPT\,H_{\rm int}[CPT]^{-1} = H_{\rm int}. \tag{5.8.9}$$

Also, in any theory CPT commutes with the free-particle Hamiltonian H_0. Thus the operator CPT, which has been defined here by its operation on free-particle operators, acts on 'in' and 'out' states in the way described in Section 3.3. The physical consequences of this symmetry principle have already been discussed in Sections 3.3 and 3.6.

5.9 Massless Particle Fields

Up to this point we have dealt only with the fields of massive particles. For some of these fields, such as the scalar and Dirac fields discussed in Sections 5.2 and 5.5, there is no special problem in passing to the limit of zero mass. On the other hand, we saw in Section

5.3 that there is a difficulty in taking the zero-mass limit of the vector field for a particle of spin one: at least one of the polarization vectors blows up in this limit. In fact, we shall see in this section that the creation and annihilation operators for physical massless particles of spin $j\geq 1$ cannot be used to construct all of the irreducible (A,B) fields that can be constructed for finite mass. This peculiar limitation on field types will lead us naturally to the introduction of gauge invariance.

Just as we did for massive particles, let us attempt to construct a general free field for a massless particle as a linear combination of the annihilation operators $a_{\mathbf{p},\sigma}$ for particles of momentum $\mathbf{p}$ and helicity σ, and the corresponding creation operators $a^{c\dagger}_{\mathbf{p},\sigma}$ for the antiparticles:[51]

$$\psi_\ell(x) = (2\pi)^{-3/2}\int d^3p\sum_\sigma\left[\kappa a_{\mathbf{p},\sigma}u_\ell(\mathbf{p},\sigma)e^{ip\cdot x} + \lambda a^{c\dagger}_{\mathbf{p},\sigma}v_\ell(\mathbf{p},\sigma)e^{-ip\cdot x}\right] \tag{5.9.1}$$

where now $p^0\equiv|\mathbf{p}|$. The creation operators transform just like the one-particle states in Eq. (2.5.42)

$$U(\Lambda)a^\dagger_{\mathbf{p},\sigma}U^{-1}(\Lambda) = \sqrt{\frac{(\Lambda p)^0}{p^0}}\exp(i\sigma\theta(p,\Lambda))a^\dagger_{\mathbf{p}_\Lambda,\sigma}, \tag{5.9.2}$$

$$U(\Lambda)a^{c\dagger}_{\mathbf{p},\sigma}U^{-1}(\Lambda) = \sqrt{\frac{(\Lambda p)^0}{p^0}}\exp(i\sigma\theta(p,\Lambda))a^{c\dagger}_{\mathbf{p}_\Lambda,\sigma}, \tag{5.9.3}$$

and hence also

$$U(\Lambda)a_{\mathbf{p},\sigma}U^{-1}(\Lambda) = \sqrt{\frac{(\Lambda p)^0}{p^0}}\exp(-i\sigma\theta(p,\Lambda))a_{\mathbf{p}_\Lambda,\sigma}, \tag{5.9.4}$$

where $p_\Lambda\equiv\Lambda p$, and θ is the angle defined by Eqs. (2.5.43). Hence if we want the field to transform according to some representation $D(\Lambda)$ of the homogeneous Lorentz group

$$U(\Lambda)\psi_\ell(x)U^{-1}(\Lambda) = \sum_{\bar\ell}D_{\ell\bar\ell}(\Lambda^{-1})\psi_{\bar\ell}(\Lambda x), \tag{5.9.5}$$

then we must take the coefficient functions u and v to satisfy the relations

$$u_{\bar\ell}(\mathbf{p}_\Lambda,\sigma)\exp(i\sigma\theta(p,\Lambda)) = \sqrt{\frac{p^0}{(\Lambda p)^0}}\sum_\ell D_{\bar\ell\ell}(\Lambda)u_\ell(\mathbf{p},\sigma), \tag{5.9.6}$$

$$v_{\bar\ell}(\mathbf{p}_\Lambda,\sigma)\exp(-i\sigma\theta(p,\Lambda)) = \sqrt{\frac{p^0}{(\Lambda p)^0}}\sum_\ell D_{\bar\ell\ell}(\Lambda)v_\ell(\mathbf{p},\sigma) \tag{5.9.7}$$

[51] We deal here with only a single species of particle, and drop the species label n. Also, κ and λ are constant coefficients to be determined by the requirement of causality with some convenient choice of normalization of the coefficient functions u_ℓ and v_ℓ.

in place of Eqs. (5.1.19) and (5.1.20). (Again, $p_\Lambda\equiv\Lambda p$.) As in the massive particle case, we can satisfy these requirements by setting (in place of Eqs. (5.1.21) and (5.1.22))

$$u_\ell(\mathbf{p},\sigma) = \sqrt{\frac{|\mathbf{k}|}{p^0}}\sum_{\bar\ell}D_{\ell\bar\ell}(\mathcal{L}(p))u_{\bar\ell}(\mathbf{k},\sigma), \tag{5.9.8}$$

$$v_\ell(\mathbf{p},\sigma) = \sqrt{\frac{|\mathbf{k}|}{p^0}}\sum_{\bar\ell}D_{\ell\bar\ell}(\mathcal{L}(p))v_{\bar\ell}(\mathbf{k},\sigma), \tag{5.9.9}$$

where $\mathbf{k}$ is a standard momentum, say $(0,0,k)$, and $\mathcal{L}(p)$ is a standard Lorentz transformation that takes a massless particle from momentum $\mathbf{k}$ to momentum $\mathbf{p}$. Also, in place of Eqs. (5.1.23) and (5.1.24), the coefficient functions at the standard momentum must satisfy

$$u_{\bar\ell}(\mathbf{k},\sigma)\exp(i\sigma\theta(k,W)) = \sum_\ell D_{\bar\ell\ell}(W)u_\ell(\mathbf{k},\sigma), \tag{5.9.10}$$

$$v_{\bar\ell}(\mathbf{k},\sigma)\exp(-i\sigma\theta(k,W)) = \sum_\ell D_{\bar\ell\ell}(W)v_\ell(\mathbf{k},\sigma) \tag{5.9.11}$$

where $W^\mu{}_\nu$ is an arbitrary element of the 'little group' for four-momentum $k=(\mathbf{k},|\mathbf{k}|)$, i.e., an arbitrary Lorentz transformation that leaves this four-momentum invariant.

We can extract the content of Eqs. (5.9.10) and (5.9.11) by considering separately the two kinds of little-group elements in Eq. (2.5.28). For a rotation $R(\theta)$ by an angle θ around the z-axis, given by Eq. (2.5.27),

$$R^\mu{}_\nu(\theta) = \begin{bmatrix}\cos\theta & -\sin\theta & 0 & 0\\ \sin\theta & \cos\theta & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix},$$

we find from Eqs. (5.9.10) and (5.9.11)

$$u_{\bar\ell}(\mathbf{k},\sigma)e^{i\sigma\theta} = \sum_\ell D_{\bar\ell\ell}(R(\theta))u_\ell(\mathbf{k},\sigma), \tag{5.9.12}$$

$$v_{\bar\ell}(\mathbf{k},\sigma)e^{-i\sigma\theta} = \sum_\ell D_{\bar\ell\ell}(R(\theta))v_\ell(\mathbf{k},\sigma). \tag{5.9.13}$$

For combined rotations and boosts $S(\alpha,\beta)$ in the $x-y$ plane, given by (2.5.26),

$$S^\mu{}_\nu(\alpha,\beta) = \begin{bmatrix}1 & 0 & -\alpha & \alpha\\ 0 & 1 & -\beta & \beta\\ \alpha & \beta & 1-\gamma & \gamma\\ \alpha & \beta & -\gamma & 1+\gamma\end{bmatrix}, \qquad \gamma\equiv(\alpha^2+\beta^2)/2,$$

Eqs. (5.9.10) and (5.9.11) give

$$u_{\bar\ell}(\mathbf{k},\sigma) = \sum_\ell D_{\bar\ell\ell}(S(\alpha,\beta))u_\ell(\mathbf{k},\sigma), \tag{5.9.14}$$

$$v_{\bar\ell}(\mathbf{k},\sigma) = \sum_\ell D_{\bar\ell\ell}(S(\alpha,\beta))v_\ell(\mathbf{k},\sigma). \tag{5.9.15}$$

Eqs. (5.9.12)–(5.9.15) are the conditions that determine the coefficient functions u and v at the standard momentum $\mathbf{k}$; Eqs. (5.9.8) and (5.9.9) then give them at arbitrary momenta. The equations for v are just the complex conjugates of the equations for u, so with a suitable adjustment of the constants κ and λ we may normalize the coefficient functions so that

$$v_\ell(\mathbf{p},\sigma) = u_\ell(\mathbf{p},\sigma)^*. \tag{5.9.16}$$

The problem is that we cannot find a u_ℓ that satisfies Eq. (5.9.14) for general representations of the homogeneous Lorentz group, even for those representations for which it is possible to construct fields for particles of a given helicity in the case $m\neq 0$.

To see what goes wrong here, let's try to construct the four-vector $[(\frac{1}{2},\frac{1}{2})]$ field for a massless particle of helicity ±1. In the four-vector representation, we have simply

$$D^\mu{}_\nu(\Lambda) = \Lambda^\mu{}_\nu.$$

It is conventional to write the coefficient function u_μ here in terms of a 'polarization vector' e_μ:

$$u_\mu(\mathbf{p},\sigma) \equiv (2p^0)^{-1/2}e_\mu(\mathbf{p},\sigma), \tag{5.9.17}$$

so that Eq. (5.9.8) gives

$$e^\mu(\mathbf{p},\sigma) = \mathcal{L}(p)^\mu{}_\nu e^\nu(\mathbf{k},\sigma). \tag{5.9.18}$$

Also, Eqs. (5.9.12) and (5.9.14) read here

$$e^\mu(\mathbf{k},\sigma)e^{i\sigma\theta} = R(\theta)^\mu{}_\nu e^\nu(\mathbf{k},\sigma), \tag{5.9.19}$$

$$e^\mu(\mathbf{k},\sigma) = S(\alpha,\beta)^\mu{}_\nu e^\nu(\mathbf{k},\sigma). \tag{5.9.20}$$

Eq. (5.9.19) requires that (up to a constant which can be absorbed into the coefficients κ and λ),

$$e^\mu(\mathbf{k},\pm1) = (1,\pm i,0,0)/\sqrt{2}. \tag{5.9.21}$$

But then Eq. (5.9.20) would require also that $\alpha\pm i\beta=0$, which is impossible for general real α,β. We therefore cannot satisfy the fundamental requirement (5.9.14) or (5.9.10); instead, we have here

$$D^\mu{}_\nu(W(\theta,\alpha,\beta))e^\nu(\mathbf{k},\pm1) = S^\mu{}_\lambda(\alpha,\beta)R^\lambda{}_\nu(\theta)e^\nu(\mathbf{k},\pm1) = \exp(\pm i\theta)\left\{e^\mu(\mathbf{k},\pm1) + \frac{\alpha\pm i\beta}{\sqrt{2}|\mathbf{k}|}k^\mu\right\}. \tag{5.9.22}$$

We have thus come to the conclusion that no four-vector field can be constructed from the annihilation and creation operators for a particle of mass zero and helicity ±1.

Let's temporarily close our eyes to this difficulty, and go ahead anyway, using Eqs. (5.9.18) and (5.9.21) to define a polarization vector for arbitrary momentum, and take the field as

$$a_\mu(x) = \int d^3p\,(2\pi)^{-3/2}(2p^0)^{-1/2} \cdot \sum_{\sigma=\pm1}\left[e_\mu(\mathbf{p},\sigma)e^{ip\cdot x}a_{\mathbf{p},\sigma} + e_\mu(\mathbf{p},\sigma)^*e^{-ip\cdot x}a^{c\dagger}_{\mathbf{p},\sigma}\right]. \tag{5.9.23}$$

We will come back later to consider how such a field can be used as an ingredient in a physical theory.

The field (5.9.23) of course satisfies

$$\Box a^\mu(x) = 0. \tag{5.9.24}$$

Other properties of the field follow from those of the polarization vector. (We shall need these properties of the polarization vector later when we come to quantum electrodynamics.) Note that the Lorentz transformation $\mathcal{L}(p)$ that takes a massless particle momentum from $\mathbf{k}$ to $\mathbf{p}$ may be written as a 'boost' $\mathcal{B}(|\mathbf{p}|)$ along the z-axis which takes the particle from energy $|\mathbf{k}|$ to energy $|\mathbf{p}|$, followed by a standardized rotation $R(\hat{\mathbf{p}})$ that takes the z-axis into the direction of $\mathbf{p}$. Since $e^\nu(\mathbf{k},\pm1)$ is a purely spatial vector with only x and y components, it is unaffected by the boost along the z-axis, and so

$$e^\mu(\mathbf{p},\pm1) = R(\hat{\mathbf{p}})^\mu{}_\nu e^\nu(\mathbf{k},\pm1). \tag{5.9.25}$$

In particular, $e^0(\mathbf{k},\pm1)=0$ and $\mathbf{k}\cdot\mathbf{e}(\mathbf{k},\pm1)=0$ so

$$e^0(\mathbf{p},\pm1) = 0 \tag{5.9.26}$$

and

$$\mathbf{p}\cdot\mathbf{e}(\mathbf{p},\pm1) = 0. \tag{5.9.27}$$

It follows that

$$a^0(x) = 0 \tag{5.9.28}$$

and

$$\nabla\cdot\mathbf{a}(x) = 0. \tag{5.9.29}$$

As we shall see in Chapter 9, these are the conditions satisfied by the vacuum vector potential of electrodynamics in what is called Coulomb or radiation gauge.

The fact that a^0 vanishes in all Lorentz frames shows vividly that a^μ cannot be a four-vector. Instead, Eq. (5.9.22) shows that for a general momentum $\mathbf{p}$ and a general Lorentz transformation Λ, in place of Eq. (5.9.6) we have

$$e^\mu(\mathbf{p}_\Lambda,\pm1)\exp(\pm i\theta(p,\Lambda)) = D^\mu{}_\nu(\Lambda)e^\nu(\mathbf{p},\pm1) + p^\mu\Omega_\pm(p,\Lambda), \tag{5.9.30}$$

so that under a general Lorentz transformation

$$U(\Lambda)a_\mu(x)U^{-1}(\Lambda) = \Lambda^\nu{}_\mu a_\nu(\Lambda x) + \partial_\mu\Omega(x,\Lambda), \tag{5.9.31}$$

where $\Omega(x,\Lambda)$ is a linear combination of annihilation and creation operators, whose precise form will not concern us here. As we will see in more detail in Chapter 8, we will be able to use a field like $a^\mu(x)$ as an ingredient in Lorentz-invariant physical theories if the couplings of $a^\mu(x)$ are not only formally Lorentz-invariant (that is, invariant under formal

Lorentz transformations under which $a^\mu\to\Lambda^\mu{}_\nu a^\nu$), but are also invariant under the 'gauge' transformations $a_\mu\to a_\mu+\partial_\mu\Omega$. This is accomplished by taking the couplings of a_μ to be of the form $a_\mu j^\mu$, where j^μ is a four-vector current with $\partial_\mu j^\mu=0$.

Although there is no ordinary four-vector field for massless particles of helicity ±1, there is no problem in constructing an antisymmetric tensor field for such particles. From Eq. (5.9.22) and the invariance of k^μ under the little group we see immediately that

$$D^\mu{}_\rho(W(\theta,\alpha,\beta))D^\nu{}_\sigma(W(\theta,\alpha,\beta))\left(k^\rho e^\sigma(\mathbf{k},\pm1) - k^\sigma e^\rho(\mathbf{k},\pm1)\right) = e^{\pm i\theta}\left(k^\mu e^\nu(\mathbf{k},\pm1) - k^\nu e^\mu(\mathbf{k},\pm1)\right). \tag{5.9.32}$$

This shows that the coefficient function that satisfies Eq. (5.9.6) for the antisymmetric tensor representation of the homogeneous Lorentz group is (with an appropriate choice of normalization)

$$u_{\mu\nu}(\mathbf{p},\pm1) = i(2\pi)^{-3/2}(2p^0)^{-1/2}\left[p_\mu e_\nu(\mathbf{p},\pm1) - p_\nu e_\mu(\mathbf{p},\pm1)\right], \tag{5.9.33}$$

where $e^\mu(\mathbf{p},\pm1)$ is given by Eq. (5.9.25). Using this together with Eq. (5.9.23) gives the general antisymmetric tensor field for massless particles of helicity ±1 in the form

$$f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu. \tag{5.9.34}$$

Note that this is a tensor even though a^μ is not a four-vector, because the extra term in Eq. (5.9.31) drops out in Eq. (5.9.34). Note also that Eqs. (5.9.34), (5.9.24), (5.9.28), and (5.9.29) show that $f^{\mu\nu}$ satisfies the vacuum Maxwell equations:

$$\partial_\mu f^{\mu\nu} = 0, \tag{5.9.35}$$
$$\epsilon^{\rho\sigma\mu\nu}\partial_\sigma f_{\mu\nu} = 0. \tag{5.9.36}$$

To calculate the commutation relations for the tensor fields we need sums over helicities of the bilinears $e^\mu e^{\nu*}$. The explicit formula (5.9.21) gives

$$\sum_{\sigma=\pm1}e^i(\mathbf{k},\sigma)e^j(\mathbf{k},\sigma)^* = \delta_{ij} - \frac{k^ik^j}{|\mathbf{k}|^2}$$

and so, using Eq. (5.9.25),

$$\sum_{\sigma=\pm1}e^i(\mathbf{p},\sigma)e^j(\mathbf{p},\sigma)^* = \delta_{ij} - \frac{p^ip^j}{|\mathbf{p}|^2}. \tag{5.9.37}$$

A straightforward calculation gives then

$$[f_{\mu\nu}(x), f_{\rho\sigma}(y)^\dagger] = (2\pi)^{-3}\left[-\eta_{\mu\rho}\partial_\nu\partial_\sigma + \eta_{\nu\rho}\partial_\mu\partial_\sigma + \eta_{\mu\sigma}\partial_\nu\partial_\rho - \eta_{\nu\sigma}\partial_\mu\partial_\rho\right] \cdot \int d^3p\,(2p^0)^{-1}\left[|\kappa|^2e^{ip\cdot(x-y)} - |\lambda|^2e^{-ip\cdot(x-y)}\right]. \tag{5.9.38}$$

This clearly vanishes for $x^0=y^0$ if and only if

$$|\kappa|^2 = |\lambda|^2 \tag{5.9.39}$$

in which case since $f_{\mu\nu}$ is a tensor the commutator also vanishes for all space-like separations. Eq. (5.9.39) also implies that the commutator of the a^μ vanishes at equal times, and as we shall see in Chapter 8 this is enough to yield a Lorentz-invariant S-matrix. The relative phase of the creation and annihilation operators can be adjusted so that $\kappa=\lambda$; the fields are then Hermitian if the particles are their own charge-conjugates, as is the case for the photon.

Why should we want to use fields like $a^\mu(x)$ in constructing theories of massless particles of spin one, rather than being content with fields like $f^{\mu\nu}(x)$ with simple Lorentz transformation properties? The presence of the derivatives in Eq. (5.9.34) means that an interaction density constructed solely from $f_{\mu\nu}$ and its derivatives will have matrix elements that vanish more rapidly for small massless particle energy and momentum than one that uses the vector field a_μ. Interactions in such a theory will have a correspondingly rapid fall-off at large distances, faster than the usual inverse-square law. This is perfectly possible, but gauge-invariant theories that use vector fields for massless spin one particles represent a more general class of theories, including those that are actually realized in nature.

Parallel remarks apply to gravitons, massless particles of helicity ±2. From the annihilation and creation operators for such particles we can construct a tensor $R_{\mu\nu\rho\sigma}$ with the algebraic properties of the Riemann–Christoffel curvature tensor: antisymmetric within the pairs μ,ν and ρ,σ, and symmetric between the pairs. However, in order to incorporate the usual inverse-square gravitational interactions we need to introduce a field $h_{\mu\nu}$ that transforms as a symmetric tensor, up to gauge transformations of the sort associated in general relativity with general coordinate transformations. Thus in order to construct a theory of massless particles of helicity ±2 that incorporates long-range interactions, it is necessary for it to have a symmetry something like general covariance. As in the case of electromagnetic gauge invariance, this is achieved by coupling the field to a conserved 'current' $\theta^{\mu\nu}$, now with two spacetime indices, satisfying $\partial_\mu\theta^{\mu\nu}=0$. The only such conserved tensor is the energy-momentum tensor, aside from possible total derivative terms that do not affect the long-range behavior of the force produced.[52] The fields of massless particles of spin $j\geq3$ would have to couple to conserved tensors with three or more spacetime indices, but aside from total derivatives there are none, so *high-spin massless particles cannot produce long-range forces.*

* * *

The problems we have encountered in constructing four-vector fields for helicities ±1 or symmetric tensor fields for helicity ±2 are just special cases of a more general limitation. To see this, let's consider how to construct fields for massless particles belonging to arbitrary representations of the homogeneous Lorentz group. As we saw in Section 5.6, any representation $D(\Lambda)$ of the homogeneous Lorentz group can be decomposed into $(2A+1)(2B+1)$-dimensional representations (A,B), for which the generators of the homogeneous Lorentz

[52] If $\theta^{\mu_1\cdots\mu_N}$ is a tensor current satisfying $\partial_{\mu_1}\theta^{\mu_1\cdots\mu_N}=0$, then $\int d^3x\,\theta^{0\mu_2\cdots\mu_N}$ is a conserved quantity that transforms like a tensor of rank $N-1$. The only such conserved tensors are the scalar 'charges' associated with various continuous symmetries, and the energy-momentum four-vector. The conservation of any other four-vector, or any tensor of higher rank, would forbid all but forward collisions.

group are represented by

$$(\mathcal{J}_{ij})_{a'b',ab} = \epsilon_{ijk}\left[(J^{(A)}_k)_{a'a}\delta_{b'b} + (J^{(B)}_k)_{b'b}\delta_{a'a}\right],$$

$$(\mathcal{J}_{i0})_{a'b',ab} = -i\left[(J^{(A)}_i)_{a'a}\delta_{b'b} - (J^{(B)}_i)_{b'b}\delta_{a'a}\right],$$

where $\mathbf{J}^{(j)}$ are the angular-momentum matrices for spin j. For θ infinitesimal, $D(R(\theta)) = 1+i\mathcal{J}_{21}\theta$, so Eqs. (5.9.12) and (5.9.13) give

$$\sigma u_{ab}(\mathbf{k},\sigma) = (a+b)u_{ab}(\mathbf{k},\sigma),$$

$$-\sigma v_{ab}(\mathbf{k},\sigma) = (a+b)v_{ab}(\mathbf{k},\sigma),$$

and so $u_{ab}(\mathbf{k},\sigma)$ and $v_{ab}(\mathbf{k},\sigma)$ must vanish unless $\sigma=a+b$ and $\sigma=-a-b$, respectively. Also, letting α and β become infinitesimal in Eq. (5.9.14) gives

$$0 = (\mathcal{J}_{31}+\mathcal{J}_{01})_{ab,a'b'}u_{a'b'}(\mathbf{k},\sigma) = (J^{(A)}_2+iJ^{(A)}_1)_{aa'}u_{a'b}(\mathbf{k},\sigma) + (J^{(B)}_2-iJ^{(B)}_1)_{bb'}u_{ab'}(\mathbf{k},\sigma),$$

$$0 = (\mathcal{J}_{32}+\mathcal{J}_{02})_{ab,a'b'}u_{a'b'}(\mathbf{k},\sigma) = (-J^{(A)}_1+iJ^{(A)}_2)_{aa'}u_{a'b}(\mathbf{k},\sigma) + (-J^{(B)}_1-iJ^{(B)}_2)_{bb'}u_{ab'}(\mathbf{k},\sigma),$$

or more simply

$$(J^{(A)}_1-iJ^{(A)}_2)_{aa'}u_{a'b}(\mathbf{k},\sigma) = 0,$$
$$(J^{(B)}_1+iJ^{(B)}_2)_{bb'}u_{ab'}(\mathbf{k},\sigma) = 0.$$

These require that $u_{ab}(\mathbf{k},\sigma)$ vanishes unless

$$a=-A, \qquad b=+B \tag{5.9.40}$$

and the same is obviously also true of $v_{ab}(\mathbf{k},\sigma)$. Putting this together, we see that a field of type (A,B) can be formed only from the annihilation operators for a massless particle of helicity σ and the creation operators for the antiparticle of helicity $-\sigma$, where

$$\sigma = B-A. \tag{5.9.41}$$

For instance, the $(\frac{1}{2},0)$ and $(0,\frac{1}{2})$ parts of the Dirac field for a massless particle can only destroy particles of helicity $-\frac{1}{2}$ and $+\frac{1}{2}$ respectively, and create antiparticles of helicity $+\frac{1}{2}$ and $-\frac{1}{2}$, respectively. In the 'two-component' theory of the neutrino, there is only a $(\frac{1}{2},0)$ field and its adjoint, so neutrinos have helicity $-\frac{1}{2}$ and antineutrinos helicity $+\frac{1}{2}$ in this theory.

By the same methods as in Section 5.7, it can be shown that the $(j,0)$ and $(0,j)$ fields for massless particles of spin j (i.e., helicity $\mp j$) commute with each other and their adjoints at space-like separations if the coefficients of the annihilation and creation terms in Eq. (5.9.1) satisfy Eq. (5.9.39). The relative phase of the annihilation and creation operators may then be adjusted so that these coefficients are equal. It is easy to see that

the fields for a massless particle of spin j of type $(A,A+j)$ or $(B+j,B)$ are just the $2A$th or $2B$th derivatives of fields of type $(0,j)$ or $(j,0)$, respectively, so these more general fields do not need to be considered separately here.

We can now see why it was impossible to construct a vector field for massless particles of helicity ±1. A vector field transforms according to the $(\frac{1}{2},\frac{1}{2})$ representation, and hence according to Eq. (5.9.19) can only describe helicity zero. (It is, of course, possible to construct a vector field for helicity zero—just take the derivative $\partial_\mu\phi$ of a massless scalar field ϕ.) The simplest covariant massless field for helicity ±1 has the Lorentz transformation type $(1,0)\oplus(0,1)$; that is, it is an antisymmetric tensor $f_{\mu\nu}$. Similarly, the simplest covariant massless field for helicity ±2 has the Lorentz transformation type $(2,0)\oplus(0,2)$: a fourth rank tensor which like the Riemann–Christoffel curvature tensor is antisymmetric within each pair of indices and symmetric between the two pairs.

The discussion of the inversions P,C,T given in the previous section can be carried over to the case of zero mass with only obvious modifications.

Problems

1. Show that if the zero-momentum coefficient functions satisfy the conditions (5.1.23) and (5.1.24), then the coefficient functions (5.1.21) and (5.1.22) for arbitrary momentum satisfy the defining conditions Eqs. (5.1.19) and (5.1.20).
2. Consider a free field $\psi^\mu_\ell(x)$ which annihilates and creates a self-charge-conjugate particle of spin $\frac{3}{2}$ and mass $m\neq0$. Show how to calculate the coefficient functions $u^\mu_\ell(\mathbf{p},\sigma)$, which multiply the annihilation operators $a_{\mathbf{p},\sigma}$ in this field, in such a way that the field transforms under Lorentz transformations like a Dirac field ψ_ℓ with an extra four-vector index μ. What field equations and algebraic and reality conditions does this field satisfy? Evaluate the matrix $P^{\mu\nu}_{\ell m}(p)$, defined (for $p^2=-m^2$) by
   $$\sum_\sigma u^\mu_\ell(\mathbf{p},\sigma)u^{\nu*}_m(\mathbf{p},\sigma) = (2p^0)^{-1}P^{\mu\nu}_{\ell m}(p).$$
   What are the commutation relations of this field? How does the field transform under the inversions P,C,T?
3. Consider a free field $h^{\mu\nu}(x)$ satisfying $h^{\mu\nu}(x)=h^{\nu\mu}(x)$ and $h^\mu{}_\mu(x)=0$, which annihilates and creates a particle of spin two and mass $m\neq0$. Show how to calculate the coefficient functions $u^{\mu\nu}(\mathbf{p},\sigma)$, which multiply the annihilation operators $a_{\mathbf{p},\sigma}$ in this field, in such a way that the field transforms under Lorentz transformations like a tensor. What field equations does this field satisfy? Evaluate the function $P^{\mu\nu,\kappa\lambda}(p)$, defined by
   $$\sum_\sigma u^{\mu\nu}(\mathbf{p},\sigma)u^{\kappa\lambda*}(\mathbf{p},\sigma) = (2p^0)^{-1}P^{\mu\nu,\kappa\lambda}(p).$$
   What are the commutation relations of this field? How does the field transform under the inversions P,C,T?
4. Show that the fields for a massless particle of spin j of type $(A,A+j)$ or $(B+j,B)$ are the $2A$th or $2B$th derivatives of fields of type $(0,j)$ or $(j,0)$, respectively.

5. Work out the transformation properties of fields of transformation type $(j,0)+(0,j)$ for massless particles of helicity $\pm j$ under the inversions P,C,T.
6. Consider a generalized Dirac field ψ that transforms according to the $(j,0)+(0,j)$ representation of the homogeneous Lorentz group. List the tensors that can be formed from products of the components of ψ and $\psi^\dagger$. Check your result against what we found for $j=\frac{1}{2}$.
7. Consider a general field ψ_{ab} describing particles of spin j and mass $m\neq0$, that transforms according to the (A,B) representation of the homogeneous Lorentz group. Suppose it has an interaction Hamiltonian of the form
   $$H_{\rm int} = \int d^3x\left[\psi_{ab}(x)J^{ab}(x) + J^{ab*}(x)\psi^\dagger_{ab}(x)\right],$$
   where J^{ab} is an external c-number current. What is the asymptotic behavior of the matrix element for emitting these particles for energy $E\gg m$ and definite helicity? (Assume that the Fourier transform of the current has values for different a,b that are of the same order of magnitude, and that do not depend strongly on E.)

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6 The Feynman Rules

In previous chapters the use of covariant free fields in the construction of the Hamiltonian density has been motivated by the requirement that the S-matrix satisfy Lorentz invariance and cluster decomposition conditions. With the Hamiltonian density constructed in this way, it makes no difference which form of perturbation theory we use to calculate the S-matrix; the results will automatically satisfy these invariance and clustering conditions in each order in the interaction density. Nevertheless, there are obvious practical advantages in using a version of perturbation theory in which the Lorentz invariance and cluster decomposition properties of the S-matrix are kept manifest at every stage in the calculation. This was not true for the perturbation theory used in the 1930s, now known as 'old-fashioned perturbation theory,' described at the beginning of Section 3.5. The great achievement of Feynman, Schwinger, and Tomonaga in the late 1940s was to develop perturbative techniques for calculating the S-matrix, in which Lorentz invariance and cluster decomposition properties are transparent throughout. This chapter will outline the diagrammatic calculational technique first described by Feynman at the Poconos Conference in 1948. Feynman was led to these diagrammatic rules in part through his development of a path-integral approach, which will be the subject of Chapter 9. In this chapter, we shall use the approach described by Dyson [1] in 1949, which until the 1970s was the basis of almost all analyses of perturbation theory in quantum field theory, and still provides a particularly transparent introduction to the Feynman rules.

6.1 Derivation of the Rules

Our starting point is a formula for the S-matrix, obtained by putting together the Dyson series (3.5.10) with expression (4.2.2) for the free-particle states:

$$S_{\mathbf{p}'_1\sigma'_1n'_1;\mathbf{p}'_2\sigma'_2n'_2;\cdots,\mathbf{p}_1\sigma_1n_1;\mathbf{p}_2\sigma_2n_2;\cdots} = \sum_{N=0}^\infty\frac{(-i)^N}{N!}\int d^4x_1\cdots d^4x_N\,(\Omega|\cdots a_{\mathbf{p}'_2\sigma'_2n'_2}a_{\mathbf{p}'_1\sigma'_1n'_1} \cdot T\{\mathcal{H}_I(x_1)\cdots\mathcal{H}_I(x_N)\}a^\dagger_{\mathbf{p}_1\sigma_1n_1}a^\dagger_{\mathbf{p}_2\sigma_2n_2}\cdots|\Omega). \tag{6.1.1}$$

As a reminder: $\mathbf{p}$, σ, and n label particle momenta, spin, and species; primes denote labels for particles in the final state; $|\Omega)$ is the free-particle vacuum state; a and $a^\dagger$ are annihilation and creation operators; T indicates a time-ordering, which puts the $\mathcal{H}_I(x)$ in an order in which the arguments x^0 decrease from left to right; and $\mathcal{H}_I(x)$ is the interaction Hamiltonian density, taken as a polynomial in the fields and their adjoints

$$\mathcal{H}_I(x) = \sum_i g_i\mathcal{H}_{I,i}(x), \tag{6.1.2}$$

each term $\mathcal{H}_{I,i}$ being a product of definite numbers of fields and field adjoints of each type. The field of a particle of species n that transforms under a particular representation of the

homogeneous Lorentz group (with or without space inversions) is given by

$$\psi_\ell(x) = \sum_\sigma(2\pi)^{-3/2}\int d^3p\left[u_\ell(\mathbf{p},\sigma,n)a_{\mathbf{p},\sigma,n}e^{ip\cdot x} + v_\ell(\mathbf{p},\sigma,n)a^\dagger_{\mathbf{p},\sigma,n^c}e^{-ip\cdot x}\right]. \tag{6.1.3}$$

Here n^c denotes the antiparticle of the species n, and $\exp(\pm ip\cdot x)$ is calculated with p^0 set equal to $\sqrt{\mathbf{p}^2+M_n^2}$. The coefficient functions u_ℓ and v_ℓ depend on the Lorentz transformation properties of the field and the spin of the particle it describes; they were calculated in Chapter 5. (For instance, in the scalar field the u_ℓ for a particle of energy E is simply $(2E)^{-1/2}$, while in a Dirac field u_ℓ and v_ℓ are the normalized Dirac spinors introduced in Section 5.5.) The index ℓ on the field should here be understood to indicate the particle type and the representation of the Lorentz group by which the field transforms, as well as including a running index labelling the components in this representation. There is no need to deal separately with interactions that involve derivatives of fields; from our point of view, the derivative of a field (6.1.3) is just another field described by (6.1.3), with different u_ℓ and v_ℓ. We will here make a distinction between some particle species that we arbitrarily call 'particles,' for instance electrons, protons, etc., and those we call 'antiparticles,' such as positrons and antiprotons. The field operators that destroy particles and create antiparticles are called simply 'fields'; their adjoints, which destroy antiparticles and create particles, are called 'field adjoints.' Of course, some particle species like the photon and π^0 are their own antiparticles; for these the field adjoints are proportional to the fields.

We now proceed to move all annihilation operators to the right in Eq. (6.1.1), repeatedly using for this purpose the commutation or anticommutation relations:

$$a_{\mathbf{p}\sigma n}a^\dagger_{\mathbf{p}'\sigma'n'} = \pm a^\dagger_{\mathbf{p}'\sigma'n'}a_{\mathbf{p}\sigma n} + \delta^3(\mathbf{p}'-\mathbf{p})\delta_{\sigma'\sigma}\delta_{n'n}, \tag{6.1.4}$$
$$a_{\mathbf{p}\sigma n}a_{\mathbf{p}'\sigma'n'} = \pm a_{\mathbf{p}'\sigma'n'}a_{\mathbf{p}\sigma n}, \tag{6.1.5}$$
$$a^\dagger_{\mathbf{p}\sigma n}a^\dagger_{\mathbf{p}'\sigma'n'} = \pm a^\dagger_{\mathbf{p}'\sigma'n'}a^\dagger_{\mathbf{p}\sigma n} \tag{6.1.6}$$

(and likewise for antiparticles), the $\pm$ sign on the right being $-$ if both particles n,n' are fermions, and $+$ if either or both are bosons. Whenever an annihilation operator appears on the extreme right (or a creation operator on the extreme left), the corresponding contribution to Eq. (6.1.1) vanishes, because these operators annihilate the vacuum state:

$$a_{\mathbf{p}\sigma n}|\Omega) = 0, \tag{6.1.7}$$
$$(\Omega|a^\dagger_{\mathbf{p}\sigma n} = 0. \tag{6.1.8}$$

The remaining contributions to Eq. (6.1.1) are those arising from the delta function terms on the right-hand side of Eq. (6.1.4), with every creation and annihilation operator in the initial or final states or in the interaction Hamiltonian density paired in this way with some other annihilation or creation operator.

In this way, the contribution to Eq. (6.1.1) of a given order in each of the terms $\mathcal{H}_{I,i}$ in the polynomial $\mathcal{H}_I(\psi(x),\psi^\dagger(x))$ is given by a sum, over all ways of pairing creation and annihilation operators [2], of the integrals of products of factors, as follows:

(a) Pairing of a final particle having quantum numbers $\mathbf{p}',\sigma',n'$ with a field adjoint $\psi^\dagger_\ell(x)$ in $\mathcal{H}_{I,i}(x)$ yields a factor

$$\left[a_{\mathbf{p}'\sigma'n'},\psi^\dagger_\ell(x)\right]_\mp = (2\pi)^{-3/2}e^{-ip'\cdot x}u^*_\ell(\mathbf{p}',\sigma',n'). \tag{6.1.9}$$

(b) Pairing of a final antiparticle having quantum numbers $\mathbf{p}',\sigma',n'^c$ with a field $\psi_\ell(x)$ in $\mathcal{H}_{I,i}(x)$ yields a factor

$$\left[a_{\mathbf{p}'\sigma'n'^c},\psi_\ell(x)\right]_\mp = (2\pi)^{-3/2}e^{-ip'\cdot x}v_\ell(\mathbf{p}',\sigma',n'). \tag{6.1.10}$$

(c) Pairing of an initial particle having quantum numbers $\mathbf{p},\sigma,n$ with a field $\psi_\ell(x)$ in $\mathcal{H}_{I,i}(x)$ yields a factor

$$\left[\psi_\ell(x),a^\dagger_{\mathbf{p}\sigma n}\right]_\mp = (2\pi)^{-3/2}e^{ip\cdot x}u_\ell(\mathbf{p},\sigma,n). \tag{6.1.11}$$

(d) Pairing of an initial antiparticle having quantum numbers $\mathbf{p},\sigma,n^c$ with a field adjoint $\psi^\dagger_\ell(x)$ in $\mathcal{H}_{I,i}(x)$ yields a factor

$$\left[\psi^\dagger_\ell(x),a^\dagger_{\mathbf{p}\sigma n^c}\right]_\mp = (2\pi)^{-3/2}e^{ip\cdot x}v^*_\ell(\mathbf{p},\sigma,n). \tag{6.1.12}$$

(e) Pairing of a final particle (or antiparticle) having numbers $\mathbf{p}',\sigma',n'$ with an initial particle (or antiparticle) having quantum numbers $\mathbf{p},\sigma,n$ yields a factor

$$\left[a_{\mathbf{p}'\sigma'n'},a^\dagger_{\mathbf{p}\sigma n}\right]_\mp = \delta^3(\mathbf{p}'-\mathbf{p})\delta_{\sigma'\sigma}\delta_{n'n}. \tag{6.1.13}$$

(f) Pairing of a field $\psi_\ell(x)$ in $\mathcal{H}_{I,i}(x)$ with a field adjoint $\psi^\dagger_m(y)$ in $\mathcal{H}_{I,j}(y)$ yields a factor[53]

$$\theta(x-y)\left[\psi^{(+)}_\ell(x),\psi^{(+)\dagger}_m(y)\right]_\mp \pm \theta(y-x)\left[\psi^{(-)\dagger}_m(y),\psi^{(-)}_\ell(x)\right]_\mp \equiv -i\Delta_{\ell m}(x,y), \tag{6.1.14}$$

where $\psi^{(+)}$ and $\psi^{(-)}$ are the terms in ψ that destroy particles and create antiparticles, respectively:

$$\psi^{(+)}_\ell(x) = (2\pi)^{-3/2}\int d^3p\sum_\sigma u_\ell(\mathbf{p}\sigma n)e^{ip\cdot x}a_{\mathbf{p}\sigma n}, \tag{6.1.15}$$

$$\psi^{(-)}_\ell(x) = (2\pi)^{-3/2}\int d^3p\sum_\sigma v_\ell(\mathbf{p}\sigma n)e^{-ip\cdot x}a^\dagger_{\mathbf{p}\sigma n^c}. \tag{6.1.16}$$

Recall that $\theta(x-y)$ is a step function, equal to $+1$ for $x^0>y^0$ and zero for $x^0<y^0$. These step functions appear in Eq. (6.1.14) because of the time-ordering in Eq. (6.1.1); we can encounter a pairing of an annihilation field $\psi^{(+)}(x)$ in $\mathcal{H}_I(x)$ with a creation field $\psi^{(+)\dagger}(y)$ in $\mathcal{H}_I(y)$ only if $\mathcal{H}_I(x)$ was initially to the left of $\mathcal{H}_I(y)$ in Eq. (6.1.1), i.e., if $x^0>y^0$;

[53] If the interaction $\mathcal{H}_I(x)$ is written in the normal-ordered form, as in Eq. (5.1.33), then there is no pairing of fields and field adjoints in the same interaction. Otherwise some sort of regularization is needed to give meaning to $\Delta_{\ell m}(0)$.

similarly, we encounter a pairing of an annihilation field $\psi^{(-)\dagger}(y)$ in $\mathcal{H}_I(y)$ with a creation field $\psi^{(-)}(x)$ in $\mathcal{H}_I(x)$ only if $\mathcal{H}_I(y)$ was initially to the left of $\mathcal{H}_I(x)$ in Eq. (6.1.1), i.e., if $y^0>x^0$. (The $\pm$ sign in the second term in (6.1.14) will be explained a little later.) The quantity (6.1.14) is known as a *propagator*; it is calculated in the following section.

The S-matrix is obtained by multiplying these factors together, along with additional numerical factors to be discussed below, then integrating over $x_1\cdots x_N$, then summing over all pairings, and then over the numbers of interaction of each type. Before filling in all the details, it will be convenient first to describe a diagrammatic formalism for keeping track of all these pairings.

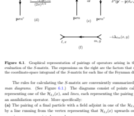


Figure 6.1. Graphical representation of pairings of operators arising in the coordinate-space evaluation of the S-matrix. The expressions on the right are the factors that must be included in the coordinate-space integrand of the S-matrix for each line of the Feynman diagram.

The rules for calculating the S-matrix are conveniently summarized in terms of *Feynman diagrams*. (See Figure 6.1.) The diagrams consist of points called *vertices*, each representing one of the $\mathcal{H}_{I,i}(x)$, and *lines*, each representing the pairing of a creation with an annihilation operator. More specifically:

(a) The pairing of a final particle with a field adjoint in one of the $\mathcal{H}_{I,i}(x)$ is represented by a line running from the vertex representing that $\mathcal{H}_{I,i}(x)$ upwards out of the diagram, carrying an arrow pointed upwards.

(b) The pairing of a final antiparticle with a field in one of the $\mathcal{H}_{I,i}(x)$ is also represented

by a line running from the vertex representing that $\mathcal{H}_{I,i}(x)$ upwards out of the diagram, but carrying an arrow pointed downwards. (Arrows are omitted throughout for particles like γ, π^0, etc. that are their own antiparticles.)

(c) The pairing of an initial particle with a field in one of the $\mathcal{H}_{I,i}(x)$ is represented by a line running into the diagram from below, ending in the vertex representing that $\mathcal{H}_{I,i}(x)$, carrying an arrow pointed upwards.

(d) The pairing of an initial antiparticle with a field adjoint in one of the $\mathcal{H}_{I,i}(x)$ is also represented by a line running into the diagram from below, ending in the vertex representing that $\mathcal{H}_{I,i}(x)$, but carrying an arrow pointed downwards.

(e) The pairing of a final particle or antiparticle with an initial particle or antiparticle is represented by a line running clear through the diagram from bottom to top, not touching any vertex, with an arrow pointed upwards or downwards for particles or antiparticles, respectively.

(f) The pairing of a field in $\mathcal{H}_{I,i}(x)$ with a field adjoint in $\mathcal{H}_{I,j}(y)$ is represented by a line joining the vertices representing $\mathcal{H}_{I,i}(x)$ and $\mathcal{H}_{I,j}(y)$, carrying an arrow pointing from y to x.

Note that arrows always point in the direction a particle is moving, and opposite to the direction an antiparticle is moving. (As mentioned above, arrows should be omitted for particles like photons that are their own antiparticles.) The arrow direction indicated in rule (f) is consistent with this convention because a field adjoint in $\mathcal{H}_{I,j}(y)$ can either create a particle destroyed by a field in $\mathcal{H}_{I,i}(x)$, or destroy an antiparticle created by a field in $\mathcal{H}_{I,i}(x)$. Note also that since every field or field adjoint in $\mathcal{H}_{I,i}(x)$ must be paired with something, the total number of lines at a vertex of type i, corresponding to a term $\mathcal{H}_{I,i}(x)$ in Eq. (6.1.2), is just equal to the total number of field or field adjoint factors in $\mathcal{H}_{I,i}(x)$. Of these lines, the number with arrows pointed into the vertex or out of it equals the number of fields or field adjoints respectively in the corresponding interaction term.

To calculate the contribution to the S-matrix for a given process, of a given order N_i in each of the interaction terms $\mathcal{H}_{I,i}(x)$ in Eq. (6.1.2), we must carry out the following steps:

(i) Draw all Feynman diagrams containing N_i vertices of each type i, and containing a line coming into the diagrams from below for each particle or antiparticle in the initial state, and a line going upwards out of the diagram for every particle or antiparticle in the final state, together with any number of internal lines running from one vertex to another, as required to give each vertex the proper number of attached lines. The lines carry arrows as described above, each of which may point upwards or downwards. Each vertex is labelled with an interaction type i and spacetime coordinate x^μ. Each internal or external line is labelled at the end where it runs into a vertex with a field type ℓ (corresponding to the field $\psi_\ell(x)$ or $\psi^\dagger_\ell(x)$ that creates or destroys the particle or antiparticle at that vertex), and each external line where it enters or leaves the diagram is labelled with the quantum numbers $\mathbf{p},\sigma,n$ or $\mathbf{p}',\sigma',n'$ of the initial or final particle (or antiparticle).

(ii) For each vertex of type i, include a factor $-i$ (from the $(-i)^N$ in Eq. (6.1.1)) and a factor g_i (the coupling constant multiplying the product of fields in $\mathcal{H}_{I,i}(x)$). For each line running upwards out of the diagram, include a factor (6.1.9) or (6.1.10), depending on whether the arrow is pointing up or down. For each line running from below into the

In some cases, there are additional combinatoric factors or signs that must be included in the contribution of individual Feynman diagrams:

(v) Suppose that an interaction $\mathcal{H}_{I,i}(x)$ contains (among other fields and field adjoints) M factors of the same field. Suppose that each of these fields is paired with a field adjoint in a different interaction (different for each one), or in the initial or final state. The first of these field adjoints

Figure 6.2. Example of a graph requiring extra combinatoric factors in the S-matrix. For an interaction involving, say, three factors of some field (as well as other fields) we usually include a factor 1/3! in the interaction Hamiltonian, to cancel factors arising from sums over ways of pairing these fields with their adjoints in other interactions. But in this diagram there are two such factors of 1/3! and only 3! different pairings, so we are left with an extra factor of 1/3!.

can be paired with any one of the M identical fields in $\mathcal{H}_{I,i}(x)$; the second with any one of the remaining $M-1$ identical fields; and so on, yielding an extra factor of $M!$. To compensate for this, it is conventional to define the coupling constants g_i so that an explicit factor $1/M!$ appears in any $\mathcal{H}_{I,i}(x)$ containing M identical fields (or field adjoints.) For instance, the interaction of Mth order in a scalar field $\phi(x)$ would be written $g\phi^M/M!$. (More generally, one often also displays an explicit factor of $1/M!$ when the interaction involves a sum of M factors of fields from the same symmetry multiplet, or when for this or any other reason the coupling coefficient is totally symmetric or antisymmetric under permutations of M boson or fermion fields.)

with N corners. (See Figure 6.3.) There are only $(N-1)!$ different diagrams here because a permutation of labels that moves each label to the next vertex around the ring yields the same diagram. Hence such a graph is accompanied with a factor

$$\frac{(N-1)!}{N!}=\frac{1}{N}. \tag{6.1.17}$$

Figure 6.3. An eighth-order graph for the vacuum-to-vacuum amplitude with particles interacting only with an external field. In this diagram the external field is represented by wiggly lines. There are 7! such diagrams, differing only by relabelling the vertices, and not counting as different those labellings that simply rotate the ring. The factor 1/8! from the Dyson formula (6.1.1) is therefore not entirely cancelled here, leaving us with an extra factor 1/8.

(vi) In theories involving fermion fields, the use of Eqs. (6.1.4)–(6.1.6) to move annihilation and creation operators to the right and left introduces minus signs into the contribution of various pairings. To be specific, we get a minus sign wherever the permutation of the

$$\mathcal{H}_I(x)=\sum_{\ell mk} g_{\ell mk}\psi^\dagger_\ell(x)\psi_m(x)\phi_k(x), \tag{6.1.18}$$

where $g_{\ell mk}$ are general constants, $\psi_\ell(x)$ are a set of complex fermion fields, and $\phi_m(x)$ are a set of real bosonic (but not necessarily scalar) fields. (Not only quantum electrodynamics, but the whole 'standard model' of weak, electromagnetic, and strong interactions, has fermionic interactions that can all be put in this form.) Let us first take up the process of fermion–fermion scattering, $12\to1'2'$, to second order in $\mathcal{H}_I$. The fermion operators in the second-order term in Eq. (6.1.1) appear in the order (with obvious abbreviations)

$$a_{2'}a_{1'}\psi^\dagger(x)\psi(x)\psi^\dagger(y)\psi(y)a^\dagger_1a^\dagger_2. \tag{6.1.19}$$

There are two connected diagrams to this order, corresponding to the pairings

$$[a_{2'}\psi^\dagger(x)]\,[a_{1'}\psi^\dagger(y)]\,[\psi(y)a^\dagger_1]\,[\psi(x)a^\dagger_2] \tag{6.1.20}$$

and

$$[a_{1'}\psi^\dagger(x)]\,[a_{2'}\psi^\dagger(y)]\,[\psi(y)a^\dagger_1]\,[\psi(x)a^\dagger_2]. \tag{6.1.21}$$

(See Figure 6.4.) To go from (6.1.19) to (6.1.20) requires an even permutation of fermionic operators. (For instance, move $\psi(x)$ past three operators to the right, and then move $a_{1'}$ past one operator to the right.) Thus there is no extra minus sign in the contribution of the pairing (6.1.20). This in itself is not so important; the *overall* sign of the S-matrix does not matter in transition rates, and in any case depends on sign conventions for the initial and final states. What is important is that the contributions of pairings (6.1.20) and (6.1.21) have opposite sign, as can be seen most easily by noting that the only difference between these two pairings is the interchange of two fermionic operators, $a_{1'}$ and $a_{2'}$. In fact, this relative minus sign is just what is required by Fermi statistics: it makes the scattering amplitude antisymmetric under the interchange of particles $1'$ and $2'$ (or 1 and 2).

[illegible]

Figure 6.5. The connected second-order diagrams for fermion–antifermion scattering in a theory with interaction (6.1.18). Here straight lines represent fermions or antifermions, depending on the arrow direction; dotted lines are neutral bosons. There is again a minus sign difference in the contributions of these two diagrams, arising from an extra interchange of fermion operators in the pairings represented by the second diagram.

(6.1.18), the fermion lines in general Feynman diagrams form either chains of lines that pass through the diagram with arbitrary numbers of interactions with the boson fields, as in Figure 6.6, or else fermionic *loops*, like that shown in Figure 6.7. Consider the effect of adding a fermionic loop with M corners to the Feynman diagram for any process. This corresponds to the pairing of fermionic operators

$$[\psi(x_1)\psi^\dagger(x_2)][\psi(x_2)\psi^\dagger(x_3)]\cdots[\psi(x_M)\psi^\dagger(x_1)]. \tag{6.1.25}$$

On the other hand, these operators appear in Eq. (6.1.1) in the order

$$\psi^\dagger(x_1)\psi(x_1)\psi^\dagger(x_2)\psi(x_2)\cdots\psi^\dagger(x_M)\psi(x_M). \tag{6.1.26}$$

To go from (6.1.26) to (6.1.25) requires an odd permutation of fermionic operators (move $\psi^\dagger(x_1)$ to the right past $2M-1$ operators) so the contribution of each such fermionic loop is accompanied with a minus sign.

These rules yield the full S-matrix, including contributions from processes in which various clusters of particles interact in widely separated regions of spacetime. As discussed in Chapter 4, to calculate the part of the S-matrix that excludes such contributions, we should include only *connected* Feynman diagrams. In particular, this excludes lines passing clean through the diagram without interacting, which would yield the factors (6.1.13).

To make the Feynman rules perfectly clear, we will calculate the low-order contributions to the S-matrix for particle scattering in two different theories.

Theory I

Consider the theory of fermions and self-charge-conjugate bosons with interaction (6.1.18). The lowest-order connected diagrams for fermion–boson scattering are shown in

Figure 6.6. The connected second-order diagrams for boson–fermion scattering in a theory with interaction (6.1.18). Straight lines are fermions; dashed lines are neutral bosons.

[illegible]

yield the S-matrix element

[illegible]

with the last term indicating subtraction of the preceding term with interchange of particles $1'$ and $2'$ (or equivalently 1 and 2). There are no second-order graphs for boson–boson scattering in this theory; the lowest-order graphs are of fourth order, such as that shown in Figure 6.7. More specific examples of formulas like Eqs. (6.1.27) and (6.1.28) will be given in Section 6.3, after we have had a chance to evaluate the propagators and go over to momentum space.

In the interaction (6.1.18) of the preceding example, the three fields are all different. It is instructive also to look at an example with a trilinear interaction in which all three fields are the same, or at least enter into the interaction in a symmetric way.

Theory II

Now take the interaction density to be a sum of terms that are trilinear in a set of *real* [illegible]

[illegible]

[illegible]

$$\Omega(t) \equiv e^{iHt}e^{-iH_0t}, \tag{6.4.5}$$

and $|\beta;\text{in}\rangle$ and $|\beta;\text{out}\rangle$ are "in" and "out" eigenstates of the full Hamiltonian H, respectively.

Here is the proof. From Eq. (3.5.10), we see immediately that the left-hand side of Eq. (6.4.3) is

$$\left[\frac{\delta^r S[\epsilon]}{\delta\epsilon_{a_1}(x_1)\cdots\delta\epsilon_{a_r}(x_r)}\right]_{\epsilon=0} = \sum_{N=0}^{\infty}\frac{(-i)^{N+r}}{N!}\int_{-\infty}^{\infty} d\tau_1\cdots d\tau_N \langle\beta|T\{H_I(\tau_1)\cdots H_I(\tau_N)o_{a_1}(x_1)\cdots o_{a_r}(x_r)\}|\alpha\rangle. \tag{6.4.6}$$

For definiteness, suppose that $x_1^0 \geq x_2^0 \geq \cdots \geq x_r^0$. Then we can denote by $\tau_{01}\cdots\tau_{0N_0}$ all τs greater than x_1^0; by $\tau_{11}\cdots\tau_{1N_1}$ all τs between x_1^0 and x_2^0, and so on; finally denoting by $\tau_{r1}\cdots\tau_{rN_r}$ all τs that are less than x_r^0. Eq. (6.4.6) then becomes:

$$\begin{aligned}\left[\frac{\delta^r S[\epsilon]}{\delta\epsilon_{a_1}(x_1)\cdots\delta\epsilon_{a_r}(x_r)}\right]_{\epsilon=0} &= \sum_{N=0}^{\infty}\frac{(-i)^{N+r}}{N!}\sum_{N_0,N_1,\dots,N_r}\frac{N!\,\delta_{N,N_0+N_1+\cdots+N_r}}{N_0!N_1!\cdots N_r!}\\ &\cdot\int_{x_1^0}^{\infty}d\tau_{01}\cdots d\tau_{0N_0}\int_{x_2^0}^{x_1^0}d\tau_{11}\cdots d\tau_{1N_1}\cdots\int_{-\infty}^{x_r^0}d\tau_{r1}\cdots d\tau_{rN_r}\\ &\cdot\langle\beta|T\{H_I(\tau_{01})\cdots H_I(\tau_{0N_0})\}o_{a_1}(x_1)T\{H_I(\tau_{11})\cdots H_I(\tau_{1N_1})\}o_{a_2}(x_2)\cdots\\ &\cdots o_{a_r}(x_r)T\{H_I(\tau_{r1})\cdots H_I(\tau_{rN_r})\}|\alpha\rangle.\end{aligned}$$

The factor $N!/N_0!N_1!\cdots N_r!$ is the number of ways of sorting N τs into $r+1$ subsets, each containing $N_0, N_1, \cdots, N_r$ of these τs. Instead of summing over $N_0, N_1, \cdots, N_r$, subject to the condition $N_0+N_1+\cdots+N_r = N$, and then summing over N, we can just sum independently over $N_0, N_1, \cdots, N_r$, setting N where it appears in $(-i)^N$ equal to $N_0+N_1+\cdots+N_r$. This gives

$$\left[\frac{\delta^r S[\epsilon]}{\delta\epsilon_{a_1}(x_1^0)\cdots\delta\epsilon_{a_r}(x_r^0)}\right]_{\epsilon=0} = (-i)^r\langle\beta|U(\infty,x_1^0)o_{a_1}(x_1)U(x_1^0,x_2^0)o_{a_2}(x_2)U(x_2^0,x_3^0)\cdots o_{a_r}(x_r)U(x_r^0,-\infty)|\alpha\rangle. \tag{6.4.7}$$

where

$$U(t',t) = \sum_{N=0}^{\infty}\frac{(-i)^N}{N!}\int_t^{t'}d\tau_1\cdots d\tau_N\,T\{H_I(\tau_1)\cdots H_I(\tau_N)\}. \tag{6.4.8}$$

The operator $U(t',t)$ satisfies the differential equation

$$\frac{d}{dt'}U(t',t) = -iH_I(t')U(t',t) \tag{6.4.9}$$

with the obvious initial condition

$$U(t,t) = \mathbf{1}. \tag{6.4.10}$$

This has the solution

$$U(t',t) = \exp(iH_0t')\exp\{-iH(t'-t)\}\exp(-iH_0t) = \Omega^{-1}(t')\Omega(t) \tag{6.4.11}$$

with Ω given by Eq. (6.4.5). Inserting Eq. (6.4.11) in Eq. (6.4.7) and using Eq. (6.4.4), we have

$$\left[\frac{\delta S[\epsilon]}{\delta\epsilon_{a_1}(x_1)\cdots\delta\epsilon_{a_r}(x_r)}\right]_{\epsilon=0} = (-i)^r\langle\beta|\Omega_\infty^\dagger O_{a_1}(x_1)\cdots O_{a_r}(x_r)\Omega(-\infty)|\alpha\rangle. \tag{6.4.12}$$

In deriving this result we supposed that $x_1^0 \geq x_2^0 \geq \cdots \geq x_r^0$, so we could just as well replace the product of operators on the right-hand-side with the time-ordered product:

$$\left[\frac{\delta S[\epsilon]}{\delta\epsilon_{a_1}(x_1)\cdots\delta\epsilon_{a_r}(x_r)}\right]_{\epsilon=0} = (-i)^r\langle\beta|\Omega_\infty^\dagger T\{O_{a_1}(x_1)\cdots O_{a_r}(x_r)\}\Omega(-\infty)|\alpha\rangle. \tag{6.4.13}$$

But now both sides are entirely symmetric (or for fermions antisymmetric) in the as and xs, so this relation holds whatever the order of the times $x_1^0 \dots x_r^0$. Also, we saw in Section 3.1 that (in the sense of Eq. (3.1.12))

$$|\beta\rangle_{\text{in/out}} = \Omega(\mp\infty)|\beta\rangle. \tag{6.4.14}$$

Hence Eq. (6.4.13) is the desired result (6.4.3).

Problems

1. Consider the theory of a real scalar field ϕ, with interaction (in the interaction picture) $H_I = g\int d^3x\,\phi(x)^3/3!$. Calculate the connected S-matrix element for scalar–scalar scattering to second order in g, doing all integrals. Use the results to calculate the differential cross-section for scalar–scalar scattering in the center-of-mass system.

2. Consider a theory involving a neutral scalar field $\phi(x)$ for a boson B and a complex Dirac field $\psi(x)$ for a fermion F, with interaction (in the interaction picture) $H_I = ig\int d^3x\,\bar\psi(x)\gamma_5\psi(x)\phi(x)$. Draw all the connected order-g^2 Feynman diagrams and calculate the corresponding S-matrix elements for the processes $F^c+B \to F^c+B$, $F+F^c \to F+F^c$, and $F^c+F \to B+B$ (where F^c is the antiparticle of F). Do all integrals.

3. Consider the theory of a real scalar field $\phi(x)$, with interaction $H_I = g\int d^3x\,\phi(x)^4/4!$. Calculate the S-matrix for scalar–scalar scattering to order g, and use the result to calculate the differential scattering cross-section. Calculate the correction terms in the S-matrix for scalar–scalar scattering to order g^2, expressing the result as an integral over a single four-momentum, but do all x-integrals.

4. What is the contribution in Feynman diagrams from the contraction of the derivative $\partial_\mu\psi_\ell(x)$ of a Dirac field with the adjoint $\bar\psi_m(y)$ of the field?

5. Use the theorem of Section 6.4 to give expressions for the vacuum expectation values of Heisenberg picture operators $\langle\Omega|\Phi(x)|\Omega\rangle$ and $\langle\Omega|T\{\Phi(x),\Phi(y)\}|\Omega\rangle$ in the theory of Problem 1, to orders g and g^2, respectively.

References

1. F. J. Dyson, *Phys. Rev.* **75**, 486, 1736 (1949).
2. The formal statement of this result is known as *Wick's theorem*; see G. C. Wick, *Phys. Rev.* **80**, 268 (1950).
3. I do not know who first proved this theorem. It was known in the early 1950s to several theorists, including M. Gell-Mann and F. E. Low.
4. J. Schwinger, *Phys. Rev.* **82**, 914 (1951).

7 The Canonical Formalism

Ever since the birth of quantum field theory in the papers of Born, Dirac, Fermi, Heisenberg, Jordan, and Pauli in the late 1920s, its development has been historically linked to the canonical formalism, so much so that it seems natural to begin any treatment of the subject today by postulating a Lagrangian and applying to it the rules of canonical quantization. This is the approach used in most books on quantum field theory. Yet historical precedent is not a very convincing reason for using this formalism. If we discovered a quantum field theory that led to a physically satisfactory S-matrix, would it bother us if it could not be derived by the canonical quantization of some Lagrangian?

To some extent this question is moot because, as we shall see in Section 7.1, all of the most familiar quantum field theories furnish canonical systems, and these can easily be put in a Lagrangian form. However, there is no proof that every conceivable quantum field theory can be formulated in this way. And even if it can, this does not in itself explain why we should *prefer* to use the Lagrangian formalism as a starting point in constructing various quantum field theories.

The point of the Lagrangian formalism is that it makes it easy to satisfy Lorentz invariance and other symmetries: a classical theory with a Lorentz-invariant Lagrangian density will when canonically quantized lead to a Lorentz-invariant quantum theory. That is, we shall see here that such a theory allows the construction of suitable quantum mechanical operators that satisfy the commutation relations of the Poincaré algebra, and therefore leads to a Lorentz-invariant S-matrix.

This is not so trivial. We saw in the previous chapter that in theories with derivative couplings or spins $j \geq 1$, it is not enough to take the interaction Hamiltonian as the integral over space of a scalar interaction density; we also need to add non-scalar terms to the interaction density to compensate for non-covariant terms in the propagators. The canonical formalism with a scalar Lagrangian density will automatically provide these extra terms. Later, when we come to non-Abelian gauge theories in Volume II, this extra convenience will become a necessity; it would be just about hopeless to try to guess at the form of the Hamiltonian in such theories without starting with a Lorentz-invariant and gauge-invariant Lagrangian density.

7.1 Canonical Variables

In this section we shall show that various quantum field theories that we have constructed so far satisfy the commutation rules and equations of motion of the Hamiltonian version of the canonical formalism. It is the Hamiltonian formalism that is needed to calculate the S-matrix (whether by operator or path-integral methods) but it is not always easy to choose Hamiltonians that yield a Lorentz-invariant S-matrix. In the balance of this chapter we shall take the Lagrangian version of the canonical formalism as our starting point, and use it to derive physically satisfactory Hamiltonians. The purpose of the present section is to identify the canonical fields and their conjugates in various field theories, to tell us how to separate the free-field terms in the Lagrangian, and incidentally to reassure us that the canonical formalism is indeed applicable to physically realistic theories.

We first show that the free fields constructed in Chapter 5 automatically provide a system of quantum operators $q^n(\mathbf{x},t)$ and canonical conjugates $p_n(\mathbf{x},t)$ that satisfy the familiar canonical commutation or anticommutation relations:

$$\begin{aligned}[q^n(\mathbf{x},t),p_{\bar n}(\mathbf{y},t)] &= i\delta^3(\mathbf{x}-\mathbf{y})\delta^n_{\bar n}, &&\text{if either particle is a boson,}\\ \{q^n(\mathbf{x},t),p_{\bar n}(\mathbf{y},t)\} &= i\delta^3(\mathbf{x}-\mathbf{y})\delta^n_{\bar n}, &&\text{if both particles are fermions.}\end{aligned} \tag{7.1.1}$$

$$\begin{aligned}[q^n(\mathbf{x},t),q^{\bar n}(\mathbf{y},t)] &= 0, &&\text{if either particle is a boson,}\\ \{q^n(\mathbf{x},t),q^{\bar n}(\mathbf{y},t)\} &= 0, &&\text{if both particles are fermions.}\end{aligned} \tag{7.1.2}$$

$$\begin{aligned}[p_n(\mathbf{x},t),p_{\bar n}(\mathbf{y},t)] &= 0, &&\text{if either particle is a boson,}\\ \{p_n(\mathbf{x},t),p_{\bar n}(\mathbf{y},t)\} &= 0, &&\text{if both particles are fermions.}\end{aligned} \tag{7.1.3}$$

In each equation, the commutator line applies if either of the particles created and destroyed by the two operators are bosons, and the anticommutator line applies if both particles are fermions. For instance, the real scalar field $\phi(x)$ for a self-charge-conjugate particle of zero spin was found in Section 5.2 to obey the commutation relation

$$[\phi(x),\phi(y)] = \Delta(x-y),$$

where Δ is the function

$$\Delta(x-y) \equiv \int\frac{d^3k}{2k^0(2\pi)^3}\left[e^{ik\cdot(x-y)} - e^{-ik\cdot(x-y)}\right]$$

with $k^0 = \sqrt{\mathbf{k}^2+m^2}$. We note that

$$\Delta(\mathbf{x},0) = 0, \qquad \dot\Delta(\mathbf{x},0) = -i\delta^3(\mathbf{x}).$$

(A dot denotes the derivative with respect to the time x^0.) It is easy then to see that the field and its time-derivative $\dot\phi$ obey the equal-time commutation relations:

$$[\phi(\mathbf{x},t),\dot\phi(\mathbf{y},t)] = i\delta^3(\mathbf{x}-\mathbf{y}), \tag{7.1.4}$$

$$[\phi(\mathbf{x},t),\phi(\mathbf{y},t)] = 0, \tag{7.1.5}$$

$$[\dot\phi(\mathbf{x},t),\dot\phi(\mathbf{y},t)] = 0. \tag{7.1.6}$$

Therefore we may define canonical variables

$$q(\mathbf{x},t) \equiv \phi(\mathbf{x},t), \qquad p(\mathbf{x},t) \equiv \dot\phi(\mathbf{x},t) \tag{7.1.7}$$

which satisfy the canonical commutation relations (7.1.1)–(7.1.3).

For the complex scalar field of a particle of spin zero with a distinct antiparticle, the commutation relations are

$$[\phi(x),\phi^\dagger(y)] = \Delta(x-y), \qquad [\phi(x),\phi(y)] = 0.$$

We may therefore define the free-particle canonical variables as the complex operators

$$q(\mathbf{x},t) \equiv \phi(\mathbf{x},t), \tag{7.1.8}$$

$$p(\mathbf{x},t) \equiv \dot\phi^\dagger(\mathbf{x},t). \tag{7.1.9}$$

Equivalently, writing $\phi \equiv (\phi_1+i\phi_2)/\sqrt2$ with ϕ_k Hermitian for $k=1,2$, we have canonical variables

$$q^k(\mathbf{x},t) = \phi_k(\mathbf{x},t), \tag{7.1.10}$$

$$p_k(\mathbf{x},t) = \dot\phi_k(\mathbf{x},t), \tag{7.1.11}$$

and these satisfy the commutation relations (7.1.1)–(7.1.3).

For the real vector field of a particle of spin one, the commutation relations are given by Section 5.3 as

$$[v^\mu(x),v^\nu(y)] = \left[\eta^{\mu\nu} - \frac{\partial^\mu\partial^\nu}{m^2}\right]\Delta(x-y).$$

(We are using v^μ rather than V^μ for the vector field because we want to reserve upper case letters here for the fields in the Heisenberg picture.) Here the free-particle canonical variables may be taken as

$$q^i(\mathbf{x},t) = v^i(\mathbf{x},t), \tag{7.1.12}$$

$$p_i(\mathbf{x},t) = \dot v^i(\mathbf{x},t) + \frac{\partial v^0(\mathbf{x},t)}{\partial x^i}, \tag{7.1.13}$$

with $i=1,2,3$. The reader may check that (7.1.12) and (7.1.13) satisfy the commutation relations (7.1.1)–(7.1.3). The field equations (5.3.36) and (5.3.38) together with Eq. (7.1.13) allow us to express v^0 in terms of the other variables as

$$v^0 = m^{-2}\nabla\cdot\mathbf{p}, \tag{7.1.14}$$

so v^0 is not regarded as one of the qs. The extension of these results to complex vector fields may be handled just as for complex scalar fields.

For the Dirac field of a non-Majorana spin $\frac12$ particle, Section 5.6 shows that the anticommutator is

$$\{\psi_\ell(x),\psi^\dagger_{\ell'}(y)\} = [(-\gamma^\mu\partial_\mu+m)\beta]_{\ell\ell'}\Delta(x-y)$$

and

$$\{\psi_\ell(x),\psi_{\ell'}(y)\} = 0.$$

Here it would be inconsistent to take ψ_ℓ and $\psi_\ell^\dagger$ to be independent canonical variables, because their anticommutator does not vanish at equal times. It is conventional instead to define

$$q^\ell(x) = \psi_\ell(x), \tag{7.1.15}$$

$$p_\ell(x) = i\psi_\ell^\dagger(x). \tag{7.1.16}$$

It is easy then to see that (7.1.15) and (7.1.16) satisfy the canonical anticommutation relations (7.1.1)–(7.1.3).

For any system of operators that satisfy commutation or anticommutation relations like (7.1.1)–(7.1.3), we may define a quantum mechanical functional derivative: for an arbitrary bosonic functional $F[q(t),p(t)]$ of $q^n(\mathbf{x},t)$ and $p_n(\mathbf{x},t)$ at a fixed time t, we define[58]

$$\frac{\delta F[q(t),p(t)]}{\delta q^n(\mathbf{x},t)} = i[p_n(\mathbf{x},t),F[q(t),p(t)]], \tag{7.1.17}$$

$$\frac{\delta F[q(t),p(t)]}{\delta p_n(\mathbf{x},t)} = i[F[q(t),p(t)],q^n(\mathbf{x},t)]. \tag{7.1.18}$$

This definition is motivated by the fact that if $F[q(t),p(t)]$ is written with all qs to the left of all ps, then (7.1.17) and (7.1.18) are respectively just the left- and right-derivatives with respect to q^n and p_n. That is, for an arbitrary c-number[59] variation δq and δp of the qs and ps, we have

$$\delta F[q(t),p(t)] = \int d^3x\sum_n\left(\delta q^n(\mathbf{x},t)\frac{\delta F[q(t),p(t)]}{\delta q^n(\mathbf{x},t)} + \frac{\delta F[q(t),p(t)]}{\delta p_n(\mathbf{x},t)}\delta p_n(\mathbf{x},t)\right).$$

For more general functionals we need the definitions (7.1.17) and (7.1.18) to pin down various signs and equal-time commutators that may appear.

In particular, H_0 is the generator of time-translations on free-particle states in the sense that:

$$q^n(\mathbf{x},t) = \exp(iH_0t)q^n(\mathbf{x},0)\exp(-iH_0t), \tag{7.1.19}$$

$$p_n(\mathbf{x},t) = \exp(iH_0t)p_n(\mathbf{x},0)\exp(-iH_0t), \tag{7.1.20}$$

so the free-particle operators have the time-dependence

$$\dot q^n(\mathbf{x},t) = i[H_0,q^n(\mathbf{x},t)] = \frac{\delta H_0}{\delta p_n(\mathbf{x},t)}, \tag{7.1.21}$$

$$\dot p_n(\mathbf{x},t) = -i[p_n(\mathbf{x},t),H_0] = -\frac{\delta H_0}{\delta q^n(\mathbf{x},t)}. \tag{7.1.22}$$

We recognize these as the familiar dynamical equations in the Hamiltonian formalism.

The free-particle Hamiltonian is given as always by

$$H_0 = \sum_{n,\sigma}\int d^3k\,a^\dagger_{\mathbf{k},\sigma,n}a_{\mathbf{k},\sigma,n}\sqrt{\mathbf{k}^2+m_n^2}. \tag{7.1.23}$$

This H_0 may be rewritten in terms of the qs and ps at time t. For instance, it is easy to see that for a real scalar field, Eq. (7.1.23) is equal up to a constant term to the functional

$$H_0 = \int d^3x\left[\frac12p^2 + \frac12(\nabla q)^2 + \frac12m^2q^2\right]. \tag{7.1.24}$$

[58] We are here using a notation that will be adopted henceforth; if $f(x,y)$ is a function of two classes of variables collectively called x and y, then $F[f(y)]$ indicates a functional that depends on the values of $f(x,y)$ for all x at fixed y. By a bosonic functional we mean one in which each term contains only even numbers of fermionic fields.

[59] Where q^n and p_n are bosonic or fermionic, δq^n and δp_n are understood to commute or anticommute with all fermionic operators, respectively, and to commute with all bosonic operators.

To be more precise, using (7.1.7) and the Fourier representation of the scalar field ϕ, we find that Eq. (7.1.24) becomes:

$$\begin{aligned}H_0 &= \frac12\int d^3k\,k^0\{a_{\mathbf{k}},a^\dagger_{\mathbf{k}}\}\\ &= \int d^3k\,k^0\left(a^\dagger_{\mathbf{k}}a_{\mathbf{k}} + \frac12\delta^3(\mathbf{k}-\mathbf{k})\right).\end{aligned} \tag{7.1.25}$$

This is the same as Eq. (7.1.23), except for the infinite constant term. Such terms only affect the zero of energy, and have no physical significance in the absence of gravity.[60] Explicit forms for H_0 as a functional of the q and p variables for other fields will be given in Section 7.5.

It is usual in textbooks on quantum field theory to derive Eq. (7.1.25) as a consequence of Eq. (7.1.24), which in turn is derived from a Lagrangian density. This seems to me backward, for Eq. (7.1.25) *must* hold; if some assumed free-particle Lagrangian did not give Eq. (7.1.25) up to a constant term, we would conclude that it was the wrong Lagrangian. Rather, we should ask what free-field Lagrangian gives Eq. (7.1.25) for spinless particles, or more generally, gives the free-particle Hamiltonian (7.1.23). This question may be answered by the well-known Legendre transformation from the Hamiltonian to the Lagrangian; the free-field Lagrangian is given by

$$L_0[q(t),\dot q(t)] \equiv \sum_n\int d^3x\,p_n(\mathbf{x},t)\dot q^n(\mathbf{x},t) - H_0, \tag{7.1.26}$$

it being understood that p_n is replaced everywhere by its expression in terms of q^n and $\dot q^n$ (and, as we shall see, perhaps some auxiliary fields as well). For instance, from the Hamiltonian (7.1.24) and (7.1.7) we can derive the free-field Lagrangian for a scalar field:

$$\begin{aligned}L_0 &= \int d^3x\left[p\dot q - \frac12p^2 - \frac12(\nabla q)^2 - \frac12m^2q^2\right]\\ &= \int d^3x\left[-\frac12\partial_\mu\phi\partial^\mu\phi - \frac12m^2\phi^2\right].\end{aligned} \tag{7.1.27}$$

Whatever we suppose the complete Lagrangian of the scalar field may be, this is the term that must be separated out and treated as a term of zeroth order in perturbation theory. A similar exercise may be carried out for the other canonical systems described in this section, but from now on we shall content ourselves with guessing the form of the free-field Lagrangian and then confirming that it gives the correct free-particle Hamiltonian.

We have seen that various free-field theories can be formulated in canonical terms. It is then a short step to show that the same is true of the interacting fields. We can introduce canonical variables in what is called the 'Heisenberg picture', defined by

$$Q^n(\mathbf{x},t) = \exp(iHt)q^n(\mathbf{x},0)\exp(-iHt), \tag{7.1.28}$$

[60] However, changes in such terms due to changes in the boundary conditions for the fields, as for instance quantizing in the space between parallel plates rather than in infinite space, are physically significant, and have even been measured [1].

$$P_n(\mathbf{x},t) = \exp(iHt)p_n(\mathbf{x},0)\exp(-iHt), \tag{7.1.29}$$

where H is the full Hamiltonian. Because this is a similarity transformation that commutes with H, the total Hamiltonian is the same functional of the Heisenberg picture operators as it was of the qs and ps:

$$H[Q,P] = e^{iHt}H[q,p]e^{-iHt} = H[q,p].$$

Also, because Eqs. (7.1.28)–(7.1.29) define a similarity transformation, the Heisenberg picture operators again satisfy the canonical commutation or anticommutation relations:

$$\begin{aligned}[Q^n(\mathbf{x},t),P_{\bar n}(\mathbf{y},t)] &= i\delta^3(\mathbf{x}-\mathbf{y})\delta^n_{\bar n}, &&\text{if either particle is a boson,}\\ \{Q^n(\mathbf{x},t),P_{\bar n}(\mathbf{y},t)\} &= i\delta^3(\mathbf{x}-\mathbf{y})\delta^n_{\bar n}, &&\text{if both particles are fermions.}\end{aligned} \tag{7.1.30}$$

$$\begin{aligned}[Q^n(\mathbf{x},t),Q^{\bar n}(\mathbf{y},t)] &= 0, &&\text{if either particle is a boson,}\\ \{Q^n(\mathbf{x},t),Q^{\bar n}(\mathbf{y},t)\} &= 0, &&\text{if both particles are fermions.}\end{aligned} \tag{7.1.31}$$

$$\begin{aligned}[P_n(\mathbf{x},t),P_{\bar n}(\mathbf{y},t)] &= 0, &&\text{if either particle is a boson,}\\ \{P_n(\mathbf{x},t),P_{\bar n}(\mathbf{y},t)\} &= 0, &&\text{if both particles are fermions.}\end{aligned} \tag{7.1.32}$$

However, they now have the time-dependence

$$\dot Q^n(\mathbf{x},t) = i[H,Q^n(\mathbf{x},t)] = \frac{\delta H}{\delta P_n(\mathbf{x},t)}, \tag{7.1.33}$$

$$\dot P_n(\mathbf{x},t) = -i[P_n(\mathbf{x},t),H] = -\frac{\delta H}{\delta Q^n(\mathbf{x},t)}. \tag{7.1.34}$$

For instance, we might take the Hamiltonian for a real scalar field as the free-particle term (7.1.24) plus the integral of a scalar interaction density $\mathcal{H}_I$, so that in terms of Heisenberg-picture variables

$$H = \int d^3x\left[\frac12P^2 + \frac12(\nabla Q)^2 + \frac12m^2Q^2 + \mathcal{H}_I(Q)\right]. \tag{7.1.35}$$

In this case the canonical conjugate to Q is given by the same formula as for free fields:

$$P = \dot Q. \tag{7.1.36}$$

However, as we shall see, the relation between the canonical conjugates $P_n(x)$ and the field variables and their time-derivatives is in general not the same as for the free-particle operators, but must be inferred from Eqs. (7.1.33) and (7.1.34).

7.2 The Lagrangian Formalism

Having seen that various realistic theories may be cast in the canonical formalism, we must now face the question of how to choose the Hamiltonian. As we will see in the next section, the easiest way to enforce Lorentz invariance and other symmetries is to choose a suitable Lagrangian and use it to derive the Hamiltonian. There is not much loss of generality in this; given a realistic Hamiltonian, we can generally reconstruct a Lagrangian from which

it could be derived, by reversing the process that we are going to describe here of deriving Hamiltonians from Lagrangians. (The derivation of Eq. (7.1.26) gives one example of this reconstruction.) But although we can go from Hamiltonians to Lagrangians or Lagrangians to Hamiltonians, it is easier to explore physically satisfactory theories by listing possible Lagrangians, rather than Hamiltonians.

The Lagrangian is, in general, a functional[61] $L[\Psi(t),\dot\Psi(t)]$ of a set of generic fields $\Psi^\ell(\mathbf{x},t)$ and their time-derivatives $\dot\Psi^\ell(\mathbf{x},t)$. The conjugate fields $\Pi_\ell(\mathbf{x},t)$ are defined as the variational derivatives[62]

$$\Pi_\ell(\mathbf{x},t) \equiv \frac{\delta L[\Psi(t),\dot\Psi(t)]}{\delta\dot\Psi^\ell(\mathbf{x},t)}. \tag{7.2.1}$$

The equations of motion are

$$\dot\Pi_\ell(\mathbf{x},t) = \frac{\delta L[\Psi(t),\dot\Psi(t)]}{\delta\Psi^\ell(\mathbf{x},t)}. \tag{7.2.2}$$

These field equations can be usefully reformulated as a variational principle. We define a functional of $\Psi^\ell(x)$ over all spacetime, known as the *action*

$$I[\Psi] \equiv \int_{-\infty}^{\infty}dt\,L[\Psi(t),\dot\Psi(t)]. \tag{7.2.3}$$

Under an arbitrary variation of $\Psi(x)$, the change in $I[\Psi]$ is

$$\delta I[\Psi] = \int_{-\infty}^{\infty}dt\int d^3x\left[\frac{\delta L}{\delta\Psi^\ell(x)}\delta\Psi^\ell(x) + \frac{\delta L}{\delta\dot\Psi^\ell(x)}\delta\dot\Psi^\ell(x)\right].$$

Assuming that $\delta\Psi^\ell(x)$ vanishes for $t \to \pm\infty$, we may integrate by parts, and write

$$\delta I[\Psi] = \int d^4x\left[\frac{\delta L}{\delta\Psi^\ell(x)} - \frac{d}{dt}\frac{\delta L}{\delta\dot\Psi^\ell(x)}\right]\delta\Psi^\ell(x). \tag{7.2.4}$$

We see that the action is stationary with respect to all variations $\delta\Psi^\ell$ that vanish at $t \to \pm\infty$ if and only if the fields satisfy the field equations (7.2.2).

Because the field equations are determined by the functional $I[\Psi]$, it is natural in trying to construct a Lorentz-invariant theory to make $I[\Psi]$ a scalar functional. In particular, since $I[\Psi]$ is a time-integral of $L[\Psi(t),\dot\Psi(t)]$, we guess that L should itself be a space-integral of an ordinary scalar function of $\Psi(x)$ and $\partial\Psi(x)/\partial x^\mu$, known as the *Lagrangian density* $\mathcal{L}$:

$$L[\Psi(t),\dot\Psi(t)] = \int d^3x\,\mathcal{L}\big(\Psi(\mathbf{x},t),\nabla\Psi(\mathbf{x},t),\dot\Psi(\mathbf{x},t)\big), \tag{7.2.5}$$

[61] Recall that in the notation we use for functionals, a functional like L in which we display the variable t is understood to depend on the fields $\Psi^\ell(\mathbf{x},t)$ and $\dot\Psi^\ell(\mathbf{x},t)$, with the undisplayed variables ℓ and $\mathbf{x}$ running over all their values at a fixed value of the displayed variable t. We use upper case Ψs and Πs to indicate that these are interacting rather than free fields.

[62] Because the Ψs and $\dot\Psi$s do not in general satisfy simple commutation or anticommutation relations, we cannot give a simple definition of the functional derivatives occurring here as we did for functional derivatives with respect to the Qs and Ps in the previous section. Instead, we will simply specify that the variational derivatives are what they would be for c-number variables, with minus signs and equal-time commutators or anticommutators supplied as needed to make the formulas correct quantum-mechanically. As far as I know, no important issues hinge on the details here.

so that the action is

$$I[\Psi] = \int d^4x\,\mathcal{L}\big(\Psi(x),\partial\Psi(x)/\partial x^\mu\big). \tag{7.2.6}$$

All field theories used in current theories of elementary particles have Lagrangians of this form.

Varying $\Psi^\ell(x)$ by an amount $\delta\Psi^\ell(x)$, and integrating by parts, we find a variation in L:

$$\begin{aligned}\delta L &= \int d^3x\left[\frac{\partial\mathcal{L}}{\partial\Psi^\ell}\delta\Psi^\ell + \frac{\partial\mathcal{L}}{\partial(\nabla\Psi^\ell)}\nabla\delta\Psi^\ell + \frac{\partial\mathcal{L}}{\partial\dot\Psi^\ell}\delta\dot\Psi^\ell\right]\\ &= \int d^3x\left[\left(\frac{\partial\mathcal{L}}{\partial\Psi^\ell} - \nabla\cdot\frac{\partial\mathcal{L}}{\partial(\nabla\Psi^\ell)}\right)\delta\Psi^\ell + \frac{\partial\mathcal{L}}{\partial\dot\Psi^\ell}\delta\dot\Psi^\ell\right],\end{aligned}$$

so (with obvious arguments suppressed)

$$\frac{\delta L}{\delta\Psi^\ell} = \frac{\partial\mathcal{L}}{\partial\Psi^\ell} - \nabla\cdot\frac{\partial\mathcal{L}}{\partial(\nabla\Psi^\ell)}, \tag{7.2.7}$$

$$\frac{\delta L}{\delta\dot\Psi^\ell} = \frac{\partial\mathcal{L}}{\partial\dot\Psi^\ell}. \tag{7.2.8}$$

The field equations (7.2.1) then read

$$\frac{\partial}{\partial x^\mu}\frac{\partial\mathcal{L}}{\partial(\partial\Psi^\ell/\partial x^\mu)} = \frac{\partial\mathcal{L}}{\partial\Psi^\ell}. \tag{7.2.9}$$

These are known as the *Euler–Lagrange equations*. As expected, if $\mathcal{L}$ is a scalar then these equations are Lorentz-invariant.

In addition to being Lorentz-invariant, the action I is required to be *real*. This is because we want just as many field equations as there are fields. By breaking up any complex fields into their real and imaginary parts, we can always think of I as being a functional only of a number of *real* fields, say N of them. If I were complex, with independent real and imaginary parts, then the real and imaginary parts of the conditions that I be stationary (the Euler–Lagrange equations) would yield $2N$ field equations for N fields, too many to be satisfied except in special cases. We will see in the next section that the reality of the action also ensures that the generators of various symmetry transformations are Hermitian operators.

Although the Lagrangian formalism makes it easy to construct theories that will satisfy Lorentz invariance and other symmetries, to calculate the S-matrix we need a formula for the interaction Hamiltonian. In general, the Hamiltonian is given by the *Legendre transformation*

$$H = \sum_\ell\int d^3x\,\Pi_\ell(\mathbf{x},t)\dot\Psi^\ell(\mathbf{x},t) - L[\Psi(t),\dot\Psi(t)]. \tag{7.2.10}$$

Although Eq. (7.2.1) does not in general allow $\dot\Psi^\ell$ to be expressed uniquely in terms of Ψ^ℓ and Π_ℓ, it is easy to see that Eq. (7.2.10) has vanishing variational derivative with respect

to $\dot\Psi^\ell$ for any $\dot\Psi^\ell$ satisfying Eq. (7.2.1), so in general it is a functional only of Ψ^ℓ and Π_ℓ. Its variational derivatives with respect to these variables are

$$\begin{aligned}\left.\frac{\delta H}{\delta\Psi^\ell(\mathbf{x},t)}\right|_\Pi &= \int d^3y\sum_{\ell'}\Pi_{\ell'}(\mathbf{y},t)\left.\frac{\delta\dot\Psi^{\ell'}(\mathbf{y},t)}{\delta\Psi^\ell(\mathbf{x},t)}\right|_\Pi - \left.\frac{\delta L}{\delta\Psi^\ell(\mathbf{x},t)}\right|_{\dot\Psi}\\ &\quad - \int d^3y\sum_{\ell'}\left.\frac{\delta L}{\delta\dot\Psi^{\ell'}(\mathbf{y},t)}\right|_\Psi\left.\frac{\delta\dot\Psi^{\ell'}(\mathbf{y},t)}{\delta\Psi^\ell(\mathbf{x},t)}\right|_\Pi,\\ \left.\frac{\delta H}{\delta\Pi_\ell(\mathbf{x},t)}\right|_\Psi &= \dot\Psi^\ell(\mathbf{x},t) + \int d^3y\sum_{\ell'}\Pi_{\ell'}(\mathbf{y},t)\left.\frac{\delta\dot\Psi^{\ell'}(\mathbf{y},t)}{\delta\Pi_\ell(\mathbf{x},t)}\right|_\Psi\\ &\quad - \int d^3y\sum_{\ell'}\left.\frac{\delta L}{\delta\dot\Psi^{\ell'}(\mathbf{y},t)}\right|_\Psi\left.\frac{\delta\dot\Psi^{\ell'}(\mathbf{y},t)}{\delta\Pi_\ell(\mathbf{x},t)}\right|_\Psi,\end{aligned}$$

where subscripts denote the quantities held fixed in these variational derivatives. Using the defining equation (7.2.1) for Π_ℓ, this simplifies to

$$\left.\frac{\delta H}{\delta\Psi^\ell(\mathbf{x},t)}\right|_\Pi = -\left.\frac{\delta L}{\delta\Psi^\ell(\mathbf{x},t)}\right|_{\dot\Psi}, \tag{7.2.11}$$

and

$$\left.\frac{\delta H}{\delta\Pi_\ell(\mathbf{x},t)}\right|_\Psi = \dot\Psi^\ell(\mathbf{x},t). \tag{7.2.12}$$

The equations of motion (7.2.2) are then equivalent to

$$\left.\frac{\delta H}{\delta\Psi^\ell(\mathbf{x},t)}\right|_\Pi = -\dot\Pi_\ell(\mathbf{x},t). \tag{7.2.13}$$

It is tempting now to identify the generic field variables Ψ^ℓ and their conjugates Π_ℓ with the canonical variables Q^n and P_n of the previous section, and impose on them the same canonical commutation relations (7.1.30)–(7.1.32), so that Eqs. (7.2.12) and (7.2.13) are the same as the Hamiltonian equations of motion (7.1.33) and (7.1.34). This is indeed the case for the simple example of the real scalar field Φ with non-derivative coupling. Consider the Lagrangian density[63]

$$\mathcal{L} = -\frac12\partial_\mu\Phi\partial^\mu\Phi - \frac{m^2}{2}\Phi^2 - \mathcal{H}_I(\Phi), \tag{7.2.14}$$

which can be obtained by adding a real function $-\mathcal{H}_I(\Phi)$ of Φ to the free-field Lagrangian density found in the previous section. The Euler–Lagrange equations here are

$$(\Box - m^2)\Phi = \mathcal{H}_I'(\Phi). \tag{7.2.15}$$

[63] We do not include a free constant factor in the term $-\frac12\partial_\mu\Phi\partial^\mu\Phi$, because any such constant if positive can be absorbed into the normalization of Φ. As we shall see, a negative constant here would lead to a Hamiltonian that is not bounded below. The constant m is known as the bare mass. The most general Lagrangian that satisfies the principle of renormalizability (discussed in Chapter 12) is of this form, with $\mathcal{H}_I(\Phi)$ a quartic polynomial in Φ.

From this Lagrangian density, we calculate a canonical conjugate to Φ:

$$\Pi = \frac{\partial\mathcal{L}}{\partial\dot\Phi} = \dot\Phi, \tag{7.2.16}$$

which is the same as Eq. (7.1.36) if we identify Φ and Π with the canonical variables Q and P. The Hamiltonian is now given by Eq. (7.2.10) as[64]

$$\begin{aligned}H &= \int d^3x\,(\Pi\dot\Phi - \mathcal{L})\\ &= \int d^3x\left[\frac12\Pi^2 + \frac12(\nabla\Phi)^2 + \frac12m^2\Phi^2 + \mathcal{H}_I(\Phi)\right],\end{aligned} \tag{7.2.17}$$

which we recognize as the Hamiltonian (7.1.35). This little exercise should not be regarded as another derivation of this Hamiltonian, but rather as a validation of the Lagrangian (7.2.14) as a possible theory of scalar fields.

Matters are not always so simple. We have already seen in the previous section that there are field variables, such as the time component of a vector field or the Hermitian conjugate of a Dirac field, that are not canonical field variables Q^n and do not have canonical conjugates; yet Lorentz invariance dictates that these must appear in the Lagrangians for the vector and Dirac fields.

From the point of view of the Lagrangian formalism, the special character of field variables like the time component of a vector field or the Hermitian conjugate of a Dirac field arises from the fact that although they appear in the Lagrangian, their time-derivatives do not. We shall denote the field variables Ψ^ℓ whose time-derivatives do not appear in the Lagrangian as C^r; the remaining independent field variables are the canonical variables Q^n. The Q^n have canonical conjugates

$$P_n(\mathbf{x},t) = \frac{\delta L[Q(t),\dot Q(t),C(t)]}{\delta\dot Q^n(\mathbf{x},t)}, \tag{7.2.18}$$

and satisfy the commutation relations (7.1.30)–(7.1.32), but there are no canonical conjugates for the C^r. Because $\delta L/\delta\dot C^r = 0$, the Hamiltonian (7.2.10) is in general

$$H = \sum_n\int d^3x\,P_n\dot Q^n - L[Q(t),\dot Q(t),C(t)], \tag{7.2.19}$$

but this is not yet useful until we express the C^r and $\dot Q^\ell$ in terms of the Qs and Ps. The equations of motion of the C^r involve only fields and their first time-derivatives

$$0 = \frac{\delta L[Q(t),\dot Q(t),C(t)]}{\delta C^r(\mathbf{x},t)}. \tag{7.2.20}$$

In the simple cases to be discussed in this chapter, these equations together with Eq. (7.2.18) can be solved to give the C^r and $\dot Q^\ell$ in terms of the Qs and Ps. Section 7.6 shows how in

[64] In order for H to be interpreted as an energy, it should be bounded below. The positivity of the first two terms shows that we guessed correctly as to the sign in the first term in Eq. (7.2.14). The remaining condition is that $\frac12m^2\Phi^2 + \mathcal{H}_I(\Phi)$ must be bounded below as a function of Φ.

such cases one can avoid the task of actually solving for the C^r and $\dot Q^\ell$. In gauge theories like electrodynamics other methods must be used: either choosing a particular gauge, as in Chapter 8, or the more modern covariant methods to be discussed in Volume II.

Once we have derived a Hamiltonian as a functional of the Heisenberg picture Qs and Ps, to use perturbation theory we must make a transition to the interaction picture. The Hamiltonian is time-independent, so it can be written in terms of the P_n and Q^n at $t=0$, which are equal to the corresponding operators p_n and q^n in the interaction picture at $t=0$. The Hamiltonian derived in this way may then be expressed in terms of the qs and ps of the interaction picture, and split into two parts, a suitable free-particle term H_0 and an interaction H_{int}. Finally, the time-dependence equations (7.1.21) and (7.1.22) and the commutation or anticommutation relations (7.1.1)–(7.1.2) are used to express the qs and ps in $H_I(t)$ as linear combinations of annihilation and creation operators.

We shall present a number of examples of this procedure in Section 7.5; for the moment we will give only one example of the simplest type, the scalar field with Hamiltonian (7.2.17). We split H into a free-particle term and an interaction

$$H = H_0 + H_{\text{int}}, \tag{7.2.21}$$

$$H_0 = \int d^3x\left[\frac12\Pi^2 + \frac12(\nabla\Phi)^2 + \frac12m^2\Phi^2\right], \tag{7.2.22}$$

$$H_{\text{int}} = \int d^3x\,\mathcal{H}_I(\Phi). \tag{7.2.23}$$

Here Φ and Π are taken at the same time t, and H is independent of t, though H_0 and H_{int} usually are not.

We now pass to the interaction representation. Taking $t=0$ in Eqs. (7.2.22) and (7.2.23), we can simply replace Φ,Π with the interaction picture variables ϕ,π, since they are defined by Eqs. (7.1.28) and (7.1.29) to be equal at that time. To calculate the interaction $H_I(t)$ in the interaction picture, we apply the similarity transformation (3.5.5)

$$H_I(t) = \exp(iH_0t)H_{\text{int}}\exp(-iH_0t) = \int d^3x\,\mathcal{H}_I(\phi(\mathbf{x},t)). \tag{7.2.24}$$

The same transformation applied to H_0 leaves it constant:

$$\begin{aligned}H_0 &= \exp(iH_0t)H_0\exp(-iH_0t)\\ &= \int d^3x\left[\frac12\pi^2(\mathbf{x},t) + \frac12(\nabla\phi(\mathbf{x},t))^2\right].\end{aligned} \tag{7.2.25}$$

The relation between π and $\dot\phi$ is dictated by Eq. (7.1.21)

$$\dot\phi(\mathbf{x},t) = \frac{\delta H_0}{\delta\pi(\mathbf{x},t)} = \pi(\mathbf{x},t). \tag{7.2.26}$$

(This happens to be the same relation as in Eq. (7.2.16), but as we shall see this is not to be expected in general.) Also, the equation of motion for ϕ is dictated by Eq. (7.1.22):

$$\dot\pi(\mathbf{x},t) = -\frac{\delta H_0}{\delta\phi(\mathbf{x},t)} = +\nabla^2\phi(\mathbf{x},t) - m^2\phi(\mathbf{x},t), \tag{7.2.27}$$

which together with Eq. (7.2.26) yields the field equations

$$(\Box - m^2)\phi(x) = 0. \tag{7.2.28}$$

The general real solution may be expressed as

$$\phi(x) = (2\pi)^{-3/2}\int d^3p\,(2p^0)^{-1/2}\left[e^{ip\cdot x}a_{\mathbf{p}} + e^{-ip\cdot x}a^\dagger_{\mathbf{p}}\right] \tag{7.2.29}$$

with $p^0 = \sqrt{\mathbf{p}^2+m^2}$ understood, and $a_{\mathbf{p}}$ some as-yet-unknown operator function of $\mathbf{p}$. Eq. (7.2.26) then gives the canonical conjugate as

$$\pi(x) = -i(2\pi)^{-3/2}\int d^3p\,(p^0/2)^{1/2}\left[e^{ip\cdot x}a_{\mathbf{p}} - e^{-ip\cdot x}a^\dagger_{\mathbf{p}}\right]. \tag{7.2.30}$$

In order to get the desired commutation relations,

$$[\phi(\mathbf{x},t),\pi(\mathbf{y},t)] = i\delta^3(\mathbf{x}-\mathbf{y}), \tag{7.2.31}$$

$$[\phi(\mathbf{x},t),\phi(\mathbf{y},t)] = 0, \tag{7.2.32}$$

$$[\pi(\mathbf{x},t),\pi(\mathbf{y},t)] = 0, \tag{7.2.33}$$

we must take the as to satisfy the familiar commutation relations

$$[a_{\mathbf{p}},a^\dagger_{\mathbf{p}'}] = \delta^3(\mathbf{p}-\mathbf{p}'), \tag{7.2.34}$$

$$[a_{\mathbf{p}},a_{\mathbf{p}'}] = 0. \tag{7.2.35}$$

Also, we have already shown in the previous section that using these expansions in Eq. (7.2.25) gives the usual formula (4.2.11) for the free-particle Hamiltonian, up to an inconsequential additive constant. As remarked before, these results should not be regarded so much as an alternative derivation of Eqs. (7.2.29), (7.2.34), and (7.2.35) (which were obtained in Chapter 5 on quite other grounds) but rather as a validation of the first two terms of Eq. (7.2.14) as the correct free-particle Lagrangian for a real scalar field. We can now proceed to use perturbation theory to calculate the S-matrix, taking (7.2.24) as $H_I(t)$, with the field $\phi(x)$ given by Eq. (7.2.29).

The procedures illustrated here will be carried out for examples that are more complicated and more interesting in Section 7.5.

* * *

In considering the various possible Lagrangian densities for physical theories it is common to apply integration by parts, treating as equivalent any Lagrangian densities that differ only by total derivatives $\partial_\mu\mathcal{F}^\mu$. It is obvious that such total derivative terms do not contribute to the action and hence do not affect the field equations. It is also obvious that a space-derivative term $\nabla\cdot\mathcal{F}$ in the Lagrangian density does not contribute to the Lagrangian and hence does not affect the quantum theory defined by the Lagrangian.[65] What is less

[65] This is under the usual assumption, that the fields vanish at infinity. These results do not necessary apply when we allow fields of different topology, as discussed in Volume II.

obvious and worth noting here is that a time-derivative $\partial_0\mathcal{F}^0$ in the Lagrangian density also does not affect the quantum structure of the theory. To see this, let's first consider the effect of adding a term to the Lagrangian of the more general form

$$\Delta L(t) = \int d^3x\,D_{n,\mathbf{x}}[Q(t)]\dot Q^n(\mathbf{x},t), \tag{7.2.36}$$

where D is an arbitrary n- and $\mathbf{x}$-dependent functional of the values of Q at a given time. This changes the formula for the conjugate variables $P(t)$ as functionals of $Q(t)$ and $\dot Q(t)$ by the amount

$$\Delta P_n(\mathbf{x},t) = \frac{\delta\Delta L(t)}{\delta\dot Q^n(\mathbf{x},t)} = D_{n,\mathbf{x}}[Q(t)]. \tag{7.2.37}$$

It follows that there is no change in the Hamiltonian as expressed as a functional of the $Q(t)$ and $\dot Q(t)$:

$$\int d^3x\,\Delta P_n(\mathbf{x},t)\dot Q^n(\mathbf{x},t) - \Delta L(t) = 0. \tag{7.2.38}$$

Hence also there is no change in the Hamiltonian as expressed as a functional of the old canonical variables Q^n and P_n. However, the Hamiltonian is *not* the same functional of the *new* canonical variables Q^n and $P_n + \Delta P_n$ as it was of the Q^n and P_n, and in a theory described by the new Lagrangian $\mathcal{L}+\Delta\mathcal{L}$ it is the new canonical variables Q^n and $P_n+\Delta P_n$ rather than the Q^n and P_n that would satisfy the canonical commutation relations. The commutators of the Q^n with each other and of the Q^n with the P_m are given by the usual canonical relations, but the commutators of the P_n with each other are now

$$\begin{aligned}[P_n(\mathbf{x},t),P_m(\mathbf{y},t)] &= [P_n(\mathbf{x},t)+\Delta P_n(\mathbf{x},t),P_m(\mathbf{y},t)+\Delta P_m(\mathbf{y},t)]\\ &\quad - [\Delta P_n(\mathbf{x},t),P_m(\mathbf{y},t)+\Delta P_m(\mathbf{y},t)]\\ &\quad - [P_n(\mathbf{x},t)+\Delta P_n(\mathbf{x},t),\Delta P_m(\mathbf{y},t)]\\ &\quad + [\Delta P_n(\mathbf{x},t),\Delta P_m(\mathbf{y},t)]\\ &= -i\frac{\delta D_{n,\mathbf{x}}[Q(t)]}{\delta Q^m(\mathbf{y},t)} + i\frac{\delta D_{m,\mathbf{y}}[Q(t)]}{\delta Q^n(\mathbf{x},t)}.\end{aligned} \tag{7.2.39}$$

In general this doesn't vanish, but if the added term in the Lagrangian is a total time-derivative

$$\Delta L = \frac{dG}{dt} = \int d^3x\,\frac{\delta G[Q(t)]}{\delta Q^n(\mathbf{x},t)}\dot Q^n(\mathbf{x},t), \tag{7.2.40}$$

then D in Eq. (7.2.36) is of the special form

$$D_{n,\mathbf{x}}[Q] = \frac{\delta G[Q(t)]}{\delta Q^n(\mathbf{x},t)}. \tag{7.2.41}$$

In this case the commutator (7.2.39) vanishes, so the variables Q^n and P_n satisfy the usual commutation relations. We have seen that a change of the form (7.2.36) in the Lagrangian does not change the form of the Hamiltonian as a functional of the Q^n and P_n, and since, as we have now shown, the commutation relations of these variables are also unchanged, the addition to the Lagrangian of the term (7.2.36) has no effect on the quantum structure of the theory. Different Lagrangian densities obtained from each other by partial integration may therefore be regarded as equivalent in quantum as well as classical field theory.

7.3 Global Symmetries

We now come to the real point of the Lagrangian formalism, that it provides a natural framework for the quantum mechanical implementation of symmetry principles. This is because the dynamical equations in the Lagrangian formalism take the form of a variational principle, the principle of stationary action. Consider any infinitesimal transformation of the fields

$$\Psi^\ell(x) \longrightarrow \Psi^\ell(x) + i\epsilon\mathcal{F}^\ell(x) \tag{7.3.1}$$

that leaves the action (7.2.3) invariant:

$$0 = \delta I = i\epsilon\int d^4x\,\frac{\delta I[\Psi]}{\delta\Psi^\ell(x)}\mathcal{F}^\ell(x). \tag{7.3.2}$$

(With ϵ a constant, such symmetries are known as *global symmetries*. In general, $\mathcal{F}^\ell$ depends on the fields and their derivatives at x.) Of course Eq. (7.3.2) is automatically satisfied for *all* infinitesimal variations of the fields if the fields satisfy the dynamical equations; by an infinitesimal symmetry transformation we mean one that leaves the action invariant even when the dynamical equations are *not* satisfied. If we now consider the same transformation with ϵ an arbitrary function of position in spacetime:

$$\Psi^\ell(x) \longrightarrow \Psi^\ell(x) + i\epsilon(x)\mathcal{F}^\ell(x), \tag{7.3.3}$$

then, in general, the variation of the action will not vanish, but it will have to be of the form

$$\delta I = -\int d^4x\,J^\mu(x)\frac{\partial\epsilon(x)}{\partial x^\mu} \tag{7.3.4}$$

in order that it should vanish when $\epsilon(x)$ is constant. If we *now* take the fields in $I[\Psi]$ to satisfy the field equations then I is stationary with respect to arbitrary field variations that vanish at large spacetime distances, including variations of the form (7.3.3), so in this case (7.3.4) should vanish. Integrating by parts, we see that $J^\mu(x)$ must satisfy a conservation law:

$$0 = \frac{\partial J^\mu(x)}{\partial x^\mu}. \tag{7.3.5}$$

It follows immediately that

$$0 = \frac{dF}{dt}, \tag{7.3.6}$$

where

$$F \equiv \int d^3x\,J^0. \tag{7.3.7}$$

There is one such conserved current J^μ and one constant of the motion F for each independent infinitesimal symmetry transformation. This represents a general feature of the canonical formalism, often referred to as Noether's theorem: *symmetries imply conservation laws*.

Many symmetry transformations leave the Lagrangian and not just the action invariant. This is the case, for instance, for translations and rotations in space and also isospin transformations and other internal symmetry transformations, though not for general Lorentz

transformations. When the Lagrangian is invariant we can go further, and write an explicit formula for the conserved quantities F. Consider a field variation (7.3.3) in which $\epsilon(x)$ depends on t but not $\mathbf{x}$. In this case the variation in the action is

$$\begin{aligned}\delta I = i\int dt\int d^3x\Bigg[&\frac{\delta L[\Psi(t),\dot\Psi(t)]}{\delta\Psi^\ell(\mathbf{x},t)}\epsilon(t)\mathcal{F}^\ell(\mathbf{x},t)\\ &+ \frac{\delta L[\Psi(t),\dot\Psi(t)]}{\delta\dot\Psi^\ell(\mathbf{x},t)}\frac{d}{dt}\big(\epsilon(t)\mathcal{F}^\ell(\mathbf{x},t)\big)\Bigg].\end{aligned} \tag{7.3.8}$$

The requirement that the Lagrangian be invariant under this transformation when ϵ is a constant yields

$$0 = \int d^3x\left[\frac{\delta L[\Psi(t),\dot\Psi(t)]}{\delta\Psi^\ell(\mathbf{x},t)}\mathcal{F}^\ell(\mathbf{x},t) + \frac{\delta L[\Psi(t),\dot\Psi(t)]}{\delta\dot\Psi^\ell(\mathbf{x},t)}\frac{d}{dt}\mathcal{F}^\ell(\mathbf{x},t)\right], \tag{7.3.9}$$

so for general fields (whether or not the field equations are satisfied) the variation in the action is

$$\delta I = i\int dt\int d^3x\,\frac{\delta L[\Psi(t),\dot\Psi(t)]}{\delta\dot\Psi^\ell(\mathbf{x},t)}\dot\epsilon(t)\mathcal{F}^\ell(\mathbf{x},t). \tag{7.3.10}$$

Comparing this with Eq. (7.3.4) gives

$$F = -i\int d^3x\,\frac{\delta L[\Psi(t),\dot\Psi(t)]}{\delta\dot\Psi^\ell(\mathbf{x},t)}\mathcal{F}^\ell(\mathbf{x},t). \tag{7.3.11}$$

Using the symmetry condition (7.3.9), the reader can easily check that this F is indeed time-independent for any fields that satisfy the dynamical equations (7.2.2).

Other symmetry transformations such as isospin rotations leave not only the action and the Lagrangian invariant but also the Lagrangian density. In such cases we can go even further, and write an explicit formula for the current $J^\mu(x)$. Writing the action as in Eq. (7.2.6) as the integral of the Lagrangian density, its variation under the transformation (7.3.3) with a general infinitesimal parameter $\epsilon(x)$ is

$$\begin{aligned}\delta I[\Psi] = i\int d^4x\Bigg[&\frac{\partial\mathcal{L}(\Psi(x),\partial_\mu\Psi(x))}{\partial\Psi^\ell(x)}\mathcal{F}^\ell(x)\epsilon(x)\\ &+ \frac{\partial\mathcal{L}(\Psi(x),\partial_\mu\Psi(x))}{\partial(\partial_\mu\Psi^\ell(x))}\partial_\mu\big(\mathcal{F}^\ell(x)\epsilon(x)\big)\Bigg].\end{aligned} \tag{7.3.12}$$

The invariance of the Lagrangian density when ϵ is a constant requires that

$$0 = \frac{\partial\mathcal{L}(\Psi(x),\partial_\mu\Psi(x))}{\partial\Psi^\ell(x)}\mathcal{F}^\ell(x) + \frac{\partial\mathcal{L}(\Psi(x),\partial_\mu\Psi(x))}{\partial(\partial_\mu\Psi^\ell(x))}\partial_\mu\mathcal{F}^\ell(x), \tag{7.3.13}$$

so for arbitrary fields the variation of the action is

$$\delta I[\Psi] = i\int d^4x\,\frac{\partial\mathcal{L}(\Psi(x),\partial_\mu\Psi(x))}{\partial(\partial_\mu\Psi^\ell(x))}\partial_\mu\epsilon(x). \tag{7.3.14}$$

Comparison with Eq. (7.3.4) shows that

$$J^\mu = -i\frac{\partial\mathcal{L}}{\partial(\partial\Psi^\ell/\partial x^\mu)}\mathcal{F}^\ell. \tag{7.3.15}$$

Using the symmetry condition (7.3.13), it is easy to see directly that $\partial_\mu J^\mu$ vanishes when the fields satisfy the Euler–Lagrange equations (7.2.9). Note also that the integral of the time component of the current (7.3.15) has the previously derived value (7.3.11).

So far everything we have said would apply to classical as well as quantum mechanical field theories. The quantum properties of the conserved quantities F are most easily seen for symmetries of the Lagrangian (not necessarily the Lagrangian density) that transform the canonical fields $Q^n(\mathbf{x},t)$ (that is, those of the Ψ^ℓ whose time derivatives appear in the Lagrangian) into $\mathbf{x}$-dependent functionals of themselves at the same time. For such transformations, we have

$$\mathcal{F}^n(\mathbf{x},t) = \mathcal{F}^n[Q(t);\mathbf{x}]. \tag{7.3.16}$$

As we shall see, infinitesimal spatial translations and rotations as well as all infinitesimal internal symmetry transformations are of the form (7.3.1), (7.3.16), with $\mathcal{F}^n$ a linear functional of the Q^m, but we will not need to assume here that the symmetry is linear. For all such symmetries the operator F is not only conserved; it also acts in quantum mechanics as a *generator* of this symmetry.

To see this, note first that when Ψ^ℓ is a canonical field Q^n, the functional derivative $\delta L/\delta\dot\Psi^\ell$ is equal to the canonical conjugate P_n, while when Ψ^ℓ is an auxiliary field C^r, this functional derivative vanishes; hence we may rewrite Eq. (7.3.11) in the form

$$F = -i\int d^3x\,P_n(\mathbf{x},t)\mathcal{F}^n(\mathbf{x},t) = -i\int d^3x\int d^3y\,P_n(\mathbf{x},t)\mathcal{F}^n[Q(t);\mathbf{x}]. \tag{7.3.17}$$

To calculate the commutator (not anticommutator) of F with a canonical field $Q^m(\mathbf{x},t)$ at an arbitrary time t, we can invoke Eq. (7.3.6) to evaluate F as a functional of the Qs and Ps at the time t, and then use the equal-time canonical commutation relations (7.1.30)–(7.1.32) to obtain[66]

$$[F,Q^m(\mathbf{x},t)] = -\mathcal{F}^m(\mathbf{x},t). \tag{7.3.18}$$

It is in this sense that F is the generator of the transformation with Eq. (7.3.16). Eq. (7.3.17) and the canonical commutation rules give also

$$[F,P_n(\mathbf{x},t)] = \int d^3y\,P_m(\mathbf{y},t)\frac{\delta\mathcal{F}^m[Q(t);\mathbf{y}]}{\delta Q^n(\mathbf{x},t)}. \tag{7.3.19}$$

Where $\mathcal{F}^m$ is linear, Eq. (7.3.19) tells us that P_n transforms contragrediently to Q^n.

As a first example, consider the symmetry transformation of spacetime translation:

$$\Psi^\ell(x) \longrightarrow \Psi^\ell(x+\epsilon) = \Psi^\ell(x) + \epsilon^\mu\partial_\mu\Psi^\ell(x). \tag{7.3.20}$$

[66] We are here assuming that for Q^n bosonic or fermionic the variation $\mathcal{F}^n$ is also respectively bosonic or fermionic, so that F is bosonic. The only exceptions are certain symmetries known as supersymmetries, for which F is fermionic and (7.3.18) is an anticommutator if Q^n is also fermionic.

This is of the form (7.3.1), with four independent parameters ϵ^μ and four corresponding transformation functions

$$\mathcal{F}^\ell_\mu = -i\partial_\mu\Psi^\ell. \tag{7.3.21}$$

In consequence we have four independent conserved currents, conventionally grouped together in the *energy–momentum tensor* $T^\mu{}_\nu$:

$$\partial_\mu T^\mu{}_\nu = 0 \tag{7.3.22}$$

from which we can derive time-independent quantities as the spatial integrals of the time components of the translation 'currents' (not to be confused with the canonical conjugate field variables $P_n(\mathbf{x},t)$):

$$P_\nu = \int d^3x\,T^0{}_\nu, \tag{7.3.23}$$

$$\frac{d}{dt}P_\nu = 0. \tag{7.3.24}$$

The Lagrangian is invariant under spatial translations, so in accordance with the above general results we can conclude that the spatial components of P_ν take the form

$$\mathbf{P} = -\int d^3x\,P_n(\mathbf{x},t)\nabla Q^n(\mathbf{x},t). \tag{7.3.25}$$

Using the equal-time commutation relations (7.1.30)–(7.1.32), we also find the commutator of this operator with the canonical fields and conjugates:

$$[\mathbf{P},Q^n(\mathbf{x},t)] = i\nabla Q^n(\mathbf{x},t), \tag{7.3.26}$$

$$[\mathbf{P},P_n(\mathbf{x},t)] = i\nabla P_n(\mathbf{x},t). \tag{7.3.27}$$

It follows that for any function $\mathcal{G}$ of Qs and Ps that does not also depend explicitly on $\mathbf{x}$, we have

$$[\mathbf{P},\mathcal{G}(\mathbf{x})] = i\nabla\mathcal{G}(\mathbf{x}). \tag{7.3.28}$$

These results show that the operator $\mathbf{P}$ is indeed the generator of space translations.

In contrast, time-translations do not leave the Lagrangian $L(t)$ invariant. However, we already know the generator of time-translations; it is the Hamiltonian $P^0 \equiv H$, which as we know satisfies the commutation relation

$$[H,\mathcal{G}(\mathbf{x},t)] = -i\dot{\mathcal{G}}(\mathbf{x},t) \tag{7.3.29}$$

for any function $\mathcal{G}$ of Heisenberg picture operators.

If we further assume that the Lagrangian is the integral of a Lagrangian density, then we may also obtain an explicit formula for the energy–momentum tensor $T^\mu{}_\nu$. However, the Lagrangian density $\mathcal{L}(x)$ is not invariant under spacetime translations, so we cannot use Eq. (7.3.15) here. Instead, note that the change in the action under a spacetime-dependent translation

$$\Psi^\ell(x) \longrightarrow \Psi^\ell(x+\epsilon(x)) = \Psi^\ell(x) + \epsilon^\mu(x)\partial_\mu\Psi^\ell(x) \tag{7.3.30}$$

is

$$\delta I[\Psi] = \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \Psi^\ell} \epsilon^\mu \partial_\mu \Psi^\ell + \frac{\partial \mathcal{L}}{\partial(\partial_\nu \Psi^\ell)} \partial_\nu [\epsilon^\mu \partial_\mu \Psi^\ell] \right). \tag{7.3.31}$$

The Euler–Lagrange equations (7.2.9) show that the terms proportional to ϵ add up to $\epsilon^\mu \partial_\mu \mathcal{L}$, so

$$\delta I[\Psi] = \int d^4x \left(\frac{\partial \mathcal{L}}{\partial x^\mu} \epsilon^\mu + \frac{\partial \mathcal{L}}{\partial(\partial_\nu \Psi^\ell)} \partial_\mu \Psi^\ell \, \partial_\nu \epsilon^\mu \right). \tag{7.3.32}$$

Integrating by parts, we see that this takes the form of Eq. (7.3.4)

$$\delta I = -\int d^4x \, T^\nu{}_\mu \partial_\nu \epsilon^\mu \tag{7.3.33}$$

with 'currents'

$$T^\nu{}_\mu = \delta^\nu{}_\mu \mathcal{L} - \frac{\partial \mathcal{L}}{\partial(\partial_\nu \Psi^\ell)} \partial_\mu \Psi^\ell. \tag{7.3.34}$$

As a check, we may note that the spatial components of Eq. (7.3.23) are the same as our previous formula (7.3.25) for $\mathbf{P}$, while for $\mu = 0$ Eq. (7.3.23) gives the usual formula for the Hamiltonian:

$$H \equiv -P_0 = \int d^3x \left[\sum_n P_n \dot{Q}^n - \mathcal{L} \right]. \tag{7.3.35}$$

(A warning: the tensor $T^{\mu\nu}$ obtained by raising the second index in Eq. (7.3.34) is not in general symmetric, and therefore cannot be used as the right-hand side of the field equations of general relativity. The correct energy–momentum tensor to use as the source of the gravitational field is the symmetric tensor $\Theta^{\mu\nu}$ introduced in the next section.)

In many theories there are also one or more symmetry principles that state the invariance of the action, under a set of linear coordinate-independent transformations of the canonical fields

$$Q^n(x) \longrightarrow Q^n(x) + i\epsilon^a (t_a)^n{}_m Q^m(x) \tag{7.3.36}$$

together with a set of suitable transformations on any auxiliary fields C^r:

$$C^r(x) \longrightarrow C^r(x) + i\epsilon^a (\tau_a)^r{}_s C^s(x). \tag{7.3.37}$$

Here t_a and τ_a are sets of Hermitian matrices furnishing some representations of the Lie algebra of the symmetry group, and we sum over repeated group indices a, b, etc. (For instance, in electrodynamics there is such a symmetry, for which the one matrix $t^n{}_m$ is diagonal, with the charges carried by each field on the main diagonal.) From any such symmetry, we can infer the existence of another set of conserved currents J^μ_a:

$$\partial_\mu J^\mu_a = 0, \tag{7.3.38}$$

whose time components are the densities of a set of time-independent operators

$$T_a \equiv \int d^3x \, J^0_a. \tag{7.3.39}$$

When the Lagrangian as well as the action is invariant under the transformation (7.3.36), Eq. (7.3.11) provides an explicit formula for the T_a:

$$T_a = -i \int d^3x \, P_n(\mathbf{x}, t)(t_a)^n{}_m Q^m(\mathbf{x}, t). \tag{7.3.40}$$

The equal-time commutation relations here give

$$[T_a, Q^n(x)] = -(t_a)^n{}_m Q^m(x), \tag{7.3.41}$$

$$[T_a, P_n(x)] = +(t_a)^m{}_n P_m(x). \tag{7.3.42}$$

(Where t_a is diagonal, this tells us that Q^n and P_n respectively lower and raise the value of T_a by an amount equal to the nth diagonal element of t_a.) Using these results, we can calculate the commutator of T_a with another generator T_b:

$$[T_a, T_b] = i \int d^3x \left[-P_m (t_a)^m{}_n (t_b)^n{}_k Q^k + P_n (t_b)^n{}_k (t_a)^k{}_m Q^m \right]. \tag{7.3.43}$$

Thus, if the matrices t_a form a Lie algebra with structure constants $f_{ab}{}^c$,

$$[t_a, t_b] = i f_{ab}{}^c t_c, \tag{7.3.44}$$

then so do the quantum operators T_a:

$$[T_a, T_b] = i f_{ab}{}^c T_c. \tag{7.3.45}$$

This confirms that the quantities (7.3.40) are correctly normalized to qualify as generators of the symmetry group.

Where the Lagrangian is the integral of a Lagrangian density which is invariant under (7.3.36) and (7.3.37) we can go further, and use Eq. (7.3.15) to provide an explicit formula for the currents associated with these global symmetries:

$$J^\mu_a = -i \frac{\partial \mathcal{L}}{\partial(\partial Q^n/\partial x^\mu)} (t_a)^n{}_m Q^m - i \frac{\partial \mathcal{L}}{\partial(\partial C^r/\partial x^\mu)} (\tau_a)^r{}_s C^s. \tag{7.3.46}$$

As an illustration, suppose we have two real scalar fields of equal mass, with Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\mu \Phi_1 \partial^\mu \Phi_1 - \frac{1}{2}m^2 \Phi_1^2 - \frac{1}{2}\partial_\mu \Phi_2 \partial^\mu \Phi_2 - \frac{1}{2}m^2 \Phi_2^2 - \mathcal{H}_I(\Phi_1^2 + \Phi_2^2). \tag{7.3.47}$$

This is invariant under a linear transformation like (7.3.36):

$$\delta\Phi_1 = -\epsilon \Phi_2, \qquad \delta\Phi_2 = +\epsilon \Phi_1,$$

so there is a conserved current (7.3.46):

$$J^\mu = \Phi_2 \partial^\mu \Phi_1 - \Phi_1 \partial^\mu \Phi_2.$$

The explicit formula (7.3.46) for the current can be used to derive other useful commutation relations. In particular, since the Lagrangian density does not involve time-derivatives of the auxiliary fields, we have

$$J^0_a = -i P_n (t_a)^n{}_m Q^m. \tag{7.3.48}$$

We can then derive the equal-time commutators of general fields not only with the symmetry generators T_a, but also with the densities J^0_a:

$$[J^0_a(\mathbf{x}, t), Q^n(\mathbf{y}, t)] = -\delta^3(\mathbf{x} - \mathbf{y})(t_a)^n{}_m Q^m(\mathbf{x}, t), \tag{7.3.49}$$

$$[J^0_a(\mathbf{x}, t), P_m(\mathbf{y}, t)] = \delta^3(\mathbf{x} - \mathbf{y})(t_a)^n{}_m P_n(\mathbf{x}, t). \tag{7.3.50}$$

If the auxiliary fields are constructed as local functions of the Ps and Qs in such a way that they transform according to a representation of the symmetry algebra with generators τ_a, then also

$$[J^0_a(\mathbf{x}, t), C^r(\mathbf{y}, t)] = -\delta^3(\mathbf{x} - \mathbf{y})(\tau_a)^r{}_s C^s(\mathbf{x}, t). \tag{7.3.51}$$

We often summarize Eqs. (7.3.49) and (7.3.51) in the single commutation relation

$$[J^0_a(\mathbf{x}, t), \Psi^\ell(\mathbf{y}, t)] = -\delta^3(\mathbf{x} - \mathbf{y})(t_a)^\ell{}_{\ell'} \Psi^{\ell'}(\mathbf{x}, t). \tag{7.3.52}$$

Commutation relations like (7.3.49)–(7.3.51) will be used in Chapter 10 to derive relations called Ward identities for matrix elements involving the current J^μ.

7.4 Lorentz Invariance

We are now going to show that the Lorentz invariance of the Lagrangian density implies the Lorentz invariance of the S-matrix. Consider an infinitesimal Lorentz transformation

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \tag{7.4.1}$$

$$\omega_{\mu\nu} = -\omega_{\nu\mu}. \tag{7.4.2}$$

According to the analysis of the previous section, the invariance of the action under such transformations tells us immediately that there are a set of conserved 'currents' $\mathcal{M}^{\rho\mu\nu}$:

$$\partial_\rho \mathcal{M}^{\rho\mu\nu} = 0, \tag{7.4.3}$$

$$\mathcal{M}^{\rho\mu\nu} = -\mathcal{M}^{\rho\nu\mu}, \tag{7.4.4}$$

one current for each independent component of $\omega_{\mu\nu}$. The integrals of the time-components of these 'currents' then provide us with a set of time-independent tensors:

$$J^{\mu\nu} \equiv \int d^3x \, \mathcal{M}^{0\mu\nu}, \tag{7.4.5}$$

$$\frac{d}{dt} J^{\mu\nu} = 0. \tag{7.4.6}$$

The $J^{\mu\nu}$ will turn out to be the generators of the homogeneous Lorentz group.

We would like to have an explicit formula for the tensor $\mathcal{M}^{\rho\mu\nu}$, but Lorentz transformations act on the coordinates and hence cannot leave the Lagrangian density invariant, so we

cannot immediately use the results of the previous section. However, translation invariance allows us to formulate Lorentz invariance as a symmetry of the Lagrangian density under a set of transformations on the fields and field-derivatives alone. The fields undergo the matrix transformation

$$\delta\Psi^\ell = \frac{i}{2}\omega^{\mu\nu} (\mathcal{J}_{\mu\nu})^\ell{}_m \Psi^m, \tag{7.4.7}$$

where $\mathcal{J}_{\mu\nu}$ are a set of matrices satisfying the algebra of the homogeneous Lorentz group

$$[\mathcal{J}_{\mu\nu}, \mathcal{J}_{\rho\sigma}] = i\mathcal{J}_{\sigma\mu}\eta_{\nu\rho} - i\mathcal{J}_{\sigma\nu}\eta_{\mu\rho} - i\mathcal{J}_{\rho\mu}\eta_{\nu\sigma} + i\mathcal{J}_{\rho\nu}\eta_{\mu\sigma}. \tag{7.4.8}$$

For example, for a scalar field ϕ we have $\delta\phi = 0$, so $\mathcal{J}_{\mu\nu} = 0$, while for an irreducible field of type (A, B) we have

$$\mathcal{J}_{ij} = \epsilon_{ijk}(\mathcal{A}_k + \mathcal{B}_k), \qquad \mathcal{J}_{k0} = -i(\mathcal{A}_k - \mathcal{B}_k),$$

where $\mathcal{A}$ and $\mathcal{B}$ are spin matrices for spin A and B, respectively. We specially note that for a covariant vector field, $\delta V_\kappa = \omega_\kappa{}^\lambda V_\lambda$, so here

$$(\mathcal{J}_{\mu\nu})_\kappa{}^\lambda = -i\eta_{\mu\kappa}\delta_\nu{}^\lambda + i\eta_{\nu\kappa}\delta_\mu{}^\lambda.$$

The derivative of a field that transforms as in Eq. (7.4.7) transforms like another such field, but with an extra vector index

$$\delta(\partial_\kappa \Psi^\ell) = \frac{i}{2}\omega^{\mu\nu}(\mathcal{J}_{\mu\nu})^\ell{}_m \partial_\kappa \Psi^m + \omega_\kappa{}^\lambda \partial_\lambda \Psi^\ell. \tag{7.4.9}$$

The Lagrangian density is assumed to be invariant under the combined transformations Eqs. (7.4.7) and (7.4.9), so

$$0 = \frac{\partial \mathcal{L}}{\partial \Psi^\ell} \frac{i}{2}\omega^{\mu\nu}(\mathcal{J}_{\mu\nu})^\ell{}_m \Psi^m + \frac{\partial \mathcal{L}}{\partial(\partial_\kappa \Psi^\ell)} \frac{i}{2}\omega^{\mu\nu}(\mathcal{J}_{\mu\nu})^\ell{}_m \partial_\kappa \Psi^m + \frac{\partial \mathcal{L}}{\partial(\partial_\kappa \Psi^\ell)} \omega_\kappa{}^\lambda \partial_\lambda \Psi^\ell.$$

Setting the coefficient of $\omega^{\mu\nu}$ equal to zero gives

$$0 = \frac{i}{2}\frac{\partial \mathcal{L}}{\partial \Psi^\ell}(\mathcal{J}_{\mu\nu})^\ell{}_m \Psi^m + \frac{i}{2}\frac{\partial \mathcal{L}}{\partial(\partial_\kappa \Psi^\ell)}(\mathcal{J}_{\mu\nu})^\ell{}_m \partial_\kappa \Psi^m + \frac{1}{2}\frac{\partial \mathcal{L}}{\partial(\partial_\kappa \Psi^\ell)}(\eta_{\mu\kappa}\partial_\nu - \eta_{\nu\kappa}\partial_\mu)\Psi^\ell.$$

Using the Euler–Lagrange equations (7.2.9), and our formula (7.3.34) for the energy–momentum tensor $T_{\mu\nu}$, we may write this as

$$0 = \partial_\kappa \left[\frac{i}{2}\frac{\partial \mathcal{L}}{\partial(\partial_\kappa \Psi^\ell)}(\mathcal{J}_{\mu\nu})^\ell{}_m \Psi^m \right] - \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu}). \tag{7.4.10}$$

This immediately suggests the definition of a new energy–momentum tensor, known as the *Belinfante tensor* [2]:

$$\Theta^{\mu\nu} = T^{\mu\nu} - \frac{i}{2}\partial_\kappa \left[\frac{\partial \mathcal{L}}{\partial(\partial_\kappa \Psi^\ell)}(\mathcal{J}^{\mu\nu})^\ell{}_m \Psi^m - \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Psi^\ell)}(\mathcal{J}^{\kappa\nu})^\ell{}_m \Psi^m - \frac{\partial \mathcal{L}}{\partial(\partial_\nu \Psi^\ell)}(\mathcal{J}^{\kappa\mu})^\ell{}_m \Psi^m \right]. \tag{7.4.11}$$

The quantity in square brackets is manifestly antisymmetric in μ and κ, so $\Theta^{\mu\nu}$ satisfies the same conservation law as $T^{\mu\nu}$,

$$\partial_\mu \Theta^{\mu\nu} = 0. \tag{7.4.12}$$

For the same reason, when we set $\mu = 0$ in Eq. (7.4.11) the index κ runs over space components only, so the derivative term here drops out when we integrate over all space

$$\int \Theta^{0\nu} d^3x = \int T^{0\nu} d^3x = P^\nu, \tag{7.4.13}$$

where $P^0 \equiv H$. Thus $\Theta^{\mu\nu}$ can be regarded as *the* energy–momentum tensor, just as well as $T^{\mu\nu}$. However, Eq. (7.4.10) tells us that, unlike $T^{\mu\nu}$, the Belinfante tensor $\Theta^{\mu\nu}$ is not only conserved but also *symmetric*:

$$\Theta^{\mu\nu} = \Theta^{\nu\mu}. \tag{7.4.14}$$

It is $\Theta^{\mu\nu}$ rather than $T^{\mu\nu}$ that acts as a source of the gravitational field [3]. In consequence of the symmetry of $\Theta^{\mu\nu}$, we may construct one more conserved tensor density:

$$\mathcal{M}^{\lambda\mu\nu} \equiv x^\mu \Theta^{\lambda\nu} - x^\nu \Theta^{\lambda\mu}. \tag{7.4.15}$$

This is conserved, in the sense that

$$\partial_\lambda \mathcal{M}^{\lambda\mu\nu} = \Theta^{\mu\nu} - \Theta^{\nu\mu} = 0. \tag{7.4.16}$$

Thus Lorentz invariance allows us to define one more time-independent tensor

$$J^{\mu\nu} = \int \mathcal{M}^{0\mu\nu} d^3x = \int d^3x (x^\mu \Theta^{0\nu} - x^\nu \Theta^{0\mu}). \tag{7.4.17}$$

The rotation generator $J_k = \epsilon_{ijk} J^{ij}/2$ is not only time-independent, but also has no explicit time-dependence, so it commutes with the Hamiltonian

$$[H, \mathbf{J}] = 0. \tag{7.4.18}$$

Also, applying Eq. (7.3.28) to the function $\Theta^{0\nu}$, we have

$$[P_j, J_i] = \frac{1}{2}\epsilon_{ikl}[P_j, J^{kl}] = \frac{i}{2}\epsilon_{ikl} \int d^3x \left(x^k \frac{\partial}{\partial x^j}\Theta^{0l} - x^l \frac{\partial}{\partial x^j}\Theta^{0k} \right) = -i\epsilon_{ijk} \int d^3x \, \Theta^{0k}$$

and therefore

$$[P_j, J_i] = -i\epsilon_{ijk} P_k. \tag{7.4.19}$$

On the other hand, the 'boost' generator $K_k \equiv J^{k0}$, though time-independent, does explicitly involve the time coordinate

$$K_k = \int d^3x (x^k \Theta^{00} - x^0 \Theta^{0k}),$$

or more explicitly

$$\mathbf{K} = -t\mathbf{P} + \int d^3x \, \mathbf{x}\Theta^{00}(\mathbf{x}, t). \tag{7.4.20}$$

Since this is a constant, we have $0 = \dot{\mathbf{K}} = -\mathbf{P} + i[H, \mathbf{K}]$, and therefore

$$[H, \mathbf{K}] = -i\mathbf{P}. \tag{7.4.21}$$

Also, applying Eq. (7.3.28) again gives

$$[P_j, K_k] = i \int d^3x \, x^k \frac{\partial}{\partial x^j}\Theta^{00} = -i\delta_{jk} \int d^3x \, \Theta^{00}$$

and therefore

$$[P_j, K_k] = -i\delta_{jk} H. \tag{7.4.22}$$

For any reasonable Lagrangian density, the operator (7.4.20) will be 'smooth' in the sense used in Section 3.3, i.e., the interaction terms in $e^{iH_0 t} \int d^3x \, \mathbf{x}\Theta^{00}(\mathbf{x}, 0) e^{-iH_0 t}$ vanish[67] for $t \to \pm\infty$. (Note that the interaction terms in $e^{iH_0 t} \int d^3x \, \Theta^{00}(\mathbf{x}, 0) e^{-iH_0 t}$ must vanish for $t \to \pm\infty$ in order to allow the introduction of 'in' and 'out' states and the S-matrix.) With this smoothness assumption and the commutation relation (7.4.21) in hand, we can repeat the arguments of Section 3.3, and conclude that the S-matrix is Lorentz-invariant.

* * *

The same arguments were also used in Section 3.3 to verify that the remaining commutation relations of the Lorentz group, those of the $J^{\mu\nu}$ with each other, take the proper form. This can also be shown directly for the commutators of the rotation generators, which here take the form

$$J^{ij} \equiv \int d^3x \frac{\partial \mathcal{L}}{\partial \dot{\Psi}^\ell} \left(x^i \partial_j \Psi^\ell - x^j \partial_i \Psi^\ell - i(\mathcal{J}^{ij})^\ell{}_m \Psi^m \right). \tag{7.4.23}$$

Since the Lagrangian density does not depend on the time-derivatives of the auxiliary fields, and the rotation generators do not mix canonical and auxiliary fields, this can also be written as a sum over canonical fields alone:

$$J^{ij} = \int d^3x \, P_n \left(x^i \partial_j Q^n - x^j \partial_i Q^n - i(\mathcal{J}^{ij})^n{}_{n'} Q^{n'} \right). \tag{7.4.24}$$

It follows immediately then from the canonical commutation relations that

$$[J^{ij}, Q^n(x)] = -i(x_i \partial_j - x_j \partial_i) Q^n(x) - (\mathcal{J}^{ij})^n{}_{n'} Q^{n'}(x), \tag{7.4.25}$$

$$[J^{ij}, P_n(x)] = i(x_i \partial_j - x_j \partial_i) P^n(x) + (\mathcal{J}^{ij})^{n'}{}_n P_{n'}(x). \tag{7.4.26}$$

These results can be used to derive the usual commutation relations of the J^{ij} with each other and other generators.[68] If there are no auxiliary fields then the same arguments may

[67] When we say that some interaction-picture operator vanishes for $t \to \pm\infty$, we mean that its matrix elements between states that are smooth superpositions of energy eigenstates vanish in this limit.

[68] Also, since J^{ij} commutes with H and $P_n \dot{Q}^n$, it commutes with L. The commutator of J^{ij} with the auxiliary fields must thus be consistent with the rotational invariance of L.

be applied to the 'boost' generators to complete the demonstration that the P^μ and $J^{\mu\nu}$ satisfy the commutation relations of the inhomogeneous Lorentz group. However the 'boost' matrices $\mathcal{J}^{i0}$ will in general mix canonical and auxiliary fields (such as the components V^i and V^0 of a vector field), so the direct proof of the commutation relations of the J^{i0} with each other has to be given on a case by case basis. Fortunately, this is not needed for the proof of the Lorentz invariance of the S-matrix given in Section 3.3.

7.5 Transition to Interaction Picture: Examples

At the end of Section 7.2 we showed how to use the Lagrangian of a simple scalar field theory to derive the structure of the interaction and the free fields it contains in the interaction picture. We will now turn to somewhat more complicated and revealing examples.

Scalar Field, Derivative Coupling

First let's consider a neutral scalar field, but now with derivative coupling. We take the Lagrangian as

$$\mathcal{L} = -\frac{1}{2}\partial_\mu \Phi \partial^\mu \Phi - \frac{1}{2}m^2 \Phi^2 - J^\mu \partial_\mu \Phi - \mathcal{H}_I(\Phi), \tag{7.5.1}$$

where J^μ is either a c-number external current (unrelated to currents J^μ introduced earlier), or a functional of various fields other than Φ (in which case terms involving these other fields need to be added to (7.5.1)). The canonical conjugate to Φ is now

$$\Pi = \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = \dot{\Phi} - J^0 \tag{7.5.2}$$

and the Hamiltonian is

$$H = \int d^3x \, [\Pi \dot{\Phi} - \mathcal{L}] = \int d^3x \left[\Pi(\Pi + J^0) + \frac{1}{2}(\nabla\Phi)^2 - \frac{1}{2}(\Pi + J^0)^2 + \frac{1}{2}m^2\Phi^2 + \mathbf{J} \cdot \nabla\Phi + J^0(\Pi + J^0) + \mathcal{H}_I(\Phi) \right].$$

Collecting terms, we can write this as

$$H = H_0 + H_{\text{int}}, \tag{7.5.3}$$

$$H_0 = \int d^3x \left[\frac{1}{2}\Pi^2 + \frac{1}{2}(\nabla\Phi)^2 + \frac{1}{2}m^2\Phi^2 \right], \tag{7.5.4}$$

$$H_{\text{int}} = \int d^3x \left[\Pi J^0 + \mathbf{J} \cdot \nabla\Phi + \frac{1}{2}(J^0)^2 + \mathcal{H}_I(\Phi) \right]. \tag{7.5.5}$$

As explained in Section 7.2, we can pass to the interaction picture by simply replacing Π and Φ with π and ϕ (and likewise for any fields in the current J^μ, though we will not bother to indicate this explicitly):

$$H_0 = \int d^3x \left[\frac{1}{2}\pi^2(\mathbf{x}, t) + \frac{1}{2}(\nabla\phi(\mathbf{x}, t))^2 + \frac{1}{2}m^2\phi^2(\mathbf{x}, t) \right], \tag{7.5.6}$$

$$H_I(t) = \int d^3x \Big[\pi(\mathbf{x}, t) J^0(\mathbf{x}, t) + \mathbf{J}(\mathbf{x}, t) \cdot \nabla\phi(\mathbf{x}, t) + \frac{1}{2}\left[J^0(\mathbf{x}, t)\right]^2 + \mathcal{H}_I(\phi(\mathbf{x}, t)) \Big]. \tag{7.5.7}$$

The free-particle Hamiltonian is just the same as Eq. (7.2.25), and leads as in Section 7.2 to Eqs. (7.2.26)–(7.2.35). Indeed, whatever the total Hamiltonian may be, we *must* take (7.5.6) as the part we split off and call the free-particle part, with the remainder called the interaction, because as we have seen it is this form of the free-particle Hamiltonian that leads to the correct expansion (7.2.29) of the scalar field in terms of creation and annihilation operators that satisfy the commutation relations (7.2.34), (7.2.35). The last step is to replace π in the interaction Hamiltonian with its value $\dot{\phi}$ in the interaction picture (*not* its value $\dot{\Phi} - J^0$ in the Heisenberg picture):

$$H_I(t) = \int d^3x \left[J^\mu(\mathbf{x}, t)\partial_\mu \phi(\mathbf{x}, t) + \frac{1}{2}\left[J^0(\mathbf{x}, t)\right]^2 + \mathcal{H}_I(\phi(\mathbf{x}, t)) \right]. \tag{7.5.8}$$

The extra non-invariant term in Eq. (7.5.8) is just what we saw in Section 6.2 is needed to cancel a non-invariant term in the propagator of $\partial_\mu \phi$.

Vector Field, Spin One

Similar results are obtained in the canonical quantization of the vector field V_μ for a particle of spin one. Let's here keep an open mind, and write the Lagrangian density in a fairly general form

$$\mathcal{L} = -\frac{1}{2}\alpha \, \partial_\mu V_\nu \partial^\mu V^\nu - \frac{1}{2}\beta \, \partial_\mu V_\nu \partial^\nu V^\mu - \frac{1}{2}m^2 V_\mu V^\mu - J_\mu V^\mu, \tag{7.5.9}$$

where α, β, and m^2 are so far arbitrary constants, and J_μ is either a c-number external current, or an operator depending on fields other than V^μ, in which case additional terms involving these fields must be added to $\mathcal{L}$. The Euler–Lagrange field equations for V_μ read

$$\alpha \Box V^\nu + \beta \partial^\nu(\partial_\mu V^\mu) + m^2 V^\nu = -J^\nu.$$

Taking the divergence gives

$$(\alpha + \beta)\Box \partial_\lambda V^\lambda + m^2 \partial_\lambda V^\lambda = -\partial_\lambda J^\lambda. \tag{7.5.10}$$

This is the equation for an ordinary scalar field with mass $m^2/(\alpha + \beta)$ and source $\partial_\lambda J^\lambda/(\alpha + \beta)$. We want to describe a theory containing only particles of spin one, not spin zero, so to avoid the appearance of $\partial_\lambda V^\lambda$ as an independently propagating scalar field, we take $\alpha = -\beta$, in which case $\partial_\lambda V^\lambda$ can be expressed in terms of an external current or other fields, as $-\partial_\lambda J^\lambda/m^2$. The constant α can be absorbed in the definition of V_μ, so we can take $\alpha = -\beta = 1$, and therefore

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}m^2 V_\mu V^\mu - J_\mu V^\mu, \tag{7.5.11}$$

where

$$F_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu. \tag{7.5.12}$$

The derivative of the Lagrangian with respect to the time-derivative of the vector field is

$$\frac{\partial \mathcal{L}}{\partial \dot{V}^\mu} = -F^{0\mu}. \tag{7.5.13}$$

This is non-vanishing for μ a spatial index i, so the V^i are canonical fields, with conjugates

$$\Pi^i = F^{i0} = \dot{V}^i + \partial_i V^0. \tag{7.5.14}$$

On the other hand $F^{00} = 0$, so $\dot{V}^0$ does not appear in the Lagrangian, and V^0 is therefore an auxiliary field. This causes no serious difficulty: the fact that $\partial\mathcal{L}/\partial\dot{V}^0$ vanishes means that the field equation for V^0 involves no second time-derivatives, and can therefore be used as a constraint that eliminates a field variable. Specifically, the Euler–Lagrange equation for $\nu = 0$ is

$$\partial_i F^{i0} = m^2 V^0 + J^0 \tag{7.5.15}$$

or using Eq. (7.5.14)

$$V^0 = \frac{1}{m^2}(\nabla \cdot \mathbf{\Pi} - J^0). \tag{7.5.16}$$

Now let us calculate the Hamiltonian $H = \int d^3x (\mathbf{\Pi} \cdot \dot{\mathbf{V}} - \mathcal{L})$ for this theory. Eq. (7.5.14) allows us to write $\dot{\mathbf{V}}$ in terms of $\mathbf{\Pi}$ and J^0:

$$\dot{\mathbf{V}} = -\nabla V^0 + \mathbf{\Pi} = \mathbf{\Pi} - \frac{1}{m^2}\nabla(\nabla \cdot \mathbf{\Pi} - J^0),$$

so

$$H = \int d^3x \Big[\mathbf{\Pi}^2 + m^{-2}(\nabla \cdot \mathbf{\Pi})(\nabla \cdot \mathbf{\Pi} - J^0) - \frac{1}{2}\mathbf{\Pi}^2 + \frac{1}{2}(\nabla \times \mathbf{V})^2 + \frac{1}{2}m^2\mathbf{V}^2 - \frac{1}{2}m^{-2}(\nabla \cdot \mathbf{\Pi} - J^0)^2 + \mathbf{J} \cdot \mathbf{V} - m^{-2}J^0(\nabla \cdot \mathbf{\Pi} - J^0) \Big].$$

Again, we split this up into a free-particle term H_0 and interaction H_{int}:

$$H = H_0 + H_{\text{int}}, \tag{7.5.17}$$

and pass to the interaction picture by replacing the Heisenberg-picture quantities $\mathbf{V}$ and $\mathbf{\Pi}$ with their interaction-picture counterparts $\mathbf{v}$ and $\boldsymbol{\pi}$ (and, though not shown explicitly, likewise for whatever fields and conjugates are present in J^μ):

$$H_0 = \int d^3x \left[\frac{1}{2}\boldsymbol{\pi}^2 + \frac{1}{2m^2}(\nabla \cdot \boldsymbol{\pi})^2 + \frac{1}{2}(\nabla \times \mathbf{v})^2 + \frac{m^2}{2}\mathbf{v}^2 \right], \tag{7.5.18}$$

$$H_I(t) = \int d^3x \left[\mathbf{J} \cdot \mathbf{v} - m^{-2}J^0 \nabla \cdot \boldsymbol{\pi} + \frac{1}{2m^2}(J^0)^2 \right]. \tag{7.5.19}$$

The relation between $\boldsymbol{\pi}$ and $\mathbf{v}$ is then

$$\dot{\mathbf{v}} = \frac{\delta H_0(\mathbf{v}, \boldsymbol{\pi})}{\delta \boldsymbol{\pi}} = \boldsymbol{\pi} - m^{-2}\nabla(\nabla \cdot \boldsymbol{\pi}) \tag{7.5.20}$$

and the 'field equation' is

$$\dot{\boldsymbol{\pi}} = -\frac{\delta H_0(\mathbf{v}, \boldsymbol{\pi})}{\delta \mathbf{v}} = +\nabla^2 \mathbf{v} - \nabla(\nabla \cdot \mathbf{v}) - m^2 \mathbf{v}. \tag{7.5.21}$$

Since V^0 is not an independent field variable, it is not related by a similarity transformation to any interaction-picture object v^0. Instead, we can *invent* a quantity

$$v^0 \equiv m^{-2}\nabla \cdot \boldsymbol{\pi}. \tag{7.5.22}$$

Eq. (7.5.20) then allows us to write $\boldsymbol{\pi}$ as

$$\boldsymbol{\pi} = \dot{\mathbf{v}} + \nabla v^0. \tag{7.5.23}$$

Inserting this in Eqs. (7.5.22) and (7.5.21) gives our field equations in the form

$$\nabla^2 v^0 + \nabla \cdot \dot{\mathbf{v}} - m^2 v^0 = 0,$$

$$\nabla^2 \mathbf{v} - \nabla(\nabla \cdot \mathbf{v}) - \ddot{\mathbf{v}} - \nabla \dot{v}^0 - m^2 \mathbf{v} = 0.$$

These can be combined in the covariant form

$$\Box v^\mu - \partial^\mu \partial_\nu v^\nu - m^2 v^\mu = 0. \tag{7.5.24}$$

Taking the divergence gives

$$\partial_\mu v^\mu = 0 \tag{7.5.25}$$

and hence

$$(\Box - m^2)v^\mu = 0. \tag{7.5.26}$$

A real vector field satisfying Eqs. (7.5.25) and (7.5.26) can be expressed as a Fourier transform

$$v^\mu(x) = (2\pi)^{-3/2} \sum_\sigma \int d^3p \, (2p^0)^{-1/2} \left\{ e^\mu(\mathbf{p}, \sigma) a_{\mathbf{p},\sigma} e^{ip\cdot x} + e^{\mu*}(\mathbf{p}, \sigma) a^\dagger_{\mathbf{p},\sigma} e^{-ip\cdot x} \right\}, \tag{7.5.27}$$

where $p^0 = \sqrt{\mathbf{p}^2 + m^2}$; the $e^\mu(\mathbf{p}, \sigma)$ for $\sigma = +1, 0, -1$ are three independent vectors satisfying

$$p_\mu e^\mu(\mathbf{p}, \sigma) = 0 \tag{7.5.28}$$

and normalized so that

$$\sum_\sigma e^\mu(\mathbf{p}, \sigma) e^{\nu*}(\mathbf{p}, \sigma) = \eta^{\mu\nu} + \frac{p^\mu p^\nu}{m^2}; \tag{7.5.29}$$

and the $a_{\mathbf{p},\sigma}$ are operator coefficients. It is straightforward using Eqs. (7.5.23), (7.5.27), and (7.5.29) to calculate that $\mathbf{v}$ and $\boldsymbol{\pi}$ satisfy the correct commutation relations

$$[v^i(\mathbf{x}, t), \pi^j(\mathbf{y}, t)] = i\delta_{ij}\delta^3(\mathbf{x} - \mathbf{y}),$$
$$[v^i(\mathbf{x}, t), v^j(\mathbf{x}, t)] = [\pi^i(\mathbf{x}, t), \pi^j(\mathbf{x}, t)] = 0, \tag{7.5.30}$$

provided that $a_{\mathbf{p},\sigma}$ and $a^\dagger_{\mathbf{p},\sigma}$ satisfy the commutation relations

$$[a_{\mathbf{p},\sigma}, a^\dagger_{\mathbf{p}',\sigma'}] = \delta^3(\mathbf{p}' - \mathbf{p})\delta_{\sigma'\sigma}, \tag{7.5.31}$$

$$[a_{\mathbf{p},\sigma}, a_{\mathbf{p}',\sigma'}] = 0. \tag{7.5.32}$$

We already know that the vector field for a spin one particle must take the form (7.5.27), so our derivation of these results serves to verify that Eq. (7.5.18) gives the correct free-particle Hamiltonian for a massive particle of spin one. It is easy to check also that Eq. (7.5.18) may be written (up to a constant term) in the standard form of a free-particle energy, as $\sum_\sigma \int d^3p \, p^0 a^\dagger_{\mathbf{p},\sigma} a_{\mathbf{p},\sigma}$. Finally, using Eq. (7.5.22) in Eq. (7.5.19) yields the interaction in the interaction picture

$$H_I(t) = \int d^3x \left[J_\mu v^\mu + \frac{1}{2m^2}(J^0)^2 \right]. \tag{7.5.33}$$

The extra non-invariant term in Eq. (7.5.33) is just what we found in Chapter 6 is needed to cancel a non-invariant term in the propagator of the vector field.

Dirac field, Spin One Half

For the Dirac field of a particle of spin 1/2, we tentatively take the Lagrangian as

$$\mathcal{L} = -\bar{\Psi}(\gamma^\mu \partial_\mu + m)\Psi - \mathcal{H}_I(\bar{\Psi}, \Psi) \tag{7.5.34}$$

with $\mathcal{H}_I$ a real function of $\bar{\Psi}$ and Ψ. This is not real, but the action is, because

$$\bar{\Psi}\gamma^\mu \partial_\mu \Psi - (\bar{\Psi}\gamma^\mu \partial_\mu \Psi)^\dagger = \bar{\Psi}\gamma^\mu \partial_\mu \Psi + (\partial_\mu \bar{\Psi})\gamma^\mu \Psi = \partial_\mu(\bar{\Psi}\gamma^\mu \Psi).$$

Hence the field equations obtained by requiring the action to be stationary with respect to $\bar{\Psi}$ are the adjoints of those obtained by requiring the action to be stationary with respect to Ψ, as necessary if we are to avoid having too many field equations. The canonical conjugate to Ψ is

$$\Pi = \frac{\partial \mathcal{L}}{\partial \dot{\Psi}} = -\bar{\Psi}\gamma^0, \tag{7.5.35}$$

so we should not regard $\bar{\Psi}$ as a field like Ψ, but rather as proportional to the canonical conjugate of Ψ. The Hamiltonian is

$$H = \int d^3x \, [\Pi \dot{\Psi} - \mathcal{L}] = \int d^3x \, [\Pi \gamma^0(\boldsymbol{\gamma} \cdot \nabla + m)\Psi + \mathcal{H}_I].$$

We write this as

$$H = H_0 + H_{\text{int}}, \tag{7.5.36}$$

where

$$H_0 = \int d^3x \, \Pi \gamma^0(\boldsymbol{\gamma} \cdot \nabla + m)\Psi, \tag{7.5.37}$$

$$H_{\text{int}} = \int d^3x \, \mathcal{H}_I(\bar{\Psi}, \Psi). \tag{7.5.38}$$

We now pass to the interaction picture. Since Eq. (7.5.35) does not involve the time, the similarity transformation (7.1.28), (7.1.29) yields immediately

$$\pi = -\bar{\psi}\gamma^0. \tag{7.5.39}$$

Likewise, H_0 and $H_I(t)$ can be calculated by replacing Ψ and Π with ψ and π in Eqs. (7.5.37) and (7.5.38). This gives the equation of motion

$$\dot{\psi} = \frac{\delta H_0}{\delta \pi} = \gamma^0(\boldsymbol{\gamma} \cdot \nabla + m)\psi \tag{7.5.40}$$

or more neatly

$$(\gamma^\mu \partial_\mu + m)\psi = 0. \tag{7.5.41}$$

(The other equation of motion, $\dot{\pi} = -\delta H_0/\delta\psi$, yields just the adjoint of this one.) Any field satisfying Eq. (7.5.41) can be written as a Fourier transform

$$\psi(x) = (2\pi)^{-3/2} \int d^3p \sum_\sigma \left\{ u(\mathbf{p}, \sigma) e^{ip\cdot x} a_{\mathbf{p},\sigma} + v(\mathbf{p}, \sigma) e^{-ip\cdot x} b^\dagger_{\mathbf{p},\sigma} \right\}, \tag{7.5.42}$$

where $p^0 = \sqrt{\mathbf{p}^2 + m^2}$; $a_{\mathbf{p},\sigma}$ and $b^\dagger_{\mathbf{p},\sigma}$ are operator coefficients; and $u(\mathbf{p}, \pm\frac{1}{2})$ are the two independent solutions of

$$(i\gamma^\mu p_\mu + m)u(\mathbf{p}, \sigma) = 0 \tag{7.5.43}$$

and likewise

$$(-i\gamma^\mu p_\mu + m)v(\mathbf{p}, \sigma) = 0 \tag{7.5.44}$$

normalized so that[69]

$$\sum_\sigma u(\mathbf{p}, \sigma)\bar{u}(\mathbf{p}, \sigma) = \frac{-i\gamma^\mu p_\mu + m}{2p^0}, \tag{7.5.45}$$

$$\sum_\sigma v(\mathbf{p}, \sigma)\bar{v}(\mathbf{p}, \sigma) = -\frac{i\gamma^\mu p_\mu + m}{2p^0}. \tag{7.5.46}$$

In order to obtain the desired anticommutators

$$\{\psi_\alpha(\mathbf{x}, t), \bar{\psi}_\beta(\mathbf{y}, t)\} = \{\psi_\alpha(\mathbf{x}, t), \pi_\gamma(\mathbf{y}, t)\}(\gamma^0)_{\gamma\beta} = i(\gamma^0)_{\alpha\beta}\delta^3(\mathbf{x} - \mathbf{y}), \tag{7.5.47}$$

$$\{\psi_\alpha(\mathbf{x}, t), \psi_\beta(\mathbf{y}, t)\} = 0, \tag{7.5.48}$$

we must adopt the anticommutation relations

$$\{a_{\mathbf{p},\sigma}, a^\dagger_{\mathbf{p}',\sigma'}\} = \{b_{\mathbf{p},\sigma}, b^\dagger_{\mathbf{p}',\sigma'}\} = \delta^3(\mathbf{p}' - \mathbf{p})\delta_{\sigma'\sigma}, \tag{7.5.49}$$

$$\{a_{\mathbf{p},\sigma}, a_{\mathbf{p}',\sigma'}\} = \{b_{\mathbf{p},\sigma}, b_{\mathbf{p}',\sigma'}\} = \{a_{\mathbf{p},\sigma}, b_{\mathbf{p}',\sigma'}\} = \{a_{\mathbf{p},\sigma}, b^\dagger_{\mathbf{p}',\sigma'}\} = 0, \tag{7.5.50}$$

[69] The matrix $i\gamma^\mu p_\mu$ has eigenvalues $\pm m$, so $\sum u\bar{u}$ and $\sum v\bar{v}$ must be proportional to the projection matrices $(-i\gamma^\mu p_\mu + m)/2m$ and $(i\gamma^\mu p_\mu + m)/2m$, respectively. The proportionality factor may be adjusted up to a sign by absorbing it in the definition of u and v. The overall sign is determined by positivity; $\text{Tr} \sum u\bar{u}\beta = \sum u^\dagger u$ and $\text{Tr} \sum v\bar{v}\beta = \sum v^\dagger v$ must be positive.

and their adjoints. These agree with the results obtained in Chapter 5, thus verifying that (7.5.37) is the correct free-particle Hamiltonian for spin 1/2. In terms of the as and bs, this Hamiltonian is

$$H_0 = \sum_\sigma \int d^3p \, p^0 \left(a^\dagger_{\mathbf{p},\sigma} a_{\mathbf{p},\sigma} - b_{\mathbf{p},\sigma} b^\dagger_{\mathbf{p},\sigma} \right). \tag{7.5.51}$$

We can rewrite this as a more conventional free-particle Hamiltonian, plus another infinite c-number[70]

$$H_0 = \sum_\sigma \int d^3p \, p^0 \left[a^\dagger_{\mathbf{p},\sigma} a_{\mathbf{p},\sigma} + b^\dagger_{\mathbf{p},\sigma} b_{\mathbf{p},\sigma} - \delta^3(\mathbf{p} - \mathbf{p}) \right]. \tag{7.5.52}$$

The c-number term in Eq. (7.5.52) is only important if we worry about gravitational phenomena; otherwise here, as for the scalar field, we can throw it away, since it only affects the zero of energy with respect to which all energies are measured. With this understanding, H_0 is a positive operator, just as for bosons.

7.6 Constraints and Dirac Brackets

The chief obstacle to deriving the Hamiltonian from the Lagrangian is the occurrence of constraints. The standard analysis of this problem is that of Dirac [5], whose terminology we will follow here. Dirac's analysis is not really needed for the simple theories discussed in this chapter, where it is easy to identify the unconstrained canonical variables. We shall use the theory of a real massive vector field for illustration here, returning to Dirac's approach in the next chapter, where it will be actually useful.

Primary constraints are either imposed on the system (as when in the next chapter we choose a gauge for the electromagnetic field) or arise from the structure of the Lagrangian itself. For an example of the latter type, consider the Lagrangian (7.5.11) of a massive vector field V^μ interacting with a current J_μ:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}m^2 V_\mu V^\mu - J_\mu V^\mu \tag{7.6.1}$$

where

$$F_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu. \tag{7.6.2}$$

Suppose we try to treat all four components of V^μ on the same basis. We should then define the conjugates

$$\Pi_\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 V^\mu)} = -F^{0\mu}. \tag{7.6.3}$$

We immediately find the primary constraint:

$$\Pi_0 = 0. \tag{7.6.4}$$

More generally, we encounter primary constraints whenever the equations $\Pi_\ell = \partial L/\partial \dot{\Psi}^\ell$ cannot be solved to give all the $\partial_0 \Psi^\ell$ (at least locally) in terms of Π_ℓ and Ψ^ℓ. This will be the case if and only if the matrix $\partial^2 L/\partial(\partial_0 \Psi^\ell)\partial(\partial_0 \Psi^m)$ has vanishing determinant. Such Lagrangians are called *irregular*.

[70] Note the negative sign of the c-number term. The conjectured symmetry known as supersymmetry [4] connects the numbers of boson and fermion fields, in such a way that the c-numbers in H_0 all cancel.

Then there are *secondary constraints*, which arise from the requirement that the primary constraints be consistent with the equations of motion. For the massive vector field, this is just the Euler–Lagrange equation (7.5.16) for V^0:

$$\partial_i \Pi_i = m^2 V^0 + J^0. \tag{7.6.5}$$

Here we are finished, but in other theories we might encounter further constraints by requiring consistency of the secondary constraints with the field equations, and so on. The distinction between primary, secondary, etc. constraints is not important; we will treat them all together here.

There is another distinction between certain types of constraint that is more important. The constraints we have found for the massive vector field are of a type known as *second class*, for which there is a universal prescription for the commutation relations. To explain the distinction between first and second class constraints, and the prescription used to deal with second class constraints, it is useful first to recall the definition of the Poisson brackets of classical mechanics.

Consider any Lagrangian $L(\Psi, \dot{\Psi})$ that depends on a set of variables $\Psi^a(t)$ and their time-derivatives $\dot{\Psi}^a(t)$. (The Lagrangians of quantum field theory are a special case, with the index a running over all pairs of ℓ and $\mathbf{x}$.) We can define canonical conjugates for *all* of these variables by

$$\Pi_a \equiv \frac{\partial L}{\partial \dot{\Psi}^a}. \tag{7.6.6}$$

The Πs and Ψs will in general not be independent variables, but may instead be related by various constraint equations, both primary and secondary. The Poisson bracket is then defined by

$$[A, B]_P \equiv \frac{\partial A}{\partial \Psi^a}\frac{\partial B}{\partial \Pi_a} - \frac{\partial B}{\partial \Psi^a}\frac{\partial A}{\partial \Pi_a} \tag{7.6.7}$$

with the constraints ignored in calculating the derivatives with respect to Ψ^a and Π_a. In particular, we always have $[\Psi^a, \Pi_b]_P = \delta^a_b$. (Here and below all fields are taken at the same time, and time arguments are everywhere dropped.) These brackets have the same algebraic properties as commutators:

$$[A, B]_P = -[B, A]_P, \tag{7.6.8}$$

$$[A, BC]_P = [A, B]_P C + B[A, C]_P, \tag{7.6.9}$$

including the Jacobi identity

$$[A, [B, C]_P]_P + [B, [C, A]_P]_P + [C, [A, B]_P]_P = 0. \tag{7.6.10}$$

If we could adopt the usual commutation relations $[\Psi^a, \Pi_b] = i\delta^a_b$, $[\Psi^a, \Psi^b] = [\Pi_a, \Pi_b] = 0$, then the commutator of any two functions of the Ψs and Πs would be just $[A, B] = i[A, B]_P$. But the constraints do not always allow this.

The constraints may in general be expressed in the form $\chi_N = 0$, where the χ_N are a set of functions of the Ψs and Πs. Because we are including secondary constraints along

with the primary constraints, the set of all the constraints is necessarily consistent with the equations of motion $\dot{A} = [A, H]_P$, and therefore

$$[\chi_N, H]_P = 0 \tag{7.6.11}$$

when the constraint equations $\chi_N = 0$ are satisfied.

We call a constraint *first class* if its Poisson bracket with all the other constraints vanishes when (*after* calculating the Poisson brackets) we impose the constraints. We shall see a simple example of such a constraint in the quantization of the electromagnetic field in the next chapter, where the first class constraint arises from a symmetry of the action, electromagnetic gauge invariance. In fact, the set of first class constraints $\chi_N = 0$ is always associated with a group of symmetries, under which an arbitrary quantity A undergoes the infinitesimal transformation

$$\delta_N A \equiv \sum_N \epsilon_N [\chi_N, A]_P. \tag{7.6.12}$$

(In field theory these are local transformations, because the index N contains a spacetime coordinate.) Eq. (7.6.11) shows that this transformation leaves the Hamiltonian invariant, and for first class constraints it also respects all other constraints. Such first class constraints can be eliminated by a choice of gauge, or treated by gauge-invariant methods described in Volume II.

After all of the first class constraints have been eliminated, the remaining constraint equations $\chi_N = 0$ are such that no linear combination $\sum_N u_N [\chi_N, \chi_M]_P$ of the Poisson brackets of these constraints with each other vanishes. It follows that the matrix of the Poisson brackets of the remaining constraints is non-singular:

$$\text{Det}\, C \neq 0, \tag{7.6.13}$$

where

$$C_{NM} \equiv [\chi_N, \chi_M]_P. \tag{7.6.14}$$

Constraints of this sort are called *second class*. Note that there must always be an even number of second class constraints, because an antisymmetric matrix of odd dimensionality necessarily has vanishing determinant.

As we have seen, in the case of the massive real vector field the constraints are

$$\chi_{1\mathbf{x}} = \chi_{2\mathbf{x}} = 0, \tag{7.6.15}$$

where

$$\chi_{1\mathbf{x}} = \Pi_0(\mathbf{x}), \qquad \chi_{2\mathbf{x}} = \partial_i \Pi_i(\mathbf{x}) - m^2 V^0(\mathbf{x}) - J^0(\mathbf{x}). \tag{7.6.16}$$

The Poisson bracket of these constraints is

$$C_{1\mathbf{x},2\mathbf{y}} = -C_{2\mathbf{y},1\mathbf{x}} = [\chi_{1\mathbf{x}}, \chi_{2\mathbf{y}}]_P = m^2\delta^3(\mathbf{x} - \mathbf{y}) \tag{7.6.17}$$

and, of course,

$$C_{1\mathbf{x},1\mathbf{y}} = C_{2\mathbf{x},2\mathbf{y}} = 0. \tag{7.6.18}$$

This 'matrix' is obviously non-singular, so the constraints (7.6.15) are second class.

Dirac suggested that when all constraints are second class, the commutation relations will be given by

$$[A, B] = i[A, B]_D, \tag{7.6.19}$$

where $[A, B]_D$ is a generalization of the Poisson bracket known as the *Dirac bracket*:

$$[A, B]_D \equiv [A, B]_P - [A, \chi_N]_P (C^{-1})^{NM} [\chi_M, B]_P. \tag{7.6.20}$$

(Here N and M are compound indices including the position in space, taking values like $1, \mathbf{x}$ and $2, \mathbf{x}$ in the vector field example.) He noted that the Dirac bracket like the Poisson bracket satisfies the same algebraic relations as the commutators

$$[A, B]_D = -[B, A]_D, \tag{7.6.21}$$

$$[A, BC]_D = [A, B]_D C + B[A, C]_D, \tag{7.6.22}$$

$$[A, [B, C]_D]_D + [B, [C, A]_D]_D + [C, [A, B]_D]_D = 0, \tag{7.6.23}$$

and also the relations

$$[\chi_N, B]_D = 0 \tag{7.6.24}$$

which make the commutation relations (7.6.19) consistent with the constraints $\chi_N = 0$. Also, the Dirac brackets are unchanged if we replace the χ_N with any functions χ'_N for which the equations $\chi'_N = 0$ and $\chi_N = 0$ define the same submanifold of phase space. But all these agreeable properties do not prove that the commutators are actually given by Eq. (7.6.19) in terms of the Dirac brackets.

This issue is illuminated if not settled by a powerful theorem proved by Maskawa and Nakajima [6]. They showed that for any set of canonical variables Ψ^a, Π_a governed by second class constraints, it is always possible by a canonical transformation[71] to construct two sets of variables Q^n, $\mathcal{Q}^r$ and their respective conjugates P_n, $\mathcal{P}_r$, such that the constraints read $\mathcal{Q}^r = \mathcal{P}_r = 0$. Using these coordinates to calculate Poisson brackets, and redefining the constraint functions as $\chi_{1r} = \mathcal{Q}^r$, $\chi_{2r} = \mathcal{P}_r$, we have

$$C_{1r,2s} = [\mathcal{Q}^r, \mathcal{P}_s]_P = \delta^r_s,$$

$$C_{1r,1s} = [\mathcal{Q}^r, \mathcal{Q}^s]_P = 0, \qquad C_{2r,2s} = [\mathcal{P}_r, \mathcal{P}_s]_P = 0,$$

and for any functions A, B

$$[A, \chi_{1r}]_P = -\frac{\partial A}{\partial \mathcal{P}_r}, \qquad [A, \chi_{2r}]_P = \frac{\partial A}{\partial \mathcal{Q}^r}.$$

[71] Recall that by a canonical transformation, we mean a transformation from a set of phase space coordinates Ψ^a, Π_a to some other phase space coordinates $\bar{\Psi}^a$, $\bar{\Pi}_a$, such that $[\bar{\Psi}^a, \bar{\Pi}_b]_P = \delta^a_b$ and $[\bar{\Psi}^a, \bar{\Psi}^b]_P = [\bar{\Pi}_a, \bar{\Pi}_b]_P = 0$. The Poisson brackets being calculated in terms of the Ψ^a and Π_a. It follows that the Poisson brackets for any functions A, B are the same whether calculated in terms of Ψ^a and Π_a or in terms of $\bar{\Psi}^a$ and $\bar{\Pi}_a$. It also follows that if Ψ^a and Π_a satisfy the Hamiltonian equations of motion, then so do $\bar{\Psi}^a$ and $\bar{\Pi}_a$, with the same Hamiltonian. The Lagrangian is changed by a canonical transformation, but only by a time-derivative, which does not affect the action.

This C-matrix has inverse $C^{-1} = -C$, so the Dirac brackets (7.6.20) are here

$$[A, B]_D = [A, B]_P + [A, \chi_{1r}]_P[\chi_{2r}, B]_P - [A, \chi_{2r}]_P[\chi_{1r}, B]_P = [A, B]_P - \frac{\partial A}{\partial \mathcal{Q}^r}\frac{\partial B}{\partial \mathcal{P}_r} + \frac{\partial B}{\partial \mathcal{Q}^r}\frac{\partial A}{\partial \mathcal{P}_r} = \frac{\partial A}{\partial Q^n}\frac{\partial B}{\partial P_n} - \frac{\partial B}{\partial Q^n}\frac{\partial A}{\partial P_n}. \tag{7.6.25}$$

In other words, *the Dirac bracket is equal to the Poisson bracket calculated in terms of the reduced set of unconstrained canonical variables* Q^n, P_n. If we assume that these unconstrained variables satisfy the canonical commutation relations, then the commutators of general operators A, B are given by Eq. (7.6.19) in terms of the Dirac brackets.[72]

We now return to the massive vector field, to see how it can be quantized using Dirac brackets. This is a case where it is easy to express the constrained variables V^0 and Π_0 in terms of the unconstrained ones[73] V_i and Π_i: we have simply $\Pi_0 = 0$, and V^0 is given by Eq. (7.6.5). From Eqs. (7.6.17) and (7.6.18), we see that C_{NM} here has the inverse

$$(C^{-1})^{1\mathbf{x},2\mathbf{y}} = -(C^{-1})^{2\mathbf{y},1\mathbf{x}} = -m^{-2}\delta^3(\mathbf{x} - \mathbf{y}), \tag{7.6.26}$$

$$(C^{-1})^{1\mathbf{x},1\mathbf{y}} = (C^{-1})^{2\mathbf{x},2\mathbf{y}} = 0. \tag{7.6.27}$$

Therefore the Dirac prescription (7.6.19), (7.6.20) yields the equal-time commutators

$$[A, B] = i[A, B]_P + im^{-2} \int d^3z \Big([A, \Pi_0(\mathbf{z})]_P [\partial_i \Pi_i(\mathbf{z}) - m^2 V^0(\mathbf{z}) - J^0(\mathbf{z}), B]_P - A \leftrightarrow B \Big). \tag{7.6.28}$$

By definition, we have

$$[V^\mu(\mathbf{x}), \Pi_\nu(\mathbf{y})]_P = \delta^3(\mathbf{x} - \mathbf{y})\delta^\mu_\nu, \qquad [V^\mu(\mathbf{x}), V^\nu(\mathbf{y})]_P = [\Pi_\mu(\mathbf{x}), \Pi_\nu(\mathbf{y})]_P = 0. \tag{7.6.29}$$

Hence

$$[V^i(\mathbf{x}), V^j(\mathbf{y})] = [V^0(\mathbf{x}), V^0(\mathbf{y})] = 0,$$
$$[V^i(\mathbf{x}), V^0(\mathbf{y})] = -im^{-2}\partial_i \delta^3(\mathbf{x} - \mathbf{y}),$$
$$[V^i(\mathbf{x}), \Pi_j(\mathbf{y})] = i\delta^i_j \delta^3(\mathbf{x} - \mathbf{y}), \tag{7.6.30}$$
$$[V^\mu(\mathbf{x}), \Pi_0(\mathbf{y})] = [V^0(\mathbf{x}), \Pi_0(\mathbf{y})] = 0,$$
$$[\Pi^\mu(\mathbf{x}), \Pi^\nu(\mathbf{y})] = 0.$$

[72] It is still an open question whether we should adopt canonical commutation relations for the unconstrained variables Q^n, P_n constructed by the Maskawa–Nakajima canonical transformation. Ultimately, the test of such canonical commutation relations is their consistency with the free-field commutation relations derived in Chapter 5, but to apply this test we need to know what the Q^n and P_n are. In the Appendix to this chapter we display two large classes of theories in which we can identify a set of unconstrained Qs and Ps, such that the Dirac commutation relations (7.6.19) follow from the ordinary canonical commutation relations of the Qs and Ps. We shall also show that in these cases, the Hamiltonian defined in terms of the unconstrained Ψs and Πs may be written just as well in terms of the constrained variables.

[73] This is a special case of the theories discussed in Part A of the Appendix.

These are indeed just the commutation relations that we would find by assuming that the unconstrained variables satisfy the usual canonical commutation relations $[V^i(\mathbf{x}), \Pi_j(\mathbf{y})] = i\delta^i_j\delta^3(\mathbf{x} - \mathbf{y})$, $[V^i(\mathbf{x}), V^j(\mathbf{y})] = [\Pi_i(\mathbf{x}), \Pi_j(\mathbf{y})] = 0$, and using the constraints to evaluate the commutators involving Π_0 and V^0.

7.7 Field Redefinitions and Redundant Couplings[74]

Observables like masses and S-matrix elements are independent of some of the coupling parameters in any action, known as the *redundant parameters*. This is because changes in these parameters can be undone by simply redefining the field variables. A continuous redefinition of the fields, such as an infinitesimal local transformation $\Psi^\ell(x) \to \Psi^\ell(x) + \epsilon F^\ell(\Psi(x), \partial_\mu \Psi(x), \ldots)$, clearly cannot affect any *observable* of the theory,[75] though, of course, it would change the values of matrix elements of the fields themselves.

How can we tell whether some variation in the parameters of a theory can be cancelled by a field redefinition? A continuous local field redefinition will produce a change in the action of the form

$$\delta I[\Psi] = \epsilon \sum_\ell \int d^4x \frac{\delta I[\Psi]}{\delta \Psi^\ell(x)} F^\ell(\Psi(x), \partial\Psi(x), \ldots). \tag{7.7.1}$$

So any change δg_i in the coupling parameters g_i, for which the change in the action is of the form

$$\sum_i \frac{\partial I}{\partial g_i}\delta g_i = \epsilon \sum_\ell \int d^4x \frac{\delta I[\Psi]}{\delta \Psi^\ell(x)} F^\ell(\Psi(x), \partial\Psi(x), \ldots), \tag{7.7.2}$$

may be compensated by a field redefinition

$$\Psi^\ell(x) \to \Psi^\ell(x) + \epsilon F^\ell(\Psi(x), \partial_\mu \Psi(x), \ldots),$$

and therefore can have no effect on any observables. In other words, *a coupling parameter is redundant if the change in the action when we vary this parameter vanishes when we use the field equations* $\delta I/\delta \Psi^\ell = 0$.

For example, suppose we write the Lagrangian density of a scalar field theory in the form

$$\mathcal{L} = -\frac{1}{2}Z(\partial^\mu \Phi \partial_\mu \Phi + m^2\Phi^2) - \frac{1}{24}gZ^2\Phi^4.$$

The constant Z is an redundant coupling, because

$$\frac{\partial I}{\partial Z} = \frac{1}{2}\int d^4x \, \Phi \left(\Box\Phi - m^2\Phi - \frac{1}{6}gZ\Phi^3 \right),$$

and this vanishes when we use the field equation

$$\Box\Phi - m^2\Phi = \frac{1}{6}gZ\Phi^3.$$

[74] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

[75] For instance, the theorem of Section 10.2 shows that as long as we multiply by the correct field renormalization constants, S-matrix elements can be obtained from the vacuum expectation value of a time-ordered product of any operators that have non-vanishing matrix elements between the vacuum and the one-particle states of the particles participating in the reaction.

On the other hand, neither the bare mass m nor the coupling g are redundant, and no function of m and g is redundant.

In this example, the field redefinition needed to compensate for a change of Z is a simple rescaling, in which F is proportional to Φ. (For this reason Z is called a field renormalization constant.) This is the most general field transformation that leaves the general form of this action invariant. But for the more general actions considered in Sections 12.3 and 12.4, with arbitrary numbers of fields and derivatives, we would have to consider non-linear as well as linear field redefinitions, and an infinite subset of the parameters of the theory would be redundant.

Appendix Dirac Brackets from Canonical Commutators

In this Appendix we shall show, in theories of two types, that the formula giving commutators as Dirac brackets times i follows from the usual canonical commutation relations for a reduced set of variables.

A

Suppose (as in the case of a massive vector field V^μ) that the quantum variables Ψ^a and Π_a appearing in the Lagrangian L may be divided into two classes:[76] one set Q^n of independent canonical variables (like $V^i(\mathbf{x})$) with independent canonical conjugates $P_n = \partial L/\partial \dot{Q}^n$; and another set $\mathcal{Q}^r(\mathbf{x})$ (like V^0) whose time-derivatives do not appear in the Lagrangian. The primary constraints are the conditions $\chi_{1r} = 0$, where

$$\chi_{1r} = \mathcal{P}_r \tag{7.A.1}$$

are the variables conjugate to the $\mathcal{Q}^r$. The secondary constraints arise from the equations of motion $0 = \partial L/\partial \mathcal{Q}^r$ for the $\mathcal{Q}^r$; we suppose that these constraints can 'solved'—i.e., they may be written in the form $\chi_{2r} = 0$, with χ_{2r} in the form

$$\chi_{2r} = \mathcal{Q}^r - f^r(Q, P). \tag{7.A.2}$$

(An example is provided by Eq. (7.6.5), which gives V^0 in terms of the independent Ps (here, the Π_i) and Qs.) We assume that the independent Qs and Ps satisfy the usual canonical commutation rules:

$$[Q^n, P_m] = i\delta^n_m, \qquad [Q^n, Q^m] = [P_n, P_m] = 0. \tag{7.A.3}$$

The constraint $\chi_{2r} = 0$ yields the commutators involving $\mathcal{Q}^r$:

$$[\mathcal{Q}^r, Q^n] = -i\frac{\partial f^r}{\partial P_n}, \qquad [\mathcal{Q}^r, P_n] = i\frac{\partial f^r}{\partial Q^n}, \tag{7.A.4}$$

$$[\mathcal{Q}^r, \mathcal{Q}^s] = i\Gamma^{rs}. \tag{7.A.5}$$

[76] We are again using a compact notation, in which labels like a, n, and r include a space coordinate $\mathbf{x}$ as well as discrete indices. Repeated labels are summed and integrated. All quantum variables are understood to be evaluated at the same time, with the common time argument dropped everywhere. The quantities $\mathcal{Q}^r$ are the same as the C^r introduced in Section 7.2.

where Γ^{rs} is the Poisson bracket

$$\Gamma^{rs} \equiv [f^r, f^s]_P, \tag{7.A.6}$$

and, of course, all commutators involving $\mathcal{P}_r$ vanish:

$$[\mathcal{P}_r, Q^n] = [\mathcal{P}_r, P_n] = [\mathcal{P}_r, \mathcal{Q}^s] = [\mathcal{P}_r, \mathcal{P}_s] = 0. \tag{7.A.7}$$

Now let us compare these commutators with the Dirac brackets. The Poisson brackets of the constraint functions are

$$C_{1r,1s} \equiv [\chi_{1r}, \chi_{1s}]_P = 0, \tag{7.A.8}$$

$$C_{1r,2s} = -C_{2s,1r} \equiv [\chi_{1r}, \chi_{2s}]_P = -\delta^s_r, \tag{7.A.9}$$

$$C_{2r,2s} \equiv [\chi_{2r}, \chi_{2s}]_P = [f^r(Q, P), f^s(Q, P)]_P = \Gamma^{rs}. \tag{7.A.10}$$

(In the example of the massive vector field Γ^{rs} vanishes, but the discussion here will apply also for non-vanishing Γ^{rs}.) It is easy to see that the C-matrix has the inverse

$$(C^{-1})^{1r,1s} = \Gamma^{rs}, \qquad (C^{-1})^{2r,2s} = 0, \qquad (C^{-1})^{1r,2s} = -(C^{-1})^{2s,1r} = \delta^r_s. \tag{7.A.11}$$

Also, the Poisson brackets of any function A with the constraint functions are

$$[A, \chi_{1r}]_P = \frac{\partial A}{\partial \mathcal{Q}^r}, \qquad [A, \chi_{2r}]_P = -\frac{\partial A}{\partial \mathcal{P}_r} - [A, f^r(Q, P)]_P.$$

Hence the Dirac bracket is

$$[A, B]_D = [A, B]_P - \frac{\partial A}{\partial \mathcal{Q}^r}\frac{\partial B}{\partial \mathcal{P}_r} + \frac{\partial B}{\partial \mathcal{Q}^r}\frac{\partial A}{\partial \mathcal{P}_r} + \frac{\partial A}{\partial \mathcal{Q}^r}\Gamma^{rs}\frac{\partial B}{\partial \mathcal{Q}^s} - \frac{\partial A}{\partial \mathcal{Q}^r}[B, f^r]_P + [A, f^r]_P\frac{\partial B}{\partial \mathcal{Q}^r}. \tag{7.A.12}$$

Now, if A and B are both functions only of the independent canonical variables Q^n and P_n, then $\partial A/\partial \mathcal{Q}^r = \partial B/\partial \mathcal{Q}^r = 0$, so the Dirac bracket is equal to the Poisson bracket. In particular,

$$[Q^n, P_m]_D = \delta^n_m, \qquad [Q^n, Q^m]_D = [P_n, P_m]_D = 0. \tag{7.A.13}$$

Where A is $\mathcal{Q}^r$ and B is a function of Qs and Ps, it is only the fifth term on the right-hand side of Eq. (7.A.12) that contributes. In particular

$$[\mathcal{Q}^r, Q^n]_D = -\frac{\partial f^r}{\partial P_n}, \qquad [\mathcal{Q}^r, P_n]_D = +\frac{\partial f^r}{\partial Q^n}. \tag{7.A.14}$$

Where both A and B are $\mathcal{Q}$s, we have only the fourth term

$$[\mathcal{Q}^r, \mathcal{Q}^s]_D = \Gamma^{rs}. \tag{7.A.15}$$

Finally, where A is $\mathcal{P}_r$ and B is anything, we have only the first and third terms, which cancel:

$$[\mathcal{P}_r, B]_D = [\mathcal{P}_r, B]_P + \frac{\partial B}{\partial \mathcal{Q}^r} = 0. \tag{7.A.16}$$

Comparison of Eqs. (7.A.13)–(7.A.16) with Eqs. (7.A.3)–(7.A.7) shows that in all cases, the commutators are equal to the Dirac brackets times i. This is only to be expected, because as remarked in Section 7.6 all Dirac brackets involving the constraint functions vanish, so the Dirac brackets involving $\mathcal{Q}^r$ and/or $\mathcal{P}_s$ are given by using the constraint equations to express $\mathcal{Q}^r$ and/or $\mathcal{P}_s$ in terms of the independent Qs and Ps.

B

Next consider the case where the constraints take the form of conditions $\chi_{1r}(\Psi) = 0$ on the Ψ^a, which can be solved by expressing them in terms of a smaller set of unconstrained variables Q^n, and an equal number of separate conditions $\chi_{2r}(\Pi) = 0$ on the Π_a, which can be solved by expressing the Π_a in terms of a smaller set of unconstrained P_n. (We will see an example in the next chapter, where the constraints on the Ψ^a are gauge fixing conditions that are used to eliminate first class constraints, and the constraints on the Π_a are secondary constraints arising from the consistency of the first class constraints with the field equations.) We assume that the unconstrained variables satisfy the usual canonical commutation relations $[Q^n, P_m] = i\delta^n_m$, $[Q^n, Q^m] = [P_n, P_m] = 0$. The constrained and unconstrained momenta are related by

$$P_n = \frac{\partial L}{\partial \dot{Q}^n} = \frac{\partial L}{\partial \dot{\Psi}^b}\frac{\partial \dot{\Psi}^b}{\partial \dot{Q}^n} = \Pi_b \frac{\partial \Psi^b}{\partial Q^n}. \tag{7.A.17}$$

It follows that

$$[\Psi^a, \Pi_b]\frac{\partial \Psi^b}{\partial Q^n} = [\Psi^a, P_n] = i\frac{\partial \Psi^a}{\partial Q^n}$$

or, in other words,

$$\left\{[\Psi^a, \Pi_b] - i\delta^a_b\right\}\frac{\partial \Psi^b}{\partial Q^n} = 0. \tag{7.A.18}$$

Now, the constraint $\chi_{1r}(\Psi) = 0$ is satisfied for $\Psi^a = \Psi^a(Q)$ for all Q, so

$$\frac{\partial \chi_{1r}}{\partial \Psi^b}\frac{\partial \Psi^b}{\partial Q^n} = 0. \tag{7.A.19}$$

Furthermore, the vectors $(V_r)_b \equiv \partial\chi_{1r}/\partial\Psi^b$ form a complete set perpendicular to all the vectors $(U_n)^b \equiv \partial\Psi^b/\partial Q^n$, because if there were some other vector V_b with $V_b(U_n)^b = 0$ for all n, then there would be additional constraints on the Ψ^a. Hence Eq. (7.A.18) implies that

$$[\Psi^a, \Pi_b] = i\delta^a_b + ic^a_r \frac{\partial \chi_{1r}}{\partial \Psi^b} \tag{7.A.20}$$

with some unknown coefficients c^a_r. To determine these coefficients, we make use of the other constraint, that $\chi_{2r}(\Pi) = 0$. It follows that

$$0 = [\Psi^a, \chi_{2r}(\Pi)] = [\Psi^a, \Pi_b]\frac{\partial \chi_{2r}(\Pi)}{\partial \Pi_b}.$$

Using Eq. (7.A.20), we have then

$$\frac{\partial \chi_{2r}(\Pi)}{\partial \Pi_a} = -c^a_s \frac{\partial \chi_{1s}(\Psi)}{\partial \Psi^b}\frac{\partial \chi_{2r}(\Pi)}{\partial \Pi_b}. \tag{7.A.21}$$

We recognize the factor multiplying c^a_s as the Poisson bracket

$$\frac{\partial \chi_{1s}(\Psi)}{\partial \Psi^b}\frac{\partial \chi_{2r}(\Pi)}{\partial \Pi_b} = [\chi_{1s}, \chi_{2r}]_P \equiv C_{1s,2r}.$$

Also, since χ_{1s} depends only on the Ψ and χ_{2r} depends only on the Π, these are the only non-vanishing Poisson brackets of constraints, so

$$C_{1r,1s} = C_{2r,2s} = 0.$$

Thus Eq. (7.A.21) may be written

$$\frac{\partial \chi_N}{\partial \Pi_a} = -c^a_s C_{1s,N} \tag{7.A.22}$$

with N running over all constraint functions. For second class constraints, this has the unique solution

$$c^a_s = -\frac{\partial \chi_N}{\partial \Pi_a}(C^{-1})^{N,1s} = -\frac{\partial \chi_{2r}}{\partial \Pi_a}(C^{-1})^{2r,1s}. \tag{7.A.23}$$

Using this in Eq. (7.A.20) shows that

$$[\Psi^a, \Pi_b] = i\left[\delta^a_b - \frac{\partial \chi_{2r}}{\partial \Pi_a}(C^{-1})^{2r,1s}\frac{\partial \chi_{1s}}{\partial \Psi^b}\right]. \tag{7.A.24}$$

The Poisson brackets of Ψ^a and Π_b with the constraint functions are

$$[\Psi^a, \chi_{1r}]_P = 0, \qquad [\Psi^a, \chi_{2r}]_P = \frac{\partial \chi_{2r}}{\partial \Pi_a},$$
$$[\chi_{1r}, \Pi_b]_P = \frac{\partial \chi_{1r}}{\partial \Psi^b}, \qquad [\chi_{2r}, \Pi_b]_P = 0, \tag{7.A.25}$$

so the quantity in brackets on the right-hand side of Eq. (7.A.24) is the Dirac bracket

$$[\Psi^a, \Pi_b] = i[\Psi^a, \Pi_b]_D \tag{7.A.26}$$

as was to be shown. Also, we can easily see that because C^{-1} has no 11 or 22 components, the other Dirac brackets are

$$[\Psi^a, \Psi^b]_D = [\Pi_a, \Pi_b]_D = 0, \tag{7.A.27}$$

so trivially

$$[\Psi^a, \Psi^b] = i[\Psi^a, \Psi^b]_D, \qquad [\Pi_a, \Pi_b] = i[\Pi_a, \Pi_b]_D. \tag{7.A.28}$$

* * *

In addition to commutation rules, we also need an explicit formula for the Hamiltonian. The usual canonical formalism tells us to take

$$H = P_n\dot{Q}^n - L, \tag{7.A.29}$$

the sum running over independent canonical variables. In theories of both types considered in this Appendix, this Hamiltonian may be written in terms of the constrained variables as

$$H = \Pi_a\dot{\Psi}^a - L. \tag{7.A.30}$$

For theories of type A, this is trivial; the sum over a runs over values n, for which $\Psi^n = Q^n$ and $\Pi_n = P_n$ are the independent canonical variables, together with values r, for which $\Pi_r = \mathcal{P}_r = 0$. For theories of type B, we note that Eq. (7.A.17) gives

$$P_n\dot{Q}^n = \Pi_b\frac{\partial \Psi^b}{\partial Q^n}\dot{Q}^n = \Pi_b\dot{\Psi}^b$$

which again yields Eq. (7.A.30).

Problems

1. Consider the theory of a set of real scalar fields Φ^n, with Lagrangian density
$$\mathcal{L} = -\frac{1}{2}\sum_{mn}\partial_\mu\Phi^n\partial^\mu\Phi^m f_{mn}(\Phi),$$
where $f_{mn}(\Phi)$ is an arbitrary real matrix function of the field. (This is called the non-linear σ-model.) Carry out the canonical quantization of this theory. Derive the interaction $H_I[\phi(t), \dot{\phi}(t)]$ in the interaction picture.
2. Consider a theory of real scalar fields Φ^n and Dirac fields Ψ^i, with Lagrangian density $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1$, where $\mathcal{L}_0$ is the usual free-field Lagrangian density, and $\mathcal{L}_1$ is an interaction term involving Φ^n and Ψ^i, but not their derivatives. Derive an explicit expression for the symmetric energy–momentum tensor $\Theta^{\mu\nu}$.
3. In the theory described in Problem 2, suppose that the Lagrangian density is invariant under a global infinitesimal symmetry $\delta\Phi^n = i\epsilon\sum_m t^n_m\Phi^m$, $\delta\Psi^i = i\epsilon\sum_j \tau^i_j\Psi^j$. Derive an explicit expression for the conserved current associated with this symmetry.
4. Consider the theory of a complex scalar field Φ and a real vector field V^μ, with Lagrangian density
$$\mathcal{L} = -(D_\mu\Phi)^\dagger D^\mu\Phi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}m^2 V_\mu V^\mu - \mathcal{H}(\Phi^\dagger\Phi),$$
where $D_\mu \equiv \partial_\mu - igV_\mu$ and $F_{\mu\nu} \equiv \partial_\mu V_\nu - \partial_\nu V_\mu$, and $\mathcal{H}$ is an arbitrary function. Carry out the canonical quantization of this theory. Derive the interaction in the interaction picture.
5. In the theory of Problem 4, derive expressions for the symmetric energy–momentum tensor $\Theta^{\mu\nu}$ and for the conserved current associated with the symmetry under $\delta\Phi = i\epsilon\Phi$, $\delta V^\mu = 0$.
6. Prove that the Dirac bracket satisfies the Jacobi identity (7.6.23).
7. Prove that the Dirac bracket is independent of the choice of constraint functions χ_N used to describe a given submanifold of phase space.

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8 Electrodynamics

The original approach to quantum electrodynamics was to take for granted Maxwell's classical theory of electromagnetism, and quantize it. It will probably not surprise the reader that this book will follow a different path. We shall first infer the need for a principle of gauge invariance from the peculiar difficulties that arise in formulating a quantum theory of massless particles with spin, and then deduce the main features of electrodynamics from the gauge invariance principle. After that we shall follow a more conventional modern approach, in which one takes gauge invariance as the starting point and uses it to deduce the existence of a vector potential describing massless particles of unit spin.

It is too soon to tell which of these two alternatives corresponds to the logical order of nature. Most theorists have tended to take gauge invariance as a starting point, but in modern string theories [1] the argument runs the other way; one first notices a state of mass zero and unit spin among the normal modes of a string, and then from that deduces the gauge invariance of the effective field theory that describes such particles. At any rate, as we shall see, using either approach one is led to the quantized version of Maxwell's theory, still the paradigmatic example of a successful quantum field theory.

8.1 Gauge Invariance

Let's start by recalling the problems encountered in constructing covariant free fields for a massless particle of helicity ± 1. We saw in Section 5.9 that there is no difficulty in constructing an antisymmetric tensor free field $f_{\mu\nu}(x)$ for such particles. This field can be expressed in terms of the four-potential $a_\mu(x)$, given by Eq. (5.9.23), through the familiar relation

$$f_{\mu\nu}(x) = \partial_\mu a_\nu(x) - \partial_\nu a_\mu(x). \tag{8.1.1}$$

However, Eq. (5.9.23) shows that the $a_\mu(x)$ transforms as a four-vector only up to a gauge transformation

$$U_0(\Lambda)a_\mu(x)U_0^{-1}(\Lambda) = \Lambda_\mu{}^\nu a_\nu(\Lambda x) + \partial_\mu\Omega(x, \Lambda). \tag{8.1.2}$$

There is, in fact, no way to construct a true four-vector as a linear combination of the creation and annihilation operators for helicity ± 1. This is one way of understanding the presence of singularities at $m = 0$ in the propagator of a massive vector field

$$\Delta_{\mu\nu}(x, y) = (2\pi)^{-4}\int d^4q\, e^{iq\cdot(x-y)}\frac{\eta_{\mu\nu} + q_\mu q_\nu/m^2}{q^2 + m^2 - i\epsilon},$$

which prevent us from dealing with massless particles of helicity ± 1 by simply passing to the limit $m \to 0$ of the theory of a massive particle of spin one.

We could avoid these problems by demanding that all interactions involve only[77] $F_{\mu\nu}(x) \equiv \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x)$ and its derivatives, not $A_\mu(x)$, but this is not the most general possibility, and not the one realized in nature. Instead of banishing $A_\mu(x)$ from the

[77] We now use A_μ and $F_{\mu\nu}$ for the electromagnetic potential vector and the field strength tensor because these are interacting fields.

action, we shall require instead that the part of the action I_M for matter and its interaction with radiation be invariant under the general gauge transformation

$$A_\mu(x) \longrightarrow A_\mu(x) + \partial_\mu\epsilon(x) \tag{8.1.3}$$

(at least when the matter fields satisfy the field equations) so that the extra term in Eq. (8.1.2) should have no effect. The change in the matter action under the transformations (8.1.3) may be written

$$\delta I_M = \int d^4x\, \frac{\delta I_M}{\delta A_\mu(x)}\partial_\mu\epsilon(x). \tag{8.1.4}$$

Hence the Lorentz invariance of I_M requires that

$$\partial_\mu\frac{\delta I_M}{\delta A_\mu(x)} = 0. \tag{8.1.5}$$

This is trivially true if I_M involves only $F_{\mu\nu}(x)$ and its derivatives, along with matter fields. In this case

$$\frac{\delta I_M}{\delta A_\nu(x)} = 2\partial^\mu\frac{\delta I_M}{\delta F_{\mu\nu}(x)}.$$

But if I_M involves $A_\mu(x)$ itself then Eq. (8.1.5) is a non-trivial constraint on the theory.

Now, what sort of theory will provide conserved currents to which we can couple the field $A^\mu(x)$? We saw in Section 7.3 that infinitesimal internal symmetries of the action imply the existence of conserved currents. In particular, if the transformation[78]

$$\delta\Psi^\ell(x) = i\epsilon(x)q_\ell\Psi^\ell(x) \tag{8.1.6}$$

leaves the matter action invariant for a constant ϵ, then for general infinitesimal functions $\epsilon(x)$ the change in the matter action must take the form

$$\delta I_M = -\int d^4x\, J^\mu(x)\partial_\mu\epsilon(x). \tag{8.1.7}$$

When the matter fields satisfy their field equations, the matter action is stationary with respect to *any* variation of the Ψ_ℓ, so in this case (8.1.7) must vanish, and hence

$$\partial_\mu J^\mu = 0. \tag{8.1.8}$$

In particular, we saw in Section 7.3 that if I_M is the integral of a function $\mathcal{L}_M$ of Ψ^ℓ and $\partial_\mu\Psi^\ell$, then the conserved current is given by[79]

$$J^\mu = -i\sum_\ell \frac{\partial\mathcal{L}_M}{\partial(\partial_\mu\Psi^\ell)}q_\ell\Psi^\ell,$$

[78] Because the field transformation matrix is taken now to be diagonal it is not convenient here to use the summation convention for sums over field indices, so there is no sum over ℓ in Eq. (8.1.6).

[79] Here Ψ^ℓ is understood to run over all independent fields other than A^μ. We use a capital psi to indicate that these are Heisenberg-picture fields, whose time-dependence includes the effects of interactions. Of course, this Ψ^ℓ is not to be confused with a state-vector or wave function.

and this generates the transformations (8.1.6) in the sense that

$$[Q, \Psi^\ell(x)] = -q_\ell\Psi^\ell(x), \tag{8.1.9}$$

where Q is the time-independent charge operator

$$Q = \int d^3x\, J^0. \tag{8.1.10}$$

We can therefore construct a Lorentz-invariant theory by coupling the vector field A_μ to the conserved current J^μ, in the sense that $\delta I_M/\delta A_\mu(x)$ is taken to be proportional to $J^\mu(x)$. Any constant of proportionality may be absorbed into the definition of the overall scale of the charges q_ℓ, so we may simply set these quantities equal:

$$\frac{\delta I_M}{\delta A_\mu(x)} = J^\mu(x). \tag{8.1.11}$$

The conservation of electric charge only allows us to fix the values of all charges in terms of the value of any one of them, conventionally taken to be the electron charge, denoted $-e$. It is Eq. (8.1.11) that gives a definite meaning[80] to the value of e.

The requirement (8.1.11) may be restated as a principle of invariance [1a]: the matter action is invariant under the joint transformations

$$\delta A_\mu(x) = \partial_\mu\epsilon(x), \tag{8.1.12}$$
$$\delta\Psi_\ell(x) = i\epsilon(x)q_\ell\Psi_\ell(x). \tag{8.1.13}$$

A symmetry of this type with an arbitrary function $\epsilon(x)$ is called a *local symmetry*, or a gauge invariance of the second kind. A symmetry under a transformation with ϵ constant is called a *global symmetry*, or a gauge invariance of the first kind. Several exact local symmetries are now known, but the only purely global symmetries appear to be accidents enforced by other principles. (See Section 12.5.)

We have not yet said anything about the action for photons themselves. As a guess, we can take this to be the same as for massive vector fields, but with $m = 0$:

$$I_\gamma = -\frac{1}{4}\int d^4x\, F_{\mu\nu}F^{\mu\nu}. \tag{8.1.14}$$

This is the same as the action used in classical electrodynamics, but its real justification is that it is (up to a constant) the unique gauge-invariant functional that is quadratic in $F_{\mu\nu}$, without higher derivatives. Also, as we will see in the next section, it leads to a consistent quantum theory. If there are any terms in the action of A with higher derivatives and/or of higher order in $F_{\mu\nu}$ they can be lumped into what we have called the matter action. Using Eqs. (8.1.11) and (8.1.14), the field equation for electromagnetism now reads

$$0 = \frac{\delta}{\delta A_\nu}[I_\gamma + I_M] = \partial_\mu F^{\mu\nu} + J^\nu. \tag{8.1.15}$$

[80] Of course, Eq. (8.1.11) fixes the definition of e only after we have defined how we are normalizing $A_\mu(x)$. The question of electromagnetic field normalization is taken up in Section 10.4.

We recognize these as the usual inhomogeneous Maxwell equations, with current J^ν. There are also other, homogeneous, Maxwell equations

$$0 = \partial_\mu F_{\nu\rho} + \partial_\rho F_{\mu\nu} + \partial_\nu F_{\rho\mu}, \tag{8.1.16}$$

which follow directly from the definition $F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$.

In the above discussion, we have started with the existence of massless spin one particles, and have been led to infer the invariance of the matter action under a local gauge transformation (8.1.12), (8.1.13). As usually presented, the derivation runs in the opposite direction. That is, one starts with a global internal symmetry

$$\delta\Psi^\ell(x) = i\epsilon q_\ell\Psi^\ell(x) \tag{8.1.17}$$

and asks what must be done to promote this to a local symmetry

$$\delta\Psi^\ell(x) = i\epsilon(x)q_\ell\Psi^\ell(x). \tag{8.1.18}$$

If the Lagrangian density $\mathcal{L}$ depended only on fields $\Psi^\ell(x)$ and not on their derivatives then it would make no difference whether ϵ is constant or not; invariance with ϵ constant would imply invariance with ϵ a function of spacetime position. But all realistic Lagrangians do involve field derivatives, and here we have the problem that derivatives of fields transform differently from fields themselves:

$$\delta\partial_\mu\Psi^\ell(x) = i\epsilon(x)q_\ell\partial_\mu\Psi^\ell(x) + iq_\ell\Psi^\ell(x)\partial_\mu\epsilon(x). \tag{8.1.19}$$

In order to cancel the second term here, we 'invent' a vector field $A_\mu(x)$ with transformation rule

$$\delta A_\mu(x) = \partial_\mu\epsilon(x) \tag{8.1.20}$$

and require that the Lagrangian density depend on $\partial_\mu\Psi^\ell$ and A_μ only in the combination

$$D_\mu\Psi^\ell \equiv \partial_\mu\Psi^\ell - iq_\ell A_\mu\Psi^\ell, \tag{8.1.21}$$

which transforms just like Ψ^ℓ

$$\delta D_\mu\Psi^\ell(x) = i\epsilon(x)q_\ell D_\mu\Psi^\ell(x). \tag{8.1.22}$$

A matter Lagrangian density $\mathcal{L}_M(\Psi, D\Psi)$ that is formed only out of Ψ^ℓ and $D_\mu\Psi^\ell$ will be invariant under the transformations (8.1.18), (8.1.20), with $\epsilon(x)$ an arbitrary function, if it is invariant with ϵ a constant. With the Lagrangian of this form, we have

$$\frac{\delta I_M}{\delta A_\mu} = \sum_\ell \frac{\partial\mathcal{L}_M}{\partial D_\mu\Psi^\ell}(-iq_\ell\Psi^\ell) = -i\sum_\ell\frac{\partial\mathcal{L}_M}{\partial\partial_\mu\Psi^\ell}q_\ell\Psi^\ell,$$

which is the same as Eq. (8.1.11). (We could also include $F_{\mu\nu}$ and its derivatives in $\mathcal{L}_M$.) From this point of view, the masslessness of the particles described by A_μ is a consequence of gauge invariance rather than an assumption: a term $-\frac{1}{2}m^2A_\mu A^\mu$ in the Lagrangian density would violate gauge invariance.

8.2 Constraints and Gauge Conditions

There are aspects of electrodynamics that stand in the way of quantizing the theory as we did for various theories of massive particles in the previous chapter. As usual, we may define the canonical conjugates to the electromagnetic vector potential by

$$\Pi^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_0 A_\mu)}. \tag{8.2.1}$$

Quantization by the usual rules would give

$$[A_\mu(\mathbf{x}, t), \Pi^\nu(\mathbf{y}, t)] = i\delta_\mu{}^\nu\delta^3(\mathbf{x} - \mathbf{y}).$$

But this is not possible here, because A_μ and Π^ν are subject to several constraints.

The first constraint arises from the fact that the Lagrangian density is independent[81] of the time-derivative of A_0, and therefore

$$\Pi^0(x) = 0. \tag{8.2.2}$$

This is called a *primary constraint*, because it follows directly from the structure of the Lagrangian. There is also a *secondary constraint* here, which follows from the field equation for the quantity fixed by the primary constraint:[82]

$$\partial_i\Pi^i = -\partial_i\frac{\partial\mathcal{L}}{\partial F_{i0}} = -\frac{\partial\mathcal{L}}{\partial A_0} = -J^0, \tag{8.2.3}$$

the time-derivative term dropping out because $F_{00} = 0$. Even though the matter Lagrangian may generally depend on A^0, the charge density depends only on the canonical matter fields[83] Q^n and their canonical conjugates P_n:

$$J^0 = -i\sum_\ell\frac{\partial\mathcal{L}}{\partial(\partial_0\Psi^\ell)}q_\ell\Psi^\ell = -i\sum_n P_n q_n Q^n. \tag{8.2.4}$$

Hence Eq. (8.2.3) is a functional relation among canonical variables. Both Eq. (8.2.2) and Eq. (8.2.3) are inconsistent with the usual assumptions that $[A_\mu(\mathbf{x}, t), \Pi^\nu(\mathbf{y}, t)] = i\delta_\mu{}^\nu\delta^3(\mathbf{x} - \mathbf{y})$ and $[Q^n(\mathbf{x}, t), \Pi^\nu(\mathbf{y}, t)] = [P_n(\mathbf{x}, t), \Pi^\nu(\mathbf{y}, t)] = 0$.

We encountered a similar problem in the theory of the massive vector field. In that case we found two equivalent ways of dealing with it: either by the method of Dirac brackets or, more directly, by treating only A_i and Π^i as canonical variables, solving the analog of Eq. (8.2.3) to calculate A^0 in terms of these variables. It is clear that here we cannot

[81] For $\mathcal{L}_\gamma = -F_{\mu\nu}F^{\mu\nu}/4$, we have $\partial\mathcal{L}_\gamma/\partial(\partial_\mu A_\nu) = -F^{\mu\nu}$, which vanishes for $\mu = 0$ because $F^{\mu\nu}$ is antisymmetric. For matter Lagrangians $\mathcal{L}_M$ that involve only Ψ^ℓ and $D_\mu\Psi^\ell$, the prescription (8.1.21) tells us that $\mathcal{L}_M$ does not depend on any derivatives of any A^μ. Even if the matter Lagrangian depends also on $F_{\mu\nu}$, $\partial\mathcal{L}_M/\partial(\partial_\mu A_\nu)$ will be again antisymmetric in μ and ν, and therefore will vanish for $\mu = \nu = 0$.

[82] As usual, i, j, etc. run over the values 1, 2, 3.

[83] Due to exhaustion of alphabetic resources, I have had to adopt a notation here that is different from that of the previous chapter. The symbols Q^n and P_n are now reserved for the canonical matter fields and their canonical conjugates, respectively, while the canonical electromagnetic fields and canonical conjugates are A_i and Π_i.

use Dirac brackets; the constraint functions χ here are Π^0 and $\partial_i\Pi_i - J^0$ (as compared with $\partial_i\Pi_i - m^2A^0 - J^0$) and these obviously have vanishing Poisson brackets. In Dirac's terminology, the constraints (8.2.2) and (8.2.3) are *first class*. Nor can we eliminate A^0 as a dynamical variable by solving for it in terms of the other variables. Instead of giving A^0 for all time, Eq. (8.2.3) is a mere initial condition; if Eq. (8.2.3) is satisfied at one time, then it is satisfied for all times, because (using the field equations for the other fields A^i), we have

$$\partial_0\left[\partial_i\frac{\partial\mathcal{L}}{\partial F_{i0}} - J^0\right] = -\partial_i\partial_0\frac{\partial\mathcal{L}}{\partial F_{0i}} - \partial_0 J^0$$
$$= +\partial_i\partial_j\frac{\partial\mathcal{L}}{\partial F_{ji}} - \partial_i J^i - \partial_0 J^0,$$

and the current conservation condition then gives

$$\partial_0\left[\partial_i\frac{\partial\mathcal{L}}{\partial F_{i0}} - J^0\right] = 0. \tag{8.2.5}$$

It should not be surprising that we still have four components of A^μ with only three field equations, because this theory has a local gauge symmetry that makes it, in principle, impossible to infer the values of the fields at arbitrary times from their values and rates of change at any one time. Given any solution $A_\mu(\mathbf{x}, t)$ of the field equations, we can always find another solution $A_\mu(\mathbf{x}, t) + \partial_\mu\epsilon(\mathbf{x}, t)$ with the same value and time-derivative at $t = 0$ (by choosing ϵ so that its first and second derivatives vanish there) but which differs from $A_\mu(\mathbf{x}, t)$ at later times.

Because of this partial arbitrariness of $A_\mu(\mathbf{x}, t)$, it is not possible to apply the canonical quantization procedure directly to A_μ (or, as for finite mass, to $\mathbf{A}$). Of the various approaches to this difficulty, two are particularly useful. One is the modern method of gauge-invariant quantization, to be discussed in Volume II. The other, which will be followed here, is to exploit the gauge invariance of the theory, to 'choose a gauge.' That is, we make a finite gauge transformation

$$A_\mu(x) \longrightarrow A_\mu(x) + \partial_\mu\lambda(x), \qquad \Psi_\ell(x) \longrightarrow \exp(iq_\ell\lambda(x))\Psi_\ell(x)$$

to impose a condition on $A_\mu(x)$ that will allow us to apply the methods of canonical quantization. There are various gauges that have been found useful in various applications:[84]

Lorentz (or Landau) gauge:	$\partial_\mu A^\mu = 0$,
Coulomb gauge:	$\nabla\cdot\mathbf{A} = 0$,
Temporal gauge:	$A^0 = 0$,
Axial gauge:	$A^3 = 0$,
Unitarity gauge:	Φ real.

[84] Here Φ is any complex scalar field with $q \neq 0$; this gauge condition is used when the gauge symmetry is spontaneously broken by a non-vanishing vacuum expectation value of Φ.

The canonical quantization procedure works most easily in the axial or Coulomb gauge, but of course Coulomb gauge keeps manifest rotation invariance in a way that axial gauge does not, so we will adopt Coulomb gauge here [2].

To check that this is possible, note that if A^μ does not satisfy the Coulomb gauge condition, then the gauge-transformed field $A^\mu + \partial^\mu\lambda$ will, provided we choose λ so that $\nabla^2\lambda = -\nabla\cdot\mathbf{A}$. From now on, we assume that this transformation has been made, so that

$$\nabla\cdot\mathbf{A} = 0. \tag{8.2.6}$$

It will be convenient henceforth to limit ourselves to theories in which the matter Lagrangian $\mathcal{L}_M$ may depend on matter fields and their time-derivatives and also on A^μ but not on derivatives of A^μ. (The standard theories of the electrodynamics of scalar and Dirac fields have Lagrangians of this type.) Then the only term in the Lagrangian that depends on $F_{\mu\nu}$ is the kinematic term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$, and the constraint equation (8.2.3) reads

$$-\partial_i F^{i0} = J^0. \tag{8.2.7}$$

Together with the Coulomb gauge condition (8.2.6), this yields

$$-\nabla^2 A^0 = J^0, \tag{8.2.8}$$

which can be solved to give

$$A^0(\mathbf{x}, t) = \int d^3y\, \frac{J^0(\mathbf{y}, t)}{4\pi|\mathbf{x} - \mathbf{y}|}. \tag{8.2.9}$$

The remaining degrees of freedom are A^i, with $i = 1, 2, 3$, subject to the gauge condition $\nabla\cdot\mathbf{A} = 0$.

As mentioned earlier, the charge density depends only on the canonical matter fields Q^n and their canonical conjugates P_n, so Eq. (8.2.9) represents an explicit solution for the auxiliary field A^0.

8.3 Quantization in Coulomb Gauge

There is still an impediment to the canonical quantization of electrodynamics in the Coulomb gauge. Even after we use Eq. (8.2.9) to eliminate A^0 (and Π_0) from the list of canonical variables, we cannot apply the usual canonical commutation relations to A^i and Π_i, because there are two remaining constraints on these variables.[85] One of them is the Coulomb gauge condition

$$\chi_{1\mathbf{x}} \equiv \partial_i A^i(\mathbf{x}) = 0. \tag{8.3.1}$$

The other is the secondary constraint Eq. (8.2.3), which requires that

$$\chi_{2\mathbf{x}} \equiv \partial_i\Pi^i(\mathbf{x}) + J^0(\mathbf{x}) = 0. \tag{8.3.2}$$

[85] In this section i, j, etc. run over the values 1, 2, 3. We continue the practice of taking all operators at the same time, and omitting the time argument.

Neither constraint is consistent with the usual commutation relations $[A^i(\mathbf{x}), \Pi_j(\mathbf{y})] = i\delta^i_j\delta^3(\mathbf{x} - \mathbf{y})$, because operating on the right-hand side with either $\partial/\partial x^i$ or $\partial/\partial y^j$ does not give zero.

These constraints are of a type known as *second class*, for which there is a universal prescription for the commutation relations, discussed in Section 7.6. Note that the constraint functions have the Poisson brackets

$$C_{1\mathbf{x},2\mathbf{y}} = -C_{2\mathbf{y},1\mathbf{x}} \equiv [\chi_{1\mathbf{x}}, \chi_{2\mathbf{y}}]_P = -\nabla^2\delta^3(\mathbf{x} - \mathbf{y}),$$
$$C_{1\mathbf{x},1\mathbf{y}} \equiv [\chi_{1\mathbf{x}}, \chi_{1\mathbf{y}}]_P = 0, \tag{8.3.3}$$
$$C_{2\mathbf{x},2\mathbf{y}} \equiv [\chi_{2\mathbf{x}}, \chi_{2\mathbf{y}}]_P = 0,$$

where here, for any functionals U and V,

$$[U, V]_P \equiv \int d^3z\left[\frac{\delta U}{\delta A^i(\mathbf{z})}\frac{\delta V}{\delta\Pi_i(\mathbf{z})} - \frac{\delta V}{\delta A^i(\mathbf{z})}\frac{\delta U}{\delta\Pi_i(\mathbf{z})}\right].$$

The 'matrix' $C_{N\mathbf{x},M\mathbf{y}}$ is non-singular, which identifies these as second class constraints. Also, the field variables A^i may be expressed in terms of independent canonical variables, which may, for instance, be taken as $Q_{1\mathbf{x}} = A^1(\mathbf{x})$, $Q_{2\mathbf{x}} = A^2(\mathbf{x})$, with A^3 given by the solution of Eq. (8.3.1):

$$A^3(\mathbf{x}) = -\int^{x^3} ds\,[\partial_1 A^1(x^1, x^2, s) + \partial_2 A^2(x^1, x^2, s)].$$

Using Eq. (8.3.2), the canonical conjugates Π_i to A^i may likewise be expressed in terms of the canonical conjugates $P_{\mathbf{x}1}$ and $P_{\mathbf{x}2}$ to $Q_{1\mathbf{x}}$ and $Q_{2\mathbf{x}}$. In such cases, Part B of the Appendix to the previous chapter tells us that if the independent variables $Q_{1\mathbf{x}}$, $Q_{2\mathbf{x}}$, $P_{1\mathbf{x}}$, and $P_{2\mathbf{x}}$ satisfy the usual canonical commutation relations, then the commutators of the constrained variables and their canonical conjugates are given (aside from a factor i) by the corresponding Dirac brackets (7.6.20). This prescription has the great advantage that we do not have to do any explicit expressions for the dependent variables in terms of the independent ones.

To calculate the Dirac brackets, we note that the matrix C has the inverse

$$(C^{-1})_{1\mathbf{x},2\mathbf{y}} = -(C^{-1})_{2\mathbf{y},1\mathbf{x}} = -\int\frac{d^3k}{(2\pi)^3}\frac{e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})}}{\mathbf{k}^2} = -\frac{1}{4\pi|\mathbf{x} - \mathbf{y}|},$$
$$(C^{-1})_{1\mathbf{x},1\mathbf{y}} = (C^{-1})_{2\mathbf{x},2\mathbf{y}} = 0. \tag{8.3.4}$$

Also, the non-vanishing Poisson brackets of the A^i and Π_i with the constraint functions are

$$[A^i(\mathbf{x}), \chi_{2\mathbf{y}}]_P = -\frac{\partial}{\partial x^i}\delta^3(\mathbf{x} - \mathbf{y})$$

and

$$[\Pi_i(\mathbf{x}), \chi_{1\mathbf{y}}]_P = +\frac{\partial}{\partial x^i}\delta^3(\mathbf{x} - \mathbf{y}).$$

Hence according to Eqs. (7.6.19) and (7.6.20), the equal-time commutators are

$$[A^i(\mathbf{x}), \Pi_j(\mathbf{y})] = i\delta^i_j\delta^3(\mathbf{x} - \mathbf{y}) + i\frac{\partial^2}{\partial x^i\partial x^j}\left(\frac{1}{4\pi|\mathbf{x} - \mathbf{y}|}\right),$$
$$[A^i(\mathbf{x}), A^j(\mathbf{y})] = [\Pi_i(\mathbf{x}), \Pi_j(\mathbf{y})] = 0. \tag{8.3.5}$$

Note that these are consistent with the Coulomb gauge conditions (8.3.1) and (8.3.2), as is guaranteed by the general properties of the Dirac bracket.

Now, what is $\mathbf{\Pi}$ in electrodynamics? For the class of theories discussed in the previous section where only the kinematic term $-\frac{1}{4}\int d^3x\, F_{\mu\nu}F^{\mu\nu}$ in the Lagrangian depends on $\dot{\mathbf{A}}$, varying the Lagrangian with respect to $\dot{\mathbf{A}}$ without worrying about the constraint $\nabla\cdot\mathbf{A} = 0$ gives

$$\Pi_i = \frac{\delta L}{\delta\dot{A}^i(\mathbf{x})} = \dot{A}^i(\mathbf{x}) + \frac{\partial}{\partial x^i}A^0(\mathbf{x}). \tag{8.3.6}$$

But with $\mathbf{A}$ constrained by the condition $\nabla\cdot\mathbf{A} = 0$, variational derivatives with respect to $\dot{\mathbf{A}}$ are not really well defined. If the variation of L under a change $\delta\dot{\mathbf{A}}$ in $\dot{\mathbf{A}}$ is $\delta L = \int d^3x\, \boldsymbol{\mathcal{P}}\cdot\delta\dot{\mathbf{A}}$, then since $\nabla\cdot\delta\dot{\mathbf{A}} = 0$, we also have $\delta L = \int d^3x\, [\boldsymbol{\mathcal{P}} + \nabla\mathcal{F}]\cdot\delta\dot{\mathbf{A}}$ for any scalar function $\mathcal{F}(\mathbf{x})$. Thus all we can conclude from inspection of the Lagrangian is that $\mathbf{\Pi}$ equals $\dot{\mathbf{A}}(\mathbf{x}) + \nabla A^0(\mathbf{x})$ plus the gradient of some scalar. This ambiguity is removed by condition (8.3.2), which requires that $\nabla\cdot\mathbf{\Pi} = -J^0 = \nabla^2 A^0$. Because $\nabla\cdot\mathbf{A} = 0$, we conclude that Eq. (8.3.6) does indeed give the correct formula for Π^i.

Although the commutation relations (8.3.5) are reasonably simple, we must face the complication that $\mathbf{\Pi}$ does not commute with matter fields and their canonical conjugates. If F is any functional of these matter degrees of freedom, then its Dirac bracket with $\mathbf{A}$ vanishes, but its Dirac bracket with $\mathbf{\Pi}$ is

$$[F, \mathbf{\Pi}(\mathbf{z})]_D = -\int d^3x\, d^3y\, [F, \chi_{2\mathbf{x}}]_P\frac{1}{4\pi|\mathbf{x} - \mathbf{y}|}[\chi_{1\mathbf{y}}, \mathbf{\Pi}(\mathbf{z})]_P$$
$$= -\int d^3x\, d^3y\, [F, J^0(\mathbf{x})]_P\frac{1}{4\pi|\mathbf{x} - \mathbf{y}|}\nabla\delta^3(\mathbf{y} - \mathbf{z})$$
$$= -\int d^3y\, [F, A^0(\mathbf{y})]_P\nabla\delta^3(\mathbf{y} - \mathbf{z})$$
$$= [F, \nabla A^0(\mathbf{z})]_P = [F, \nabla A^0(\mathbf{z})]_D.$$

In order to facilitate the transition to the interaction picture, instead of expressing the Hamiltonian in terms of $\mathbf{A}$ and $\mathbf{\Pi}$, we shall write it in terms of $\mathbf{A}$ and $\mathbf{\Pi}_\perp$, where $\mathbf{\Pi}_\perp$ is the solenoidal part of $\mathbf{\Pi}$:

$$\mathbf{\Pi}_\perp \equiv \mathbf{\Pi} - \nabla A^0 = \dot{\mathbf{A}}, \tag{8.3.7}$$

for which $[F, \mathbf{\Pi}_\perp(\mathbf{z})]$ vanishes. By using the facts that $\mathbf{\Pi}_\perp(\mathbf{x})$ commutes with $\mathbf{\Pi}(\mathbf{y}) - \mathbf{\Pi}_\perp(\mathbf{y}) = \nabla A^0(\mathbf{y})$ and that $\partial_i A^0(\mathbf{x})$ commutes with $\partial_j A^0(\mathbf{y})$, it is easy to see that $\mathbf{\Pi}_\perp(\mathbf{x})$ satisfies the same commutation relations (8.3.5) as $\mathbf{\Pi}(\mathbf{x})$, and also the simple constraint

$$\nabla\cdot\mathbf{\Pi}_\perp = 0. \tag{8.3.8}$$

Now we need to construct a Hamiltonian. According to the general results of the Appendix to Chapter 7, we can apply the usual relation between the Hamiltonian and Lagrangian using the constrained variables $\mathbf{A}$ and $\mathbf{\Pi}_\perp$, without first having explicitly to write the Hamiltonian in terms of the unconstrained Qs and Ps. In electrodynamics, this gives

$$H = \int d^3x\, [\Pi_{\perp i}\dot{A}^i + P_n\dot{Q}^n - \mathcal{L}], \tag{8.3.9}$$

where, as mentioned earlier, Q^n and P_n are to be understood as the matter canonical fields and their canonical conjugates. (We can use $\mathbf{\Pi}_\perp$ in place of $\mathbf{\Pi}$ in Eq. (8.3.9) because $\nabla\cdot\dot{\mathbf{A}} = 0$.)

To be specific, consider a theory with a Lagrangian density of the form

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + J_\mu A^\mu + \mathcal{L}_{\text{matter}}, \tag{8.3.10}$$

where J_μ is a current that does not involve A^μ, and $\mathcal{L}_{\text{matter}}$ is the Lagrangian for whatever other fields do appear in J^μ, aside from their electromagnetic interactions, which are given explicitly by the term $J_\mu A^\mu$ in Eq. (8.3.10). (The electrodynamics of spin $\frac{1}{2}$ particles has a Lagrangian of this form, but the electrodynamics of spinless particles is more complicated.) Replacing $\dot{\mathbf{A}}$ everywhere with $\mathbf{\Pi}_\perp$, this gives a Hamiltonian (8.3.9) of the form

$$H = \int d^3x\left[\mathbf{\Pi}_\perp^2 + \frac{1}{2}(\nabla\times\mathbf{A})^2 - \frac{1}{2}(\mathbf{\Pi}_\perp + \nabla A^0)^2 - \mathbf{J}\cdot\mathbf{A} + J^0A^0\right] + H_M,$$

where H_M is the Hamiltonian for matter fields, excluding their electromagnetic interactions

$$H_M \equiv \int d^3x\, (P_n\dot{Q}^n - \mathcal{L}_{\text{matter}}).$$

Using the solution (8.2.9) for A^0, this is

$$H = \int d^3x\left[\frac{1}{2}\mathbf{\Pi}_\perp^2 + \frac{1}{2}(\nabla\times\mathbf{A})^2 - \mathbf{J}\cdot\mathbf{A} + \frac{1}{2}J^0A^0\right] + H_M. \tag{8.3.11}$$

The term $\frac{1}{2}J^0A^0$ may look peculiar, but this is nothing but the familiar Coulomb energy

$$V_{\text{Coul}} = \frac{1}{2}\int d^3x\, J^0A^0$$
$$= \frac{1}{2}\int d^3x\int d^3y\, \frac{J^0(\mathbf{x})J^0(\mathbf{y})}{4\pi|\mathbf{x} - \mathbf{y}|}. \tag{8.3.12}$$

The reader can verify, using the commutation relations (8.3.5), that the rate of change of any operator function F of $\mathbf{A}$ and $\mathbf{\Pi}$ is given by $i\dot{F} = [F, H]$, as it should be.

8.4 Electrodynamics in the Interaction Picture

We now break up the Hamiltonian (8.3.11) into a free-particle term H_0 and an interaction H_{int}

$$H = H_0 + H_{\text{int}}, \tag{8.4.1}$$

$$H_0 = \int d^3x\left[\frac{1}{2}\mathbf{\Pi}_\perp^2 + \frac{1}{2}(\nabla\times\mathbf{A})^2\right] + H_{\text{matter},0}, \tag{8.4.2}$$

$$H_{\text{int}} = -\int d^3x\, \mathbf{J}\cdot\mathbf{A} + V_{\text{Coul}} + H_{\text{int,matter}}, \tag{8.4.3}$$

where $H_{\text{matter},0}$ and $H_{\text{int,matter}}$ are the free-particle and interaction terms in H_{matter}, and V_{Coul} is the Coulomb interaction (8.3.12). The total Hamiltonian (8.4.1) is time-independent, so Eqs. (8.4.2) and (8.4.3) can be evaluated at any time we like (as long as both are evaluated at the same time), in particular at $t = 0$. As in Chapter 7, the transition to the interaction picture is made by applying the similarity transformation

$$H_I(t) = \exp(iH_0t)\, H_{\text{int}}[\mathbf{A}, \mathbf{\Pi}_\perp, Q, P]_{t=0}\exp(-iH_0t)$$
$$= H_{\text{int}}[\mathbf{a}(t), \boldsymbol{\pi}(t), q(t), p(t)], \tag{8.4.4}$$

where P here denotes the canonical conjugates to the matter fields Q, and any operator $O(\mathbf{x}, t)$ in the interaction picture is related to its value $O(\mathbf{x}, 0)$ in the Heisenberg picture at $t = 0$ by

$$O(\mathbf{x}, t) = \exp(iH_0t)O(\mathbf{x}, 0)\exp(-iH_0t), \tag{8.4.5}$$

so that

$$i\dot{O}(\mathbf{x}, t) = [O(\mathbf{x}, t), H_0]. \tag{8.4.6}$$

(We are dropping the subscript $\perp$ on $\boldsymbol{\pi}(\mathbf{x})$.) Since Eq. (8.4.5) is a similarity transformation, the equal-time commutation relations are the same as in the Heisenberg picture:

$$[a^i(\mathbf{x}, t), \pi^j(\mathbf{y}, t)] = i\left[\delta_{ij}\delta^3(\mathbf{x} - \mathbf{y}) + \frac{\partial^2}{\partial x^i\partial x^j}\frac{1}{4\pi|\mathbf{x} - \mathbf{y}|}\right], \tag{8.4.7}$$
$$[a^i(\mathbf{x}, t), a^j(\mathbf{y}, t)] = 0, \tag{8.4.8}$$
$$[\pi^i(\mathbf{x}, t), \pi^j(\mathbf{y}, t)] = 0, \tag{8.4.9}$$

and likewise for the matter fields and their conjugates. For the same reason, the constraints (8.2.6) and (8.3.8) still apply

$$\nabla\cdot\mathbf{a} = 0, \tag{8.4.10}$$
$$\nabla\cdot\boldsymbol{\pi} = 0. \tag{8.4.11}$$

To establish the relation between $\boldsymbol{\pi}$ and $\dot{\mathbf{a}}$, we must use Eq. (8.4.6) to evaluate $\dot{\mathbf{a}}$:

$$i\dot{a}_i(\mathbf{x}, t) = [a_i(\mathbf{x}, t), H_0]$$
$$= i\int d^3y\left[\delta_{ij}\delta^3(\mathbf{x} - \mathbf{y}) + \frac{\partial^2}{\partial x^i\partial x^j}\frac{1}{4\pi|\mathbf{x} - \mathbf{y}|}\right]\pi_j(\mathbf{y}, t).$$

We can replace $\partial/\partial x^j$ in the second term with $-\partial/\partial y^j$, integrate by parts, and use Eq. (8.4.11), yielding

$$\dot{\mathbf{a}} = \boldsymbol{\pi} \tag{8.4.12}$$

just as in the Heisenberg picture. The field equation is likewise determined by

$$i\dot{\pi}_i(\mathbf{x}, t) = [\pi_i(\mathbf{x}, t), H_0]$$
$$= -i\int d^3y\left[\delta_{ij}\delta^3(\mathbf{x} - \mathbf{y}) + \frac{\partial^2}{\partial x^i\partial x^j}\frac{1}{4\pi|\mathbf{x} - \mathbf{y}|}\right](\nabla\times\nabla\times\mathbf{a}(\mathbf{y}, t))_j,$$

which (using Eqs. (8.4.10) and (8.4.12)) just yields the usual wave equation

$$\Box\mathbf{a} = 0. \tag{8.4.13}$$

Since A^0 is not an independent Heisenberg-picture field variable, but rather a functional (8.2.9) of the matter fields and their canonical conjugates that vanishes in the limit of zero charges, we do not introduce any corresponding operator a^0 in the interaction picture, but rather take

$$a^0 = 0. \tag{8.4.14}$$

The most general real solution of Eqs. (8.4.10), (8.4.13), and (8.4.14) may be written

$$a^\mu(x) = (2\pi)^{-3/2}\int\frac{d^3p}{\sqrt{2p^0}}\sum_\sigma\left[e^{ip\cdot x}e^\mu(\mathbf{p}, \sigma)a_{\mathbf{p},\sigma}\right.$$
$$\left.+ e^{-ip\cdot x}e^{\mu*}(\mathbf{p}, \sigma)a^\dagger_{\mathbf{p},\sigma}\right], \tag{8.4.15}$$

where $p^0 \equiv |\mathbf{p}|$; $e^\mu(\mathbf{p}, \sigma)$ are any two independent 'polarization vectors' satisfying

$$\mathbf{p}\cdot\mathbf{e}(\mathbf{p}, \sigma) = 0, \tag{8.4.16}$$
$$e^0(\mathbf{p}, \sigma) = 0, \tag{8.4.17}$$

and $a_{\mathbf{p},\sigma}$ are a pair of operator coefficients, with σ a two-valued index. By adjusting the normalization of $a_{\mathbf{p},\sigma}$, we can normalize the $e^\mu(\mathbf{p}, \sigma)$ so that the completeness relation reads

$$\sum_\sigma e^i(\mathbf{p}, \sigma)e^j(\mathbf{p}, \sigma)^* = \delta_{ij} - \frac{p_ip_j}{|\mathbf{p}|^2}. \tag{8.4.18}$$

For instance, we could take the $e(\mathbf{p}, \sigma)$ to be the same polarization vectors that we encountered in Section 5.9:

$$e^\mu(\mathbf{p}, \pm 1) = R(\hat{\mathbf{p}})\begin{bmatrix}0\\ 1/\sqrt{2}\\ \pm i/\sqrt{2}\\ 0\end{bmatrix}, \tag{8.4.19}$$

where $R(\hat{\mathbf{p}})$ is a standard rotation that carries the three-axis into the direction of $\mathbf{p}$. Using Eqs. (8.4.18) and (8.4.12), we can easily see that the commutation relations (8.4.7)–(8.4.9) are satisfied if (and in fact only if) the operator coefficients in Eq. (8.4.15) satisfy

$$[a_{\mathbf{p},\sigma}, a^\dagger_{\mathbf{p}',\sigma'}] = \delta^3(\mathbf{p} - \mathbf{p}')\delta_{\sigma\sigma'}, \tag{8.4.20}$$
$$[a_{\mathbf{p},\sigma}, a_{\mathbf{p}',\sigma'}] = 0. \tag{8.4.21}$$

As remarked before for massive particles, this result should be regarded not so much as an alternative derivation of Eqs. (8.4.20) and (8.4.21), but rather as a verification that Eq. (8.4.2) gives the correct Hamiltonian for free massless particles of helicity ± 1. In the same spirit one can also use Eqs. (8.4.12) and (8.4.15) in Eq. (8.4.2) to calculate the free-photon Hamiltonian

$$H_0 = \int d^3p\sum_\sigma\frac{1}{2}p^0[a_{\mathbf{p},\sigma}, a^\dagger_{\mathbf{p},\sigma}]_+$$
$$= \int d^3p\sum_\sigma p^0\left(a^\dagger_{\mathbf{p},\sigma}a_{\mathbf{p},\sigma} + \frac{1}{2}\delta^3(\mathbf{p} - \mathbf{p})\right) \tag{8.4.22}$$

which (aside from an inconsequential infinite c-number term) is just what we should expect.

Finally, we record that the interaction (8.4.4) in the interaction picture is

$$H_I(t) = -\int d^3x\, j_\mu(\mathbf{x}, t)a^\mu(\mathbf{x}, t) + V_{\text{Coul}}(t) + H_{I,\text{matter}}(t), \tag{8.4.23}$$

where in terms of the current J in the Heisenberg picture

$$j_\mu(\mathbf{x}, t) \equiv \exp(iH_0t)J_\mu(\mathbf{x}, 0)\exp(-iH_0t), \tag{8.4.24}$$

while $V_{\text{Coul}}(t)$ is the Coulomb term

$$V_{\text{Coul}}(t) = \exp(iH_0t)V_{\text{Coul}}\exp(-iH_0t)$$
$$= \frac{1}{2}\int d^3x\, d^3y\, \frac{j^0(\mathbf{x}, t)j^0(\mathbf{y}, t)}{4\pi|\mathbf{x} - \mathbf{y}|} \tag{8.4.25}$$

and $H_{I,\text{matter}}(t)$ is the non-electromagnetic part of the matter field interaction in the interaction picture:

$$H_{I,\text{matter}}(t) = \exp(iH_0t)H_{\text{int,matter}}\exp(-iH_0t). \tag{8.4.26}$$

We have written $j_\mu a^\mu$ instead of $\mathbf{j}\cdot\mathbf{a}$ in Eq. (8.4.23), but these are equal because a^μ has been defined to have $a^0 = 0$.

8.5 The Photon Propagator

The general Feynman rules described in Chapter 6 dictate that an internal photon line in a Feynman diagram contributes a factor to the corresponding term in the S-matrix, given by the propagator:

$$-i\Delta_{\mu\nu}(x - y) \equiv \langle\Omega|T\{a_\mu(x), a_\nu(y)\}|\Omega\rangle, \tag{8.5.1}$$

where T as usual denotes a time-ordered product. Inserting our formula (8.4.15) for the electromagnetic potential then yields

$$-i\Delta_{\mu\nu}(x - y) = \int\frac{d^3p}{(2\pi)^3 2|\mathbf{p}|}P_{\mu\nu}(\mathbf{p})\left[e^{ip\cdot(x-y)}\theta(x^0 - y^0) + e^{ip\cdot(y-x)}\theta(y^0 - x^0)\right], \tag{8.5.2}$$

where

$$P_{\mu\nu}(\mathbf{p}) \equiv \sum_{\sigma=\pm 1}e_\mu(\mathbf{p}, \sigma)e_\nu(\mathbf{p}, \sigma)^*, \tag{8.5.3}$$

and p^0 in the exponentials is taken with $p^0 = |\mathbf{p}|$. We recall from Eqs. (8.4.18) and (8.4.17) that

$$P_{ij}(\mathbf{p}) = \delta_{ij} - \frac{p^ip^j}{|\mathbf{p}|^2},$$
$$P_{0i}(\mathbf{p}) = P_{i0}(\mathbf{p}) = P_{00}(\mathbf{p}) = 0. \tag{8.5.4}$$

As we saw in Chapter 6, the theta functions in Eq. (8.5.2) may be expressed as integrals over an independent time-component q^0 of an off-shell four-momentum q^μ, so that Eq. (8.5.2) may be rewritten

$$\Delta_{\mu\nu}(x - y) = (2\pi)^{-4}\int d^4q\, \frac{P_{\mu\nu}(\mathbf{q})}{q^2 - i\epsilon}e^{iq\cdot(x-y)}. \tag{8.5.5}$$

Thus in using the Feynman rules in momentum space, the contribution of an internal photon line carrying four-momentum q that runs between vertices where the photon is created and destroyed by fields a^μ and a^ν is

$$\frac{-i}{(2\pi)^4}\frac{P_{\mu\nu}(\mathbf{q})}{q^2 - i\epsilon}. \tag{8.5.6}$$

It will be very useful (though apparently perverse) to rewrite Eq. (8.5.4) as

$$P_{\mu\nu}(\mathbf{q}) = \eta_{\mu\nu} + \frac{q^0q_\mu n_\nu + q^0q_\nu n_\mu - q_\mu q_\nu + q^2n_\mu n_\nu}{|\mathbf{q}|^2}, \tag{8.5.7}$$

where $n^\mu = (1, 0, 0, 0)$ is a fixed time-like vector, q^2 as usual is $|\mathbf{q}|^2 - (q^0)^2$, but q^0 is here entirely arbitrary. We shall choose q^0 in Eq. (8.5.7) to be given by four-momentum conservation: it is the difference of the matter p^0s flowing in and out of the vertex where the photon line is created. The terms proportional to q_μ and/or q_ν then do not contribute to the S-matrix, because the factors q_μ or q_ν act like derivatives ∂_μ and ∂_ν, and the photon fields a_μ and a_ν are coupled to currents j^μ and j^ν that satisfy the conservation condition $\partial_\mu j^\mu = 0$.[86] The term proportional to $n_\mu n_\nu$ contains a factor q^2 that cancels the q^2 in the denominator of the propagator, yielding a term that is the same as would be produced by a term in the action:

$$-i\frac{1}{2}\int d^4x\int d^4y\,[-ij^0(x)][-ij^0(y)]\frac{-i}{(2\pi)^4}\int\frac{d^4q}{|\mathbf{q}|^2}e^{iq\cdot(x-y)}.$$

The integral over q^0 here yields a delta function in time, so this is equivalent to a correction to the interaction Hamiltonian $H_I(t)$, of the form

$$-\frac{1}{2}\int d^3x\int d^3y\, \frac{j^0(\mathbf{x}, t)j^0(\mathbf{y}, t)}{4\pi|\mathbf{x} - \mathbf{y}|}.$$

This is just right to cancel the Coulomb interaction (8.4.25). Our result is that the photon propagator can be taken effectively as the covariant quantity

$$\Delta^{\text{eff}}_{\mu\nu}(x - y) = (2\pi)^{-4}\int d^4q\, \frac{\eta_{\mu\nu}}{q^2 - i\epsilon}e^{iq\cdot(x-y)} \tag{8.5.8}$$

[86] This argument as given here is little better than hand-waving. The result has been justified by a detailed analysis of Feynman diagrams [3], but the easiest way to treat this problem is by path-integral methods, as discussed in Section 9.6.

with the Coulomb interaction dropped from now on. We see that the apparent violation of Lorentz invariance in the instantaneous Coulomb interaction is cancelled by another apparent violation of Lorentz invariance, that as noted in Section 5.9 the fields $a^\mu(x)$ are not four-vectors, and therefore have a non-covariant propagator. From a practical point of view, the important point is that in the momentum space Feynman rules, the contribution of an internal photon line is simply given by

$$\frac{-i}{(2\pi)^4}\frac{\eta_{\mu\nu}}{q^2 - i\epsilon} \tag{8.5.9}$$

and the Coulomb interaction is dropped.

8.6 Feynman Rules for Spinor Electrodynamics

We are now in a position to state the Feynman rules for calculating the S-matrix in quantum electrodynamics. For definiteness, we will consider the electrodynamics of a single species of spin $\frac{1}{2}$ particles of charge $q = -e$ and mass m. We will call these fermions electrons, but the same formalism applies to muons and other such particles. The simplest gauge- and Lorentz-invariant Lagrangian for this theory is[87]

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{\Psi}(\gamma^\mu[\partial_\mu + ieA_\mu] + m)\Psi. \tag{8.6.1}$$

The electric current four-vector is then simply

$$J^\mu = \frac{\partial\mathcal{L}}{\partial A_\mu} = -ie\bar{\Psi}\gamma^\mu\Psi. \tag{8.6.2}$$

The interaction (8.4.23) in the interaction picture is here

$$H_I(t) = +ie\int d^3x\,(\bar{\psi}(\mathbf{x}, t)\gamma^\mu\psi(\mathbf{x}, t))a_\mu(\mathbf{x}, t) + V_{\text{Coul}}(t). \tag{8.6.3}$$

(There is no $H_{I,\text{matter}}$ here.) As we have seen, the Coulomb term $V_{\text{Coul}}(t)$ just serves to cancel a part of the photon propagator that is non-covariant and local in time.

Following the general results of Section 6.3, we can state the momentum space Feynman rules for the connected part of the S-matrix in this theory as follows:

(i) Draw all Feynman diagrams with up to some given number of vertices. The diagrams consist of electron lines carrying arrows and photon lines without arrows, with the lines joined at vertices, at each of which there is one incoming and one outgoing electron line and one photon line. There is one external line coming into the diagram from below or going upwards out of the diagram for each particle in the initial or final states, respectively; electrons are represented by external lines carrying arrows pointing upwards into or out of the diagram, while positrons are represented by lines carrying arrows pointing downwards into or out of the diagram. There are also as many internal lines as are needed to give each

[87] In Chapter 12 we will discuss reasons why more complicated terms are excluded from the Lagrangian density.

vertex the required number of attached lines. Each internal line is labelled with an off-mass-shell four-momentum flowing in a definite direction along the line (taken conventionally to flow along the direction of the arrow for electron lines). Each external line is labelled with the momentum and spin z-component or helicity of the electron or photon in the initial and final states.

(ii) Associate factors with the components of the diagram as follows:

Vertices

Label each vertex with a four-component Dirac index α at the electron line with its arrow coming into the vertex, a Dirac index β at the electron line with its arrow going out of the vertex, and a spacetime index μ at the photon line. For each such vertex, include a factor

$$(2\pi)^4e(\gamma^\mu)_{\beta\alpha}\delta^4(k - k' + q), \tag{8.6.4}$$

where k and k' are the electron four-momenta entering and leaving the vertex, and q is the photon four-momentum entering the vertex (or minus the photon momentum leaving the vertex).

External lines:

Label each external line with the three-momentum $\mathbf{p}$ and spin z-component or helicity σ of the particle in the initial or final state. For each line for an electron in the final state running out of a vertex carrying a Dirac label β on this line, include a factor[88]

$$\frac{\bar{u}_\beta(\mathbf{p}, \sigma)}{(2\pi)^{3/2}}. \tag{8.6.5}$$

For each line for a positron in the final state running into a vertex carrying a Dirac label α on this line, include a factor

$$\frac{v_\alpha(\mathbf{p}, \sigma)}{(2\pi)^{3/2}}. \tag{8.6.6}$$

For each line for an electron in the initial state running into a vertex carrying a Dirac label α on this line, include a factor

$$\frac{u_\alpha(\mathbf{p}, \sigma)}{(2\pi)^{3/2}}. \tag{8.6.7}$$

For each line for a positron in the initial state running out of a vertex carrying a Dirac label β on this line, include a factor

$$\frac{\bar{v}_\beta(\mathbf{p}, \sigma)}{(2\pi)^{3/2}}. \tag{8.6.8}$$

The us and vs are the four-component spinors discussed in Section 5.5. For each line for a photon in the final state connected to a vertex carrying a spacetime label μ on this line, include a factor

$$\frac{e^*_\mu(\mathbf{p}, \sigma)}{(2\pi)^{3/2}\sqrt{2p^0}}. \tag{8.6.9}$$

[88] A matrix β has been extracted from the interaction in (8.6.4), so that $\bar{u}$ and $\bar{v}$ appear instead of $u^\dagger$ and $v^\dagger$.

For each line for a photon in the initial state connected to a vertex carrying a spacetime label μ on this line, include a factor

$$\frac{e_\mu(\mathbf{p},\sigma)}{(2\pi)^{3/2}\sqrt{2p^0}}. \tag{8.6.10}$$

The e_μ are the photon polarization four-vectors described in the previous section.

Internal lines:

For each internal electron line carrying a four-momentum k and running from a vertex carrying a Dirac label β to another vertex carrying a Dirac label α, include a factor

$$\frac{-i}{(2\pi)^4}\frac{[-i\not{k}+m]_{\alpha\beta}}{k^2+m^2-i\epsilon}. \tag{8.6.11}$$

(We are here using the very convenient 'Dirac slash' notation: for any four-vector v^μ, $\not{v}$ denotes $\gamma_\mu v^\mu$.) For each internal photon line carrying a four-momentum q that runs between two vertices carrying spacetime labels μ and ν include a factor

$$\frac{-i}{(2\pi)^4}\frac{\eta_{\mu\nu}}{q^2-i\epsilon}. \tag{8.6.12}$$

(iii) Integrate the product of all these factors over the four-momenta carried by the internal lines, and sum over all Dirac and spacetime indices.

(iv) Add up the results obtained in this way from each Feynman diagram.

Additional combinatoric factors and fermionic signs may need to be included, as described in parts (v) and (vi) of Section 6.1.

The difficulty of evaluating Feynman diagrams increases rapidly with the number of internal lines and vertices, so it is important to have some idea of what numerical factors tend to suppress the contributions of the more complicated diagrams. We shall estimate these numerical factors including not only the factors of the electronic charge e associated with vertices, but also the factors of 2 and π from vertices, propagators, and momentum space integrals.

Consider a connected Feynman diagram with V vertices, I internal lines, E external lines, and L loops. These quantities are not independent, but are subject to relations already used in Section 6.3:

$$L=I-V+1, \qquad 2I+E=3V.$$

There is a factor $e(2\pi)^4$ from each vertex, a factor $(2\pi)^{-4}$ from each internal line, and a four-dimensional momentum space integral for each loop. The volume element in four-dimensional Euclidean space in terms of a radius parameter κ is $\pi^2\kappa^2 d\kappa^2$, so each loop contributes a factor π^2. Thus the diagram will contain a factor

$$(2\pi)^{4V}e^V(2\pi)^{-4I}\pi^{2L}=(2\pi)^4 e^{E-2}\left(\frac{e^2}{16\pi^2}\right)^L.$$

The number E of external lines is fixed for a given process, so we see that the expansion parameter that governs the suppression of Feynman graphs for each additional loop is

$$\frac{e^2}{16\pi^2}=\frac{\alpha}{4\pi}=5.81\cdot10^{-4}.$$

Fortunately this is small enough that good accuracy can usually be obtained from Feynman diagrams with at most a few loops.

* * *

We must say a little more about the spin states of photons and electrons in realistic experiments, where not every particle in the initial and final states has a definite known helicity or spin z-component. This consideration is especially important for photons, which in practice are often characterized by a state of transverse or elliptical polarization rather than helicity. As we saw in the previous section, for photons of helicity ±1, the polarization vectors are

$$e^\mu(\mathbf{p},\pm1)=R(\hat{\mathbf{p}})\begin{bmatrix}0\\1/\sqrt{2}\\\pm i/\sqrt{2}\\0\end{bmatrix},$$

where $R(\hat{\mathbf{p}})$ is the standard rotation that takes the z-axis to the $\mathbf{p}$ direction. These are not the only possible photon states; in general, a photon state can be a linear combination of helicity states $|\mathbf{p},\pm1\rangle$

$$\alpha_+|\mathbf{p},+1\rangle+\alpha_-|\mathbf{p},-1\rangle \tag{8.6.13}$$

which is properly normalized if

$$|\alpha_+|^2+|\alpha_-|^2=1. \tag{8.6.14}$$

To calculate the S-matrix element for absorbing or emitting such a photon, we simply replace $e_\mu(\mathbf{p},\pm1)$ in the Feynman rules with

$$e_\mu(\mathbf{p})=\alpha_+e_\mu(\mathbf{p},+1)+\alpha_-e_\mu(\mathbf{p},-1). \tag{8.6.15}$$

The polarization vectors for definite helicity satisfy the normalization condition

$$e^*_\mu(\mathbf{p},\lambda')e^\mu(\mathbf{p},\lambda)=\delta_{\lambda'\lambda} \tag{8.6.16}$$

and therefore in general

$$e^*_\mu(\mathbf{p})e^\mu(\mathbf{p})=1. \tag{8.6.17}$$

The two extreme cases are *circular polarization*, for which $\alpha_-=0$ or $\alpha_+=0$, and *linear polarization*, for which $|\alpha_+|=|\alpha_-|=1/\sqrt{2}$. For linear polarization, by an adjustment of the overall phase of the state (8.6.13), we can make α_+ and α_- complex conjugates, so that they can be expressed as

$$\alpha_\pm=\exp(\mp i\phi)/\sqrt{2}. \tag{8.6.18}$$

Then in the Feynman rules we should use a polarization vector

$$e_\mu(\mathbf{p})=R(\hat{\mathbf{p}})\begin{bmatrix}0\\\cos\phi\\\sin\phi\\0\end{bmatrix}. \tag{8.6.19}$$

That is, ϕ is the azimuthal angle of the photon polarization in the plane perpendicular to $\mathbf{p}$. Note that the photon polarization vector here is real, which is only possible for linear polarization. In between the extremes of circular and linear polarization are the states of *elliptic polarization*, for which $|\alpha_+|$ and $|\alpha_-|$ are non-zero and unequal.

More generally, an initial photon may be prepared in a statistical mixture of spin states. In the most general case, an initial photon may have any number of possible polarization vectors $e^{(r)}_\mu(\mathbf{p})$, each with probability P_r. The rate for absorbing such a photon in a given process will then be of the form

$$\Gamma=\sum_r P_r|e^{(r)}_\mu(\mathbf{p})M^\mu|^2=M^{\mu*}M^\nu\rho_{\nu\mu}, \tag{8.6.20}$$

where ρ is the *density matrix*

$$\rho_{\nu\mu}\equiv\sum_r P_r e^{(r)}_\nu(\mathbf{p})e^{(r)*}_\mu(\mathbf{p}). \tag{8.6.21}$$

Since ρ is obviously a Hermitian positive matrix of unit trace (because $\sum_r P_r=1$) with $\rho_{\mu0}=\rho_{0\mu}=0$ and $\rho_{\nu\mu}p^\nu=\rho_{\nu\mu}p^\mu=0$, it may be written as

$$\rho_{\nu\mu}=\sum_{s=1,2}\lambda_s e_\nu(\mathbf{p};s)e^*_\mu(\mathbf{p};s), \tag{8.6.22}$$

where $e_\mu(\mathbf{p};s)$ are the two orthonormal eigenvectors of ρ with

$$e_0(\mathbf{p};s)=e_\mu(\mathbf{p};s)p^\mu=0 \tag{8.6.23}$$

and λ_s are the corresponding eigenvalues, with

$$\lambda_s\geq0,\qquad\sum_{s=1,2}\lambda_s=1.$$

We may then write the rate for the photon absorption process as

$$\Gamma=\sum_{s=1,2}\lambda_s|e_\nu(\mathbf{p};s)M^\nu|^2. \tag{8.6.24}$$

Thus any statistical mixture of initial photon states is always equivalent to having just two orthonormal polarizations $e_\nu(\mathbf{p};s)$ with probabilities λ_s.

In particular, if we know nothing whatever about the initial photon polarization, then the two probabilities λ_s for the polarization vectors $e_\nu(\mathbf{p};s)$ are equal, so that $\lambda_1=\lambda_2=\frac12$,

and the density matrix (and hence the absorption rate) is an average over initial polarizations

$$\rho_{ij}=\frac12\sum_{s=1,2}e_i(\mathbf{p};s)e^*_j(\mathbf{p};s)=\frac12(\delta_{ij}-\hat{p}_i\hat{p}_j). \tag{8.6.25}$$

Fortunately, this result does not depend on the particular pair of polarization vectors $e_i(\mathbf{p};s)$ over which we average; for unpolarized photons we can average the absorption rate over any pair of orthonormal polarization vectors. Similarly, if we make no attempt to measure the polarization of a photon in the final state, then the rate may be calculated by summing over any pair of orthonormal final photon polarization vectors.

The same remarks apply to electrons and positrons; if (as is usually the case) we make no attempt to prepare an electron or positron so that some spin states are more likely than others, then the rate is to be calculated by *averaging* over any two orthonormal initial spin states, such as those with spin z-component $\sigma=\pm\frac12$; if we make no attempt to measure a final electron's or positron's spin state, then we must *sum* the rate over any two orthonormal initial spin states, such as those with spin z-component $\sigma=\pm\frac12$. Such sums may be performed using the relations (5.5.37) and (5.5.38):

$$\sum_\sigma u_\alpha(\mathbf{p},\sigma)\bar{u}_\beta(\mathbf{p},\sigma)=\left(\frac{-i\not{p}+m}{2p^0}\right)_{\alpha\beta}, \tag{8.6.26}$$

$$\sum_\sigma v_\alpha(\mathbf{p},\sigma)\bar{v}_\beta(\mathbf{p},\sigma)=\left(\frac{-i\not{p}-m}{2p^0}\right)_{\alpha\beta}, \tag{8.6.27}$$

where $p^0=\sqrt{\mathbf{p}^2+m^2}$. For instance, if the initial state contains an electron with momentum $\mathbf{p}$ and spin z-component σ, and a positron with momentum $\mathbf{p}'$ and spin z-component σ', then the S-matrix element for the process will be of the form $\bar{v}_\alpha(\mathbf{p}',\sigma')\mathcal{M}_{\alpha\beta}u_\beta(\mathbf{p},\sigma)$. Hence if neither electron nor positron spins are observed, the rate will be proportional to

$$\frac14\sum_{\sigma',\sigma}|\bar{v}_\alpha(\mathbf{p}',\sigma')\mathcal{M}_{\alpha\beta}u_\beta(\mathbf{p},\sigma)|^2=\frac14\mathrm{Tr}\left\{\beta\mathcal{M}^\dagger\beta\left(\frac{-i\not{p}'-m}{2p'^0}\right)\mathcal{M}\left(\frac{-i\not{p}+m}{2p^0}\right)\right\}.$$

Techniques for the calculation of such traces are described in the Appendix to this chapter.

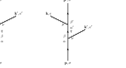


Figure 8.1. The two lowest-order Feynman diagrams for Compton scattering. Straight lines are electrons; wavy lines are photons.

8.7 Compton Scattering

As an example of the methods described in this chapter, we shall consider here the scattering of a photon by an electron (or other particle of spin $\frac12$ and charge $-e$), to lowest order in e. We label the initial and final photon momenta and polarization vectors by k^μ, e^μ and k'^μ, e'^μ, where $k^0=|\mathbf{k}|$ and $k'^0=|\mathbf{k}'|$. Also, the initial and final electron momenta and spin z-components are labelled p^μ,σ and p'^μ,σ', where $p^0=\sqrt{\mathbf{p}^2+m^2}$ and $p'^0=\sqrt{\mathbf{p}'^2+m^2}$, with m the electron mass. The lowest order Feynman diagrams for this process are shown in Figure 8.1. Using the rules outlined in the previous section, the corresponding S-matrix element is

$$\begin{aligned}&S(\mathbf{p},\sigma+\mathbf{k},e\to\mathbf{p}',\sigma'+\mathbf{k}',e')\\&=\frac{\bar{u}(\mathbf{p}',\sigma')_{\beta'}}{(2\pi)^{3/2}}\frac{e'^*_\nu}{(2\pi)^{3/2}\sqrt{2k'^0}}\frac{u(\mathbf{p},\sigma)_\alpha}{(2\pi)^{3/2}}\frac{e_\mu}{(2\pi)^{3/2}\sqrt{2k^0}}\int d^4q\left[\frac{-i}{(2\pi)^4}\right]\left[\frac{-i\not{q}+m}{q^2+m^2-i\epsilon}\right]_{\alpha'\beta}\\&\cdot\Big\{[e(2\pi)^4\gamma^\nu_{\beta'\alpha'}\delta^4(q-p'-k')][e(2\pi)^4\gamma^\mu_{\beta\alpha}\delta^4(q-p-k)]\\&+[e(2\pi)^4\gamma^\mu_{\beta'\alpha'}\delta^4(q+k-p')][e(2\pi)^4\gamma^\nu_{\beta\alpha}\delta^4(q+k'-p)]\Big\}.\end{aligned} \tag{8.7.1}$$

Performing the (trivial) q-integral, collecting factors of i and 2π, and rewriting the result in matrix notation, we have more simply

$$S=\frac{-ie^2\delta^4(p'+k'-p-k)}{(2\pi)^2\sqrt{2k'^0\,2k^0}}\bar{u}(\mathbf{p}',\sigma')\left[\not{e}'^*\frac{-i(\not{p}+\not{k})+m}{(p+k)^2+m^2}\not{e}+\not{e}\frac{-i(\not{p}-\not{k}')+m}{(p-k')^2+m^2}\not{e}'^*\right]u(\mathbf{p},\sigma). \tag{8.7.2}$$

(Here $\not{e}'^*$ means $e'^*_\mu\gamma^\mu$, not $(\not{e}')^*$. Also, we drop the $-i\epsilon$, because the denominators here do not vanish.) Because $p^2=-m^2$ and $k^2=k'^2=0$, the denominators can be simplified

$$(p+k)^2+m^2=2p\cdot k, \tag{8.7.3}$$

$$(p-k')^2+m^2=-2p\cdot k'. \tag{8.7.4}$$

Also, the 'Feynman amplitude' M is defined in general by Eq. (3.3.2), which (because some scattering is assumed to take place) here reads

$$S=-2\pi i\delta^4(p'+k'-p-k)M, \tag{8.7.5}$$

so

$$M=\frac{e^2}{4(2\pi)^3\sqrt{k^0k'^0}}\bar{u}(\mathbf{p}',\sigma')\Big\{\not{e}'^*[-i(\not{p}+\not{k})+m]\not{e}/(p\cdot k)-\not{e}[-i(\not{p}-\not{k}')+m]\not{e}'^*/(p\cdot k')\Big\}u(\mathbf{p},\sigma). \tag{8.7.6}$$

The differential cross-section is given in terms of M by Eq. (3.4.15), which here reads

$$d\sigma=(2\pi)^4u^{-1}|M|^2\delta^4(p'+k'-p-k)d^3p'\,d^3k'. \tag{8.7.7}$$

Since one of the particles here is massless, Eq. (3.4.17) for the initial velocity gives

$$u=|p\cdot k|/p^0k^0. \tag{8.7.8}$$

To go further, it will be convenient to adopt a specific coordinate frame. Since electrons in atoms move non-relativistically, the laboratory frame for high-energy (X ray or gamma ray) photon–electron scattering experiments is usually (though not always) one in which the initial electron can be taken to be at rest. We will adopt this frame here, so that

$$\mathbf{p}=0,\qquad p^0=m. \tag{8.7.9}$$

The velocity (8.7.8) is then simply

$$u=1. \tag{8.7.10}$$

To save writing, we denote the photon energies by

$$\omega\equiv k^0=|\mathbf{k}|=-p\cdot k/m, \tag{8.7.11}$$

$$\omega'\equiv k'^0=|\mathbf{k}'|=-p\cdot k'/m. \tag{8.7.12}$$

The three-momentum delta function in Eq. (8.7.7) just serves to eliminate the differential d^3p', setting $\mathbf{p}'=\mathbf{k}-\mathbf{k}'$. This leaves the remaining energy delta function

$$\delta(p'^0+k'^0-p^0-k^0)=\delta\left(\sqrt{(\mathbf{k}-\mathbf{k}')^2+m^2}+\omega'-m-\omega\right). \tag{8.7.13}$$

This fixes ω' to satisfy

$$\sqrt{\omega^2-2\omega\omega'\cos\theta+\omega'^2+m^2}=\omega+m-\omega',$$

where θ is the angle between $\mathbf{k}$ and $\mathbf{k}'$. Squaring both sides and cancelling ω'^2 terms gives[89, 90]

$$\omega'=\omega\frac{m}{m+\omega(1-\cos\theta)}\equiv\omega_c(\theta). \tag{8.7.14}$$

The energy delta function (8.7.13) can be written

$$\begin{aligned}\delta(p'^0+k'^0-p^0-k^0)&=\frac{\delta(\omega'-\omega_c(\theta))}{|\partial[\sqrt{\omega^2-2\omega\omega'\cos\theta+\omega'^2+m^2}+\omega']/\partial\omega'|}\\&=\frac{\delta(\omega'-\omega_c(\theta))}{|(\omega'-\omega\cos\theta)/p'^0+1|}=\frac{p'^0\omega'}{m\omega}\delta(\omega'-\omega_c(\theta)).\end{aligned} \tag{8.7.15}$$

Also, the differential d^3k' can be written

$$d^3k'=\omega'^2d\omega'd\Omega, \tag{8.7.16}$$

where $d\Omega$ is the solid angle into which the final photon is scattered. The final delta function in Eq. (8.7.15) just serves to eliminate the differential $d\omega'$ in Eq. (8.7.16), leaving us with a differential cross-section

$$d\sigma=(2\pi)^4|M|^2\frac{p'^0\omega'^3}{m\omega}d\Omega \tag{8.7.17}$$

with $p'^0=m+\omega-\omega'$, and ω' given by Eq. (8.7.14).

Usually we do not measure the spin z-component of the initial or final electron. In such cases, we must sum over σ' and average over σ, or in other words take half the sum over σ and σ':

$$d\bar{\sigma}(\mathbf{p}+\mathbf{k},e\to\mathbf{p}'+\mathbf{k}',e')\equiv\frac12\sum_{\sigma,\sigma'}d\sigma(\mathbf{p},\sigma+\mathbf{k},e\to\mathbf{p}',\sigma'+\mathbf{k}',e'). \tag{8.7.18}$$

To calculate this, we use the standard formula

$$\sum_\sigma u_\alpha(\mathbf{p},\sigma)\bar{u}_\beta(\mathbf{p},\sigma)=\frac{(-i\not{p}+m)_{\alpha\beta}}{2p^0} \tag{8.7.19}$$

and likewise for the sum over σ'. It follows that for an arbitrary $4\cdot4$ matrix A

$$\begin{aligned}\sum_{\sigma,\sigma'}|\bar{u}(\mathbf{p}',\sigma')Au(\mathbf{p},\sigma)|^2&=\sum_{\sigma,\sigma'}(\bar{u}'Au)(\bar{u}\beta A^\dagger\beta u')\\&=\mathrm{Tr}\left\{A\left(\frac{-i\not{p}+m}{2p^0}\right)\beta A^\dagger\beta\left(\frac{-i\not{p}'+m}{2p'^0}\right)\right\}.\end{aligned} \tag{8.7.20}$$

[89] Equivalently, there is an increase in wavelength

$$\frac{1}{\omega'}-\frac{1}{\omega}=\frac{1-\cos\theta}{m}.$$

[90] The verification of this formula in the scattering of X rays by electrons by A. H. Compton in 1922–3 played a key role in confirming Einstein's 1905 proposal of a quantum of light, which soon after Compton's experiments came to be known as the photon.

Recalling that $\beta\gamma^\dagger_\mu\beta=-\gamma_\mu$, Eq. (8.7.6) gives now

$$\begin{aligned}\sum_{\sigma,\sigma'}|M|^2=\frac{e^4}{64(2\pi)^6\omega\omega'p^0p'^0}\mathrm{Tr}\Bigg[&\left\{\not{e}'^*\frac{[-i(\not{p}+\not{k})+m]}{p\cdot k}\not{e}-\not{e}\frac{[-i(\not{p}-\not{k}')+m]}{p\cdot k'}\not{e}'^*\right\}(-i\not{p}+m)\\&\cdot\left\{\not{e}^*\frac{[-i(\not{p}+\not{k})+m]}{p\cdot k}\not{e}'-\not{e}'\frac{[-i(\not{p}-\not{k}')+m]}{p\cdot k'}\not{e}^*\right\}(-i\not{p}'+m)\Bigg].\end{aligned} \tag{8.7.21}$$

(Recall again that $\not{e}^*$ means $e^*_\mu\gamma^\mu$, not $(e_\mu\gamma^\mu)^*$, and likewise for $\not{e}'^*$.) We work in a 'gauge' in which

$$e\cdot p=e^*\cdot p=e'\cdot p=e'^*\cdot p=0, \tag{8.7.22}$$

such as for instance Coulomb gauge in the laboratory frame, where $e^0=e'^0=0$ and $\mathbf{p}=0$. This implies that

$$[-i\not{p}+m]\not{e}[-i\not{p}+m]=\not{e}[i\not{p}+m][-i\not{p}+m]=\not{e}(p^2+m^2)=\not{e}(p_\mu p^\mu+m^2)=0$$

and likewise for $\not{e}^*$, $\not{e}'$, and $\not{e}'^*$. Eq. (8.7.21) can therefore be written in the greatly simplified form

$$\begin{aligned}\sum_{\sigma,\sigma'}|M|^2=\frac{-e^4}{64(2\pi)^6\omega\omega'p^0p'^0}\mathrm{Tr}\Bigg[&\left\{\frac{\not{e}'^*\not{k}\not{e}}{p\cdot k}+\frac{\not{e}\not{k}'\not{e}'^*}{p\cdot k'}\right\}(-i\not{p}+m)\\&\cdot\left\{\frac{\not{e}^*\not{k}\not{e}'}{p\cdot k}+\frac{\not{e}'\not{k}'\not{e}^*}{p\cdot k'}\right\}(-i\not{p}'+m)\Bigg].\end{aligned} \tag{8.7.23}$$

The trace of any product of an odd number of gamma matrices vanishes, so this breaks up into terms of zeroth and second order in m:

$$\begin{aligned}\sum_{\sigma,\sigma'}|M|^2=\frac{e^4}{64(2\pi)^6\omega\omega'p^0p'^0}\Bigg(&\frac{T_1}{(p\cdot k)^2}+\frac{T_2}{(p\cdot k)(p\cdot k')}+\frac{T_3}{(p\cdot k)(p\cdot k')}+\frac{T_4}{(p\cdot k')^2}\\&-\frac{m^2t_1}{(p\cdot k)^2}-\frac{m^2t_2}{(p\cdot k)(p\cdot k')}-\frac{m^2t_3}{(p\cdot k)(p\cdot k')}-\frac{m^2t_4}{(p\cdot k')^2}\Bigg),\end{aligned} \tag{8.7.24}$$

where

$$T_1=\mathrm{Tr}\{\not{e}'^*\not{k}\not{e}\not{p}\not{e}^*\not{k}\not{e}'\not{p}'\}, \tag{8.7.25}$$

$$T_2=\mathrm{Tr}\{\not{e}'^*\not{k}\not{e}\not{p}\not{e}'\not{k}'\not{e}^*\not{p}'\}, \tag{8.7.26}$$

$$T_3=\mathrm{Tr}\{\not{e}\not{k}'\not{e}'^*\not{p}\not{e}^*\not{k}\not{e}'\not{p}'\}, \tag{8.7.27}$$

$$T_4=\mathrm{Tr}\{\not{e}\not{k}'\not{e}'^*\not{p}\not{e}'\not{k}'\not{e}^*\not{p}'\}, \tag{8.7.28}$$

$$t_1=\mathrm{Tr}\{\not{e}'^*\not{k}\not{e}\not{e}^*\not{k}\not{e}'\}, \tag{8.7.29}$$

$$t_2=\mathrm{Tr}\{\not{e}'^*\not{k}\not{e}\not{e}'\not{k}'\not{e}^*\}, \tag{8.7.30}$$

$$t_3=\mathrm{Tr}\{\not{e}\not{k}'\not{e}'^*\not{e}^*\not{k}\not{e}'\}, \tag{8.7.31}$$

$$t_4=\mathrm{Tr}\{\not{e}\not{k}'\not{e}'^*\not{e}'\not{k}'\not{e}^*\}. \tag{8.7.32}$$

The Appendix to this chapter shows how to calculate any trace $\mathrm{Tr}\{\not{a}\not{b}\not{c}\not{d}\cdots\}$ as a sum of products of scalar products of the four-vectors $a,b,c,d,\ldots$. In general, traces of products of 6 or 8 gamma matrices like the t_k or T_k would be given by a sum of 15 or 105 terms, respectively, but fortunately here most scalar products vanish; in addition to Eq. (8.7.22), we also have $k\cdot k=k'\cdot k'=0$. (Furthermore, $e\cdot e^*=e'\cdot e'^*=1$.) To simplify the calculation further, let us specialize to the case of *linear* polarization, where e^μ and e'^μ are real. Dropping the asterisks in Eqs. (8.7.25)–(8.7.32), we have then

$$T_1=\mathrm{Tr}\{\not{e}'\not{k}\not{e}\not{p}\not{e}\not{k}\not{e}'\not{p}'\}.$$

Since $e^\mu p_\mu=0$ and $e^\mu e_\mu=1$, we have

$$\not{e}\not{p}\not{e}=-\not{p}\not{e}\not{e}=-\not{p},$$

so

$$T_1=-\mathrm{Tr}\{\not{e}'\not{k}\not{p}\not{k}\not{e}'\not{p}'\}.$$

Also, $k^\mu k_\mu=0$, so

$$\not{k}\not{p}\not{k}=-\not{k}\not{k}\not{p}+2\not{k}\,p\cdot k=2\not{k}\,p\cdot k$$

and hence

$$T_1=-2p\cdot k\,\mathrm{Tr}\{\not{e}'\not{k}\not{e}'\not{p}'\}.$$

Using Eq. (8.A.6), this is

$$T_1=-8p\cdot k[2e'\cdot k\,e'\cdot p'-k\cdot p'].$$

It is convenient to make the substitutions

$$e'\cdot p'=e'\cdot[p+k-k']=e'\cdot k,$$

$$k\cdot p'=-\tfrac12(p'-k)^2-\tfrac12m^2=-\tfrac12(p-k')^2-\tfrac12m^2=p\cdot k',$$

so

$$T_1=-16p\cdot k(e'\cdot k)^2+8p\cdot k\,p\cdot k'. \tag{8.7.33}$$

A similar (though more lengthy) calculation gives

$$\begin{aligned}T_2=T_3=&-8(e\cdot k')^2(p\cdot k)+16(e\cdot e')^2p\cdot k'\,p\cdot k+8(e\cdot e')^2k\cdot k'm^2\\&-8(e\cdot e')m^2(k\cdot e')(k'\cdot e)+8(e'\cdot k)^2p\cdot k'-4(k\cdot p)^2\\&+4(k\cdot k')(p\cdot p')-4(k\cdot p')(p\cdot k'),\end{aligned} \tag{8.7.34}$$

$$T_4=-16p\cdot k'(e\cdot k')^2+8(p\cdot k)(p\cdot k'), \tag{8.7.35}$$

$$t_1=t_4=0, \tag{8.7.36}$$

$$t_2=t_3=-8(e\cdot e')(k\cdot e')(k'\cdot e)+8(p\cdot k')(e\cdot e')^2-4(k\cdot k'). \tag{8.7.37}$$

Combining all these terms in Eq. (8.7.24) gives

$$\sum_{\sigma,\sigma'}|M|^2=\frac{e^4}{64(2\pi)^6\omega\omega'p^0p'^0}\left[\frac{8(k\cdot k')^2}{(k\cdot p)(k'\cdot p)}+32(e\cdot e')^2\right]. \tag{8.7.38}$$

All this applies in any Lorentz frame. In the laboratory frame, we have the special results

$$k\cdot k'=\omega\omega'(\cos\theta-1)=m\omega\omega'\left(\frac{1}{\omega}-\frac{1}{\omega'}\right),\qquad p\cdot k=-m\omega,\quad p\cdot k'=-m\omega'.$$

Combining Eq. (8.7.38) with Eq. (8.7.17), the laboratory frame cross-section is

$$\frac12\sum_{\sigma,\sigma'}d\sigma(\mathbf{p},\sigma+\mathbf{k},e\to\mathbf{p}',\sigma'+\mathbf{k}',e')=\frac{e^4\omega'^2d\Omega}{64\pi^2m^2\omega^2}\left[\frac{\omega}{\omega'}+\frac{\omega'}{\omega}-2+4(e\cdot e')^2\right]. \tag{8.7.39}$$

This is the celebrated formula derived (using old-fashioned perturbation theory) by O. Klein and Y. Nishina [4] in 1929.

As discussed in Section 8.6, if the incoming photon is (as usual) not prepared in a state with any particular polarization, then we must average over two orthonormal values of $\mathbf{e}$. This average gives

$$\frac12\sum_e e_ie_j=\frac12(\delta_{ij}-\hat{k}_i\hat{k}_j)$$

and the differential cross-section is then

$$\frac14\sum_{e,\sigma,\sigma'}d\sigma(\mathbf{p},\sigma+\mathbf{k},e\to\mathbf{p}',\sigma'+\mathbf{k}',e')=\frac{e^4\omega'^2d\Omega}{64\pi^2m^2\omega^2}\left[\frac{\omega}{\omega'}+\frac{\omega'}{\omega}-2(\hat{\mathbf{k}}\cdot e')^2\right]. \tag{8.7.40}$$

We see that the scattered photon is preferentially polarized in a direction perpendicular to the incident as well as the final photon direction, i.e., perpendicular to the plane in which the scattering takes place. This is a well-known result, responsible among other things for the polarization of light from eclipsing binary stars.[91]

To calculate the cross-section for experiments in which the final photon polarization is not measured, we must sum Eq. (8.7.40) over e', using

$$\sum_{e'}e'_ie'_j=\delta_{ij}-\hat{k}'_i\hat{k}'_j.$$

This gives

$$\frac14\sum_{e,e',\sigma,\sigma'}d\sigma(\mathbf{p},\sigma+\mathbf{k},e\to\mathbf{p}',\sigma'+\mathbf{k}',e')=\frac{e^4\omega'^2d\Omega}{32\pi^2m^2\omega^2}\left[\frac{\omega}{\omega'}+\frac{\omega'}{\omega}-1+\cos^2\theta\right], \tag{8.7.41}$$

where θ is the angle between $\hat{\mathbf{k}}$ and $\hat{\mathbf{k}}'$. In the non-relativistic case, $\omega\ll m$, Eq. (8.7.41) gives

$$\frac14\sum_{e,e',\sigma,\sigma'}d\sigma=\frac{e^4d\Omega}{32\pi^2m^2}(1+\cos^2\theta). \tag{8.7.42}$$

[91] The light from one of the stars is polarized when it is scattered by free electrons in the outer atmosphere of the other, cooler, star when both are along the same line of sight. This polarization is normally undetectable because it cancels when the astronomer adds up light from all parts of the star's disk. The polarization has been observed in eclipsing binary stars at times when the cooler star blocks the light from just one side of the hotter star.

The solid angle integral is

$$\int[1+\cos^2\theta]d\Omega=\int_0^{2\pi}d\phi\int_0^\pi[1+\cos^2\theta]\sin\theta\,d\theta=\frac{16\pi}{3},$$

giving a total cross-section for $\omega\ll m$:

$$\sigma_T=\frac{e^4}{6\pi m^2}. \tag{8.7.43}$$

This is often written $\sigma_T=8\pi r_0^2/3$, where $r_0=e^2/4\pi m=2.818\cdot10^{-13}$ cm is known as the *classical electron radius*. Expression (8.7.43) is called the *Thomson cross-section*, after J. J. Thomson, the discoverer of the electron. Eqs. (8.7.42) and (8.7.43) were originally derived using classical mechanics and electrodynamics, by calculating the reradiation of light by a non-relativistic point charge in a plane wave electromagnetic field.

8.8 Generalization: p-Form Gauge Fields[92]

The antisymmetric field strength tensor $F_{\mu\nu}$ of electromagnetism is a special case of a general class of tensors of special importance in physics and mathematics. A p-form is an antisymmetric covariant tensor of rank p. From a p-form $t_{\mu_1\mu_2\cdots\mu_p}$ one may construct a $(p+1)$-form called the *exterior derivative*[93] dt by taking the derivative and then antisymmetrizing with respect to all indices:

$$\begin{aligned}(dt)_{\mu_1\mu_2\cdots\mu_{p+1}}&=\partial_{[\mu_1}t_{\mu_2\mu_3\cdots\mu_{p+1}]}\\&=\partial_{\mu_1}t_{\mu_2\cdots\mu_{p+1}}-\partial_{\mu_2}t_{\mu_1\mu_3\cdots\mu_{p+1}}+\cdots+(-1)^p\partial_{\mu_{p+1}}t_{\mu_1\mu_2\cdots\mu_p},\end{aligned} \tag{8.8.1}$$

with square brackets indicating antisymmetrization with respect to the indices within the brackets. Because derivatives commute, repeated exterior derivatives vanish

$$d(dt)=0. \tag{8.8.2}$$

A p-form whose exterior derivative vanishes is called *closed*, while a p-form that is itself an exterior derivative is called *exact*. From Eq. (8.8.2) it follows that any exact p-form is closed; a famous theorem of Poincaré [6] states that in a simply connected region, any closed p-form is exact.[94] For instance, the homogeneous Maxwell equations (8.1.16) tell us that the electromagnetic field strength two-form $F_{\mu\nu}$ is closed; Poincaré's theorem then shows that it is also exact, so that it can be written as an exterior derivative, i.e., as $F_{\mu\nu}=\partial_\mu A_\nu-\partial_\nu A_\mu$.

[92] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

[93] Exterior derivatives and p-forms play a special role in general relativity, in part because the exterior derivative of a tensor transforms like a tensor even though it is calculated using ordinary rather than covariant derivatives [5].

[94] In multiply connected spaces closed forms are not necessarily exact; although it is possible to write a closed p-form as an exterior derivative locally, this cannot in general be done smoothly throughout the space. The set of closed p-forms, modulo exact p-forms, makes up what is called the pth de Rham cohomology group of the space. There is a deep relation between the de Rham cohomology groups of a space and its topology, which will be discussed further in Volume II.

Again using Eq. (8.8.2), we see that the two-form $F_{\mu\nu}$ is invariant if A_μ is changed by an exterior derivative, i.e., by a gauge transformation $\delta A_\mu=\partial_\mu\Omega$.

The formalism of p-forms and exterior derivatives makes it natural to consider the possibility of massless particles described by p-form field gauge fields[95] $A_{\mu_1\cdots\mu_p}$, with an invariance under gauge transformations

$$\delta A=d\Omega \tag{8.8.3}$$

or in more detail

$$\delta A_{\mu_1\cdots\mu_p}=\partial_{[\mu_1}\Omega_{\mu_2\cdots\mu_p]},$$

where $\Omega_{\mu_1\cdots\mu_{p-1}}$ is an arbitrary $(p-1)$-form. From such a p-form gauge field, we can construct a gauge-invariant field strength tensor given by

$$F=dA \tag{8.8.4}$$

or in detail

$$F_{\mu_1\cdots\mu_{p+1}}=\partial_{[\mu_1}A_{\mu_2\cdots\mu_{p+1}]}. \tag{8.8.5}$$

(Alternatively, we can start with a $(p+1)$-form F, and from an assumed condition $dF=0$ infer the existence of a p-form A with $F=dA$.) By analogy with electrodynamics, we might expect the Lagrangian density for A to take the form

$$\mathcal{L}=-\frac{1}{2(p+1)}F_{\mu_1\cdots\mu_{p+1}}F^{\mu_1\cdots\mu_{p+1}}+J^{\mu_1\cdots\mu_p}A_{\mu_1\cdots\mu_p}, \tag{8.8.6}$$

where J is an antisymmetric tensor current (either a c-number, or a function of fields other than A) that in order to make the action gauge-invariant must satisfy the conservation condition

$$\partial_{\mu_1}J^{\mu_1\cdots\mu_p}=0. \tag{8.8.7}$$

The Euler–Lagrange equations are then

$$\partial_{\mu_1}F^{\mu_1\cdots\mu_{p+1}}=-J^{\mu_2\cdots\mu_{p+1}}. \tag{8.8.8}$$

These p-form gauge fields play an important role in theories with more than four spacetime dimensions. For instance, in the simplest string theories in 26 spacetime dimensions, there is a normal mode of the string represented at low energies by a two-form gauge field $A_{\mu\nu}$. But in four spacetime dimensions, p-forms offer no new possibilities.

To see this, note first that in D spacetime dimensions there are no antisymmetric tensors with more than D indices, so in general we must take $p+1\leq D$. Like any other $(p+1)$-form with $p+1\leq D$, the field strength F may be expressed in terms of a dual $(D-p-1)$-form $\mathcal{F}$, as

$$F^{\mu_1\cdots\mu_{p+1}}=\epsilon^{\mu_1\cdots\mu_D}\mathcal{F}_{\mu_{p+2}\cdots\mu_D}. \tag{8.8.9}$$

[95] We are speaking loosely in calling $A_{\mu_1\cdots\mu_p}$ a p-form, because for $F=dA$ to be a tensor it is only necessary for A to be a tensor up to a gauge transformation. In fact, we have already seen that in four spacetime dimensions, it is not possible to construct a four-vector field from the creation and annihilation operators of physical massless particles of helicity ±1, so we have to deal with an $A^\mu(x)$ that according to Eq. (8.1.2) transforms as a four-vector only up to a gauge transformation.

Likewise, the p-form current J may be expressed in terms of a dual $(D-p)$-form current $\mathcal{J}$, as

$$J^{\mu_1\cdots\mu_p}=\epsilon^{\mu_1\cdots\mu_D}\mathcal{J}_{\mu_{p+1}\cdots\mu_D}. \tag{8.8.10}$$

The field equation (8.8.8) and conservation condition (8.8.7) then read simply

$$d\mathcal{F}=\mathcal{J},\qquad d\mathcal{J}=0. \tag{8.8.11}$$

Because the dual current $\mathcal{J}$ is closed, it may be written in terms of a $(D-p-1)$-form $\mathcal{S}$ as

$$\mathcal{J}=d\mathcal{S}. \tag{8.8.12}$$

Eqs. (8.8.11) and (8.8.12) tell us that the difference of $\mathcal{F}$ and $\mathcal{S}$ is closed, and therefore according to Poincaré's theorem, may be written as

$$\mathcal{F}=\mathcal{S}+d\phi, \tag{8.8.13}$$

with ϕ a $(D-p-2)$-form. There is an exception for the case $p=D-1$, where $\mathcal{F}$ and $\mathcal{S}$ are zero-forms, i.e., scalars, and the condition $d\mathcal{F}=d\mathcal{S}$ simply tells us that $\mathcal{F}$ and $\mathcal{S}$ differ only by a constant. In this case the gauge field describes no degrees of freedom at all. We may therefore concern ourselves only with the cases $p\leq D-2$.

For $p\leq D-2$, the homogeneous 'Maxwell' equations $dF=0$ read

$$\partial_{\mu_1}\mathcal{F}^{\mu_1\cdots\mu_{D-p}}=0, \tag{8.8.14}$$

which with Eq. (8.8.13) yields the field equation for ϕ:

$$\partial_{\mu_1}(d\phi)^{\mu_1\cdots\mu_{D-p}}=-(d\mathcal{S})^{\mu_2\cdots\mu_{D-p}}. \tag{8.8.15}$$

This is invariant under a new set of gauge transformations $\phi\to\phi+d\omega$, except that where $D-p-2=0$ the gauge transformation that leaves F invariant is $\phi\to\phi+c$, where c is an arbitrary constant. *We see that in D spacetime dimensions, the theory of a p-form gauge field A with current J is equivalent to the theory of a $(D-p-2)$-form gauge field ϕ, with current $-d\mathcal{S}$.*

We can now understand why p-form gauge fields offer no new possibilities in four spacetime dimensions. As we have seen, we need only consider the cases $p\leq D-2$, or $p=0$, 1, or 2. A zero-form gauge field is a scalar S, for which Eq. (8.8.5) reads $F_\mu=\partial_\mu S$, and the field equations (8.8.8) read simply $\Box S=-J$. The gauge invariance here is invariance under a shift $S\to S+c$, with c a constant. This is just the theory of a massless scalar field with only derivative interactions. A one-form gauge field is a four-vector $A^\mu(x)$ coupled to a conserved four-vector current, just as in electrodynamics. Finally, according to the general result quoted above, a two-form gauge field in four spacetime dimensions is equivalent to a zero-form gauge field, which as we have seen is equivalent to a massless derivatively coupled scalar field.

Appendix Traces

In calculating S-matrix elements and transition rates for processes involving particles of spin $\frac12$, we often encounter traces of products of Dirac gamma matrices. It will therefore be useful to give formulas for these traces that can be used in all such calculations.

For products of even numbers of gamma matrices, the trace is given by

$$\mathrm{Tr}\{\gamma_{\mu_1}\gamma_{\mu_2}\cdots\gamma_{\mu_{2N}}\}=4\sum_{\text{pairings}}\delta_P\prod_{\text{pairs}}\eta_{\text{paired }\mu\text{s}}. \tag{8.A.1}$$

Here the sum is over all different ways of pairing the indices $\mu_1,\ldots,\mu_{2N}$. A pairing can be regarded as a permutation of the integers $1,2,\ldots,2N$ into some order $P1,P2,\ldots,P\cdot(2N)$, in which we pair μ_{P1} with μ_{P2}, μ_{P3} with μ_{P4}, and so on. Permuting pairs or permuting μs within a pair yields the same pairing, so the number of different pairings is

$$\frac{(2N)!}{N!\,2^N}=(2N-1)(2N-3)\cdots1\equiv(2N-1)!!. \tag{8.A.2}$$

We can avoid summing over equivalent pairings by requiring that

$$P1<P2,\quad P3<P4,\quad\ldots,\quad P\cdot(2N-1)<P\cdot(2N) \tag{8.A.3}$$

and

$$P1<P3<P5<\cdots. \tag{8.A.4}$$

With this convention, the factor δ_P is $+1$ or -1 according to whether the pairing involves an even or odd permutation of indices. The product in Eq. (8.A.1) is over all N pairs, the nth pair contributing a factor $\eta_{\mu_{P\cdot(2n-1)}\mu_{P\cdot(2n)}}$. For instance (writing $\mu,\nu,\rho,\sigma,\ldots$ in place of $\mu_1,\mu_2,\mu_3,\mu_4,\ldots$), for $N=1$, 2, and 3 we have[96]

$$\mathrm{Tr}\{\gamma_\mu\gamma_\nu\}=4\eta_{\mu\nu}, \tag{8.A.5}$$

$$\mathrm{Tr}\{\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma\}=4[\eta_{\mu\nu}\eta_{\rho\sigma}-\eta_{\mu\rho}\eta_{\nu\sigma}+\eta_{\mu\sigma}\eta_{\nu\rho}], \tag{8.A.6}$$

$$\begin{aligned}\mathrm{Tr}\{\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma\gamma_\kappa\gamma_\eta\}=4\big[&\eta_{\mu\nu}\eta_{\rho\sigma}\eta_{\kappa\eta}-\eta_{\mu\nu}\eta_{\rho\kappa}\eta_{\sigma\eta}+\eta_{\mu\nu}\eta_{\rho\eta}\eta_{\sigma\kappa}\\&-\eta_{\mu\rho}\eta_{\nu\sigma}\eta_{\kappa\eta}+\eta_{\mu\rho}\eta_{\nu\kappa}\eta_{\sigma\eta}-\eta_{\mu\rho}\eta_{\nu\eta}\eta_{\sigma\kappa}+\eta_{\mu\sigma}\eta_{\nu\rho}\eta_{\kappa\eta}-\eta_{\mu\sigma}\eta_{\nu\kappa}\eta_{\rho\eta}\\&+\eta_{\mu\sigma}\eta_{\nu\eta}\eta_{\rho\kappa}-\eta_{\mu\kappa}\eta_{\nu\rho}\eta_{\sigma\eta}+\eta_{\mu\kappa}\eta_{\nu\sigma}\eta_{\rho\eta}-\eta_{\mu\kappa}\eta_{\nu\eta}\eta_{\rho\sigma}+\eta_{\mu\eta}\eta_{\nu\rho}\eta_{\sigma\kappa}\\&-\eta_{\mu\eta}\eta_{\nu\sigma}\eta_{\rho\kappa}+\eta_{\mu\eta}\eta_{\nu\kappa}\eta_{\rho\sigma}\big].\end{aligned} \tag{8.A.7}$$

For an odd number of gamma matrices, the result is much simpler

$$\mathrm{Tr}\{\gamma_{\mu_1}\gamma_{\mu_2}\cdots\gamma_{\mu_{2N+1}}\}=0. \tag{8.A.8}$$

The proof of Eq. (8.A.1) is by mathematical induction. First note that

$$\mathrm{Tr}\{\gamma_\mu\gamma_\nu\}=-\mathrm{Tr}\{\gamma_\nu\gamma_\mu\}+2\,\mathrm{Tr}\{\eta_{\mu\nu}\mathbf{1}\}=-\mathrm{Tr}\{\gamma_\mu\gamma_\nu\}+8\eta_{\mu\nu},$$

[96] There are now computer programs [7] available for the calculation of traces of products of large numbers of Dirac matrices.

so $\mathrm{Tr}\{\gamma_\mu\gamma_\nu\}=4\eta_{\mu\nu}$, in agreement with Eq. (8.A.1). Next, suppose that Eq. (8.A.1) is true for $N\leq M-1$. We then have

$$\begin{aligned}\mathrm{Tr}\{\gamma_{\mu_1}\gamma_{\mu_2}\cdots\gamma_{\mu_{2M}}\}&=2\eta_{\mu_1\mu_2}\mathrm{Tr}\{\gamma_{\mu_3}\cdots\gamma_{\mu_{2M}}\}-\mathrm{Tr}\{\gamma_{\mu_2}\gamma_{\mu_1}\gamma_{\mu_3}\cdots\gamma_{\mu_{2M}}\}\\&=2\eta_{\mu_1\mu_2}\mathrm{Tr}\{\gamma_{\mu_3}\cdots\gamma_{\mu_{2M}}\}-2\eta_{\mu_1\mu_3}\mathrm{Tr}\{\gamma_{\mu_2}\gamma_{\mu_4}\cdots\gamma_{\mu_{2M}}\}\\&\quad+\mathrm{Tr}\{\gamma_{\mu_2}\gamma_{\mu_3}\gamma_{\mu_1}\gamma_{\mu_4}\cdots\gamma_{\mu_{2M}}\}\\&=2\eta_{\mu_1\mu_2}\mathrm{Tr}\{\gamma_{\mu_3}\cdots\gamma_{\mu_{2M}}\}-2\eta_{\mu_1\mu_3}\mathrm{Tr}\{\gamma_{\mu_2}\gamma_{\mu_4}\cdots\gamma_{\mu_{2M}}\}\\&\quad+2\eta_{\mu_1\mu_4}\mathrm{Tr}\{\gamma_{\mu_2}\gamma_{\mu_3}\gamma_{\mu_5}\cdots\gamma_{\mu_{2M}}\}-\cdots\\&\quad+2\eta_{\mu_1\mu_{2M}}\mathrm{Tr}\{\gamma_{\mu_2}\cdots\gamma_{\mu_{2M-1}}\}-\mathrm{Tr}\{\gamma_{\mu_2}\cdots\gamma_{\mu_{2M}}\gamma_{\mu_1}\}.\end{aligned}$$

All commutators have zero trace, so the last term subtracted here is the same as the left-hand side, and so

$$\begin{aligned}\mathrm{Tr}\{\gamma_{\mu_1}\gamma_{\mu_2}\cdots\gamma_{\mu_{2M}}\}&=\eta_{\mu_1\mu_2}\mathrm{Tr}\{\gamma_{\mu_3}\cdots\gamma_{\mu_{2M}}\}-\eta_{\mu_1\mu_3}\mathrm{Tr}\{\gamma_{\mu_2}\gamma_{\mu_4}\cdots\gamma_{\mu_{2M}}\}\\&\quad+\eta_{\mu_1\mu_4}\mathrm{Tr}\{\gamma_{\mu_2}\gamma_{\mu_3}\gamma_{\mu_5}\cdots\gamma_{\mu_{2M}}\}-\cdots\\&\quad+\eta_{\mu_1\mu_{2M}}\mathrm{Tr}\{\gamma_{\mu_2}\cdots\gamma_{\mu_{2M-1}}\}.\end{aligned} \tag{8.A.9}$$

If we assume that Eq. (8.A.1) correctly gives the trace of any product of $2N-2$ Dirac matrices, then Eq. (8.A.9) shows that Eq. (8.A.1) also correctly gives the trace of any product of $2N$ Dirac matrices.

The easiest way to see that the trace of an odd number of Dirac matrices vanishes is to note that $-\gamma_\mu$ is related to γ_μ by a similarity transformation, $-\gamma_\mu=\gamma^5\gamma_\mu(\gamma^5)^{-1}$. Traces are unaffected by such similarity transformations, so the trace of an odd number of Dirac matrices is equal to minus itself, and hence vanishes.

One occasionally encounters another class of traces, of the form

$$\mathrm{Tr}\{\gamma^5\gamma_{\mu_1}\gamma_{\mu_2}\cdots\gamma_{\mu_n}\}.$$

This vanishes for odd n for the same reason as given above for traces without a γ^5. It also vanishes for $n=0$ and $n=2$:

$$\mathrm{Tr}\{\gamma^5\}=0 \tag{8.A.10}$$

$$\mathrm{Tr}\{\gamma^5\gamma_\mu\gamma_\nu\}=0. \tag{8.A.11}$$

(To see this just recall that $\gamma^5\equiv i\gamma^0\gamma^1\gamma^2\gamma^3$, and note that there is no way of pairing the indices in $\mathrm{Tr}\{\gamma_0\gamma_1\gamma_2\gamma_3\}$ or in $\mathrm{Tr}\{\gamma_0\gamma_1\gamma_2\gamma_3\gamma_\mu\gamma_\nu\}$ so that the spacetime indices in each pair are equal.) For $n=4$ it is possible to pair the indices in $\mathrm{Tr}\{\gamma_0\gamma_1\gamma_2\gamma_3\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma\}$ so that the spacetime indices in each pair are equal, but only if μ,ν,ρ,σ are some permutation of 0, 1, 2, 3. Furthermore this trace must be odd under permutations of μ,ν,ρ,σ since gamma matrices with different indices anticommute. Thus the trace $\mathrm{Tr}\{\gamma^5\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma\}$ must be proportional to the totally antisymmetric tensor $\epsilon_{\mu\nu\rho\sigma}$. The constant of proportionality may be worked out by letting μ,ν,ρ,σ take the values 0, 1, 2, 3, and recalling that $\epsilon_{0123}=-1$. In this way we find

$$\mathrm{Tr}\{\gamma^5\gamma_\mu\gamma_\nu\gamma_\rho\gamma_\sigma\}=4i\epsilon_{\mu\nu\rho\sigma}. \tag{8.A.12}$$

The trace of products of γ^5 with six, eight, or more Dirac matrices may be calculated by the same methods used above to verify Eq. (8.A.1).

Problems

1. Calculate the differential and total cross-sections for the process $e^+e^-\to\mu^+\mu^-$ to lowest order in e. Assume that electron and muon spins are not observed. Use the simplest Lagrangian for the electrodynamics of electrons and muons.

2. Carry out the canonical quantization of the theory of a charged scalar field Φ and its interaction with electromagnetism, with Lagrangian density:

$$\mathcal{L}=-(D_\mu\Phi)^\dagger(D^\mu\Phi)-m^2\Phi^\dagger\Phi-\lambda(\Phi^\dagger\Phi)^2-\frac14F_{\mu\nu}F^{\mu\nu},$$

where

$$D_\mu\Phi\equiv\partial_\mu\Phi-iqA_\mu\Phi,\qquad F_{\mu\nu}\equiv\partial_\mu A_\nu-\partial_\nu A_\mu.$$

Use Coulomb gauge. Express the Hamiltonian in terms of the fields $\mathbf{A}$, Φ, and $\Phi^\dagger$ and their canonical conjugates. Evaluate the interaction $H_I(t)$ in the interaction picture in terms of the interaction-picture fields and their derivatives.

3. Use the results of Problem 2 to calculate the differential and total cross-sections for photon scattering by a massive charged scalar particle to lowest order in e.

4. Write a gauge-invariant Lagrangian for a charged massive vector field interacting with the electromagnetic field.

5. Calculate the differential cross-section for electron-electron scattering to lowest order in e. Assume that final and initial spins are not measured.

References

1. See, e.g., M. B. Green, J. H. Schwarz, and E. Witten, *Superstring Theory* (Cambridge University Press, Cambridge, 1987): Section 2.2.

1a. V. Fock, *Z. f. Phys.* **39**, 226 (1927); H. Weyl, *Z. Phys.* **56**, 330 (1929). The term 'gauge invariance' derives from an analogy with earlier speculations about scale invariance by H. Weyl, in *Raum, Zeit, Materie*, 3rd ed. (Springer-Verlag, Berlin, 1920). Also see F. London, *Z. f. Phys.* **42**, 375 (1927). This history has been reviewed by C. N. Yang, talk at City College (unpublished).

2. The use of Coulomb gauge in electrodynamics was strongly advocated by Schwinger on pretty much the same grounds as here: that we ought not to introduce photons with helicities other than ±1. See J. Schwinger, *Phys. Rev.* **78**, 1439 (1948); **127**, 324 (1962); *Nuovo Cimento* **30**, 278 (1963).

3. R. P. Feynman, *Phys. Rev.* **101**, 769 (1949): Section 8.

4. O. Klein and Y. Nishina, *Z. f. Phys.* **52**, 853 (1929); Y. Nishina, *ibid.*, 869 (1929); also see I. Tamm, *Z. f. Phys.* **62**, 545 (1930).

5. See, e.g., S. Weinberg, *Gravitation and Cosmology* (Wiley, New York, 1972): Section 4.11.

6. For a readable general introduction to the geometry and topology of p-forms, see H. Flanders, *Differential Forms* (Academic Press, New York, 1963).

7. T. West, *Comput. Phys. Commun.* **77**, 286 (1993).

9 Path-Integral Methods

In Chapters 7 and 8 we applied the canonical quantization operator formalism to derive the Feynman rules for a variety of theories. In many cases, such as the scalar field with derivative coupling or the vector field with zero or non-zero mass, the procedure though straightforward was rather awkward. The interaction Hamiltonian turned out to contain a covariant term, equal to the negative of the interaction term in the Lagrangian, plus a non-covariant term, which served to cancel non-covariant terms in the propagator. In the case of electrodynamics this non-covariant term (the Coulomb energy) turned out to be not even spatially local, though it is local in time. Yet the final results are quite simple: the Feynman rules are just those we should obtain with covariant propagators, and using the negative of the interaction term in the Lagrangian to calculate vertex contributions. The awkwardness in obtaining these simple results, which was bad enough for the theories considered in Chapters 7 and 8, becomes unbearable for more complicated theories, like the non-Abelian gauge theories to be discussed in Volume II, and also general relativity. One would very much prefer a method of calculation that goes directly from the Lagrangian to the Feynman rules in their final, Lorentz-covariant form.

Fortunately, such a method does exist. It is provided by the path-integral approach to quantum mechanics. This was first presented in the context of non-relativistic quantum mechanics in Feynman's Princeton Ph. D. thesis [1], as a means of working directly with a Lagrangian rather than a Hamiltonian. In this respect, it was inspired by earlier work of Dirac [2]. The path-integral approach played a part (along with inspired guesswork) in Feynman's later derivation of his diagrammatic rules [3]. However, although Feynman diagrams became widely used in the 1950s, most physicists (including myself) tended to derive them using the operator methods of Schwinger and Tomonaga, which were shown by Dyson in 1949 to lead to the same diagrammatic rules that had been obtained by Feynman by his own methods.

The path-integral approach was revived in the late 1960s, when Faddeev and Popov [4] and De Witt [5] showed how to apply it to non-Abelian gauge theories and general relativity. For most theorists, the turning point came in 1971, when 't Hooft [6] used path-integral methods to derive the Feynman rules for spontaneously broken gauge theories (discussed in Volume II), including in particular the theory of weak and electromagnetic interactions, in a gauge that made the high energy behavior of these theories transparent. Soon after, as also discussed in Volume II, it was discovered that the path-integral method allows us to take account of contributions to the S-matrix that have an essential singularity at zero coupling constant and therefore cannot be discovered in any finite order of perturbation theory. Since then, the path-integral methods described here have become an indispensable part of the equipment of all physicists who make use of quantum field theory.

At this point the reader may be wondering why if the path-integral method is so convenient we bothered in Chapter 7 to introduce the canonical formalism. Indeed, Feynman seems at first to have thought of his path-integral approach as a substitute for the ordinary canonical formulation of quantum mechanics. There are two reasons for starting with the canonical formalism. The first is a point of principle: although the path-integral formalism

provides us with manifestly Lorentz-invariant diagrammatic rules, it does not make clear why the S-matrix calculated in this way is unitary. As far as I know, the only way to show that the path-integral formalism yields a unitary S-matrix is to use it to reconstruct the canonical formalism, in which unitarity is obvious. There is a kind of conservation of trouble here; we can use the canonical approach, in which unitarity is obvious and Lorentz invariance obscure, or the path-integral approach, which is manifestly Lorentz-invariant but far from manifestly unitary. Since the path-integral approach is here derived from the canonical approach, we know that the two approaches yield the same S-matrix, so that the S-matrix must indeed be both Lorentz-invariant and unitary.

The second reason for introducing the canonical formalism first is more practical: there are important theories in which the simplest version of the Feynman path-integral method, in which propagators and interaction vertices are taken directly from the Lagrangian, is simply wrong. One example is the non-linear σ-model, with Lagrangian density $\mathcal{L}=-\frac12g_{k\ell}(\phi)\partial_\mu\phi^k\partial^\mu\phi^\ell$. In such theories, using the naive Feynman rules derived directly from the Lagrangian density would yield an S-matrix that is not only wrong but even non-unitary, and that also depends on the way in which we define the scalar field [7]. In this chapter we shall derive the path-integral formalism from the canonical formalism, and in this way we will see what additional sorts of vertices are needed to supplement the simplest version of the Feynman path-integral method.

9.1 The General Path-Integral Formula

We start with a general quantum mechanical system, with Hermitian operator 'coordinates' Q_a and conjugate 'momenta' P_b, satisfying the canonical commutation relations[97]:

$$[Q_a,P_b]=i\delta_{ab}, \tag{9.1.1}$$

$$[Q_a,Q_b]=[P_a,P_b]=0. \tag{9.1.2}$$

(We shall restrict ourselves in this and the next three sections to bosonic operators, which satisfy commutation rather than anticommutation relations. Our results will be generalized to include fermionic operators in Section 9.5.) In a field theory, the index a consists of a position $\mathbf{x}$ and a discrete Lorentz and species index m, and we conventionally write

$$Q_{\mathbf{x},m}\equiv Q_m(\mathbf{x}), \tag{9.1.3}$$

$$P_{\mathbf{x},m}\equiv P_m(\mathbf{x}). \tag{9.1.4}$$

Also, the Kronecker delta in Eq. (9.1.1) is interpreted in a field theory as

$$\delta_{\mathbf{x},m;\mathbf{y},n}\equiv\delta^3(\mathbf{x}-\mathbf{y})\delta_{mn}. \tag{9.1.5}$$

However, for the present it will be convenient to use the more compact notation of Eqs. (9.1.1) and (9.1.2). These are 'Schrödinger-picture' operators, taken at a fixed time (say,

[97] We are tacitly assuming here that any first class constraints are eliminated by choosing a gauge, and any remaining second class constraints are 'solved' by writing the constrained degrees of freedom in terms of the unconstrained Q_a and P_a, as in Section 7.6. The direct application of path-integral methods to constrained systems is described by Faddeev [9].

$t=0$). The time-dependent operators in the Heisenberg picture will be considered a little later.

Since the Q_a all commute, we can find a simultaneous eigenstate $|q\rangle$, with eigenvalues q_a:

$$Q_a|q\rangle=q_a|q\rangle. \tag{9.1.6}$$

(We are using lower case qs and ps here to denote eigenvalues rather than to denote operators in the interaction picture as in Chapter 7, but since we will not be using the interaction picture in this chapter no confusion should arise.) The eigenvectors can be taken to be orthonormal,

$$\langle q'|q\rangle=\prod_a\delta(q'_a-q_a)\equiv\delta(q'-q), \tag{9.1.7}$$

so that the completeness relation reads

$$\mathbf{1}=\int\prod_a dq_a\,|q\rangle\langle q|. \tag{9.1.8}$$

Similarly, we can find a complete orthonormal set of eigenstates of the P_a:

$$P_a|p\rangle=p_a|p\rangle, \tag{9.1.9}$$

$$\langle p'|p\rangle=\prod_a\delta(p'_a-p_a)\equiv\delta(p'-p), \tag{9.1.10}$$

$$\mathbf{1}=\int\prod_a dp_a\,|p\rangle\langle p|. \tag{9.1.11}$$

As usual, it follows from Eq. (9.1.1) that these two complete sets of eigenstates have the scalar product[98]

$$\langle q|p\rangle=\prod_a\frac{1}{\sqrt{2\pi}}\exp(iq_ap_a). \tag{9.1.12}$$

In the Heisenberg picture, the Q and P operators are given a time-dependence

$$Q_a(t)=\exp(iHt)Q_a\exp(-iHt), \tag{9.1.13}$$

$$P_a(t)=\exp(iHt)P_a\exp(-iHt), \tag{9.1.14}$$

where H is the total Hamiltonian. These have eigenstates $|q;t\rangle$ and $|p;t\rangle$

$$Q_a(t)|q;t\rangle=q_a|q;t\rangle, \tag{9.1.15}$$

$$P_a(t)|p;t\rangle=p_a|p;t\rangle, \tag{9.1.16}$$

given by

$$|q;t\rangle=\exp(iHt)|q\rangle, \tag{9.1.17}$$

[98] The proof follows the same lines as in the quantum mechanics of point particles. From Eq. (9.1.1), we see that P_a acts as $-i\partial/\partial q_a$ on wave functions in a q-basis. The right-hand side of Eq. (9.1.12) is then seen to be the wave function in this basis of an eigenstate of P. The factor $\prod_a1/\sqrt{2\pi}$ is fixed by the normalization requirement, Eq. (9.1.10).

$$|p;t\rangle = \exp(iHt)|p\rangle. \tag{9.1.18}$$

(Note that $|q;t\rangle$ is the eigenstate of $Q_a(t)$ with eigenvalue q_a, and *not* the result of letting the state $|q\rangle$ evolve for a time t. This is why its time-dependence is given by a factor $\exp(iHt)$ rather than $\exp(-iHt)$.) These states obviously satisfy the completeness and orthonormality conditions

$$\langle q';t|q;t\rangle = \delta(q'-q), \tag{9.1.19}$$

$$\langle p';t|p;t\rangle = \delta(p'-p), \tag{9.1.20}$$

$$\int \prod_a dq_a\, |q;t\rangle\langle q;t| = \mathbf{1}, \tag{9.1.21}$$

$$\int \prod_a dp_a\, |p;t\rangle\langle p;t| = \mathbf{1}, \tag{9.1.22}$$

and also

$$\langle q;t|p;t\rangle = \prod_a \frac{1}{\sqrt{2\pi}}\exp(iq_ap_a). \tag{9.1.23}$$

If, by measurements at time t, we find that our system is in a definite state $|q;t\rangle$, then the probability amplitude for measurements at time t' to give a state $|q';t'\rangle$ is the scalar product $\langle q';t'|q;t\rangle$. Our central dynamical problem is to calculate this scalar product.

This is easy when t' and t are infinitesimally close, say $t'=\tau+d\tau$ and $t=\tau$. Using Eq. (9.1.17), we have

$$\langle q';\tau+d\tau|q;\tau\rangle = \langle q';\tau|\exp(-iH\,d\tau)|q;\tau\rangle. \tag{9.1.24}$$

The Hamiltonian H is given as a function $H(Q,P)$, but since (9.1.13) and (9.1.14) are similarity transformations, and H commutes with itself, it can equally well be written as the *same* function of $Q(t)$ and $P(t)$

$$H \equiv H(Q,P) = e^{iHt}H(Q,P)e^{-iHt} = H\big(Q(t),P(t)\big). \tag{9.1.25}$$

This function can be written in various different forms, with different constant coefficients, by using the commutation relations (9.1.1) and (9.1.2) to move the Qs and Ps past each other. It will be convenient to adopt a standard form, in which all Qs appear to the *left* of all Ps. For instance, given a term in the Hamiltonian of form $P_aQ_bP_c$, we would rewrite it as $P_aQ_bP_c = Q_bP_aP_c - i\delta_{ab}P_c$. With this convention, the $Q_a(t)$s in the Hamiltonian in Eq. (9.1.24) may be replaced[99] with their eigenvalues q'_a. To deal with the $P(t)$, we use Eq. (9.1.23) to expand $|q;\tau\rangle$ in P-eigenstates $|p;\tau\rangle$, and find

$$\begin{aligned}\langle q';\tau+d\tau|q;\tau\rangle &= \int \prod_a dp_a\,\langle q';\tau|\exp\big(-iH(Q(\tau),P(\tau))d\tau\big)|p;\tau\rangle\langle p;\tau|q;\tau\rangle\\ &= \int\prod_a\frac{dp_a}{2\pi}\exp\left[-iH(q',p)d\tau + i\sum_a(q'_a-q_a)p_a\right],\end{aligned} \tag{9.1.26}$$

[99] This is only possible because with $d\tau$ infinitesimal, $\exp(-iH\,d\tau)$ is linear in H.

with each p_a integrated from $-\infty$ to $+\infty$.

Now let's return to the more general case of a finite time-interval. To calculate $\langle q';t'|q;t\rangle$, with $t<t'$, we break up the time-interval from t to t' into steps $t,\tau_1,\tau_2,\ldots,\tau_N,t'$, with

$$\tau_{k+1}-\tau_k = d\tau = (t'-t)/(N+1) \tag{9.1.27}$$

and sum over a complete set of states $|q;\tau_k\rangle$ at each time τ_k:

$$\langle q';t'|q;t\rangle = \int dq_1\cdots dq_N\,\langle q';t'|q_N;\tau_N\rangle\langle q_N;\tau_N|q_{N-1};\tau_{N-1}\rangle\cdots\langle q_1;\tau_1|q;t\rangle. \tag{9.1.28}$$

Inserting Eq. (9.1.26), this becomes

$$\begin{aligned}\langle q';t'|q;t\rangle = \int\left[\prod_{k=1}^N\prod_a dq_{k,a}\right]\left[\prod_{k=0}^N\prod_a\frac{dp_{k,a}}{2\pi}\right]\\ \cdot\exp\left[i\sum_{k=1}^{N+1}\left\{\sum_a(q_{k,a}-q_{k-1,a})p_{k-1,a} - H(q_k,p_{k-1})d\tau\right\}\right],\end{aligned} \tag{9.1.29}$$

where

$$q_0\equiv q,\qquad q_{N+1}\equiv q'. \tag{9.1.30}$$

Our result, Eq. (9.1.29), can be put in a much more elegant form. Define smooth interpolating functions, $q(\tau)$ and $p(\tau)$, such that

$$q_a(\tau_k)=q_{k,a},\qquad p_a(\tau_k)=p_{k,a}. \tag{9.1.31}$$

In the limit $d\tau\to 0$ (i.e., $N\to\infty$), the argument of the exponential in Eq. (9.1.29) becomes just an integral over τ

$$\begin{aligned}&\sum_{k=1}^{N+1}\left\{\sum_a(q_{k,a}-q_{k-1,a})p_{k-1,a}-H(q_k,p_{k-1})d\tau\right\}\\ &=\sum_{k=1}^{N+1}\left\{\sum_a\dot q_a(\tau_k)p_a(\tau_k)-H\big(q(\tau_k),p(\tau_k)\big)\right\}d\tau + O(d\tau^2)\\ &\longrightarrow\int_t^{t'}d\tau\left\{\sum_a\dot q_a(\tau)p_a(\tau)-H\big(q(\tau),p(\tau)\big)\right\}.\end{aligned} \tag{9.1.32}$$

Further, we may define integrals *over the functions* $q(\tau),p(\tau)$ by

$$\int\prod_{\tau,a}dq_a(\tau)\prod_{\tau,b}\frac{dp_b(\tau)}{2\pi}\cdots\equiv\lim_{d\tau\to 0}\int\prod_{k,a}dq_{k,a}\prod_{k,b}\frac{dp_{k,b}}{2\pi}\cdots. \tag{9.1.33}$$

Eq. (9.1.29) then becomes a constrained path integral

$$\begin{aligned}\langle q';t'|q;t\rangle = \int_{\substack{q_a(t)=q_a\\ q_a(t')=q'_a}}\prod_{\tau,a}dq_a(\tau)\prod_{\tau,b}\frac{dp_b(\tau)}{2\pi}\\ \cdot\exp\left[i\int_t^{t'}d\tau\left\{\sum_a\dot q_a(\tau)p_a(\tau)-H\big(q(\tau),p(\tau)\big)\right\}\right].\end{aligned} \tag{9.1.34}$$

This is called a *path integral*, because we integrate over all paths that take $q(\tau)$ from q at $\tau=t$ to q' at $\tau=t'$, as well as over all $p(\tau)$. The great advantage of writing matrix elements in this way is that, as shown in Section 9.3, the path integrals are easy to calculate when expanded in powers of the coupling constants in H.

The path-integral formalism allows us to calculate not only transition probability amplitudes like $\langle q';t'|q;t\rangle$, but also the matrix elements between states $\langle q';t'|$ and $|q;t\rangle$ of time-ordered products of general operators $\mathcal{O}(P(t),Q(t))$. It will be convenient to define these operators with (unlike H) all Ps moved to the *left* and all Qs to the *right*. Then by inserting any such operator $\mathcal{O}[P(t),Q(t)]$ in Eq. (9.1.26), we have

$$\begin{aligned}&\langle q';\tau+d\tau|\mathcal{O}\big(P(t),Q(t)\big)|q;\tau\rangle\\ &=\int\prod_a dp_a\,\langle q';\tau|\exp\big(-iH(Q(\tau),P(\tau))d\tau\big)|p;\tau\rangle\langle p;\tau|\mathcal{O}\big(P(t),Q(t)\big)|q;\tau\rangle\\ &=\int\prod_a\frac{dp_a}{2\pi}\exp\left[-iH(q',p)d\tau+i\sum_a(q'_a-q_a)p_a\right]\mathcal{O}(p,q).\end{aligned} \tag{9.1.35}$$

In order to calculate the matrix element of a product $\mathcal{O}_A(P(t_A),Q(t_A))\mathcal{O}_B(P(t_B),Q(t_B))\cdots$ of operators with $t_A>t_B>\cdots$, we can insert the $\mathcal{O}$-operators between the appropriate states in Eq. (9.1.28), and use Eq. (9.1.35). For instance, if the time t_A falls between τ_k and τ_{k+1}, then insert $\mathcal{O}_A(P(t_A),Q(t_A))$ between $\langle q_{k+1};\tau_{k+1}|$ and $|q_k;\tau_k\rangle$. Note that in Eq. (9.1.28) each successive sum over states is at a later time, so this is only possible because of our assumption that $t_A>t_B>\cdots$. Following the same steps as before, we now find the general path-integral formula

$$\begin{aligned}&\langle q';t'|\mathcal{O}_A\big(P(t_A),Q(t_A)\big)\mathcal{O}_B\big(P(t_B),Q(t_B)\big)\cdots|q;t\rangle\\ &=\int_{\substack{q_a(t)=q_a\\ q_a(t')=q'_a}}\prod_{\tau,a}dq_a(\tau)\prod_{\tau,b}\frac{dp_b(\tau)}{2\pi}\mathcal{O}_A\big(p(t_A),q(t_A)\big)\mathcal{O}_B\big(p(t_B),q(t_B)\big)\cdots\\ &\quad\cdot\exp\left[i\int_t^{t'}d\tau\left\{\sum_a\dot q_a(\tau)p_a(\tau)-H\big(q(\tau),p(\tau)\big)\right\}\right].\end{aligned} \tag{9.1.36}$$

This result is only valid if the times are ordered, with

$$t'>t_A>t_B>\cdots>t. \tag{9.1.37}$$

However, nothing on the right-hand side of Eq. (9.1.36) refers to the order of time-arguments. Hence if we are presented with a path integral like the right-hand side of Eq. (9.1.36), with $t_A,t_B,\ldots$ in arbitrary order (all between t and t', with $t<t'$), then this path integral will equal a matrix element like the left-hand side of Eq. (9.1.36), but with the operators arranged in order (from left to right) of decreasing time. That is, for

$t_A,t_B,\ldots$ in arbitrary order, we have

$$\begin{aligned}&\langle q';t'|T\left\{\mathcal{O}_A\big(P(t_A),Q(t_A)\big),\mathcal{O}_B\big(P(t_B),Q(t_B)\big),\ldots\right\}|q;t\rangle\\ &=\int_{\substack{q_a(t)=q_a\\ q_a(t')=q'_a}}\prod_{\tau,a}dq_a(\tau)\prod_{\tau,b}\frac{dp_b(\tau)}{2\pi}\mathcal{O}_A\big(p(t_A),q(t_A)\big)\mathcal{O}_B\big(p(t_B),q(t_B)\big)\cdots\\ &\quad\cdot\exp\left[i\int_t^{t'}d\tau\left\{\sum_a\dot q_a(\tau)p_a(\tau)-H\big(q(\tau),p(\tau)\big)\right\}\right],\end{aligned} \tag{9.1.38}$$

where T denotes the usual time-ordered product.

It should perhaps be stressed that the c-number functions $q_a(\tau)$, $p_a(\tau)$ in Eq. (9.1.38) are mere variables of integration, and in particular are *not* constrained to obey the equations of motion of classical Hamiltonian dynamics

$$\dot q_a(\tau)-\frac{\partial H(q(\tau),p(\tau))}{\partial p_a(\tau)}=0, \tag{9.1.39}$$

$$\dot p_a(\tau)+\frac{\partial H(q(\tau),p(\tau))}{\partial q_a(\tau)}=0. \tag{9.1.40}$$

(For this reason, the Hamiltonian $H(q(\tau),p(\tau))$ in Eq. (9.1.38) is *not* constant in τ.) Nevertheless, there is a limited sense in which path integrals do respect these equations of motion. Suppose that one of the functions in Eq. (9.1.38), say $\mathcal{O}_A(p(t_A),q(t_A))$, happens to be the left-hand side of either Eq. (9.1.39) or Eq. (9.1.40). We note that (for $t<t_A<t'$)

$$\left(\dot q_a(t_A)-\frac{\partial H(q(t_A),p(t_A))}{\partial p_a(t_A)}\right)\exp(iI[q,p]) = -i\frac{\delta}{\delta p_a(t_A)}\exp(iI[q,p]),$$

$$\left(\dot p_a(t_A)+\frac{\partial H(q(t_A),p(t_A))}{\partial q_a(t_A)}\right)\exp(iI[q,p]) = i\frac{\delta}{\delta q_a(t_A)}\exp(iI[q,p]),$$

where I is the argument of the exponential in Eq. (9.1.38):

$$I[q,p]\equiv\int_t^{t'}d\tau\left\{\sum_a\dot q_a(\tau)p_a(\tau)-H\big(q(\tau),p(\tau)\big)\right\}.$$

As long as t_A does not approach t or t', the integrations over $q_a(t_A)$ and $p_a(t_A)$ are unconstrained, and so with reasonable assumptions about the convergence of these integrals, the integral of such variational derivatives must vanish. Hence the path integral (9.1.38) vanishes if $\mathcal{O}_A(p,q)$ is taken to be the left-hand side of either of the equations of motion (9.1.39) or (9.1.40).

This simple rule applies only if the integration variables $q_a(t_A)$, $p_a(t_A)$ are independent of any of the variables $q_a(t_B)$, $p_a(t_B)$, etc. appearing in any of the other functions $\mathcal{O}_B$, $\mathcal{O}_C$, etc. in Eq. (9.1.38), and hence only if we prohibit t_A from approaching t_B,t_C, etc. as well as t or t'. When t_A approaches, say, t_B, the path integral will be found to involve a non-zero term proportional to $\delta(t_A-t_B)$ or its derivatives. These delta functions are the same as would be found in the operator formalism from time derivatives of the step functions implicit in the definition of the time-ordered product.

In evaluating the path integrals (9.1.34) and (9.1.38), we only need to know the classical Hamiltonian, the c-number function $H(q,p)$. If we were to define a theory by the path integrals, the question would naturally arise, which of many possible quantum mechanical Hamiltonians $H(Q,P)$ (differing is the order of Qs and Ps) governed the quantum theory that corresponds to these path integrals. Our derivation has provided an answer: the quantum Hamiltonian is to be taken with all Qs on the left, all Ps on the right. But it would be a mistake to give this prescription too much significance. There are a great many ways of interpreting the measure $\prod dq_a(\tau)\prod dp_b(\tau)$ appearing in path integrals like (9.1.34) or (9.1.38). Our prescription, of putting all Qs to the left of all Ps, is appropriate only if the measure is interpreted according to Eqs. (9.1.31)–(9.1.33). Other measures would lead to other prescriptions for operator ordering. The question is not an urgent one, because different prescriptions for ordering operators in the Hamiltonian just correspond to different choices of the constants that appear as coefficients of the various terms in the Hamiltonian, and we generally formulate theories with these constants left as arbitrary parameters anyway.

It is difficult to use the general path integral in Eq. (9.1.38) for numerical calculations or as a source of rigorous theorems. For these purposes it is better to use the path-integral method to calculate amplitudes in Euclidean space, where t is replaced with an imaginary quantity $-ix_4$, and the argument of the exponential in Eq. (9.1.38) is a negative real quantity. In this way, instead of jagged paths producing rapid oscillations of the integrand from one path to another, all jagged paths are exponentially suppressed. Though we shall not go into it here, quantum field theory may be formulated from the beginning in terms of Feynman amplitudes in Euclidean spacetime [8a]. Under certain plausible assumptions, it is possible to reconstruct the Feynman amplitudes in Minkowskian spacetime from their Euclidean counterparts [8b]. But we may as well stick to the Minkowski space formulation of the path integral if we are only going to use it to calculate Feynman amplitudes in perturbation theory.

9.2 Transition to the S-Matrix

As already mentioned, we can easily convert the general quantum mechanical results of Section 9.1 to a notation appropriate to quantum field theory, by letting the index a run over points $\mathbf{x}$ in space and over a spin-and-species index m, and replacing $Q_a(t)$ and $P_a(t)$ with $Q_m(\mathbf{x},t)$ and $P_m(\mathbf{x},t)$, respectively. Eq. (9.1.38) then reads[100]

$$\begin{aligned}&\langle q';t'|T\left\{\mathcal{O}_A[P(t_A),Q(t_A)],\mathcal{O}_B[P(t_B),Q(t_B)],\ldots\right\}|q;t\rangle\\ &=\int_{\substack{q_m(\mathbf{x},t)=q_m(\mathbf{x})\\ q_m(\mathbf{x},t')=q'_m(\mathbf{x})}}\prod_{\tau,\mathbf{x},m}dq_m(\mathbf{x},\tau)\prod_{\tau,\mathbf{x},m}\frac{dp_m(\mathbf{x},\tau)}{2\pi}\\ &\quad\cdot\mathcal{O}_A[p(t_A),q(t_A)]\mathcal{O}_B[p(t_B),q(t_B)]\cdots\\ &\quad\cdot\exp\left[i\int_t^{t'}d\tau\left\{\int d^3x\,\dot q_m(\mathbf{x},\tau)p_m(\mathbf{x},\tau)-H[q(\tau),p(\tau)]\right\}\right].\end{aligned} \tag{9.2.1}$$

[100] We are now writing H and the $\mathcal{O}$s with square brackets, to remind us that $H[q(t),p(t)]$ and $\mathcal{O}[p(t),q(t)]$ are functionals of $q_m(\mathbf{x},t)$ and $p_m(\mathbf{x},t)$ at a fixed time t.

However, in field theory Eq. (9.2.1) is not exactly what we want. Experimentalists do not measure probability amplitudes for transitions between eigenstates $\langle q';t'|$ and $|q;t\rangle$ of the quantum field Q, but rather S-matrix elements, the probability amplitudes for transitions between states that at $t\to-\infty$ or $t\to+\infty$ contain definite numbers of particles of various types. These are called 'in' and 'out' states, $|\alpha\rangle_{\rm in}$ and $|\beta\rangle_{\rm out}$, where α and β denote sets of particles characterized by the various particles' momenta, spin z-component (or helicity), and species. To calculate a matrix element of a time-ordered product (perhaps empty) between such states, we need to multiply Eq. (9.2.1) by the 'wave functions' $\langle\beta;{\rm out}|q';t'\rangle$ and $\langle q;t|\alpha;{\rm in}\rangle$ at any fixed times t and t', taken for convenience here to be $-\infty$ and $+\infty$, respectively, and then perform an integral over the 'arguments' $q_m(\mathbf{x})$ and $q'_m(\mathbf{x})$ of these wave functions. But instead of constraining the path integral over $q_m(\mathbf{x},\tau)$ by the conditions

$$q_m(\mathbf{x},+\infty)=q'_m(\mathbf{x}),\qquad q_m(\mathbf{x},-\infty)=q_m(\mathbf{x}), \tag{9.2.2}$$

and then integrating over $q'_m(\mathbf{x})$ and $q_m(\mathbf{x})$, we can just as well do an unconstrained integral over $q_m(\mathbf{x},\tau)$ (and also over $p_m(\mathbf{x},\tau)$), and set the arguments of the wave functions equal to the values given by Eq. (9.2.2):

$$\begin{aligned}&{}_{\rm out}\langle\beta|T\left\{\mathcal{O}_A[P(t_A),Q(t_A)],\mathcal{O}_B[P(t_B),Q(t_B)],\ldots\right\}|\alpha\rangle_{\rm in}\\ &=\int\prod_{\tau,\mathbf{x},m}dq_m(\mathbf{x},\tau)\prod_{\tau,\mathbf{x},m}\frac{dp_m(\mathbf{x},\tau)}{2\pi}\mathcal{O}_A[p(t_A),q(t_A)]\mathcal{O}_B[p(t_B),q(t_B)]\cdots\\ &\quad\cdot\exp\left[i\int_{-\infty}^{+\infty}d\tau\left\{\int d^3x\,\dot q_m(\mathbf{x},\tau)p_m(\mathbf{x},\tau)-H[q(\tau),p(\tau)]\right\}\right]\\ &\quad\cdot\langle\beta;{\rm out}|q(+\infty);+\infty\rangle\langle q(-\infty);-\infty|\alpha;{\rm in}\rangle.\end{aligned} \tag{9.2.3}$$

Incidentally, this result leads immediately[101] to Eq. (6.4.3), a theorem that we use repeatedly to relate sums of off-shell Feynman graphs to matrix elements of Heisenberg-picture operators between exact energy eigenstates.

It is necessary now to consider how to calculate the wave functions appearing as the final pair of factors in Eq. (9.2.3). Let's first consider the simplest and most important case, the vacuum. (We saw in Section 6.4 that S-matrix elements may be easily calculated from the vacuum expectation values of time-ordered products.) We assume as usual that

[101] It is only necessary to note that, for a Hamiltonian $H[P(t),Q(t)]+\sum_A\int d^3x\,\epsilon_A(\mathbf{x},t)\mathcal{O}_A(\mathbf{x},t)$, the S-matrix is given by Eq. (9.2.3) as

$$\begin{aligned}{}_{\rm out}\langle\beta|\alpha\rangle_{{\rm in},\epsilon}=\int\prod_{\tau,\mathbf{x},m}dq_m(\mathbf{x},\tau)\prod_{\tau,\mathbf{x},m}\frac{dp_m(\mathbf{x},\tau)}{2\pi}\\ \cdot\exp\left[i\int_{-\infty}^{+\infty}dt\left\{\int d^3x\,\dot q_m(\mathbf{x},t)p_m(\mathbf{x},t)-H[q(t),p(t)]-\sum_A\int d^3x\,\epsilon_A(\mathbf{x},t)\mathcal{O}_A(\mathbf{x},t)\right\}\right]\\ \cdot\langle\beta;{\rm out}|q(+\infty);+\infty\rangle\langle q(-\infty);-\infty|\alpha;{\rm in}\rangle.\end{aligned}$$

The left-hand side of Eq. (6.4.3) is the derivative of this expression with respect to ϵ_A, ϵ_B, etc., at $\epsilon=0$, which yields the right-hand side of Eq. (9.2.3), and using Eq. (9.2.3) again then immediately gives the right-hand side of Eq. (6.4.3).

for $t\to\mp\infty$, matrix elements may be calculated as if there were no interactions. The 'in' and 'out' vacua may thus be defined by the conditions

$$a_{\mathbf{p},\sigma,n,{\rm in}}|\Omega\rangle_{\rm in}=0,\qquad a_{\mathbf{p},\sigma,n,{\rm out}}|\Omega\rangle_{\rm out}=0, \tag{9.2.4}$$

where $a_{\mathbf{p},\sigma,n,{\rm in}}$ and $a_{\mathbf{p},\sigma,n,{\rm out}}$ are the operators appearing in the coefficients of $\exp(i\mathbf{p}\cdot\mathbf{x}-iEt)$ in the plane-wave expansion of the operator $Q_m(\mathbf{x},t)$ at $t\to-\infty$ and $t\to+\infty$, respectively. For instance, for the real scalar field of a neutral spinless particle, we have in effect

$$\Phi(\mathbf{x},t)\underset{t\to\mp\infty}{\longrightarrow}(2\pi)^{-3/2}\int d^3p\,(2E)^{-1/2}\left[a_{\substack{\rm in\\ \rm out}}(\mathbf{p})e^{ip\cdot x}+{\rm H.c.}\right], \tag{9.2.5}$$

$$\Pi(\mathbf{x},t)\underset{t\to\mp\infty}{\longrightarrow}\dot\Phi(\mathbf{x},t)\underset{t\to\mp\infty}{\longrightarrow}-i(2\pi)^{-3/2}\int d^3p\,(E/2)^{1/2}\left[a_{\substack{\rm in\\ \rm out}}(\mathbf{p})e^{ip\cdot x}-{\rm H.c.}\right], \tag{9.2.6}$$

where $p^0\equiv E\equiv\sqrt{\mathbf{p}^2+m^2}$, and we here use the conventional Φ and Π rather than Q and P for a scalar field, and drop the unnecessary labels m,σ,n. Inverting the Fourier transforms and taking a linear combination of the resulting expressions, we have

$$a_{\substack{\rm in\\ \rm out}}(\mathbf{p})=\lim_{t\to\mp\infty}\frac{e^{iEt}}{(2\pi)^{3/2}}\int d^3x\,e^{-i\mathbf{p}\cdot\mathbf{x}}\left[\sqrt{\frac{E}{2}}\Phi(\mathbf{x},t)+i\sqrt{\frac{1}{2E}}\Pi(\mathbf{x},t)\right]. \tag{9.2.7}$$

As mentioned in Section 9.1, the 'momentum' $\Pi(\mathbf{x},t)$ acts on wave-functions in a ϕ-basis as the variational derivative $-i\delta/\delta\phi(\mathbf{x},t)$, so in this basis the conditions (9.2.4) read

$$0=\int d^3x\,e^{-i\mathbf{p}\cdot\mathbf{x}}\left[\frac{\delta}{\delta\phi(\mathbf{x})}+E(\mathbf{p})\phi(\mathbf{x})\right]\langle\phi(\mathbf{x});\mp\infty|\Omega;{}^{\rm in}_{\rm out}\rangle. \tag{9.2.8}$$

The analogous ordinary differential equation has a well-known Gaussian solution, so let's try a Gaussian ansatz here:

$$\langle\phi(\mathbf{x});\mp\infty|\Omega;{}^{\rm in}_{\rm out}\rangle=\mathcal{N}\exp\left(-\frac12\int d^3x\,d^3y\,\mathcal{E}(\mathbf{x},\mathbf{y})\phi(\mathbf{x})\phi(\mathbf{y})\right), \tag{9.2.9}$$

with kernel $\mathcal{E}$ and constant $\mathcal{N}$ to be determined. Substituting this in Eq. (9.2.8), we see that the functional differential equation for the vacuum wave functional is satisfied if for all ϕ

$$0=\int d^3x\,e^{-i\mathbf{p}\cdot\mathbf{x}}\left[\int d^3y\,\mathcal{E}(\mathbf{x},\mathbf{y})\phi(\mathbf{y})-E(\mathbf{p})\phi(\mathbf{x})\right] \tag{9.2.10}$$

or, in other words, if

$$\int d^3x\,e^{-i\mathbf{p}\cdot\mathbf{x}}\mathcal{E}(\mathbf{x},\mathbf{y})=E(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{y}}. \tag{9.2.11}$$

The solution is easily found by inverting the Fourier transform

$$\mathcal{E}(\mathbf{x},\mathbf{y})=(2\pi)^{-3}\int d^3p\,e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}E(\mathbf{p}). \tag{9.2.12}$$

(Recall that $E(\mathbf{p})\equiv\sqrt{\mathbf{p}^2+m^2}$.) This is actually the most useful representation for the kernel $\mathcal{E}$, but we may note in passing that for $\mathbf{x}\neq\mathbf{y}$, $\mathcal{E}$ may also be written in terms of a Hankel function of negative order

$$\mathcal{E}(\mathbf{x},\mathbf{y})=\frac{m}{2\pi^2r}\frac{d}{dr}\left(\frac1rK_{-1}(imr)\right), \tag{9.2.13}$$

where $r\equiv|\mathbf{x}-\mathbf{y}|$. The constant $\mathcal{N}$ in Eq. (9.2.9) may be formally obtained from the normalization condition for the vacuum state, but we will not need this result.

According to Eq. (9.2.9), in calculating vacuum expectation values in the theory of a scalar field, the product of the last two factors in Eq. (9.2.3) is

$$\begin{aligned}&{}_{\rm out}\langle\Omega|\phi(+\infty);+\infty\rangle\langle\phi(-\infty);-\infty|\Omega\rangle_{\rm in}\\ &=|\mathcal{N}|^2\exp\left(-\frac12\int d^3x\,d^3y\,\mathcal{E}(\mathbf{x},\mathbf{y})\left[\phi(\mathbf{x},+\infty)\phi(\mathbf{y},+\infty)+\phi(\mathbf{x},-\infty)\phi(\mathbf{y},-\infty)\right]\right)\\ &=|\mathcal{N}|^2\exp\left(-\frac12\epsilon\int d^3x\,d^3y\int_{-\infty}^{+\infty}d\tau\,\mathcal{E}(\mathbf{x},\mathbf{y})\phi(\mathbf{x},\tau)\phi(\mathbf{y},\tau)e^{-\epsilon|\tau|}\right),\end{aligned} \tag{9.2.14}$$

where ϵ is a positive infinitesimal. To obtain the final expression, we have used the fact that for any reasonably smooth function $f(\tau)$,

$$f(+\infty)+f(-\infty)=\lim_{\epsilon\to 0+}\epsilon\int_{-\infty}^{+\infty}d\tau\,f(\tau)e^{-\epsilon|\tau|}. \tag{9.2.15}$$

Inserting Eq. (9.2.14) in Eq. (9.2.3) now gives

$$\begin{aligned}&{}_{\rm out}\langle\Omega|T\left\{\mathcal{O}_A[\Pi(t_A),\Phi(t_A)],\mathcal{O}_B[\Pi(t_B),\Phi(t_B)],\ldots\right\}|\Omega\rangle_{\rm in}\\ &=|\mathcal{N}|^2\int\prod_{\tau,\mathbf{x}}d\phi(\mathbf{x},\tau)\prod_{\tau,\mathbf{x}}\frac{d\pi(\mathbf{x},\tau)}{2\pi}\mathcal{O}_A[\pi(t_A),\phi(t_A)]\mathcal{O}_B[\pi(t_B),\phi(t_B)]\cdots\\ &\quad\cdot\exp\left[i\int_{-\infty}^{+\infty}d\tau\left\{\begin{aligned}&\int d^3x\,\dot\phi(\mathbf{x},\tau)\pi(\mathbf{x},\tau)-H[\phi(\tau),\pi(\tau)]\\ &+\frac12 i\epsilon\int d^3x\,d^3y\,\mathcal{E}(\mathbf{x},\mathbf{y})e^{-\epsilon|\tau|}\phi(\mathbf{x},\tau)\phi(\mathbf{y},\tau)\end{aligned}\right\}\right].\end{aligned} \tag{9.2.16}$$

We shall see in Section 9.4 that the whole effect of the last term in the argument of the exponential in Eq. (9.2.16) is to provide the $-i\epsilon$ in the denominator of the scalar field propagator in momentum space, $[p^2+m^2-i\epsilon]^{-1}$. We will not go into the corresponding details for fields of general spin, but will simply state that in general

$$\begin{aligned}&{}_{\rm out}\langle\Omega|T\left\{\mathcal{O}_A[P_A(t_A),Q(t_A)],\mathcal{O}_B[P_B(t_B),Q(t_B)],\ldots\right\}|\Omega\rangle_{\rm in}\\ &=|\mathcal{N}|^2\int\left[\prod_{\tau,\mathbf{x},m}dq_m(\mathbf{x},\tau)\right]\left[\prod_{\tau,\mathbf{x},m}\frac{dp_m(\mathbf{x},\tau)}{2\pi}\right]\mathcal{O}_A[p(t_A),q(t_A)]\mathcal{O}_B[p(t_B),q(t_B)]\cdots\\ &\quad\cdot\exp\left[i\int_{-\infty}^{+\infty}d\tau\left\{\int d^3x\sum_m\dot q_m(\mathbf{x},\tau)p_m(\mathbf{x},\tau)-H[q(\tau),p(\tau)]+i\epsilon\text{ terms}\right\}\right],\end{aligned} \tag{9.2.17}$$

where the '$i\epsilon$ terms' just have the effect of putting the correct $-i\epsilon$ in the denominators of all propagators.

This is a good place to mention that field-independent factors in Eq. (9.2.17), like the constant $|\mathcal{N}|^2$, are not important. This is because such factors contribute also to the matrix element ${}_{\rm out}\langle\Omega|\Omega\rangle_{\rm in}$. In calculating the *connected* part of vacuum expectation values of time-ordered products (or the S-matrix) we eliminate the contribution of disconnected

vacuum fluctuation subgraphs by dividing by ${}_{\rm out}\langle\Omega|\Omega\rangle_{\rm in}$, and any constant factors in the vacuum expectation values cancel in this ratio.

We could go on and calculate matrix elements between multi-particle states, by inserting the appropriate 'wave functionals' in Eq. (9.2.3). These can be calculated by applying the adjoints of annihilation operators such as (9.2.7) to the vacuum state; just as for the harmonic oscillator, these wave functionals turn out to be Hermite polynomials in the field times the vacuum Gaussian. We do not need to work all this out here, because as shown in Section 6.4, the vacuum expectation values (9.2.17) are all we need in order to be able to calculate S-matrix elements.

9.3 Lagrangian Version of the Path-Integral Formula

The integrand in the exponential in Eqs. (9.1.38) or (9.2.17) looks like the Lagrangian L associated with the Hamiltonian H. This appearance is somewhat misleading because here the 'momenta' $p_a(t)$ or $p_n(\mathbf{x},t)$ are independent variables, not yet related to $q_a(t)$ or $q_n(\mathbf{x},t)$ or their derivatives. However, there is a large and important class of theories in which the integral over the 'momenta' can be done by just replacing them with the values dictated by the canonical formalism, in which case the integrand in the exponential in the path integrals really is the Lagrangian.

These are theories with a Hamiltonian that is quadratic in the 'momenta'—in the language of field theory

$$\begin{aligned}H[Q,P]=\frac12\sum_{nm}\int d^3x\,d^3y\,A_{xn,ym}[Q]P_n(\mathbf{x})P_m(\mathbf{y})\\ +\sum_n\int d^3x\,B_{xn}[Q]P_n(\mathbf{x})+C[Q]\end{aligned} \tag{9.3.1}$$

with a 'matrix' A that is real, symmetric, positive, and non-singular. The argument of the exponential in Eq. (9.2.17) is then quadratic in the ps:

$$\begin{aligned}&\int d\tau\left\{\int d^3x\sum_n p_n(\mathbf{x},\tau)\dot q_n(\mathbf{x},\tau)-H[q(\tau),p(\tau)]\right\}\\ &=-\frac12\sum_{nm}\int d^3x\,d^3y\,d\tau\,d\tau'\,\mathcal{A}_{xn\tau,ym\tau'}[q]p_n(\mathbf{x},\tau)p_m(\mathbf{y},\tau')\\ &\quad-\sum_n\int d^3x\int d\tau\,\mathcal{B}_{xn\tau}[q]p_n(\mathbf{x},\tau)-\mathcal{C}[q],\end{aligned} \tag{9.3.2}$$

where

$$\mathcal{A}_{xn\tau,ym\tau'}[q]\equiv A_{xn,ym}[q(\tau)]\delta(\tau-\tau'), \tag{9.3.3}$$

$$\mathcal{B}_{xn\tau}[q]\equiv B_{xn}[q(\tau)]-\dot q_n(\mathbf{x},\tau), \tag{9.3.4}$$

$$\mathcal{C}[q]\equiv\int d\tau\,C[q(\tau)]. \tag{9.3.5}$$

Now, in general the integral of the exponential of a quadratic expression like (9.3.2) will be proportional to the exponential evaluated at the stationary point of its argument. For

a finite number of real variables ξ_s, this formula reads

$$\begin{aligned}&\int_{-\infty}^{\infty}\left(\prod_s d\xi_s\right)\exp\left\{-\frac i2\sum_{sr}\mathcal{A}_{sr}\xi_s\xi_r-i\sum_s\mathcal{B}_s\xi_s-i\mathcal{C}\right\}\\ &=\left({\rm Det}\left[\frac{i\mathcal{A}}{2\pi}\right]\right)^{-1/2}\exp\left\{-\frac i2\sum_{sr}\mathcal{A}_{sr}\bar\xi_s\bar\xi_r-i\sum_s\mathcal{B}_s\bar\xi_s-i\mathcal{C}\right\},\end{aligned} \tag{9.3.6}$$

where $\bar\xi$ is the stationary point

$$\bar\xi_s=-\sum_r(\mathcal{A}^{-1})_{sr}\mathcal{B}_r. \tag{9.3.7}$$

(For a proof of this formula, see the Appendix to this chapter.) Hence, as long as the $\mathcal{O}_A,\mathcal{O}_B$, etc. in Eq. (9.2.17) are independent of the ps, for such a Hamiltonian we can evaluate the path integral over the ps in Eq. (9.2.17) by setting these variables at the stationary point of the quadratic expression in the argument of the exponential. But the variational derivative of this quadratic is

$$\begin{aligned}&\frac{\delta}{\delta p_n(\mathbf{x},\tau)}\int_{-\infty}^{\infty}d\tau\left\{\int d^3x\sum_m\dot q_m(\mathbf{x},\tau)p_m(\mathbf{x},\tau)-H[q(\tau),p(\tau)]+i\epsilon\text{ terms}\right\}\\ &=\dot q_n(\mathbf{x},\tau)-\frac{\delta}{\delta p_n(\mathbf{x},\tau)}H[q(\tau),p(\tau)].\end{aligned}$$

(The $i\epsilon$ terms depend only on the qs.) Thus the stationary 'point' $\bar p_n(\mathbf{x},t)$ where this vanishes is just the value of $p_n(\mathbf{x},t)$ dictated by the canonical formula

$$\dot q_n(\mathbf{x},t)=\left[\frac{\delta H[q(t),p(t)]}{\delta p_n(\mathbf{x},t)}\right]_{p=\bar p}. \tag{9.3.8}$$

With $p_n(\mathbf{x},t)$ set equal to this value, the argument of the exponential in Eq. (9.2.17) is the ordinary Lagrangian

$$L[q(t),\dot q(t)]\equiv\int d^3x\left(\sum_n\dot q_n(\mathbf{x},t)\bar p_n(\mathbf{x},t)-H[q(t),\bar p(t)]\right) \tag{9.3.9}$$

and we can write Eq. (9.2.17) as

$$\begin{aligned}&\langle\Omega;{\rm out}|T\left\{\mathcal{O}_A[Q(t_A)],\mathcal{O}_B[Q(t_B)],\ldots\right\}|\Omega;{\rm in}\rangle\\ &=|\mathcal{N}|^2\int\prod_{\tau,\mathbf{x},n}dq_n(\mathbf{x},\tau)\left({\rm Det}[2i\pi\mathcal{A}[q]]\right)^{-1/2}\\ &\quad\cdot\mathcal{O}_A[q(t_A)]\mathcal{O}_B[q(t_B)]\cdots\\ &\quad\cdot\exp\left[i\int_{-\infty}^{\infty}d\tau\left\{L[q(\tau),\dot q(\tau)]+i\epsilon\text{ terms}\right\}\right].\end{aligned} \tag{9.3.10}$$

(We have combined the $1/2\pi$ factors in the integrals over the p_n with the determinant coming from Eq. (9.3.6).) This is the desired Lagrangian form of the path-integral formula.

In deriving Eq. (9.3.10), it was necessary to assume that the operators $\mathcal{O}_A,\mathcal{O}_B,\ldots$ were independent of the canonical 'momenta'. This is not as restrictive as it may seem. For instance, in a scalar field theory for which the canonical conjugate to Φ is $\Pi=\dot\Phi$, it is possible to calculate the matrix element of a time-ordered product of operators, one of which is $\dot\Phi(t)$, by taking the difference of matrix elements in which this operator is replaced with $\Phi(t+d\tau)$ and $\Phi(t)$, and then dividing by $d\tau$, with $d\tau\to 0$. Equivalently, as long as t is not equal to any of the other time-arguments of the operator in Eq. (9.3.10), we can simply differentiate Eq. (9.3.10) with respect to t.

The one serious remaining complication in Eq. (9.3.10) is the determinant of $\mathcal{A}[q]$. If $\mathcal{A}[q]$ is field-independent, then this is no problem; we have already noted that overall constants make no contribution to the connected parts of vacuum expectation values, in which we divide by a vacuum-vacuum amplitude proportional to the same constant factor. This is the case for instance for the theory of a set of scalar fields Φ_n with non-derivative coupling to each other and/or derivative coupling to external currents J_n. The Lagrangian density here is

$$\mathcal{L}=-\sum_n\left[\frac12\partial_\mu\Phi_n\partial^\mu\Phi_n+J_n^\mu\partial_\mu\Phi_n\right]-V(\Phi).$$

An obvious extension of the results of Section 7.5 from one to several derivatively coupled scalars shows that this Lagrangian implies the Hamiltonian

$$H=\int d^3x\sum_n\left[\frac12\Pi_n^2+\frac12(\boldsymbol{\nabla}\Phi_n)^2+\mathbf{J}_n\cdot\boldsymbol{\nabla}\Phi_n+J_n^0\Pi_n+\frac12(J_n^0)^2\right]+\int d^3x\,V(\Phi).$$

(The Φ_n are taken to be real scalars, but complex scalars can be accommodated by separating them into real and imaginary parts.) In general there is a non-trivial term that is linear in the Π_n, but the coefficient of the quadratic term is a constant, just the unit 'matrix':

$$\mathcal{A}_{xn,x'n'}=\delta^4(x-x')\delta_{nn'}.$$

The factor $\left({\rm Det}[4i\pi\mathcal{A}[q]]\right)^{-1/2}$ in Eq. (9.3.10) is here a field-independent constant, and therefore is without effect.

However, matters are not always so simple. As a second example, let us consider the so-called non-linear σ-model, with Lagrangian density

$$\mathcal{L}=-\frac12\sum_{nm}\partial_\mu\Phi_n\partial^\mu\Phi_m[\delta_{nm}+U_{nm}(\Phi)]-V(\Phi).$$

A straightforward calculation gives the Hamiltonian as

$$H=\int d^3x\left[\frac12\Pi_n(\mathbf{1}+U(\Phi))^{-1}_{nm}\Pi_m+\frac12\boldsymbol{\nabla}\Phi_n\cdot\boldsymbol{\nabla}\Phi_m(\mathbf{1}+U(\Phi))_{nm}+V(\Phi)\right].$$

Here $\mathcal{A}$ is the field-dependent quantity

$$\mathcal{A}_{nx,my}=[\mathbf{1}+U(\Phi(x))]^{-1}_{nm}\delta^4(x-y).$$

In cases of this sort, the determinant may be reexpressed as a contribution to the effective Lagrangian, using the relation ${\rm Det}\,\mathcal{A}=\exp{\rm Tr}\ln\mathcal{A}$. By replacing the continuum of spacetime positions with a discrete lattice of points surrounded by separate regions of very small spacetime volume Ω, we may interpret the delta function in $\mathcal{A}_{nx,my}$ as $\delta^4(x-y)=\Omega^{-1}\delta_{x,y}$, so that

$$(\ln\mathcal{A})_{nx,my}=\delta_{x,y}\left[\ln(\mathbf{1}+U(\Phi(x)))-\mathbf{1}\cdot\ln\Omega\right]_{nm}$$

with the logarithm of a matrix defined now by its power series expansion

$$\ln(\mathbf{1}+U)=U-\frac{U^2}{2}+\frac{U^3}{3}-\cdots.$$

To evaluate the trace, we note that $\sum_x\cdots=\Omega^{-1}\int d^4x\cdots$. The determinant factor here is then

$${\rm Det}\,\mathcal{A}\propto\exp\left[-\Omega^{-1}\int d^4x\,{\rm tr}\ln[\mathbf{1}+U(\Phi(x))]\right],$$

where 'tr' is to be understood as the trace in an ordinary matrix sense. The constant of proportionality (which arises from the $-\ln\Omega$ term) is field-independent and therefore of no present interest. We can regard this determinant as providing a correction to the effective Lagrangian density

$$\Delta\mathcal{L}=-\frac i2\Omega^{-1}\,{\rm tr}\ln[\mathbf{1}+U(\Phi(x))].$$

The factor Ω^{-1} may be written as an ultraviolet divergent integral

$$\Omega^{-1}=\delta^4(x-x)=(2\pi)^{-4}\int d^4p\cdot 1.$$

We shall not show this here, but the extra terms in the Feynman diagrams for this theory contributed by $\Delta\mathcal{L}$ could also have been derived in the canonical formalism by taking account of the equal-time-commutator terms in the propagator of time-derivatives of the scalar field [7]. Ignoring this correction would lead to a spurious dependence of the S-matrix on the way that the scalar field is defined, and would also be inconsistent with any symmetries of the Lagrangian under transformations of the scalar fields.

Even where the factor $({\rm Det}\,\mathcal{A})^{-1/2}$ in the path-integral formula (9.3.10) is field-independent, the Lagrangian in this formula may not be the same as the one with which we started. As an example, let's consider the theory of a set of real vector fields, with Lagrangian density

$$\begin{aligned}\mathcal{L}=-\sum_n\Big[&\frac14(\partial_\mu A_{n\nu}-\partial_\nu A_{n\mu})(\partial^\mu A_n^\nu-\partial^\nu A_n^\mu)\\ &+\frac12m_n^2A_{n\mu}A_n^\mu+J_n^\mu A_{n\mu}\Big],\end{aligned}$$

where the currents J_n^μ are either externally produced c-number quantities or depend on other fields (in which case terms describing these other fields are to be added to the Lagrangian).

By a simple extension of the results of Section 7.5, we see that the Hamiltonian is

$$\begin{aligned}H=\int d^3x\sum_n\Big[&\frac12\mathbf{\Pi}_n^2+\frac12(\boldsymbol{\nabla}\cdot\mathbf{A}_n)^2+\frac12m_n^2\mathbf{A}_n^2+\frac{1}{2m_n^2}(\boldsymbol{\nabla}\cdot\mathbf{\Pi}_n)^2\\ &+\mathbf{J}_n\cdot\mathbf{A}_n-\frac{1}{m_n^2}J_n^0\boldsymbol{\nabla}\cdot\mathbf{\Pi}_n+\frac{1}{2m_n^2}(J_n^0)^2\Big],\end{aligned}$$

again with the understanding that other terms must be added involving any fields that appear in J_n^μ. Here the coefficient of the quadratic term is somewhat more complicated than in our first example:

$$\mathcal{A}_{nix,mjy}=\delta_{nm}\left[\delta_{ij}\delta^4(x-y)-\frac{1}{2m_n^2}\nabla_i\nabla_j\delta^4(x-y)\right],$$

but it is field-independent, so that the factor $({\rm Det}\,\mathcal{A})^{-1/2}$ has no effect. On the other hand, the Lagrangian (9.3.9) is here not the one with which we started; it is expressed entirely in terms of $\mathbf{A}$ and its spacetime derivatives, with no dependence on any time-component A^0. For this reason, the Lorentz invariance of Eq. (9.3.10) is far from obvious.

To remedy this, we may reintroduce the auxiliary field. Suppose we add to the Hamiltonian a term

$$\Delta H=-\frac12\sum_n m_n^2\int d^3x\left[A_n^0-m_n^{-2}\boldsymbol{\nabla}\cdot\mathbf{\Pi}_n+m_n^{-2}J_n^0\right]^2$$

and integrate over the A_n^0 as well as the $\mathbf{A}_n$ and $\mathbf{\Pi}_n$. This can only introduce a field-independent overall factor, since ΔH is a quadratic in A^0 (with a field-independent coefficient in the term of second order in A^0) whose stationary value vanishes. However, suppose that we now integrate over the $\mathbf{\Pi}_n$ *before* integrating over the A_n^0. The Hamiltonian in the path integral (9.2.17) is here replaced with

$$\begin{aligned}H+\Delta H=\int d^3x\sum_n\Big[&\frac12\mathbf{\Pi}_n^2+\frac12(\boldsymbol{\nabla}\cdot\mathbf{A}_n)^2+\frac12m_n^2\mathbf{A}_n^2-\frac12m_n^2(A_n^0)^2\\ &+\mathbf{J}_n\cdot\mathbf{A}_n-J_n^0A_n^0+A_n^0\boldsymbol{\nabla}\cdot\mathbf{\Pi}_n\Big].\end{aligned}$$

This is still quadratic in $\mathbf{\Pi}_n$, with a field-independent (and somewhat simpler) coefficient of the quadratic term, so the integral over the $\mathbf{\Pi}_n$s can be done by just replacing $\mathbf{\Pi}_n$ with its value at the stationary point of the functional $\sum_n\int d^3x\,\mathbf{\Pi}_n\cdot\dot{\mathbf{A}}_n-H-\Delta H$:

$$\mathbf{\Pi}_n=\dot{\mathbf{A}}_n+\boldsymbol{\nabla}A_n^0.$$

With $\mathbf{\Pi}_n$ eliminated in this way, $\sum_n\int d^3x\,\mathbf{\Pi}_n\cdot\dot{\mathbf{A}}_n-H-\Delta H$ is just the Lorentz-invariant Lagrangian with which we started.

In order to take account of the possible need to introduce auxiliary fields like A_n^0, from now on we shall write the path-integral formula after elimination of the canonical conjugates

in terms of fields ψ_ℓ that include both canonical fields q_n and auxiliary fields c_r:

$$\begin{aligned}&\langle\Omega;{\rm out}|T\left\{\mathcal{O}_A[\Psi_A(t_A)],\mathcal{O}_B[\Psi_B(t_B)],\ldots\right\}|\Omega;{\rm in}\rangle\\ &\propto\int\prod_{\tau,\mathbf{x},n}d\psi_n(\mathbf{x},\tau)\mathcal{O}_A[\psi(t_A)]\mathcal{O}_B[\psi(t_B)]\cdots\\ &\quad\cdot\exp\left[i\int_{-\infty}^{\infty}d\tau\left\{L[\psi(\tau),\dot\psi(\tau)]+i\epsilon\text{ terms}\right\}\right],\end{aligned} \tag{9.3.11}$$

it now being understood that L includes any terms arising from a possible field-dependent factor $({\rm Det}\,\mathcal{A})^{-1/2}$.

9.4 Path-Integral Derivation of Feynman Rules

We are now ready to use the path-integral formalism to derive the Feynman rules in a wide class of theories. We will concentrate here on the vacuum expectation values of time-ordered products of field operators (and their adjoints),

$$M_{\ell_A\ell_B\cdots}(x_Ax_B\cdots)=\frac{\langle\Omega;{\rm out}|T\{\Psi_{\ell_A}(x_A),\Psi_{\ell_B}(x_B),\ldots\}|\Omega;{\rm in}\rangle}{\langle\Omega;{\rm out}|\Omega;{\rm in}\rangle}, \tag{9.4.1}$$

from which S-matrix elements may be obtained (as shown in Section 6.4) by stripping off the final propagators associated with each field, replacing them with the coefficient functions that multiply creation or annihilation operators in the corresponding free fields, and summing over the indices on these coefficient functions.

For the simpler theories whose Hamiltonian is quadratic in the Πs, Eq. (9.3.11) gives

$$M_{\ell_A\ell_B\cdots}(x_Ax_B\cdots)=\frac{\displaystyle\int\left[\prod_{x,\ell}d\psi_\ell(x)\right]\psi_{\ell_A}(x_A)\psi_{\ell_B}(x_B)\cdots e^{iI[\psi]}}{\displaystyle\int\left[\prod_{x,\ell}d\psi_\ell(x)\right]e^{iI[\psi]}}, \tag{9.4.2}$$

where $I[\psi]$ is the action

$$I[\psi]=\int_{-\infty}^{\infty}d\tau\left\{L[\psi(\tau),\dot\psi(\tau)]+i\epsilon\text{ terms}\right\} \tag{9.4.3}$$

with L now including any terms that may arise from a field-dependent determinant in Eq. (9.3.10).

Let us now suppose that the Lagrangian is the integral of a Lagrangian density, consisting of a quadratic term $\mathcal{L}_0$ which would be present in the absence of interactions, plus a Lagrangian interaction density $\mathcal{L}_1$:

$$\begin{aligned}L[\psi(t),\dot\psi(t)]=\int d^3x\big[&\mathcal{L}_0(\psi(\mathbf{x},t),\partial_\mu\psi(\mathbf{x},t))\\ &+\mathcal{L}_1(\psi(\mathbf{x},t),\partial_\mu\psi(\mathbf{x},t))\big].\end{aligned} \tag{9.4.4}$$

$$\langle p| = \langle \Omega| \exp\left(-i\sum_a p_a Q_a\right) \tag{9.5.24}$$

with scalar product (now derived by moving the Ps to the left)

$$\langle p'|p\rangle = \prod_a (p'_a - p_a). \tag{9.5.25}$$

The scalar products of these two sorts of eigenstate with each other are

$$\begin{aligned}\langle q|p\rangle &= \langle q|\exp\left(-i\sum_a Q_a p_a\right)\left(\prod_a P_a\right)|\Omega\rangle \\ &= \left(\prod_a \exp(-iq_a p_a)\right)\langle q|\left(\prod_a P_a\right)|\Omega\rangle \\ &= \left(\prod_a \exp(-iq_a p_a)\right)\langle \Omega|\left(\prod_a Q_a\right)\left(\prod_a P_a\right)|\Omega\rangle,\end{aligned}$$

and so

$$\langle q|p\rangle = \chi_N \exp\left(-i\sum_a q_a p_a\right) = \chi_N \exp\left(i\sum_a p_a q_a\right), \tag{9.5.26}$$

where χ_N is a phase that depends only on the number N of Q_a operators:

$$\chi_N \equiv \langle \Omega|\left(\prod_a Q_a\right)\left(\prod_a P_a\right)|\Omega\rangle = i^N(-1)^{N(N-1)/2}.$$

Somewhat more simply, we also find

$$\langle p|q\rangle = \prod_a \exp(-ip_a q_a). \tag{9.5.27}$$

It is easy to see that the states $|q\rangle$ are in a sense a complete set (and so also are the $|p\rangle$.) From the definitions (9.5.17), we see that the state $|a,b,\ldots\rangle$ in the general basis is (up to a phase) just the coefficient of the product $q_a q_b \cdots$ in an expansion of $|q\rangle$ in a sum of products of qs. Therefore we can write any state $|f\rangle$ in the form

$$|f\rangle = f_0|q\rangle_0 + \sum_a f_a|q\rangle_a + \sum_{a\neq b} f_{ab}|q\rangle_{ab} + \cdots,$$

where the fs are numerical coefficients, and a subscript $a, b, \ldots$ on $|q\rangle$ denotes the coefficient of $q_a q_b \cdots$ in $|q\rangle$.

In summing over states, it will be very convenient to introduce a sort of integration over fermionic variables, known as *Berezin integration* [10], that is designed to pick out the coefficients of such products of anticommuting c-numbers. For any set of such variables ξ_n (either ps or qs or both together), the most general function $f(\xi)$ (either a c-number or a state-vector like $|q\rangle$) can be put in the form

$$f(\xi) = \left(\prod_n \xi_n\right)c + \text{terms with fewer } \xi \text{ factors} \tag{9.5.28}$$

and the integral over the ξs is defined simply by

$$\int\left(\widetilde{\prod_n} d\xi_n\right) f(\xi) \equiv c \tag{9.5.29}$$

with the tilde in Eq. (9.5.29) indicating that we use the convenient convention that the differentials are written in an order *opposite* to that of the product of integration variables in Eq. (9.5.28). Since this product is antisymmetric under the interchange of any two ξs, the integral is likewise antisymmetric under the interchange of any two $d\xi$s, so these 'differentials' effectively anticommute

$$d\xi_n\, d\xi_m + d\xi_m\, d\xi_n = 0. \tag{9.5.30}$$

Also, the coefficient c may itself depend on other unintegrated c-number variables that anticommute with the ξs over which we integrate, in which case it is important to standardize the definition of c by moving all ξs to the left of c before integrating over them, as we have done in Eq. (9.5.28).

For instance, the most general function of a pair of anticommuting c-numbers ξ_1 and ξ_2 takes the form

$$f(\xi_1,\xi_2) = \xi_1\xi_2 c_{12} + \xi_1 c_1 + \xi_2 c_2 + d$$

because the squares and all higher powers of ξ_1 and ξ_2 vanish. This function has the integrals

$$\int d\xi_1\, f(\xi_1,\xi_2) = \xi_2 c_{12} + c_1, \qquad \int d\xi_2\, f(\xi_1,\xi_2) = -\xi_1 c_{12} + c_2,$$

$$\int d\xi_2\, d\xi_1\, f(\xi_1,\xi_2) = c_{12}.$$

Note that the multiple integral is the same as a repeated integral:

$$\int d\xi_2\, d\xi_1\, f(\xi_1,\xi_2) = \int d\xi_2\left[\int d\xi_1\, f(\xi_1,\xi_2)\right],$$

a result that can easily be extended to integrals over any number of fermionic variables. (It was in order to obtain this result without extra sign factors that we took the product of differentials in Eq. (9.5.29) to be in the opposite order to the product of variables in Eq. (9.5.28).) Indeed, we could have first defined the integral over a single anticommuting c-number ξ_1, and then defined multiple integrals in the usual way by iteration. The most general function of anticommuting c-numbers is linear in any one of them

$$f(\xi_1,\xi_2,\ldots) = b(\xi_2,\ldots) + \xi_1 c(\xi_2,\ldots)$$

(because $\xi_1^2 = 0$), and its integral over ξ_1 is defined as

$$\int d\xi_1\, f(\xi_1,\xi_2,\ldots) = c(\xi_2,\ldots).$$

Repeating this process leads to the same multiple integral as defined by Eqs. (9.5.28) and (9.5.29).

This definition of integration shares some other properties with multiple integrals (from $-\infty$ to $+\infty$) over ordinary real variables, but there are significant differences.

Obviously, Berezin integration is linear, in the sense that

$$\int\left(\widetilde{\prod_n} d\xi_n\right)[f(\xi)+g(\xi)] = \int\left(\widetilde{\prod_n} d\xi_n\right)f(\xi) + \int\left(\widetilde{\prod_n} d\xi_n\right)g(\xi) \tag{9.5.31}$$

and also

$$\int\left(\widetilde{\prod_n} d\xi_n\right)[f(\xi)a(\xi')] = \left[\int\left(\widetilde{\prod_n} d\xi_n\right)f(\xi)\right]a(\xi'), \tag{9.5.32}$$

where $a(\xi')$ is any function (including a constant) of any anticommuting c-numbers ξ'_m over which we are *not* integrating. However, linearity with respect to left-multiplication is not so obvious. If we are integrating over ν variables, then since ξ'_m is assumed to anticommute with all ξ_n, we have

$$a((-1)^\nu\xi')\left(\prod_n \xi_n\right) = \left(\prod_n \xi_n\right)a(\xi')$$

and so

$$\int\left(\widetilde{\prod_n} d\xi_n\right)[a((-1)^\nu\xi')f(\xi)] = a(\xi')\int\left(\widetilde{\prod_n} d\xi_n\right)f(\xi). \tag{9.5.33}$$

It is therefore very convenient (though not strictly necessary) to take the differentials $d\xi_n$ to anticommute with all anticommuting variables (including the ξ_n):

$$(d\xi_n)\xi'_m + \xi'_m(d\xi_n) = 0 \tag{9.5.34}$$

in which case Eq. (9.5.33) reads more simply

$$\int a(\xi')\left(\widetilde{\prod_n} d\xi_n\right)f(\xi) = a(\xi')\int\left(\widetilde{\prod_n} d\xi_n\right)f(\xi). \tag{9.5.35}$$

Another similarity with ordinary integration is that, for an arbitrary anticommuting c-number ξ' independent of ξ,

$$\int\left(\widetilde{\prod_n} d\xi_n\right)f(\xi+\xi') = \int\left(\widetilde{\prod_n} d\xi_n\right)f(\xi) \tag{9.5.36}$$

since shifting ξ by a constant only affects the terms in f with fewer than the total number of ξ-variables.

On the other hand, consider a change of variables

$$\xi_n \longrightarrow \xi'_n = \sum_m \mathcal{G}_{nm}\xi_m, \tag{9.5.37}$$

where $\mathcal{G}$ is an arbitrary non-singular matrix of ordinary numbers. The product of the new variables is

$$\prod_n \xi'_n = \sum_{m_1 m_2\cdots}\left(\prod_n \mathcal{G}_{nm_n}\xi_{m_n}\right).$$

But $\prod_n \xi_{m_n}$ here is just the same as the product (in the original order) $\prod_n \xi_n$, except for a sign $\epsilon[m]$ which is $+1$ or -1 according to whether the permutation $n \to m_n$ is an even or odd permutation of the original order:

$$\prod_n \xi'_n = \left[\sum_{m_1 m_2\cdots}\left(\prod_n \mathcal{G}_{nm_n}\right)\epsilon[m]\right]\prod_n \xi_n = (\operatorname{Det}\mathcal{G})\prod_n \xi_n.$$

This applies whatever order we take for the ξ_n, as long as we take the ξ'_n in the same order. It follows that the coefficient of $\prod_n \xi'_n$ in any function $f(\xi)$ is just $(\operatorname{Det}\mathcal{G})^{-1}$ times the coefficient of $\prod_n \xi_n$, a statement we write as

$$\int\left(\widetilde{\prod_n} d\xi'_n\right)f = (\operatorname{Det}\mathcal{G})^{-1}\int\left(\widetilde{\prod_n} d\xi_n\right)f. \tag{9.5.38}$$

This is the usual rule for changing variables of integration, except that $(\operatorname{Det}\mathcal{G})$ appears to the power -1 instead of $+1$. We shall use Eq. (9.5.38) and the linearity properties (9.5.31), (9.5.32), and (9.5.35) later to evaluate the integrals encountered in deriving the Feynman rules for theories with fermions.

We can now use this definition of integration to write the completeness condition as a formula for an integral over eigenvalues. As already mentioned, any state $|f\rangle$ can be expanded in a series of the states $|\Omega\rangle$, $|a\rangle$, $|a,b\rangle$, etc. and these states are (up to a phase) the coefficients of the products 1, q_a, $q_a q_b$, etc. in the Q-eigenstate $|q\rangle$. According to the definition of integration here, we can pick out the coefficient of any product $q_b q_c q_d \cdots$ in the state $|q\rangle$ by integrating the product of $|q\rangle$ with all q_a with a *not* equal to $b, c, d, \ldots$. Thus, by choosing a function $f(q)$ as a suitable sum of such products of qs, we can write any state $|f\rangle$ as an integral:

$$|f\rangle = \int\left(\widetilde{\prod_a} dq_a\right)|q\rangle f(q) = \int|q\rangle\left(\widetilde{\prod_a} dq_a\right)f(q). \tag{9.5.39}$$

(We can move $|q\rangle$ to the left of the differentials without any sign changes because the exponential in Eq. (9.5.17) used to define $|q\rangle$ involves only even numbers of fermionic quantities.) To find the function $f(q)$ for a given state-vector $|f\rangle$, take the scalar product of Eq. (9.5.39) with some bra $\langle q'|$ (with q' any fixed Q-eigenvalue). According to Eqs. (9.5.35) and (9.5.20), this is

$$\langle q'|f\rangle = \int\left(\prod_a(q_a - q'_a)\right)\left(\widetilde{\prod_b} dq_b\right)f(q).$$

Moving every factor $(q_a - q'_a)$ to the right past every differential dq_b yields a sign factor $(-)^{N^2} = (-)^N$, where N is now the total number of q_a variables, so

$$\langle q'|f\rangle = (-)^N\int\left(\widetilde{\prod_b} dq_b\right)\left(\prod_a(q_a - q'_a)\right)f(q).$$

We can rewrite $f(q)$ as $f(q' + (q - q'))$ and expand in powers of $q - q'$. All terms beyond the lowest order vanish when multiplied with the product

$\prod_a(q_a - q'_a)$, so

$$\left(\prod_a(q_a - q'_a)\right)f(q) = \left(\prod_a(q_a - q'_a)\right)f(q'), \tag{9.5.40}$$

which partly justifies our earlier remark that Eq. (9.5.20) plays the role of a delta function for integrals over the qs. Using Eq. (9.5.32), we now have

$$\langle q'|f\rangle = (-)^N\left[\int\left(\widetilde{\prod_b} dq_b\right)\left(\prod_a(q_a - q'_a)\right)f(q')\right].$$

The term in the integrand proportional to $\prod_a q_a$ has coefficient $f(q')$, so according to our definition of integration $\langle q'|f\rangle = (-)^N f(q')$. Inserting this back in Eq. (9.5.39) gives our completeness relation

$$|f\rangle = (-)^N\int|q\rangle\left(\widetilde{\prod_b} dq_b\right)\langle q|f\rangle,$$

or as an operator equation

$$\mathbf{1} = \int|q\rangle\left(\widetilde{\prod_a}(-dq_a)\right)\langle q|. \tag{9.5.41}$$

In exactly the same way, we can also show that

$$\mathbf{1} = \int|p\rangle\left(\widetilde{\prod_a} dp_a\right)\langle p|. \tag{9.5.42}$$

We are now in a position to calculate transition matrix elements. As before, we define time-dependent operators

$$Q_a(t) = \exp(iHt)Q_a\exp(-iHt) \tag{9.5.43}$$

$$P_a(t) = \exp(iHt)P_a\exp(-iHt) \tag{9.5.44}$$

and their right- and left-eigenstates

$$|q;t\rangle \equiv \exp(iHt)|q\rangle, \qquad |p;t\rangle \equiv \exp(iHt)|p\rangle, \tag{9.5.45}$$

$$\langle q;t| \equiv \langle q|\exp(-iHt), \qquad \langle p;t| \equiv \langle p|\exp(-iHt). \tag{9.5.46}$$

The scalar product between q-eigenstates defined at infinitesimally close times is then

$$\langle q';\tau+d\tau|q;\tau\rangle = \langle q'|\exp(-iH\,d\tau)|q\rangle.$$

Now insert Eq. (9.5.42) to the left of the operator $\exp(-iH\,d\tau)$. It is convenient here to define the Hamiltonian operator $H(P,Q)$ with all Ps to the left of all Qs, so that (for $d\tau$ infinitesimal)

$$\langle p|\exp[-iH(P,Q)d\tau]|q\rangle = \langle p|q\rangle\exp[-iH(p,q)d\tau].$$

(We could move the c-number $H(p,q)$ to either side of the matrix element without any sign changes because each term in the Hamiltonian is assumed to contain an even number of fermionic operators.) This gives

$$\begin{aligned}\langle q';\tau+d\tau|q;\tau\rangle &= \int\langle q'|p\rangle\left(\widetilde{\prod_a} dp_a\right)\langle p|\exp(-iH\,d\tau)|q\rangle \\ &= \int\langle q'|p\rangle\left(\widetilde{\prod_a} dp_a\right)\langle p|q\rangle\exp[-iH(p,q)d\tau].\end{aligned}$$

Using Eqs. (9.5.26) and (9.5.27), and noting that the products $p_a q_a$ and $p_a q'_a$ commute with all anticommuting c-numbers, we find

$$\langle q';\tau+d\tau|q;\tau\rangle = \int\left(\widetilde{\prod_a} i\,dp_a\right)\exp\left[i\sum_a p_a(q'_a - q_a) - iH(p,q)d\tau\right]. \tag{9.5.47}$$

The rest of the derivation follows the same lines as in Section 9.1. To calculate the matrix element $\langle q';t'|\mathcal{O}_A(P(t_A),Q(t_A))\mathcal{O}_B(P(t_B),Q(t_B))\cdots|q;t\rangle$ of a product of operators (with $t' > t_A > t_B > \cdots > t$), divide the time-interval from t to t' into a large number of very close time steps; at each time step insert the completeness relation (9.5.41); use Eq. (9.5.47) to evaluate the resulting matrix elements (with $\mathcal{O}_A, \mathcal{O}_B$, etc. inserted where appropriate); move all differentials to the left (this introduces no sign changes, because at each step we have an equal number of dps and dqs); and then introduce functions $q_a(t)$ and $p_a(t)$ that interpolate between the values of q_a and p_a at each step. We then find

$$\begin{aligned}&\langle q';t'|T\{\mathcal{O}_A(P(t_A),Q(t_A)),\mathcal{O}_B(P(t_B),Q(t_B)),\ldots\}|q;t\rangle \\ &= (-i)^N\chi_N\int_{q_a(t)=q_a,\,q_a(t')=q'_a}\left(\widetilde{\prod_{a\tau}} dq_a(\tau)\,dp_a(\tau)\right) \\ &\quad\cdot\mathcal{O}_A[p(t_A),q(t_A)]\mathcal{O}_B[p(t_B),q(t_B)]\cdots \\ &\quad\cdot\exp\left[i\int_t^{t'}d\tau\left\{\sum_a p_a(\tau)\dot q_a(\tau) - H[p(\tau),q(\tau)]\right\}\right].\end{aligned} \tag{9.5.48}$$

The symbol T here denotes the ordinary product if the times are in the order originally assumed, $t_A > t_B > \cdots$. However, the right-hand side is totally symmetric in the $\mathcal{O}_A, \mathcal{O}_B, \ldots$ (except for minus signs where anticommuting c-numbers are interchanged) so this formula holds for general times (between t and t'), provided T is interpreted as the time-ordered product, with an overall minus sign if time-ordering the operators involves an odd number of permutations of fermionic operators.

Up to this point we have kept track of the overall phase factor $(-i)^N\chi_N$. But in fact these phases contribute only to the vacuum–vacuum transition amplitude, and hence will not be of importance to us.

The transition to quantum field theory follows along the same lines as described for bosonic fields in Section 9.2. The vacuum expectation value of a time-ordered product of

operators is given by a formula just like Eq. (9.2.17):

$$\begin{aligned}&\langle\Omega;\text{out}|T\{\mathcal{O}_A[P(t_A),Q(t_A)],\\ &\qquad\mathcal{O}_B[P(t_B),Q(t_B)],\ldots\}|\Omega;\text{in}\rangle \\ &\propto\int\left[\prod_{\tau,\mathbf{x},m} dq_m(\mathbf{x},\tau)\right]\left[\prod_{\tau,\mathbf{x},m} dp_m(\mathbf{x},\tau)\right] \\ &\quad\cdot\mathcal{O}_A[p(t_A),q(t_A)]\mathcal{O}_B[p(t_B),q(t_B)]\cdots \\ &\quad\cdot\exp\left[i\int_{-\infty}^{\infty}d\tau\left\{\begin{array}{l}\int d^3x\sum_m p_m(\mathbf{x},\tau)\dot q_m(\mathbf{x},\tau)\\ -H[q(\tau),p(\tau)] + i\epsilon\text{ terms}\end{array}\right\}\right],\end{aligned} \tag{9.5.49}$$

where the proportionality constant is the same for all operators $\mathcal{O}_A, \mathcal{O}_B$, etc., and the '$i\epsilon$ terms' again arise from the wave function of the vacuum. As before, we have replaced each discrete index like a with a space position $\mathbf{x}$ and a field index m. We are also dropping the tilde on the product of differentials, since it only affects the constant phase in the path integral.

A major difference between the fermionic and bosonic cases is that here we will not want to integrate out the ps before the qs. Indeed, in the standard model of electroweak interactions (and in other theories, such as the older Fermi theory of beta decay) the canonical conjugates p_m are auxiliary fields unrelated to the q_m, and the Lagrangian is linear in the $\dot q_m$, so that the quantity $\int d^3x\sum_m p_m\dot q_m - H$ in Eq. (9.5.49) as it stands is the Lagrangian L. Each term in the Hamiltonian for a fermionic field that carries a non-vanishing quantum number (like the electron field in quantum electrodynamics) generally contains an equal number of ps (proportional to $q_m^\dagger$) and qs. In particular, the free-particle term H_0 in the Hamiltonian is bilinear in p and q, so that

$$\begin{aligned}&\int_{-\infty}^{\infty}d\tau\left\{\int d^3x\sum_m p_m(\mathbf{x},\tau)\dot q_m(\mathbf{x},\tau) - H_0[q(\tau),p(\tau)] + i\epsilon\text{ terms}\right\} \\ &= -\sum_{mn}\int d^4x\,d^4y\,\mathcal{D}_{m\mathbf{x},n\mathbf{y}}p_m(x)q_n(y)\end{aligned} \tag{9.5.50}$$

with $\mathcal{D}$ some numerical 'matrix'. The interaction Hamiltonian $H_{\text{int}} = H - H_0$ is a sum of products of equal numbers of fermionic qs and ps (with coefficients that may depend on bosonic fields) so when we expand Eq. (9.5.49) in powers of the H_{int} we encounter a sum of fermionic integrals of the form

$$\begin{aligned}&\mathcal{I}_{m_1 n_1 m_2 n_2\cdots m_N n_N}(x_1,y_1,x_2,y_2,\ldots,x_N,y_N) \\ &\equiv\int\left[\prod_{\tau,\mathbf{x},m} dq_m(\mathbf{x},\tau)\right]\left[\prod_{\tau,\mathbf{x},m} dp_m(\mathbf{x},\tau)\right]q_{m_1}(x_1)p_{n_1}(y_1)q_{m_2}(x_2)p_{n_2}(y_2)\cdots \\ &\quad\cdot q_{m_N}(x_N)p_{n_N}(y_N) \\ &\quad\cdot\exp\left(-i\sum_{mn}\int d^4x\,d^4y\,\mathcal{D}_{m\mathbf{x},n\mathbf{y}}p_m(x)q_n(y)\right),\end{aligned} \tag{9.5.51}$$

one such term for each possible set of vertices in the Feynman diagram, with coefficients contributed by each vertex given by i times the coefficient of the product of fields in the corresponding term in the interaction.

To calculate this sort of integral, first consider a generating function for all these integrals:

$$\mathcal{J}(f,g) = \int\left[\prod_{\mathbf{x},\tau,m} dq_m(\mathbf{x},\tau)\,dp_m(\mathbf{x},\tau)\right]\cdot\exp\left\{\begin{array}{l}-i\sum_{mn}\int d^4x\,d^4y\,\mathcal{D}_{m\mathbf{x},n\mathbf{y}}p_m(x)q_n(y)\\ -i\sum_m\int d^4x\,p_m(x)f_m(x)\\ -i\sum_n\int d^4y\,g_n(y)q_n(y)\end{array}\right\}, \tag{9.5.52}$$

where $f_m(x)$ and $g_n(y)$ are arbitrary anticommuting c-number functions. We shift to new variables of integration

$$p'_m(x) = p_m(x) + \sum_n\int d^4y\,g_n(y)(\mathcal{D}^{-1})_{n\mathbf{y},m\mathbf{x}},$$

$$q'_n(y) = q_n(y) + \sum_m\int d^4x\,(\mathcal{D}^{-1})_{n\mathbf{y},m\mathbf{x}}f_m(x).$$

Using the translation invariance condition (9.5.36), we then find

$$\begin{aligned}\mathcal{J}(f,g) &= \exp\left(i\sum_{mn}\int d^4x\,d^4y\,(\mathcal{D}^{-1})_{n\mathbf{y},m\mathbf{x}}g_n(y)f_m(x)\right) \\ &\quad\cdot\int\left[\prod_{\mathbf{x},\tau,m} dq'_m(\mathbf{x},\tau)\,dp'_m(\mathbf{x},\tau)\right] \\ &\quad\cdot\exp\left(-i\sum_{mn}\int d^4x\,d^4y\,\mathcal{D}_{m\mathbf{x},n\mathbf{y}}p'_m(x)q'_n(y)\right).\end{aligned} \tag{9.5.53}$$

The integral is a constant (i.e., independent of the functions f and g) which can be shown using Eq. (9.5.38) to be proportional to $\operatorname{Det}\mathcal{D}$. Of more importance to us is the first factor. Expanding this factor in powers of gf and comparing with the direct expansion of Eq. (9.5.52), we see that

$$\begin{aligned}&\mathcal{I}_{m_1 n_1 m_2 n_2\cdots n_N m_N}(x_1,y_1,x_2,y_2,\ldots,x_N,y_N) \\ &\propto\sum_{\text{pairings}}\delta_{\text{pairing}}\prod_{\text{pairs}}\left(-i\mathcal{D}^{-1}\right)_{\text{paired }m\mathbf{x},n\mathbf{y}},\end{aligned} \tag{9.5.54}$$

with a proportionality constant that is independent of the x, y, m, or n, and also independent of the number of these variables. The sum is over all different ways of pairing ps with qs, not counting as different pairings that only differ in the order of the pairs. In other

words, we sum over the $N!$ permutations either of the ps or the qs. The sign factor δ_{pairing} is $+1$ if this permutation is even; -1 if it is odd.

This sign factor and sum over pairings are just the same as we encountered in our earlier derivation of the Feynman rules, with the sum over pairings corresponding to the sum over ways of connecting the lines associated with vertices in the Feynman diagrams, and the factors $(\mathcal{D}^{-1})_{m\mathbf{x},n\mathbf{y}}$ playing the role of the propagator for the pairing of $q_m(x)$ with $p_n(y)$. In the Dirac formalism for spin $\frac{1}{2}$, the free-particle action is

$$\begin{aligned}&\int_{-\infty}^{\infty}d\tau\left\{\int d^3x\sum_m p_m(\mathbf{x},\tau)\dot q_m(\mathbf{x},\tau) - H_0[q(\tau),p(\tau)]\right\} \\ &= -\int d^4x\,\bar\psi(x)[\gamma^\mu\partial_\mu + m]\psi(x),\end{aligned} \tag{9.5.55}$$

where in the usual notation the canonical variables here are

$$q_m(x) = \psi_m(x), \qquad p_m(x) = -[\bar\psi(x)\gamma^0]_m = i\psi_m^\dagger(x) \tag{9.5.56}$$

with m a four-valued Dirac index. Comparing this with Eq. (9.5.50), we find here

$$\begin{aligned}\mathcal{D}_{m\mathbf{x},n\mathbf{y}} &= \left\{\gamma^0\left[\gamma^\mu\frac{\partial}{\partial x^\mu} + m - i\epsilon\right]\right\}_{mn}\delta^4(x-y) \\ &= \int\frac{d^4k}{(2\pi)^4}\{\gamma^0[i\gamma^\mu k_\mu + m - i\epsilon]\}_{mn}e^{ik\cdot(x-y)}.\end{aligned} \tag{9.5.57}$$

(Though we shall not work it out in detail, the $i\epsilon$ term here arises in much the same way as for the scalar field in Section 9.2.) The propagator is then

$$(\mathcal{D}^{-1})_{m\mathbf{x},n\mathbf{y}} = \int\frac{d^4k}{(2\pi)^4}\{[i\gamma^\mu k_\mu + m - i\epsilon]^{-1}[-\gamma^0]\}_{mn}e^{ik\cdot(x-y)}, \tag{9.5.58}$$

just as we found in the operator formalism. The extra factor $-\gamma^0$ arises because this propagator is the vacuum expectation value of

$$T\{\psi_m(x), -[\bar\psi(y)\gamma^0]_n\}, \qquad \text{not} \qquad T\{\psi_m(x), \bar\psi_n(y)\}.$$

As one example of a problem that is easier to solve by path-integral than by operator methods, let us calculate the field dependence of the vacuum→vacuum amplitude for a Dirac field that interacts only with an external field. Take the Lagrangian as

$$\mathcal{L} = -\bar\psi[\gamma^\mu\partial_\mu + m + \Gamma]\psi, \tag{9.5.59}$$

where $\Gamma(x)$ is an x-dependent matrix representing the interaction of the fermion with the external field. According to Eq. (9.5.49), the vacuum persistence amplitude in the presence of this external field is

$$\begin{aligned}\langle\Omega;\text{out}|\Omega;\text{in}\rangle_\Gamma &\propto\int\left[\prod_{\tau,\mathbf{x},m} dq_m(\mathbf{x},\tau)\right]\left[\prod_{\tau,\mathbf{x},m} dp_m(\mathbf{x},\tau)\right] \\ &\quad\cdot\exp\left\{-i\int d^4x\,p^T\gamma^0[\gamma^\mu\partial_\mu + m + \Gamma - i\epsilon]q\right\}\end{aligned} \tag{9.5.60}$$

with a proportionality constant that is independent of $\Gamma(x)$. We write this as

$$\begin{aligned}\langle\Omega;\text{out}|\Omega;\text{in}\rangle_\Gamma &\propto\int\left[\prod_{\tau,\mathbf{x},m} dq_m(\mathbf{x},\tau)\right]\left[\prod_{\tau,\mathbf{x},m} dp_m(\mathbf{x},\tau)\right] \\ &\quad\cdot\exp\left\{-i\sum_{mn}\int d^4x\,d^4y\,p_m(x)q_n(y)\mathcal{K}[\Gamma]_{m\mathbf{x},n\mathbf{y}}\right\},\end{aligned} \tag{9.5.61}$$

where

$$\mathcal{K}[\Gamma]_{m\mathbf{x},n\mathbf{y}} = \{\gamma^0[\gamma^\mu\partial/\partial x^\mu + m + \Gamma(x) - i\epsilon]\}_{mn}\delta^4(x-y). \tag{9.5.62}$$

To evaluate this, we change the variables of integration $q_m(x)$ to

$$q'_m(x) = \sum_n\int d^4y\,\mathcal{K}[\Gamma]_{m\mathbf{x},n\mathbf{y}}q_n(y). \tag{9.5.63}$$

The remaining integral is now Γ-independent, so the whole dependence of the vacuum persistence amplitude is contained in the determinant arising according to Eq. (9.5.38) from the change of variables:

$$\langle\Omega;\text{out}|\Omega;\text{in}\rangle_\Gamma \propto \operatorname{Det}\mathcal{K}[\Gamma]. \tag{9.5.64}$$

To recover the results of perturbation theory, let us write

$$\mathcal{K}[\Gamma] \equiv \mathcal{D} + \mathcal{G}[\Gamma], \tag{9.5.65}$$

$$\mathcal{G}[\Gamma]_{m\mathbf{x},n\mathbf{y}} = (\gamma^0\Gamma(x))_{mn}\delta^4(x-y), \tag{9.5.66}$$

and expand in powers of $\mathcal{G}[\Gamma]$. Eq. (9.5.64) gives then

$$\begin{aligned}\langle\Omega;\text{out}|\Omega;\text{in}\rangle_\Gamma &\propto \operatorname{Det}\left(\mathcal{D}[\mathbf{1} + \mathcal{D}^{-1}\mathcal{G}[\Gamma]]\right) \\ &= [\operatorname{Det}\mathcal{D}]\exp\left(\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\operatorname{Tr}(\mathcal{D}^{-1}\mathcal{G}[\Gamma])^n\right).\end{aligned} \tag{9.5.67}$$

This is just what we should expect from the Feynman rules: the contributions from internal lines and vertices in this theory are $-i\mathcal{D}^{-1}$ and $-i\mathcal{G}[\Gamma]$; the trace of the product of n factors of $-\mathcal{D}^{-1}\mathcal{G}[\Gamma]$ thus corresponds to a loop with n vertices connected by n internal lines; $1/n$ is the usual combinatoric factor associated with such loops (see Section 6.1); the sign factor is $(-1)^{n+1}$ rather than $(-1)^n$ because an extra minus sign is associated with fermion loops; and the sum over n appears as the argument of an exponential because the vacuum persistence amplitude receives contributions from graphs with any number of disconnected loops. The Γ-independent factor $\operatorname{Det}\mathcal{D}$ is less easy to derive from the Feynman rules; it represents the contribution of any number of fermion loops that carry no vertices.

More to the point, a formula like Eq. (9.5.64) allows us to derive non-perturbative results by using topological theorems to derive information about the eigenvalues of kernels like $\mathcal{K}[\Gamma]$. This will be pursued further in Volume II.

9.6 Path-Integral Formulation of Quantum Electrodynamics

The path-integral approach to quantum field theory really comes into its own when applied to gauge theories of massless spin one particles, such as quantum electrodynamics. The derivation of the Feynman rules for quantum electrodynamics in the previous chapter involved a fair amount of hand-waving, in arguing that the terms in the photon propagator $\Delta^{\mu\nu}(q)$ proportional to q^μ or q^ν could be dropped, and that the purely time-like terms would just cancel the Coulomb term in the Hamiltonian, so that the effective photon propagator could be taken as $\eta^{\mu\nu}/q^2$. To give a real justification of this result using the methods of Chapter 8 would involve us in a complicated analysis of Feynman diagrams. But as we shall now see, the path-integral approach yields the desired form of the photon propagator, without ever having to think about the details of Feynman diagrams.

In Chapter 8 we found that in Coulomb gauge, the Hamiltonian for the interaction of photons with charged particles takes the form

$$H[\mathbf{A},\mathbf{\Pi}_\perp,\ldots] = H_M + \int d^3x\left[\frac{1}{2}\mathbf{\Pi}_\perp^2 + \frac{1}{2}(\nabla\times\mathbf{A})^2 - \mathbf{A}\cdot\mathbf{J}\right] + V_{\text{Coul}}. \tag{9.6.1}$$

Here $\mathbf{A}$ is the vector potential, subject to the Coulomb gauge condition

$$\nabla\cdot\mathbf{A} = 0, \tag{9.6.2}$$

while $\mathbf{\Pi}_\perp$ is the solenoidal part of its canonical conjugate, satisfying the same constraint

$$\nabla\cdot\mathbf{\Pi}_\perp = 0. \tag{9.6.3}$$

Also, H_M is the matter Hamiltonian and V_{Coul} is the Coulomb energy

$$V_{\text{Coul}}(t) = \frac{1}{2}\int d^3x\,d^3y\,\frac{J^0(\mathbf{x},t)J^0(\mathbf{y},t)}{4\pi|\mathbf{x}-\mathbf{y}|}. \tag{9.6.4}$$

Just as for any other Hamiltonian system, we can calculate vacuum expectation values of time-ordered products as path integrals[102]

$$\begin{aligned}\langle T\{\mathcal{O}_A\mathcal{O}_B\cdots\}\rangle_\Omega &= \int\left[\prod_{x,i} da_i(x)\prod_{x,i} d\pi_i(x)\prod_{x,\ell} d\psi_\ell(x)\right] \\ &\quad\cdot\mathcal{O}_A\mathcal{O}_B\cdots \\ &\quad\cdot\exp\left\{i\int d^4x\left[\boldsymbol{\pi}\cdot\dot{\mathbf{a}} - \frac{1}{2}\boldsymbol{\pi}^2 - \frac{1}{2}(\nabla\times\mathbf{a})^2\right.\right. \\ &\qquad\left.\left. + \mathbf{a}\cdot\mathbf{j} + \mathcal{L}_M\right] - i\int dt\,V_{\text{Coul}}\right\} \\ &\quad\cdot\left[\prod_x\delta(\nabla\cdot\mathbf{a}(x))\right]\left[\prod_x\delta(\nabla\cdot\boldsymbol{\pi}(x))\right],\end{aligned} \tag{9.6.5}$$

[102] Note that $\boldsymbol{\pi}(x)$ is the interpolating c-number field for the quantum operator $\mathbf{\Pi}_\perp$, whose commutation relations with each other and with $\mathbf{A}$ are the same as those of $\mathbf{\Pi}$, but which unlike $\mathbf{\Pi}$ commutes with all canonical matter variables.

where $\psi_\ell(x)$ are generic matter fields. In writing Eq. (9.6.5) in terms of a matter Lagrangian density, we are assuming that H_M is local and either linear in the matter πs (as in spinor electrodynamics) or quadratic with field-independent coefficients (as in scalar electrodynamics). We have inserted delta functions[103] in Eq. (9.6.5) to enforce the constraints (9.6.2) and (9.6.3).

The argument of the exponential in Eq. (9.6.5) is evidently quadratic in the independent components of $\boldsymbol{\pi}$ (say, π_1 and π_2), with field-independent coefficients in the term of second order in $\boldsymbol{\pi}$. Thus, according to Eq. (9.A.9), the integral over $\boldsymbol{\pi}$ can be done (up to a constant factor) by setting $\boldsymbol{\pi}$ equal to the stationary point of the argument of the exponential, $\boldsymbol{\pi} = \dot{\mathbf{a}}$:

$$\begin{aligned}\langle T\{\mathcal{O}_A\mathcal{O}_B\cdots\}\rangle_\Omega &= \int\left[\prod_{x,i} da_i(x)\prod_{x,\ell} d\psi_\ell(x)\right] \\ &\quad\cdot\mathcal{O}_A\mathcal{O}_B\cdots \\ &\quad\cdot\exp\left\{i\int d^4x\left[\frac{1}{2}\dot{\mathbf{a}}^2 - \frac{1}{2}(\nabla\times\mathbf{a})^2\right.\right. \\ &\qquad\left.\left. + \mathbf{a}\cdot\mathbf{j} + \mathcal{L}_M\right] - i\int dt\,V_{\text{Coul}} + i\epsilon\text{ terms}\right\} \\ &\quad\cdot\left[\prod_x\delta(\nabla\cdot\mathbf{a}(x))\right].\end{aligned} \tag{9.6.6}$$

To bring out the essential covariance of this result, we use a trick. Introduce a new variable of integration $a^0(x)$, and replace the Coulomb term $-\int dt\,V_{\text{Coul}}$ in the action with

$$\int d^4x\left[-a^0(x)j^0(x) + \frac{1}{2}(\nabla a^0(x))^2\right]. \tag{9.6.7}$$

Since (9.6.7) is quadratic in a^0, the integral over a^0 can be done (up to a constant factor) by setting $a^0(x)$ equal to the stationary point of (9.6.7), i.e., to the solution of

$$-j^0(x) - \nabla^2 a^0(x) = 0$$

or in other words, to

$$a^0(\mathbf{x},t) = \int d^3y\,\frac{j^0(\mathbf{y},t)}{4\pi|\mathbf{x}-\mathbf{y}|}. \tag{9.6.8}$$

[103] This is not strictly accurate. If we take the canonical variables to be, say, a_1, a_2 and π_1, π_2, with a_3 and π_3 regarded as functionals of these variables given by Eqs. (9.6.2) and (9.6.3), then we should insert the delta functions

$$\prod_x\delta\left(a_3(x) + \partial_3^{-1}[\partial_1 a_1(x) + \partial_2 a_2(x)]\right)\delta\left(\pi_3(x) + \partial_3^{-1}[\partial_1\pi_1(x) + \partial_2\pi_2(x)]\right).$$

However, this differs from the product of delta functions in Eq. (9.6.5) only by a factor $\operatorname{Det}\partial_3^2$, which although infinite is field-independent and hence cancels in ratios like (9.4.1).

Using this in Eq. (9.6.7) just gives the Coulomb action $-\int dt\,V_{\text{Coul}}$. Hence we can rewrite the argument of the exponential in Eq. (9.6.6) as

$$\begin{aligned}&\frac{1}{2}\dot{\mathbf{a}}^2 - \frac{1}{2}(\nabla\times\mathbf{a})^2 + \mathbf{a}\cdot\mathbf{j} + \mathcal{L}_M - a^0 j^0 + \frac{1}{2}(\nabla a^0)^2 \\ &= -\frac{1}{4}f_{\mu\nu}f^{\mu\nu} + a_\mu j^\mu + \mathcal{L}_M + \text{total derivatives}\end{aligned}$$

with $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$, and integrate over a^0 as well as over $\mathbf{a}$ and matter fields. That is, the path integral (9.6.6) is now

$$\begin{aligned}\langle T\{\mathcal{O}_A\mathcal{O}_B\cdots\}\rangle_\Omega &\propto\int\left[\prod_{x,\mu} da_\mu(x)\right]\left[\prod_{x,\ell} d\psi_\ell(x)\right]\mathcal{O}_A\mathcal{O}_B\cdots \\ &\quad\cdot\exp(iI[a,\psi])\prod_x\delta(\nabla\cdot\mathbf{a}(x)),\end{aligned} \tag{9.6.9}$$

where I is the original action

$$I[a,\psi] = \int d^4x\left[-\frac{1}{4}f_{\mu\nu}f^{\mu\nu} + a_\mu j^\mu + \mathcal{L}_M\right] + i\epsilon\text{ terms}. \tag{9.6.10}$$

Now everything is manifestly Lorentz- and gauge-invariant, except for the final product of delta functions which enforce the Coulomb gauge condition.[104] To make further progress, we shall use a simple version of a trick [4,5] that in Volume II will be used to treat the more difficult case of non-Abelian gauge theories. For simplicity, we shall deal here with the case where the operators $\mathcal{O}_A[A,\Psi]$, $\mathcal{O}_B[A,\Psi], \ldots$ as well as the action $I[a,\psi]$ and measure $[\prod da][\prod d\psi]$ are gauge-invariant.

First, replace the field variables of integration $a_\mu(x)$ and $\psi_\ell(x)$ everywhere in Eq. (9.6.9) with the new variables

$$a_{\mu\Lambda}(x) \equiv a_\mu(x) + \partial_\mu\Lambda(x), \tag{9.6.11}$$

$$\psi_{\ell\Lambda}(x) \equiv \exp(iq_\ell\Lambda(x))\psi_\ell(x) \tag{9.6.12}$$

with arbitrary finite $\Lambda(x)$. This step is a mathematical triviality, like changing an integral $\int_{-\infty}^{\infty}f(x)dx$ to read $\int_{-\infty}^{\infty}f(y)dy$, and does not require use of the postulated gauge invariance of the theory. Next, use gauge invariance to replace $a_{\mu\Lambda}(x)$ and $\psi_{\ell\Lambda}(x)$ in the action, measure, and $\mathcal{O}$-functions with the original fields $a_\mu(x)$ and $\psi_\ell(x)$, respectively. Eq. (9.6.9) then becomes

$$\begin{aligned}&\langle T\{\mathcal{O}_A[A,\Psi],\mathcal{O}_B[A,\Psi],\ldots\}\rangle_\Omega \\ &\propto\int\left[\prod_{x,\mu} da_\mu(x)\right]\left[\prod_{x,\ell} d\psi_\ell(x)\right]\mathcal{O}_A[a,\psi]\mathcal{O}_B[a,\psi]\cdots \\ &\quad\cdot\exp(iI[a,\psi])\prod_x\delta(\nabla\cdot\mathbf{a}(x) + \nabla^2\Lambda(x)).\end{aligned} \tag{9.6.13}$$

[104] Note that now $a^0(x)$ is not equal to the value (9.6.8), but is an independent variable of integration. We will not integrate over $a^0(x)$ first, which would lead back to Eq. (9.6.6), but instead will treat it in tandem with $\mathbf{a}(x)$.

Now, the function $\Lambda(x)$ was chosen at random, so despite appearances the right-hand side of Eq. (9.6.13) cannot depend on this function. We shall exploit this fact to put the path integral in a much more convenient form. Multiply Eq. (9.6.13) by the functional

$$B[\Lambda,a] = \exp\left(-\frac{1}{2}i\alpha\int d^4x\,[\partial_0 a^0 - \nabla^2\Lambda]^2\right) \tag{9.6.14}$$

(where α is an arbitrary constant), and integrate over $\Lambda(x)$. By shifting the integration variable $\Lambda(x)$, and noting the actual Λ-independence of (9.6.13), we see that the effect is simply to multiply Eq. (9.6.13) with the field-independent constant

$$\int\left[\prod_x d\Lambda(x)\right]\exp\left(-\frac{1}{2}i\alpha\int d^4x\,(\nabla^2\Lambda)^2\right). \tag{9.6.15}$$

This factor cancels out in the connected part of the vacuum expectation value, and thus has no physical effect. But (9.6.13) is only Λ-independent *after* we integrate over $a^\mu(x)$ and $\psi(x)$. We can just as well integrate over $\Lambda(x)$ *before* we integrate over $a^\mu(x)$ and $\psi(x)$, in which case the factor $\prod_x\delta(\nabla\cdot\mathbf{a}(x) + \nabla^2\Lambda)$ in Eq. (9.6.13) is replaced with

$$\begin{aligned}&\int\left[\prod_x d\Lambda(x)\right]\exp\left(-\frac{1}{2}i\alpha\int d^4x\,[\partial_0 a^0 - \nabla^2\Lambda]^2\right)\prod_x\delta(\nabla\cdot\mathbf{a}(x) + \nabla^2\Lambda) \\ &\propto\exp\left(-\frac{1}{2}i\alpha\int d^4x\,(\partial_\mu a^\mu)^2\right),\end{aligned} \tag{9.6.16}$$

where "$\propto$" again means proportional with a field-independent factor. Dropping constant factors, Eq. (9.6.9) now becomes

$$\langle T\{\mathcal{O}_A\mathcal{O}_B\cdots\}\rangle_\Omega \propto\int\left[\prod_{x,\mu} da_\mu(x)\right]\left[\prod_{x,\ell} d\psi_\ell(x)\right]\mathcal{O}_A\mathcal{O}_B\cdots\exp(iI_{\text{eff}}[a,\psi]), \tag{9.6.17}$$

where

$$I_{\text{eff}}[a,\psi] = I[a,\psi] - \frac{1}{2}\alpha\int(\partial_\mu a^\mu)^2 d^4x. \tag{9.6.18}$$

This is now manifestly Lorentz-invariant.

We consider the new term in (9.6.18) as a contribution to the unperturbed part of the action, whose photonic part now reads

$$\begin{aligned}I_0[a] &= \int d^4x\left[-\frac{1}{4}(\partial_\mu a_\nu - \partial_\nu a_\mu)(\partial^\mu a^\nu - \partial^\nu a^\mu) - \frac{1}{2}\alpha(\partial_\mu a^\mu)^2 + i\epsilon\text{ terms}\right] \\ &= -\frac{1}{2}\int d^4x\,d^4y\,a^\mu(x)a^\nu(y)\mathcal{D}_{\mu x,\nu y},\end{aligned} \tag{9.6.19}$$

where

$$\begin{aligned}\mathcal{D}_{\mu x,\nu y} &= \left[\eta_{\mu\nu}\frac{\partial^2}{\partial x^\rho\partial y_\rho} - (1-\alpha)\frac{\partial^2}{\partial x^\mu\partial y^\nu}\right]\delta^4(x-y) + i\epsilon\text{ terms} \\ &= (2\pi)^{-4}\int d^4q\,\left[\eta_{\mu\nu}q^2 - (1-\alpha)q_\mu q_\nu - i\epsilon\eta_{\mu\nu}\right]e^{iq\cdot(x-y)}.\end{aligned} \tag{9.6.20}$$

The photon propagator is then found immediately by inverting the $4\cdot 4$ matrix in the integrand of Eq. (9.6.20)

$$\Delta_{\mu x,\nu y} = (2\pi)^{-4}\int d^4q\left[\frac{\eta^{\mu\nu}}{q^2 - i\epsilon} + \frac{1-\alpha}{\alpha}\frac{q^\mu q^\nu}{(q^2 - i\epsilon)^2}\right]e^{iq\cdot(x-y)}. \tag{9.6.21}$$

We are free to choose α as seems most convenient. Two common choices are $\alpha = 1$, which yields the propagator in *Feynman gauge*:

$$\Delta^{\text{Feynman}}_{\mu x,\nu y} = (2\pi)^{-4}\int d^4q\left[\frac{\eta^{\mu\nu}}{q^2 - i\epsilon}\right]e^{iq\cdot(x-y)}, \tag{9.6.22}$$

or $\alpha = \infty$, in which case the factor (9.6.14) acts as a delta function, and we obtain the propagator in *Landau gauge* (often also called *Lorentz gauge*):

$$\Delta^{\text{Landau}}_{\mu x,\nu y} = (2\pi)^{-4}\int d^4q\left[\frac{\eta^{\mu\nu}}{q^2 - i\epsilon} - \frac{q^\mu q^\nu}{(q^2 - i\epsilon)^2}\right]e^{iq\cdot(x-y)}. \tag{9.6.23}$$

Practical calculations are made far more convenient by working with such manifestly Lorentz-invariant interactions and propagators.

9.7 Varieties of Statistics[105]

We can now take up a question raised in Chapter 4: what are the possibilities for the change of state-vectors when we interchange identical particles?

For this purpose, we will consider the preparation of the initial or final states in a scattering process. Suppose that a set of indistinguishable particles in either of these states is brought to a particular configuration with momenta $\mathbf{p}_1, \mathbf{p}_2$, etc. from a standard configuration with momenta $\mathbf{k}_1, \mathbf{k}_2$, etc., by some sort of slowly varying external fields, keeping the particles far enough apart in the process to justify the use of non-relativistic quantum mechanics. (Spin indices are not shown explicitly here; they should be understood to accompany momentum labels.) To calculate the amplitude for this process we can use the path-integral method,[106] taking the qs and ps of Section 9.1 as particle positions and momenta, rather than fields and their canonical conjugates. These always satisfy canonical commutation rather than anticommutation relations, whether or not the particles are bosons or fermions or something else, so at this point we are not committing ourselves to any particular statistics. The path-integral formula (9.1.34) gives an amplitude $\langle\mathbf{p}_1,\mathbf{p}_2,\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle_D$ as an integral over paths in which one particle is brought continuously from momentum $\mathbf{k}_1$ to momentum $\mathbf{p}_1$, another identical particle is brought continuously from momentum $\mathbf{k}_2$ to momentum $\mathbf{p}_2$, and so on. The subscript 'D' indicates

[105] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

[106] Here I am following the discussion of Laidlaw and C. DeWitt [11], except that they apply the path-integral method to the whole scattering process, rather than just the preparation of initial or final states. In a relativistic theory the possibility of particle creation and annihilation makes it necessary to apply the path-integral method to fields rather than particle orbits. For us, this is not a problem, because we limit such calculations to sufficiently early or late times, when the particles participating in a scattering process are all far apart.

that this is the amplitude we would calculate for distinguishable particles. In particular, this amplitude is symmetric under permutations of the $\mathbf{p}$s *and* simultaneous permutations of the $\mathbf{k}$s, but has no particular symmetry under separate permutations of the $\mathbf{p}$s or $\mathbf{k}$s. But if the particles are really indistinguishable, then there are other paths that are topologically distinct, but that yield the same final configuration. For space dimensionality $d \geq 3$, the only such paths[107] are those that take $\mathbf{k}_1, \mathbf{k}_2, \ldots$ into some non-trivial permutation $\mathcal{P}$ of $\mathbf{p}_1, \mathbf{p}_2, \ldots$. Hence the true amplitude should be written

$$\langle\mathbf{p}_1,\mathbf{p}_2,\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle = \sum_{\mathcal{P}}C_{\mathcal{P}}\langle\mathbf{p}_{\mathcal{P}1},\mathbf{p}_{\mathcal{P}2},\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle_D, \tag{9.7.1}$$

the sum running over all $N!$ permutations of the N indistinguishable particles in the state, and $C_{\mathcal{P}}$ a set of complex constants. These amplitudes must satisfy a composition rule appropriate for indistinguishable particles:

$$\begin{aligned}&\langle\mathbf{p}_1,\mathbf{p}_2,\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle \\ &= \frac{1}{N!}\int d^3q_1\,d^3q_2\cdots\langle\mathbf{p}_1,\mathbf{p}_2,\ldots|\mathbf{q}_1,\mathbf{q}_2,\ldots\rangle\langle\mathbf{q}_1,\mathbf{q}_2,\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle.\end{aligned} \tag{9.7.2}$$

Using Eq. (9.7.1), this is the requirement

$$\begin{aligned}&\sum_{\mathcal{P}}C_{\mathcal{P}}\langle\mathbf{p}_{\mathcal{P}1},\mathbf{p}_{\mathcal{P}2},\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle_D \\ &= \frac{1}{N!}\sum_{\mathcal{P}',\mathcal{P}''}C_{\mathcal{P}'}C_{\mathcal{P}''}\int d^3q_1\,d^3q_2\cdots\langle\mathbf{p}_{\mathcal{P}'1},\mathbf{p}_{\mathcal{P}'2},\ldots|\mathbf{q}_1,\mathbf{q}_2,\ldots\rangle_D \\ &\quad\cdot\langle\mathbf{q}_{\mathcal{P}''1},\mathbf{q}_{\mathcal{P}''2},\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle_D.\end{aligned}$$

Applying a permutation $\mathcal{P}''$ to both the initial and final states in the first amplitude on the right, this is

$$\begin{aligned}&\sum_{\mathcal{P}}C_{\mathcal{P}}\langle\mathbf{p}_{\mathcal{P}1},\mathbf{p}_{\mathcal{P}2},\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle_D \\ &= \frac{1}{N!}\sum_{\mathcal{P}',\mathcal{P}''}C_{\mathcal{P}'}C_{\mathcal{P}''}\int d^3q_1\,d^3q_2\cdots\langle\mathbf{p}_{\mathcal{P}''\mathcal{P}'1},\mathbf{p}_{\mathcal{P}''\mathcal{P}'2},\ldots|\mathbf{q}_{\mathcal{P}''1},\mathbf{q}_{\mathcal{P}''2},\ldots\rangle_D \\ &\quad\cdot\langle\mathbf{q}_{\mathcal{P}''1},\mathbf{q}_{\mathcal{P}''2},\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle_D.\end{aligned}$$

But the amplitudes $\langle\mathbf{p}_1,\mathbf{p}_2,\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle_D$ satisfy the composition rule for distinguishable particles

$$\begin{aligned}&\langle\mathbf{p}_1,\mathbf{p}_2,\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle_D \\ &= \int d^3q_1\,d^3q_2\cdots\langle\mathbf{p}_1,\mathbf{p}_2,\ldots|\mathbf{q}_1,\mathbf{q}_2,\ldots\rangle_D\langle\mathbf{q}_1,\mathbf{q}_2,\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle_D,\end{aligned} \tag{9.7.3}$$

[107] This is expressed formally in the statement that the first homotopy group of configuration space in $d \geq 3$ is the permutation group [12]. By 'configuration space' for N distinguishable particles is meant the space of N d-vectors, excluding d-vectors that coincide with (or are within an arbitrary limiting distance of) each other, and identifying configurations that differ only by a permutation of the vectors.

so the composition rule for the physical amplitudes may be written

$$\begin{aligned}&\sum_{\mathcal{P}}C_{\mathcal{P}}\langle\mathbf{p}_{\mathcal{P}1},\mathbf{p}_{\mathcal{P}2},\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle_D \\ &= \frac{1}{N!}\sum_{\mathcal{P}',\mathcal{P}''}C_{\mathcal{P}'}C_{\mathcal{P}''}\langle\mathbf{p}_{\mathcal{P}''\mathcal{P}'1},\mathbf{p}_{\mathcal{P}''\mathcal{P}'2},\ldots|\mathbf{k}_1,\mathbf{k}_2,\ldots\rangle_D,\end{aligned}$$

which will be satisfied if and only if

$$C_{\mathcal{P}'\mathcal{P}''} = C_{\mathcal{P}'}C_{\mathcal{P}''}. \tag{9.7.4}$$

That is, the coefficients $C_{\mathcal{P}}$ must furnish a one-dimensional representation of the permutation group. But the permutation group has only two such representations: one is the identity, with $C_{\mathcal{P}} = +1$ for all permutations, and the other is the alternating representation, with $C_{\mathcal{P}} = +1$ or $C_{\mathcal{P}} = -1$ according to whether $\mathcal{P}$ is an even or odd permutation. These two possibilities correspond to Bose or Fermi statistics, respectively.[108]

The nice feature of this argument is that it makes it clear why the case of two space dimensions is an exception. In this case there is a much richer variety of topologically distinct paths.[109] For instance, a path in which one particle circles another a definite number of times cannot be deformed into a path where it does not. In consequence, in two space dimensions it is possible to have *anyons* [15], particles with more general permutation properties than just Fermi or Bose statistics [8a].

Appendix Gaussian Multiple Integrals

We wish first to calculate the multiple integral, over a finite number of real variables ξ_r, of the exponential of a general quadratic function of ξ:

$$\mathcal{I} \equiv \int_{-\infty}^{\infty}\prod_r d\xi_r\exp\{-Q(\xi)\}, \tag{9.A.1}$$

$$Q(\xi) = \frac{1}{2}\sum_{rs}K_{rs}\xi_r\xi_s + \sum_r L_r\xi_r + M, \tag{9.A.2}$$

where K_{rs}, L_r, and M are arbitrary constants, except that the matrix K is required to be symmetric and non-singular. For this purpose, we begin by considering the case where K_{rs}, L_r, and M are all real, with K_{rs} also positive. The result in the general case can then be obtained by analytic continuation.

Any real symmetric matrix can be diagonalized by an orthogonal matrix. Therefore, there is a matrix $\mathcal{S}$ with transpose $\mathcal{S}^{\mathrm{T}} = \mathcal{S}^{-1}$ such that

$$(\mathcal{S}^{\mathrm{T}}K\mathcal{S})_{rs} = \delta_{rs}\kappa_r. \tag{9.A.3}$$

[108] There has been much discussion in the literature of possibilities other than Bose or Fermi statistics, often under the label *parastatistics*. It has been shown [13] that parastatistics theories in $d \geq 3$ space dimensions are equivalent to theories in which all particles are ordinary fermions or bosons, but carrying an extra quantum number, so that wave functions could have unusual properties under permutations of momenta and spins.

[109] This is expressed in the statement that the first homotopy group of configuration space in two space dimensions is not the permutation group, but a larger group known as the braid group [14].

Because K is assumed positive and non-singular, the eigenvalues κ_r are positive-definite. We can use the matrix $\mathcal{S}$ to perform a change of variables:

$$\xi_r = \sum_s\mathcal{S}_{rs}\xi'_s. \tag{9.A.4}$$

The Jacobian $|\operatorname{Det}\mathcal{S}|$ of this transformation is unity, so the multiple integral (9.A.1) is now given by a product of ordinary integrals:

$$\begin{aligned}\mathcal{I} &= e^{-M}\prod_r\int_{-\infty}^{\infty}d\xi'_r\exp\left\{-\frac{\kappa_r}{2}\xi_r'^2 - (\mathcal{S}^{\mathrm{T}}L)_r\xi'_r\right\} \\ &= e^{-M}\prod_r\sqrt{\frac{2\pi}{\kappa_r}}\exp\left\{\frac{1}{2\kappa_r}(\mathcal{S}^{\mathrm{T}}L)_r^2\right\}.\end{aligned} \tag{9.A.5}$$

But the determinant and the reciprocal of Eq. (9.A.3) give

$$\operatorname{Det}K = \prod_r\kappa_r, \qquad K_{rs}^{-1} = \sum_t\mathcal{S}_{rt}\mathcal{S}_{st}\kappa_t^{-1},$$

so Eq. (9.A.5) may be written

$$\mathcal{I} = \left(\operatorname{Det}\frac{K}{2\pi}\right)^{-1/2}\exp\left\{\frac{1}{2}\sum_{rs}L_r L_s K_{rs}^{-1} - M\right\}. \tag{9.A.6}$$

Eq. (9.A.1) defines a function of K_{rs}, L_r, and M that is analytic in K_{rs} in a finite region around the surface where K_{rs} is real and positive, where the integral converges, and for such K_{rs} is analytic everywhere in L_r and M. Since (9.A.6) equals (9.A.1) for K_{rs}, L_r, and M all real, with K_{rs} also positive, Eq. (9.A.6) provides an analytic continuation of Eq. (9.A.1) to the whole complex plane, with a cut required by the square root. The sign of the square root is fixed by this analytic continuation. In field theory K_{rs} is actually imaginary, except for a small real part due to the '$i\epsilon$ term'.

It is useful to express Eq. (9.A.6) in terms of the stationary point of the function (9.A.2):

$$\bar\xi_r = -\sum_s K_{rs}^{-1}L_s, \tag{9.A.7}$$

$$\frac{\partial Q(\xi)}{\partial\xi_r} = 0 \quad \text{at} \quad \xi = \bar\xi, \tag{9.A.8}$$

as

$$\mathcal{I} = \left(\operatorname{Det}\frac{K}{2\pi}\right)^{-1/2}\exp\{-Q(\bar\xi)\}. \tag{9.A.9}$$

This is the result to remember: *Gaussian integrals can be evaluated up to a determinant factor by setting the integration variable equal to the point where the argument of the exponential is stationary.*

We next wish to use this result to calculate the integrals

$$I_{r_1 r_2\cdots r_{2N}} \equiv \int\left(\prod_r d\xi_r\right)\xi_{r_1}\xi_{r_2}\cdots\xi_{r_{2N}}\exp\left\{-\frac{1}{2}\sum_{rs}K_{rs}\xi_r\xi_s\right\}. \tag{9.A.10}$$

(Integrals of this sort with an odd number of ξ-factors in the integrand obviously vanish.) From the power-series expression of $\exp(-\sum_r L_r\xi_r)$ in Eq. (9.A.1), we have the sum rule

$$\begin{aligned}&\sum_{N=0}^{\infty}\sum_{r_1 r_2\cdots r_{2N}}\frac{1}{(2N)!}I_{r_1 r_2\cdots r_{2N}}L_{r_1}L_{r_2}\cdots L_{r_{2N}} \\ &= \int\left(\prod_r d\xi_r\right)\exp\left\{-\sum_r L_r\xi_r - \frac{1}{2}\sum_{rs}K_{rs}\xi_r\xi_s\right\} \\ &= \left[\operatorname{Det}\left(\frac{K}{2\pi}\right)\right]^{-1/2}\exp\left\{\frac{1}{2}\sum_{rs}L_r L_s K_{rs}^{-1}\right\} \\ &= \left[\operatorname{Det}\left(\frac{K}{2\pi}\right)\right]^{-1/2}\sum_{N=0}^{\infty}\frac{1}{N!2^N}\left(\sum_{rs}L_r L_s K_{rs}^{-1}\right)^N.\end{aligned} \tag{9.A.11}$$

Comparing the coefficients of $L_{r_1}L_{r_2}\cdots L_{r_{2N}}$ on both sides, we see that $I_{r_1 r_2\cdots r_{2N}}$ must be proportional to a sum of products of elements of K^{-1}, which symmetry requires to take the form

$$I_{r_1 r_2\cdots r_{2N}} = c_N\sum_{\substack{\text{pairings}\\ \text{of } r_1\cdots r_{2N}}}\prod_{\text{pairs}}(K^{-1})_{\text{paired indices}}. \tag{9.A.12}$$

Here the sum is over all ways of pairing the indices $r_1\cdots r_{2N}$, with two pairings being considered the same if they differ only by the order of the pairs, or by the order of indices within a pair. To calculate the constant factor c_N, we note that the number ν_N of terms in the sum over pairings in Eq. (9.A.12) is equal to the number $(2N)!$ of permutations of the indices, divided by the number of ways $N!$ of permuting index pairs and by the number 2^N of permutations within index pairs

$$\nu_N = \frac{(2N)!}{N!2^N}. \tag{9.A.13}$$

Therefore, Eq. (9.A.12) gives

$$\sum_{r_1 r_2\cdots r_{2N}}L_{r_1}L_{r_2}\cdots L_{r_{2N}}I_{r_1 r_2\cdots r_{2N}} = \nu_N c_N\left(\sum_{rs}L_r L_s K_{rs}^{-1}\right)^N. \tag{9.A.14}$$

Comparing this with Eq. (9.A.11) shows that the factors $(2N)!$ and $N!2^N$ are cancelled by ν_N, leaving us with

$$c_N = \left[\operatorname{Det}\left(\frac{K}{2\pi}\right)\right]^{-1/2}. \tag{9.A.15}$$

For instance,

$$I_{r_1 r_2} = I_0(K^{-1})_{r_1 r_2}, \tag{9.A.16}$$

$$\begin{aligned}I_{r_1 r_2 r_3 r_4} = I_0\big[&(K^{-1})_{r_1 r_2}(K^{-1})_{r_3 r_4} + (K^{-1})_{r_1 r_3}(K^{-1})_{r_2 r_4} \\ &+ (K^{-1})_{r_1 r_4}(K^{-1})_{r_2 r_3}\big],\end{aligned} \tag{9.A.17}$$

and so on, where I_0 is the integral with no indices

$$I_0 = \int\left(\prod_r d\xi_r\right)\exp\left\{-\frac{1}{2}\sum_{rs}K_{rs}\xi_r\xi_s\right\} = \left[\operatorname{Det}\left(\frac{K}{2\pi}\right)\right]^{-1/2}. \tag{9.A.18}$$

Problems

1. Consider a non-relativistic particle of mass m, moving along the x-axis in a potential $V(x) = m\omega^2x^2/2$. Use path-integral methods to find the probability that if the particle is at x_1 at time t_1, then it is between x and $x + dx$ at time t.
2. Find the wave function in field space of a state consisting of a single spinless particle of mass $m \neq 0$. Use the result to derive the Feynman rules for emission or absorption of such a particle.
3. Find the wave function in field space of the vacuum in the theory of a neutral vector field of mass $m \neq 0$. Use the result to derive the form of the $i\epsilon$ terms in the propagator of this field.
4. The Lagrangian density of the free spin 3/2 Rarita–Schwinger field ψ^μ is

$$\mathcal{L} = -\bar\psi^\mu(\gamma^\nu\partial_\nu + m)\psi_\mu - \frac{1}{3}\bar\psi^\mu(\gamma_\mu\partial_\nu + \gamma_\nu\partial_\mu)\psi^\nu + \frac{1}{3}\bar\psi^\mu\gamma_\mu(\gamma^\nu\partial_\nu - m)\gamma^\nu\psi_\nu.$$

Use path-integral methods to find the propagator of this field.

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10 Non-Perturbative Methods

We are now going to begin our study of higher-order contributions to physical processes, corresponding to Feynman diagrams involving one or more loops. It will be very useful in this work to have available a method of deriving results valid to all orders in perturbation theory (and in some cases beyond perturbation theory). In this chapter we will exploit the field equations and commutation relations of the interacting fields in the Heisenberg picture for this purpose. The essential bridge between the Heisenberg picture and the Feynman diagrams of perturbation theory is provided by the theorem proved in Section 6.4: the sum of all diagrams for a process $\alpha \to \beta$ with extra vertices inserted corresponding to operators $O_a(x)$, $O_b(y)$, etc. is given by the matrix element of the time-ordered product of the corresponding Heisenberg-picture operators

$$_{\rm out}(\beta|T\{-iO_a(x),-iO_b(y),\dots\}|\alpha)_{\rm in}.$$

As a special case, where the operators $O_a(x)$, $O_b(x)$, etc. are elementary particle fields, this matrix element equals the sum of all Feynman diagrams with incoming lines on the mass shell corresponding to the state α, outgoing lines on the mass shell corresponding to the state β, and lines off the mass shell (including propagators) corresponding to the operators $O_a(x)$, $O_b(y)$, etc. After exploring some of the non-perturbative results that can be obtained in this way we will be in a good position to take up the perturbative calculation of radiative corrections.

10.1 Symmetries

One obvious but important use of the theorem quoted above is to extend the application of symmetry principles from S-matrix elements, where all external lines have four-momenta on the mass shell, to parts of Feynman diagrams, with some or all external lines off the mass shell.

For instance, consider the symmetry of spacetime translational invariance. This symmetry has as a consequence the existence of a Hermitian four-vector operator P^μ, with the property that, for any local function $O(x)$ of field operators and their canonical conjugates,

$$[P_\mu, O(x)] = i\frac{\partial}{\partial x^\mu}O(x). \tag{10.1.1}$$

(See Eqs. (7.3.28) and (7.3.29).) Also, the states α and β are usually chosen to be eigenstates of the four-momentum:

$$P^\mu|\alpha)_{\rm in} = p^\mu_\alpha|\alpha)_{\rm in}, \qquad P^\mu|\beta)_{\rm out} = p^\mu_\beta|\beta)_{\rm out}. \tag{10.1.2}$$

It follows that for any set of local functions $O_a(x)$, $O_b(x)$, etc. of fields and/or field derivatives

$$\begin{aligned}(p_{\beta\mu} - p_{\alpha\mu})\,_{\rm out}(\beta|T\{O_a(x_1),O_b(x_2),\dots\}|\alpha)_{\rm in}\\ = {}_{\rm out}(\beta|[P_\mu, T\{O_a(x_1),O_b(x_2),\dots\}]|\alpha)_{\rm in}\\ = i\left(\frac{\partial}{\partial x_1^\mu}+\frac{\partial}{\partial x_2^\mu}+\cdots\right){}_{\rm out}(\beta|T\{O_a(x_1),O_b(x_2),\dots\}|\alpha)_{\rm in}.\end{aligned} \tag{10.1.3}$$

This has the solution

$$\begin{aligned}{}_{\rm out}(\beta|T\{O_a(x_1),O_b(x_2),\dots\}|\alpha)_{\rm in}\\ = \exp\big(i(p_\beta - p_\alpha)\cdot x\big)F_{ab\dots}(x_1-x_2,\dots),\end{aligned} \tag{10.1.4}$$

where x is any sort of average spacetime coordinate

$$x^\mu = c_1 x_1^\mu + c_2 x_2^\mu + \cdots, \qquad c_1 + c_2 + \cdots = 1 \tag{10.1.5}$$

and F depends only on differences among the xs. (In particular, a vacuum expectation value can depend only on the coordinate differences.) We can Fourier transform Eq. (10.1.4) by integrating separately over x^μ and the coordinate differences, with the result that

$$\begin{aligned}\int d^4x_1\, d^4x_2\cdots {}_{\rm out}(\beta|T\{O_a(x_1),O_b(x_2),\dots\}|\alpha)_{\rm in}\\ \cdot \exp(-ik_1\cdot x_1 - ik_2\cdot x_2 - \cdots) \propto \delta^4(p_\beta - p_\alpha - k_1 - k_2 - \cdots).\end{aligned} \tag{10.1.6}$$

We saw in Section 6.4 that the matrix element of the time-ordered product is given by applying the usual coordinate-space Feynman rules to the sum of all graphs with incoming particles corresponding to particles in α, outgoing particles in β, and external lines that simply terminate in vertices at $x_1, x_2, \dots$. The Fourier transform (10.1.6) is correspondingly given by applying the momentum-space Feynman rules to the same sum of Feynman diagrams, with off-shell external lines carrying four-momenta $k_1, k_2, \dots$ into the diagrams. Eq. (10.1.6) is then just the statement that this sum of Feynman graphs conserves four-momentum. The result is obvious in perturbation theory, because four-momentum is conserved at every vertex, so it is not surprising to see the same result emerging without having to rely on perturbation theory.

With somewhat more effort, one can use the Lorentz transformation properties of the Heisenberg-picture fields and the 'in' and 'out' states to show that the sum of all graphs with a given set of on- and off-shell lines satisfies the same Lorentz transformation conditions as the lowest-order terms.

Similar arguments apply to the conservation of internal quantum numbers, like electric charge. As shown in Section 7.3, a field or other operator $O_a(x)$ that destroys a charge q_a (or creates a charge $-q_a$) will satisfy

$$[Q, O_a(x)] = -q_a O_a(x)$$

in the Heisenberg and interaction pictures alike. Also, if the free-particle states α and β have charges q_α and q_β, then so do the corresponding 'in' and 'out' states. We then have

$$\begin{aligned}(q_\beta - q_\alpha)\,_{\rm out}(\beta|T\{O_a(x),O_b(y),\dots\}|\alpha)_{\rm in}\\ = {}_{\rm out}(\beta|[Q, T\{O_a(x),O_b(y),\dots\}]|\alpha)_{\rm in}\\ = -(q_a + q_b + \cdots)\,_{\rm out}(\beta|T\{O_a(x),O_b(y),\dots\}|\alpha)_{\rm in}.\end{aligned}$$

Thus the amplitude $_{\rm out}(\beta|T\{O_a(x),O_b(y),\dots\}|\alpha)_{\rm in}$ vanishes unless charge is conserved

$$q_\beta = q_\alpha - q_a - q_b - \cdots. \tag{10.1.7}$$

A somewhat less trivial example is provided by the symmetry of charge-conjugation invariance. As we saw in Chapter 5, there is an operator C that interchanges electron and positron operators

$$C a_{\mathbf{p},\sigma,e^-}C^{-1} = \xi^* a_{\mathbf{p},\sigma,e^+},$$
$$C a_{\mathbf{p},\sigma,e^+}C^{-1} = \xi a_{\mathbf{p},\sigma,e^-},$$

with ξ a phase factor. For the free electron field $\psi(x)$, this gives

$$C\psi(x)C^{-1} = \xi^*\beta C\psi(x)^*,$$

where βC is a $4\cdot 4$ matrix, which (for the Dirac matrix representation we have been using, with γ_5 diagonal) takes the form

$$\beta C = \begin{bmatrix}0&0&0&1\\0&0&-1&0\\0&-1&0&0\\1&0&0&0\end{bmatrix}.$$

Applied to the free-particle electric current in spinor electrodynamics, this gives

$$C(\bar\psi\gamma^\mu\psi)C^{-1} = -\bar\psi C\gamma^{\mu T}C^{-1}\psi = -\bar\psi\gamma^\mu\psi.$$

If C is to be conserved in electrodynamics, it must then also be defined to anticommute with the free photon field

$$C(a^\mu)C^{-1} = -a^\mu.$$

In theories like electrodynamics for which C commutes with the interaction as well as H_0, it also commutes with the similarity transformation $\Omega(t)$ between the Heisenberg and interaction pictures, and so it anticommutes with the electric current of the interacting fields

$$C(\bar\Psi\gamma^\mu\Psi)C^{-1} = -\bar\Psi\gamma^\mu\Psi \tag{10.1.8}$$

and the electromagnetic field in the Heisenberg-picture

$$C(A^\mu)C^{-1} = -A^\mu. \tag{10.1.9}$$

It follows then that the vacuum expectation value of the time-ordered product of any odd number of electromagnetic currents and/or fields vanishes. Therefore the sum of all Feynman graphs with an odd number of external photon lines (off or on the photon mass shell) and no other external lines vanishes.

This result is known as Furry's theorem [1]. It can be proved perturbatively by noting that a graph consisting of electron loops ℓ, to each of which are attached n_ℓ photon lines, must have numbers I and E of internal and external photon lines related by an analog of Eq. (6.3.11):

$$2I + E = \sum_\ell n_\ell.$$

Hence if E is odd at least one of the loops must have attached an odd number of photon lines. For any such loop there is a cancellation between the two diagrams in which the electron arrows circulate around the loop in opposite directions. Hence Furry's theorem is a somewhat less trivial consequence of a symmetry principle than translation or Lorentz invariance; it is not true of individual diagrams, but rather of certain sums of diagrams. Figure ?? illustrates the application of Furry's theorem that was historically most important, its use to show that the scattering of a photon by an external electromagnetic field receives no contributions of first order (or any odd order) in the external field.

10.2 Polology

One of the most important uses of the non-perturbative methods described in this chapter is to clarify the pole structure of Feynman amplitudes as functions of the momenta carried by external lines. Often the S-matrix for a physical process can be well approximated by the contribution of a single pole. Also, an understanding of this pole structure will help us later in dealing with radiative corrections to particle propagators.

Consider the momentum-space amplitude

$$\begin{aligned}\int d^4x_1\cdots d^4x_n\, e^{-iq_1\cdot x_1}\cdots e^{-iq_n\cdot x_n}\,(T\{A_1(x_1)\cdots A_n(x_n)\})_0\\ \equiv G(q_1\cdots q_n).\end{aligned} \tag{10.2.1}$$

The As are Heisenberg-picture operators of arbitrary Lorentz type, and $(\cdots)_0$ denotes the expectation value in the true vacuum $|\Omega)_{\rm in} = |\Omega)_{\rm out} \equiv |\Omega)$. As discussed in Section 6.4, if $A_1,\dots,A_n$ are ordinary fields appearing in the Lagrangian, then (10.2.1) is a sum of the terms calculated using the ordinary Feynman rules, for all graphs with external lines corresponding to the fields $A_1,\dots,A_n$, carrying off-shell four-momenta $q_1\cdots q_n$ into the graph. However, we will not be limited to this case; the A_i may be arbitrary local functions of fields and field derivatives.

We are interested in poles of G at certain values of the invariant squares of the total four-momenta carried by various subsets of the external lines. To be definite, let's consider G as a function of q^2, where

$$q \equiv q_1 + \cdots + q_r = -q_{r+1} - \cdots - q_n \tag{10.2.2}$$

with $1 \le r \le n-1$. We will show that G has a pole at $q^2 = -m^2$, where m is the mass of any one-particle state that has non-vanishing matrix elements with the states $A_1^\dagger\cdots A_r^\dagger|\Omega)$ and $A_{r+1}\cdots A_n|\Omega)$, and that the residue at this pole is given by

$$\begin{aligned}G \longrightarrow \frac{-2i\sqrt{\mathbf{q}^2+m^2}}{q^2+m^2-i\epsilon}(2\pi)^4\delta^4(q_1+\cdots+q_n)\\ \cdot\sum_\sigma M_{0|\mathbf{q},\sigma}(q_2\cdots q_r)M_{\mathbf{q},\sigma|0}(q_{r+2}\cdots q_n),\end{aligned} \tag{10.2.3}$$

where the Ms are defined by[110]

$$\begin{aligned}\int d^4x_1\cdots d^4x_r\, e^{-iq_1\cdot x_1}\cdots e^{-iq_r\cdot x_r}(\Omega|T\{A_1(x_1)\cdots A_r(x_r)\}|\mathbf{p},\sigma)\\ = (2\pi)^4\delta^4(q_1+\cdots+q_r-p)M_{0|\mathbf{p},\sigma}(q_2\cdots q_r),\end{aligned} \tag{10.2.4}$$

$$\begin{aligned}\int d^4x_{r+1}\cdots d^4x_n\, e^{-iq_{r+1}\cdot x_{r+1}}\cdots e^{-iq_n\cdot x_n}(\mathbf{p},\sigma|T\{A_{r+1}(x_{r+1})\cdots A_n(x_n)\}|\Omega)\\ = (2\pi)^4\delta^4(q_{r+1}+\cdots+q_n+p)M_{\mathbf{p},\sigma|0}(q_{r+2}\cdots q_n)\end{aligned} \tag{10.2.5}$$

(with $p^0 = \sqrt{\mathbf{p}^2+m^2}$), and the sum is over all spin (or other) states of the particle of mass m.

Before proceeding to the proof, it will help to clarify the significance of (10.2.3) if we write it in the somewhat long-winded form

$$\begin{aligned}G(q_1\cdots q_n) \longrightarrow \int d^4k\Big[(2\pi)^4\delta^4(q_1+\cdots+q_r-k)(2\pi)^{3/2}\big(2\sqrt{\mathbf{k}^2+m^2}\big)^{1/2}M_{0|\mathbf{k},\sigma}(q_2\cdots q_r)\Big]\\ \cdot\left[\frac{-i}{(2\pi)^4}\frac{1}{k^2+m^2-i\epsilon}\right]\\ \cdot\Big[(2\pi)^4\delta^4(k+q_{r+1}+\cdots+q_n)(2\pi)^{3/2}\big(2\sqrt{\mathbf{k}^2+m^2}\big)^{1/2}M_{\mathbf{k},\sigma|0}(q_{r+2}\cdots q_n)\Big].\end{aligned} \tag{10.2.6}$$

This is just what we should expect from a Feynman diagram with a single internal line for a particle of mass m connecting the first r and the last $n-r$ external lines.[111] However, it is not necessary that the particle of mass m correspond to a field that appears in the Lagrangian of the theory. Eqs. (10.2.3) and (10.2.6) apply even if this particle is a bound state of the so-called elementary particles whose fields do appear in the Lagrangian. In this case, the pole arises not from single Feynman diagrams, like Figure ??, but rather from infinite sums of diagrams, such as the one shown in Figure ??. This is the first place where the methods of this chapter take us beyond results that could be derived as properties of each order of perturbation theory.

Now to the proof. Among the $n!$ possible orderings of the times $x_1^0\cdots x_n^0$ in Eq. (10.2.1), there are $n!/r!(n-r)!$ for which the first r of the x_i^0 are all larger than the last $n-r$. Isolating the contribution of this part of the volume of integration in Eq. (10.2.1), we have

$$\begin{aligned}G(q_1\cdots q_n) = \int d^4x_1\cdots d^4x_n\, e^{-iq_1\cdot x_1}\cdots e^{-iq_n\cdot x_n}\\ \cdot\theta\big(\min[x_1^0\cdots x_r^0] - \max[x_{r+1}^0\cdots x_n^0]\big)\\ \cdot(\Omega|T\{A_1(x_1)\cdots A_r(x_r)\}T\{A_{r+1}(x_{r+1})\cdots A_n(x_n)\}|\Omega)\\ + {\rm OT},\end{aligned} \tag{10.2.7}$$

[110] Recall that in the absence of time-varying external fields, there is no distinction between 'in' and 'out' one-particle states, so that $|\mathbf{p},\sigma)_{\rm in} = |\mathbf{p},\sigma)_{\rm out} \equiv |\mathbf{p},\sigma)$.

[111] See Figure ??. The factors $(2\pi)^{3/2}(2\sqrt{\mathbf{k}^2+m^2})^{1/2}$ just serve to remove kinematic factors associated with the mass m external line in $M_{0|\mathbf{k},\sigma}$ and $M_{\mathbf{k},\sigma|0}$. Also, the sum over σ of the product of coefficient-function factors from these two matrix elements yields the numerator of the propagator associated with the internal line in Figure ??.

where 'OT' denotes the other terms arising from different time-orderings. We can evaluate the matrix element here by inserting a complete set of intermediate states between time-ordered products. Among these may be the single-particle state $|\mathbf{p},\sigma)$ of a definite species of mass m. Further isolating the contribution of these one-particle intermediate states, we have

$$\begin{aligned}G(q_1\cdots q_n) = \int d^4x_1\cdots d^4x_n\, e^{-iq_1\cdot x_1}\cdots e^{-iq_n\cdot x_n}\\ \cdot\theta\big(\min[x_1^0\cdots x_r^0] - \max[x_{r+1}^0\cdots x_n^0]\big)\sum_\sigma\int d^3p\\ \cdot(\Omega|T\{A_1(x_1)\cdots A_r(x_r)\}|\mathbf{p},\sigma)(\mathbf{p},\sigma|T\{A_{r+1}(x_{r+1})\cdots A_n(x_n)\}|\Omega)\\ + {\rm OT},\end{aligned} \tag{10.2.8}$$

where 'OT' now denotes other terms, here arising not only from other time-orderings, but also from other intermediate states. It will be convenient to shift variables of integration, so that

$$x_i = x_1 + y_i, \qquad i = 2, 3, \dots, r,$$
$$x_i = x_{r+1} + y_i, \qquad i = r+2, \dots, n,$$

and use the results of the previous section to write

$$\begin{aligned}(\Omega|T\{A_1(x_1)\cdots A_r(x_r)\}|\mathbf{p},\sigma)\\ = e^{ip\cdot x_1}(\Omega|T\{A_1(0)A_2(y_2)\cdots A_r(y_r)\}|\mathbf{p},\sigma),\end{aligned} \tag{10.2.9}$$

$$\begin{aligned}(\mathbf{p},\sigma|T\{A_{r+1}(x_{r+1})\cdots A_n(x_n)\}|\Omega)\\ = e^{-ip\cdot x_{r+1}}(\mathbf{p},\sigma|T\{A_{r+1}(0)\cdots A_n(y_n)\}|\Omega).\end{aligned} \tag{10.2.10}$$

Also, the argument of the theta function becomes

$$\begin{aligned}\min[x_1^0\cdots x_r^0] - \max[x_{r+1}^0\cdots x_n^0]\\ = x_1^0 - x_{r+1}^0 + \min[0, y_2^0\cdots y_r^0] - \max[0, y_{r+1}^0\cdots y_n^0].\end{aligned}$$

We also insert the Fourier representation (6.2.15) of the step function

$$\theta(\tau) = -\frac{1}{2\pi i}\int_{-\infty}^{\infty}\frac{d\omega\, e^{-i\omega\tau}}{\omega + i\epsilon}.$$

The integrals over x_1 and x_{r+1} now just yield delta functions:

$$\begin{aligned}G(q_1\cdots q_n) = \int d^4y_2\cdots d^4y_r\, d^4y_{r+2}\cdots d^4y_n\\ \cdot e^{-iq_2\cdot y_2}\cdots e^{-iq_r\cdot y_r}e^{-iq_{r+2}\cdot y_{r+2}}\cdots e^{-iq_n\cdot y_n}\\ \cdot\left(-\frac{1}{2\pi i}\right)\int_{-\infty}^{\infty}\frac{d\omega}{\omega+i\epsilon}\exp\big(-i\omega\,[\min[0,y_2^0\cdots y_r^0] - \max[0,y_{r+1}^0\cdots y_n^0]]\big)\\ \cdot\sum_\sigma\int d^3p\,(\Omega|T\{A_1(0)\cdots A_r(y_r)\}|\mathbf{p},\sigma)\\ \cdot(\mathbf{p},\sigma|T\{A_{r+1}(0)\cdots A_n(y_n)\}|\Omega)\\ \cdot(2\pi)^4\delta^3(\mathbf{p}-\mathbf{q}_1-\cdots-\mathbf{q}_r)\delta\Big(\sqrt{\mathbf{p}^2+m^2}+\omega-q_1^0-\cdots-q_r^0\Big)\\ \cdot(2\pi)^4\delta^3(\mathbf{q}_{r+1}+\cdots+\mathbf{q}_n+\mathbf{p})\delta\Big(q_{r+1}^0+\cdots+q_n^0+\sqrt{\mathbf{p}^2+m^2}+\omega\Big)\\ + {\rm OT}.\end{aligned} \tag{10.2.11}$$

We are interested here only in the pole that arises from the vanishing of the denominator $\omega + i\epsilon$, so for our present purposes we can set the factor $\exp(-i\omega[\min - \max])$ equal to unity. The integrals over both $\mathbf{p}$ and ω are now trivial, and yield the pole

$$\begin{aligned}G(q_1\cdots q_n) \longrightarrow i(2\pi)^7\delta^4(q_1+\cdots+q_n)\Big[q^0 - \sqrt{\mathbf{q}^2+m^2}+i\epsilon\Big]^{-1}\\ \cdot\sum_\sigma M_{0|-\mathbf{q},\sigma}(q_2\cdots q_r)M_{\mathbf{q},\sigma|0}(q_{r+2}\cdots q_n) + \cdots\end{aligned} \tag{10.2.12}$$

where now

$$q \equiv q_1 + \cdots + q_r = -q_{r+1} - \cdots - q_n,$$

$$\begin{aligned}M_{0|\mathbf{q},\sigma}(q_2\cdots q_r) \equiv \int d^4y_2\cdots d^4y_r\, e^{-iq_2\cdot y_2}\cdots e^{-iq_r\cdot y_r}\\ \cdot(\Omega|T\{A_1(0)A_2(y_2)\cdots A_r(y_r)\}|\mathbf{q},\sigma),\end{aligned} \tag{10.2.13}$$

$$\begin{aligned}M_{\mathbf{q},\sigma|0}(q_{r+2}\cdots q_n) \equiv \int d^4y_{r+2}\cdots d^4y_n\, e^{iq_{r+2}\cdot y_{r+2}}\cdots e^{-iq_n\cdot y_n}\\ \cdot(\mathbf{p},\sigma|T\{A_{r+1}(0)A_{r+2}(y_{r+2})\cdots A_n(y_n)\}|\Omega).\end{aligned} \tag{10.2.14}$$

and the final '$\cdots$' in Eq. (10.2.12) denotes terms that do not exhibit this particular pole. (The 'other terms' arising from other single-particle states produce poles in q at different positions, while those arising from multi-particle states produce branch points in q, and those arising from other time-orderings produce poles and branch cuts in other variables.) Using Eqs. (10.2.9) and (10.2.10), it is easy to see that these Ms are the same as defined by Eqs. (10.2.4) and (10.2.5). Also, near the pole we can write

$$\frac{1}{q^0 - \sqrt{\mathbf{q}^2+m^2}+i\epsilon} = \frac{-q^0 - \sqrt{\mathbf{q}^2+m^2}+i\epsilon}{-(q^0)^2 + (\sqrt{\mathbf{q}^2+m^2}-i\epsilon)^2} \longrightarrow \frac{-2\sqrt{\mathbf{q}^2+m^2}}{q^2+m^2-i\epsilon}.$$

(We again redefine ϵ by a positive factor $2\sqrt{\mathbf{q}^2+m^2}$, which is permissible since ϵ stands for any positive infinitesimal.) Eq. (10.2.12) is thus the same as the desired result (10.2.3).

This result has a classic application to the theory of nuclear forces. Let $\Phi_a(x)$ be any real field or combination of fields (for instance, proportional to a quark-antiquark bilinear $\bar q\gamma_5\tau_a q$) that has a non-vanishing matrix element between a one-pion state of isospin a and the vacuum, normalized so that

$$(\Omega|\Phi_a(0)|\pi_b,\mathbf{p}) = (2\pi)^{-3/2}(2p^0)^{-1/2}\delta_{ab}. \tag{10.2.15}$$

The matrix element of Φ_a between one-nucleon states with four-momenta p, p' then has a pole at $(p-p')^2 \to -m_\pi^2$, which isospin and Lorentz invariance (including space inversion invariance) dictate must take the form[112]

$$(N',\sigma',\mathbf{p}'|\Phi_a(0)|N,\sigma,\mathbf{p}) \longrightarrow i(2\pi)^{-3}G_\pi\frac{\bar u'\gamma_5\tau_a u}{(p-p')^2+m_\pi^2}, \tag{10.2.16}$$

where u and u' are the initial and final nucleon spinor coefficient functions, including the nucleon wave functions in isospin space, and τ_a with $a = 1, 2, 3$ are the $2\cdot 2$ Pauli isospin matrices. The constant G_π is known as the *pion–nucleon coupling constant*. This pole is not actually in the physical region for the matrix element (10.2.16), for which $(p-p')^2 \ge 0$, but it can be reached by analytic extension of this matrix element, for instance by considering the off-shell matrix element

$$\int d^4x\, d^4x'\, e^{-ip\cdot x}e^{ip'\cdot x'}(\Omega|T\{\Phi_a(0)\bar N(x)N'(x')\}|\Omega),$$

where N and N' are appropriate components of a field operator or product of field operators with non-vanishing matrix elements between one-nucleon states and the vacuum. The theorem proved above in this section shows then that exchange of a pion in the scattering of two nucleons with initial four-momenta p_1, p_2, and final four-momenta p_1', p_2' yields a pole at $(p_1-p_1')^2 = (p_2-p_2')^2 \to -m_\pi^2$.

$$\begin{aligned}S_{N_1'N_2',N_1N_2} \longrightarrow -i(2\pi)^4\delta^4(p_1'+p_2'-p_1-p_2)\frac{G_\pi^2}{(p_1-p_1')^2+m_\pi^2}\\ \cdot(2\pi)^{-3}[\bar u_{1'}\gamma_5\tau_a u_1]\cdot(2\pi)^{-3}[\bar u_{2'}\gamma_5\tau_a u_2].\end{aligned} \tag{10.2.17}$$

(The easiest way to get the phases and numerical factors right in such formulas is to use Feynman diagrams; our theorem just says that the pole structure is the same as would be

[112] Lorentz and isospin invariance requires this matrix to take the form $(\bar u'\Gamma\tau_a u)$, where Γ is a $4\cdot 4$ matrix for which the bilinear $(\bar\psi\Gamma\psi)$ transforms as a pseudoscalar. Like any $4\cdot 4$ matrix, Γ can be expanded as a sum of terms proportional to the Dirac matrices 1, γ_μ, $[\gamma_\mu,\gamma_\nu]$, $\gamma_5\gamma_\mu$, and γ_5. The coefficients must be respectively pseudoscalar, pseudovector, pseudotensor, vector, and scalar. Out of the two momenta p and p' it is possible to construct no pseudoscalars or pseudovectors; just one pseudotensor, proportional to $\epsilon^{\mu\nu\rho\sigma}p_\rho p_\sigma'$; two independent vectors, proportional to p_μ or p_μ'; and a scalar proportional to unity, in each case with a proportionality factor depending on the only independent scalar variable, $(p-p')^2$. By using the momentum-space Dirac equations for u and u', it is easy to see that the tensor and pseudovector matrices in Γ give contributions proportional to γ_5.

found in a field theory in which the Lagrangian involved an elementary pion field.) Again, this pion pole is not actually in the physical region for scattering of nucleons on the mass shell, for which $(p_1-p_1')^2 \ge 0$, but it can be reached by analytic extension of the S-matrix element, for instance by considering the off-shell matrix element

$$\int d^4x_1\, d^4x_2\, d^4x_1'\, d^4x_2'\, e^{-ip_1\cdot x_1}e^{-ip_2\cdot x_2}e^{ip_1'\cdot x_1'}e^{ip_2'\cdot x_2'}$$
$$\cdot(\Omega|T\{\bar N_1(x_1),\bar N_2(x_2),N_1'(x_1'),N_2'(x_2')\}|\Omega).$$

Although this pole is not in the physical region for nucleon–nucleon scattering, the pion mass is small enough so that the pole is quite near the physical region, and under some circumstances may dominate the scattering amplitude, as for instance for large ℓ in the partial wave expansion.

Interpreted in coordinate space, a pole like this at $(p_1-p_1')^2 = (p_2-p_2')^2 \to -m_\pi^2$ implies a force of range $1/m_\pi$. For instance, in Yukawa's original theory [2] of nuclear force the exchange of mesons (then assumed scalar rather than pseudoscalar) produced a local potential of the form $\exp(-m_\pi r)/4\pi r$, which in the first Born approximation yields an S-matrix for non-relativistic nucleon scattering proportional to the Fourier transform:

$$\begin{aligned}\int d^3\mathbf{x}_1 d^3\mathbf{x}_2 d^3\mathbf{x}_1' d^3\mathbf{x}_2'\, e^{-i\mathbf{x}_1\cdot\mathbf{p}_1}e^{-i\mathbf{x}_2\cdot\mathbf{p}_2}e^{i\mathbf{x}_1'\cdot\mathbf{p}_1'}e^{i\mathbf{x}_2'\cdot\mathbf{p}_2'}\\ \cdot\frac{\exp(-m_\pi|\mathbf{x}_1-\mathbf{x}_2|)}{4\pi|\mathbf{x}_1-\mathbf{x}_2|}\delta^3(\mathbf{x}_2-\mathbf{x}_2')\delta^3(\mathbf{x}_1-\mathbf{x}_1')\\ = -(2\pi)^3\delta^3(\mathbf{p}_1+\mathbf{p}_2-\mathbf{p}_1'-\mathbf{p}_2')\frac{1}{(\mathbf{p}_1-\mathbf{p}_1')^2+m^2}.\end{aligned}$$

The factor $1/[(\mathbf{p}_1-\mathbf{p}_1')^2+m_\pi^2]$ is just the non-relativistic limit of the propagator $1/[(p_1-p_1')^2+m_\pi^2]$ in (10.2.17). (In (10.2.17) the energy transfer $p_1^0-p_1'^0$ for $|\mathbf{p}_1| \ll m_N$ and $|\mathbf{p}_1'| \ll m_N$ equals $[\mathbf{p}_1^2-\mathbf{p}_1'^2]/2m_N$, which is negligible compared with the magnitude $|\mathbf{p}_1-\mathbf{p}_1'|$ of the momentum transfer.) When Yukawa's theory was first proposed, it was generally supposed that this sort of momentum-dependence arises from the appearance of a meson field in the theory. It was not until the 1950s that it became generally understood that the existence of a pole at $(p_1-p_1')^2 \to -m_\pi^2$ follows from the existence of a pion *particle* and has nothing to do with whether this is an elementary particle with its own field in the Lagrangian.

10.3 Field and Mass Renormalization

We will now use a special case of the result of the previous section to clarify the treatment of radiative corrections in the internal and external line of general processes.

The special case that concerns us here is the one in which the four-momentum of a single external line approaches the mass shell. (In the notation of the previous section, this corresponds to taking $r = 1$.) We will consider a function

$$\begin{aligned}G_\ell(q_1q_2\cdots) = \int d^4x_1\, d^4x_2\cdots e^{-iq_1\cdot x_1}e^{-iq_2\cdot x_2}\cdots\\ \cdot(\Omega|T\{\mathcal{O}_\ell(x_1), A_2(x_2),\dots\}|\Omega),\end{aligned} \tag{10.3.1}$$

where $\mathcal{O}_\ell(x)$ is a Heisenberg-picture operator, with the Lorentz transformation properties of some sort of free field ψ_ℓ belonging to an irreducible representation of the homogeneous Lorentz group (or the Lorentz group including space inversion for theories that conserve parity), as labelled by the subscript ℓ, and A_2, A_3, etc. are arbitrary Heisenberg-picture operators. Suppose there is a one-particle state $|q_1,\sigma)$ that has non-vanishing matrix elements with the states $\mathcal{O}_\ell^\dagger|\Omega)$ and with $A_2A_3\cdots|\Omega)$. Then according to the theorem proved in the previous section, G_ℓ has a pole at $q_1^2 = -m^2$, with

$$\begin{aligned}G_\ell(q_1q_2\cdots) \to \frac{-2i\sqrt{\mathbf{q}_1^2+m^2}}{q_1^2+m^2-i\epsilon}(2\pi)^3\sum_\sigma(\Omega|\mathcal{O}_\ell(0)|q_1,\sigma)\\ \cdot\int d^4x_2\cdots e^{-iq_2\cdot x_2}\cdots(q_1,\sigma|T\{A_2(x_2)\cdots\}|\Omega).\end{aligned} \tag{10.3.2}$$

We use Lorentz invariance to write

$$(\Omega|\mathcal{O}_\ell(0)|q_1,\sigma) = (2\pi)^{-3/2}N u_\ell(\mathbf{q}_1,\sigma), \tag{10.3.3}$$

where $u_\ell(\mathbf{q},\sigma)$ is (aside from the factor $(2\pi)^{-3/2}$) the coefficient function[113] appearing in the free field ψ_ℓ with the same Lorentz transformation properties as $\mathcal{O}_\ell$, and N is a constant. (It was in order to obtain Eq. (10.3.3) with a single free constant N that we had to assume that $\mathcal{O}_\ell$ transforms irreducibly.) We also define a 'truncated' matrix element $M_{\ell'}$ by

$$\begin{aligned}\int d^4x_2\cdots e^{-iq_2\cdot x_2}\cdots(q_1,\sigma|T\{A_2(x_2)\cdots\}|\Omega)\\ \equiv N^{-1}(2\pi)^{-3/2}\sum_{\ell'}u^*_{\ell'}(\mathbf{q}_1,\sigma)M_{\ell'}(q_2\cdots).\end{aligned} \tag{10.3.4}$$

Eq. (10.3.2) then reads, for $q_1^2 \to -m^2$,

$$G_\ell \to \frac{-2i\sqrt{\mathbf{q}_1^2+m^2}}{q_1^2+m^2-i\epsilon}\sum_{\sigma,\ell'}u_\ell(\mathbf{q}_1,\sigma)u^*_{\ell'}(\mathbf{q}_1,\sigma)M_{\ell'}. \tag{10.3.5}$$

According to Eqs. (6.2.2) and (6.2.18), the quantity multiplying $M_{\ell'}$ in (10.3.5) is the momentum space matrix propagator $-i\Delta_{\ell\ell'}(q_1)$ for the free field with the Lorentz transformation properties of $\mathcal{O}_\ell$ (or at least its limiting behavior for $q_1^2 \to -m^2$), so (10.3.5) allows us to identify $M_{\ell'}$ as the sum of all graphs with external lines carrying momenta $q_1, q_2, \dots$ corresponding to the operators $\mathcal{O}_\ell, A_2, \dots$, but with the final propagator for the $\mathcal{O}_\ell$ line stripped away. Eq. (10.3.4) is then just the usual prescription for how to calculate the matrix element for emission of a particle from the sum of Feynman diagrams: strip away the particle propagator, and contract with the usual external line factor $(2\pi)^{-3/2}u^*_{\ell'}$. The only discrepancy with the usual Feynman rules is the factor N.

The above theorem is a famous result due to Lehmann, Symanzik, and Zimmerman [3], known as the *reduction formula*, which we have proved here by a somewhat different method that has allowed us easily to generalize this result to the case of arbitrary spin. One

[113] For instance, for a conventionally normalized free scalar field, $u_\ell(\mathbf{q}_1,\sigma) = [2\sqrt{\mathbf{q}_1^2+m^2}]^{-1/2}$.

important aspect of this result is that it applies to any sort of operator; $\mathcal{O}_\ell$ need not be some field that actually appears in the Lagrangian, and the particle it creates may be a bound state composed of those particles whose fields do occur in the Lagrangian. It provides an important lesson even where $\mathcal{O}_\ell$ is some field Ψ_ℓ in the Lagrangian: if we are to use the usual Feynman rules to calculate S-matrix elements, then we should first redefine the normalization of the fields by a factor $1/N$, so that (with apologies for the multiple use of the symbol Ψ):

$$(\Omega|\Psi_\ell(0)|q,\sigma) = (2\pi)^{-3/2}u_\ell(\mathbf{q},\sigma). \tag{10.3.6}$$

A field normalized as in Eq. (10.3.6) is called a *renormalized field*.

The field renormalization constant N shows up in another place. Suppose that there is just one of the operators $A_2, A_3, \dots$ in Eq. (10.3.1), and take it to be the adjoint of a member of the same field multiplet as $\mathcal{O}_\ell$. Then Eq. (10.3.2) reads

$$\begin{aligned}\int d^4x_1\int d^4x_2\, e^{-iq_1\cdot x_1}e^{-iq_2\cdot x_2}(\Omega|T\{\mathcal{O}_\ell(x_1)\mathcal{O}^\dagger_{\bar\ell}(x_2)\}|\Omega)\\ \xrightarrow{q_1^2\to -m^2}\frac{-2i\sqrt{\mathbf{q}_1^2+m^2}(2\pi)^3}{q_1^2+m^2-i\epsilon}\sum_\sigma(\Omega|\mathcal{O}_\ell(0)|q_1,\sigma)\\ \cdot\int d^4x_2\, e^{-iq_2\cdot x_2}e^{-iq_1\cdot x_2}(q_1,\sigma|\mathcal{O}^\dagger_{\bar\ell}(0)|\Omega)\\ = \frac{-2i|N|^2\sqrt{\mathbf{q}_1^2+m^2}}{q_1^2+m^2-i\epsilon}\sum_\sigma u_\ell(\mathbf{q}_1,\sigma)u^*_{\bar\ell}(\mathbf{q}_1,\sigma)(2\pi)^4\delta^4(q_1+q_2).\end{aligned}$$

This is just the usual behavior of a propagator (the sum of all graphs with two external lines) near its pole, except for the factor $|N|^2$. According to Eq. (10.3.6), this factor is absent in the propagator of the renormalized field Ψ_ℓ. Thus *a renormalized field is one whose propagator has the same behavior near its pole as for a free field, and the renormalized mass is defined by the position of the pole.*

To see how this works in practice, consider the theory of a real self-interacting scalar field Φ_B, the subscript B being added here to remind us that so far this is a 'bare' (i.e., unrenormalized) field. The Lagrangian density is taken as usual as

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\Phi_B\partial^\mu\Phi_B - \frac{1}{2}m_B^2\Phi_B^2 - V_B(\Phi_B). \tag{10.3.7}$$

In general there would be no reason to expect that the field Φ_B would satisfy condition (10.3.6), nor that the pole in q^2 would be at $-m_B^2$, so let us introduce a renormalized field and mass

$$\Phi \equiv Z^{-1/2}\Phi_B, \tag{10.3.8}$$

$$m^2 \equiv m_B^2 + \delta m^2, \tag{10.3.9}$$

with Z to be chosen so that Φ does satisfy Eq. (10.3.6), and δm^2 chosen so that the pole of the propagator is at $q^2 = -m^2$. (The use of the symbol Z in this context has become conventional; there is a different Z for each field in the Lagrangian.) The Lagrangian density (10.3.7) may then be rewritten

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1, \tag{10.3.10}$$

$$\mathcal{L}_0 = -\frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi - \frac{1}{2}m^2\Phi^2, \tag{10.3.11}$$

$$\mathcal{L}_1 = -\frac{1}{2}(Z-1)[\partial_\mu\Phi\partial^\mu\Phi + m^2\Phi^2] + \frac{1}{2}Z\delta m^2\Phi^2 - V(\Phi), \tag{10.3.12}$$

where

$$V(\Phi) \equiv V_B(\sqrt{Z}\Phi).$$

In calculating the corrections to the complete momentum space propagator of the renormalized scalar field, conventionally called $\Delta'(q)$, it is convenient to consider separately the *one-particle-irreducible* graphs: those connected graphs (excluding a graph consisting of a single scalar line) that cannot be disconnected by cutting through any one internal scalar line. An example is shown in Figure ??. It is conventional to write the sum of all such graphs, with the two external line propagator factors $-i(2\pi)^{-4}(q^2+m^2-i\epsilon)^{-1}$ omitted, as $i(2\pi)^4\Pi^*(q^2)$, with the asterisk to remind us that these are one-particle-irreducible graphs. Then the corrections to the complete propagator are given by a sum of chains of one, two, or more of these one-particle-irreducible subgraphs connected with the usual uncorrected propagator factors:

$$\begin{aligned}\frac{-i}{(2\pi)^4}\Delta'(q) = \frac{-i}{(2\pi)^4}\frac{1}{q^2+m^2-i\epsilon}\\ + \left[\frac{-i}{(2\pi)^4}\frac{1}{q^2+m^2-i\epsilon}\right][i(2\pi)^4\Pi^*(q^2)]\left[\frac{-i}{(2\pi)^4}\frac{1}{q^2+m^2-i\epsilon}\right]\\ + \left[\frac{-i}{(2\pi)^4}\frac{1}{q^2+m^2-i\epsilon}\right][i(2\pi)^4\Pi^*(q^2)]\left[\frac{-i}{(2\pi)^4}\frac{1}{q^2+m^2-i\epsilon}\right]\\ \cdot[i(2\pi)^4\Pi^*(q^2)]\left[\frac{-i}{(2\pi)^4}\frac{1}{q^2+m^2-i\epsilon}\right] + \cdots\end{aligned} \tag{10.3.13}$$

or more simply

$$\begin{aligned}\Delta'(q) = [q^2+m^2-i\epsilon]^{-1} + [q^2+m^2-i\epsilon]^{-1}\Pi^*(q^2)[q^2+m^2-i\epsilon]^{-1}\\ + [q^2+m^2-i\epsilon]^{-1}\Pi^*(q^2)[q^2+m^2-i\epsilon]^{-1}\Pi^*(q^2)[q^2+m^2-i\epsilon]^{-1} + \cdots.\end{aligned} \tag{10.3.14}$$

Summing the geometric series, this gives

$$\Delta'(q) = [q^2+m^2-\Pi^*(q^2)-i\epsilon]^{-1}. \tag{10.3.15}$$

In calculating Π^*, we encounter a tree graph arising from a single insertion of vertices corresponding to the terms in Eq. (10.3.12) proportional to $\partial_\mu\Phi\partial^\mu\Phi$ and Φ^2, plus a term $\Pi^*_{\rm LOOP}$ arising from loop graphs like that in Figure ??(a):

$$\Pi^*(q^2) = -(Z-1)[q^2+m^2] + Z\delta m^2 + \Pi^*_{\rm LOOP}(q^2). \tag{10.3.16}$$

The condition that m^2 is the true mass of the particle is that the pole of the propagator should be at $q^2 = -m^2$, so that

$$\Pi^*(-m^2) = 0. \tag{10.3.17}$$

Also, the condition that the pole of the propagator at $q^2 = -m^2$ should have a unit residue (like the uncorrected propagator) is that

$$\left[\frac{d}{dq^2}\Pi^*(q^2)\right]_{q^2=-m^2} = 0. \tag{10.3.18}$$

These conditions allow us to evaluate Z and δm^2:

$$Z\delta m^2 = -\Pi^*_{\rm LOOP}(0), \tag{10.3.19}$$

$$Z = 1 + \left[\frac{d}{dq^2}\Pi^*_{\rm LOOP}(q^2)\right]_{q^2=-m^2}. \tag{10.3.20}$$

This incidentally shows that $Z\delta m^2$ and $Z-1$ are given by a series of terms containing one or more coupling constant factors, justifying the treatment of the first two terms in Eq. (10.3.12) as part of the interaction $\mathcal{L}_1$.

In actual calculations it is simplest just to say that from the loop terms $\Pi^*_{\rm LOOP}(q^2)$ we must subtract a first-order polynomial in q^2 with coefficients chosen so that the difference satisfies Eqs. (10.3.17) and (10.3.18). As we shall see, this subtraction procedure incidentally cancels the infinities that arise from the momentum space integrals in $\Pi^*_{\rm LOOP}$. However, as this discussion should make clear, *the renormalization of masses and fields has nothing directly to do with the presence of infinities, and would be necessary even in a theory in which all momentum space integrals were convergent.*

An important consequence of the conditions (10.3.17) and (10.3.18) is that it is not necessary to include radiative corrections in external lines on the mass shell. That is,

$$[\Pi^*(q^2)[q^2+m^2-i\epsilon]^{-1} + \Pi^*(q^2)[q^2+m^2-i\epsilon]^{-1}\Pi^*(q^2)[q^2+m^2-i\epsilon]^{-1} + \cdots]_{q^2\to -m^2} = 0. \tag{10.3.21}$$

Similar remarks apply to particles of arbitrary spin. For instance, for the 'bare' Dirac field the Lagrangian is

$$\mathcal{L} = -\bar\Psi_B[\not\partial + m_B]\Psi_B - V_B(\bar\Psi_B\Psi_B). \tag{10.3.22}$$

We introduce renormalized fields and masses

$$\Psi \equiv Z_2^{-1/2}\Psi_B, \tag{10.3.23}$$

$$m \equiv m_B + \delta m. \tag{10.3.24}$$

(The subscript 2 on Z_2 is conventionally used to distinguish the renormalization constant of a fermion field.) The Lagrangian density is then rewritten

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1, \tag{10.3.25}$$

$$\mathcal{L}_0 = -\bar\Psi[\not\partial + m]\Psi, \tag{10.3.26}$$

$$\mathcal{L}_1 = -(Z_2 - 1)[\bar\Psi[\not\partial + m]\Psi] + Z_2\delta m\bar\Psi\Psi - V_B(Z_2\bar\Psi\Psi). \tag{10.3.27}$$

Let $i(2\pi)^4\Sigma^*(\not k)$ be the sum of all connected graphs, with one fermion line coming in with four-momentum k and one going out with the same four-momentum, that cannot be disconnected by cutting through any single internal fermion line, and with external line propagator factors $-i(2\pi)^{-4}$ and $[i\not k + m - i\epsilon]^{-1}$ omitted. (Lorentz invariance is being used to justify writing Σ^* as an ordinary function of the Lorentz scalar matrix $\not k = k_\mu\gamma^\mu$.) Then the complete fermion propagator is

$$\begin{aligned}S'(k) = [i\not k + m - i\epsilon]^{-1} + [i\not k + m - i\epsilon]^{-1}\Sigma^*(\not k)[i\not k + m - i\epsilon]^{-1}\\ + [i\not k + m - i\epsilon]^{-1}\Sigma^*(\not k)[i\not k + m - i\epsilon]^{-1}\Sigma^*(\not k)[i\not k + m - i\epsilon]^{-1} + \cdots\\ = [i\not k + m - \Sigma^*(\not k) - i\epsilon]^{-1}.\end{aligned} \tag{10.3.28}$$

In calculating $\Sigma^*(\not k)$ we take into account the tree graphs from the terms in Eq. (10.3.27) proportional to $\bar\Psi\not\partial\Psi$ and $\bar\Psi\Psi$ as well as loop contributions:

$$\Sigma^*(\not k) = -(Z_2 - 1)[i\not k + m] + Z_2\delta m + \Sigma^*_{\rm LOOP}(\not k). \tag{10.3.29}$$

The condition that the complete propagator has a pole at $k^2 = -m^2$ with the same residue as the uncorrected propagator is then that

$$\Sigma^*(im) = 0, \tag{10.3.30}$$

$$\left.\frac{\partial\Sigma^*(\not k)}{\partial\not k}\right|_{\not k = im} = 0, \tag{10.3.31}$$

and hence

$$Z_2\delta m = -\Sigma^*_{\rm LOOP}(im), \tag{10.3.32}$$

$$Z_2 = 1 - i\left.\frac{\partial\Sigma^*_{\rm LOOP}(\not k)}{\partial\not k}\right|_{\not k = im}. \tag{10.3.33}$$

Just as for scalars, the vanishing of $[i\not k + m]^{-1}\Sigma^*(\not k)$ in the limit $\not k \to im$ tells us that radiative corrections may be ignored in external fermion lines. Corresponding results for the photon propagator will be derived in Section 10.5.

10.4 Renormalized Charge and Ward Identities

The use of the commutation and conservation relations of Heisenberg-picture operators allows us to make a connection between the charges (or other similar quantities) in the Lagrangian density and the properties of physical states. Recall that the invariance of the Lagrangian density with respect to global gauge transformations $\Psi_\ell \to \exp(iq_\ell\alpha)\Psi_\ell$ (with α an arbitrary constant phase) implies the existence of a current

$$J^\mu = -i\sum_\ell\frac{\partial\mathcal{L}}{\partial(\partial_\mu\Psi_\ell)}q_\ell\Psi_\ell, \tag{10.4.1}$$

satisfying the conservation condition

$$\partial_\mu J^\mu = 0. \tag{10.4.2}$$

This implies that the space-integral of the time component of J^μ is time-independent:

$$i\frac{d}{dt}Q = [Q, H] = 0, \tag{10.4.3}$$

where

$$Q \equiv \int d^3x\, J^0. \tag{10.4.4}$$

(There is a very important possible exception here, that the integral (10.4.4) may not exist if there are long-range forces due to massless scalars in the system. We will return to this point when we consider broken symmetries in Volume II.) Also, since it is a space-integral, Q is manifestly translation-invariant

$$[\mathbf{P}, Q] = 0 \tag{10.4.5}$$

and since J^μ is a four-vector, Q is invariant with respect to homogeneous Lorentz transformations

$$[J^{\mu\nu}, Q] = 0. \tag{10.4.6}$$

It follows that Q acting on the true vacuum $|\Omega)$ must be another Lorentz-invariant state of zero energy and momentum, and hence (assuming no vacuum degeneracy) must be proportional to $|\Omega)$ itself. But the proportionality constant must vanish, because Lorentz invariance requires that $(\Omega|J_\mu|\Omega)$ vanish. Hence

$$Q|\Omega) = 0. \tag{10.4.7}$$

Also, Q acting on any one-particle state $|\mathbf{p},\sigma,n)$ must be another state with the same energy, momentum, and Lorentz transformation properties, and thus (assuming no degeneracy of one-particle states) must be proportional to the same one-particle state

$$Q|\mathbf{p},\sigma,n) = q_{(n)}|\mathbf{p},\sigma,n). \tag{10.4.8}$$

The Lorentz invariance of Q ensures that the eigenvalue $q_{(n)}$ is independent of $\mathbf{p}$ and σ, depending only on the species of the particle. This eigenvalue is what is known as the electric charge (or whatever other quantum number of which J^μ may be the current) of the one-particle state. To relate this to the q_ℓ parameters in the Lagrangian, we note that the canonical commutation relations give

$$[J^0(\mathbf{x},t), \Psi_\ell(\mathbf{y},t)] = -q_\ell\Psi_\ell(\mathbf{y},t)\delta^3(\mathbf{x}-\mathbf{y}), \tag{10.4.9}$$

or integrating over $\mathbf{x}$:

$$[Q, \Psi_\ell(y)] = -q_\ell\Psi_\ell(y). \tag{10.4.10}$$

The same is true of any local function $F(y)$ of the fields and field derivatives and their adjoints, containing definite numbers of each:

$$[Q, F(y)] = -q_F F(y), \tag{10.4.11}$$

where q_F is the sum of the q_ℓ for all fields and field derivatives in $F(y)$, minus the sum of the q_ℓ for all field adjoints and their derivatives. Taking the matrix element of this equation between a one-particle state and the vacuum, and using Eqs. (10.4.7) and (10.4.8), we have

$$(\Omega|F(y)|\mathbf{p},\sigma,n)(q_F - q_{(n)}) = 0. \tag{10.4.12}$$

Hence we must have

$$q_{(n)} = q_F \tag{10.4.13}$$

as long as

$$(\Omega|F(y)|\mathbf{p},\sigma,n) \ne 0. \tag{10.4.14}$$

As we saw in the previous section, Eq. (10.4.14) is the condition that assures that momentum space Greens functions involving F have poles corresponding to the one-particle state $|\mathbf{p},\sigma,n)$. For a one-particle state corresponding to one of the fields in the Lagrangian we could take $F = \Psi_\ell$, in which case $q_F = q_\ell$, but our results here apply to general one-particle states, whether or not their fields appear in the Lagrangian.

This almost, but not quite, tells us that despite all the possible high-order graphs that affect the emission and absorption of photons by charged particles, the physical electric charge is just equal to a parameter q_ℓ appearing in the Lagrangian (or to a sum of such parameters, like q_F.) The qualification that has to be added here is that the requirement, that the Lagrangian be invariant under the transformations $\Psi_\ell \to \exp(iq_\ell\alpha)\Psi_\ell$, does nothing to fix the over-all scale of the quantities q_ℓ. The physical electric charges are those that determine the response of matter fields to a given *renormalized* electromagnetic field A^μ. That is, the scale of the q_ℓ is fixed by requiring that the renormalized electromagnetic field appears in the matter Lagrangian $\mathcal{L}_M$ in the linear combinations $[\partial_\mu - iq_\ell A_\mu]\Psi_\ell$, so that the current J^μ is

$$J^\mu = \frac{\delta\mathcal{L}_M}{\delta A_\mu}. \tag{10.4.15}$$

But A^μ and q_ℓ are not the same as the 'bare electromagnetic field' A^μ_B and 'bare charges' $q_{B\ell}$ that appear in the Lagrangian when we write it in its simplest form

$$\begin{aligned}\mathcal{L} = -\frac{1}{4}(\partial_\mu A_{B\nu} - \partial_\nu A_{B\mu})(\partial^\mu A^\nu_B - \partial^\nu A^\mu_B)\\ + \mathcal{L}_M\big(\Psi_\ell, [\partial_\mu - iq_{B\ell}A_{B\mu}]\Psi_\ell\big).\end{aligned} \tag{10.4.16}$$

The renormalized electromagnetic field (defined to have a complete propagator whose pole at $p^2 = 0$ has unit residue) is conventionally written in terms of A^μ_B as

$$A^\mu = Z_3^{-1/2}A^\mu_B, \tag{10.4.17}$$

so in order for the charge q_ℓ to characterize the response of the charged particles to a given renormalized electromagnetic field, we should define the renormalized charges by

$$q_\ell = \sqrt{Z_3}\,q_{B\ell}. \tag{10.4.18}$$

We see that the physical electric charge q of any particle is just proportional to a parameter q_B related to those appearing in the Lagrangian, with a proportionality constant

$Z_3^{-1/2}$ that is the same for all particles. This helps us to understand how a particle like the proton, that is surrounded by a cloud of virtual mesons and other strongly interacting particles, can have the same charge as the positron, whose interactions are all much weaker. It is only necessary to assume that for some reason the charges $q_{B\ell}$ in the Lagrangian are equal and opposite for the electron and for those particles (two u quarks and one d quark) that make up the proton; the effect of higher-order corrections then appears solely in the common factor $Z_3^{1/2}$.

In order for charge renormalization to arise only from radiative corrections to the photon propagator, there must be cancellations among the great variety of other radiative corrections to the propagators and electromagnetic vertices of the charged particles. We can see a little more deeply into the nature of these cancellations by making use of the celebrated relations between these charged particle propagators and vertices known as the *Ward identities*.

For instance, consider the Greens function for an electric current $J^\mu(x)$ together with a Heisenberg-picture Dirac field $\Psi_n(y)$ of charge q and its covariant adjoint $\bar\Psi_m(z)$. We define the electromagnetic vertex function Γ^μ of the charged particle by

$$\begin{aligned}\int d^4x\, d^4y\, d^4z\, e^{-ip\cdot x}e^{-ik\cdot y}e^{+i\ell\cdot z}(\Omega|T\{J^\mu(x)\Psi_n(y)\bar\Psi_m(z)\}|\Omega)\\ \equiv -iqS'_{nn'}(k)\Gamma^\mu_{n'm'}(k,\ell)S'_{m'm}(\ell)\delta^4(p+k-\ell),\end{aligned} \tag{10.4.19}$$

where

$$-iS'_{nm}(k)\delta^4(k-\ell) \equiv \int d^4y\, d^4z\,(\Omega|T\{\Psi_n(y)\bar\Psi_m(z)\}|\Omega)e^{-ik\cdot y}e^{+i\ell\cdot z}. \tag{10.4.20}$$

According to the theorem of Section 6.4, Eq. (10.4.20) gives the sum of all Feynman graphs with one incoming and one outgoing fermion line, i.e., the complete Dirac propagator. Also, Eq. (10.4.19) gives the sum of all such graphs with an extra photon line attached, so Γ^μ is the sum of "vertex" graphs with one incoming Dirac line, one outgoing Dirac line, and one photon line, but with the complete Dirac external line propagators and the bare photon external line propagator stripped away. To make the normalization of S' and Γ^μ perfectly clear, we mention that in the limit of no interactions, these functions take the values

$$S'(k) \to [i\gamma_\mu k^\mu + m - i\epsilon]^{-1}, \qquad \Gamma^\mu(k,\ell) \to \gamma^\mu.$$

The one-loop diagrams that provide corrections to these limiting values are shown in Figure ??.

We can derive a relation between Γ^μ and S' by use of the identity

$$\begin{aligned}\frac{\partial}{\partial x^\mu}T\{J^\mu(x)\Psi_n(y)\bar\Psi_m(z)\} = T\{\partial_\mu J^\mu(x)\Psi_n(y)\bar\Psi_m(z)\}\\ + \delta(x^0-y^0)T\{[J^0(x),\Psi_n(y)]\bar\Psi_m(z)\}\\ + \delta(x^0-z^0)T\{\Psi_n(y)[J^0(x),\bar\Psi_m(z)]\}.\end{aligned} \tag{10.4.21}$$

where the delta functions arise from time-derivatives of step functions. The conservation condition (10.4.2) tells us that the first term vanishes, while the second and third terms can be calculated using the commutation relations (10.4.9), which here give

$$[J^0(\mathbf{x},t), \Psi_n(\mathbf{y},t)] = -q\Psi_n(\mathbf{y},t)\delta^3(\mathbf{x}-\mathbf{y}) \tag{10.4.22}$$

and its adjoint

$$[J^0(\mathbf{x},t), \bar\Psi_n(\mathbf{y},t)] = q\bar\Psi_n(\mathbf{y},t)\delta^3(\mathbf{x}-\mathbf{y}). \tag{10.4.23}$$

Eq. (10.4.21) then reads

$$\begin{aligned}\frac{\partial}{\partial x^\mu}T\{J^\mu(x)\Psi_n(y)\bar\Psi_m(z)\} = -q\delta^4(x-y)T\{\Psi_n(y)\bar\Psi_m(z)\}\\ + q\delta^4(x-z)T\{\Psi_n(y)\bar\Psi_m(z)\}.\end{aligned} \tag{10.4.24}$$

Inserting this in the Fourier transform (10.4.19) gives

$$(\ell-k)_\mu S'(k)\Gamma^\mu(k,\ell)S'(\ell) = iS'(\ell) - iS'(k)$$

or in other words

$$(\ell-k)_\mu\Gamma^\mu(k,\ell) = iS'^{-1}(k) - iS'^{-1}(\ell). \tag{10.4.25}$$

This is known as the *generalized Ward identity*, first derived (by these methods) by Takahashi [4]. The original Ward identity, derived earlier by Ward [5] from a study of perturbation theory, can be obtained from Eq. (10.4.25) by letting ℓ approach k. In this limit, Eq. (10.4.25) gives

$$\Gamma^\mu(k,k) = -i\frac{\partial}{\partial k_\mu}S'^{-1}(k). \tag{10.4.26}$$

The fermion propagator is related to the self-energy insertion $\Sigma^*(k)$ by Eq. (10.3.28)

$$S'^{-1}(k) = i\not k + m - \Sigma^*(k),$$

so Eq. (10.4.26) may be written

$$\Gamma^\mu(k,k) = \gamma^\mu + i\frac{\partial}{\partial k_\mu}\Sigma^*(k). \tag{10.4.27}$$

For a renormalized Dirac field, Eqs. (10.3.31) and (10.4.27) tell us that on the mass shell

$$\bar u_k\Gamma^\mu(k,k)u_k = \bar u_k\gamma^\mu u_k, \tag{10.4.28}$$

where $[i\gamma_\mu k^\mu + m]u_k = [i\gamma_\mu k^\mu + m]u_k = 0$. Thus the renormalization of the fermion field ensures that the radiative corrections to the vertex function Γ_μ cancel when a fermion on the mass shell interacts with an electromagnetic field with zero momentum transfer, as is the case when we set out to measure the fermion's electric charge. If we had not used a renormalized fermion field then the corrections to the vertex function would have just cancelled the corrections due to radiative corrections to the external fermion lines, leaving the electric charge again unchanged.

10.5 Gauge Invariance

The conservation of electric charge may be used to prove a useful result for the quantities

$$\begin{aligned}M^{\mu\mu'\cdots}_{\beta\alpha}(q,q',\dots) \equiv \int d^4x\int d^4x'\cdots e^{-iq\cdot x}e^{-iq'\cdot x'}\cdots\\ \cdot\,_{\rm out}(\beta|T\{J^\mu(x),J^{\mu'}(x'),\dots\}|\alpha)_{\rm in}.\end{aligned} \tag{10.5.1}$$

In theories like spinor electrodynamics in which the electromagnetic interaction is linear in the field A^μ, this is the matrix element for emission (and/or absorption) of on- or off-shell photons having four-momenta q, q', etc. (and/or $-q$, $-q'$, etc.), with external line photon coefficient functions or propagators omitted, in an arbitrary transition $\alpha \to \beta$. Our result is that Eq. (10.5.1) vanishes when contracted with any one of the photon four-momenta:

$$q_\mu M^{\mu\mu'\cdots}_{\beta\alpha}(q,q',\dots) = q'_{\mu'}M^{\mu\mu'\cdots}_{\beta\alpha}(q,q',\dots) = \cdots = 0. \tag{10.5.2}$$

Since M is defined symmetrically with respect to the photon lines, it will be sufficient to show the vanishing of the first of these quantities.

For this purpose, note that by an integration by parts

$$\begin{aligned}q_\mu M^{\mu\mu'\cdots}_{\beta\alpha}(q,q',\dots) = -i\int d^4x\int d^4x'\cdots e^{-iq\cdot x}e^{-iq'\cdot x'}\cdots\\ \cdot\,_{\rm out}(\beta|\frac{\partial}{\partial x^\mu}T\{J^\mu(x),J^{\mu'}(x'),\dots\}|\alpha)_{\rm in}.\end{aligned} \tag{10.5.3}$$

The electric current $J^\mu(x)$ is conserved, but this does not immediately imply that Eq. (10.5.3) vanishes, because we still have to take account of the x^0-dependence contained in the theta functions that appear in the definition of the time-ordered product. For instance, for just two currents

$$T\{J^\mu(x)J^\nu(y)\} = \theta(x^0-y^0)J^\mu(x)J^\nu(y) + \theta(y^0-x^0)J^\nu(y)J^\mu(x)$$

so, taking account of the conservation of $J^\mu(x)$:

$$\begin{aligned}\frac{\partial}{\partial x^\mu}T\{J^\mu(x)J^\nu(y)\} = \delta(x^0-y^0)J^0(x)J^\nu(y) - \delta(y^0-x^0)J^\nu(y)J^0(x)\\ = \delta(x^0-y^0)[J^0(x), J^\nu(y)].\end{aligned} \tag{10.5.4}$$

With more than two currents, we get an equal-time commutator like this (inside the time-ordered product) for each current aside from $J^\mu(x)$ itself. To evaluate this commutator, we recall that (as shown in the previous section) for any product F of field operators and their adjoints and/or derivatives

$$[J^0(\mathbf{x},t), F(\mathbf{y},t)] = -q_F F(\mathbf{x},t)\delta^3(\mathbf{x}-\mathbf{y}),$$

where q_F is the sum of the q_ℓs for the fields and field derivatives in F, minus the sum of the q_ℓs for the field adjoints and their derivatives. For the electric current, q_J is zero; $J^\nu(y)$ is itself an electrically neutral operator. It follows that

$$[J^0(\mathbf{x},t), J^\nu(\mathbf{y},t)] = 0 \tag{10.5.5}$$

[illegible]

2. Derive the generalized Ward identity that governs the electromagnetic vertex function of a charged scalar field.
3. What is the most general form of the matrix element $\langle p_2, \sigma_2 | J^\mu(x) | p_1, \sigma_1 \rangle$ of the electromagnetic current $J^\mu(x)$ between two spin $\frac{1}{2}$ one-particle states of *different* masses m_1 and m_2 and equal parity? What if the parities were opposite? (Assume parity conservation throughout.)
4. Derive the spectral (Källén–Lehmann) representation for the vacuum expectation value $\langle \Omega | T\{J^\mu(x) J^\nu(y)\} | \Omega \rangle$, where $J^\mu(x)$ is a complex conserved current.
5. Derive the spectral (Källén–Lehmann) representation for the vacuum expectation value $\langle \Omega | T\{\psi_n(x) \bar{\psi}_m(y)\} | \Omega \rangle$, where $\psi(x)$ is a Dirac field.
6. Without using any assumptions about the asymptotic behavior of the scattering amplitude or cross-sections, show that it is impossible for forward photon scattering amplitudes to satisfy unsubtracted dispersion relations.
7. Derive the spectral (Källén–Lehmann) representation for a complex scalar field by using the methods of dispersion theory.
8. Use dispersion theory and the results of Section 8.7 to calculate the amplitude for forward photon–electron scattering in the electron rest frame to order e^4.

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11 One-Loop Radiative Corrections in Quantum Electrodynamics

In this chapter we shall proceed to carry out some of the classic one-loop calculations in the theory of charged leptons—massive spin $\frac{1}{2}$ particles that interact only with the electromagnetic field. There are three known species or "flavors" of leptons: the electron and muon, and the heavier, more recently discovered tauon. For definiteness we shall refer to the charged particles in our calculations here as "electrons," though most of our calculations will apply equally to muons and tauons. After some generalities in Section 11.1, we will move on to the calculation of the vacuum polarization in Section 11.2, the anomalous magnetic moment of the electron in Section 11.3, and the electron self-energy in Section 11.4. Along the way, we will introduce a number of the mathematical techniques that prove useful in such calculations, including the use of Feynman parameters, Wick rotation, and both the dimensional regularization of 't Hooft and Veltman and the older regularization method of Pauli and Villars. Although we shall encounter infinities, it will be seen that the final results are finite if expressed in terms of the renormalized charge and mass. In the next chapter we shall extend what we have learned here about renormalization to general theories in arbitrary orders of perturbation theory.

11.1 Counterterms

The Lagrangian density for electrons and photons is taken in the form[121]

$$\mathcal{L} = -\frac{1}{4}F_B^{\mu\nu}F_{B\,\mu\nu} - \bar\psi_B\left[\gamma_\mu(\partial^\mu + ie_B A_B^\mu) + m_B\right]\psi_B, \tag{11.1.1}$$

where $F_B^{\mu\nu} \equiv \partial^\mu A_B^\nu - \partial^\nu A_B^\mu$; A_B^μ and ψ_B are the bare (i.e., unrenormalized) fields of the photon and electron, and $-e_B$ and m_B are the bare charge and mass of the electron. As described in the previous chapter, we introduce renormalized fields and charge and mass:

$$\psi \equiv Z_2^{-1/2}\psi_B, \tag{11.1.2}$$
$$A^\mu \equiv Z_3^{-1/2}A_B^\mu, \tag{11.1.3}$$
$$e \equiv Z_3^{+1/2}e_B, \tag{11.1.4}$$
$$m \equiv m_B + \delta m, \tag{11.1.5}$$

with the constants Z_2, Z_3, and δm adjusted so that the propagators of the renormalized fields have poles in the same position and with the same residues as the propagators of the free fields in the absence of interactions. The Lagrangian may then be written in terms of renormalized quantities, as

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1 + \mathcal{L}_2, \tag{11.1.6}$$

where

$$\mathcal{L}_0 = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - \bar\psi\left[\gamma_\mu\partial^\mu + m\right]\psi, \tag{11.1.7}$$

[121] In this chapter we will not be making transformations between Heisenberg- and interaction-picture operators, so we shall return to a conventional notation, in which an upper case A and a lower case ψ are used to denote the photon and charged particle fields, respectively.

$$\mathcal{L}_1 = -ieA_\mu\bar\psi\gamma^\mu\psi, \tag{11.1.8}$$

and $\mathcal{L}_2$ is a sum of "counterterms"

$$\mathcal{L}_2 = -\frac{1}{4}(Z_3-1)F^{\mu\nu}F_{\mu\nu} - (Z_2-1)\bar\psi\left[\gamma_\mu\partial^\mu + m\right]\psi + Z_2\delta m\,\bar\psi\psi - ie(Z_2-1)A_\mu\bar\psi\gamma^\mu\psi. \tag{11.1.9}$$

It will turn out that all of the terms in $\mathcal{L}_2$ are of second order and higher order in e, and that these terms just suffice to cancel the ultraviolet divergences that arise from loop graphs.

11.2 Vacuum Polarization

We now begin our first calculation of a radiative correction involving loop graphs, the so-called vacuum polarization effect, consisting of the corrections to the propagator associated with an internal photon line. Vacuum polarization produces measurable shifts in the energy levels of hydrogen, and makes an important correction to the energies of muons bound in atomic orbits around heavy nuclei. Also, as we shall see in Volume II, the calculation of the vacuum polarization provides a key element in the calculation of the high energy behavior of electrodynamics and other gauge theories.

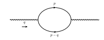


Figure 11.1. The one-loop diagram for the vacuum polarization in quantum electrodynamics. Here wavy lines represent photons; lines carrying arrows represent electrons.

As in Section 10.5, we define $i(2\pi)^4\Pi^{*\mu\nu}(q)$ as the sum of all connected graphs with two external photon lines with polarization indices μ and ν and carrying four-momentum q into and out of the diagram, not including photon propagators for the two external lines, and with the asterisk indicating that we exclude diagrams that can be disconnected by cutting through some internal photon line. The complete photon propagator $\Delta'^{\mu\nu}(q)$ is given by Eq. (10.5.13):

$$\Delta' = \Delta\left[\mathbf{1} - \Pi^*\Delta\right]^{-1}, \tag{11.2.1}$$

where $\Delta^{\mu\nu}(q)$ is the photon propagator without radiative corrections. Our task here is to calculate the leading contributions to $\Pi^{*\mu\nu}(q)$.

In lowest order there is a one-loop contribution to Π^*, corresponding to the diagram in Figure 11.1:

$$i(2\pi)^4\Pi^{*\rho\sigma}_{1\,\text{loop}}(q) = -\int d^4p\,\text{Tr}\left\{\left[\frac{-i}{(2\pi)^4}\frac{-i\not p + m}{p^2+m^2-i\epsilon}\right]\left[(2\pi)^4e\gamma^\rho\right]\cdot\left[\frac{-i}{(2\pi)^4}\frac{-i(\not p - \not q)+m}{(p-q)^2+m^2-i\epsilon}\right]\left[(2\pi)^4e\gamma^\sigma\right]\right\}, \tag{11.2.2}$$

with the first minus sign on the right required by the presence of a fermion loop. More simply, this is

$$\Pi^{*\rho\sigma}_{1\,\text{loop}}(q) = \frac{-ie^2}{(2\pi)^4}\int d^4p\,\frac{\text{Tr}\{[-i\not p + m]\gamma^\rho[-i(\not p - \not q)+m]\gamma^\sigma\}}{(p^2+m^2-i\epsilon)((p-q)^2+m^2-i\epsilon)}. \tag{11.2.3}$$

The first step in doing this integral is to use a trick introduced by Feynman [1]. We use the elementary formula

$$\frac{1}{AB} = \int_0^1\frac{dx}{[(1-x)A + xB]^2} \tag{11.2.4}$$

to write the product of scalar propagators in Eq. (11.2.3) as

$$\begin{aligned}\frac{1}{(p^2+m^2-i\epsilon)((p-q)^2+m^2-i\epsilon)} &= \int_0^1\left[(p^2+m^2-i\epsilon)(1-x) + ((p-q)^2+m^2-i\epsilon)x\right]^{-2}dx\\ &= \int_0^1[p^2+m^2-i\epsilon-2p\cdot qx+q^2x]^{-2}\,dx\\ &= \int_0^1[(p-qx)^2+m^2-i\epsilon+q^2x(1-x)]^{-2}\,dx.\end{aligned}$$

(This is a special case of a class of integrals given in the Appendix to this chapter.) We can now shift the variable of integration in momentum space[122]

$$p \longrightarrow p + qx,$$

so that Eq. (11.2.3) becomes

$$\Pi^{*\rho\sigma}_{1\,\text{loop}}(q) = \frac{-ie^2}{(2\pi)^4}\int_0^1 dx\int d^4p\,[p^2+m^2-i\epsilon+q^2x(1-x)]^{-2}\cdot\text{Tr}\{[-i(\not p+\not q x)+m]\gamma^\rho[-i(\not p - \not q(1-x))+m]\gamma^\sigma\}. \tag{11.2.5}$$

Using the results of the Appendix to Chapter 8, the trace here can easily be calculated as

$$\begin{aligned}&\text{Tr}\{[-i(\not p+\not q x)+m]\gamma^\rho[-i(\not p-\not q(1-x))+m]\gamma^\sigma\}\\ &= 4\left[-(p+qx)^\rho(p-q(1-x))^\sigma + (p+qx)\cdot(p-q(1-x))\eta^{\rho\sigma} - (p+qx)^\sigma(p-q(1-x))^\rho + m^2\eta^{\rho\sigma}\right].\end{aligned} \tag{11.2.6}$$

Our next step is called a Wick rotation [2]. As long as $-q^2 < 4m^2$, the quantity $m^2 + q^2x(1-x)$ is positive for all x between 0 and 1, so the poles in the integrand of Eq. (11.2.5) are at $p^0 = \pm\sqrt{\mathbf{p}^2+m^2+q^2x(1-x)-i\epsilon}$, i.e., just above the negative real axis and just below the positive real axis. (See Figure 11.2.) We can rotate the contour of integrations of p^0 counterclockwise without crossing either of these poles, so that instead of integrating p^0 on the real axis from $-\infty$ to $+\infty$, we integrate it on the imaginary axis from $-i\infty$ to $+i\infty$. That is, we can write $p^0 = ip^4$, with p^4 integrated over real values from $-\infty$ to $+\infty$. (If an $i\epsilon$ instead of $-i\epsilon$ had appeared in the denominator of the propagator, then we would have been setting $p^0 = -ip^4$, with p^4 again integrated over real values from $-\infty$ to $+\infty$. The effect would be a change of sign of $\Pi^{*\rho\sigma}_{1\,\text{loop}}(q)$.)

[122] Strictly speaking, this step is only valid in convergent integrals. In principle, in order to justify the shift of variables, we should introduce some regulator scheme to make all integrals converge, such as the dimensional regularization scheme discussed below.

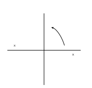


Figure 11.2. Wick rotation of the p^0 contour of integration. Small x's mark the poles in the p^0 complex plane; the arrow indicates the direction of rotation of the contour of integration, from the real to the imaginary p^0-axes.

Equation (11.2.5) now becomes

$$\Pi^{*\rho\sigma}_{1\,\text{loop}}(q) = \frac{4e^2}{(2\pi)^4}\int_0^1 dx\int(d^4p)_E[p^2+m^2+q^2x(1-x)]^{-2}\cdot\left[-(p+qx)^\rho(p-q(1-x))^\sigma + (p+qx)\cdot(p-q(1-x))\eta^{\rho\sigma} - (p+qx)^\sigma(p-q(1-x))^\rho + m^2\eta^{\rho\sigma}\right], \tag{11.2.7}$$

where

$$(d^4p)_E = dp^1dp^2dp^3dp^4$$

and all scalar products are evaluated using the Euclidean norm

$$a\cdot b = a^1b^1 + a^2b^2 + a^3b^3 + a^4b^4$$

with the understanding that $q^4 \equiv -iq^0$. Also, $\eta^{\rho\sigma}$ can be taken as either the Kronecker delta, with the indices running over 1, 2, 3, 4, or as the usual Minkowski tensor, with the indices running over 0, 1, 2, 3.

The integral (11.2.7) is badly divergent. Eventually all infinities will cancel, but to see this it is necessary at intermediate stages of the calculation to use some sort of regularization technique that makes the integrals finite. It would not do simply to cut off the integrals at some maximum momentum Λ, integrating only over p^μ with $p^2 < \Lambda^2$, because this would amount to introducing a step function $\theta(\Lambda^2 - p^2)$ into the electron propagator, and the Ward identity (10.4.25) shows that in order to maintain gauge invariance, any modification of the electron propagator must be accompanied with a modification of the electron–photon vertex. In fact, with an ordinary cutoff Λ, radiative corrections would induce a photon mass, a clear violation of the requirements of gauge invariance.

Experience has shown that the most convenient method for regulating divergent integrals without impairing gauge invariance is the dimensional regularization technique introduced by 't Hooft and Veltman [3] in 1972, based on a continuation from four to an arbitrary number d of spacetime dimensions. This amounts to carrying out angular averages in integrals like (11.2.7) by dropping all terms that are odd in p, and replacing the terms that have even numbers of p-factors with[123]

$$p^\mu p^\nu \longrightarrow p^2\eta^{\mu\nu}/d, \tag{11.2.8}$$
$$p^\mu p^\nu p^\rho p^\sigma \longrightarrow (p^2)^2\left[\eta^{\mu\nu}\eta^{\rho\sigma} + \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}\right]/d(d+2). \tag{11.2.9}$$

Also, after writing the integrand in this way as a function only of p^2, the volume element d^4p_E is to be replaced with $\Omega_d\kappa^{d-1}d\kappa$, where $\kappa \equiv \sqrt{p^2}$, and Ω_d is the area of a unit sphere in d dimensions

$$\Omega_d = 2\pi^{d/2}/\Gamma(d/2). \tag{11.2.10}$$

The integral (11.2.7) now converges for complex spacetime dimensionality d. We can continue the integral through complex d-values to $d = 4$, the infinities then reappearing as factors $(d-4)^{-1}$.

For the integral (11.2.7), dimensional regularization gives

$$\Pi^{*\rho\sigma}_{1\,\text{loop}}(q) = \frac{4e^2\Omega_d}{(2\pi)^4}\int_0^1 dx\int_0^\infty\kappa^{d-1}d\kappa\,[\kappa^2+m^2+q^2x(1-x)]^{-2}\cdot\left[-\frac{2\kappa^2}{d}\eta^{\rho\sigma} + 2q^\rho q^\sigma x(1-x) + (\kappa^2 - q^2x(1-x))\eta^{\rho\sigma} + m^2\eta^{\rho\sigma}\right].$$

The integrals over κ can be carried out for any complex d (or for any real d, aside from the even integers). We use the well-known formulas (given in greater generality in the Appendix to this chapter):

$$\int_0^\infty\kappa^{d-1}[\kappa^2+v^2]^{-2}d\kappa = \frac{1}{2}(v^2)^{d/2-2}\Gamma(d/2)\Gamma(2-d/2), \tag{11.2.11}$$
$$\int_0^\infty\kappa^{d+1}[\kappa^2+v^2]^{-2}d\kappa = \frac{1}{2}(v^2)^{d/2-1}\Gamma(1+d/2)\Gamma(1-d/2). \tag{11.2.12}$$

and find

$$\Pi^{*\rho\sigma}_{1\,\text{loop}}(q) = \frac{2e^2\Omega_d}{(2\pi)^4}\int_0^1 dx\left[\left(1-\frac{2}{d}\right)\eta^{\rho\sigma}(m^2+q^2x(1-x))^{d/2-1}\Gamma(1+d/2)\Gamma(1-d/2) + \left(2q^\rho q^\sigma x(1-x) - q^2\eta^{\rho\sigma}x(1-x) + m^2\eta^{\rho\sigma}\right)(m^2+q^2x(1-x))^{d/2-2}\cdot\Gamma(d/2)\Gamma(2-d/2)\right].$$

The two terms in the integrand can be combined, using

$$(1-2/d)\Gamma(1+d/2)\Gamma(1-d/2) = -\Gamma(d/2)\Gamma(2-d/2).$$

[123] These expressions may most easily be derived by noting that their form is dictated by Lorentz invariance and the symmetry among the indices μ, ν, ρ, etc., while the factors may be found by requiring that both sides give the same result when contracted with η's.

We find

$$\Pi^{*\rho\sigma}_{1\,\text{loop}}(q) = \frac{4e^2\Omega_d}{(2\pi)^4}\Gamma(d/2)\Gamma(2-d/2)(q^\rho q^\sigma - q^2\eta^{\rho\sigma})\cdot\int_0^1 dx\,x(1-x)(m^2+q^2x(1-x))^{d/2-2}. \tag{11.2.13}$$

We note the very important result that this contribution to $\Pi^{*\rho\sigma}$ satisfies the relation

$$q_\rho\Pi^{*\rho\sigma}_{1\,\text{loop}}(q) = 0 \tag{11.2.14}$$

that was derived in Section 10.5 on the basis of the conservation and neutrality of the electric current. It was precisely to achieve this result that we adopted the dimensional regularization scheme. The reason that dimensional regularization gives this result is that the conservation of current does not depend on the dimensionality of spacetime.

The gamma function $\Gamma(2-d/2)$ in Eq. (11.2.13) blows up for $d \to 4$. Fortunately, as we saw in Section 11.1, there is another term that must be added to $\Pi^{*\rho\sigma}(q)$, arising from the term $-\frac{1}{4}(Z_3-1)F_{\mu\nu}F^{\mu\nu}$ in the interaction Lagrangian. This term has a structure like Eq. (11.2.13)

$$\Pi^{*\rho\sigma}_{\mathcal{L}_2}(q) = -(Z_3-1)(q^2\eta^{\rho\sigma} - q^\rho q^\sigma), \tag{11.2.15}$$

so to order e^2, the complete Π^* has the form

$$\Pi^{*\rho\sigma}(q) = (q^2\eta^{\rho\sigma} - q^\rho q^\sigma)\pi(q^2), \tag{11.2.16}$$

with

$$\pi(q^2) = -\frac{4e^2\Omega_d}{(2\pi)^4}\Gamma(d/2)\Gamma(2-d/2)\int_0^1 dx\,x(1-x)(m^2+q^2x(1-x))^{d/2-2} - (Z_3-1). \tag{11.2.17}$$

As we saw in Section 10.5, the definition of the renormalized electromagnetic field requires that $\pi(0) = 0$ (in order that the residue of the pole in the complete photon propagator at $q^2 = 0$ should be the same as for the bare propagator, aside from gauge-dependent terms). Therefore, to order e^2,

$$Z_3 = 1 - \frac{4e^2\Omega_d}{(2\pi)^4}\Gamma(d/2)\Gamma(2-d/2)(m^2)^{d/2-2}\int_0^1 x(1-x)\,dx, \tag{11.2.18}$$

so that, to order e^2,

$$\pi(q^2) = -\frac{4e^2\Omega_d}{(2\pi)^4}\Gamma(d/2)\Gamma(2-d/2)\int_0^1 dx\,x(1-x)\cdot\left[(m^2+q^2x(1-x))^{d/2-2} - (m^2)^{d/2-2}\right]. \tag{11.2.19}$$

Now we can remove the regularization, allowing d to approach its physical value $d = 4$. As mentioned before, there is an infinity in the one-loop contribution, arising from the limiting behavior of the Gamma function

$$\Gamma(2-d/2) \longrightarrow \frac{1}{2-d/2} - \gamma,$$

where γ is the Euler constant, $\gamma = 0.5772157$. The infinite part of $Z_3 - 1$ is given by using $1/(2-d/2)$ for $\Gamma(2-d/2)$, and replacing d everywhere else by 4:

$$(Z_3-1)_\infty = -\frac{4e^2\cdot 2\pi^2}{6(2\pi)^4}\frac{1}{2-d/2} = \frac{e^2}{6\pi^2}\frac{1}{d-4}. \tag{11.2.20}$$

We shall see in Volume II that this result may be used to derive the leading term in the renormalization group equation for the electric charge.

The poles at $d = 4$ obviously cancel in $\pi(q^2)$, because for $d = 4$ both $(m^2+q^2x(1-x))^{d/2-2}$ and $(m^2)^{d/2-2}$ have the same limit, unity. For the same reason, the term $-\gamma$ in $\Gamma(2-d/2)$ cancels in the total $\pi(q^2)$, though it does make a finite contribution to $Z_3 - 1$. There are other finite contributions to $Z_3 - 1$, that arise from the product of the pole in $\Gamma(2-d/2)$ with the linear terms in the expansion of $\Omega_d\Gamma(d/2)$ around $d = 4$, but these also cancel in the total $\pi(q^2)$. Indeed, in carrying out our dimensional regularization, we might have replaced $(2\pi)^{-4}$ with $(2\pi)^{-d}$, and the factor $\text{Tr}\,\mathbf{1} = 4$ might have been replaced with the dimensionality $2^{d/2}$ of gamma matrices in arbitrary even spacetime dimensionalities d, and these too would have contributed to the finite part of $Z_3 - 1$, but not of $\pi(q^2)$. Moreover, e^2 cannot be supposed to be d-independent, because as shown by inspection of Eq. (11.2.13), it has the d-dependent dimensionality $[\text{mass}]^{4-d}$. If we take $e^2 \propto \mu^{4-d}$, where μ is some quantity with the units of mass, then there are additional finite terms in $Z_3 - 1$, arising from the product of the pole in $\Gamma(2-d/2)$ with the term $(4-d)\ln\mu$ in the expansion of μ^{4-d} in powers of $4-d$, but again, these cancel between $Z_3 - 1$ and the one-loop contributions to $\pi(q^2)$.

The only terms that do contribute to $\pi(q^2)$ in the limit $d \to 4$ are those arising from the product of the pole in $\Gamma(2-d/2)$ with the linear terms in the expansion of $(m^2+q^2x(1-x))^{d/2-2}$ and $(m^2)^{d/2-2}$ in powers of $d-4$:

$$(m^2+q^2x(1-x))^{d/2-2} - (m^2)^{d/2-2} \longrightarrow \left(\frac{d}{2}-2\right)\ln\left(1+\frac{q^2x(1-x)}{m^2}\right). \tag{11.2.21}$$

This gives at last

$$\pi(q^2) = \frac{e^2}{2\pi^2}\int_0^1 x(1-x)\ln\left(1+\frac{q^2x(1-x)}{m^2}\right)dx. \tag{11.2.22}$$

The physical significance of the vacuum polarization can be explored by considering its effect on the scattering of two charged particles of spin $\frac{1}{2}$. The Feynman diagrams of Figure 11.3 make contributions to the scattering S-matrix element of the form

$$S_a(1,2\to 1',2') = (2\pi)^{-12/2}\delta^4(p_{1'}+p_{2'}-p_1-p_2)\left[e_1(2\pi)^4\bar u_{1'}\gamma^\mu u_1\right]\cdot\left[-i(2\pi)^{-4}\frac{1}{q^2}\right]\left[e_2(2\pi)^4\bar u_{2'}\gamma_\mu u_2\right],$$

$$S_b(1,2\to 1',2') = (2\pi)^{-12/2}\delta^4(p_{1'}+p_{2'}-p_1-p_2)\left[e_1(2\pi)^4\bar u_{1'}\gamma^\mu u_1\right]\cdot\left[-i(2\pi)^{-4}\frac{1}{q^2}\right]^2\left[i(2\pi)^4(q^2\eta_{\mu\nu} - q_\mu q_\nu)\pi(q^2)\right]\left[e_2(2\pi)^4\bar u_{2'}\gamma^\nu u_2\right],$$

where e_1 and e_2 are the charges of the two particles being scattered; $\pi(q^2)$ is calculated using for e in Eq. (11.2.22) the magnitude of the charge of the particle circulating in the loop in Figure 11.3; and q^μ is the momentum transfer $q = p_1 - p_{1'} = p_{2'} - p_2$. Using the conservation property $q_\mu\bar u_{1'}\gamma^\mu u_1 = 0$ the two diagrams together yield an S-matrix element:

$$S_{a+b}(1,2\to 1',2') = \frac{-ie_1e_2}{4\pi^2q^2}[1+\pi(q^2)]\delta^4(p_{1'}+p_{2'}-p_1-p_2)\cdot[\bar u_{1'}\gamma^\mu u_1][\bar u_{2'}\gamma_\mu u_2]. \tag{11.2.23}$$

In the non-relativistic limit, $\bar u_{1'}\gamma^0u_1 \simeq -i\delta_{\sigma_{1'}\sigma_1}$ while $\bar u_{1'}\gamma^iu_1 \simeq 0$, and likewise for particle 2. Also, in this limit q^0 is negligible compared with $|\mathbf{q}|$. Equation (11.2.23) in this limit becomes

$$S_{a+b}(1,2\to 1',2') = \frac{-ie_1e_2}{4\pi^2\mathbf{q}^2}[1+\pi(\mathbf{q}^2)]\delta^4(p_{1'}+p_{2'}-p_1-p_2)\delta_{\sigma_{1'}\sigma_1}\delta_{\sigma_{2'}\sigma_2}. \tag{11.2.24}$$

This may be compared with the S-matrix in the Born approximation due to a local spin-independent central potential $V(r)$:

$$S_{\text{Born}}(1,2\to 1',2') = -2\pi i\delta(E_{1'}+E_{2'}-E_1-E_2)T_{\text{Born}}(1,2\to 1',2'). \tag{11.2.25}$$

$$T_{\text{Born}}(1,2\to 1',2') = \delta_{\sigma_{1'}\sigma_1}\delta_{\sigma_{2'}\sigma_2}\int d^3x_1\int d^3x_2\,V(|\mathbf{x}_1-\mathbf{x}_2|)\cdot(2\pi)^{-12/2}e^{-i\mathbf{p}_{1'}\cdot\mathbf{x}_1}e^{-i\mathbf{p}_{2'}\cdot\mathbf{x}_2}e^{i\mathbf{p}_1\cdot\mathbf{x}_1}e^{i\mathbf{p}_2\cdot\mathbf{x}_2}. \tag{11.2.26}$$

Setting $\mathbf{x}_1 = \mathbf{x}_2 + \mathbf{r}$, this gives

$$S_{\text{Born}} = \frac{-i}{4\pi^2}\delta^4(p_{1'}+p_{2'}-p_1-p_2)\delta_{\sigma_{1'}\sigma_1}\delta_{\sigma_{2'}\sigma_2}\int d^3r\,V(r)e^{-i\mathbf{q}\cdot\mathbf{r}}. \tag{11.2.27}$$

Comparing this with Eq. (11.2.23) shows that in the non-relativistic limit the diagrams of Figure 11.3 yield the same S-matrix element as a potential $V(r)$ such that

$$\int d^3r\,V(r)e^{-i\mathbf{q}\cdot\mathbf{r}} = e_1e_2\frac{1+\pi(\mathbf{q}^2)}{\mathbf{q}^2}$$

or, inverting the Fourier transform,

$$V(r) = \frac{e_1e_2}{(2\pi)^3}\int d^3q\,e^{i\mathbf{q}\cdot\mathbf{r}}\left[\frac{1+\pi(\mathbf{q}^2)}{\mathbf{q}^2}\right]. \tag{11.2.28}$$

Equation (11.2.28) is to first order in the radiative correction the same potential energy that would be produced by the electrostatic interaction of two extended charge distributions $e_1\eta(\mathbf{x})$ and $e_2\eta(\mathbf{y})$ at a distance $\mathbf{r}$:

$$V(|\mathbf{r}|) = e_1e_2\int d^3x\int d^3y\,\frac{\eta(\mathbf{x})\eta(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}+\mathbf{r}|}, \tag{11.2.29}$$

where

$$\eta(\mathbf{r}) = \delta^3(\mathbf{r}) + \frac{1}{2(2\pi)^3}\int d^3q\,\pi(\mathbf{q}^2)e^{i\mathbf{q}\cdot\mathbf{r}}. \tag{11.2.30}$$

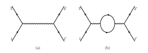


Figure 11.3. Two diagrams for the scattering of charged particles. Here lines carrying arrows are charged particles; wavy lines are photons. Diagram (b) represents the lowest-order vacuum polarization correction to the tree approximation graph (a).

Note that

$$\int d^3r\,\eta(\mathbf{r}) = 1 + \frac{1}{2}\pi(0) = 1, \tag{11.2.31}$$

so the total charges of particles 1 and 2, as determined from the long-range part of the Coulomb potential, are the same constants e_1 and e_2 that govern the interactions of the renormalized electromagnetic field.

For $|\mathbf{r}| \neq 0$ the integral (11.2.30) can be carried out by a straightforward contour integration:

$$\eta(\mathbf{r}) = -\frac{e^2}{8\pi^3r^3}\int_0^1 x(1-x)\,dx\left[1+\frac{mr}{\sqrt{x(1-x)}}\right]\exp\left(\frac{-mr}{\sqrt{x(1-x)}}\right).$$

This expression is negative everywhere. However, we have seen that the integral of $\eta(\mathbf{r})$ over all $\mathbf{r}$ equals +1. Therefore, $\eta(\mathbf{r})$ must contain a term $(1+L)\delta^3(\mathbf{r})$ that is singular at $\mathbf{r} = 0$, with L chosen to satisfy Eq. (11.2.31):

$$L = \frac{e^2}{8\pi^3}\int\frac{d^3r}{r^3}\int_0^1 x(1-x)\,dx\left[1+\frac{mr}{\sqrt{x(1-x)}}\right]\exp\left(\frac{-mr}{\sqrt{x(1-x)}}\right). \tag{11.2.32}$$

The complete expression for the charge distribution function is then

$$\eta(\mathbf{r}) = (1+L)\delta^3(\mathbf{r}) - \frac{e^2}{8\pi^3r^3}\int_0^1 x(1-x)\,dx\cdot\left[1+\frac{mr}{\sqrt{x(1-x)}}\right]\exp\left(\frac{-mr}{\sqrt{x(1-x)}}\right). \tag{11.2.33}$$

The physical interpretation of this result is that a bare point charge attracts particles of charge of opposite sign out of the vacuum, repelling their antiparticles to infinity, so that the bare charge is partially shielded, yielding a renormalized charge smaller by a factor $1/(1+L)$. As a check, we may note that if we cut off the divergent integral (11.2.32) by taking the integral to extend only over $r \geq a$, we find that the part that is divergent for $a \to 0$ is

$$L_\infty = \frac{e^2}{12\pi^2}\ln a^{-1}. \tag{11.2.34}$$

Hence if we identify the momentum space cutoff Λ with a^{-1}, the divergent part of L is related to the divergent part of $Z_3 - 1$ by

$$(Z_3-1)_\infty = -2L_\infty, \tag{11.2.35}$$

because to order e^2 the renormalized charge (10.4.18) is given by

$$e_r = Z_3^{1/2}e_{\text{Br}} \simeq \left(1+\frac{1}{2}(Z_3-1)\right)e_{\text{Br}} \simeq (1+L)^{-1}e_{\text{Br}}. \tag{11.2.36}$$

Equation (11.2.35) is confirmed below.

Vacuum polarization has a measurable effect on muonic atomic energy levels. As we shall see in Chapter 14, the effect of Feynman graph (b) in Figure 11.3 is to shift the energy of an atomic state with wave function $\psi(\mathbf{r})$ by

$$\Delta E = \int d^3r\,\Delta V(\mathbf{r})|\psi(\mathbf{r})|^2, \tag{11.2.37}$$

where $\Delta V(\mathbf{r})$ is the perturbation in the potential (11.2.28):

$$\Delta V(\mathbf{r}) = \frac{e_1e_2}{(2\pi)^3}\int d^3q\,e^{i\mathbf{q}\cdot\mathbf{r}}\left[\frac{\pi(\mathbf{q}^2)}{\mathbf{q}^2}\right]. \tag{11.2.38}$$

This perturbation falls off exponentially for $r \gg m^{-1}$. On the other hand, the wave function of electrons in ordinary atoms will generally be confined within a much larger radius $a \gg m^{-1}$; for instance, for hydrogenic orbits of electrons around a nucleus of charge Ze we have $a = 137/Zm$ (where here $m = m_e$). The energy shift will then depend only on the behavior of the wave function for $r \ll a$. For orbital angular momentum ℓ, the wave function behaves like r^ℓ for $r \ll a$, so Eq. (11.2.37) gives ΔE proportional to a factor $(ma)^{-2\ell-3}$. The effect of vacuum polarization is therefore very much larger for $\ell = 0$ than for higher orbital angular momenta. For $\ell = 0$ the wave function is approximately equal to the constant $\psi(0)$ for r less than or of the order of m^{-1}, so Eq. (11.2.37) becomes

$$\Delta E = |\psi(0)|^2\int d^3r\,\Delta V(\mathbf{r}). \tag{11.2.39}$$

Using Eqs. (11.2.38) and (11.2.22), the integral of the shift in the potential (for $e_1e_2 = -Ze^2$) is

$$\int d^3r\,\Delta V(\mathbf{r}) = -Ze^2\pi'(0) = -\frac{4Z\alpha^2}{15m^2}. \tag{11.2.40}$$

Also, in states of hydrogenic atoms with $\ell = 0$ and principal quantum number n the wave function at the origin is

$$\psi(0) = \frac{2}{\sqrt{4\pi}}\left(\frac{Z\alpha m}{n}\right)^{3/2}, \tag{11.2.41}$$

so the energy shift (11.2.39) is

$$\Delta E = -\frac{4Z^4\alpha^5m}{15\pi n^3}. \tag{11.2.42}$$

For instance, in the 2s state of hydrogen this energy shift is $-1.122\cdot 10^{-7}$ eV, corresponding to a frequency shift $\Delta E/2\pi\hbar$ of -37.13 MHz. This is sometimes called the Uehling effect [4]. As discussed in Chapter 1, such tiny energy shifts became measurable because in the absence of various radiative corrections the pure Dirac theory would predict exact degeneracy of the 2s and 2p states of hydrogen. As we shall see in Chapter 14, most of the +1058 MHz "Lamb shift" between the 2s and the 2p states comes from other radiative corrections, but the agreement between theory and experiment is good enough to verify the presence of the -37.13 MHz shift due to vacuum polarization.

Although vacuum polarization contributes only a small part of the radiative corrections in ordinary atoms, it dominates the radiative corrections in muonic atoms, in which a muon takes the place of the orbiting electron. This is because most radiative corrections give energy shifts in muonic atoms that on dimensional grounds are proportional to m_μ, while the integrated vacuum polarization energy $\int d^3r\,\Delta V$ due to an electron loop is still proportional to m_e^{-2} as in Eq. (11.2.40), giving an energy shift proportional to $m_\mu^3m_e^{-2} = (210)^2m_\mu$. However, in this case the muonic atomic radius is not much larger than the electron Compton wavelength, so the approximate result (11.2.39) only gives the order of magnitude of the energy shift due to vacuum polarization.

* * *

For the purposes of comparison with later calculations, note that if we had cut off the integral (11.2.7) at $\kappa = \Lambda$, then in place of Eq. (11.2.20) we would have encountered an integral of the form

$$(Z_3-1)_\infty = -\frac{e^2}{6\pi^2}\int_\mu^\Lambda\kappa^{d-5}\,d\kappa = \frac{e^2}{6\pi^2}\frac{\mu^{d-4}-\Lambda^{d-4}}{d-4},$$

where μ is an infrared effective cutoff of the order of the mass of the charged particle circulating in the loop of Figure 11.1. (The easiest way to find the constant factor here is to require that the limit of this expression for $d < 3$ and $\Lambda \to \infty$ matches Eq. (11.2.20).) With such an ultraviolet cutoff in place, we can pass to the limit $d \to 4$, and obtain

$$(Z_3-1)_\infty = -\frac{e^2}{6\pi^2}\ln(\Lambda/\mu). \tag{11.2.43}$$

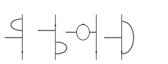


Figure 11.4. One-loop diagrams for the photon-lepton vertex function Γ^μ. Here wavy lines represent photons; other lines represent electrons or muons. Diagrams (a) and (b) are cancelled by lepton field renormalization terms; diagram (c) arises from the vacuum polarization calculated in Section 11.2; and (d) is the term calculated in Section 11.3.

11.3 Anomalous Magnetic Moments and Charge Radii

For our next example, we shall calculate the shift in the magnetic moment and the charge radius of an electron or muon due to lowest-order radiative corrections. The one-loop graphs and renormalization corrections for the photon–lepton vertex are shown in Figure 11.4. Of these graphs, those involving insertions in incoming or outgoing lepton lines vanish because the lepton is on the mass shell, as discussed in Section 10.3. The graph involving an insertion in the external photon line is the vacuum polarization effect, discussed in the previous section. This leaves one one-loop graph (the last in Figure 11.4) that needs to be calculated here:

$$\Gamma^\mu_{1\,\text{loop}}(p',p) = \int d^4k\left[e\gamma^\rho(2\pi)^4\right]\left[\frac{-i}{(2\pi)^4}\frac{-i(\not p'-\not k)+m}{(p'-k)^2+m^2-i\epsilon}\right][\gamma^\mu]\cdot\left[\frac{-i}{(2\pi)^4}\frac{-i(\not p-\not k)+m}{(p-k)^2+m^2-i\epsilon}\right]\left[e\gamma_\rho(2\pi)^4\right]\left[\frac{-i}{(2\pi)^4}\frac{1}{k^2-i\epsilon}\right], \tag{11.3.1}$$

where p' and p are the final and initial lepton four-momenta, respectively. (The contribution of the vertex connecting the external photon line and the internal lepton line is taken as γ^μ, because a factor $e(2\pi)^4$ was extracted in defining Γ^μ.)

This integral has an obvious ultraviolet divergence, roughly like $\int d^4k/(k^2)^2$. Unlike the case of the vacuum polarization, here we do not need a fancy regularization procedure like dimensional regularization to maintain the structure required by gauge invariance, because the photon is a neutral particle and so the integral may be rendered finite by suitable modifications of the photon propagator (for instance by including a factor $M^2/(k^2+M^2)$ with a large cutoff mass M), without having to introduce modifications elsewhere to maintain gauge invariance. In any case, as we shall see, the anomalous magnetic moment and charge radii can be calculated without encountering any ultraviolet divergences at all. In what follows we shall leave the integrals for the vertex function in their infinite form,

with it being understood that if necessary any divergent integrals can be expressed in terms of a cutoff mass M.

We start by combining denominators, using a repeated version of the Feynman trick described in the Appendix to this chapter

$$\frac{1}{ABC} = 2\int_0^1 dx\int_0^x dy\,[Ay + B(x-y) + C(1-x)]^{-3}. \tag{11.3.2}$$

Applied to the denominators in Eq. (11.3.1), this gives

$$\begin{aligned}&\frac{1}{(p'-k)^2+m^2-i\epsilon}\,\frac{1}{(p-k)^2+m^2-i\epsilon}\,\frac{1}{k^2-i\epsilon}\\ &= 2\int_0^1 dx\int_0^x dy\left[((p'-k)^2+m^2-i\epsilon)y + ((p-k)^2+m^2-i\epsilon)(x-y) + (k^2-i\epsilon)(1-x)\right]^{-3}\\ &= 2\int_0^1 dx\int_0^x dy\left[(k-p'y-p(x-y))^2 + m^2x^2 + q^2y(x-y) - i\epsilon\right]^{-3},\end{aligned} \tag{11.3.3}$$

where $q \equiv p - p'$ is the momentum transferred to the photon. Shifting the variable of integration

$$k \longrightarrow k + p'y + p(x-y)$$

the integral (11.3.1) becomes

$$\Gamma^\mu_{1\,\text{loop}}(p',p) = \frac{2ie^2}{(2\pi)^4}\int_0^1 dx\int_0^x dy\int\frac{d^4k}{[k^2+m^2x^2+q^2y(x-y)-i\epsilon]^3}\cdot\gamma^\rho\left[-i(\not p'(1-y) - \not k - \not p(x-y)) + m\right]\gamma^\mu\cdot\left[-i(\not p(1-x+y) - \not k - \not p'y) + m\right]\gamma_\rho. \tag{11.3.4}$$

Our next step is a Wick rotation. As explained in the previous section, the $-i\epsilon$ in the denominator dictates that when we rotate the k^0 contour of integration to the imaginary axis we must rotate counterclockwise, so that the integral over k^0 from $-\infty$ to $+\infty$ is replaced with an integral over imaginary values from $-i\infty$ to $+i\infty$, or equivalently over real values of $k^4 \equiv -ik^0$ from $-\infty$ to $+\infty$. We also exploit the rotational symmetry of the denominator in Eq. (11.3.4); we drop terms in the denominator of odd order in k, replace $k^\mu k^\nu$ with $\eta^{\mu\nu}\kappa^2/4$, and replace the volume element $d^4k = i\,dk^1dk^2dk^3dk^4$ with $2i\pi^2\kappa^3d\kappa$, where κ is the Euclidean length of the four-vector k. Putting this all together, Eq. (11.3.4) now becomes

$$\Gamma^\mu_{1\,\text{loop}}(p',p) = \frac{-4\pi^2e^2}{(2\pi)^4}\int_0^1 dx\int_0^x dy\int_0^\infty\kappa^3d\kappa\left\{-\frac{\kappa^2}{4}\gamma^\rho\gamma^\sigma\gamma^\mu\gamma_\sigma\gamma_\rho + \gamma^\rho\left[-i(\not p'(1-y) - \not p(x-y)) + m\right]\gamma^\mu\left[-i(\not p(1-x+y) - \not p'y) + m\right]\gamma_\rho\right\}\cdot[\kappa^2+m^2x^2+q^2y(x-y)]^{-3}. \tag{11.3.5}$$

We are interested here only in the matrix element $\bar u'\Gamma^\mu u$ of the vertex function between Dirac spinors that satisfy the relations

$$\bar u'[i\not p' + m] = 0, \qquad [i\not p + m]u = 0.$$

We can therefore simplify this expression by using the anticommutation relations of the Dirac matrices to move all factors $\not p'$ to the right and all factors $\not p$ to the left, replacing them when they arrive on the right or left with im. After a straightforward but tedious calculation, Eq. (11.3.5) then becomes

$$\bar u'\Gamma^\mu_{\text{one loop}}(p',p)u = \frac{-4\pi^2e^2}{(2\pi)^4}\int_0^1 dx\int_0^x dy\int_0^\infty\kappa^3d\kappa\,\bar u'\Big\{\gamma^\mu\left[-\kappa^2+2m^2(x^2-4x+2) + 2q^2(y(x-y)+1-x)\right] + 4imp'^\mu(y-x+xy) + 4imp^\mu(x^2-xy-y)\Big\}u\cdot[\kappa^2+m^2x^2+q^2y(x-y)]^{-3}. \tag{11.3.6}$$

We next exploit the symmetry of the final factor under the reflection $y \longrightarrow x - y$. Under this reflection, the functions $y - x + xy$ and $x^2 - xy - y$ that multiply p'^μ and p^μ are interchanged, so both may be replaced with their average:

$$\tfrac{1}{2}(y-x+xy) + \tfrac{1}{2}(x^2-xy-y) = -\tfrac{1}{2}x(1-x).$$

This gives finally

$$\bar u'\Gamma^\mu_{\text{one loop}}(p',p)u = \frac{-4\pi^2e^2}{(2\pi)^4}\int_0^1 dx\int_0^x dy\int_0^\infty\kappa^3d\kappa\cdot\bar u'\Big\{\gamma^\mu\left[-\kappa^2+2m^2(x^2-4x+2)+2q^2(y(x-y)+1-x)\right] - 2im(p'^\mu+p^\mu)x(1-x)\Big\}u[\kappa^2+m^2x^2+q^2y(x-y)]^{-3}. \tag{11.3.7}$$

Note that p^μ and p'^μ now enter only in the combination $p^\mu + p'^\mu$, as required by current conservation.

There are other diagrams that need to be taken into account. Of course, there is the zeroth-order term γ^μ in Γ^μ. The term proportional to $Z_2 - 1$ in the correction term (11.1.9) yields a term in Γ^μ

$$\Gamma^\mu_{\mathcal{L}_2} = (Z_2-1)\gamma^\mu. \tag{11.3.8}$$

Also, the effect of insertions of corrections to the external photon propagator is a term:

$$\Gamma^\mu_{\text{vac pol}}(p',p) = \frac{1}{(p'-p)^2-i\epsilon}\Pi^{\mu\nu}(p'-p)\gamma_\nu. \tag{11.3.9}$$

The form of each of these terms is in agreement with the general result (10.6.10) (with $H(q^2) = 0$)

$$\bar u'\Gamma^\mu(p',p)u = \bar u'\left[\gamma^\mu F(q^2) - \frac{i}{2m}(p+p')^\mu G(q^2)\right]u. \tag{11.3.10}$$

To order e^2, the form factors are

$$F(q^2) = Z_2 + \pi(q^2) + \frac{4\pi^2e^2}{(2\pi)^4}\int_0^1 dx\int_0^x dy\int_0^\infty\kappa^3d\kappa\,\frac{\kappa^2 - 2m^2(x^2-4x+2) - 2q^2(y(x-y)+1-x)}{[\kappa^2+m^2x^2+q^2y(x-y)]^3}, \tag{11.3.11}$$

$$G(q^2) = \frac{-4\pi^2e^2}{(2\pi)^4}\int_0^1 dx\int_0^x dy\int_0^\infty\frac{4m^2x(1-x)\kappa^3d\kappa}{[\kappa^2+m^2x^2+q^2y(x-y)]^3}, \tag{11.3.12}$$

where $\pi(q^2)$ is the vacuum polarization function (11.2.22).

The integral for the form factor $G(q^2)$ is finite as it stands:

$$G(q^2) = -\frac{e^2m^2}{4\pi^2}\int_0^1 dx\int_0^x dy\,\frac{x(1-x)}{m^2x^2+q^2y(x-y)}. \tag{11.3.13}$$

This makes it easy to calculate the anomalous magnetic moment. As noted in Section 10.6, it is only the γ^μ term that contributes to the magnetic moment, so the effect of radiative corrections is to multiply the Dirac value $e/2m$ of the magnetic moment by a factor $F(0)$. But the definition of e as the true lepton charge requires that

$$F(0) + G(0) = 1, \tag{11.3.14}$$

so the magnetic moment may be expressed as

$$\mu = \frac{e}{2m}(1 - G(0)). \tag{11.3.15}$$

From Eq. (11.3.13), we find

$$-G(0) = \frac{e^2}{8\pi^2} = 0.001161. \tag{11.3.16}$$

This is the famous $\alpha/2\pi$ correction first calculated by Schwinger [5].

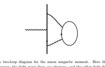


Figure 11.5. A two-loop diagram for the muon magnetic moment. Here the heavy straight line represents a muon; the light wavy lines are photons; and the other light lines are electrons. This diagram makes a relatively large contribution to the fourth-order muon gyromagnetic ratio, proportional to $\ln(m_\mu/m_e)$.

Of course, this is only the first term in the radiative corrections to the magnetic moment. Even in just the next order, fourth order in e, there are so many terms that the calculations become quite complicated. However, because of the large muon–electron mass ratio, there is one fourth-order term in the magnetic moment of the muon that is somewhat larger than any of the others. It arises from the insertion of an electron loop in the virtual photon line of the second-order diagram, as shown in Figure 11.5. The effect of this electron loop is to change the photon propagator $1/k^2$ in Eq. (11.3.1) to $(1+\pi_e(k^2))/k^2$, where $\pi_e(k^2)$ is given by Eq. (11.2.22), but with the mass m taken as the electron mass:

$$\pi_e(k^2) = \frac{e^2}{2\pi^2}\int_0^1 x(1-x)\ln\left(1+\frac{k^2x(1-x)}{m_e^2}\right)dx.$$

Inspection of Eq. (11.3.12) shows that in calculating the muon magnetic moment the effective cutoff on the virtual photon momentum k is m_μ. The ratio m_μ/m_e is so large that for k^2 of order m_μ^2 we may approximate

$$\pi_e(k^2) \simeq \frac{e^2}{2\pi^2}\int_0^1 x(1-x)\ln(m_\mu^2/m_e^2) = \frac{e^2}{12\pi^2}\ln(m_\mu^2/m_e^2), \tag{11.3.17}$$

with the neglected terms having coefficients of order unity in place of $\ln(m_\mu^2/m_e^2)$. Since this is a constant, the change in $-G(0)$ produced by adding an electron loop in the virtual photon line is simply given by multiplying our previous result (11.3.16) for $-G(0)$ by Eq. (11.3.17), so that now

$$\mu_\mu = \frac{e}{2m_\mu}\left(\frac{e^2}{8\pi^2} + \frac{e^4}{96\pi^4}\left[\ln\frac{m_\mu^2}{m_e^2} + O(1)\right]\right). \tag{11.3.18}$$

(As we shall see in Volume II, this argument is a primitive version of the method of the renormalization group.) The result (11.3.18) may be compared with the full fourth-order result [6]

$$\mu_\mu = \frac{e}{2m_\mu}\left\{\frac{e^2}{8\pi^2} + \frac{e^4}{96\pi^4}\left[\ln\frac{m_\mu^2}{m_e^2} - \frac{25}{6} + \frac{197}{24} + \frac{\pi^2}{2} + \frac{9\zeta(3)}{2} - 3\pi^2\ln 2 + O\left(\left(\frac{m_e}{m_\mu}\right)^2\right)\right]\right\}. \tag{11.3.19}$$

It turns out that the $O(1)$ terms multiplying $e^4/96\pi^4$ add up to -6.137, which is not very much smaller than $\ln(m_\mu^2/m_e^2) = 10.663$, so the approximation (11.3.18) gives the fourth-order terms only to a factor of order 2. The correct fourth-order result (11.3.19) gives $\mu_\mu = 1.00116546e/2m_\mu$, in comparison with the second-order result $\mu_\mu = 1.001161e/2m_\mu$ and the current experimental value [7], $\mu_\mu = 1.001165923(8)e/2m_\mu$.

Now let us turn to the other form factor. The integral in Eq. (11.3.11) for $F(q^2)$ has an ultraviolet divergence. However, in order to satisfy the charge-non-renormalization condition (11.3.14), it is necessary that Z_2 take the value

$$Z_2 = 1 + \frac{e^2}{8\pi^2} - \frac{4\pi^2e^2}{(2\pi)^4}\int_0^1 dx\int_0^x dy\int_0^\infty\kappa^3d\kappa\,\frac{\kappa^2 - 2m^2(x^2-4x+2)}{[\kappa^2+m^2x^2]^3}. \tag{11.3.20}$$

(Recall that $\pi(0) = 0$.) This is itself ultraviolet divergent, with an infinite part

$$(Z_2-1)_\infty = -\frac{e^2}{8\pi^2}\int^\infty\frac{d\kappa}{\kappa}. \tag{11.3.21}$$

Inserting Eq. (11.3.20) back into Eq. (11.3.11) gives

$$F(q^2) = 1 + \frac{e^2}{8\pi^2} + \pi(q^2) + \frac{4\pi^2e^2}{(2\pi)^4}\int_0^1 dx\int_0^x dy\int_0^\infty\kappa^3d\kappa\cdot\left\{\frac{\kappa^2 - 2m^2(x^2-4x+2) - 2q^2[y(x-y)+1-x]}{[\kappa^2+m^2x^2+q^2y(x-y)]^3} - \frac{\kappa^2 - 2m^2(x^2-4x+2)}{[\kappa^2+m^2x^2]^3}\right\}. \tag{11.3.22}$$

The integral over κ is now convergent:

$$F(q^2) = 1 + \frac{e^2}{8\pi^2} + \pi(q^2) + \frac{2\pi^2e^2}{(2\pi)^4}\int_0^1 dx\int_0^x dy\left\{\frac{-m^2[x^2-4x+2] - q^2[y(x-y)+1-x]}{m^2x^2+q^2y(x-y)} + \frac{x^2-4x+2}{x^2} - \ln\left[\frac{m^2x^2+q^2y(x-y)}{m^2x^2}\right]\right\}. \tag{11.3.23}$$

However, we see that the integral over x and y now diverges logarithmically at $x = 0$ and $y = 0$, because there are two powers of x and/or y in the denominators, and just two differentials $dx\,dy$ in the numerator. This divergence can be traced to the vanishing of the denominator $[\kappa^2+m^2x^2+q^2y(x-y)]^3$ in Eq. (11.3.11) at $x = 0$, $y = 0$, and $\kappa = 0$. Because this infinity comes from the region of small rather than large κ, it is termed an *infrared divergence* rather than an ultraviolet divergence.

We shall give a comprehensive treatment of the infrared divergences in Chapter 13. It will be shown there that infrared divergences in the cross-section for processes like electron–electron scattering, such as those that are introduced by the infrared divergence in the electron form factor $F(q^2)$, are cancelled when we include the emission of low-energy photons as well as elastic scattering. Also, as we shall see in Chapter 14, when we calculate radiative corrections to atomic energy levels the infrared divergence in $F(q^2)$ is cut off because the bound electron is not exactly on the free-particle mass shell. For the present we shall continue our calculation by simply introducing a fictitious photon mass μ to cut off the infrared divergence in $F(q^2)$, leaving it for Chapter 14 to see how to use this result.

With a photon mass μ, the denominator $k^2 - i\epsilon$ in Eq. (11.3.1) would be replaced with $k^2 + \mu^2 - i\epsilon$. The effect would than to be add a term $\mu^2(1-x)$ to the cubed quantity in the denominators of Eqs. (11.3.3)–(11.3.7), (11.3.11), (11.3.20), and (11.3.22). Eq. (11.3.23) then is replaced with

$$F(q^2) = 1 + \frac{e^2}{8\pi^2} + \pi(q^2) + \frac{2\pi^2e^2}{(2\pi)^4}\int_0^1 dx\int_0^x dy\left\{\frac{-m^2[x^2-4x+2] - q^2[y(x-y)+1-x]}{m^2x^2+q^2y(x-y)+\mu^2(1-x)} + \frac{m^2[x^2-4x+2]}{m^2x^2+\mu^2(1-x)} - \ln\left[\frac{m^2x^2+q^2y(x-y)+\mu^2(1-x)}{m^2x^2+\mu^2(1-x)}\right]\right\}. \tag{11.3.24}$$

This integral is now completely convergent. It can be expressed in terms of Spence functions, but the result is not particularly illuminating. For our purposes in Chapter 14, it will be sufficient to calculate the behavior of $F(q^2)$ for small q^2. We already know from the Ward identity that $F(0) = 1 - G(0) = 1 + e^2/8\pi^2$, so let us consider the first derivative $F'(q^2)$ at $q^2 = 0$. According to Eq. (11.3.24), this is

$$F'(0) = \pi'(0) + \frac{2\pi^2e^2}{(2\pi)^4}\int_0^1 dx\int_0^x dy\left\{\frac{2y(x-y)+1-x}{m^2x^2+\mu^2(1-x)} + \frac{m^2[x^2-4x+2]y(x-y)}{[m^2x^2+\mu^2(1-x)]^2}\right\}. \tag{11.3.25}$$

The vacuum polarization contribution is given by Eq. (11.2.22) as

$$\pi'(0) = \frac{e^2}{60\pi^2m^2}. \tag{11.3.26}$$

Dropping all terms proportional to powers of μ/m in Eq. (11.3.25), we then have[124]

$$F'(0) = \frac{e^2}{24\pi^2m^2}\left[\ln\left(\frac{\mu^2}{m^2}\right) + \frac{2}{5} + \frac{1}{4}\right] \tag{11.3.27}$$

with the term 2/5 the contribution of vacuum polarization. On the other hand, Eq. (11.3.13) shows that $G(q^2)$ has a finite derivative at $q^2 = 0$,

$$G'(0) = \frac{e^2}{48\pi^2m^2}. \tag{11.3.28}$$

These results are most conveniently expressed in terms of the charge form factor $F_1(q^2)$, defined by the alternative representation (10.6.15) of the vertex function

$$\bar u(p',\sigma')\Gamma^\mu(p',p)u(p,\sigma) = \bar u(p',\sigma')\left[\gamma^\mu F_1(q^2) + \tfrac{1}{2}i[\gamma^\mu,\gamma^\nu](p'-p)_\nu F_2(q^2)\right]u(p,\sigma). \tag{11.3.29}$$

According to Eqs. (10.6.17) and (10.6.18),

$$F_1(q^2) = F(q^2) + G(q^2). \tag{11.3.30}$$

For $|q^2| \ll m^2$, this form factor is approximately

$$F_1(q^2) \simeq 1 + \frac{e^2}{24\pi^2}\frac{q^2}{m^2}\left[\ln\left(\frac{\mu^2}{m^2}\right) + \frac{2}{5} + \frac{3}{4}\right]. \tag{11.3.31}$$

This may be expressed in terms of a *charge radius* a, defined by the limiting behavior of the charge form factor for $q^2 \longrightarrow 0$:

$$F_1(q^2) \longrightarrow 1 - q^2a^2/6. \tag{11.3.32}$$

[124] The y-integral is trivial. The x-integral is most easily calculated in the limit $\mu \ll m$ by dividing the range of integration into two parts, one from 0 to s, where $\mu/m \ll s \ll 1$, and the second from s to 1.

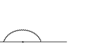


Figure 11.6. The one-loop diagram for the electron self-energy function. As usual, the straight line represents an electron; the wavy line is a photon.

(This definition is motivated by the fact that the average of $\exp(i\mathbf{q}\cdot\mathbf{x})$ over a spherical shell of radius a goes as $1 - \mathbf{q}^2a^2/6$ for $\mathbf{q}^2a^2 \ll 1$.) We see that the charge radius of the electron is given by

$$a^2 = -\frac{e^2}{4\pi^2m^2}\left[\ln\left(\frac{\mu^2}{m^2}\right) + \frac{2}{5} + \frac{3}{4}\right]. \tag{11.3.33}$$

We will see in Chapter 14 that for electrons in atoms the role of the photon mass is played by an effective infrared cutoff that is much less than m, so the logarithm here is large and negative, yielding a positive value for a^2.

11.4 Electron Self-Energy

We conclude this chapter with a calculation of the electron self-energy function. This by itself does not have any direct experimental implications, but some of the results here will be useful in Chapter 14 and Volume II.

As in Section 10.3, we define $i(2\pi)^4[\Sigma^*(p)]_{\beta\alpha}$ as the sum of all graphs with one incoming and one outgoing electron line carrying momenta p and Dirac indices α and β respectively, with the asterisk indicating that we exclude diagrams that can be disconnected by cutting through some internal electron line, and with propagators omitted for the two external lines. The complete electron propagator is then given by the sum

$$[-i(2\pi)^{-4}S'(p)] = [-i(2\pi)^{-4}S(p)] + [-i(2\pi)^{-4}S(p)][i(2\pi)^4\Sigma^*(p)][-i(2\pi)^{-4}S(p)] + \cdots, \tag{11.4.1}$$

where

$$S(p) = \frac{-i\not p + m_e}{p^2+m_e^2-i\epsilon}. \tag{11.4.2}$$

The sum is trivial, and gives

$$S'(p) = [i\not p + m_e - \Sigma^*(p) - i\epsilon]^{-1}. \tag{11.4.3}$$

In lowest order there is a one-loop contribution to Σ^*, given by Figure 11.6:

$$i(2\pi)^4\Sigma^*_{1\,\text{loop}}(p) = \int d^4k\left[\frac{-i}{(2\pi)^4}\frac{\eta_{\rho\sigma}}{k^2-i\epsilon}\right]\cdot[(2\pi)^4e\gamma^\rho]\cdot\left[\frac{-i}{(2\pi)^4}\frac{-i\not p + i\not k + m_e}{(p-k)^2+m_e^2-i\epsilon}\right][(2\pi)^4e\gamma^\sigma]$$

or more simply

$$\Sigma^*_{1\,\text{loop}}(p) = \frac{ie^2}{(2\pi)^4}\int d^4k\left[\frac{1}{k^2-i\epsilon}\right]\left[\frac{\gamma^\rho(-i\not p + i\not k + m_e)\gamma_\rho}{(p-k)^2+m_e^2-i\epsilon}\right]. \tag{11.4.4}$$

(This is in Feynman gauge; amplitudes with charged particles off the mass shell are not gauge-invariant.) For use in our calculation of the Lamb shift, it will be convenient to use a method of regularization introduced by Pauli and Villars [8]. We replace the photon propagator $(k^2 - i\epsilon)^{-1}$ with

$$\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon}$$

so that the electron self-energy function is

$$\Sigma^*_{1\ \mathrm{loop}}(p) = \frac{ie^2}{(2\pi)^4}\int d^4k \left[\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2+\mu^2-i\epsilon}\right]\cdot\left[\frac{\gamma^\rho(-i\not p + i\not k + m_e)\gamma_\rho}{(p-k)^2 + m_e^2 - i\epsilon}\right]. \quad (11.4.5)$$

Later we can drop the regulator by letting the regulator mass μ go to infinity. In Chapter 14 we will also be interested in the case where $\mu \ll m_e$.

We again use the Feynman trick to combine denominators, and recall that $\gamma^\rho\gamma^\kappa\gamma_\rho = -2\gamma^\kappa$ and $\gamma^\rho\gamma_\rho = 4$. This gives

$$\Sigma^*_{1\ \mathrm{loop}}(p) = \frac{ie^2}{(2\pi)^4}\int d^4k [2i(\not p - \not k) + 4m_e] \cdot \int_0^1 dx \left[\frac{1}{\{(k-px)^2 + p^2x(1-x) + m_e^2 x - i\epsilon\}^2} - \frac{1}{\{(k-px)^2 + p^2x(1-x) + m_e^2 x + \mu^2(1-x) - i\epsilon\}^2}\right]. \quad (11.4.6)$$

Shifting the variable of integration $k \to k + px$ and rotating the contour of integration gives

$$\Sigma^*_{1\ \mathrm{loop}}(p) = \frac{-2\pi^2 e^2}{(2\pi)^4}\int_0^1 dx\, [2i(1-x)\not p + 4m_e]\int_0^\infty d\kappa\,\kappa^3 \cdot\left[\frac{1}{\{\kappa^2 + p^2x(1-x) + m_e^2 x\}^2} - \frac{1}{\{\kappa^2 + p^2x(1-x) + m_e^2x + \mu^2(1-x)\}^2}\right]. \quad (11.4.7)$$

The κ-integral is trivial:

$$\Sigma^*_{1\ \mathrm{loop}}(p) = \frac{-\pi^2e^2}{(2\pi)^4}\int_0^1 dx\, [2i(1-x)\not p + 4m_e]\ln\left(\frac{p^2x(1-x) + m_e^2x + \mu^2(1-x)}{p^2x(1-x) + m_e^2 x}\right). \quad (11.4.8)$$

The interaction (11.1.9) also contributes a renormalization counterterm $-(Z_2-1)(i\not p + m_e) + Z_2\delta m_e$ in $\Sigma^*(p)$, with Z_2 and δm_e determined by the condition that the complete propagator $S'(p)$ regarded as a function of $i\not p$ should have a pole at $i\not p = -m_e$ with residue unity. (As we shall see in the next chapter, this makes Σ^* finite as $\mu \to \infty$ to all orders in e.) In lowest order, this gives

$$\delta m_e = -\Sigma^*_{1\ \mathrm{loop}}\big|_{i\not p = -m_e} = \frac{2m_e\pi^2e^2}{(2\pi)^4}\int_0^1 dx\,[1+x]\ln\left(\frac{m_e^2x^2 + \mu^2(1-x)}{m_e^2x^2}\right), \quad (11.4.9)$$

$$Z_2 - 1 = -i\frac{\partial\Sigma^*_{1\ \mathrm{loop}}}{\partial \not p}\bigg|_{i\not p = -m_e} = \frac{-2\pi^2e^2}{(2\pi)^4}\int_0^1 dx\left\{(1-x)\ln\left(\frac{m_e^2x^2 + \mu^2(1-x)}{m_e^2x^2}\right) - \frac{2\mu^2(1-x)^2(1+x)}{x[m_e^2x^2 + \mu^2(1-x)]}\right\}. \quad (11.4.10)$$

(To this order, we do not distinguish between δm_e and $Z_2\delta m_e$.) Dropping terms that vanish for $\mu^2 \to \infty$, Eqs. (11.4.8)–(11.4.10) yield

$$\Sigma^*_{1\ \mathrm{loop}}(p) = \frac{-\pi^2e^2}{(2\pi)^4}\int_0^1 dx\,[2i(1-x)\not p + 4m_e]\ln\left(\frac{\mu^2(1-x)}{p^2x(1-x) + m_e^2 x}\right), \quad (11.4.11)$$

$$\delta m_e = \frac{2m_e\pi^2e^2}{(2\pi)^4}\int_0^1 dx\,[1+x]\ln\left(\frac{\mu^2(1-x)}{m_e^2x^2}\right), \quad (11.4.12)$$

$$Z_2 - 1 = \frac{-2\pi^2e^2}{(2\pi)^4}\int_0^1 dx\left\{(1-x)\ln\left(\frac{\mu^2(1-x)}{m_e^2x^2}\right) - \frac{2(1-x^2)}{x}\right\}. \quad (11.4.13)$$

Inspection then shows that in the complete self-energy function the $\ln\mu^2$ terms cancel, leaving us with

$$\Sigma^*_{\mathrm{order}\ e^2}(p) = \Sigma^*_{1\ \mathrm{loop}}(p) - (Z_2-1)(i\not p + m_e) + Z_2\delta m_e = -\frac{2\pi^2e^2}{(2\pi)^4}\int_0^1 dx\left\{[i(1-x)\not p + 2m_e]\ln\left(\frac{m_e^2(1-x)}{p^2x(1-x) + m_e^2x}\right) - m_e[1+x]\ln\left(\frac{1-x}{x^2}\right) - (i\not p + m_e)\left[(1-x)\ln\left(\frac{1-x}{x^2}\right) - \frac{2(1-x^2)}{x}\right]\right\}. \quad (11.4.14)$$

There is still a divergence from the behavior of the last term as $x \to 0$, which can be traced to the singular behavior of the integral over the photon momentum k in Eq. (11.4.5) at $k^2 = 0$, when we take p^2 at the point $p^2 = -m_e^2$ where we evaluated $Z_2 - 1$. Such infrared divergences will be discussed in detail in Chapter 13. For the present, the point that concerns us is that the ultraviolet divergence has cancelled.

* * *

The result (11.4.9) for δm_e is of some interest in itself. Note that $\delta m_e/m_e > 0$, as we would expect for the electromagnetic self-energy due to the interaction of a charge with its own field. But unlike the classical estimates of electromagnetic self-energy by Poincaré, Abraham, and others [9], Eq. (11.4.9) is only logarithmically divergent in the limit $\mu \to \infty$ where the cutoff is removed. In this limit:

$$\delta m_e \to \frac{6m_e\pi^2e^2}{(2\pi)^4}\ln\left(\frac{\mu}{m_e}\right). \quad (11.4.15)$$

In our calculation of the Lamb shift in Section 14.3 we will be interested in the opposite limit, $\mu \ll m_e$. Here Eq. (11.4.9) gives

$$\delta m_e \to \frac{e^2\mu}{8\pi}\left[1 - \frac{\mu}{2\pi m_e} + \cdots\right]. \quad (11.4.16)$$

References

1. R. P. Feynman, *Phys. Rev.* **76**, 769 (1949).
2. G. C. Wick, *Phys. Rev.* **80**, 268 (1950).
3. G. 't Hooft and M. Veltman, *Nucl. Phys.* **B44**, 189 (1972).
4. E. A. Uehling, *Phys. Rev.* **48**, 55 (1935). The one-loop function $\pi(q^2)$ was first given for $q^2 \neq 0$ by J. Schwinger, *Phys. Rev.* **75**, 651 (1949).
5. J. Schwinger, *Phys. Rev.* **73**, 416 (1948).
6. This is calculated (including terms that vanish for $m_e \ll m_\mu$) by H. Suura and E. Wichmann, *Phys. Rev.* **105**, 1930 (1967); A. Petermann, *Phys. Rev.* **105**, 1931 (1957); H. H. Elend, *Phys. Lett.* **20**, 682 (1966); **21**, 720 (1966); G. W. Erickson and H. H. T. Liu, UCD-CNL-81 report (1968).
7. J. Bailey *et al.* (CERN–Mainz–Daresbury Collaboration), *Nucl. Phys.* **B150**, 1 (1979). These experiments are done by observing the precession of the muon spin in a storage ring.
8. W. Pauli and F. Villars, *Rev. Mod. Phys.* **21**, 434 (1949). Also see J. Rayski, *Phys. Rev.* **75**, 1961 (1949).
9. See, e.g., A. I. Miller, *Theory of Relativity — Emergence (1905) and Early Interpretation (1905–1911)* (Addison-Wesley, Reading, MA, 1981): Chapter 1.

the electron that provides the most useful limit on M. The calculated value of the magnetic moment of the muon agrees with experiment to within terms of order $10^{-8}e/2m_e$, so M must be greater than about $\sqrt{8\cdot 10^8}\,m_\mu = 3\cdot 10^3$ GeV. In any case, if M is anywhere near as large as 10^{18} GeV, then we are certainly justified in neglecting any non-renormalizable interactions that might appear in quantum electrodynamics.

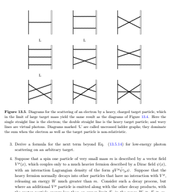
These considerations help us to cope with some of the puzzles associated with higher-derivative terms in the Lagrangian. For instance, in the general theory of a real scalar field ϕ, we would expect to find terms in the Lagrangian density of the form $\phi\Box^n\phi$. Any one such term would make a direct contribution to the scalar self-energy function $\Pi^*(q^2)$ proportional to $(q^2)^n$. If we were to include this contribution to all orders, but ignore all other effects of non-renormalizable interactions, then the propagator $\Delta'(q^2) = 1/(q^2+m^2-\Pi^*(q^2))$ would not have the simple pole in q^2 at negative q^2 expected from the general arguments of Section 10.7, but n such poles (some of which may coincide), generally at complex values of q^2. But if the non-renormalizable term $\phi\Box^n\phi$ has a coefficient of order $M^{-2(n-1)}$, where $M \gg m$, then the extra poles are at q^2 of order M^2, where it is illegitimate to ignore the infinite number of other non-renormalizable interactions that must also appear in the Lagrangian. Thus the appearance of higher-derivative terms in a general non-renormalizable Lagrangian is not in conflict with the general principles underlying quantum field theory that were used in Section 10.7. But by the same token, we also cannot use higher-derivative terms to avoid ultraviolet divergences altogether, as has been repeatedly proposed. A term $M^{-2(n-1)}\phi\Box^n\phi$ in the Lagrangian density provides a cutoff at momenta $q^2 \approx M^2$, but at these momenta we cannot ignore all the other non-renormalizable interactions that must be present.

Although highly suppressed, non-renormalizable interactions may be detectable if they have effects that would otherwise be forbidden. For instance, we will see in Section 12.5 that the symmetries of charge-conjugation and space-inversion invariance are an automatic consequence of the structure of the electromagnetic interactions that is imposed by gauge invariance, Lorentz invariance, and renormalizability, but we can easily imagine non-renormalizable terms that would violate these symmetries, such as an electron electric dipole moment term $\bar\psi\gamma_5[\gamma_\mu,\gamma_\nu]\psi F^{\mu\nu}$, or the Fermi interaction $\bar\psi\gamma_5\gamma_\mu\psi\bar\psi\gamma^\mu\psi$. It is widely believed today that the conservation of baryon and lepton number is violated by very small effects of highly suppressed non-renormalizable interactions. Another example of a detectable non-renormalizable interaction is provided by gravitation. As mentioned before, gravitons have no renormalizable interactions at all. But, of course, we detect gravitation, because it has the special property that the gravitational fields of all the particles in a macroscopic body add up coherently.

Although non-renormalizable theories involve an infinite number of free parameters, they retain considerable predictive power: [12] they allow us to calculate the non-analytic parts of Feynman amplitudes, like the $\ln q$ and $q\ln q$ terms in the one-dimensional examples at the beginning of the previous section. Such calculations just reproduce the results required by the axiom of S-matrix theory, that the S-matrix has only those singularities required by unitarity.

Paradoxically, it is just in the case where symmetry principles forbid renormalizable interactions that non-renormalizable quantum field theories prove the most useful. In such

[135] The first factor here is the usual factor of i accompanying the constants in the interaction Lagrangian of the heavy charged particle in the Feynman rules.



14 Bound States in External Fields

In our calculations of radiative corrections in Chapter 11 we went just one step beyond the lowest order in perturbation theory. However, there is a very important class of problems where even the simplest calculation requires that from the beginning we consider classes of Feynman diagrams of arbitrarily high order in coupling constants like e. These problems are those involving bound states—in electrodynamics, either ordinary atoms and molecules, or such exotic atoms as positronium or muonium.

It is easy to see that such problems necessarily involve a breakdown of ordinary perturbation theory. Consider for instance the amplitude for electron–proton scattering as a function of the center-of-mass energy E. As shown in Section 10.3, the existence of a bound state like the ground state of hydrogen implies the existence of a pole in this amplitude at $E = m_p + m_e - 13.6$ eV. However, no single term in the perturbation series for electron–proton scattering has such a pole. The pole therefore can only arise from a divergence of the sum over all diagrams at center-of-mass energies near $m_p + m_e$.

The reason for this divergence of the perturbation series is also easy to see, especially if for the moment we consider the time-ordered diagrams of old-fashioned perturbation theory instead of Feynman diagrams. Suppose that in the center-of-mass system the electron and proton both have momenta of magnitude $q \ll m_e$, and consider an intermediate state in which the electron and proton momenta are different but also of order q. The energy denominator factor contributed by this state will be of order $[q^2/m_e]^{-1}$. Each such state will also contribute a matrix element of the Coulomb interaction of order e^2/q^2 (the Fourier transform of e^2/r), and the corresponding momentum space integration will contribute a factor of order q^3. Putting this together, we see that each additional Coulomb interaction contributes an overall factor of order

$$[q^2/m_e]^{-1}[e^2/q^2][q^3] = e^2 m_e/q .$$

Thus the perturbation theory should break down when q is less than or of the order of $e^2 m_e$, or in other words when the kinetic and potential energies, which are of order q^2/m_e, are no larger than about $e^4 m_e$, which, of course, is of the order of the binding energy of hydrogen.

Our problem here is to learn how to use perturbation theory to evaluate radiative corrections in bound-state problems, summing to all orders those diagrams that need to be summed to all orders, and keeping only a finite number of those that don't.

14.1 The Dirac Equation

We shall limit ourselves in this chapter to problems in which bound states arise because of the Coulomb interaction of electrons (or muons) with heavy charged particles such as atomic nuclei. As shown in Section 13.6, this interaction may be taken into account by adding to

potential $\mathcal{A}^\mu(x)$:

$$\mathcal{L}_A = -ie\bar{\Psi}\gamma^\mu\Psi\mathcal{A}_\mu - \frac{1}{4}(Z_3-1)(\partial^\mu\mathcal{A}^\nu - \partial^\nu\mathcal{A}^\mu)(\partial_\mu A_\nu - \partial_\nu A_\mu) - ie(Z_2-1)\mathcal{A}_\mu\bar{\Psi}\gamma^\mu\Psi, \quad (14.1.1)$$

which is obtained by replacing the quantum vector potential A^μ with $A^\mu + \mathcal{A}^\mu$ in the interaction part of Eq. (11.1.6). For instance, for a single heavy particle of charge Ze at the origin,

$$\mathcal{A}^0(x) = \frac{Ze}{4\pi|\mathbf{x}|}, \qquad \mathbf{A}(x) = 0. \quad (14.1.2)$$

It is the interaction (14.1.1) that must be taken into account to all orders. In this section we will consider the theory with only this interaction, leaving radiative corrections to subsequent sections.

Physicists learn in kindergarten to approach this sort of problem by solving the wave equation of Dirac in the presence of the external field. It might seem unnecessary to derive this equation here, but as emphasized in Chapter 1, Dirac's original motivation for this equation as a sort of relativistic Schrödinger equation does not stand up to inspection. Also, in the course of our derivation we will discover the normalization conditions that have to be imposed on the solutions of the Dirac equation, which seemed somewhat *ad hoc* in Dirac's approach. The solutions of the Dirac equation discussed here will be important ingredients in our treatment of radiative corrections in the next section.

We will work here in a version of the Heisenberg picture, in which the time-dependence of operators is determined by a Hamiltonian including the external field interaction (14.1.1), but no other interactions. The electron field $\psi(x)$ in this picture satisfies the field equation

$$\left[\gamma^\lambda\frac{\partial}{\partial x^\lambda} + m + ie\gamma^\lambda\mathcal{A}_\lambda(x)\right]\psi(x) = 0. \quad (14.1.3)$$

This is not the Dirac equation in the original sense of Dirac [1], because $\psi(x)$ here is not a c-number wave function but a quantum operator. The c-number Dirac wave functions are defined by

$$u_N(x) \equiv \langle\Omega|\psi(x)|N\rangle, \quad (14.1.4)$$

$$v_N(x) \equiv \langle N|\psi(x)|\Omega\rangle, \quad (14.1.5)$$

where the $|N\rangle$ are a complete orthonormal set of state-vectors, with $|\Omega\rangle$ the vacuum. It follows immediately from Eq. (14.1.3) that these functions satisfy the homogeneous Dirac equation

$$\left[\gamma^\lambda\frac{\partial}{\partial x^\lambda} + m + ie\gamma^\lambda\mathcal{A}_\lambda(x)\right]u_N(x) = \left[\gamma^\lambda\frac{\partial}{\partial x^\lambda} + m + ie\gamma^\lambda\mathcal{A}_\lambda(x)\right]v_N(x) = 0. \quad (14.1.6)$$

We can also derive a normalization condition from the equal-time anticommutation relations for the Dirac field. These are unaffected by the interaction (14.1.1), and therefore take the same form as for the free fields:

$$\{\psi(\mathbf{x},t), \bar{\psi}(\mathbf{y},t)\} = i\gamma^0\delta^3(\mathbf{x}-\mathbf{y}). \quad (14.1.7)$$

Taking the vacuum expectation value and inserting a sum over the states $|N\rangle$, we find

$$\sum_N u_N(\mathbf{x},t)u_N^\dagger(\mathbf{y},t) + \sum_N v_N(\mathbf{x},t)v_N^\dagger(\mathbf{y},t) = \delta^3(\mathbf{x}-\mathbf{y}), \quad (14.1.8)$$

it being understood that the sum over N includes an integral over continuum states as well as a sum over any discrete bound states.

We are chiefly interested in the case of a time-independent external field, like (14.1.2). In such cases, the states $|N\rangle$ may be taken as eigenstates of the Hamiltonian (including the interaction (14.1.1)) with energies E_N. Time-translation invariance then tells us that the $u_N(x)$ and $v_N(x)$ have the time-dependence:

$$u_N(\mathbf{x},t) = e^{-iE_Nt}u_N(\mathbf{x}), \qquad v_N(\mathbf{x},t) = e^{+iE_Nt}v_N(\mathbf{x}). \quad (14.1.9)$$

The homogeneous Dirac equations (14.1.6) then become

$$i\gamma^0\left[\gamma\cdot\nabla + m + ie\gamma^\lambda\mathcal{A}_\lambda(\mathbf{x})\right]u_N(\mathbf{x}) = E_Nu_N(\mathbf{x}), \quad (14.1.10)$$

$$i\gamma^0\left[\gamma\cdot\nabla + m + ie\gamma^\lambda\mathcal{A}_\lambda(\mathbf{x})\right]v_N(\mathbf{x}) = -E_Nv_N(\mathbf{x}). \quad (14.1.11)$$

The minus sign on the right-hand side of Eq. (14.1.11) shows that the v_N are the famous 'negative-energy' solutions of Dirac. As shown by Eq. (14.1.8), these negative-energy solutions are needed to make up a complete set of wave functions. Of course, for moderate external fields there are no negative-energy states in the theory, so all E_N are positive, but there is still an important difference between the states with non-vanishing u_N or v_N: the definitions (14.1.4) and (14.1.5) show that a state can have $u_N \neq 0$ or $v_N \neq 0$ only if it has charge $-e$ or $+e$, respectively. It is in this sense that negative-energy solutions of the Dirac equation have something to do with the existence of antiparticles. However, this argument has nothing to do with the details of the Dirac equation, or even with the spin of the electron.

From the Dirac wave equations (14.1.10) and (14.1.11) we can easily see that wave functions of different energy are orthogonal. That is,

$$(E_M - E_N^*)(u_N^\dagger u_M) = \nabla\cdot(u_N^\dagger i\gamma^0\gamma u_M),$$

so if $|\mathbf{x}|^2(u_N^\dagger i\gamma^0\gamma u_M)$ remains bounded as $|\mathbf{x}| \to 0$ and $|\mathbf{x}| \to \infty$, then

$$\int d^3x\,(u_N^\dagger(\mathbf{x})u_M(\mathbf{x})) = 0 \quad \text{if} \quad E_N \neq E_M^*. \quad (14.1.12)$$

With similar boundary conditions for the v_N, we find in the same way that

$$\int d^3x\,(v_N^\dagger(\mathbf{x})v_M(\mathbf{x})) = 0 \quad \text{if} \quad E_N \neq E_M^*, \quad (14.1.13)$$

$$\int d^3x\,(u_N^\dagger(\mathbf{x})v_M(\mathbf{x})) = 0 \quad \text{if} \quad E_N \neq -E_M^*. \quad (14.1.14)$$

Taking $N = M$, Eqs. (14.1.12) and (14.1.13) tell us that the energies are all real. Dropping the complex conjugation of E_M in Eqs. (14.1.12)–(14.1.14), we see that us of different

energy are orthogonal, vs of different energy are orthogonal, and (as long as the potential is not strong enough to produce negative energy states) all us are orthogonal to all vs. By a suitable choice of the discrete quantum numbers that characterize the states along with the energy, we can then always arrange that

$$\int d^3x\,(u_N^\dagger(\mathbf{x})u_M(\mathbf{x})) = 0 \quad \text{if} \quad N \neq M, \quad (14.1.15)$$

$$\int d^3x\,(v_N^\dagger(\mathbf{x})v_M(\mathbf{x})) = 0 \quad \text{if} \quad N \neq M, \quad (14.1.16)$$

$$\int d^3x\,(u_N^\dagger(\mathbf{x})v_M(\mathbf{x})) = 0. \quad (14.1.17)$$

Multiplying Eq. (14.1.8) on the right with $u_M(\mathbf{y})$ or $v_M(\mathbf{y})$, we find then that these wave functions must satisfy the normalization conditions

$$\int d^3y\,(u_N^\dagger(\mathbf{y})u_M(\mathbf{y})) = \int d^3y\,(v_N^\dagger(\mathbf{y})v_M(\mathbf{y})) = \delta_{NM}, \quad (14.1.18)$$

where δ_{NM} is a product of Kronecker deltas and momentum space delta functions, with normalization adapted to that used in defining $\sum_N$, in such a way that $\sum_N \delta_{NM} = 1$. These normalization conditions have nothing directly to do with any probabilistic interpretation of the Dirac wave functions, but arise instead from the anticommutation relations (14.1.7) for the fields.

Let us now specialize to the case of a pure electrostatic external field with $\mathbf{A} = 0$. In our standard representation of the Dirac matrices, we have

$$\gamma = i\begin{pmatrix} 0 & -\sigma \\ \sigma & 0 \end{pmatrix}, \qquad i\gamma^0 = \beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

where σ is the usual three-vector of $2\cdot 2$ Pauli matrices, and '1' and '0' here are the $2\cdot 2$ unit and zero matrices. We introduce two-component wave functions f_N and g_N by setting

$$u_N = \frac{1}{\sqrt{2}}\begin{pmatrix} f_N + ig_N \\ f_N - ig_N \end{pmatrix}. \quad (14.1.19)$$

The energy eigenvalue condition (14.1.10) then takes the form:

$$(\sigma\cdot\nabla)f_N = (E_N + e\mathcal{A}^0 + m)g_N, \quad (14.1.20)$$

$$(\sigma\cdot\nabla)g_N = -(E_N + e\mathcal{A}^0 - m)f_N. \quad (14.1.21)$$

In the non-relativistic case where $e\mathcal{A}^0 r \approx Z\alpha \ll 1$ the binding energy $m - E_N$ is of order $Z^2\alpha^2 m$, while the gradient operator is of the order of $Z\alpha m$, so g_N is smaller than f_N by a factor of order $Z\alpha$. (To find the positron wave functions v_N we replace E_N everywhere by $-E_N$, so in this case f_N is smaller than g_N by the same factor.) We shall return to this non-relativistic case at the end of this section.

Physical states may be classified as even or odd under space inversion:

$$P|N\rangle = \eta_N|N\rangle, \quad (14.1.22)$$

where η_N is a sign factor, ± 1. Recall that with the intrinsic parity of the electron defined to be $+1$, the Dirac field has the space-inversion property

$$P\psi(\mathbf{x},t)P^{-1} = \beta\psi(-\mathbf{x},t),$$

so Eqs. (14.1.4) and (14.1.22) show that the Dirac wave functions satisfy the parity condition

$$u_N(\mathbf{x}) = \eta_N\beta u_N(-\mathbf{x}), \quad (14.1.23)$$

or equivalently

$$f_N(\mathbf{x}) = \eta_Nf_N(-\mathbf{x}), \qquad g_N(\mathbf{x}) = -\eta_Ng_N(-\mathbf{x}). \quad (14.1.24)$$

Note that the parity of the state is the same as the parity of $f_N(\mathbf{x})$, not $g_N(\mathbf{x})$.

Where the potential $\mathcal{A}^0$ is rotationally invariant, the solutions of the wave equations here may be classified according to their total angular momentum j and parity η. For a given j, the components f and g may be expanded in spherical harmonics with orbital angular momentum $\ell = j + \frac{1}{2}$ and $\ell = j - \frac{1}{2}$, but for a definite parity $\eta = (-1)^{j\pm\frac{1}{2}}$, Eqs. (14.1.24) show that we can only have $\ell = j \mp \frac{1}{2}$ in f and $\ell = j \pm \frac{1}{2}$ in g. The usual rules of angular-momentum addition then show that for a state of total angular momentum j, total angular-momentum z-component μ, and parity $(-1)^{j\mp\frac{1}{2}}$, the 'large' two-component wave function f has the form

$$f(\mathbf{x}) = \begin{pmatrix} C_{j\mp\frac{1}{2}\,\frac{1}{2}}(j\,\mu;\mu-\frac{1}{2},\frac{1}{2})Y^{\mu-\frac{1}{2}}_{j\mp\frac{1}{2}}(\hat{\mathbf{x}}) \\ C_{j\mp\frac{1}{2}\,\frac{1}{2}}(j\,\mu;\mu+\frac{1}{2},-\frac{1}{2})Y^{\mu+\frac{1}{2}}_{j\mp\frac{1}{2}}(\hat{\mathbf{x}}) \end{pmatrix}F(|\mathbf{x}|), \quad (14.1.25)$$

where C and Y are the usual Clebsch–Gordan coefficients and spherical harmonics [2]. Also, given any wave function of definite total angular momentum and parity we can construct another wave function with the same j and μ but opposite parity by applying the operator $\sigma\cdot\hat{\mathbf{x}}$, so the 'small' components may be put in the form

$$g(\mathbf{x}) = (\sigma\cdot\hat{\mathbf{x}})\begin{pmatrix} C_{j\mp\frac{1}{2}\,\frac{1}{2}}(j\,\mu;\mu-\frac{1}{2},\frac{1}{2})Y^{\mu-\frac{1}{2}}_{j\mp\frac{1}{2}}(\hat{\mathbf{x}}) \\ C_{j\mp\frac{1}{2}\,\frac{1}{2}}(j\,\mu;\mu+\frac{1}{2},-\frac{1}{2})Y^{\mu+\frac{1}{2}}_{j\mp\frac{1}{2}}(\hat{\mathbf{x}}) \end{pmatrix}G(|\mathbf{x}|). \quad (14.1.26)$$

It is conventional to define the orbital angular momentum ℓ of the state as the orbital angular momentum of the 'large' components $f(\mathbf{x})$,

$$\ell = j \mp \frac{1}{2}, \quad (14.1.27)$$

so that the parity is always $(-1)^\ell$.

Inserting Eqs. (14.1.25) and (14.1.26) in Eqs. (14.1.20) and (14.1.21) yields the coupled differential equations

$$\frac{dG}{dr} + \frac{k+1}{r}G + (E + e\mathcal{A}^0 - m)F = 0, \quad (14.1.28)$$

$$\frac{dF}{dr} - \frac{k-1}{r}F - (E + e\mathcal{A}^0 + m)G = 0, \quad (14.1.29)$$

where for parity $\eta = (-1)^{j\pm\frac{1}{2}}$,

$$k = \pm\left(j + \frac{1}{2}\right). \quad (14.1.30)$$

Let us now concentrate on the simple Coulomb field (14.1.2), for which $e\mathcal{A}^0 = Z\alpha/r$. The treatment of the Dirac equation in this case [3] is familiar, so it will be summarized briefly here just for completeness. It is easy to see that the solutions near the origin go as r^{s-1}, with $s^2 = k^2 - Z^2\alpha^2$. (Note that $k^2 \geq 1$, so the exponent s is real for $Z\alpha \leq 1$.) We must reject the solutions with $s < 0$ as being inconsistent with the normalization condition (14.1.18). The condition that the wave functions do not blow up for $r \to \infty$ then fixes the allowed values of the energy eigenvalues:

$$E_{n,j} = m\left[1 + \frac{Z^2\alpha^2}{\left[n - j - \frac{1}{2} + \sqrt{(j+\frac{1}{2})^2 - Z^2\alpha^2}\right]^2}\right]^{-1/2}, \quad (14.1.31)$$

where n is a 'principal quantum number' with

$$j + \frac{1}{2} \leq n. \quad (14.1.32)$$

It is noteworthy that these energies do not depend on the parity or ℓ, but only on n and j. For each n and j there are two solutions, corresponding to the two signs of k or the two possible parities, except that for $n = j + \frac{1}{2}$ we only have $k > 0$ and parity $(-1)^{j-\frac{1}{2}}$, so that $\ell = j - \frac{1}{2}$. With Eq. (14.1.32), this is the same as the familiar non-relativistic restriction that $\ell \leq n - 1$.

For light atoms with $Z\alpha \ll 1$, Eq. (14.1.31) yields the power series

$$E = m\left[1 - \frac{Z^2\alpha^2}{2n^2} - \frac{Z^4\alpha^4}{n^4}\left(\frac{3}{8} - \frac{n}{2j+1}\right) + \cdots\right]. \quad (14.1.33)$$

The first two terms, of course, just represent the rest energy and the binding energy as given by the non-relativistic Schrödinger equation. The leading term that depends on j as well as n is the third term, the first relativistic correction. For $n = 1$ there is only one value of the total angular momentum, $j = \frac{1}{2}$, and since here $n = j + \frac{1}{2}$ there is also only one parity, $(-1)^{j-\frac{1}{2}} = +1$, corresponding to $\ell = 0$. It is therefore difficult to see the effects of the relativistic corrections in Eq. (14.1.33) in the $n = 1$ states of hydrogenic atoms, though as we shall see in Section 14.3, this has recently become possible. On the other hand, for $n = 2$ we have a $j = \frac{1}{2}$ state with both parities (i.e., $2s_{1/2}$ and $2p_{1/2}$) as well as a $2p_{3/2}$ state with $j = \frac{3}{2}$ and negative parity. Eq. (14.1.33) gives the splitting between the p states in hydrogen as

$$E(2p_{3/2}) - E(2p_{1/2}) = \frac{\alpha^4 m_e}{32} = 4.5283\cdot 10^{-5}\ \text{eV}. \quad (14.1.34)$$

Such relativistic line splitting is known as the *fine structure* of the atomic state. From the beginning it was known that this prediction is in good agreement with the observed fine structure. On the other hand, the Dirac equation does not yield any energy difference in the

$2s_{1/2}$ and $2p_{1/2}$ states, so this is a good place to look for the effects of further corrections, to be considered in Section 14.3.

Before closing this section we shall consider the approximate forms for the wave functions and matrix elements in the non-relativistic case for a general electrostatic potential $\mathcal{A}^0$. (For a Coulomb potential, this is the limit $Z\alpha \ll 1$.) Since here $E_N + m \simeq 2m \gg |e\mathcal{A}^0|$, the 'small' components of the electron wave function are given approximately in terms of the large components by

$$g_N \simeq (\sigma\cdot\nabla)f_N/2m_e. \quad (14.1.35)$$

Eq. (14.1.21) then becomes just the non-relativistic Schrödinger equation

$$\left[-\frac{\nabla^2}{2m_e} - e\mathcal{A}^0\right]f_N \simeq (E_N - m)f_N. \quad (14.1.36)$$

Since there is no longer any coupling between spin and orbital degrees of freedom in the equation for f_N, we may find a complete set of solutions of this equation in the form

$$f_N = \chi_N\psi_N(\mathbf{x}),$$

where χ_N is a two-component constant spinor, and $\psi_N(\mathbf{x})$ is an ordinary one-component solution of the Schrödinger equation. However, we often work with states that have definite values of the total angular momentum j, for which f_N is (for non-vanishing orbital angular momentum) a sum of such terms.

In the non-relativistic approximation, the four-component Dirac wave function takes the form

$$u_N \simeq \frac{1}{\sqrt{2}}\begin{bmatrix} (1 + i\sigma\cdot\nabla/2m)f_N \\ (1 - i\sigma\cdot\nabla/2m)f_N \end{bmatrix} \quad (14.1.37)$$

and Eq. (14.1.18) gives the normalization condition

$$\int d^3x\,(f_N^\dagger f_M) \simeq \delta_{NM} - \frac{1}{4}(v^2)_{NM}, \quad (14.1.38)$$

where

$$(v^2)_{NM} \equiv -\frac{1}{m_e^2}\int d^3x\,f_N^\dagger(\mathbf{x})\nabla^2 f_M(\mathbf{x}).$$

In relating matrix elements in an external field to free-particle matrix elements, it is useful to note that the momentum space wave function in an energy eigenstate N may be written

$$u_N(\mathbf{p}) \equiv (2\pi)^{-3/2}\int d^3p\,e^{-i\mathbf{p}\cdot\mathbf{x}}u_N(\mathbf{x}) \simeq \sum_\sigma u(\mathbf{p},\sigma)[f_N(\mathbf{p})]_\sigma, \quad (14.1.39)$$

where $u(\mathbf{p},\sigma)$ is the free-particle Dirac spinor

$$u(\mathbf{p},\sigma) \simeq \frac{1}{\sqrt{2}}\begin{bmatrix} (1 - \mathbf{p}\cdot\sigma/2m_e)\chi_\sigma \\ (1 + \mathbf{p}\cdot\sigma/2m_e)\chi_\sigma \end{bmatrix}, \qquad \chi_{+\frac{1}{2}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi_{-\frac{1}{2}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and $f_N(\mathbf{p})$ is the Fourier transform of the two-component Schrödinger wave function

$$f_N(\mathbf{p}) \equiv (2\pi)^{-3/2}\int d^3p\,e^{-i\mathbf{p}\cdot\mathbf{x}}f_N(\mathbf{x}).$$

In closing, as an aid in calculating the effects of various perturbations, we note that the leading terms in the electron matrix elements of the sixteen independent $4\cdot 4$ matrices are

$$(\bar{u}_Mu_N) \simeq (f_M^\dagger f_N) - \frac{1}{4m_e^2}(\nabla f_M^\dagger\cdot\sigma\,\sigma\cdot\nabla f_N), \quad (14.1.40)$$

$$i(\bar{u}_M\gamma^0u_N) \simeq (f_M^\dagger f_N) + \frac{1}{4m_e^2}(\nabla f_M^\dagger\cdot\sigma\,\sigma\cdot\nabla f_N), \quad (14.1.41)$$

$$(\bar{u}_M\gamma^iu_N) \simeq \frac{1}{2m_e}\left[(\nabla_i f_M^\dagger\cdot\sigma\,\sigma_i f_N) - (f_M^\dagger\sigma_i\sigma\cdot\nabla f_N)\right], \quad (14.1.42)$$

$$(\bar{u}_M[\gamma^0,\gamma^i]u_N) \simeq \frac{i}{m_e}\left[(\nabla f_M^\dagger\cdot\sigma\,\sigma_i f_N) + (f_M^\dagger\sigma_i\sigma\cdot\nabla f_N)\right], \quad (14.1.43)$$

$$(\bar{u}_M[\gamma^i,\gamma^j]u_N) \simeq 2i\epsilon_{ijk}(f_M^\dagger\sigma_k f_N), \quad (14.1.44)$$

$$(\bar{u}_M\gamma^5\gamma^iu_N) \simeq -i(f_M^\dagger\sigma_i f_N), \quad (14.1.45)$$

$$(\bar{u}_M\gamma^5\gamma^0u_N) \simeq \frac{1}{2m_e}\left[(\nabla f_M^\dagger\cdot\sigma f_N) - (f_M^\dagger\sigma\cdot\nabla f_N)\right], \quad (14.1.46)$$

$$(\bar{u}_M\gamma^5u_N) \simeq \frac{1}{2m_e}\left[(\nabla f_M^\dagger\cdot\sigma f_N) + (f_M^\dagger\sigma\cdot\nabla f_N)\right]. \quad (14.1.47)$$

14.2 Radiative Corrections in External Fields

We now consider radiative corrections to the results of the previous section, due to the interaction of electrons with the quantum electromagnetic field as well as the external field of the heavy charged particles. These radiative corrections can be calculated using Feynman diagrams of the usual sort, with the whole effect of the external field being to modify the propagator of the electron field in the presence of the external electromagnetic field (and to supply the external-field-dependent renormalization counterterms shown in Eq. (14.1.1)). To be specific, the effect of inserting any number of vertices corresponding to the first term of the interaction (14.1.1) in an internal electron line of any graph is to replace the bare coordinate space propagator $-iS(x-y)$ with a corrected propagator

$$-iS_{\mathcal{A}}(x,y) \equiv -iS(x-y) + (-i)^2\int d^4z_1\,S(x-z_1)e\gamma^\mu\mathcal{A}_\mu(z_1)S(z_1-y) + (-i)^3\int d^4z_1\int d^4z_2\,S(x-z_1)e\gamma^\mu\mathcal{A}_\mu(z_1)S(z_1-z_2)e\gamma^\nu\mathcal{A}_\nu(z_2)S(z_2-y) + \cdots, \quad (14.2.1)$$

where as usual

$$S(x-y) \equiv \frac{1}{(2\pi)^4}\int d^4p\,\frac{-i\not{p} + m}{p^2 + m^2 - i\epsilon}e^{ip\cdot(x-y)}.$$

(We must write $S_{\mathcal{A}}$ as a function of x and y rather than of $x - y$, because the external field invalidates translation invariance.) The theorem proved in Section 6.4 tells us that Eq. (14.2.1) is the same as

$$-iS_{\mathcal{A}}(x,y) = {}_{\mathcal{A}}\langle\Omega|T\{\psi(x),\bar{\psi}(y)\}|\Omega\rangle_{\mathcal{A}}, \quad (14.2.2)$$

with the subscript $\mathcal{A}$ on the right indicating that the vacuum state $|\Omega\rangle_{\mathcal{A}}$ and electron field $\psi(x)$ are to be defined in a Heisenberg picture in which the only interaction taken into account is the interaction (14.1.1) with the external field. Inserting a complete set of intermediate states $|N\rangle_{\mathcal{A}}$ in Eq. (14.2.2) yields an expression for the propagator in terms of the Dirac wave functions u_N and v_N introduced in the previous section

$$-iS_{\mathcal{A}}(x,y) = \theta(x^0 - y^0)\sum_N u_N(x)\bar{u}_N(y) - \theta(y^0 - x^0)\sum_M v_M(x)\bar{v}_M(y). \quad (14.2.3)$$

It is also possible to obtain the propagator (14.2.2) as the solution of the *inhomogeneous Dirac equation*:

$$\left[\gamma^\mu\frac{\partial}{\partial x^\mu} + m + ie\gamma^\mu\mathcal{A}_\mu(x)\right]S_{\mathcal{A}}(x,y) = \delta^4(x-y), \quad (14.2.4)$$

which follows from the field equations (14.1.3) and anticommutation relations (14.1.7), or formally from the perturbation series (14.2.1). Also Eq. (14.2.3) tells us that the propagator satisfies boundary conditions; its Fourier decomposition contains only 'positive frequency terms' proportional to $\exp\{-iE(x^0 - y^0)\}$ with $E > 0$ for $x^0 - y^0 \to \infty$, and only 'negative frequency terms' proportional to $\exp\{+iE(x^0 - y^0)\}$ with $E > 0$ for $x^0 - y^0 \to -\infty$. The inhomogeneous Dirac equation with these boundary conditions may be used to obtain a numerical solution [4] for this propagator even in cases where the external field is too strong to allow the use of the perturbation series (14.2.1). Once the propagator $S_{\mathcal{A}}(x,y)$ has been calculated, the amplitudes for scattering in an external field can be calculated using ordinary Feynman diagrams, but with $S_{\mathcal{A}}(x,y)$ in place of $S(x-y)$ (and with $\mathcal{A}$-dependent renormalization counterterms inserted where appropriate).

Now let us see how to use the perturbation series with this corrected propagator to calculate the shifts of bound-state energy levels. Consider the full electron propagator $S'_{\mathcal{A}}(x,y)$, involving interactions of the electron with the quantum electromagnetic field as well as the external field:

$$-iS'_{\mathcal{A}}(x,y) = {}_{\mathcal{A}}\langle\Omega|T\{\Psi(x),\bar{\Psi}(y)\}|\Omega\rangle_{\mathcal{A}} \quad (14.2.5)$$

with $\Psi(x)$ the electron field in a Heisenberg picture including all interactions, and $|\Omega\rangle$ the vacuum eigenstate of the full Hamiltonian. For a time-independent external potential we can find a complete orthonormal set $|N\rangle$ of eigenstates of the full Hamiltonian with energies E'_N. Inserting a sum over these states in the operator product in Eq. (14.2.5), we find

$$-iS'_{\mathcal{A}}(x,y) = \theta(x^0 - y^0)e^{-iE'_N(x^0-y^0)}\sum_N U_N(\mathbf{x})\bar{U}_N(\mathbf{y}) - \theta(y^0 - x^0)e^{-iE'_N(y^0-x^0)}\sum_N V_N(\mathbf{x})\bar{V}_N(\mathbf{y}), \quad (14.2.6)$$

where

$$\langle\Omega|\Psi(\mathbf{x},t)|N\rangle \equiv e^{-iE'_Nt}U_N(\mathbf{x}), \quad (14.2.7)$$

$$\langle N|\bar{\Psi}(\mathbf{x},t)|\Omega\rangle \equiv e^{+iE'_Nt}\bar{V}_N(\mathbf{x}). \quad (14.2.8)$$

(The sum includes an integral over continuum states as well as a sum over discrete bound states. As before, U_N and V_N are non-zero only if the state $|N\rangle$ has charge $-e$ or $+e$, respectively.) We can redefine the propagator as a function of energy rather than time

$$S'_{\mathcal{A}}(\mathbf{x},\mathbf{y};E) \equiv \int_{-\infty}^{\infty}dx^0\,e^{iE(x^0-y^0)}S'_{\mathcal{A}}(x,y). \quad (14.2.9)$$

(Time-translation invariance dictates that $S'_{\mathcal{A}}(x,y)$ is a function of $x^0 - y^0$ but not of x^0 and y^0 separately.) From Eq. (14.2.6) we see that

$$S'_{\mathcal{A}}(\mathbf{x},\mathbf{y};E) = \sum_N\frac{U_N(\mathbf{x})\bar{U}_N(\mathbf{y})}{E'_N - E - i\epsilon} - \sum_N\frac{V_N(\mathbf{x})\bar{V}_N(\mathbf{y})}{E'_N + E - i\epsilon}. \quad (14.2.10)$$

In particular, $S'_{\mathcal{A}}(\mathbf{x},\mathbf{y};E)$ has a pole at any electron bound-state energy, and also at the negative of any positron bound-state energy. (Of course, positrons do not have bound states in the Coulomb field of an ordinary positively charged nucleus.)

Let us now consider the lowest-order radiative corrections to the complete propagator. The Feynman rules here give the complete propagator to this order as $S'_{\mathcal{A}} = S_{\mathcal{A}} + \delta S_{\mathcal{A}}$, with a correction term

$$\delta S_{\mathcal{A}}(x,y) = \int d^4z\int d^4w\,S_{\mathcal{A}}(x,z)\Sigma^*_{\mathcal{A}}(z,w)S_{\mathcal{A}}(w,y), \quad (14.2.11)$$

where $i\Sigma^*_{\mathcal{A}}$ is the sum of all one-loop diagrams with one incoming and one outgoing electron line (excluding final electron propagators) calculated using $S_{\mathcal{A}}(x,y)$ in place of $S(x-y)$ for internal electron lines, plus second-order renormalization counterterms. Using energy variables in place of time variables, this is

$$\delta S_{\mathcal{A}}(\mathbf{x},\mathbf{y};E) = \int d^3z\int d^3w\,S_{\mathcal{A}}(\mathbf{x},\mathbf{z};E)\Sigma^*_{\mathcal{A}}(\mathbf{z},\mathbf{w};E)S_{\mathcal{A}}(\mathbf{w},\mathbf{y};E), \quad (14.2.12)$$

where

$$\Sigma^*_{\mathcal{A}}(\mathbf{z},\mathbf{w};E) \equiv \int dz^0\,e^{iE(z^0-w^0)}\Sigma^*_{\mathcal{A}}(z,w). \quad (14.2.13)$$

The effect of these radiative corrections is to change the wave functions to $U_N = u_N + \delta u_N$ and $V_N = v_N + \delta v_N$ and the bound-state energies to $E'_N = E_N + \delta E_N$, so that the complete propagator is

$$S'_{\mathcal{A}}(\mathbf{x},\mathbf{y};E) \simeq S_{\mathcal{A}}(\mathbf{x},\mathbf{y};E) + \sum_N\frac{\delta u_N(\mathbf{x})\bar{u}_N(\mathbf{y}) + u_N(\mathbf{x})\delta\bar{u}_N(\mathbf{y})}{E_N - E} - \sum_N\frac{\delta v_N(\mathbf{x})\bar{v}_N(\mathbf{y}) + v_N(\mathbf{x})\delta\bar{v}_N(\mathbf{y})}{E_N + E} - \sum_N\frac{u_N(\mathbf{x})\bar{u}_N(\mathbf{y})\delta E_N}{(E_N - E)^2} + \sum_N\frac{v_N(\mathbf{x})\bar{v}_N(\mathbf{y})\delta E_N}{(E_N + E)^2}. \quad (14.2.14)$$

(We are dropping the $i\epsilon$ terms because we are now not taking E to lie in the continuum of scattering states.) We see that the shift δE_N of an electron bound-state energy is given by

the coefficient of $-u_N(\mathbf{x})\bar{u}_N(\mathbf{y})/(E_N - E)^2$ in the complete propagator. To calculate this, we note that Eq. (14.2.3) gives

$$S_{\mathcal{A}}(\mathbf{x},\mathbf{y};E) = \sum_N\frac{u_N(\mathbf{x})\bar{u}_N(\mathbf{y})}{E_N - E - i\epsilon} - \sum_N\frac{v_N(\mathbf{x})\bar{v}_N(\mathbf{y})}{E_N + E - i\epsilon}. \quad (14.2.15)$$

Inserting this in Eq. (14.2.12) gives

$$\delta S_{\mathcal{A}}(\mathbf{x},\mathbf{y};E) = \sum_{N,M}\frac{u_N(\mathbf{x})\bar{u}_M(\mathbf{y})}{(E_N - E)(E_M - E)}\cdot\int d^3z\int d^3w\,\bar{u}_N(\mathbf{z})\Sigma^*_{\mathcal{A}}(\mathbf{z},\mathbf{w};E)u_M(\mathbf{w}) + \cdots, \quad (14.2.16)$$

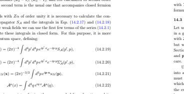


Figure 14.1. Lowest-order Feynman diagrams for the electron self-energy function $\Sigma^*_{\mathcal{A}}(x,y)$ in the presence of an external field. Here double straight lines represent electron propagators $S_{\mathcal{A}}$ that include effects of the external field; single straight lines are the incoming and outgoing electron lines; wavy lines are virtual photons; the cross represents renormalization counterterms.

where the dots denote additional terms involving at least one negative-energy pole. Comparing the coefficients of $(E_N - E)^{-2}$ here and in Eq. (14.2.14), we find

$$\delta E_N = -\int d^3x\int d^3y\,\bar{u}_N(\mathbf{x})\Sigma^*_{\mathcal{A}}(\mathbf{x},\mathbf{y};E_N)u_N(\mathbf{y}). \quad (14.2.17)$$

The u_N are solutions of the homogeneous Dirac equation satisfying the normalization condition (14.1.8), so this is very much like ordinary first-order perturbation theory, but with $-\Sigma^*$ in place of a perturbation to the Hamiltonian.

Generally δE_N turns out to be complex. This is simply a consequence of the instability of atomic energy levels to radiative decay to lower levels; we saw in Chapter 3 that an unstable state of energy E and decay rate Γ produces poles in various amplitudes at the complex energy $E - i\Gamma/2$. The imaginary part of Eq. (14.2.17) therefore equals $-\Gamma/2$, while its real part gives the energy shift.

The Feynman diagrams for Σ^* are shown in Figure 14.1. (Note that there are new photon tadpole diagrams here because the external field breaks the Lorentz invariance and

charge-conjugation invariance that forbids such diagrams in the ordinary Feynman rules.) Application of the position space Feynman rules to these diagrams gives

$$i\Sigma^*_{\mathcal{A}}(x,y) = [e\gamma^\mu][-iS_{\mathcal{A}}(x,y)][e\gamma_\mu][-iD(x-y)] - [e\gamma^\mu\delta^4(x-y)]\int d^4z\,[-iD(x-z)]\text{Tr}\{[-iS_{\mathcal{A}}(z,z)][e\gamma_\mu]\} - i(Z_2-1)(\gamma^\mu\partial_\mu + m)\delta^4(x-y) + i\delta m\,\delta^4(x-y) + e\gamma^\mu(Z_2-1)\delta^4(x-y)\mathcal{A}_\mu(x) + i(Z_3-1)[e\gamma_\mu]\delta^4(x-y)\int d^4z\,[-iD(x-z)]\partial_\nu\{\partial^\nu\mathcal{A}^\mu(z) - \partial^\mu\mathcal{A}^\nu(z)\}. \quad (14.2.18)$$

with the renormalization constants (Z_2-1), (Z_3-1), and δm calculated to second order in e. (The minus sign in the second term is the usual one that accompanies closed fermion loops.)

For strong external fields with $Z\alpha$ of order unity it is necessary to calculate the configuration space electron propagator $S_{\mathcal{A}}$ and the integrals in Eqs. (14.2.17) and (14.2.18) numerically [4]. However, for weak fields we can use the first few terms of the series (14.2.1) in Eq. (14.2.18) and calculate these integrals in closed form. For this purpose, it is more convenient to work in momentum space, defining:

$$S_{\mathcal{A}}(x,y) = (2\pi)^{-4}\int d^4p'\,d^4p\,e^{ip'\cdot x}e^{-ip\cdot y}S_{\mathcal{A}}(p',p), \quad (14.2.19)$$

$$\Sigma^*_{\mathcal{A}}(x,y) = (2\pi)^{-4}\int d^4p'\,d^4p\,e^{ip'\cdot x}e^{-ip\cdot y}\Sigma^*_{\mathcal{A}}(p',p), \quad (14.2.20)$$

$$u_N(\mathbf{x}) = (2\pi)^{-3/2}\int d^3p\,e^{i\mathbf{p}\cdot\mathbf{x}}u_N(\mathbf{p}), \quad (14.2.21)$$

$$\mathcal{A}^\mu(x) = \int d^4q\,e^{iq\cdot x}\mathcal{A}^\mu(q). \quad (14.2.22)$$

(We are here committing the impropriety of using the same symbol for a function and its Fourier transform, leaving it to the displayed arguments of the functions to indicate which is which.) Then Eqs. (14.2.1) and (14.2.18) become

$$S_{\mathcal{A}}(p',p) = \frac{-i\not{p}' + m}{p'^2 + m^2 - i\epsilon} - ie\frac{-i\not{p}' + m}{p'^2 + m^2 - i\epsilon}\not{\mathcal{A}}(p'-p)\frac{-i\not{p} + m}{p^2 + m^2 - i\epsilon} + \cdots, \quad (14.2.23)$$

and

$$\Sigma^*_{\mathcal{A}}(p',p) = \frac{ie^2}{(2\pi)^4}\int\frac{d^4k}{k^2 - i\epsilon}\gamma^\mu S_{\mathcal{A}}(p'-k,p-k)\gamma_\mu + [-(Z_2-1)(i\not{p} + m) + Z_2\delta m]\delta^4(p'-p) - ie(Z_2-1)\not{\mathcal{A}}(p'-p) - \frac{ie^2\gamma^\mu}{(2\pi)^4[(p-p')^2 - i\epsilon]}\int d^4q\,\text{Tr}\{S_{\mathcal{A}}(q,q+p'-p)\gamma_\mu\} + \frac{ie(Z_3-1)}{(p-p')^2 - i\epsilon}\left[(p-p')^2\not{\mathcal{A}}(p'-p) - (\not{p} - \not{p}')(p-p')\cdot\mathcal{A}(p'-p)\right]. \quad (14.2.24)$$

Because the external field is time-independent, $S_{\mathcal{A}}(x,y)$ and $\Sigma^*_{\mathcal{A}}(x,y)$ can depend on x^0 and y^0 only through the difference $x^0 - y^0$, so $S_{\mathcal{A}}(p',p)$ and $\Sigma^*_{\mathcal{A}}(p',p)$ as well as $\mathcal{A}^\mu(p'-p)$ must be proportional to $\delta(p'^0 - p^0)$:

$$\mathcal{A}^\mu(p'-p) = \delta(p'^0 - p^0)\mathcal{A}^\mu(\mathbf{p}' - \mathbf{p}), \quad (14.2.25)$$

$$S_{\mathcal{A}}(p',p) = \delta(p'^0 - p^0)S_{\mathcal{A}}(\mathbf{p}',\mathbf{p};p^0), \quad (14.2.26)$$

$$\Sigma^*_{\mathcal{A}}(p',p) = \delta(p'^0 - p^0)\Sigma^*_{\mathcal{A}}(\mathbf{p}',\mathbf{p};p^0). \quad (14.2.27)$$

The energy shift is then given by Eqs. (14.2.17) and (14.2.13) as

$$\delta E_N = -\int d^3p'\int d^3p\,\bar{u}_N(\mathbf{p}')\Sigma^*_{\mathcal{A}}(\mathbf{p}',\mathbf{p};E_N)u_N(\mathbf{p}), \quad (14.2.28)$$

with $\Sigma^*_{\mathcal{A}}(\mathbf{p}',\mathbf{p};E_N)$ given by Eqs. (14.2.23), (14.2.24), and (14.2.27). This is the master formula we shall use in the next section to calculate energy shifts in weak external fields.

14.3 The Lamb Shift in Light Atoms

Let us now consider radiative corrections to the energy levels of a non-relativistic electron in a general electrostatic field, such as an electron in the Coulomb field of a light nucleus with $Z\alpha \ll 1$. It is natural in this limit to treat the Coulomb field as a weak perturbation, but we will see that this would lead to an infrared divergence, related to that found in Section 11.3. The infrared divergence is really fictitious, because the four-momenta $\mathbf{p}, E_N$ and $\mathbf{p}', E_N$ are not on the electron mass shell, but it does force us to proceed with some care.

Usually this problem is dealt with by dividing the integral over virtual photon energies into a low-energy range, within which we can treat the electrons non-relativistically but must include effects to all orders in the external field, and a high-energy range, within which we have to include relativistic effects but may include only effects of lowest order in the external field. Instead, we shall here introduce a fictitious photon mass μ, chosen to be much larger than typical electron kinetic energies, but much less than typical electron momenta. For a Coulomb field, this amounts to the requirement that

$$(Z\alpha)^2m_e \ll \mu \ll Z\alpha m_e. \quad (14.3.1)$$

We write the photon propagator in the first two terms of Eq. (14.2.24) (including the formulas for the counterterms $Z_2 - 1$ and $Z_2\delta m$) in the form

$$\frac{1}{k^2 - i\epsilon} = \left[\frac{1}{k^2 + \mu^2 - i\epsilon}\right] + \left[\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon}\right]. \quad (14.3.2)$$

The energy shift is correspondingly a sum of two terms, a 'high-energy' and a 'low-energy' term: The high-energy term is calculated by using the first term in the photon propagator (14.3.2) in the first three terms in Eq. (14.2.24), and adding the result to the last two terms (the vacuum polarization terms) in Eq. (14.2.24) which are not infrared divergent anyway; the low-energy term is calculated using the second term in Eq. (14.3.2) in the first three

terms of Eq. (14.2.24). One advantage of this procedure is that we shall be able to use the results of the relativistic calculations of Sections 11.3 and 11.4 directly, without the rather tricky conversion from a photon mass to an infrared energy cutoff. Of course, in the end we shall have to check that the dependence on the photon mass μ in the high-energy and low-energy contributions to the energy shift cancel, leaving the total energy shift μ-independent.

A High-Energy Term

Because μ is taken much larger than the atomic binding energies we can here keep only terms of lowest order in the external field. The one-loop radiative correction to atomic energy levels in a general time-independent external vector potential $\mathcal{A}^\mu(\mathbf{x})$ is given in momentum space by Eq. (14.2.28), with the self-energy insertion $\Sigma_{\mathcal{A}}(p',p)$ given by Eqs. (14.2.24) and (14.2.23). The terms of zeroth order in the external field simply cancel: the δm term cancels the first term with $\mathcal{A} = 0$; the $Z_2 - 1$ term cancels the third term with $\mathcal{A} = 0$; and the $Z_2 - 1$ terms vanish because $u(p)$ satisfies the Dirac equation. The term in $\Sigma_{\mathcal{A}}(p',p)$ that is of first order in $\mathcal{A}^\mu$ may be put in the form

$$\Sigma_{\mathcal{A}1}(p',p) = -ie\mathcal{A}_\mu(p'-p)\Gamma^\mu_1(p',p) \quad (14.3.3)$$

with

$$\Gamma^\mu_1(p',p) = \frac{ie^2}{(2\pi)^4}\int\frac{d^4k}{k^2 + \mu^2 - i\epsilon}\gamma^\nu\left[\frac{-i(\not{p}' - \not{k}) + m_e}{(p'-k)^2 + m_e^2 - i\epsilon}\right]\gamma^\mu\left[\frac{-i(\not{p} - \not{k}) + m_e}{(p-k)^2 + m_e^2 - i\epsilon}\right]\gamma_\nu + (Z_2 - 1)\gamma^\mu - \frac{ie^2\gamma_\nu}{(2\pi)^4[(p-p')^2 - i\epsilon]}\int d^4l\,\text{Tr}\left\{\left[\frac{-i\not{l} + m_e}{l^2 + m_e^2}\right]\gamma^\mu\left[\frac{-i\not{l} - i\not{p}' + i\not{p} + m_e}{(l + p' - p)^2 + m_e^2}\right]\gamma^\nu\right\} - \frac{Z_3 - 1}{(p'-p)^2 - i\epsilon}\left[(p-p')^2\eta_{\mu\nu} - (p'-p)_\mu(p'-p)_\nu\right]\gamma^\nu. \quad (14.3.4)$$

Comparison of the first two terms with Eqs. (11.3.1) and (11.3.8) and the second two terms with Eqs. (11.3.9), (11.2.3), and (11.2.15) reveals that $\Gamma^\mu_1(p',p)$ is the complete one-loop vertex function, including vacuum polarization and all counterterms, whose mass-shell matrix elements we have already calculated in Section 11.3. Using Eqs. (14.2.26) and (14.2.25), this contribution to the energy shift (14.2.28) is given by

$$[\delta E_N]_{\text{high energy}} = ie\int d^3p'\int d^3p\,(\bar{u}_N(\mathbf{p}')\Gamma^\mu_1(\mathbf{p}',E_N,\mathbf{p},E_N)u_N(\mathbf{p}))\mathcal{A}_\mu(\mathbf{p}' - \mathbf{p}). \quad (14.3.5)$$

(This could have been guessed at, by simply replacing the γ^μ in the interaction of the electron with the external field with Γ^μ_1.) As discussed in Section 14.1, because $Z\alpha \ll 1$ we can approximate the Dirac wave function u_N in Eq. (14.3.5) as

$$[u_N(\mathbf{p})]_\alpha = \sum_\sigma u_\alpha(\mathbf{p},\sigma)[f_N(\mathbf{p})]_\sigma, \quad (14.3.6)$$

where f_N is the non-relativistic two-component wave function of an electron in the external Coulomb field, and $u(\mathbf{p},\sigma)$ is the four-component normalized solution of the momentum space Dirac equation

$$[i\gamma_\mu p^\mu + m_e]u(\mathbf{p},\sigma) = 0 \quad (14.3.7)$$

for spin z-component σ. Since $u_N(\mathbf{p})$ approximately satisfies the free-particle Dirac equation, Eq. (10.6.15) gives the general form of the matrix element of Γ^μ_1 as

$$\bar{u}_N(\mathbf{p}')[\gamma^\mu + \Gamma^\mu_1(p',p)]u_N(\mathbf{p}) = \bar{u}_N(\mathbf{p}')\left[\gamma^\mu F_1(q^2) + \frac{1}{2}[\gamma^\mu,\gamma^\nu]q_\nu F_2(q^2)\right]u_N(\mathbf{p}), \quad (14.3.8)$$

where $q \equiv p' - p$. The wave functions $u_N(\mathbf{p})$ fall off very rapidly for $|\mathbf{p}| \gg Z\alpha m_e$, so we only need $F_1(q^2)$ and $F_2(q^2)$ in the limit $|q^2| \ll m_e^2$. In this limit, Eqs. (11.3.31), (10.6.18), and (11.3.16) give

$$F_1(q^2) \simeq 1 + \frac{e^2}{24\pi^2}\left(\frac{q^2}{m_e^2}\right)\left[\ln\left(\frac{\mu^2}{m_e^2}\right) + \frac{2}{5} + \frac{3}{4}\right], \quad (14.3.9)$$

$$F_2(q^2) \simeq \frac{e^2}{16m_e\pi^2}. \quad (14.3.10)$$

Let us first consider the contribution of the F_1 term in Eq. (14.3.8), which makes by far the largest contribution to the energy shift, and raises the most interesting problems in its calculation. For a pure electrostatic field with $\mathcal{A} = 0$, Eqs. (14.3.5), (14.3.8), and (14.3.9) give

$$[\delta E_N]_{F_1} = -\frac{e^2}{24\pi^2m_e^2}\left[\ln\left(\frac{\mu^2}{m_e^2}\right) + \frac{2}{5} + \frac{3}{4}\right]\int d^3p'\int d^3p\,\bar{u}_N(\mathbf{p}')(-ie\mathcal{A}^0(\mathbf{p}' - \mathbf{p}))\gamma^0(\mathbf{p}' - \mathbf{p})^2u_N(\mathbf{p}). \quad (14.3.11)$$

To calculate this contribution, we may use the leading term in the non-relativistic matrix element (14.1.41), and find

$$[\delta E_N]_{F_1} = -\frac{e^2}{24\pi^2m_e^2}\left[\ln\left(\frac{\mu^2}{m_e^2}\right) + \frac{2}{5} + \frac{3}{4}\right]\int d^3p'\int d^3p\,f_N^\dagger(\mathbf{p}')e\mathcal{A}^0(\mathbf{p}' - \mathbf{p})|\mathbf{p}' - \mathbf{p}|^2f_N(\mathbf{p}) \quad (14.3.12)$$

or in position space

$$[\delta E_N]_{F_1} = \frac{e^2}{24\pi^2m_e^2}\left[\ln\left(\frac{\mu^2}{m_e^2}\right) + \frac{2}{5} + \frac{3}{4}\right]\int d^3x\,f_N^\dagger(\mathbf{x})[e\nabla^2\mathcal{A}^0(\mathbf{x})]f_N(\mathbf{x}). \quad (14.3.13)$$

In particular, for the Coulomb potential (14.1.2), we have $e\nabla^2\mathcal{A}^0(\mathbf{x}) = -Ze^2\delta^3(\mathbf{x})$, and the label N consists of a principal quantum number n and angular-momentum quantum numbers j, m, ℓ, while (11.2.41) gives $[f_{njm\ell}(0)]_\sigma = 2(Z\alpha m_e/n)^{3/2}\delta_{\ell 0}\delta_{\sigma m}/\sqrt{4\pi}$. The energy shift (14.3.11) is then

$$[\delta E_{nj\ell}]_{F_1} = -\frac{2Z^4\alpha^5m_e}{3\pi n^3}\left[\ln\left(\frac{\mu^2}{m_e^2}\right) + \frac{2}{5} + \frac{3}{4}\right]\delta_{\ell 0}. \quad (14.3.14)$$

(The lack of dependence of δE on the total angular momentum z-component m is guaranteed by rotational invariance.) The term 2/5 in the brackets in Eqs. (14.3.12) and (14.3.13) arises from vacuum polarization, and yields just the energy shift calculated somewhat heuristically in Section 11.2.

Before going on to calculate the magnetic and low-energy contributions to the energy shift, it is worth noting that the result we have obtained so far yields a fair order-of-magnitude estimate of the Lamb shift without further work. We can anticipate that the low-energy terms will contain a term proportional to $\ln(\mu/B)$, with a coefficient such as to cancel the μ-dependence in Eq. (14.3.12). The constant B here is an energy which must be included to make the argument of the logarithm dimensionless; since it is the binding of the electron in the atom that will eventually provide our infrared cutoff, we may guess that B is a typical atomic binding energy, of order $B \simeq (Z\alpha)^2m_e$. The total energy shift in a state N with principal quantum number n and orbital angular momentum ℓ is therefore of the form

$$\delta E_N = \frac{2Z^4\alpha^5m_e}{3\pi n^3}\left[\ln(Z^4\alpha^4)\delta_{\ell 0} + O(1)\right]. \quad (14.3.15)$$

For the 2s state of hydrogen, the logarithmic term alone gives

$$\delta E_{2s} = -\frac{\alpha^5m_e}{12\pi}\ln(\alpha^4) = 5.5\cdot 10^{-6}\ \text{eV} = 1300\ \text{MHz}\cdot 2\pi\hbar.$$

As we shall see, the '$O(1)$' terms in Eq. (14.3.15) will lower the total energy shift by about 25%.

Now let us consider the contribution of the F_2 term in the matrix element of Γ^μ_1, which as we saw in Section 10.6 may be interpreted as a radiative correction to the magnetic moment of the electron. Using Eqs. (14.3.10), (14.3.8), and (14.3.6) in Eq. (14.3.5), we find that this term gives an energy shift

$$[\delta E_N]_{F_2} = -\frac{e^2}{32\pi^2m_e}\int d^3p'\int d^3p\,(\bar{u}_N(\mathbf{p}')[\gamma^\mu,\gamma^\nu]u_N(\mathbf{p}))e\mathcal{A}_\mu(\mathbf{p}' - \mathbf{p})(p' - p)_\nu, \quad (14.3.16)$$

or in position space

$$[\delta E_N]_{F_2} = \frac{ie^2}{64\pi^2m_e}\int d^3x\,(\bar{u}_N(\mathbf{x})[\gamma^\mu,\gamma^\nu]u_N(\mathbf{x}))e\mathcal{F}_{\mu\nu}(\mathbf{x}), \quad (14.3.17)$$

where

$$\mathcal{F}_{\mu\nu}(\mathbf{x}) = \partial_\mu\mathcal{A}_\nu(\mathbf{x}) - \partial_\nu\mathcal{A}_\mu(\mathbf{x}). \quad (14.3.18)$$

For a pure electrostatic field with $\mathcal{A} = 0$, this is

$$[\delta E_N]_{F_2} = -\frac{ie^2}{32\pi^2m_e}\int d^3x\,(\bar{u}_N(\mathbf{x})[\gamma^0,\gamma]u_N(\mathbf{x}))\cdot\nabla[e\mathcal{A}^0(\mathbf{x})]. \quad (14.3.19)$$

In the non-relativistic limit $Z\alpha \ll 1$, we can use the approximate result (14.1.43), which here reads

$$(\bar{u}_N[\gamma^0,\gamma]u_N) \simeq \frac{i}{m_e}\left[(\nabla f_N^\dagger)\cdot\sigma\,\sigma f_N + f_N^\dagger\sigma\,\sigma\cdot\nabla f_N\right] = \frac{i}{m_e}\left[\nabla(f_N^\dagger f_N) - i(\nabla f_N^\dagger\times\sigma)f_N - if_N^\dagger(\sigma\times\nabla f_N)\right]. \quad (14.3.20)$$

Using this in Eq. (14.3.8) and integrating by parts, this part of the energy shift is

$$[\delta E_N]_{F_2} = \frac{e^2}{32\pi^2m_e^2}\int d^3x\left[-|f_N(\mathbf{x})|^2\nabla^2(e\mathcal{A}^0(\mathbf{x})) + 2if_N^\dagger(\mathbf{x})\sigma\cdot(\nabla(e\mathcal{A}^0(\mathbf{x}))\times\nabla f_N(\mathbf{x}))\right]. \quad (14.3.21)$$

Combining Eqs. (14.3.12) and (14.3.21) gives the total high-energy contribution to the energy shift in an arbitrary electrostatic potential $\mathcal{A}^0$:

$$[\delta E_N]_{\text{high energy}} = \frac{e^2}{24\pi^2m_e^2}\left[\ln\left(\frac{\mu^2}{m_e^2}\right) + \frac{2}{5}\right]\int d^3x\,f_N^\dagger(\mathbf{x})[e\nabla^2\mathcal{A}^0(\mathbf{x})]f_N(\mathbf{x}) + \frac{ie^2}{16\pi^2m_e^2}\int d^3x\,f_N^\dagger(\mathbf{x})\sigma\cdot\{\nabla[e\mathcal{A}^0(\mathbf{x})]\times\nabla f_N(\mathbf{x})\}. \quad (14.3.22)$$

B Low-Energy Term

The low-energy contribution to the energy shift is obtained from the first three terms in Eq. (14.2.24), making the replacement in the photon propagator

$$\frac{1}{k^2 - i\epsilon} \longrightarrow \frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon}. \quad (14.3.23)$$

This substitution will eventually serve to cut off the integral over components of the photon four-momentum k at values of order μ, but it is not possible to see this until we carefully take mass renormalization into account, so we shall defer making any non-relativistic approximations until then. Also, we are now including photon momenta as small or smaller than the binding energies of the atomic states, so we must treat the electrostatic forces responsible for this binding to all orders.

Instead of working with the momentum space formula (14.2.24), it will be convenient to return to the configuration space formula (14.2.18). This gives the low-energy contribution to the electron self-energy function as

$$[\Sigma^*_{\mathcal{A}}(x,y)]_{\text{low energy}} = ie^2\gamma^\mu S_{\mathcal{A}}(x,y)\gamma_\mu D(x-y;\mu) + \delta m_e(\mu)\delta^4(x-y) - i(Z_2(\mu) - 1)[\gamma^\mu[\partial_\mu + ie\mathcal{A}_\mu] + m_e]\delta^4(x-y), \quad (14.3.24)$$

where $D(x-y;\mu)$ is the modified photon propagator

$$D(x-y;\mu) = \frac{1}{(2\pi)^4}\int d^4k\,e^{-ik\cdot(x-y)}\left[\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon}\right], \quad (14.3.25)$$

and the counterterms $Z_2(\mu) - 1$ and $\delta m(\mu)$ are calculated using this modified propagator. Converting from time to energy variables, the low-energy contribution to the function (14.2.13) is then

$$[\Sigma^*_{\mathcal{A}}(\mathbf{x},\mathbf{y};E)]_{\text{low energy}} = \frac{ie^2}{(2\pi)^4}\int d^4k\,\gamma^\mu S_{\mathcal{A}}(\mathbf{x},\mathbf{y};E - k^0)\gamma_\mu\left[\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon}\right]e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})} - (Z_2(\mu) - 1)(\gamma\cdot\nabla + i\gamma^0E + ie\gamma^\mu\mathcal{A}_\mu + m_e)\delta^3(\mathbf{x}-\mathbf{y}) + \delta m_e(\mu)\delta^3(\mathbf{x}-\mathbf{y}). \quad (14.3.26)$$

The low-energy contribution to the energy shift is then given by Eq. (14.2.17) as

$$[\delta E_N]_{\text{low energy}} = -\int d^3x\int d^3y\,\bar{u}_N(\mathbf{x})[\Sigma^*_{\mathcal{A}}(\mathbf{x},\mathbf{y};E_N)]_{\text{low energy}}u_N(\mathbf{y}) = -\frac{ie^2}{(2\pi)^4}\int d^4k\int d^3x\int d^3y\,\bar{u}_N(\mathbf{x})\gamma^\mu S_{\mathcal{A}}(\mathbf{x},\mathbf{y};E_N - k^0)\gamma_\mu u_N(\mathbf{y})\cdot\left[\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon}\right]e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})} - \delta m_e(\mu)\int d^3x\,\bar{u}_N(\mathbf{x})u_N(\mathbf{x}). \quad (14.3.27)$$

Note that the terms proportional to $Z_2(\mu) - 1$ have dropped out because the Dirac wave function $u_N(\mathbf{x})$ satisfies the Dirac equation Eq. (14.1.10). For the electron propagator in the presence of the Coulomb field, we use Eq. (14.2.15):

$$S_{\mathcal{A}}(\mathbf{x},\mathbf{y};E) = \sum_M\frac{u_M(\mathbf{x})\bar{u}_M(\mathbf{y})}{E_M - E - i\epsilon} - \sum_M\frac{v_M(\mathbf{x})\bar{v}_M(\mathbf{y})}{E_M + E - i\epsilon},$$

with the sums in the first and second terms running over all one-electron and one-positron states, respectively. The k^0 integrals can be done most easily by closing the contour of integration with a large semi-circle in the lower half-plane in the first term and in the upper half-plane in the second term:

$$\int dk^0\left(\frac{1}{k^2 + \mu^2 - i\epsilon}\right)\left(\frac{1}{E_M - E_N \pm k^0 - i\epsilon}\right) = \frac{i\pi}{\sqrt{\mathbf{k}^2 + \mu^2}}\left(\frac{1}{E_M \mp E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon}\right),$$

and likewise if μ is replaced with zero. The energy shift (14.3.27) now becomes

$$[\delta E_N]_{\text{low energy}} = -\frac{e^2}{2(2\pi)^3}\int d^3k\sum_M\left[\Gamma^\rho_{MN}(\mathbf{k})^*\Gamma_{\rho MN}(\mathbf{k})\left(\frac{1}{|\mathbf{k}|(E_M - E_N + |\mathbf{k}| - i\epsilon)} - \frac{1}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M - E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)}\right) - \bar{\Gamma}^\rho_{MN}(\mathbf{k})^*\bar{\Gamma}_{\rho MN}(\mathbf{k})\left(\frac{1}{|\mathbf{k}|(E_M + E_N + |\mathbf{k}| - i\epsilon)} - \frac{1}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M + E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)}\right)\right] - \delta m_e(\mu)\int d^3x\,\bar{u}_N(\mathbf{x})u_N(\mathbf{x}), \quad (14.3.28)$$

where

$$\Gamma^\rho_{MN}(\mathbf{k}) \equiv \int d^3y\,e^{-i\mathbf{k}\cdot\mathbf{y}}\bar{u}_M(\mathbf{y})\gamma^\rho u_N(\mathbf{y}), \quad (14.3.29)$$

$$\bar{\Gamma}^\rho_{MN}(\mathbf{k}) \equiv \int d^3y\,e^{-i\mathbf{k}\cdot\mathbf{y}}\bar{v}_M(\mathbf{y})\gamma^\rho u_N(\mathbf{y}). \quad (14.3.30)$$

(Of course, the 'sum' over M in Eq. (14.3.28) receives contributions only from electron states in the first term and positron states in the second term.) Eq. (14.3.28) could have been derived more directly from old-fashioned perturbation theory; the energy denominators

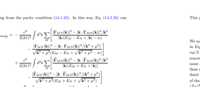


Figure 14.2. Old-fashioned perturbation theory diagrams for the low-energy part of the electron energy shift. Here solid lines are electrons; wavy lines are photons; the dashed line cuts through particle lines corresponding to the intermediate states appearing in the first two terms in (14.3.28).

$E_M - E_N + \omega$ and $E_M + E_N + \omega$ are the result of subtracting the energy E_N of the initial state from the energy of an intermediate state consisting of either an electron of energy E_M and a photon of energy ω, or else a positron of energy E_M, a photon of energy ω, and both the final and the initial electron. (See Figure 14.2.)

Before making any approximations to Eq. (14.3.28), it will be convenient to express the time components of the matrix elements Γ^ρ_{MN} and $\bar{\Gamma}^\rho_{MN}$ in terms of the corresponding space components, using relations[153] derived from the conservation of the electric current:

$$k_i\Gamma^i_{MN}(\mathbf{k}) = (E_N - E_M)\Gamma^0_{MN}(\mathbf{k}), \quad (14.3.31)$$

$$k_i\bar{\Gamma}^i_{MN}(\mathbf{k}) = (E_N + E_M)\bar{\Gamma}^0_{MN}(\mathbf{k}). \quad (14.3.32)$$

Furthermore, by using the completeness relation (14.1.8) it is straightforward to show that

$$\sum_M\left[|\Gamma^0_{MN}(\mathbf{k})|^2 + |\bar{\Gamma}^0_{MN}(\mathbf{k})|^2\right] = 1 \quad (14.3.33)$$

and

$$\sum_M\left[|\Gamma^0_{MN}(\mathbf{k})|^2(E_M - E_N) - |\bar{\Gamma}^0_{MN}(\mathbf{k})|^2(E_M + E_N)\right] = \sum_M\left[-\Gamma^{0*}_{MN}(\mathbf{k})\mathbf{k}\cdot\Gamma_{MN}(\mathbf{k}) - \bar{\Gamma}^{0*}_{MN}(\mathbf{k})\mathbf{k}\cdot\bar{\Gamma}_{MN}(\mathbf{k})\right] = -i\mathbf{k}\cdot\int d^3x\,\bar{u}_N(\mathbf{x})\gamma u_N(\mathbf{x}) = 0, \quad (14.3.34)$$

[153] To derive Eq. (14.3.31), note that

$$k_i\Gamma^i_{MN}(\mathbf{k}) = -i\int d^3x\,e^{-i\mathbf{k}\cdot\mathbf{x}}\nabla\cdot(\bar{u}_M(\mathbf{x})\gamma u_N(\mathbf{x})) = i\int d^3x\,e^{-i\mathbf{k}\cdot\mathbf{x}}\partial_0\left[\bar{u}_M(\mathbf{x})\gamma^0u_N(\mathbf{x})e^{-i(E_N - E_M)t}\right]_{t=0} = (E_N - E_M)\Gamma^0_{MN}(\mathbf{k}).$$

Eq. (14.3.32) is derived in the same way.

the last step following from the parity condition (14.1.23). In this way, Eq. (14.3.28) can be rewritten

$$[\delta E_N]_{\text{low energy}} = -\frac{e^2}{2(2\pi)^3}\int d^3k\sum_M\left[\frac{|\Gamma_{MN}(\mathbf{k})|^2 - |\mathbf{k}\cdot\Gamma_{MN}(\mathbf{k})|^2/\mathbf{k}^2}{|\mathbf{k}|(E_M - E_N + |\mathbf{k}| - i\epsilon)} - \frac{|\Gamma_{MN}(\mathbf{k})|^2 - |\mathbf{k}\cdot\Gamma_{MN}(\mathbf{k})|^2/(\mathbf{k}^2 + \mu^2)}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M - E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)}\right] + \frac{e^2}{2(2\pi)^3}\int d^3k\sum_M\left[\frac{|\bar{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k}\cdot\bar{\Gamma}_{MN}(\mathbf{k})|^2/\mathbf{k}^2}{|\mathbf{k}|(E_M + E_N + |\mathbf{k}|)} - \frac{|\bar{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k}\cdot\bar{\Gamma}_{MN}(\mathbf{k})|^2/(\mathbf{k}^2 + \mu^2)}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M + E_N + \sqrt{\mathbf{k}^2 + \mu^2})}\right] - \frac{e^2}{(2\pi)^3}\int d^3k\sum_M|\bar{\Gamma}^0_{MN}(\mathbf{k})|^2\left(\frac{1}{\mathbf{k}^2} - \frac{1}{\mathbf{k}^2 + \mu^2}\right) + \frac{1}{2}\alpha\mu - \delta m_e(\mu)\int d^3x\,\bar{u}_N(\mathbf{x})u_N(\mathbf{x}). \quad (14.3.35)$$

In the next-to-last term, we have used the elementary integral

$$\int d^3k\left(\frac{1}{\mathbf{k}^2} - \frac{1}{\mathbf{k}^2 + \mu^2}\right) = 2\mu\pi^2.$$

So far, this has been an exact rewriting of Eq. (14.3.28). We must now invoke several approximations. First, consider mass renormalization. We have already calculated $\delta m_e(\mu)$ to order α in Section 11.4; it is

$$\delta m_e(\mu) = \frac{2m_e\pi^2e^2}{(2\pi)^4}\int_0^1 dx\,(1 + x)\ln\left(\frac{m_e^2x^2 + \mu^2(1-x)}{m_e^2x^2}\right). \quad (14.3.36)$$

Although in Section 11.4 we regarded μ as a regulator mass, to be taken much larger than m_e, we can just as well use Eq. (14.3.36) to provide a value for $\delta m_e(\mu)$ in the case that interests us here, $\mu \ll m_e$. In this limit Eq. (14.3.36) gives

$$\delta m_e(\mu) \longrightarrow \frac{\alpha\mu}{2}\left[1 - \frac{3\mu}{2\pi m_e} + \cdots\right]. \quad (14.3.37)$$

We also recall that for $Z\alpha \ll 1$, $u_N(\mathbf{x})$ is given by Eq. (14.1.37) as

$$u_N(\mathbf{x}) = \frac{1}{\sqrt{2}}\begin{bmatrix} (1 - \sigma\cdot\mathbf{v}/2 + \cdots)f_N(\mathbf{x}) \\ (1 + \sigma\cdot\mathbf{v}/2 + \cdots)f_N(\mathbf{x}) \end{bmatrix}, \quad (14.3.38)$$

where dots indicate terms of higher order in $Z\alpha$; $\mathbf{v}$ is the non-relativistic velocity operator $-i\nabla/m_e$; and $f_N(\mathbf{x})$ is a two-component spinor solution of the non-relativistic Schrödinger equation, normalized according to Eq. (14.1.38) so that

$$\int|f_N(\mathbf{x})|^2 = 1 - \frac{1}{4}(\mathbf{v}^2)_{NN} + \cdots. \quad (14.3.39)$$

This gives the coefficient of $\delta m_e(\mu)$ in Eq. (14.3.35) as

$$\int d^3x\,\bar{u}_N(\mathbf{x})u_N(\mathbf{x}) = 1 - \frac{1}{2}(\mathbf{v}^2)_{NN} + \cdots. \quad (14.3.40)$$

We note immediately that the leading term in $-\delta m_e(\mu)\int d^3x\,\bar{u}_Nu_N$ cancels the term $\alpha\mu/2$ in Eq. (14.3.35). Indeed, we could have anticipated this cancellation, because the term $\alpha\mu/2$ in Eq. (14.3.35) survives in the limit $Z\alpha \to 0$, and the definition of $m_e(\mu)$ as the renormalized electron mass implies that there must be no energy shift in this limit. By the same argument we can anticipate that the term of order $\alpha\mu^2/m_e$ in $\delta m_e(\mu)$ (which is larger than of order $\alpha(Z\alpha)^4m_e$, and therefore cannot simply be neglected) cancels the second and third terms in Eq. (14.3.35), which are also of this order.[154] On the other hand, the product of the $\alpha\mu^2/m_e$ term in δm_e with the second term in the matrix element (14.3.40) is of order $(Z\alpha)^2\alpha\mu^2/m_e \ll \alpha(Z\alpha)^4m_e$, and may therefore be neglected. To order $\alpha(Z\alpha)^4m_e$, the only remaining effect of mass renormalization is to leave us with the product of the leading term in $\delta m_e(\mu)$ with the term of order $(Z\alpha)^2$ in $\int d^3x\,\bar{u}_Nu_N$:

$$-\left[\frac{e^2\mu}{8\pi}\right]\left[-\frac{1}{2}(\mathbf{v}^2)_{NN}\right] = \frac{e^2\mu}{16\pi}(\mathbf{v}^2)_{NN}.$$

(This is the effect of mass renormalization on the electron kinetic energy, mentioned in Section 1.3.) It turns out that this is just the negative of what the first term in Eq. (14.3.35) would be if we neglected the difference between energy levels. To see this, note that the

[154] The cancellation can be shown as follows. We anticipate that the second and third terms in Eq. (14.3.35) are small enough so that they can be evaluated using the extreme nonrelativistic approximation $\beta u_N(\mathbf{x}) = u_N(\mathbf{x})$ for the Dirac equation satisfied by $u_N(\mathbf{x})$. On the other hand, although the Coulomb force may be neglected in the positron wave functions $v_M(\mathbf{x})$, the sum over M in the third term receives important contributions from relativistic positrons, so we use the approximation $v_{\mathbf{p},\sigma}(\mathbf{x}) = (2\pi)^{-3/2}v(\mathbf{p},\sigma)e^{i\mathbf{p}\cdot\mathbf{x}}$, where $v(\mathbf{p},\sigma)$ is the positron spinor introduced in Section 5.5, normalized so that $\bar{v}(\mathbf{p},\sigma')v(\mathbf{p},\sigma) = \delta_{\sigma'\sigma}$. Then the sums over M in the second and third terms of Eq. (14.3.35) are approximately given by

$$\frac{1}{2}\left[\sum_M\bar{\Gamma}^i_{MN}(\mathbf{k})\bar{\Gamma}^j_{MN}(\mathbf{k}) + (i\leftrightarrow j)\right] = \delta_{ij}\left(\frac{\sqrt{\mathbf{k}^2 + m_e^2} + m_e}{2\sqrt{\mathbf{k}^2 + m_e^2}}\right),$$

$$\sum_M|\bar{\Gamma}^0_{MN}(\mathbf{k})|^2 \simeq \left(\frac{\sqrt{\mathbf{k}^2 + m_e^2} - m_e}{2\sqrt{\mathbf{k}^2 + m_e^2}}\right).$$

To leading order in μ/m_e, the second and third terms in Eq. (14.3.35) are then, respectively,

$$-\frac{e^2}{4m_e(2\pi)^3}\int d^3k\left[\frac{2}{k} - \frac{3 - k^2/(k^2 + \mu^2)}{\sqrt{k^2 + \mu^2}}\right]\left(\frac{\sqrt{k^2 + m_e^2} + m_e}{2\sqrt{k^2 + m_e^2}}\right) = \frac{\alpha\mu^2}{4\pi m_e},$$

$$-\frac{e^2}{2(2\pi)^3}\int d^3k\left(\frac{1}{k^2} - \frac{1}{k^2 + \mu^2}\right)\left(\frac{\sqrt{k^2 + m_e^2} - m_e}{2\sqrt{k^2 + m_e^2}}\right) = -\frac{\alpha\mu^2}{\pi m_e}.$$

(Eq. (14.3.32) rules out the possibility that a relativistic correction to the latter expression might not be suppressed by the factor k^2/m_e^2 that appears in this expression for $|\mathbf{k}|^2 \ll m_e^2$.) These two terms are cancelled by the term $+3\mu^2/4\pi m_e$ in $-\delta m_e(\mu)\int d^3x\,\bar{u}_N(\mathbf{x})u_N(\mathbf{x})$. Finally, we note that relativistic corrections to the above estimates for the sums over positron states would involve additional factors of $v^2/c^2 \simeq (Z\alpha)^2$, yielding contributions of order $\alpha(Z\alpha)^2\mu^2/m_e \ll \alpha(Z\alpha)^4m_e$, which justifies the non-relativistic approximations used here.

integral in this term is effectively cut off at $|\mathbf{k}| \sim p \ll Zm\alpha$, so we can evaluate the matrix element $\mathbf{T}_{nm}(\mathbf{k})$ in the limit $\mathbf{k} \to 0$. To lowest order in $Z\alpha$, Eq. (14.1.42) gives

$$\mathbf{T}(\mathbf{x})\Omega = (\mathbf{v}/\alpha)\,\Omega \tag{14.5.41}$$

and using the completeness of the solutions $|\psi_n\rangle$ of the non-relativistic Schrödinger equation, we then have

$$\sum_n \langle\psi_m|\mathbf{k}\mathbf{T}|\psi_n\rangle\langle\psi_n|\mathbf{k}\cdot\mathbf{v}\,|\psi_p\rangle = (\mathbf{v}^2/\alpha)\,\psi_{mp}, \tag{14.5.42}$$

so to this order

$$\frac{\alpha^3}{\pi m^2}\int d^3\mathbf{k}\sum_{n,p}\left(\frac{|\mathbf{T}_{mp}(\mathbf{k})|^2}{E_p}+\frac{\mathbf{k}\cdot\mathbf{T}_{mn}(\mathbf{k})\mathbf{T}^*(\mathbf{k})\cdot\mathbf{k}}{E_n E_p}\right)\frac{|\mathbf{T}_{np}(\mathbf{k})|^2+|\mathbf{k}\cdot\mathbf{T}_{np}(\mathbf{k})|^2/\mathbf{k}^2}{\mathbf{k}^2+p^2}$$
$$\approx \frac{\alpha^3}{\pi m^2}\sum_{n,p}\int d^3\mathbf{k}\left(\frac{1}{E_p}+\frac{1\cdot\mathbf{k}^2/3\mathbf{k}^2}{E_n E_p}\right)\frac{1+\mathbf{k}^2/3\mathbf{k}^2}{\mathbf{k}^2+p^2}\,|\psi_{np}|^2 = \frac{\alpha^3}{m^2}\sum_{n,p}|\psi_{np}|^2\,\frac{4m}{3}.$$

Thus after mass renormalization we are left with just the first term in Eq. (14.5.35), less the same with energy difference $E_1 - E_2$ dropped:

$$[\delta E(n)]_{\rm one\text{-}loop} = \frac{\alpha^3}{3\pi m}\int d^3\mathbf{k}\sum_p\left[|\psi_{np}-E_1|\left(\frac{\mathbf{T}_{np}(\mathbf{k})\mathbf{T}^*\cdot\mathbf{k}}{\mathbf{k}\cdot(E_p-E_1+\mathbf{k})}+\frac{\mathbf{k}\cdot\mathbf{T}_{np}(\mathbf{k})\mathbf{T}^*\mathbf{k}}{\mathbf{k}^2}\right)\frac{1}{\mathbf{k}^2+p^2}\right.$$
$$\left.-|\psi_{np}-E_2|\left(\frac{\mathbf{T}_{np}(\mathbf{k})\mathbf{T}^*\cdot\mathbf{k}}{\mathbf{k}\cdot(E_p-E_2+\mathbf{k})}+\frac{\mathbf{k}\cdot\mathbf{T}_{np}(\mathbf{k})\mathbf{T}^*\mathbf{k}}{\mathbf{k}^2}\right)\frac{1}{\mathbf{k}^2+p^2}\right]. \tag{14.5.43}$$

Again using Eq. (14.5.42), this is

$$[\delta E_n]_{\rm one\text{-}loop} = \frac{\alpha^3}{3\pi m}\sum_p|\psi_{np}-E_n|\psi_{np}|^2\int d^3\mathbf{k}\left(\frac{1}{\mathbf{k}\cdot(E_p-E_n+\mathbf{k})}-\frac{1}{\mathbf{k}^2+p^2}\right)$$
$$= \frac{\alpha^3}{3\pi m}\sum_p|\psi_{np}-E_n|\psi_{np}|^2\left[\ln\frac{\Lambda^2+2m(E_p-E_n)}{2m(E_p-E_n)}-\frac{\Lambda}{\sqrt{\Lambda^2+p^2}}+1\right]. \tag{14.5.44}$$

Even though typical values of the electron momentum are much larger than typical atomic energy differences, the integral is still dominated by the region where the two terms in the integrand would be infrared divergent if we did not keep the $E_p - E_n$ terms in the denominator. The integral in Eq. (14.5.43) may be evaluated in the limit $p \gg |E_p - E_n| \sim |Z\alpha|^2 m$, by dividing the range of integration of $\mathbf{k}$ into two segments, from zero to λ and from λ to infinity, with λ chosen so that $|E_p - E_n| \ll \lambda \ll p$ but otherwise arbitrary. In this way we find

$$\int_0^\lambda d^3\mathbf{k}\left[\frac{1}{\mathbf{k}\cdot(E_p-E_n+\mathbf{k})}-\frac{1}{p^2}\right] = \frac{1}{p^2}\frac{4\pi|\mathbf{p}|^3}{3}\frac{\lambda}{p}$$
$$+\int_0^\lambda d^3\mathbf{k}\left[\frac{1}{\mathbf{k}\cdot(E_p-E_n+\mathbf{k})}-\frac{1}{p^2}\right] = 4\pi\int_0^\lambda k^2\,dk\left[\frac{1}{k^2+2k(E_p-E_n)}-\frac{1}{p^2}\right]$$
$$= 2\left[\ln\left(\frac{\lambda^2+2\lambda(E_p-E_n)}{2|E_p-E_n|\lambda}\right)+\frac{\lambda}{p}+o(E_p-E_n)\right]. \tag{14.5.45}$$

The imaginary term here reflects the possibility of decay of the atom in state N to states M of lower energy. This term contributes to the decay rate, given by the imaginary part of the energy shift. We are interested here in the real part of the energy shift, and so will drop this imaginary term in what follows. Eq. (14.5.44) now gives

$$[\delta E(n)]_{\rm one\text{-}loop} = \frac{\alpha^3}{3\pi m}\sum_p|E_p-E_n|\psi_{np}|^2\left[\ln\left(\frac{m\alpha^2}{2|E_p-E_n|}\right)+\frac{3\pi}{4}\right]. \tag{14.5.46}$$

C Total Energy Shift

We need to make a correction between the sum in Eq. (14.5.35) and the matrix element in the Schrödinger equation (14.5.34). For this purpose, let's first see what value the sum in Eq. (14.5.45) would have if we could ignore the logarithm. We note that $|E_p - E_n|\psi_{np} = |p\cdot\partial|\psi_{np}$, so

$$\sum_p|E_p-E_n|\psi_{np}|^2 = \frac{1}{2}\sum_p\int d^3\mathbf{x}\,\psi^*_{np}(\mathbf{x})\,|\mathbf{p}\cdot\nabla|^2\psi_{np}(\mathbf{x})$$
$$= \frac{1}{2m}\int d^3\mathbf{x}\,\langle\mathbf{x}|\mathbf{p}\cdot\mathbf{p}|\mathbf{x}\rangle = \frac{1}{2m}\langle\mathbf{p}\cdot\mathbf{p}\rangle_{nn}.$$

The only term in the non-relativistic Hamiltonian H that does not commute with the momentum operator $\mathbf{p}$ is the potential term $-\alpha/|\mathbf{x}|$, so this gives

$$\sum_p|E_p-E_n|\psi_{np}|^2 = \frac{m\alpha^2}{2}\langle\mathbf{x}^{-2}\rangle_{nn}. \tag{14.5.48}$$

Inspection of Eqs. (14.5.45) and (14.5.33) now shows that the term proportional to $\ln p$ in the high-energy term is cancelled by the term proportional to $\ln p$ in the low-energy term:

$$\delta E_n \equiv [\delta E(n)]_{\rm one\text{-}loop} + [\delta E(n)]_{\rm low\text{-}energy}$$
$$= \frac{\alpha^3}{3\pi m}\sum_p|E_p-E_n|\psi_{np}|^2\left[\ln\left(\frac{m\alpha^2}{2|E_p-E_n|}\right)+\frac{3}{4}\right]+\frac{\alpha^3}{6\pi}\left\langle\frac{1}{|\mathbf{x}|}\right\rangle_{nn}$$
$$\frac{\alpha^3}{6\pi m}\langle\mathbf{p}\cdot\nabla|\mathbf{x}|\mathbf{p}\cdot\nabla\rangle_{nn}+\frac{\alpha^3}{6\pi}\langle\mathbf{x}^{-1}\rangle_{nn}. \tag{14.5.47}$$

So far, this has been for a general electrostatic field, $\mathcal{A}^0(\mathbf{x})$. Let us now specialize to a pure Coulomb field, with

$$\mathcal{A}^0(\mathbf{x}) = Z\alpha/|\mathbf{x}|. \tag{14.5.49}$$

In this case, Eq. (14.5.46) reads

$$\sum_p|E_p-E_n|\psi_{np}|^2 = \frac{Z^2\alpha^2}{2m}\langle\mathbf{x}^{-2}\rangle_{nn} + \frac{Z\alpha^2}{2m}\langle\delta(\mathbf{x})\rangle_{nn}. \tag{14.5.49}$$

This is non-vanishing only for $\ell = 0$. Also, the matrix element in the last term in Eq. (14.5.47) has the value

$$\langle\mathbf{x}\cdot\nabla|\mathbf{x}|\mathbf{x}\cdot\nabla\rangle = \langle\mathbf{x}|_{nn} = -2\alpha^{-1}\left\langle\frac{\mathbf{p}\cdot\mathbf{x}}{|\mathbf{x}|}\right\rangle_{nn}, \tag{14.5.50}$$

which is non-vanishing only for $\ell \neq 0$. It is therefore useful at this point to divide our consideration between the two cases, $\ell = 0$ and $\ell \neq 0$.

i. $\ell = 0$. It is convenient here to define a mean excitation energy ΔE_{nP}

$$\sum_p|\psi_{np}|^2(E_p-E_n)\ln|E_p-E_n| \equiv \ln\Delta E_{nP}\sum_p|\psi_{np}|^2(E_p-E_n)$$
$$= \frac{Z\alpha^2}{2m}\ln\Delta E_{nP}\langle\delta(\mathbf{x})\rangle_{nn}. \tag{14.5.51}$$

For n-wave hydrogenic states, the label N consists of a principal quantum number n and spin z-component m, and $\langle\delta(\mathbf{x})\rangle_{nn} = |\mathcal{R}(0)|^2\chi^\dagger_m\chi_m = 1/\pi n^3$ so

$$\langle\delta(\mathbf{x})\rangle_{nn} = \frac{1}{\pi}\left(\frac{Z\alpha m}{n}\right)^3. \tag{14.5.52}$$

Using Eqs. (14.5.51) and (14.5.52) in Eq. (14.5.47) gives the energy shift in these states as

$$[\delta E]_{\ell=0} = \frac{Z\alpha^2}{3\pi}\frac{(Z\alpha m)^3}{n^3}\left[\ln\left(\frac{m\alpha^2}{2\Delta E_{nP}}\right)+\frac{19}{30}\right]. \tag{14.5.53}$$

ii. $\ell \neq 0$. For non-vanishing orbital angular momentum the sum (14.5.48) vanishes, so the definition (14.5.51) is inappropriate. Instead, it is conventional here to define a mean excitation energy $\Delta E_{\rm kl}$ by

$$\sum_p|E_p-E_n|\psi_{np}|^2\ln|E_p-E_n| \equiv \frac{Z\alpha^2}{2m}\ln\left(\frac{m\alpha^2}{2\Delta E_{\rm kl}}\right)\langle\delta(\mathbf{x})\rangle_{nn}. \tag{14.5.54}$$

(Because Eq. (14.5.49) vanishes, it makes no difference what units are used to measure $E_p - E_n$ in Eq. (14.5.54).) Also, in a state of total angular momentum j and orbital angular momentum ℓ the scalar product $\mathbf{x}\cdot\mathbf{L}$ has the constant value $j(j+1) - \ell(\ell+1) - 2$, and for principal quantum number n the operator $\mathbf{x}\cdot\mathbf{L}$ has the eigenvalue zero, so

$$\int d^3\mathbf{x}\,|\psi_{n\ell}|^2\mathbf{x}^{-1} = \frac{(Z\alpha m)^2}{n(j(j+1)-\ell(\ell+1)-\frac{3}{4})}. \tag{14.5.55}$$

Putting this all together in Eq. (14.5.47), we have for $\ell \neq 0$

$$[\delta E]_{j\ell} = \frac{\alpha^3}{6\pi}\frac{(Z\alpha)^2 m}{n}\left[\frac{\ln\left(\frac{m\alpha^2}{2\Delta E_{nP}}\right)}{\pi}\right.$$
$$\left.+\frac{(Z\alpha)^2 m}{2\pi}\frac{j(j+1)-\ell(\ell+1)-\frac{3}{4}}{\ell(\ell+1)(2\ell+1)}\right]. \tag{14.5.56}$$

It only remains to use these results to give numerical value for the energy shifts. The mean excitation energies have must be calculated numerically, using non-relativistic hydrogen wave functions, they have the values [3]

$$\Delta E_{1s} = 19.7676981972(8)\,{\rm Ry},$$
$$\Delta E_{2s} = 19.4395448921(9)\,{\rm Ry},$$
$$\Delta E_{2p} = 0.0784293582(2)\,{\rm Ry}.$$

where $1\,{\rm Ry} = m\alpha^2/2 = 13.6057\,{\rm eV}$. Eq. (14.5.53) then gives

$$[\delta E]_{1s} = \frac{\alpha^3}{3\pi}\frac{(Z\alpha)^4 m}{n^3}\left[\ln\left(\frac{m\alpha^2}{2\Delta E_{nP}}\right)+\frac{19}{30}\right] = 5.3653\cdot 10^{-5}\,{\rm eV}$$
$$= 2\pi\hbar\cdot 8137.2\,{\rm MHz}, \tag{14.5.57}$$

$$[\delta E]_{2s} = \frac{\alpha^3}{3\pi}\frac{(Z\alpha)^4 m}{n^3}\left[\ln\left(\frac{m\alpha^2}{2\Delta E_{nP}}\right)+\frac{19}{30}\right] = 6.3963\cdot 10^{-6}\,{\rm eV}$$
$$= 2\pi\hbar\cdot 1559.55\,{\rm MHz}, \tag{14.5.58}$$

while Eq. (14.5.56) gives

$$[\delta E]_{2p,j=\frac{1}{2}} = \frac{\alpha^3}{6\pi}\frac{(Z\alpha)^4 m}{n}\left[\frac{1}{\pi}-\frac{1}{3}\right] = -5.2467\cdot 10^{-6}\,{\rm eV}$$
$$= 2\pi\hbar\cdot(-1246)\,{\rm MHz}. \tag{14.5.59}$$

The classic Lamb shift is the energy difference between the $2s$ and $2p_{1/2}$ states of the hydrogen atom, states that would be degenerate in the absence of radiative corrections. Our calculation has given

$$[\delta E]_{\rm Lamb} = [\delta E]_{2s} - [\delta E]_{2p,j=\frac{1}{2}} = 4.23152\cdot 10^{-5}\,{\rm eV} = 2\pi\hbar\cdot 10231.19\,{\rm MHz}.$$

This is numerically close (though analytically not identical) to the old result of Kroll and Lamb [3] and Pesnell and Weinhaul [7], which they obtained using the techniques of old-fashioned perturbation theory. Earlier in this section we made a crude estimate of 12becMHz by considering only the high-energy contribution to the $2s$ energy shift, with an infrared cutoff at the energy $\Delta E_{2P} = 10.534\,{\rm Ry}$ of the $2p$ state. The agreement to within a factor of unity among nearly fans the fact that the true value of the effective infrared cutoff $\Delta E_{\rm kl} = 19.64\,{\rm Ry}$ is considerably larger than we had guessed. On the other hand, as described in Section 1.2, in 1947 Hans Bethe [3] was able to make a rather good estimate of the Lamb shift, 1040MHz, by considering only the low-energy contribution to the $2s$ energy shift, with a cutoff energy that he estimated to be m. (Bethe made the first estimate of the ordinary energy, $\Delta E_{\rm kl} = 17.8\,{\rm Ry}$.)

The calculation of the Lamb shift described here has been improved by the inclusion of higher-order radiative corrections and nuclear size and recoil effects. At present the greatest uncertainty in the is a doubt about the correct value of the one charge radius r_p of the proton. Eides, Grotch, and Shelyuto [8] give the value 1057.845MHz for the Lamb shift of other 1057.873MHz, or 1057.813MHz, while another [10] gives either 1057.881MHz or 1057.803 MHz. Given the uncertainty in proton radius, the agreement is excellent with the present experimental value [11], 1057.842(9)MHz. The accuracy of this experimental value is limited chiefly by the ~ 100 MHz natural linewidth of the $2p$ state in hydrogen, so further experimental progress is to be hoped for.

In the last few years there has been an important improvement in measurements of the energy shift of the $1s$ ground state itself, by direct comparison of the frequency of the $1s$–$2s$

resonance with line times the frequencies of the $2s$ and $2s$–$4d$ two-photon resonances. The $1s$ Lamb shift determined in this way is about 8 MHz smaller than the theoretical value we've calculated more accurately than the classic Lamb shift. For a brief while it seemed that there was a discrepancy here between theory and experiment. Calculations [12][13] showed that for a proton radius $r_p = 0.862(11)\cdot 10^{-13}\,{\rm cm}$ or $0.862(11)\cdot 10^{-3}\,{\rm am}$, the inclusion of proton size and other corrections increases the theoretical $1s$ energy shift from the value we've calculated by 8192.2 MHz to 8192.2 MHz. This value for r_p is close to the result for a proton radius $r_p = 0.862(11)\cdot 10^{-13}\,{\rm cm}$, which is believed to be more reliable, was not quite in agreement with the measured value [12] of 8173.86(5)MHz. But a later calculation [14] that adopts this proton radius and includes terms of order $(Z\alpha)^6 m$ yields energy shifts for the $1s$, $2s$, and $4s$ states in agreement with experiment. So apparently quantum electrodynamics wins again.

Problems

1. Consider a charged scalar particle of mass $m \neq 0$, described by a field $\phi(x)$ whose only interaction is with an external (non-independent electromagnetic field $\mathcal{A}^\mu(x)$. Let $|N\rangle$ be a complete set of normalized one-boson or one-antiboson states with energies E_n, and define

$$u_N(\mathbf{x})e^{-iE_N t} = \langle 0|\phi(\mathbf{x},t)|N\rangle, \qquad v_N(\mathbf{x})e^{iE_N t} = \langle N|\phi(\mathbf{x},t)|0\rangle,$$

where $|0\rangle$ is the vacuum. Show that the u_N and v_N together form a complete set, and give formulas for the coefficients of u and v in the expansion of a general function $f(\mathbf{x})$.

2. Suppose we include radiative corrections in the theory of Problem 1. Let $iS'(x,y)$ be the sum of all diagrams with one incoming and one outgoing charged scalar line (excluding final scalar propagators) to order α. Derive a formula for the shift in the energy E_n of the nth boson states due to these radiative corrections, in terms of $u_N(\mathbf{x})$ and E_N or $v_N(\mathbf{x})$ and E_N.

3. Use the results of Section 14.3 to calculate the radiative decay rate of the $2p$ states of hydrogen.

4. Suppose that the electron has an interaction with a light scalar field ϕ, of the form $g\phi\bar{\psi}\psi$. Suppose that the scalar mass m_ϕ is in the range $|Z\alpha|^2 m_e \ll m_\phi \ll Zm_e$. Calculate the change in the energy of the $1s$ state of hydrogenic atoms due to this interaction.

5. Carry out the calculation of Problem 4 for $m_\phi = 0$.

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